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បក្សីឆ្មេសក៍ព្យាបាលគ្លីនិក
CLINICAL PRACTICE GUIDELINES

ផ្នែក
FOR

វេជ្ជសាស្ត្រទូទៅ
MEDICINE

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នាយកដ្ឋានសេវាសុខភាព
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Kingdom of Cambodia
Nation Religion King



CLINICAL PRACTICE GUIDELINES

FOR

MEDICINE

Part 3

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Ministry of Health

CLINICAL PRACTICE GUIDELINES FOR MEDICINE

Part
3

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CHAPTERVII NEUROLOGY

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21. Alzheimer disease (Demencia syndrome)
22. Movement disorders

VERTIGO AND BENIGN PAROXYSMAL POSITIONAL VERTIGO

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I. DEFINITION:

- Vertigo is an illusion of movement. Vertigo is a sensation of motion or spinning that is often described as dizziness. People with vertigo feel as though they are actually spinning or moving, or that the world is spinning around them.
- Vertigo is a symptom, not a disease.
- Positional vertigo is defined as a spinning sensation produced by changes in head position relative to gravity.
- BPPV (Benign Paroxysmal Positional Vertigo) is defined as a disorder of the inner ear characterized by repeated episodes of positional vertigo.

II. EPIDEMIOLOGY:

- Accounts for 5.6 million clinic visits in the US per year.
- Prevalence: 11 – 64 per 100,000 .
- 17 - 42% of patients with vertigo ultimately receive a diagnosis of BPPV.
- 20 – 30% of the population may have experienced vertigo or dizziness in their lifetime.
- BPPV have a life time prevalence of 2.4%.
- Vertigo affects both men and women but is about two to three times more common in women than men.

III. ETIOLOGY:

Causes of vertigo may include:

- Peripheral vestibular vertigo
- Benign paroxysmal positioning vertigo
- Vestibular Neuritis
- Meniere's disease
- Cholesteatoma
- Otosclerosis
- Perilymphatic fistula (PLF)
- Cervicogenic vertigo
- Central Vestibular Syndromes
- Brain tumors (Cerebellopontine angle schwannoma, Meningioma, Metastases)
- Acoustic neuroma
- Strokes
- Multiple sclerosis
- Basilar/Vestibular migraine
- Uncertain 10%

IV. PATHOPHYSIOLOGY:

- Asymmetry in the vestibular system may result from damage or dysfunction in the peripheral system (the vestibular labyrinth or vestibular nerve or a central disturbance in the brainstem or cerebellum).
- Schucknecht in 1969 was the first to provide a pathophysiological concept of BPPV and proposed the theory of “cupulolithiasis”. The cupula, which became heavy due to attached otolithic debris, could be deflected by position changes. Reorientation of the canal relative to gravity deflects the cupula, exciting or inhibiting the ampullary organ, thus evoking nystagmus.
- Hall proposed in 1979, the concept of “canalolithiasis”, which states that otolithic debris from the utricular macule migrates into the semicircular canal via the non-ampullary portion, causing vertigo and nystagmus by moving freely inside the semicircular canal and inducing endolymph flow during positional changes. This concept of canalolithiasis became the theoretical basis of the canalith repositioning maneuvers (CRMs) to treat BPPV.

V. DIAGNOSIS

1. Clinic

- Base on the history and physical examinations.
- Self-motion or motion of the environment?
- Spinning? Whirling? Swaying? Tilting?
- Non-vertigo-dizziness or Vertigo dizziness?
- Non-vertigo-dizziness (Presyncope, disequilibrium, non-specific dizziness)
- Nearly blacking out? Nearly fainting?
- Lightheadedness? Feeling of warmth? Diaphoresis? Nausea? Visual blurring?
- Time course (single, episode, recurrent, continuous)? Lasting (seconds, minutes, hours, days)? Prolonged? Severe episodes?
- Provoking factor: spontaneously? Change head position? Coughing? Sneezing? Lying down, rolling over in bed? Bending the neck back to look up?
- Worse by moving the head?

2. Investigation

- Deafness? Tinnitus?
- Headache? Nausea? Vomiting? Shortness of breath? Palpitation? Sweating?
- Past medical history: Head trauma? Migraine? DM? Hypertension? Smoking? Vascular disease? Family history of vertigo? Medications?
- Dizziness: non specific
- “Non-vertigo-dizziness” and “Vertigo-dizziness”
- Asking open-ended questions
- The spinning quality of vertiginous sensations is unreliable
- **Time course:** never continuous for more than a few weeks, lasting months is usually psychogenic; Vertigo can occur as single or recurrent episodes; last seconds, hours, days (best clues). Recurrent and lasting < 1min (BPPV); Single and lasting several min to hours (Vertiginous Migraine, TIA of the labyrinth or brainstem; Recurrent and last hours (Meniere disease); More prolonged, severe episodes (Vestibular neuritis, MS or infarction of the brainstem or cerebellum).
- **Provoking factor:** spontaneously, change head position, change middle ear pressure (eg. Coughing, sneezing, or Valsalva maneuvers) (Perilymphatic fistula), lying down, rolling over in bed, and bending the neck back to look up; Head trauma; Barotrauma; Middle ear surgery; Recent hyperextension injury to the neck (Vertebral artery dissection); Recent viral symptom (Vestibular neuritis).
- **Aggravating factors:** worse by moving the head.

- **Nausea and vomiting.**
- **Nystagmus:** dizziness is vertigo. Some are only seen after a provocative maneuver (eg. Dix-Hallpike)
- **The Dix-Hallpike maneuver is the gold standard test for the diagnosis of posterior canal BPPV.**
- **Postural instability.**
- **Hearing loss and tinnitus:** peripheral cause.
- **Brain stem signs:** ataxic gait, vomiting, headache, double vision, visual loss, slurred speech, numbness of the face or body, weakness, clumsiness, or incoordination, cranial nerve abnormalities, Horner syndrome, motor or sensory changes, dysmetria, or abnormal reflexes.
- **Headache, photophobia, and sonophobia:** Vestibular Migraine.
- **Shortness of breath, palpitations, and sweating** (panic attack).
- **PMH:** Migraine, Stroke risk factor (Hypertension, T2DM, smoking, vascular disease), family history of vertigo (Hereditary channelopathy), medications (aminoglycosides, phenytoin).

VI. DIFFERENTIAL DIAGNOSIS

- Epileptic seizure
- Cardiac syncope (cardiac dysrhythmias, coronary artery disease, chronic heart failure)
- Psychiatric disorder (Major depression, General anxiety disorder, panic disorder, alcohol dependence, personality disorder)
- Orthostatic hypotension, anemia
- Disequilibrium (peripheral neuropathy, musculoskeletal disorder, cervical spondylosis, Parkinson's disease)
- Hypoglycemia
- Antidepressants and anticholinergics side effect or abrupt drug withdrawal
- Meningoencephalitis

VII. MANAGEMENT

1. Objectives

- Dependent upon the etiology
- Medications treatment (Pharmacologic treatment)
- The Epley maneuver (Nonpharmacologic treatments)
- Surgery (Stroke and brain tumors)

2. Critical sign

- “Red flag” symptoms: persistent, worsening vertigo or disequilibrium, atypical “non-peripheral” vertigo such as vertigo movement; severe headache; diplopia; cranial nerve palsies; dysarthria; ataxia, or other cerebellar signs; and papilledema.

3. Mortality risk factor

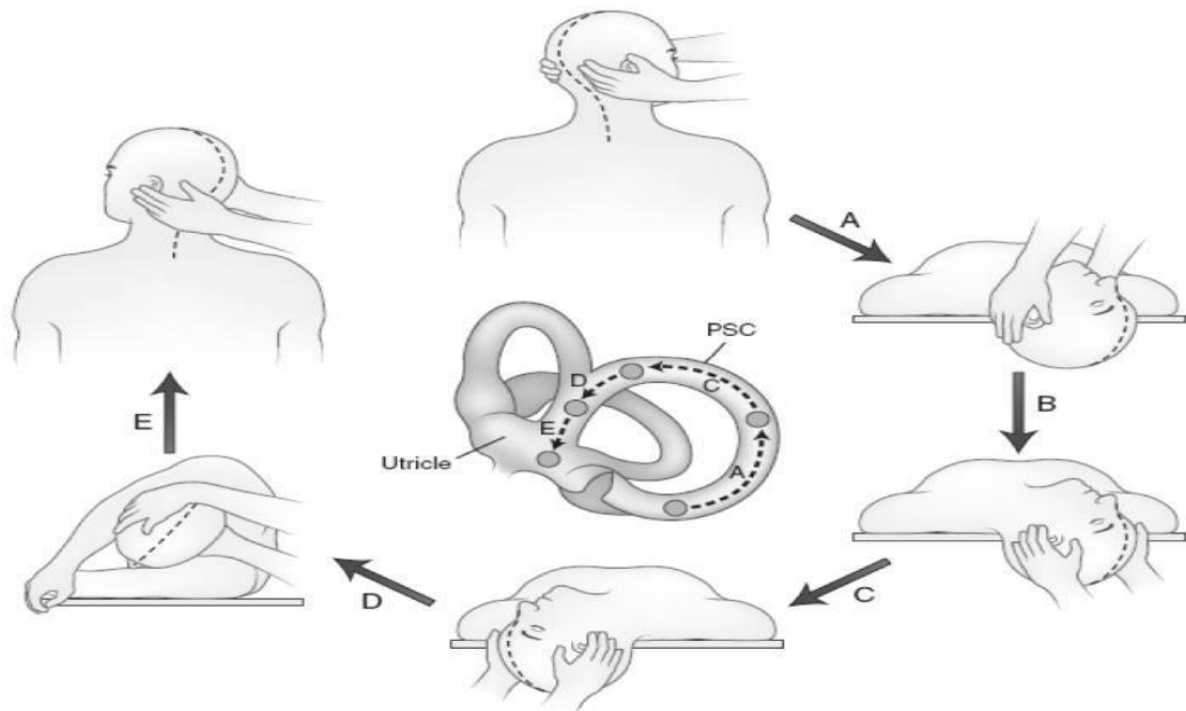
- Central causes of vertigo include neoplasm, infection, and cerebrovascular accidents.
- Central etiologies of vertigo should always be considered in the differential. Ischemic or hemorrhagic strokes, particularly involving the cerebellum or vertebrobasilar system, are life-threatening and must be ruled out by history, physical and other diagnostic tests if warranted.
- Other more serious central causes include tumors, particularly those arising from the cerebellopontine angle.

4. Outpatient management

- History taking and physical examination include neurological examinations
- Investigation: blood test and heart tests
- Brain imaging needed only in some cases and the modality of choice is MRI and MRA.

5. First-line treatment

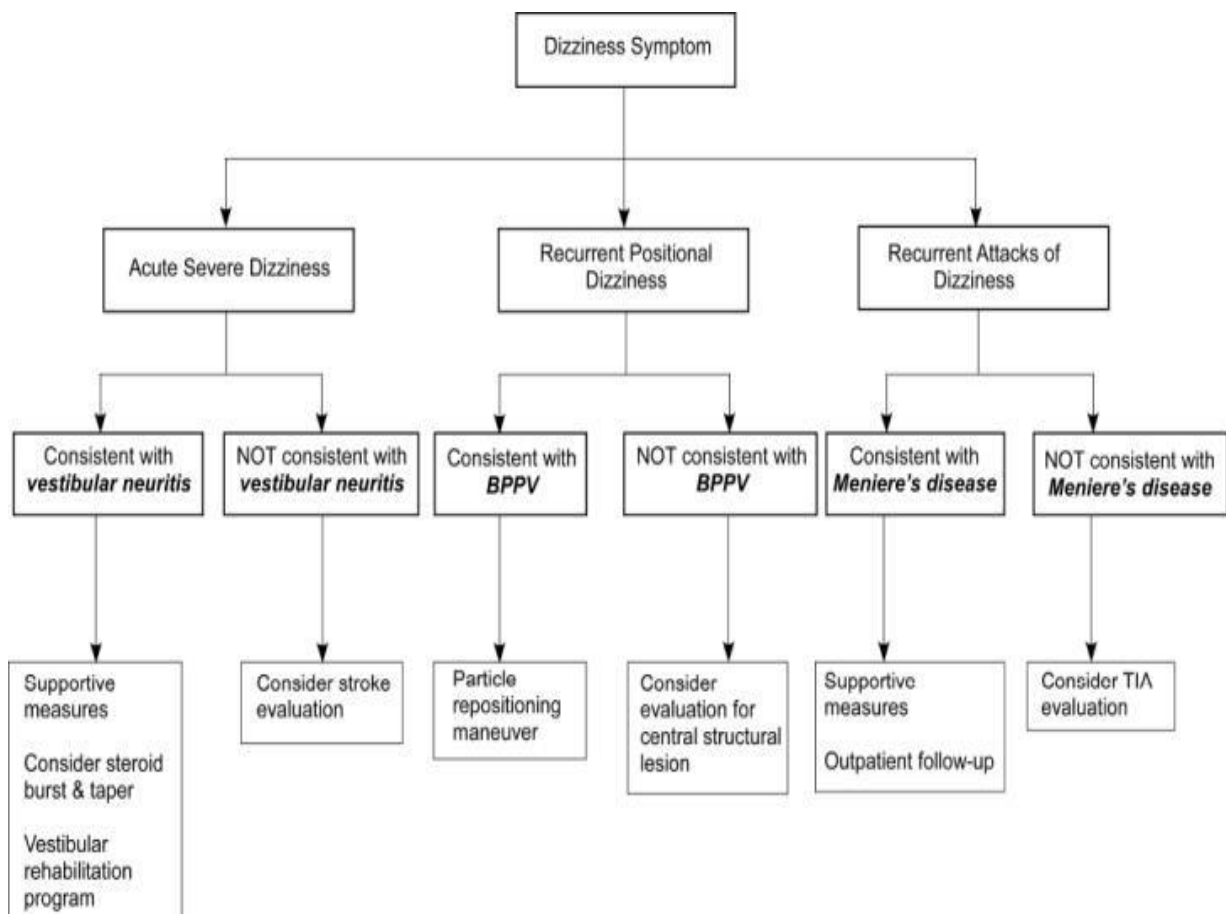
- Treatment of vertigo is dependent upon the etiology.
- Physical therapy with vestibular rehabilitation for vestibular dysfunction.
- Corticosteroids are recommended in the acute vestibular neuritis. Antiviral medication should not be included in the treatment of vestibular neuronitis.
- Ménière disease: lifestyle adjustments (avoid high salt diet, caffeine, and alcohol), medications (diuretics), and vestibular rehabilitations. Hydrochlorothiazide 50mg/tab once a day with potassium supplementation and monitoring of serum K level.
- BPPV (Benign Paroxysmal Positional Vertigo): focused on head rotation maneuvers (Epley maneuvers) is that the patient at home can perform it.
- To perform a modified Epley maneuver (as shown in the image), instruct the patient to position themselves upright on a bed with their head turned 45 degrees to the left and a pillow behind them. The pillow should be positioned so that when supine, the pillow is directly under their shoulders. Once the patient is in position, they should lie back quickly onto the pillow, so the head is reclined onto the bed. They should hold this position for 30 seconds. Without raising their head, they should then turn their head 90 degrees to the opposite side (right) and hold this position for another 30 seconds. After 30 seconds, they should turn their body and head another 90 degrees to the right and wait for another 30 seconds. Finally, they should sit up on the right side of the bed. This maneuver can be repeated starting on the opposite side and should be performed at least three times a day until the patient has no further episodes of positional vertigo for 24 hours.
- Postural restriction after reposition maneuvers among patients with BPPV is not recommended.

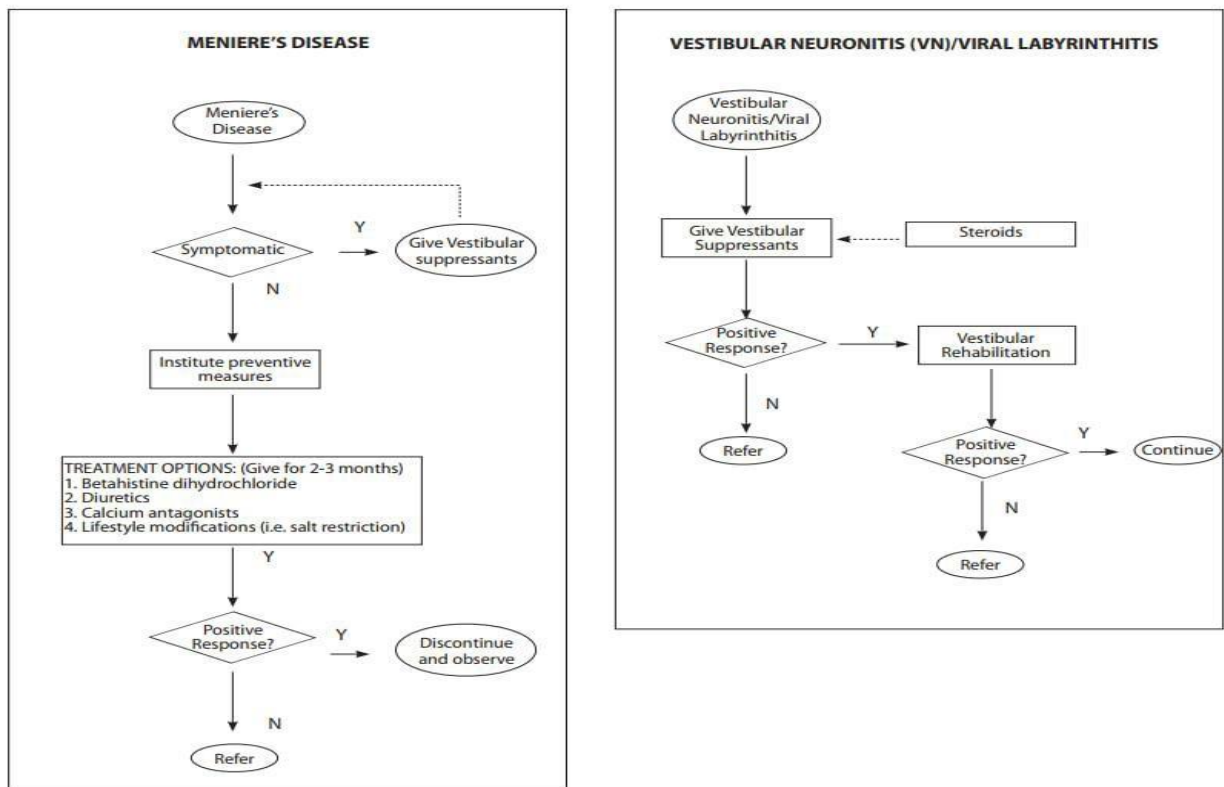
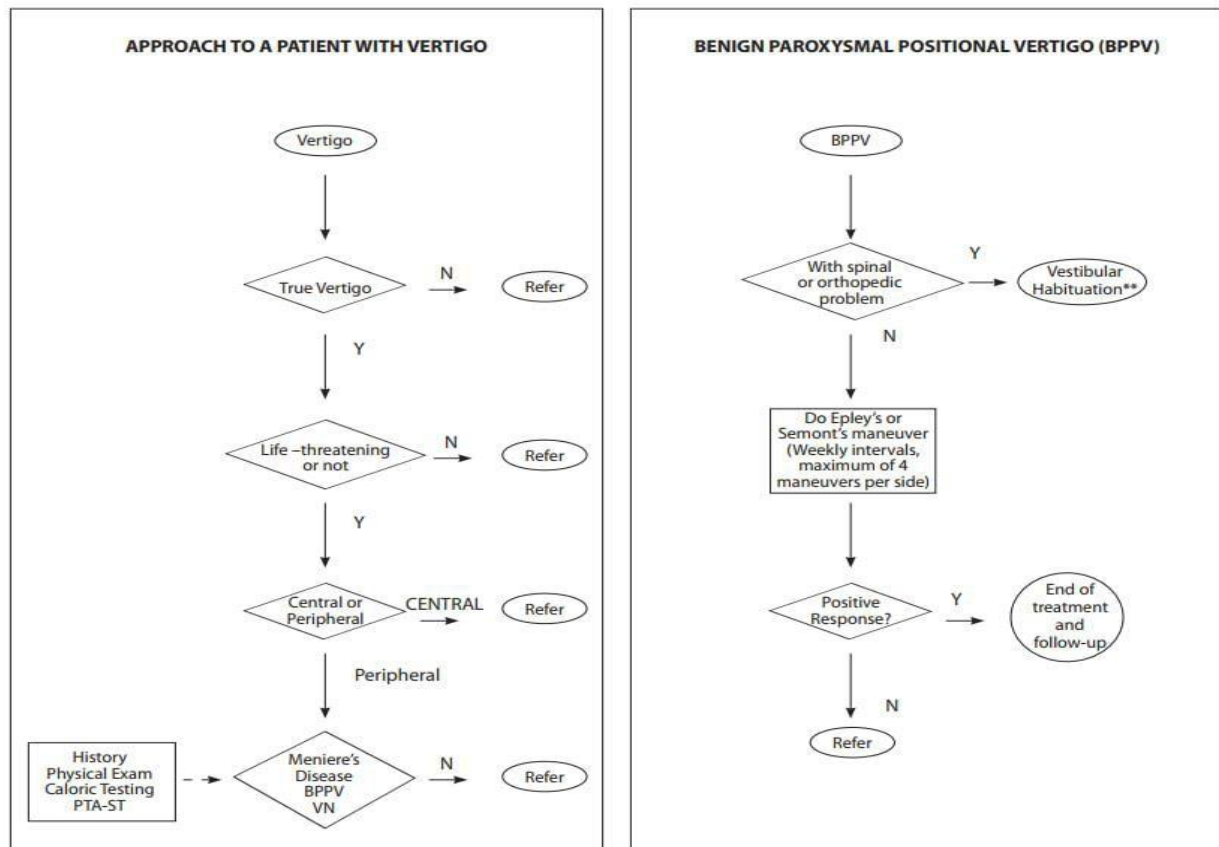


6. Treatment adjustment

- Vestibular suppressants (e.g. benzodiazepines) and antivertigo drugs (e.g. H3 receptor antagonists) may be given for symptomatic relief of patients with BPPV.
- Diazepam - widely used as a vestibular suppressant because of its additional tranquilizing effect.
- Dimenhydrinate
- Diphenhydramine
- Meclizine
- Anti-vertiginous medications, anti-emetics, sedatives, reported to be beneficial in reducing the severity of vertigo and vegetative symptoms and in improving tolerance of Meniere's symptoms.
- Calcium antagonists, e.g. nimodipine and cinnarizine: nimodipine and cinnarizine, were found to be equally effective in the symptomatic treatment of vestibular vertigo. The dosages were at 30 mg nimodipine tablet taken TID and 150 mg cinnarizine tablet taken OD for 12 weeks. Both had similar safety profiles.
- Betahistine dihydrochloride, at 24 mg BID or 16 mg tablet TID for 2-3 months, is recommended for patients diagnosed with Meniere's disease.
- Oral flunarizine at 5mg tablet daily in a single dose was taken together with physical exercises. Flunarizine appears to be useful in the treatment of vertigo caused by vestibular neuritis.

7. Algorithm





VIII. SPECIAL CASES

- Drug-induced 'dizziness' and vestibular symptoms: drug-induced orthostatic hypotension, sedation, or cerebellar intoxication. These medicines are alpha blockers, chlorpromazine, tricyclic antidepressants, antiparkinsonian dopaminergic drugs, benzodiazepine, antipsychotic, antihistamines, alcohol, carbamazepine, valproate,

antiepileptic used for neuropathic pain (pregabalin, gabapentin), lithium, aminoglycosides, chemotherapy agents.

- The 'gold standard' of diagnosis of orthostatic hypotension is to measure the drop in systolic blood pressure of more than 20 mmHg at a time of subjective 'dizziness'.
- Cardiovascular causes of dizziness are mediated by episodes of hypotension causing presyncopal symptoms.
- Following recovery from the acute phase of TBI (e.g. recovery from coma and/or other life-threatening organ injuries) the commonest chronic symptoms following TBI is headache and dizziness.
- Addisonian crises may present with postural dizziness (e.g. due to cardiovascular insufficiency). Patients with chronic hypo-adrenalism do not typically complain of postural dizziness unless there are additional aggravating factors (since cardiovascular effects of hypoadrenalism are only seen in acute crises).
- Patients with renal insufficiency are more at risk of drug-related dizziness due to problems of renal clearance.
- Patients on renal dialysis experience significant fluid shifts that may lead to postural hypotension.

IX. EVOLUTION AND PROGNOSIS

- Benign paroxysmal positional vertigo recurrence rates are 50% at 5 years. There is the persistence of dizziness related to anxiety in almost a third of patients 1 year after vestibular neuritis.
- As with vertigo, loss of hearing is highest in the early years of the disease and stabilizes in later years. Usually, there is no recovery from hearing loss.
- The acute vertigo of labyrinthitis should resolve within a couple of days.

X. COMPLICATIONS

- Differentiate vertigo from other causes of dizziness or imbalance and distinguish central from peripheral cause of vertigo.

XI. PREVENTION AND RECOMMENDATION:

- Patients with vertigo should be encouraged to attempt mobilizing as soon as possible, as this is believed to help with vestibular compensation and prognosis.
- Patients who receive a prolonged course of benzodiazepines and/or antihistamines to treat their vertigo appear to have delayed vestibular recovery.
- In the cases of BPPV, it is best to explain that it is a non-life-threatening condition. Its favorable prognosis reassures patients that it is not a severe condition.
- Patients should also be warned that recurrences are very common even after successful management with repositioning maneuvers so that further treatment may be necessary.

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MIGRAINE, MIGRAINE ATTACK

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I. DEFINITION

Migraine is a common, chronic, disorder that is typically characterized by recurrent disabling attacks of headache and accompanying symptoms, including aura.

Migraines can be classified into subtypes according to the headache classification committee of the International Headache Society. These subtypes are:

- **Migraine without aura** is a recurrent headache attack of 4 to 72 hours; typically, unilateral in location, pulsating in quality, moderate to severe in intensity, aggravated by physical activity, and associated with nausea and light and sound sensitivity (photophobia and phonophobia).
- **Migraine with aura** has recurrent fully reversible attacks, lasting minutes, typically one or more of these unilateral symptoms: visual, sensory, speech and language, motor, brainstem, and retinal, usually followed by headache and migraine symptoms.
- **Chronic migraine** is a headache that occurs on 15 or more days in a month for more than three months and has migraine features on at least eight or more days in a month.

II. EPIDEMIOLOGY

Migraine is highly prevalent, affecting 12% of the population, attacking up to 17% of women and 6% of men yearly. Among children, it tends to happen more in girls than boys. The adjusted prevalence of migraine is highest in North America, followed by South America, Central America, Europe, Asia, and Africa. It is ranked as the second leading cause of disability worldwide. Migraine tends to run in families. There is a reported risk of 40% if one parent has a history of migraine, which increases to 75% when both parents have a migraine history.

It is consistently the fourth or fifth most common reason for emergency visits accounting for an annual 3% of all emergency visits. Its prevalence increases in puberty but continues to increase until 35 to 39 years of age, decreasing later in life, especially after menopause. Moreover, it is considered the second major cause of disability after back pain with respect to years of life lived with disability.

III. ETIOLOGY

-Genetics and Inheritance

-Migraine has a strong genetic component. The risk of migraines in ill relatives is three times greater than that of relatives of non-ill subjects, but no inheritance pattern was identified.

-Identifying these genes in an individual with migraines could predict the targeted prophylactic treatment.

-Familial Hemiplegic Migraine

-Hemiplegic migraine can occur in families or sporadically (one individual, as the first member of the family to have a hemiplegic migraine). Channelopathies cause the primary three types:

- Type 1 is caused by mutations in the CACNA1A gene (calcium voltage-gated channel alpha 1A subunit) on chromosome 19p13.[7]
- Type 2 is caused by mutations in the ATP1A2 gene (ATPase, Na⁺/K⁺ transporting alpha two subunit) on chromosome 1q23.[8]
- Type 3 is caused by mutations in the SCN1A gene (sodium voltage-gated channel Type 1 alpha subunit).

-MELAS

-It is a syndrome of mitochondrial encephalomyopathy, lactic acidosis, and stroke-like episodes, a multisystemic disorder by maternal inheritance that can present recurrent migraine headaches.

-CADASIL

-Cerebral autosomal dominant arteriopathy with subcortical infarcts and leukoencephalopathy) angiopathy by autosomal dominant inheritance, caused by mutations in the NOTCH3 gene (notch receptor 3) on chromosome 19 that can present migraine with aura (prodrome in 80%) in nearly 50% of carriers.

-RVCL

-Retinal vasculopathy with cerebral leukodystrophy is angiopathy by C-terminal frame-shift mutations in TREX1 (three prime repair exonuclease 1) presents almost 60% of the cases.

-HIHRATL

-Hereditary infantile hemiparesis, retinal arteriolar tortuosity, and leukoencephalopathy

-HERNS

-Hereditary endotheliopathy with retinopathy, nephropathy, and stroke

-Precipitating and exacerbating factors

- Emotional stress
- Hormones in females
- Not eating
- Weather
- Sleep disturbances
- Odors
- Neck pain
- Lights
- Alcohol
- Smoke
- Sleeping late
- Heat

- Food
- Exercise
- Sexual activity

IV. PHYSIOPATHOLOGY

Although not fully understood, pathogenesis involves multiple components of both the peripheral and central nervous systems.

According to the older vascular theory of migraine, headache is produced by vasodilation and aura by vasoconstriction, but this theory is no longer viable. Nowadays, the suggestions pose that multiple primary neuronal impairments lead to a series of intracranial and extracranial changes that cause migraines.

There is activation of trigeminal afferents by neuronal pannexin-1 mega channel opening and subsequent activation of caspase-1, followed by the release of proinflammatory mediators, activation of NF- κ B (nuclear factor kappa-B), and spreading of this inflammatory signal to trigeminal nerve fibers around vessels of the pia mater. This causes a series of cortical, meningeal, and brainstem events, provoking inflammation in the pain-sensitive meninges and resulting in headaches through central and peripheral mechanisms. This pathway can, therefore, explain the cortical depression (which establishes the aura) and the latter prolonged activation of trigeminal nociception (which leads to headache).

The cortical spreading depression of Leão, a propagating wave of neuronal and glial depolarization that initiates a cascade, is hypothesized to cause the aura, activate trigeminal afferents, and alter the hematoencephalic barrier permeability by activating brain matrix metalloproteinases. In migraine without aura, the suggestions are that cortical depression may occur in areas where depolarization is not consciously perceived, such as the cerebellum.

The anterior structures are most innervated by the ophthalmic division of the trigeminal nerve, which could explain the pain in the anterior region of the head. There is a convergence of fibers from the upper cervical roots, which originate from trigeminal nerve neurons along with the trigeminal ganglion and the trigeminal nerve at the trigeminal nucleus caudalis, which can explain the anterior to the posterior distribution of pain, from where the fibers ascend to the thalamus and the sensory cortex.

Based on vasodilation, edema, and plasma protein extravasation, neurogenic inflammation results from nociceptor activation, in this case, the trigeminal system. It is associated with the release of substance P, calcitonin gene-related peptide, and neurokinin a, all vasoactive neuropeptides liberated by trigeminal ganglion stimulation. Elevated levels of these neuropeptides have been found in the spinal fluid of chronic migraine patients. Neurogenic

inflammation can lead to sensitization, which is the process where neurons tend to become more responsive to stimulation. This can explain some clinical symptoms of the pain and the conversion from episodic migraine to chronic ones.

Following neuropeptides are believed to play a role in pathogenesis:

Serotonin, released from the brainstem serotonergic nuclei, may play a role in migraine; however, the exact role of its mechanisms remains a matter of controversy. Most likely, serotonin levels are low between attacks because it may cause a deficiency in the serotonin pain inhibition system, therefore helping the activation of the trigeminal system. It could mediate by acting directly over the cranial vessels, in central pain control pathways, or by cortical projections of brainstem serotonergic nuclei.

Calcitonin gene-related peptide (CGRP) is abundant in trigeminal ganglion neurons. It is released from the peripheral and central nerve terminals and secreted within the trigeminal ganglion. When released from the peripheral terminals, it initiates an increased synthesis of nitric oxide and later sensitization of trigeminal nerves. It is a strong vasodilator of cerebral and dura mater vessels, therefore, a component of neurogenic inflammation, and it also mediates trigeminal pain transmission from vessels to the central nervous system.

Pituitary adenylate cyclase-activating polypeptide (PACAP) may also play an important role in mediating migraine attacks as its concentration is elevated during the attacks, and its infusion may trigger migraine in susceptible patients.

V.DIAGNOSIS

1. Clinic

Four Phases of Migraine Attacks

-Prodrome:

- Around 77% of patients suffer prodromic symptoms for up to 24 to 48 hours before headache onset. It is more common in females than males.

- Frequent symptoms are yawning, mood change, lethargy, neck symptoms, light sensitivity, restlessness, difficulties in focusing vision, feeling cold, craving, sound sensitivity, sweating, excess energy, thirst, and edema.

-Aura:

- It can precede the headache, or it can present simultaneously.

- They are typically gradual, less than 60 minutes in duration, more often visual, and have positive and negative symptoms.

- Positive symptoms are caused by active release from central nervous system neurons (bright lines or shapes, tinnitus, noises, paresthesias, allodynia, or rhythmic movements).

-Negative symptoms indicate a lack or loss of function (reduction or loss of vision, hearing, sensation, or motion).

-Headache:

-Often unilateral, generally with a pulsatile or throbbing feature and increasing intensity within the first hours.

-The intensity can correlate to nausea, vomiting, photophobia, phonophobia, rhinorrhea, lachrymation, allodynia, and osmophobia.

-It can take place over hours to days.

-Patients may seek relief in dark places, as the pain usually resolves in sleep.

-Postdrome:

Common symptoms can be exhaustion, dizziness, difficulty concentrating, and euphoria.

2. Criteria to diagnosis

The diagnosis of migraine is based on the history. According to diagnostic criteria established by the International Headache Society

Migraine without aura

- A. At least five attacks¹ fulfilling criteria B-D
- B. Headache attacks lasting 4-72 h (untreated or unsuccessfully treated)
- C. Headache has at least two of the following four characteristics:
 - unilateral location
 - pulsating quality
 - moderate or severe pain intensity
 - aggravation by or causing avoidance of routine physical activity (eg, walking or climbing stairs)
- D. During headache at least one of the following:
 - nausea and/or vomiting
 - photophobia and phonophobia
- E. Not better accounted for by another ICHD-3 diagnosis.

Migraine with aura

- A. At least two attacks fulfilling criteria B and C
- B. One or more of the following fully reversible aura symptoms:
 - Visual
 - Sensory
 - Speech and/or language
 - Motor
 - Brainstem
 - Retinal

- C. At least three of the following six characteristics:
- At least one aura symptom spreads gradually over ≥ 5 minutes
 - Two or more aura symptoms occur in succession
 - Each individual aura symptom lasts 5-60 minutes¹
 - At least one aura symptom is unilateral²
 - At least one aura symptom is positive³
 - The aura is accompanied, or followed within 60 minutes, by headache
- D. Not better accounted for by another ICHD-3 diagnosis.

-Chronic Migraine

1. Headache (tension-type-like or migraine-like) on 15 or more days per month for more than three months and fulfilling criteria B and C
2. It occurs in a patient who has had at least five attacks fulfilling the following criteria for migraine without aura (B1) or migraine with aura (B2)

3. Investigation

-Neuroimaging (computed tomographic scan, magnetic resonance imaging, magnetic resonance angiography, or magnetic resonance venography) is indicated in the following cases:

- Acute severe headache, especially if it is the first or worst episode (discard subarachnoid hemorrhage)
- Abnormal neurologic examination, especially if there are unexplained symptoms or signs (confusion, stiff neck, papilledema, epilepsy)
- Non-typical characteristics
- Changes in the patient's typical features or patterns
- New episodes in older (> 50 years of age) or immunosuppressed patients
- Systemic or meningeal signs or symptoms (fever, weight loss, fatigue)
- New neurological symptoms or signs on examination
- Headache not responding to treatment
- Headache lasting > 72 hours
- A significant change in the frequency, pattern, or severity of headaches
- Severe headache pain ("worst headache of life") or headache causing awakening from sleep
- New-onset headache in patients with HIV infection or cancer
- Associated symptoms or signs suggestive of meningitis or stroke

The commonly used acronym "SNOOP" can be used to aid in the determination of neuroimaging indications:

- "S" for systemic signs or symptoms and secondary risk factors
- "N" for neurologic signs or symptoms
- "O" for onset
- "O" for older
- "P" for position-dependent intensity changes, the prior pattern changes, papilledema, and precipitated by Valsalva maneuvers.

- Cerebrospinal fluid analysis and electroencephalogram are not typically performed unless seizure activity of infectious etiology has to be excluded.

VI. DIFFERENTIAL DIAGNOSIS

- Tension-type headache
- Cluster headache
- Cerebral aneurysms
- Chronic paroxysmal hemicranias
- Dissection syndromes
- Encephalitis
- Subarachnoid/intracranial hemorrhage
- Meningitis
- Temporal/giant cell arteritis

VII. MANAGEMENT

Treatment options are based on the onset scenarios: acute or chronic.

Acute or Abortive Treatment

Acute treatment aims to stop the progression of a headache. It has to be treated quickly and with a large single dose. Oral agents can be ineffective in patients with migraine-induced gastric stasis. Therefore, parenteral medication could be the rule for some patients, especially the ones with nausea or vomiting. Therapy consists of stratified options:

- **NSAIDs** (nonsteroidal anti-inflammatory drugs): Ibuprofen (400 to 600 mg), naproxen (275 to 825 mg), diclofenac (65 mg), aspirin (900 to 1000 mg), or acetaminophen (1000 mg). Usually, in mild to moderate attacks without nausea or vomiting. If one NSAID is ineffective, another class of drug should be used.

- **Triptans** (the first line in patients with allodynia): Sumatriptan (administered as a subcutaneous injection of 6 mg, a nasal spray of 20 to 40 mg over 24 hours, a nasal

powder of 10 to 30 mg over 24 hours, or orally 50 to 100 mg once), zolmitriptan 10 mg per 24 hours (nasal 2.5 to 5 mg as a single dose and oral 2.5 mg as a single dose), eletriptan, rizatriptan, almotriptan. They might be used with or without naproxen for moderate to severe attacks. Unlike NSAIDs, patients who do not respond well to one triptan may respond to another; therefore, therapy may be individualized.

- To avoid medication overuse, triptans should be limited to less than ten days of use within a month.

- Because of the activation of the 5-HT (1B) and 5-HT (1D) receptors on coronary arteries and cerebral vessels, there are recommendations against its use in patients with ischemic stroke, ischemic heart disease, poor-controlled hypertension, and angina, and pregnancy, hemiplegic or basilar migraine. In these patients with cardiovascular risks, the best-suited medication is a selective serotonin 1F receptor agonist that does not produce vasoconstriction; lasmiditan.

- It is recommended to monitor therapy if the patient takes selective serotonin reuptake inhibitors or selective serotonin-noradrenaline reuptake inhibitors because of the risk of serotonin syndrome.

- The combination of nonsteroidal anti-inflammatory drugs (NSAID) with triptan appears more effective than using either drug class alone. In a single tablet, sumatriptan succinate 85 mg and naproxen sodium 500 mg may be prescribed as initial therapy.

Antiemetics: Metoclopramide, chlorpromazine, or prochlorperazine. They are generally used as adjunctive therapy with NSAIDs or triptans to decrease nausea and vomiting, especially in the emergency department. Diphenhydramine can also be added to prevent dystonic reactions (mostly metoclopramide).

Calcitonin-gene-related peptide antagonists: Rimegepant (75 mg as a single dose) or ubrogepant could be considered in patients who don't respond to conventional treatment or those with coronary artery disease.

Selective serotonin 1F receptor agonist: US Food and Drug Administration (FDA) approved lasmiditan oral tablets in October 2019 for the acute treatment of migraine in adults. The initial dose of lasmiditan is 50 or 100 mg, particularly effective for patients who cannot use triptans due to cardiovascular risks. The dose may be increased to 100 or 200 mg per requirement, but no more than one dose should be taken over 24 hours. However, one significant side effect is dizziness. So patients should not engage in potentially hazardous activities or drive a motor vehicle for at least eight hours after each dose of lasmiditan.

Ergots: Ergotamine and dihydroergotamine (intravenous (IV), intramuscular (IM), subcutaneous, and intranasal use) are recommended for acute attacks as a parenteral administration and effective as bridge therapy for medication overuse headache and status migrainosus. Ergotamine has not demonstrated particular effectiveness yet and can present significant side effects, including cerebrovascular, cardiovascular, and peripheral ischemic complications.

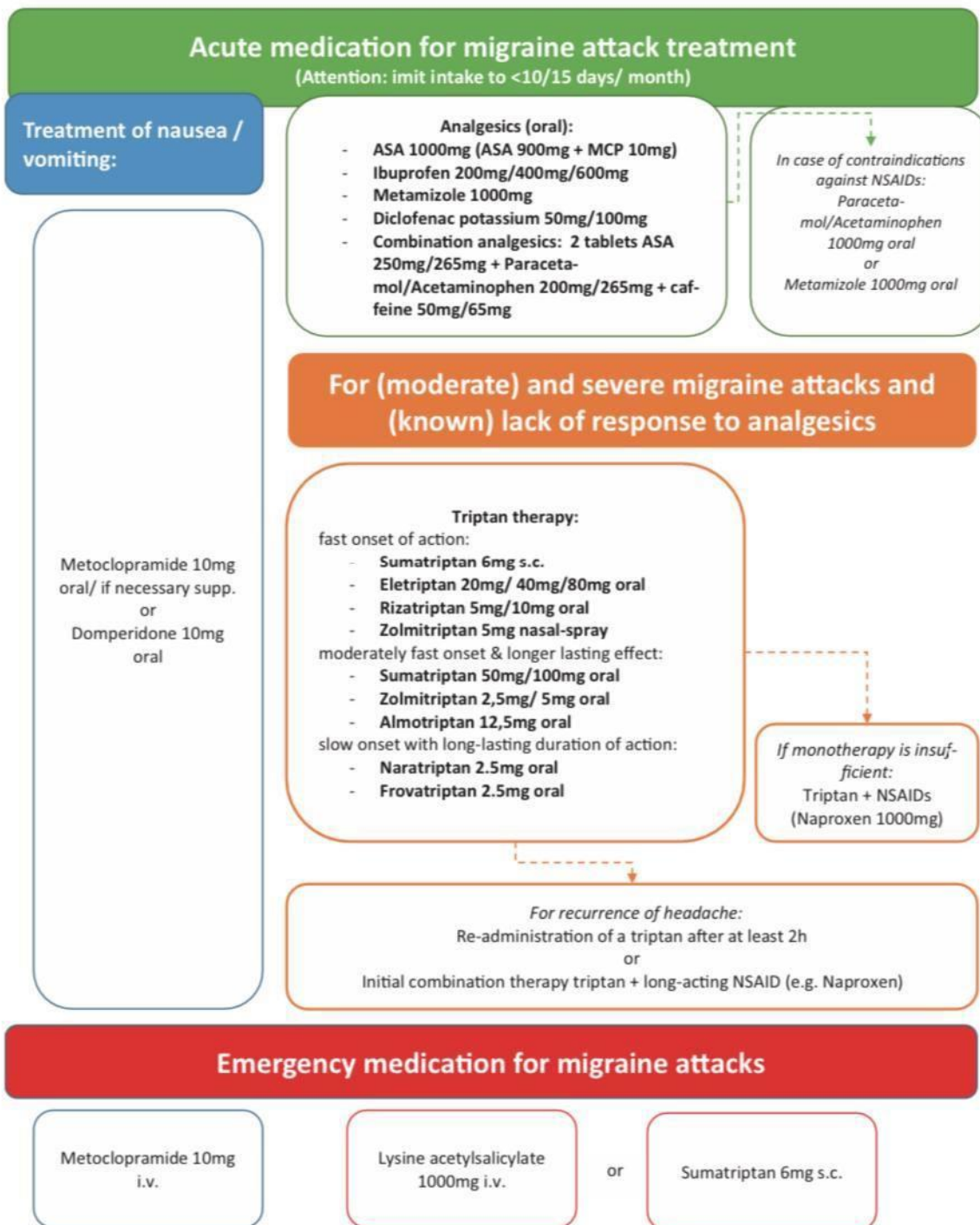
Dexamethasone can reduce the recurrence of early headaches but does not provide immediate relief.

Transcutaneous supraorbital nerve stimulation can reduce intensity.

Transcranial magnetic stimulation is proved effective as a second-line treatment, with no serious side effects. It can also be offered as an option to treat chronic migraines. It is contraindicated in patients with epilepsy.

Nonpainful remote electric neurostimulation could be considered a first-line treatment in some patients.

Peripheral nerve blocking (occipital plexus and sphenopalatine ganglion) may also be tried.



Prophylactic or Preventive Treatment

Preventive treatment aims to reduce attack frequency, improve responsiveness to acute attacks' severity and duration, and reduce disability. Migraine triggers have to be documented by each individual to reduce them in the future.

Indications for preventive treatment are:

- Frequent or long-lasting headaches
- Attacks that cause significant disability and reduced quality of life
- Contraindications or failure to acute therapies
- Significant adverse effects of abortive therapies
- Risk of medication overuse headache
- Menstrual migraine (along with short-term premenstrual prophylaxis)
- Hemiplegic migraine
- Brainstem aura migraine
- Persistent aura without infarction
- Migrainous infarction

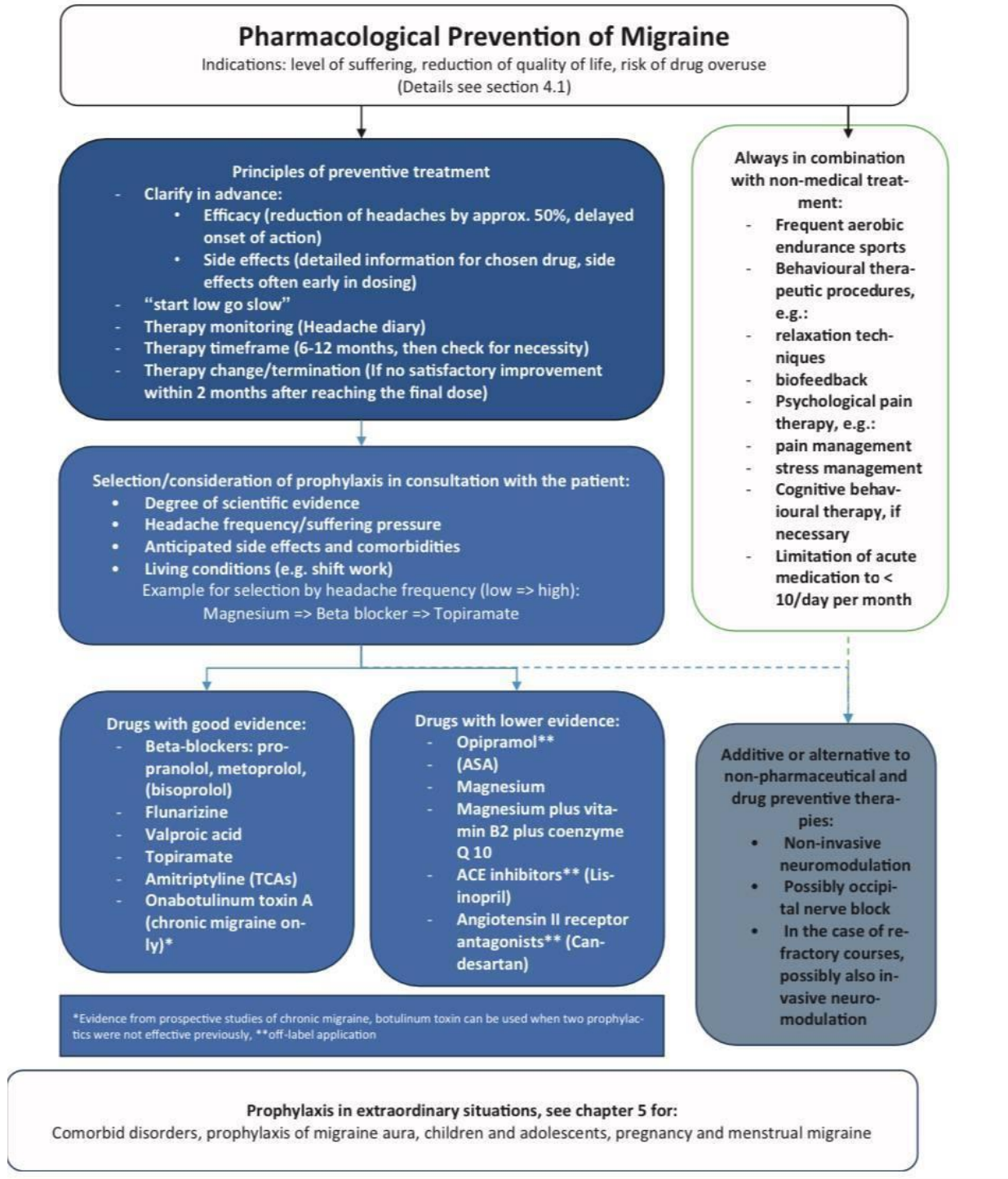
Preventive treatment agents are the following:

- Beta-blockers: Metoprolol and propranolol - especially in hypertensive and non-smoker patients
- Antidepressants: Amitriptyline and venlafaxine - especially in patients with depression or anxiety disorders and insomnia
- Anticonvulsants: Valproate acid and topiramate - especially in epileptic patients
- Calcium channel blockers: Verapamil and flunarizine - especially in women of childbearing age or patients with Raynaud's phenomenon
- Calcitonin gene-related peptide antagonists: Erenumab, fremanezumab, and galcanezumab

Alternative Treatment

-Lifestyle changes; must be a commitment from the patient; however, social support is of great importance to improve mental health to help the patient's involvement.

- Regular exercise
- Yoga
- Relaxation training
- Cognitive-behavioral therapy
- Biofeedback
- Reduction of triggers
- Detoxification
- Butterbur
- Melatonin



Drug Class	Drug	Dose Range	Adverse Effects/Contraindications
Beta-adrenoceptor blockers	• Propranolol • Metoprolol • Timolol	• 80-240 mg • 50-150 mg • 10-20 mg	• Contraindicated in asthma, syncope, heart block
Antidepressants	• Amitriptyline • Nortriptyline • Venlafaxine	• 10-150 mg • 25-100 mg • 37.5-150 mg	• Somnolence • Insomnia, hypertension
Calcium-channel blockers	• Verapamil	• 180-480 mg	• Constipation, hypotension, edema
Antiepileptic drugs	• Divalproex sodium • Topiramate • Gabapentin	• 200-1500 mg • 25-150 mg • 300-1800 mg	• Weight gain, thrombocytopenia, tremor • Renal calculi, amnesia, glaucoma, dysequilibrium, weight loss
CGRP monoclonal antibodies	• Erenumab • Galcanezumab	• 70 to 140 mg every four weeks, subcutaneously • 120 mg monthly subcut	• Injection site reactions, muscle spasms, hypersensitivity reaction • Injection site reactions, vertigo, pruritus, constipation

VIII. SPECIAL CASES:

Treatment of migraine attacks in children

The use of ASA is not recommended before the age of 12. If anti-emetics are necessary, domperidone should be used and not metoclopramide, due to the elevated risk of acute extrapyramidal dystonia with metoclopramide.

Sumatriptan 10 mg and zolmitriptan 5 mg as nasal spray are approved for the treatment of migraine in adolescents age 12 or older. Sufficient data are available to justify the use of triptans (sumatriptan 10–20 mg as nasal spray, zolmitriptan 2.5–5 mg tablets, rizatriptan 5–10 mg tablets and almotriptan 12.5 mg tablets) in adolescents in case of inadequate response to acute therapy with analgesics.

Treatment of migraine attacks during pregnancy

Migraine attacks can be treated between the first and second trimester of pregnancy with ASA or ibuprofen. These substances should be avoided in the third trimester. Paracetamol should only be given if there are contraindications for ASA. Triptans are not approved for use in pregnancy. Sumatriptan should only be taken by pregnant women when the expected benefit for the mother outweighs a possible risk for the child. Ergots are contraindicated during pregnancy.

Treatment of migraine attacks in menstrually associated migraine

By definition, menstrually associated migraine is a migraine in which the attacks occur exclusively in the time from 2 days before up to 3 days after the onset of bleeding in at least 2 or 3 cycles. If attacks also occur independent of menstruation, the term is menstrually associated migraine. Menstrually associated attacks are considered particularly severe and long-lasting with

poor response to acute therapy and increased occurrence of recurrent headache. Menstrually associated migraine attacks last longer and, after initially successful therapy, tend more frequently to lead to recurrent headache.

The acute therapy does not differ from the general therapy of migraine attacks. Superiority over placebo has been shown for all triptans in acute therapy. This applies also for the combination of sumatriptan and naproxen. In comparison studies, within the triptan groups, frovatriptan (2.5 mg) had a lower recurrence rate than 10 mg rizatriptan and 12.5 mg almotriptan with equally good effectiveness. A further study showed that the combination of 10 mg rizatriptan with 4 mg dexamethasone was more effective but also associated with more side effects than the administration of rizatriptan alone. Dexamethasone alone was inferior to monotherapy with rizatriptan and can therefore not be recommended. If the response of menstrually associated migraine to the usual acute therapy is inadequate, the indication for short-term prevention should be considered.

Emergency treatment of migraine attacks

The treatment of first choice is the intravenous administration of 1000 mg ASA with or without metoclopramide. In a dose of 10–40 mg, metoclopramide i.v. was found to provide an independent analgesic effect. If there are no contraindications, sumatriptan 6 mg can also be given subcutaneously. Triptans act better at the start of an attack than during its course or with repeated administration within an attack. Sumatriptan s.c. is slightly more effective than ASA i.v., but it has significantly more side effects. The two substances do not differ with respect to the occurrence or onset of recurrent headache. The intravenous administration of 1000 mg metamizole is significantly more effective than placebo but can lead to drop in blood pressure and allergic reactions. The intravenous administration of 1000 mg paracetamol was not superior to placebo in acute migraine attacks. There is evidence that the intravenous administration of valproic acid in a dose of 300 mg or 800 mg is also effective in the treatment of acute migraine attacks. Valproic acid is not approved for the treatment of migraine attacks. Opioids cannot be recommended for the therapy of acute migraine attacks. They are inferior to other acute medications, have a high side effect potential and lead very frequently to recurrence of headache.

Based on expert consensus, therapy of a status migrainosus is recommended with a single administration of 50-100 mg prednisone or 4–8 mg dexamethasone. This is confirmed by a survey of studies on the therapy of migraine attacks with corticosteroids. The data indicate a reduction in headache intensity and a reduction of recurrent headache.

IX. ROGNOSIS

A migraine is a chronic condition that can revert to episodic migraine in 26 to 70% of patients. Prolonged remissions are common; however, some patients have a pattern of leaving and returning to chronic states. The severity and frequency of attacks can diminish with age. Episodes increase during puberty but continue to climb until 35 to 39 years of age, decreasing later in life, especially after menopause.

X. COMPLICATION

- Status migrainosus, a debilitating migraine attack lasting over 72 hours
- Migrainous infarction
- Migraine aura-triggered seizure
- Persistent aura without infarction
- Work disability
- Loss of work

XI. PREVENTION AND RECOMMENDATION

Timely diagnosis and management of migraine headaches are essential, as they can sometimes be debilitating and affect the quality of life. Patients should be educated about the different phases of migraine headaches and the benefits of abortive and prevention therapy. Patients should be educated about lifestyle changes, which can help in reducing the frequency and severity of migraine attacks. Patients should also be instructed to go to a doctor in case of worsening symptoms or the occurrence of new neurological symptoms, which might warrant neuroimaging.

Drug Class/Drug	Drug	Dose Range	Notes
Nonsteroidal anti-inflammatory drugs	• Aspirin • Ibuprofen • Naproxen • Diclofenac • Diclofenac epolamine • Tolfenamic acid • Celecoxib • Dexlpropofol	• 900-100 mg • 400-600 mg • 275-825 mg • 50-100 mg • 65 mg • 200 mg • 120 mg • 50 mg	All NSAIDs have similar efficacy
Non-opioid analgesic	• Acetaminophen	• 1000-3000 g	Acute-life threatening hepatotoxicity at > 4 g/d

Serotonin 1b/1d agonists (triptans) (Sumatriptan)	- Sumatriptan (oral) - Sumatriptan (intranasal solution) - Sumatriptan (intranasal powder) - Sumatriptan (spray) - Sumatriptan (subcutaneous)	50-100 mg as a single dose, maximum dose: 200 mg/d 20 mg as a single dose in 1 nostril; if symptoms persist, may repeat dose after ≥ 2 hours, maximum dose: 40 mg/d 22 mg as a single dose, may repeat dose after ≥ 2 hours if symptoms persist or return, maximum dose: 44 mg/d 10 mg as a single dose in 1 nostril. Repeat dose after ≥ 1 hour if symptoms persist or return; maximum dose: 30 mg/d 6 mg as a single dose, may repeat dose (usually same as the first dose) after ≥ 1 hour if symptoms persist or return, or lesser dosage if 6 mg was not tolerated, maximum dose: 6 mg/dose; 12 mg/d	All formulations of triptans are contraindicated in severe hepatic impairment. Contraindicated in patients with cardiovascular illness as prolonged QT interval on ECG and subsequent ventricular arrhythmias, including torsades de pointes (TdP) and ventricular fibrillation, are reported. It may also cause dizziness, lethargy, tremor, vertigo, akathisia, dystonia, and pathological laughter. Other vasospasm-related events include peripheral ischemia, ischemic colitis, splenic infarction, and Raynaud disease. It should be avoided in patients with uncontrolled hypertension and pregnancy. Ocular side effects include transient and permanent blindness and significant partial vision loss. The use of concomitant serotonergic drugs may cause serotonin syndrome. Unpleasant taste is lower with intranasal zolmitriptan as compared to intranasal sumatriptan. Patients who do not respond to one triptan may respond to another. Naratriptan and frovatriptan have a slower onset and lower efficacy.
Serotonin 1b/1d agonists (triptans) (Naratriptan)	Naratriptan	2.5 mg as a single dose; may repeat dose after ≥ 4 hours; maximum dose: 2.5 mg/dose; 5 mg/d.	Use within 24 hours of an ergotamine preparation or a different triptan is not advised. Contraindicated with severe renal impairment (CrCl < 15 mL/minute).

Serotonin 1b/1d agonists (triptans) (Zolmitriptan)	- Zolmitriptan (oral) - Zolmitriptan (intranasal)	2.5 mg as a single dose; may repeat dose after ≥ 2 hours; maximum dose: 5 mg/dose; 10 mg/d 2.5 to 5 mg as a single dose; may repeat dose after ≥ 2 hours; maximum: 5 mg/dose; 10 mg/d	Refer to the section on sumatriptan above for the side effects of triptans.
Serotonin 1b/1d agonists (triptans) (Frovatriptan)	Frovatriptan	2.5 mg as a single dose; may repeat dose after ≥ 2 hours if needed; maximum dose: 2.5 mg/dose; 5 mg/d	Slower onset
Serotonin 1b/1d agonists (triptans) (Almotriptan)	Almotriptan	12.5 mg as a single dose; may repeat dose after ≥ 2 hours when needed; maximum dose: 12.5 mg/dose; 25 mg/d	Reduce dose to half with hepatic impairment
Serotonin 1b/1d agonists (triptans) (Rizatriptan)	Rizatriptan	5 to 10 mg as a single dose; may repeat dose after ≥ 2 hours if needed; maximum dose: 20-30 mg/d	Propranolol increases rizatriptan levels by 70%. So the dose of rizatriptan must be adjusted downward in these patients.
Serotonin 1b/1d agonists (triptans) (Eletriptan)	Eletriptan	40 mg as a single dose; may repeat dose after ≥ 2 hours if needed; maximum dose: 40 mg/dose; 80 mg/d	Primarily metabolized by cytochrome P-450 enzyme CYP3A4. Not advised within at least 72 hours of treatment with other drugs that are potent CYP3A4 inhibitors : itraconazole, ketoconazole, clarithromycin, nefazodone, troleanomycin, ritonavir, and nelfinavir.
Antiemetics	- Metoclopramide (IV, IM, oral) - Prochlorperazine (IV, IM)	10-20 mg as a single dose 10 mg as a single dose	IV route is preferred for metoclopramide. Pretreat with diphenhydramine to prevent akathisia and other acute dystonic reactions .
Calcitonin-gene-related peptide (CGRP) antagonists	Rimegepant	75 mg every other day; maximum dose: 75 mg/d	Administration early in the course of a migraine attack may improve response to treatment. Second-line therapy when triptans are contraindicated, poorly tolerated, or ineffective. More studies are needed to establish efficacy and safety.

	• Ubrogepant	• 50 to 100 mg as a single dose; may repeat dose after ≥ 2 hours if needed; maximum dose: 200 mg/d	
Serotonin 5-HT _{1F} receptor agonist	• Lasmiditan	• 50 to 100 mg as a single dose; may increase to 100 or 200 mg as a single dose if needed; repeat doses have not established efficacy.	Administration early in the course of a migraine attack may improve response to treatment. Second-line therapy when triptans are contraindicated, poorly tolerated, or ineffective. A major side effect is dizziness (9% to 17%). Wait for at least 8 hours between dosing and driving or operating heavy machinery. It may enhance the CNS depressant effect of alcohol.
Ergot derivative	• Dihydroergotamine	• IM: 1 mg as a single dose; may repeat hourly as required; maximum dose: 3 mg/d, 6 mg/week • IV: 1 mg as a single dose; may repeat hourly as required; maximum dose: 2 mg/d, 6 mg/week • SUBQUT: 1 mg as a single dose; may repeat every 2 hours as required; maximum dose: 3 mg/d, 6 mg/week • Intranasal: 0.5 mg per spray; 1 spray (0.5 mg) into each nostril; repeat after 15 minutes (total of 4 sprays per dose); maximum dose: 4 sprays (1 dose)/d	Use is contraindicated in severe hepatic or renal impairment and pregnancy or breastfeeding. Also contraindicated in patients with hypertension or ischemic heart disease. It should not be used within 24 hours of triptans or ergot-like agents. Use with potent inhibitors of CYP3A4 (including azole antifungals, protease inhibitors, and some macrolide antibiotics) is also avoided.

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ACUTE AND CHRONIC HEADACHE

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I. DEFINITION

A headache is characterized by pain or discomfort in the head or face. The location and intensity of the pain can vary widely, as well as the frequency with which headaches occur. While brain tissue itself lacks pain-sensitive nerve fibers and does not perceive pain, other areas of the head can contribute to the sensation of a headache, including:

- A network of nerves that extends over the scalp
- Certain nerves in the face, mouth, and throat
- Muscles of the head, neck, and shoulders
- Blood vessels found along the surface and at the base of the brain

II. EPIDEMIOLOGY

The lifetime prevalence of headaches is 96%, with a higher occurrence in females. The global active prevalence of tension-type headaches is about 40%, while migraines affect approximately 10% of the population. Migraines are most prevalent among individuals aged 25 to 55 and are three times more common in women. Despite causing significant disability, migraines are often underdiagnosed and inadequately treated. Trigeminal autonomic cephalalgias are much rarer than migraines and tension-type headaches, with cluster headache being the most common type. Its prevalence in the general population is around 0.1%, with a male-to-female ratio of 3.5 to 7:1.

Chronic daily headache refers to experiencing headaches daily or nearly every day for extended periods, often reported in the literature, though it is not officially recognized as a diagnosis in the International Classification of Headache Disorders. Long-lasting chronic daily headaches may include chronic migraine, chronic tension-type headache, hemicrania continua, and new daily persistent headache. The worldwide prevalence of chronic daily headache consistently falls between 3% and 5%, with most cases likely representing chronic migraine.

III. ETIOLOGY

Detailed history provides the most useful information in evaluating acute headache. Aspects to focus on include the temporal profile, precipitating factors, associated symptoms, and medical and family history. The following are the most common diagnostic clues, although not all headaches present with typical symptoms.

1. Temporal features

Sudden onset

- Subarachnoid hemorrhage, stroke, venous sinus thrombosis, meningitis, hypertensive encephalopathy, pituitary apoplexy, acute angle-closure glaucoma

Duration of hours to days

- Tension headache, migraine
- Episodic, 2 to 10 episodes per day, lasting seconds to minutes
- Cluster headache, Trigeminal neuralgia
- Awaken from sleep/morning headache
- Intracranial mass, obstructive sleep apnea
- Aligned with menstrual cycle
- Menstrual headache

2. Precipitating factors

- Trauma, increasing severity and frequency, recent high-altitude changes, fatigue or stress, Valsalva (coughing, sneezing, bending, or exertion). Aligned with menstrual cycle, medication changes/withdrawal (caffeine, opiates, pseudoephedrine)

3. Associated symptoms

- Fever, vomiting, seizure, dizziness, ischemic or hemorrhagic infarct (posterior fossa), migraine. Warning signs: visual, auditory, gustatory disturbance, migraine with aura, neck pain, facial pain, eye area: glaucoma, sinusitis, visual disturbance, lacrimation, rhinorrhea

4. Medical history

- HIV/AIDS or immunocompromised, age >50 years with new-onset headache
- COPD, connective tissue disorders (e.g., systemic lupus erythematosus), cancer
- Hypertension, family history of migraines

IV. PATHOPHYSIOLOGY

The pathophysiology of headaches involves complex interactions between various biological systems, including neurovascular, inflammatory, and genetic components. Here's an overview of the mechanisms involved:

1. Neurovascular Mechanisms:

Headaches, especially migraines, involve changes in the brain's vascular system. This includes the dilation and constriction of cerebral blood vessels, which can lead to the release of inflammatory mediators. The trigeminal nerve, a key component in the head's sensory system, plays a crucial role in headache pain. Activation of this nerve can lead to the release of neurotransmitters such as CGRP (calcitonin gene-related peptide) and substance P, contributing to inflammation and pain.

2. Inflammatory Processes:

In conditions like migraine, there is an inflammatory response that occurs in the meninges (the protective membranes covering the brain), which can lead to sensitization of pain pathways.

3. Cortical Spreading Depression (CSD):

This phenomenon is particularly associated with migraines. It refers to a wave of depolarization followed by a period of inhibition, spreading across the cerebral cortex. CSD can trigger the release of neuropeptides that contribute to headache symptoms.

4. Genetic Factors:

Certain genetic predispositions can influence an individual's susceptibility to headaches. For instance, mutations in genes related to neurotransmitter systems may affect pain perception and headache occurrence.

5. Central Sensitization:

This concept relates to the increased responsiveness of neurons in the central nervous system to stimuli, leading to heightened pain sensitivity.

6. Psychological and Environmental Triggers:

Stress, sleep disturbances, hormonal changes, and diet are known triggers that can exacerbate or provoke headaches. Understanding the pathophysiology of headaches is essential for developing effective treatment strategies and managing different headache types in clinical practice.

V. DIAGNOSIS

1. Classification

The International Classification of Headache Disorders (ICHD) was first released in 1988 and has since undergone two revisions, the latest in 2013. Typically, headache classification is determined by the characteristics of the headache itself, specifically in the year before diagnosis, rather than focusing on the individual experiencing the headache. However, individual-specific features can assist in distinguishing between two similar diagnostic categories. The ICHD is regularly reviewed and continues to adapt over time.

Table 1: The International Classification of Headache Disorders, 3rd edition (beta version). (8)

Part 1: The primary headaches
1. Migraine
2. Tension-type headache
3. Trigeminal autonomic cephalgia
4. Other primary headache disorders
Part 2: The secondary headaches—Headache (or facial pain) attributed to:
5. Trauma or injury to the head and/or neck
6. Cranial or cervical vascular disease
7. Nonvascular intracranial disorder
8. A substance or its withdrawal
9. Infection
10. Disorder of homeostasis
11. Disorder of the cranium, neck, eyes, ears, nose, sinuses, teeth, mouth, or other facial or cervical structure
12. Psychiatric disorder
Part 3: Painful cranial neuropathies, other facial pains, and other headaches
13. Painful cranial neuropathies and other facial pain
14. Other headache disorders

2. Clinical presentation

In primary care, the majority of headaches are primary headaches, which lack identifiable associated pathology, and are usually tension headaches or migraines. Most primary care physicians possess considerable experience in evaluating and managing these types of headaches. However, it's important to note that an individual patient can experience more than one type of headache, and overusing analgesics can result in increased headache frequency and/or a lack of response to standard treatments. Certain characteristics may indicate a secondary cause for headaches. Depending on the level of concern, patients may need further investigation and/or a referral to secondary care, particularly in situations such as:

- Worsening headache with fever
- Sudden onset headache reaching maximum intensity within 5 minutes (Thunderclap headache)

- New onset neurological deficit, new onset of cognitive dysfunction
- Change in personality, impaired level of consciousness
- Recent (typically within the past 3 months) head trauma
- Headache triggered by cough, Valsalva or sneeze (triggered, not worsened)
- Headache triggered by exercise (triggered, not worsened), orthostatic headache (headache that changes with posture), symptoms suggestive of giant cell arteritis (particularly jaw claudication), symptoms and signs of acute narrow angle glaucoma, and a substantial change in the characteristics of their headache

Or, in patients who:

- Have compromised immunity (caused for example by HIV or immunosuppressive drugs), are aged under 20 years and a history of malignancy, a history of malignancy known to metastasize to the brain (e.g. lung, breast, melanoma, renal, colorectal) or vomiting without any other cause.

3. Key diagnostic factors

The accurate diagnosis of headaches relies on a comprehensive evaluation of several key factors, including the history of the present illness, associated symptoms, potential triggers, medical history, and a thorough physical examination. Recognizing specific features and red flags is crucial for differentiating between primary headaches, such as migraines and tension-type headaches, and secondary headaches that may signify more serious underlying conditions.

4. Investigations

When evaluating headaches, certain investigations may be necessary to rule out secondary causes and to confirm a diagnosis. The decision to pursue specific tests is based on the clinical presentation, the presence of any red flags, and the patient's overall medical history.

Imaging Studies

Magnetic Resonance Imaging (MRI):

- Preferred for detailed imaging of the brain's structure.
- Useful for identifying tumors, vascular abnormalities like aneurysms, Chiari malformations, or demyelinating diseases.

Computed Tomography (CT) Scan:

- Often used in emergency settings to quickly assess for acute issues such as hemorrhage, skull fractures, or large masses.
- Can be particularly useful if there's suspicion of a subarachnoid hemorrhage following a sudden, severe headache.

Lumbar Puncture:

- Performed if there is a suspicion of conditions such as meningitis, encephalitis, or a subarachnoid hemorrhage, especially when CT/MRI results are inconclusive.

Blood Tests:

- Complete Blood Count (CBC): Can help detect infections or systemic inflammation.
- Erythrocyte Sedimentation Rate (ESR)/C-Reactive Protein (CRP): Useful for identifying systemic inflammation, potentially indicative of conditions like temporal arteritis.
- Thyroid Function Tests: To rule out thyroid disorders which might contribute to headache onset.

Electroencephalogram (EEG)

- Typically, not a standard investigation for headaches unless there is suspicion of seizure activity.

Angiography

- Magnetic Resonance Angiography (MRA)/CT Angiography (CTA): Useful for visualizing the blood vessels in the brain, helping to identify vascular malformations or aneurysms. Provides detailed images of the blood vessel architecture, which can be crucial if vascular issues are suspected.

Ophthalmologic Examination

- Especially indicated if there is suspicion of increased intracranial pressure or optic nerve abnormalities, such as papilledema.

VI. DIFFERENTIAL DIAGNOSIS

Table 2: Differential diagnosis of headaches

Common	Uncommon
- Tension headache - Migraine - Acute sinusitis - Otitis media - Menstrual headache - Medication withdrawal - Medication overuse headache - Cervical paraspinal muscle tenderness - Dental caries/wisdom tooth impaction - Temporomandibular disorders - Acute hydrocephalus	- Idiopathic intracranial hypertension - Brain tumor, Hypertensive encephalopathy - Eclampsia/pre-eclampsia, Pituitary apoplexy - Venous sinus thrombosis, Epidural hematoma, Subdural hematoma, Subarachnoid hemorrhage (SAH) - Meningitis, Brain abscess, Carbon monoxide poisoning, Concussive syndrome/trauma - Acute mountain sickness/hypoxia - Cluster headaches, Paroxysmal hemicrania - Trigeminal neuralgia, Acute angle-closure glaucoma, Giant cell arteritis, Zika virus infection, Chikungunya, Dengue fever, Stroke

VII. MANAGEMENT

1. Objectives

The treatment strategy for many secondary headaches prioritizes addressing the underlying cause, such as treating a sinus infection. The treatment of some secondary headaches, such as posttraumatic headache, may default to the treatment of migraine because the majority of post-traumatic headaches have the phenotype of migraine (13,14).

There is a clear distinction between abortive and preventive medications in headache management. Abortive medications are intended to treat a specific headache attack, while preventive medications aim to decrease the frequency and intensity of attacks, ultimately reducing disability.

2. Critical Sign

Headache “Red Flags” that could indicate need for evaluation

- New headache in older patients
- Abnormal neurologic examination including papilledema and change in mental status
- New change in headache pattern or progressive headache

- New headache in the setting of HIV risk factors, cancer, or immunocompromised status
- Signs of a systemic illness (eg, fever, stiff neck, rash)
- Triggered by cough, exertion, Valsalva maneuver
- Headache in pregnancy/postpartum period, and first or worst headache

3. Mortality risk factors

While headaches themselves are typically not directly associated with mortality, certain underlying conditions or risk factors related to headaches can increase the risk of serious health issues, potentially contributing to higher mortality rates. Here are some mortality risk factors that may be associated with headaches:

Underlying Medical Conditions:

- Cardiovascular Disease
- Stroke
- Tumors

Type of Headache:

- Migraines
- Chronic Daily Headaches

Age and Gender:

- Older adults and males may have an increased risk of serious underlying conditions associated with headaches, leading to higher mortality rates.

Mental Health:

- Depression and Anxiety Disorders

Lifestyle Factors:

- Obesity
- Substance Abuse

Delayed Diagnosis:

- Failure to recognize serious underlying causes of headaches, such as subarachnoid hemorrhage or infections like meningitis, can lead to delayed treatment and increased mortality risk.

4. Outpatient Management

Outpatient management of headaches involves several strategies tailored to the type of headache, its frequency, severity, and the individual patient's needs. Here's an overview of key components for managing headaches in an outpatient setting:

Initial Assessment

- Comprehensive history and physical examination

Education and Lifestyle Modifications

- Trigger Identification: Help patients identify and avoid headache triggers (e.g., stress, certain foods, dehydration, lack of sleep).

Lifestyle Changes: Encourage:

- Regular exercise, adequate hydration, consistent sleep patterns. Stress management techniques (e.g., mindfulness, yoga, relaxation techniques).

Medication Management

- Acute Medications (abortive treatment)

- Preventive Medications: Consider for patients with frequent or severe headaches (e.g., migraines occurring more than 4-6 days per month).

Follow-Up and Monitoring

- Schedule regular follow-up appointments to assess the effectiveness of the treatment plan, monitor for side effects, and make adjustments as needed.

5. First-line treatment

Acute and chronic headache in primary headaches A-

Tension headache

The characteristics of a tension headache are:

- Bilateral, pressure or “tight band”, mild to moderate in intensity, no nausea, not aggravated by physical activity

Acute treatment

- Consider Aspirin, Paracetamol or Ibuprofen for the acute treatment of a tension-type, headache, taking into account the person’s preference, co-morbidities and risk of adverse effects.

Prophylactic treatment

- Consider tricyclic antidepressants, for example, Amitriptyline (10 mg to 50mg per day) or Nortriptyline which is less sedating.

B- Migraine

Acute treatment

Treatment should be selected according to severity and frequency of attacks, symptoms, age, patient’s medical history and history of treatment. Caution is needed when recommending over the counter products containing codeine preparations (typically containing paracetamol 500 mg and codeine phosphate 8 mg). Products containing caffeine (typically containing Aspirin 300 mg, Paracetamol 200 mg and Caffeine 45 mg) are also prone to cause this complication.

Second choice treatment if previous attacks have not been controlled using simple analgesics:

Triptans are slightly more effective than simple analgesics when combined with an antiemetic. Triptans are contraindicated in patients with ischemic heart disease, myocardial infarction, coronary vasospasm or uncontrolled hypertension.

- Sumatriptan 50-100mg tablets, s/c 6mg/0.5ml
- Almotriptan 12.5mg tablets
- Frovatriptan 2.5mg tablets
- Zolmitriptan nasal spray, 5mg/0.1ml spray

Prophylactic agents for migraine

- **Propranolol**: Start at 80 mg of Propranolol SR at night. Increase to 160 mg after 4 weeks. Maximum dose usually 240 to 320 mg per day.
- **Topiramate**: Start 25 mg nocte. Increase by 25 mg every fortnight, in twice daily dosage, aiming for initial maintenance dose of 50 mg BD. Some patients require more, with a maximum dose of 250 mg BD (depending on tolerability).
- **Candesartan**: Start at 4 mg daily and increase in weekly steps of 4 mg, aiming for a therapeutic dose of 16 mg daily.
- **Zonisamide**: Start 25 mg nocte. Increase by 25 mg per week, to a twice daily (BD) regimen. Initial potential therapeutic dose of 50 mg BD. If patients experience partial benefit, further increases to a maximum of 200 mg BD can be considered.

- **Botox:** Botox has now been approved for use by NICE.
- **Amitriptyline:** Starting dose is 10 mg nocte; side effects include sedation, so it should be taken 12 hours prior to planned waking time. Slowly (fortnightly) increases to 25 mg and to around 1 mg/kg. 10 mg dose rarely effective.
- **Nortriptyline:** Use at similar doses to Amitriptyline.
- **Sodium valproate:** Start 200 mg BD, increasing to 400 mg or 600 mg BD as tolerated over 2-to-4-week intervals.
- **Gabapentin:** Some physicians still use Gabapentin as a (generally) well tolerated agent.

C- Cluster Headache

Cluster headaches are typically shorter in duration (30 to 60 minutes). Daily frequency can range from every other day to 8 per day. Patients are typically restless during the headache (unlike migraines where patients usually rest/lie down).

Acute treatment

Subcutaneous injection Sumatriptan is the first-choice treatment for acute attacks. Nasal Sumatriptan (unlicensed indication) or Zolmitriptan (unlicensed indication) for people who cannot tolerate subcutaneous Sumatriptan can be an option. High flow oxygen via a non-rebreathing mask often very effective and well tolerated.

Prophylaxis

Verapamil recommended for prophylaxis, potentially requiring high doses (240 mg BD and more). ECG checks are recommended during dose titrations.

D- Medication overuse headache (MOH)

MOH can also be called “rebound” or “painkiller” headaches and is typically described as a headache which is present chronically (usually for 15 days or more per month) and which has developed or worsened while taking regular symptomatic medication. \

Acute and chronic headache in secondary headaches A-

Thunderclap headache

Described as a high-intensity headache of rapid onset reaching maximum intensity, usually in seconds, more rarely a few minutes. It may be primary or secondary, with subarachnoid hemorrhage being the commonest secondary cause. Arrange admission to hospital for emergency assessment. Patients typically require urgent CT head imaging for diagnosis. Further investigations may be planned depending on CT results.

B- Headache and associated symptoms following head injury

Consider referral to services for neurology and neurosurgery assessment.

6. Treatment adjustment

Adjusting treatment for headaches involves reassessing the current management plan and making modifications to better address the patient’s needs. This process is individualized, depending on the type of headache, patient response, and any changes in symptoms.

7. Algorithm

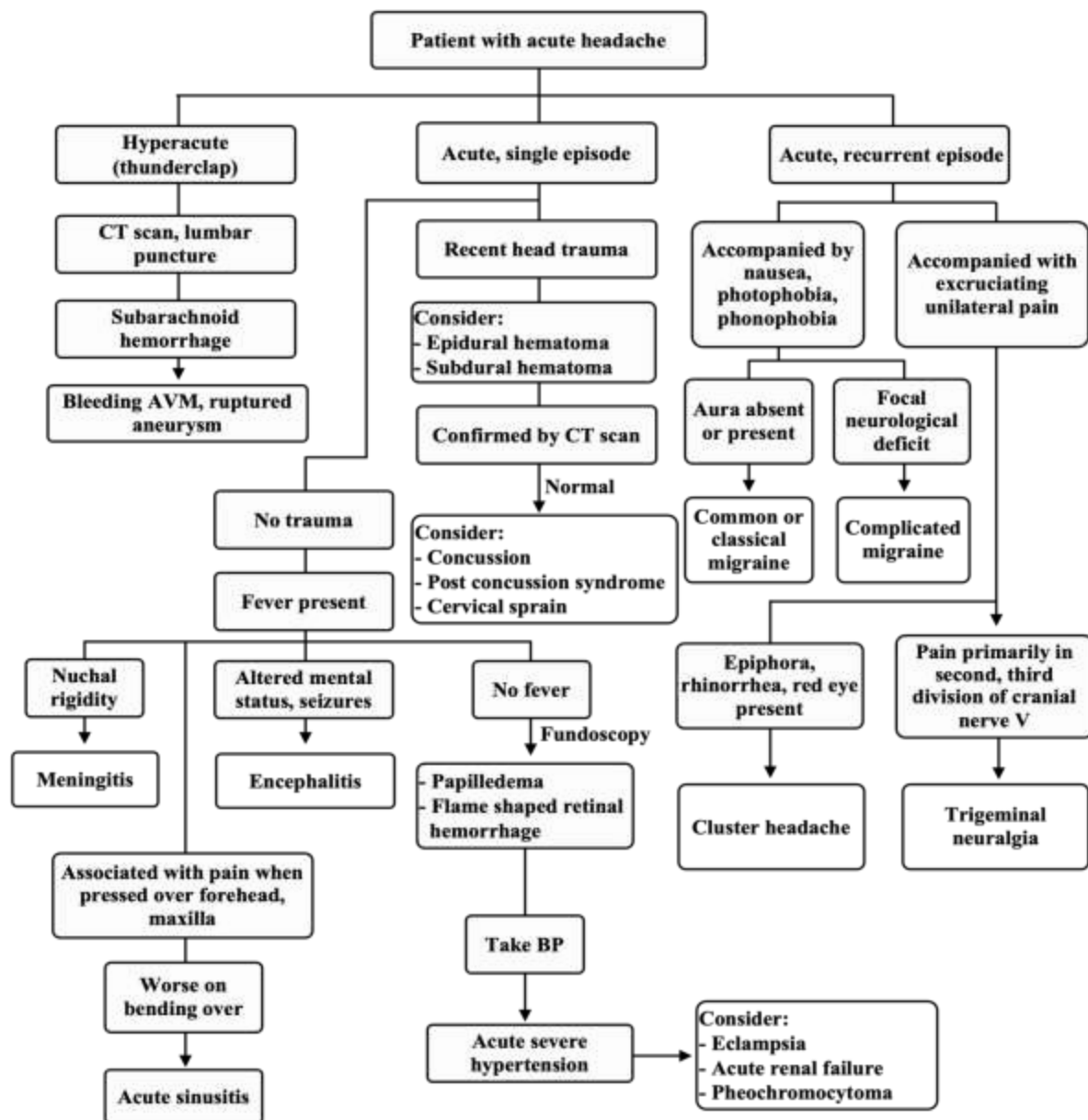


Figure 1: Clinical Instruments for Diagnosing and Screening of Headache

VIII. SPECIAL CASE

A 36-year-old male who presents to the outpatient clinic with complaints of recurrent headaches that have been increasingly bothersome over the past four months. He reports that the headaches began as occasional episodes occurring once or twice per month but have escalated to nearly every other day in the past month. The headaches began approximately four months ago, with the initial episodes being infrequent and mild. He describes the pain as primarily bilateral, with a sensation of tightness across his forehead and temples. He characterizes the pain as a dull, pressure-like sensation, rating it usually as a 5 out of 10 on the pain scale and lasts for 1 to 6 hours.

IX. EVOLUTION AND PROGNOSIS

Evolution of Acute and Chronic Headaches

Primary vs. Secondary Headaches:

- Primary Headaches: These include migraine, tension-type headache, and cluster headache. They tend to evolve over time due to various factors, such as changes in hormonal levels, stress, environmental factors, and lifestyle.
- Secondary Headaches: These result from underlying medical conditions (e.g., infections, trauma, vascular disorders). The evolution of secondary headaches largely depends on the treatment and management of the underlying condition.

Headache Frequency:

- Many individuals with primary headaches may experience infrequent episodes, which can evolve into more frequent attacks due to increased triggers or stressors. Tension-type headaches can become chronic with regular stress or exacerbating factors, while migraines may become more frequent or severe with time.

Prognosis of Headaches

Primary Headaches:

- Migraine: the prognosis can vary widely.
- Tension-type headaches: generally, have a favorable prognosis.
- Cluster headaches: prognosis varies, and some individuals may have remissions, while others experience chronic patterns.

Secondary Headaches:

- The prognosis largely depends on the underlying cause.

X. COMPLICATIONS

Long-Term Outcomes:

- Many individuals with primary headaches can achieve good control and quality of life through appropriate management, including medication and lifestyle modifications.

XI. PREVENTION AND RECOMMENDATION

Preventing headaches involves lifestyle changes, identifying triggers, and possibly using medications. Here are key prevention strategies and recommendations:

Lifestyle modifications

- Aim for 7-9 hours of consistent sleep, drink plenty of water throughout the day to avoid dehydration, maintain a balanced diet and avoid known trigger foods, and engage in regular physical activity to reduce stress and tension.

Identify and avoid triggers

- Headache Diary: Track headaches and potential triggers to identify patterns.
- Environment: Adjust your surroundings to minimize exposure to headache triggers.

Medication Management and regular medical check-ups

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TRIGEMINAL NEURALGIA

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I. DEFINITION

The trigeminal nerve is the largest of all cranial nerves. It has three branches that provide the main sensory innervation of the anterior two-thirds of the head and face. Trigeminal neuralgia (TN) is characterized by touch-evoked unilateral brief shock-like paroxysmal pain in one or more divisions of the trigeminal nerve.

It is a highly debilitating disorder that impacts on basic human functions such as talking, eating, drinking and touching the face, thereby resulting in a poor quality of life.

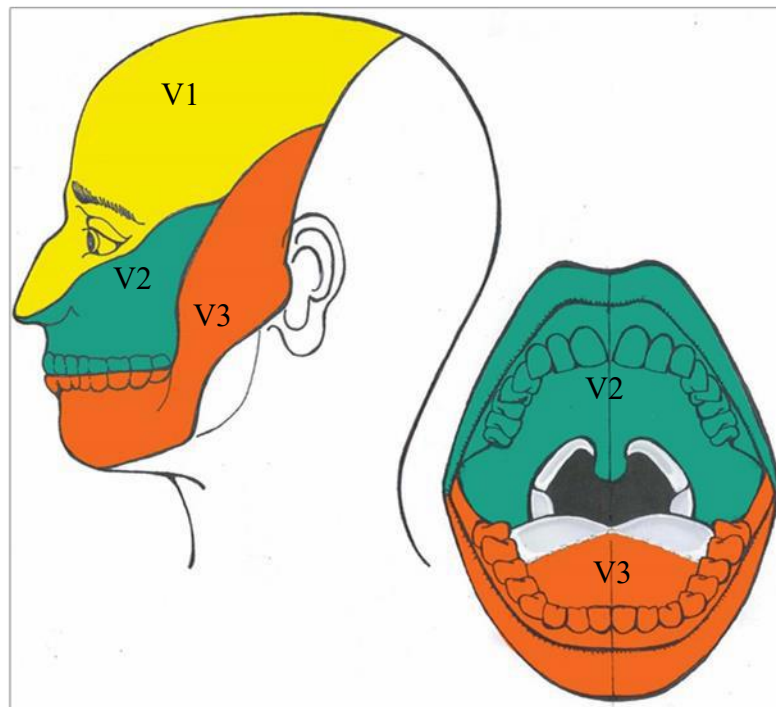


Figure 1: Innervation territories of the trigeminal nerve

II. EPIDEMIOLOGIE

Trigeminal neuralgia is rare, and hence it is difficult to obtain high quality epidemiological data. The TN is the most frequently occur in women (3 women for 2 men) and its peak incidence lies between age 50 and 60 years, with prevalence increasing with age that 75% of all case occur with population 50 years and above.

III. ETIOLOGIE AND CLASSIFICATION

Trigeminal neuralgia is further classified into classical trigeminal neuralgia (CTN), secondary trigeminal neuralgia (STN) and idiopathic trigeminal neuralgia (ITN), depending on the underlying cause.

The classical type, which is the most common and accounts for 75% of cases, is diagnosed when there is trigeminal neurovascular compression with morphological changes (distortion, indentation, atrophy), ipsilateral to the side of the pain, demonstrated either on MR imaging with appropriate trigeminal sequences or during surgery.

The secondary type, accounting for approximately 15% of cases, is attributable to an identifiable underlying neurological disease (except trigeminal neurovascular compression) that is known to cause TN, such as cerebellopontine angle tumour, arteriovenous malformation and multiple sclerosis.

The idiopathic type, accounting for approximately 10% of cases, is diagnosed when no apparent cause for TN can be found.

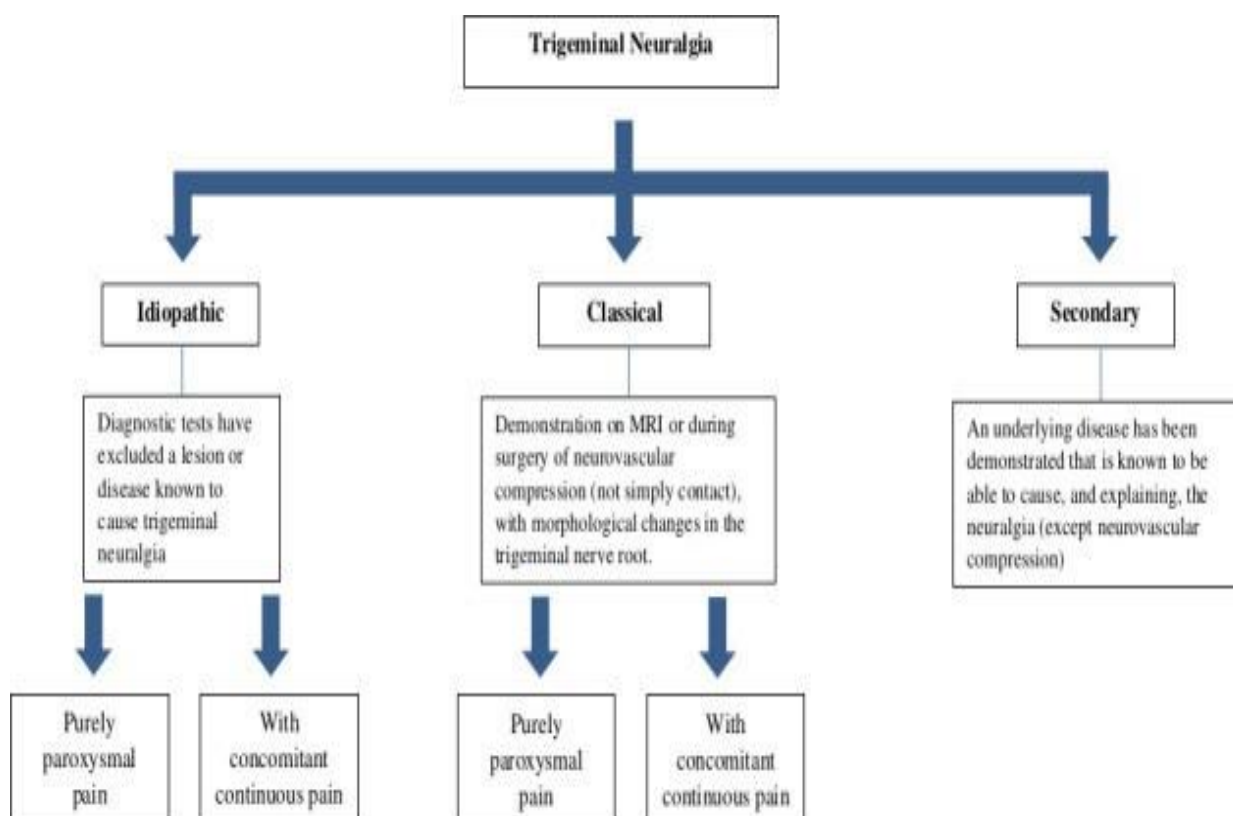


Figure 2 : Classification of trigeminal neuralgia

IV. PATHOPHYSIOLOGIE

Demyelination of primary sensory trigeminal afferents in the root entry zone is the predominant pathophysiological mechanism. Demyelination paves the way of generation of ectopic impulses and ephaptic crosstalk. The demyelination is caused by a neurovascular conflict with morphological changes such as compression of the trigeminal root.

However, there are also other unknown etiological factors, as only half of the CTN patients have morphological changes (distortion, indentation, atrophy). Vascular compression by an aberrant loop of an artery or vein accounts for 80-90%

of idiopathic TN. Other compressive causes are benign tumors of the posterior fossa such as acoustic neuroma, meningioma and epidermoid cyst.

V. DIAGNOSIS

1. Clinic

The diagnosis of trigeminal neuralgia is based on clinical features and his history. The key feature is a sudden and severe lancinating pain, usually unilateral, precipitated by touching facial zones. This pain occurs in paroxysms, within the trigeminal nerve distribution, typically involving the maxillary nerve (V2) or mandibular nerve (V3) distribution and lasts for a fraction of a second to 2 minutes.

Pain character of trigeminal neuralgia

Site

- unilateral
- bilateral : 3% of patients
- pain active in both sides at the same time (rarely)

Periodicity

- Episodic
- Sudden onset
- Lasting a few seconds to minutes and stopping suddenly
- Many attacks a day
- Refractory period between each attack
- Remission: for weeks or months
- Pain free intervals: gradually shorten between episodes

Character

- Electric shock-like
- Sharp
- Shooting

Severity

- Serious
- Milder after treatment

Factors affecting pain

- light touch to the face, talking
- eating, drinking, chewing, shaving
- cold winds

2. Investigation

Biochemical investigation

No laboratory or electrophysiologic test is routinely indicated for the diagnosis of TN, as patients with characteristic history and normal neurological examination may be treated without further workup. However, under treatment, the antiepileptic drugs issues related to reductions in white blood cell count, elevation in liver transaminases and hyponatraemia are well documented. It is recommended to check full blood count , electrolytes, liver enzymes, serum calcium and alkaline phosphatase every two to five years.

Imagery investigation

Advances in neuroimaging techniques will provide further insight into the causes of trigeminal neuralgia and its pathophysiology.

Magnetic Resonance Imaging (MRI) using specific sequences should be a part of the diagnostic workup to detect a possible neurovascular contact and exclude secondary causes. It can detect vascular contact with the trigeminal nerve that is best seen with specialised T2 and T1 volumetric acquisition techniques with fine slicing in all three plane. Demonstration of a neurovascular contact should not be used to confirm a diagnosis but rather to facilitate surgical decision making.

MRI is very sensitive in identifying the following features:

- Sinusitis
- Posterior fossa tumors: acoustic neuroma, meningioma, epidermoid cyst
- Extracranial enhancement of the trigeminal nerve that could indicate perineural spread of malignancy
- Cavernous sinus masses
- Arteriovenous malformation
- Demyelination plaques that might indicated multiple sclerosis
- Intrinsic brain lesions in the thalamus or trigeminal brain stem pathways such as lacunar infarctions
- Cerebellopontine angle mass lesions such as tumor, epidermoid, dermoid or arachnoid cyst and arteriovenous malformation

Posterior fossa tumours are the tumours most likely to be associated with trigeminal neuralgia pain characteristics and are most commonly vestibular schwannomas, meningiomas or epidermoid cyst.

VI. DIFFERENTIAL DIAGNOSIS

Differential diagnosis include dental causes (dental caries, pulpitis, cracked tooth, alveolar osteitis), maxillary sinusitis, salivary stone, temporomandibular joint causes, post-herpetic neuralgia, Burning mouth syndrome, trigeminal autonomic cephalalgias, Cluster headache, post-traumatic pain and other facial pains.

VII. MANAGEMENT

1. Objectives

The mainstay of managing trigeminal neuralgia is conservative treatment (antiepileptic drug) and surgical treatment which can decompress the trigeminal root from offending vessels, aiming at a permanent cure with no or little sensory deficit.

2. Critical Sign

Diagnostic criteria for trigeminal neuralgia

International Classification of Headache Disorders edition 3 (ICHD-3) diagnostic criteria for trigeminal neuralgia :

- A- Recurrent paroxysms of unilateral facial pain in the distribution of one or more divisions of the trigeminal nerve, with no radiation beyond, and fulfilling criteria B and C.
- B- Pain has all of the following characteristics:
 - 1- Lasting from a fraction of a second to 2 min.

- 2- Severe intensity.
- 3- Electric shock-like shooting, stabbing or sharp in quality.
- C- Precipitated by innocuous stimuli within the affected trigeminal distribution.
- D- Not better accounted for by another ICHD-3 diagnosis.

3. Mortality risk factor

Severity : Serious, milder after treatment

Factors affecting pain: light touch to the face, talking, eating, drinking, chewing, shaving, cold winds, vibration

4. Out patient management

OPD (Out Patient Department) patients frequently refer from dentist to consult neurologist after persist pain under dentistry treatment. Most patients with TN will be prescribed medicine to help control their pain, although surgery may be considered for the long term in cases where medicine is ineffective or causes too many side effects.

TN patient is still treated on an out patient basis, unless neurosurgical intervention is required.

5. First-line treatment

First-line treatment is prophylactic medication with antiepileptic drug and second line treatment is neurosurgical intervention.

Most patients respond well to medical treatment, carbamazepine and oxcarbazepine are first line therapy, while lamotrigine and baclofen are considered second line treatments. Other drug such as topiramate, levetiracetam, gabapentin, pregabalin and botulinum toxin-A are alternative treatments.

All the drugs should be started at low doses and gradually escalated every three to seven days to determine the best pain control with lowest side effects. Careful monitoring is needed initially and if drug doses become high.

However, many patients will have side effects, mainly affecting the central nervous system such as tiredness, poor concentration and high risk of drug interaction.

The periods of remission tend to get shorter with time and the attacks of pain often get longer. 65% of patients newly diagnosed with trigeminal neuralgia will have a second episode within five years, and 77% within 10 years. 50% of patients will have remission of at least six months duration. Patients can have as few as three or four attacks a day, but they could have as many as 70 per day. At times the attacks of pain come so quickly that it is impossible to determine whether there is a gap between each attacks. Pain can occur at night in a third of patients,

Opioid do not control this pain, but Lidocaine injected into trigger points is much more effective in patients who develop severe unremitting pain.

However, up to 10% of patients will not respond to antiepileptic drugs. If quality of life becomes impaired and symptoms are uncontrolled with drug treatment, patients should be referred to a neurosurgeon for consideration of surgical management.

Table 1: Evidence-based pharmacotherapy for Trigeminal Neuralgia

Table 3. Evidence-Based Pharmacotherapy for Trigeminal Neuralgia			
Drug	Dosage Regimen	Usual Dose Range	Common Side Effects
<i>First-line therapy</i>			
Carbamazepine ^a	100 mg twice daily; double after 3 days to 200 mg twice daily; increase by 100-200 mg twice daily every 3 days, then 200 mg 4 times daily	800-1,200 mg daily	• Sedation, dizziness, blurred vision, nausea, unsteady lethargy, double vision, headache • May cause hyponatremia, skin rashes, pancytopenia • Risk of osteoporosis with long-term use • May reduce oral contraception efficacy
Oxcarbazepine ^a	150 mg twice daily, double to 300 mg twice daily after 3 days; increase by 150-300 mg twice daily every 3 days, then 300 mg 4 times daily	1,200-1,800 mg daily	• Drowsiness, dizziness, diplopia, confusion, nausea, abdominal pain, headache, depression, diarrhea • High risk of hyponatremia • Chronic use risks osteoporosis • May reduce oral contraception efficacy
<i>Second-line therapy</i>			
Lamotrigine	Start 25 mg a day for 7 days, then 25 mg twice daily for 7 days, then 50 mg twice daily for another week; subsequent dose increments of 50 mg every 7 days, up to 10 weeks	200 mg twice daily	• Blurred vision, agitation, aggression, unsteadiness, dizziness, nausea, dry mouth, insomnia, joint pains • Risk of skin rashes and Stevens-Johnson syndrome with rapid dose escalation • Probably safe for pregnant women at a dose of 100 mg twice daily
Baclofen	5 mg 3 times daily for 3 days, then increase to 10 mg 3 times daily for 3 days; increase by 10 mg 3 times daily every 3 days until maximum dose	40-80 mg daily	Anxiety, depression, agitation, unsteadiness, headache, sedation, tremor, skin rash, blurred vision, dry mouth, abdominal pain, withdrawal symptoms if stopped too rapidly
Gabapentin	100 mg 3 times daily day 1, 200 mg 3 times daily day 2 and 300 mg 3 times daily day 3. Increase by 1-300 mg 3 times daily every 3 days to maximum dose; start 100 mg 3 times daily or 300 mg at bedtime; can increase dose up to 300 to 600 mg 3 times daily	900-3,600 mg daily	Amnesia, confusion, dizziness, vertigo, drowsiness, depression, nausea, blurred vision, peripheral edema, constipation, abdominal bloating, weight gain
Pregabalin	25 mg twice daily; increase by 25-50 mg twice daily every 3 days	600 mg daily	Confusion, drowsiness, constipation, blurred vision, dizziness, nausea, peripheral edema, increased appetite, weight gain
^a Carbamazepine and oxcarbazepine are available in liquid form. Dosage ranges vary due to lack of high-quality trials.			

6. Treatment adjustment

Surgical interventions are reserved for the patients with incapacitating symptoms of TN despite a trial of at least 3 drugs in sufficient dosage or when medications caused unacceptable adverse effects. Relapse of symptoms might also be another strong factor in opting for surgical treatment.

Two main techniques are suggested: thermocoagulation of Gasser's ganglion (Rhizolysis) and neurovascular decompression.

Rhizolysis

The most common technique is thermocoagulation of Gasser's ganglion by transcutaneous route. This technique is effective for a dozen years and can be repeated two or three times in a patient's lifetime. Following this, there is a risk of denervation pain, which can be equally debilitating.

Neurovascular decompression

In the case of neurovascular compression, a surgical approach can be carried out to eliminate the compression, and usually involves inserting a material (Teflon plate) between the vessel and the root.

Among all surgical procedures, microvascular decompression (MVD) is the most invasive surgery, but offers the highest success rate of pain-free status in patients with CTN. MVD is the first line surgery in medically refractory patients. The most common complication was aseptic meningitis (11%), follow by long-term hearing loss (10%) and sensory loss (7%). Major adverse events were noted in 4% if patients in the form of cerebrospinal fluid leaks, infarction or hematoma. Although complication rates were low, this surgical option has a small risk of death (around 0,2-0,4%).

Surgical decompression of neurovascular conflict results in separation of demyelinated fibers with possible remyelination of fibers and rapid pain relief.

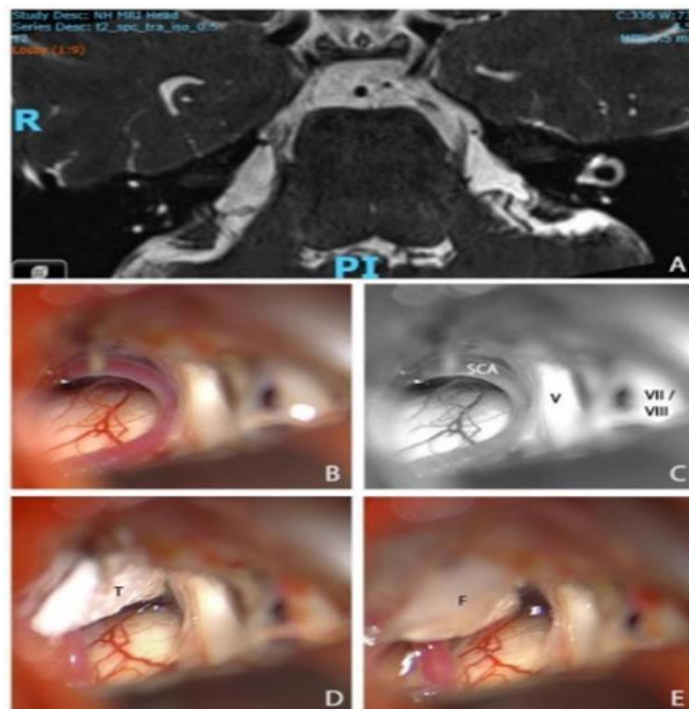


Figure 3: MRI of the trigeminal nerve and intraoperative pictures during MDR in patient with CTN

7. Algorithm

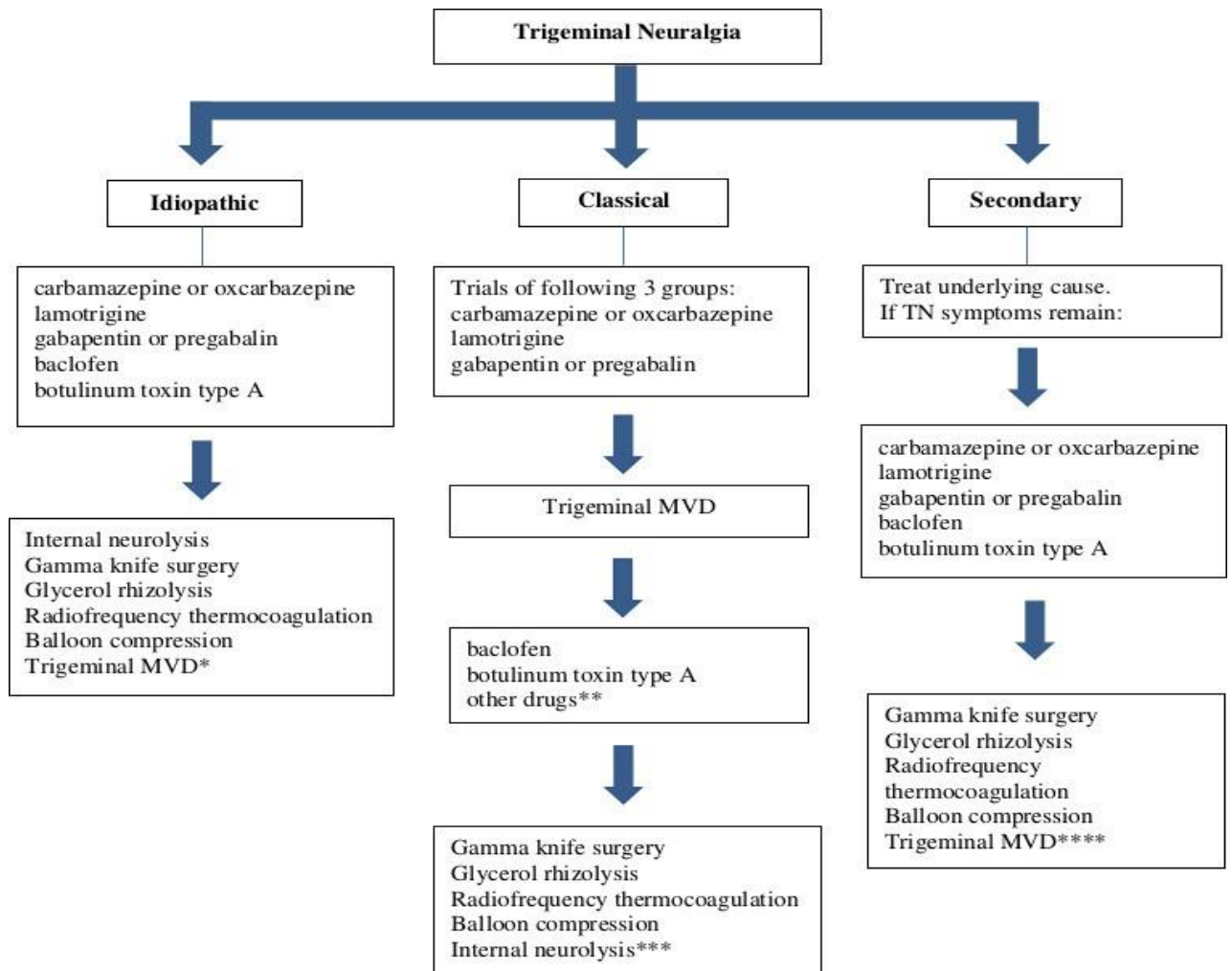


Figure 4 : Algorithm for the treatment of trigeminal neuralgia

VIII. SPECIAL CASES

Primary bilateral trigeminal neuralgia is a relatively rare disease accounting for 0,3%-0,6% of TN cases, which vascular compression is a common cause and microvascular decompression (MVD) is a safe and effective treatment.

Although, there is rare case of familial TN that patients are more likely to have bilateral disease, to present with earlier onset, and to become refractory to medical therapy and may require more aggressive approaches such as Stereotactic Radiosurgery (Neuroablative therapy).

IX. EVOLUTION AND PROGNOSIS

In case no treatment, trigeminal neuralgia progresses in a discontinuous manner towards worsening with both increasingly short remissions and frequent discharges during painful periods.

Untreated or poorly treated trigeminal neuralgia could result in weight loss, depression and can lead to suicide, thus referral to multidisciplinary teams with an interest conditions (neurosurgeons, neurologists pain specialist, psychologist).

X. COMPLICATION

The pain in trigeminal neuralgia is so severe and debilitating that the patients can develop depression if not adequately treated. Patients with severe pain associated with facial twitches can become socially withdrawn due to embarrassment and fear of an impending attack.

Many patients experience a refractory period after a paroxysmal attack where new attacks cannot be elicited. The pathophysiological mechanism of this phenomenon is unknown.

Microvascular decompression implies delicate posterior fossa surgery with partial exposure of the cerebellum, brainstem and cranial nerves, and therefore carries a risk of complications such as cranial nerve dysfunction, aggravation of pain, meningitis, leakage of cerebrospinal fluid or stroke.

XI. PREVENTION AND RECOMMENDATION

Prevention

Unfortunately, trigeminal neuralgia is not preventable, but patients can prevent painful episodes by avoiding certain activities that cause severe, intense pain. Applying a warm compress or heated bean bag may help to relieve pain.

Recommendation

Careful history taking should detect signs of symptomatic trigeminal neuralgia. If there are secondary causes of trigeminal neuralgia, MRI should be used. Carbamazepine or Oxcarbazepine are highly effective but need to be titrated up slowly because they cause significant side effects to the central nervous system. Surgical treatments are generally reserved for patients with debilitating pain refractory to pharmacological treatments.

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BELL'S PALSY

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I. DEFINITION

Bell's palsy is the term used to refer to cases of isolated peripheral facial paralysis that are idiopathic. A viral etiology (ie, activation of the herpes simplex virus) is suspected in most cases of Bell's palsy, although there is no established or widely available method of confirming a viral mechanism in clinical practice.

II. EPIDEMIOLOGY

The estimated annual incidence rate is between 12 and 50 cases per 100,000 population. There is no race, geographic, or sex predilection.

The risk is three times greater during pregnancy, especially in the third trimester and in the first postpartum week

Microvascular damage and inflammation associated with diabetes may account for approximately 5 to 10 percent of patients with Bell's palsy.

III. ETIOLOGY

The etiology of Bell's palsy is unclear, and this affect its treatment. The diagnosis of idiopathic Bell's palsy is based on exclusion. In many cases, herpes simplex virus activation is the suspected cause of Bell's palsy. Nevertheless, such patients with peripheral facial palsy are labeled as having Bell's palsy because there is no established or widely available method of confirming herpes simplex virus as the mechanism in clinical practice.

However, inflammatory changes that cause facial paralysis may also result from many identifiable causes, including infectious sources. When facial paralysis occurs due to a specific cause or is associated with additional signs or symptoms, the syndrome is named according to the etiology or symptom complex.

IV. PATHOPHYSIOLOGY

Bell's palsy occurs when inflammation at the facial nerve impairs peripheral function. The facial nerve becomes thickened with an edematous perineurium and diffuse infiltrates of small, round, inflammatory cells between nerve bundles and around intraneural blood vessels. Myelin sheaths undergo degeneration. These changes may be seen throughout the bony course of the facial nerve, although nerve damage is maximal in the narrow labyrinthine part of the facial canal where edema causes nerve compression, and the tenuous blood supply adds to the damage. The most common finding is a unilateral facial weakness that includes the muscles of the forehead.

Several suspected sources of inflammation at the facial nerve have been implicated in the pathogenesis of Bell's palsy.

- Herpes simplex – Herpes simplex virus activation has become widely accepted as the most frequent cause of Bell's palsy, though the evidence is not entirely conclusive. A herpes simplex-mediated viral inflammatory/immune mechanism was the subject of controversy for years but was suspected based upon serologic evidence.

- Other viral associations – An inflammatory response to other viral infections such as herpes zoster has also been associated with facial palsy. Other potential infectious sources of acute peripheral facial palsy include cytomegalovirus, Epstein-Barr virus, adenovirus, rubella virus, mumps, influenza, coxsackievirus, and severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2).

- Noninfectious mechanisms – Alternate postulated mechanisms of Bell's palsy include facial nerve ischemia.

An immune-mediated inflammatory response to some vaccines have been associated with Bell's palsy. However, the causal role of vaccines is uncertain as the reported case rate may be similar to the background rate of Bell's palsy.

A genetic predisposition has been implicated in some cases of Bell's palsy.

V. DIAGNOSIS

1. Clinic

Bell's palsy should be suspected in patients with acute-onset isolated unilateral facial weakness. The clinical diagnosis is made in patients with typical features:

- Weakness of the face that is unilateral and involves the lower (mouth, cheek) and upper (eye, forehead) muscles, with or without loss of taste on the anterior two-thirds of the tongue or altered secretion of the lacrimal and salivary glands. Classic findings include the eyebrow sagging, inability to close the eye, disappearance of the nasolabial fold, and drooping at the affected corner of the mouth, which is drawn to the unaffected side.
- Acute onset of symptoms, within one to three days. The course is progressive, reaching maximal clinical weakness/paralysis within three weeks or less from the first day of visible weakness. Recovery of some degree of function occurs within four months.

Features that are atypical for Bell's palsy include:

- Atypical clinical features
 - Bilateral acute facial weakness
 - Additional cranial neuropathies or other neurologic signs
 - Vesicles in ear canal
 - Systemic signs (eg, fever, prominent headache, stiff neck)
- Atypical temporal pattern
 - Sudden onset of symptoms at maximal severity (ie, no progression)
 - Insidious onset of symptoms (eg, over weeks to months)
 - Continued worsening of symptoms beyond three weeks
 - No improvement in symptoms within four months of onset

Patients with atypical features require diagnostic evaluation to evaluate for other causes for symptoms.

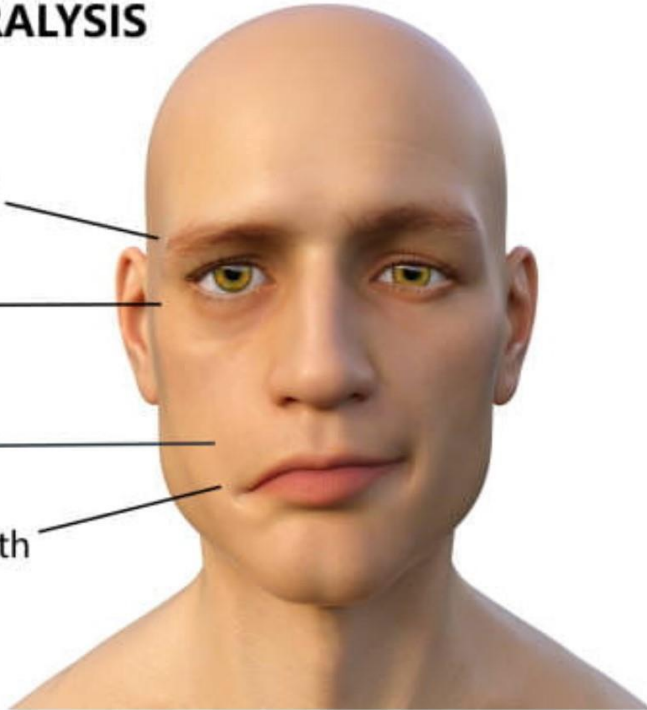
FACIAL NERVE PARALYSIS

Inability to wrinkle brow

Drooping eyelid;
inability to close eye

Inability to puff cheek;
asymmetrical smile

Drooping corner of mouth



2. Investigations

Diagnostic testing for atypical presentations — Diagnostic evaluation is warranted to exclude alternative causes for patients with features that are atypical for Bell's palsy. Testing is targeted to the specific clinical features and most likely diagnosis and may include neuroimaging, laboratory studies, cerebrospinal fluid analysis, and electrodiagnostic studies.

A. Imaging studies

- a. Magnetic resonance imaging – MRI of the brain with sequences to assess the facial nerve and parotid gland can delineate the soft tissue structures and identify inflammation and edema associated with Bell's palsy. It can also identify alternative causes to symptoms such as cerebral infarction or hemorrhage and neoplasm in the brain, facial nerve, or parotid gland. Herpes zoster can be identified by enhancement of the inner ear structures even without a vesicular eruption.
- b. Computed tomography – High-resolution computed tomography (CT) of the head and temporal bone with contrast may be performed as an alternative study for patients unable to undergo MRI or when bony pathology is suspected.

Repeat imaging – For patients with acute onset of facial paralysis and negative initial imaging studies who have continued complete flaccid paralysis at seven months, repeat imaging is warranted to evaluate for slow-growing neoplasm.

- B. Laboratory testing and other investigations — Other investigations, including laboratory testing, are typically reserved for patients with atypical presentations of facial weakness

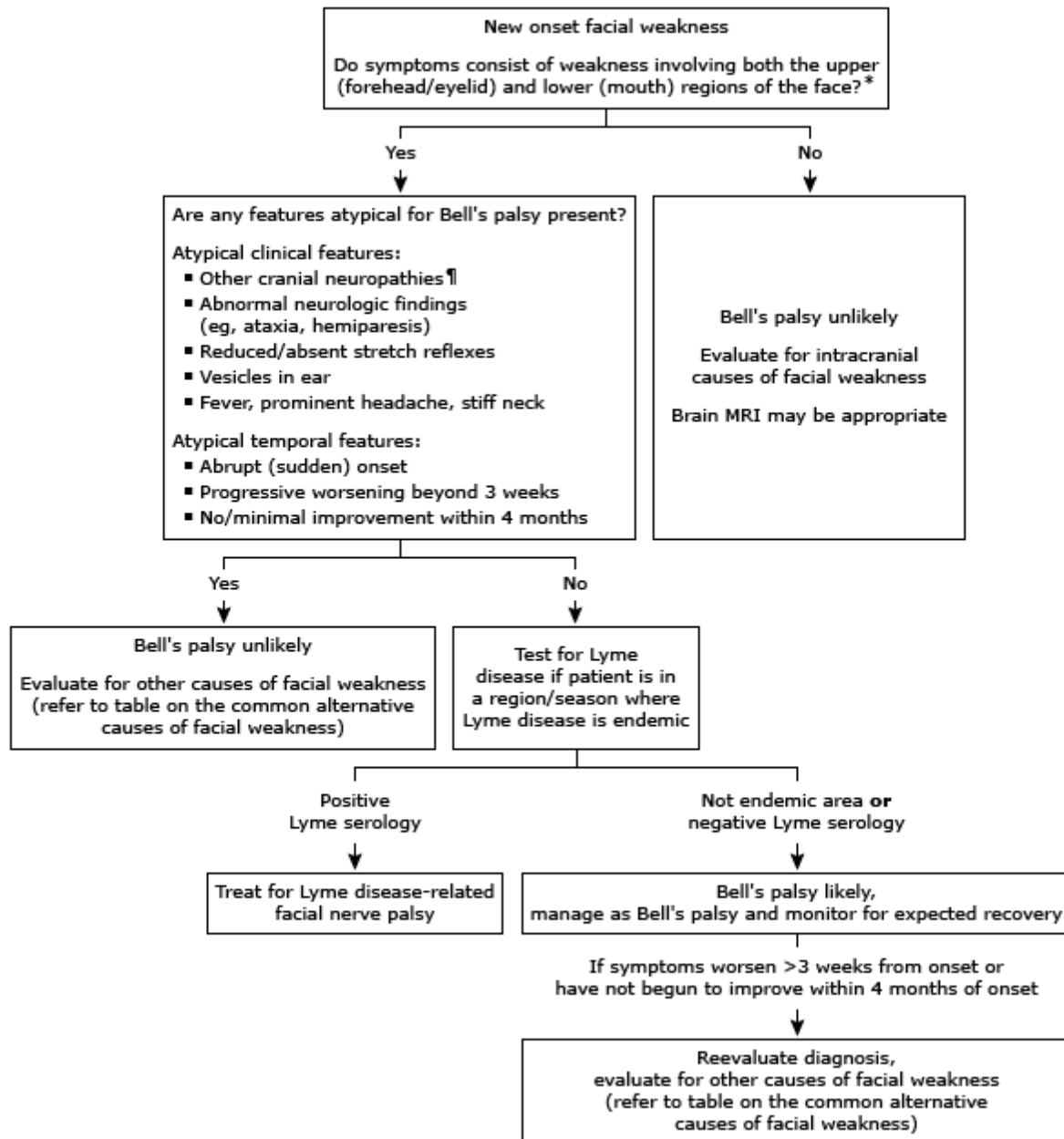
when imaging is nondiagnostic and for those who have features suggestive of an alternative cause to symptoms.

- a. Laboratory investigations for infectious and/or inflammatory conditions are guided by history and examination findings and may include erythrocyte sedimentation rate, high-sensitivity C-reactive protein, and blood cultures.
- b. Lumbar puncture for cerebrospinal fluid analysis may be used to identify meningitis, Guillain-Barré syndrome, or other inflammatory conditions such as sarcoidosis and herpes zoster.
- c. Parotid gland biopsy is warranted for patients with facial paralysis when there is no recovery by seven months and initial and repeat imaging studies have been nondiagnostic.

C. Electrodiagnostic studies

Electrodiagnostic studies are rarely performed for patients diagnosed with Bell's palsy but may be used to assess the severity of facial nerve injury to help determine the prognosis. They may also be used as part of the diagnostic evaluation of patients with facial weakness who also have signs or symptoms of polyneuropathy such as patients with Guillain-Barré syndrome. Incomplete return of facial motor function and synkinesis are predicted by a lack of early clinical improvement in complete facial paralysis and by results of electroneurography.

Evaluation of suspected Bell's palsy



VI. DIFFERENTIAL DIAGNOSIS

An acute isolated facial nerve palsy may be caused by a variety of disorders with a presentation similar to Bell's palsy. These include other conditions that can cause peripheral nerve dysfunction such as herpes zoster infection, Lyme disease, otitis media, Guillain-Barré syndrome (GBS), human immunodeficiency virus (HIV) infection, sarcoidosis, Sjögren's disease, or primary and metastatic tumors. In addition, stroke or other central (brain) conditions may produce a "peripheral" pattern of facial weakness mimicking Bell's palsy.

VII. MANAGEMENT

1. Objectives

Aim to maximize recovery speed of Bell's palsy and reduce the complications such as synkinesis and corneal complications.

2. Critical sign

Severity of Bell's palsy is measured by House Brackmann facial paralysis scale.

Electrodiagnostic studies can help in evaluation of severity after 1 week of onset of Bell's palsy.

3. Mortality risk factors

Typical Bell's palsy usually has good prognosis, more than two thirds of the patients have complete spontaneous recovery.

4. Outpatient management

Bell's palsy can be treated as outpatient basis after making accurate clinical diagnosis and rule out differential diagnosis for atypical clinical presentations.

5. First line treatment

The mainstay of pharmacologic therapy for Bell's palsy or facial nerve palsy is early short-term oral glucocorticoid treatment. In severe acute cases, combining antiviral therapy with glucocorticoids may improve outcomes. Eye care is essential for patients with incomplete eye closure.

Glucocorticoids for all patients — A short-term course of oral glucocorticoids is recommended for all patients with new-onset Bell's palsy. Treatment should ideally begin within three days of symptom onset, as this is the setting in which glucocorticoids have been studied and shown to be beneficial. We typically treat patients who are within seven days of symptom onset. Our suggested regimen is prednisone (60 to 80 mg/day) for one week or oral prednisolone 60 mg daily for five days followed by a five-day taper by 10 mg per day

Antiviral therapy for severe symptoms — Our suggested regimen for antiviral therapy in patients with severe Bell's palsy is valacyclovir 1000 mg three times daily for one week or Acyclovir 400 mg five times daily for 10 days.

Absent better data, we suggest coadministration of oral valacyclovir or acyclovir along with glucocorticoids for patients with severe facial palsy, defined as House-Brackmann (H-B) grade IV or higher. In patients with milder presentations, we use glucocorticoids alone, although some clinicians may reasonably choose to treat these patients as well. Antiviral therapy alone (without glucocorticoids) is not recommended.

Physical therapy – We do not routinely prescribe physical therapy for newly diagnosed Bell's palsy, since most patients will recover function spontaneously.

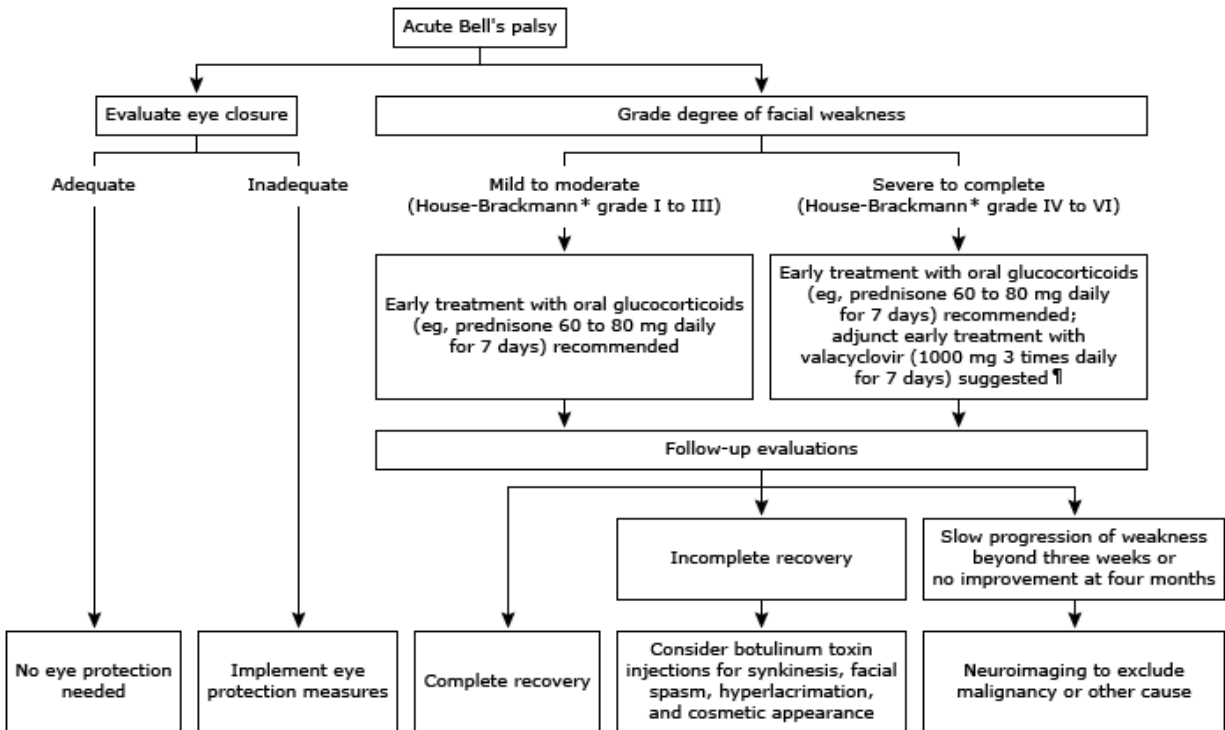
Surgical decompression (not recommended)

6. Treatment adjustment

In children, the dose of prednisone and antiviral agents must be adjusted for weight. The same treatment can be given during pregnancy, although the safety of antiviral agents in pregnancy has not been established. There is no need to taper the dose of prednisone after only one week of treatment.

Prednisone should be used cautiously in patients with diabetes, peptic ulcer disease, renal or hepatic dysfunction, or severe hypertension.

7. Algorithm



VIII. SPECIAL CASES

However, atypical presentations may also be seen in some patients with facial weakness.

- Lower facial weakness that spares the eyelid and forehead muscles is usually due to a central lesion but may also be due to a distal peripheral cause. As an example, a partial lesion of the distal facial nerve at the pes anserinus (between the deep and superficial lobes of the parotid gland) that spares the temporal branch to the frontalis muscle results in lower facial paralysis, but the patient is still able to wrinkle the forehead. Nonetheless, cases of facial weakness with forehead sparing should stimulate further evaluation of a possible central etiology.
- An ischemic stroke in the ventrolateral pons that involves the ipsilateral facial nerve tract or facial nucleus and distal projections can affect all peripheral facial nerve fibers and may result in upper and lower facial weakness.

IX. EVOLUTION AND PROGNOSIS

New or worsening symptoms — The weakness in patients with Bell's palsy typically evolves quickly to reach a nadir within one to two days of the onset of symptoms. The timing and pace of early recovery is variable and may wax and wane slightly, but the general course should be one of stability by three weeks and gradual improvement by two to three months.

Worsening facial weakness beyond three weeks of presentation and/or development of any new neurologic symptoms (eg, diplopia, facial numbness, ataxia) should prompt reevaluation for alternative etiologies.

Incomplete recovery by three to four months — Patients with incomplete recovery of facial function following Bell's palsy can have varying degrees of facial weakness,

hypertonia, and synkinesis, with functional problems related to incomplete eye closure, brow ptosis, and nasal valve collapse. Follow-up is essential for eye care, psychologic support, and management of long-term sequelae.

Recovery of facial nerve function — The prognosis of Bell's palsy is very good. Approximately 70 percent of patients are expected to recover spontaneously by three to six months, and rates of complete recovery increase to approximately 80 to 85 percent with glucocorticoid treatment. At the 6-month follow-up visit, we reevaluated the H-B grade in all patients and defined favorable outcome as an H-B grade of I or II and an H-B grade III or higher as an unfavorable outcome.

The main risk factor for incomplete recovery is the severity of weakness at presentation using the House-Brackmann (H-B) grading scale.

Nerve conduction studies (NCS) and electromyography (EMG) of the facial nerve can be used to help predict recovery in patients with severe weakness.

House-Brackmann Classification of Facial Nerve Dysfunction

Grade	Characteristics
I. Normal	Normal function in all areas
II. Mild dysfunction	Gross: - Slight weakness noticeable on close inspection - May have slight synkinesis - Normal symmetry and tone at rest Motion: - Forehead: Moderate to good function - Eye: Complete closure with minimal effort - Mouth: Slight asymmetry
III. Moderate dysfunction	Gross: - Obvious but not disfiguring difference between the two sides - Noticeable but not severe synkinesis, contracture, or hemifacial spasm - Normal symmetry and tone at rest Motion: - Forehead: Slight to moderate movement - Eye: Complete closure with effort - Mouth: Slightly weak with maximum effort
IV. Moderately severe dysfunction	Gross: - Obvious weakness and/or disfiguring asymmetry - Normal symmetry and tone at rest Motion: - Forehead: None - Eye: Incomplete closure - Mouth: Asymmetric with maximum effort
V. Severe dysfunction	Gross: - Only barely perceptible motion - Asymmetry at rest

	Motion: - Forehead: None - Eye: Incomplete closure - Mouth: Slight movement
VI. Total paralysis	No movement

X. COMPLICATIONS

Complications of regeneration — Synkinesis and contractures of the face develop in approximately 15 percent of patients as a late complication of Bell's palsy due to disorganized and misdirected nerve regrowth.

Most synkinesis after Bell's palsy is mild and may not be noticed by the affected individual. More severe or bothersome synkinesis can be treated with botulinum toxin injections.

Psychological impact — Facial disfigurement can cause psychosocial distress resulting in decreased quality of life and increased risk for depression and anxiety. Patients may suffer from poor self-confidence, phobic avoidance, and social isolation.

Risk of recurrent Bell's palsy — Recurrent attacks of Bell's palsy on either the ipsilateral or contralateral side have been observed in 7 to 15 percent of patients.

Pregnancy may be a risk factor for recurrence of Bell's palsy. Additional risk factors are not well defined. Some patients with recurrent attacks have a family history of multiple attacks, suggesting a genetic predisposition to Bell's palsy.

XI. PREVENTION AND RECOMMENDATION

There is no way to prevent Bell's palsy.

If the symptom of acute facial palsy occurs, the first step in evaluating patients is to determine whether the paralysis is central or peripheral. If it is peripheral without any apparent cause (and is thus Bell's palsy), and is diagnosed within one week after the onset of symptoms, as in the patient described in the vignette, no tests are indicated unless other cranial-nerve deficits develop (indicating more widespread disease), there is no recovery three to six weeks after the onset of symptoms, or a facial twitch or spasm preceded Bell's palsy (indicating continuous facial-nerve irritation suggestive of a tumor)

The collective findings of facial-nerve swelling, MRI changes consistent with inflammation, and available data regarding clinical outcomes support the use of a short course of prednisone within 2 to 14 days after the onset of symptoms.

Because Bell's palsy is associated with HSV infection, antiviral treatment may help, although data are lacking to show that such treatment speeds recovery or leads to a better long-term outcome.

Clinical recommendation	Evidence rating	Comments
Patients with Bell palsy should be prescribed oral corticosteroids.	A	Meta-analysis with high degree of certainty
Combination therapy with oral corticosteroids and antivirals should be considered in patients with Bell palsy to reduce rates of synkinesis.	B	Meta-analysis with moderate degree of certainty
Patients with Bell palsy should not be treated with antivirals alone.	A	Meta-analysis with high degree of certainty
Physical therapy should be offered to patients with severe paralysis (House-Brackmann grade V or VI) or persistent paralysis (more than three months).	B	Cochrane review of lower-quality studies; one high-quality randomized trial

A = consistent, good-quality patient-oriented evidence; **B** = inconsistent or limited-quality patient-oriented evidence; **C** = consensus, disease-oriented evidence, usual practice, expert opinion, or case series. For information about the SORT evidence rating system.

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NEUROPATHIC PAIN

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I. DEFINITION:

Neuropathic pain is now defined by the International Association for the Study of Pain (IASP) as ‘pain caused by a lesion or disease of the somatosensory nervous system’. This replaces the older definition of ‘pain initiated or caused by a primary lesion, dysfunction or transitory perturbation of the peripheral or central nervous system.

II. EPIDEMIOLOGY:

Neuropathic pain is associated with impaired quality of life, and is often poorly managed. Around 7–8% of adults have pain with neuropathic characteristics. A quarter of people with diabetes and 35% of people with HIV have neuropathic pain.

III. ETIOLOGY:

Classification of Neuropathic Pain According to site of Major Pathology

Table 1: Pathology Classification

Pathology	Peripheral	Spinal	Brain
Genetic	Fabry neuropathy	Syringomyelia	Syringobulbia
Metabolic	Painful diabetic neuropathy	B12 myelopathy	
Traumatic	Nerve injury	Spinal cord injury	Multiple sclerosis
Vascular	Vasculitic neuropathy	Spinal cord stroke	Brain stroke
Neoplastic	Tumor compression neuropathy	Tumor compression	Tumor compression
Immunological	Guillain-Barré syndrome	Multiple sclerosis	Multiple sclerosis
Infectious	HIV, Borreliosis	Infectious myelitis	Encephalitis
Toxic	Chemotherapy neuropathy		

III. PATHOPHYSIOLOGY:

Peripheral mechanisms

After an injury in the peripheral nervous system, chemicals are released from damaged cells and inflammatory cells. A partial list of these chemicals includes noradrenaline, bradykinin, histamine, prostaglandins, potassium, cytokines, NE, 5HT, and neuropeptides. These cellular mediators act to sensitize nociceptors to further neural input. This produces changes in the number and location of ion channels especially.

Central mechanisms

Peripheral mechanisms alone cannot explain all of the characteristics of neuropathic pain; central mechanisms play a critical role as well. A key process known as central

sensitization, which is the abnormal hyperexcitability of central nociceptor neurons, is thought to occur in the spinal cord because of peripheral injury and the release of tachykinins and neurotransmitters. Tachykinins include the neuropeptide substance P and neurokinin A.

V. DIAGNOSIS

1. Clinic

- Hyperesthesia, dysesthesia
- hyperpathia
- hyperalgesia or analgesic
- allodynia, alloesthesia
- electric-current and tingling
- Burn sensation
- Dysynchiria
- Pricking
- Pressing
- Throbbing
- Numbness

DN4 (Douleur Neuropathique en 4 Questions) is a screening tool for neuropathic pain consisting of interview questions (DN4-interview) and physical tests. Scores $\geq 4/10$ indicate neuropathic pain.

DN4 – QUESTIONNAIRE

To estimate the probability of neuropathic pain, please answer yes or no for each item of the following four questions.

INTERVIEW OF THE PATIENT

QUESTION 1:

Does the pain have one or more of the following characteristics?	YES	NO
Burning	☐	☐
Painful cold	☐	☐
Electric shocks	☐	☐

QUESTION 2:

Is the pain associated with one or more of the following symptoms in the same area?	YES	NO
Tingling	☐	☐
Pins and needles	☐	☐
Numbness	☐	☐
Itching	☐	☐

EXAMINATION OF THE PATIENT

QUESTION 3:

Is the pain located in an area where the physical examination may reveal one or more of the following characteristics?	YES	NO
Hypoesthesia to touch	☐	☐
Hypoesthesia to pinprick	☐	☐

QUESTION 4:

In the painful area, can the pain be caused or increased by:	YES	NO
Brushing?	☐	☐

YES = 1 point
NO = 0 points

Patient's Score: /10

2. Investigation

Evaluating patients with neuropathy involves a detailed history and physical, including a review of current and past medications. Although there are no standard laboratory or imaging studies to test for peripheral neuropathies, the following studies may aid in the diagnosis and

help narrow down the underlying cause of the neuropathy (eg, inflammatory, infectious, metabolic):

- Complete blood count (macrocytic anemia may clue the clinician to vitamin B12 or folate deficiency, or even alcohol abuse)
- Metabolic panels such as basic metabolic panel or complete metabolic panel (look for electrolyte imbalances contributing to neuropathy and renal failure as uremia can also lead to neuropathy).
- Hemoglobin A1c Testing (Diabetes is a common cause of neuropathy)
- Testing for vitamin and mineral deficiencies such as copper, thiamine, pyridoxine, folate, B12, and vitamin E play fundamental roles in nervous system development and maintenance.
- Heavy metal toxicities such as mercury, lead, and arsenic are known to cause peripheral nerve toxicities along with CNS disturbances
- Infectious workup for Lyme disease, Epstein-Barr virus, hepatitis C, HIV, and syphilis as a long-standing disease may manifest with peripheral neuropathies and paresthesias
- Thyroid function testing
- Antibody testing for specific autoimmune diseases known to cause peripheral neuropathies, such as Sjögren syndrome, lupus, and rheumatoid arthritis
- Nerve conduction study and electromyography
- MRI or CT scans in cases where compression of the nerve is of concern
- Nerve biopsy
- Genetic testing (for inherited neuropathies)
- Urine Test (looking for Bence-Jones proteins as multiple myeloma and its treatments can cause peripheral neuropathy)

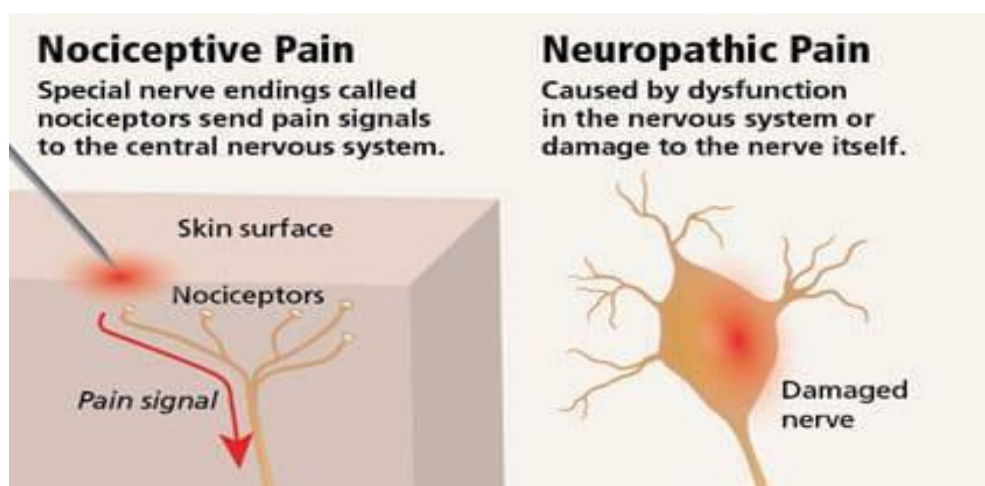
VI. DIFFERENTIAL DIAGNOSIS

Pain can be categorized along a variety of dimensions, including one of the most important divisions, nociceptive versus neuropathic pain (NP).

Nociceptive pain results from activity in neural pathways secondary to actual tissue damage or potentially tissue-damaging stimuli.

NP is chronic pain that is initiated by nervous system lesions or dysfunction and can be maintained by a number of different mechanisms.

Three common conditions that are often associated with acute and chronic NP are painful diabetic peripheral neuropathy (DPN), painful postherpetic neuralgia (PHN), and cancer.



VII. MANAGEMENT

1. Objectives

- Relieving nerve pain
- Improve balance and mobility
- Delay disease progression to disability
- Manage complications and restore function

2. Critical sign

- Failure to achieve adequate control of dysaesthesia
- Difficulty with balance and coordination.
- Acute onset
- Progressive symptoms and signs (days – weeks)
- Sphincter disturbance (cord pathology likely)
- Associated respiratory compromise

3. Mortality risk factor

- Neuropathic pain is rarely fatal. However, it can cause serious complications if left untreated. These complications may affect a person's life expectancy.
- Risk of developing potentially serious complications, such as a foot ulcer that becomes infected. This can lead to gangrene if untreated, and in severe cases may mean the foot has to be amputated.

4. Outpatient management

- History of patient symptoms, lifestyle, and social circumstances.
- Examination of the foot, the status of the skin (e.g. absence of sweating and presence of ulceration) immobility of joints, gait and footwear should be noted.
- Blood test
- Simple tests should be performed to assess peripheral sensation, including sensation to pinprick, light touch, vibration, pressure, and ankle reflexes should be checked.
- Requires periodic reevaluation to rule out other treatable underlying medical conditions, patient education, and reassurance.
- Lifestyle changes including dietary assistance, smoking cessation, weight loss, and regular exercise programs should be suggested.

5. First-line treatment

Table 2: Firstline Medications

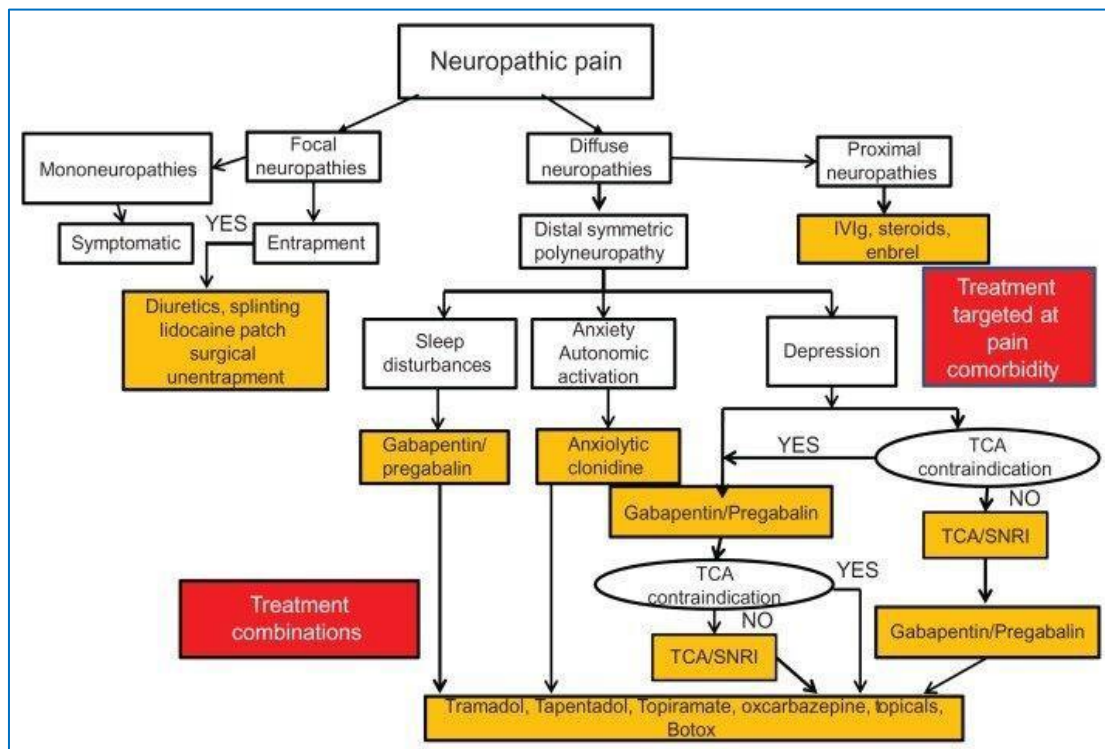
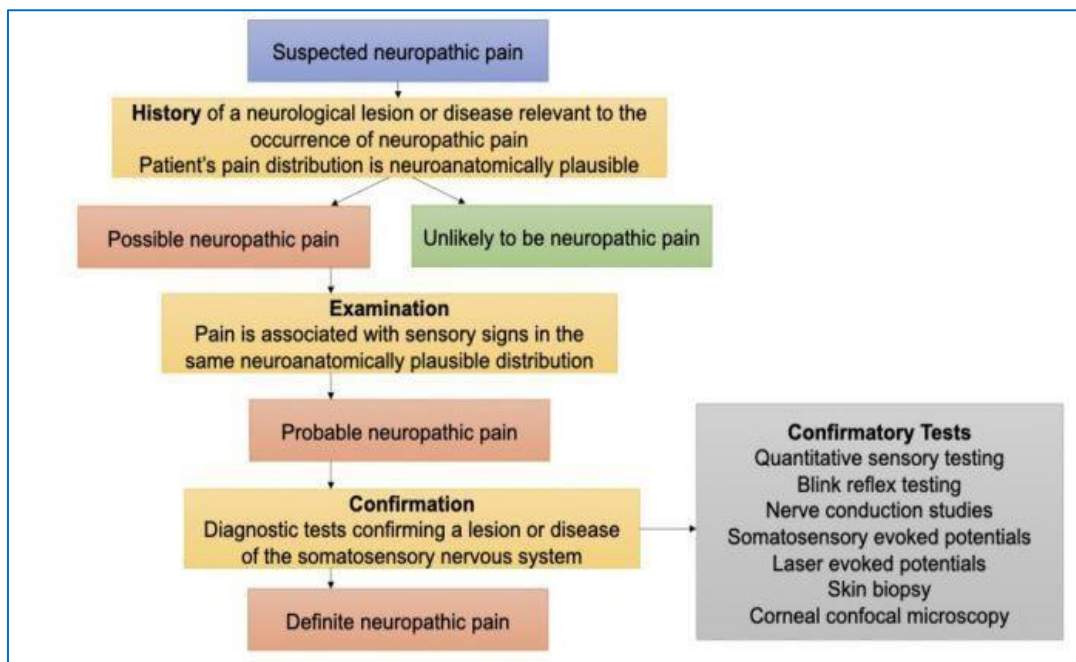
Drug Class	Drug	Recommendations	Cautions
Gabapentinoids	Gabapentin	Slow titration up to 600 mg PO TID. Max daily dose = 3600 mg.	Reduce dose for renal impairment
	Pregabalin	Start at 150 mg PO BID or TID. Max daily dose = 600 mg.	Reduce dose for renal impairment

Serotonin and norepinephrine reuptake inhibitors	Duloxetine	Start at 30 mg PO daily. Max daily dose = 60 mg.	Renal or liver disease
	Venlafaxine	Start at 37.5 mg PO daily. Max daily dose = 225 mg.	Renal or liver disease
Tricyclic antidepressants	Nortriptyline	Start at 10–25 mg PO QHS. Max daily dose = 150 mg.	Autonomic neuropathy, urinary retention, glaucoma
	Amitriptyline	Start at 10–25 mg PO QHS. Max daily dose = 150 mg.	Taking SNRI, SSRI, MAOI, and/or tramadol
Topicals (focal neuropathic pain)	5% lidocaine	Available in cream or patch. Apply to site of pain 12 hours on, 12 hours off. Max of three patches at one time.	Avoid in diabetic peripheral neuropathy
	8% capsaicin	Apply for 60 minutes under supervision of a physician.	Avoid in diabetic peripheral neuropathy
Combination therapy	Gabapentinoid + TCA	Only use if single agent provides inadequate relief and no adverse effects.	Avoid in elderly
	Gabapentinoid + SNRI	Titrate as indicated for single agent. Aim for lower doses of both.	Avoid in elderly

6. Treatment adjustment

- Psychotherapy using cognitive-behavioral approaches
- Physiotherapy
- Occupation therapy
- Surgical therapy can release of distal lower limb nerves is not recommended for Pain diabetic sensorimotor polyneuropathy and can result in severe complications of foot and ankle surgery. If mononeuropathy due to entrapment is present, surgical intervention is a consideration following neurological consultation and/or nerve conduction studies.
- Noninvasive Cortical Stimulation: Transcranial magnetic stimulation (TMS), Transcranial direct current stimulation (tDCS), Transcranial random noise stimulation (tRNS), Cranial electrotherapy (CES), Transcranial low-intensity focused ultrasound (T-LIFU) or subablative sonication.
- Invasive Cortical Stimulation
- Deep Brain Stimulation
- Spinal cord stimulation
- Neurofeedback
- Transcutaneous electrical nerve stimulation
- Acupuncture
- Others

7. Algorithm



VIII. SPECIAL CASES

- Painful Somatosensory Seizures: Pain, as an aura or a symptom of epileptic seizures, appears to be rare (0.2% in a two recent large series).
- Parkinson's Disease (PD), Dystonia, and Corticobasal Syndrome: It affects the body side with predominant motor symptoms but also the unaffected side.
- Poststroke Shoulder Pain: typically develops 2–3 months after stroke and persists for more than 1 year in 65% of the patients. Its etiology is unknown, but local factors

(spasticity, glenohumeral subluxation, capsular inflammation), peripheral neuropathy, central poststroke pain (CPSP), and autonomic dysfunction have been related.

- Creutzfeldt–Jakob Disease: which affects younger subjects, presents usually with predominantly rapidly progressive psychiatric symptoms followed initially by sensory complaints like pain and dysesthesia and later by rapidly progressive neurological symptoms and signs.
- Burning Mouth Syndrome: characterized by spontaneous burning pain arising from a visibly intact oral mucosa, normal findings in clinical examination, and no identifiable medical or local dental cause.
- Headache Associated with Pontine Infarction: lesions of pain-processing: structures and descending pain pathways may play a role in the development of acute stroke-associated headache similar to other Central post-stroke pain.

IX. VOLUTION AND PROGNOSIS

In those diseases where peripheral nerves suffer damage through Wallerian or axonal degeneration, the prognosis is poorer, as the recovery of the nerve is more challenging. The axon must regenerate itself and reinnervate the affected muscle or organ for clinical improvement.

The prognosis of diseases that occur secondary to segmental demyelination is more favorable because remyelination is achieved more quickly, allowing the return of function of the axon.

X. COMPLICATIONS

Complications of peripheral neuropathies include pain, altered sensation, muscle atrophy, and weakness. Diabetic peripheral neuropathy is infamous for complications, including foot ulcers, which can lead to gangrenous digits and limbs, sometimes progressing to amputation.

XI. PREVENTION AND RECOMMANDATION

- Always wear shoes
- Keep floor clean
- Check temperature before bathing with elbow
- Don't stay in one position for long period
- Massage,
- Yoga, meditations, exercises etc
- Avoiding alcohol and smoking
- Controlling blood sugar level

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PERIPHERAL NEUROPATHY

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I. DEFINITION

Peripheral neuropathy happens when the nerves that are located outside of the brain and spinal cord (peripheral nerves) are damaged. This condition often causes weakness, numbness and pain, usually in the hands and feet. It also can affect other areas and body functions including digestion and urination.

II. EPIDEMIOLOGY

The prevalence of peripheral neuropathy in the general population ranges from 1% to 7%, with higher rates among those older than 50 years. The most common identifiable causes of peripheral neuropathy include diabetes mellitus, nerve compression or injury, alcohol use, toxin exposure, hereditary diseases, and nutritional deficiencies.

Peripheral neuropathy affects 25% to 50% of patients with diabetes, depending on factors such as the patient's age, number of years with diabetes, and level of diabetes control. Peripheral neuropathy is idiopathic in 25% to 46% of cases, and this is more common with increasing patient age.

Risk factor

- Diabetes, especially if your sugar levels are not controlled well.
- Alcohol misuse.
- Low levels of vitamins in the body, especially vitamin B-12.
- Infections, such as Lyme disease, shingles, hepatitis B and C, and HIV.
- Autoimmune diseases, such as rheumatoid arthritis and lupus, in which the immune system attacks your own tissues.
- Kidney, liver or thyroid disorders.
- Exposure to toxins.
- Repetitive motion, such as those performed for certain jobs.
- Family history of neuropathy.

III. ETIOLOGY

Most peripheral neuropathies fall into three etiologic categories, namely hereditary, toxic/metabolic, and those associated with systemic disease.

1. Hereditary

This is a large group of disorders in which the onset of symptoms is insidious and progression is indolent over years or decades. Three of these hereditary neuropathies will be discussed:

- **Charcot-Marie-Tooth Disease** (Hereditary Sensory-Motor Neuropathy [HSMN I]).
This is the most common hereditary neuropathy that has an autosomal dominant pattern

of inheritance. Phenotypic expression is often variable, such that affected family members of a proband may have no symptoms and minimal neurologic findings. Characteristic clinical findings include striking atrophy of the calves, resulting in an inverted "champagne-bottle" appearance to the lower extremities. Peripheral nerves are often palpably enlarged. Large fiber sensory loss is present, with a marked reduction in vibratory perception and proprioception. Ankle jerk reflexes are lost. Since this is a demyelinating polyneuropathy, nerve conduction velocity measurements are characteristically slow, at approximately 50% of normal values.

- **Dejerine-Sottas Disease** (HSMN II). This is a rare pediatric disorder with autosomal recessive inheritance that causes severe weakness and numbness, markedly enlarged peripheral nerves with "onion-bulb" formation and markedly slowed conduction velocities.

- **Refsum's Disease** (HSMN III). This autosomal recessive disorder is caused by an enzymatic defect that results in accumulation of phytanic acid. The clinical triad includes peripheral neuropathy, retinitis pigmentosa and dry, scaly skin. Treatment includes dietary restriction of phytanic acid and plasmapheresis.

2. Toxic/Metabolic

Numerous drugs and toxins can cause peripheral neuropathy. A partial list follows:

- **Drugs:** amiodarone, cis-platinum, dapsone, INH, phenytoin, pyridoxine, vincristine, nitrofurantoin, dDI, ddC.

- **Toxins:** heavy metals including mercury, arsenic, lead, zinc and thallium; alcohol; and the organophosphates.

3. Neuropathy Associated with Systemic Diseases

- Numerous systemic diseases are associated with neuropathy. Among the most common systemic disorders are: uremia; porphyria; vitamin B12 deficiency; amyloidosis; hypothyroidism; lymphoma, including Hodgkins' disease; multiple myeloma; cryoglobulinemia; vasculitis, including systemic lupus erythematosus (SLE), rheumatoid arthritis and polyarteritis nodosa; sarcoidosis.

- A purely sensory neuropathy can be seen with several carcinomas, especially oat cell carcinoma of the lung.

- Four systemic infections have a high incidence of neuropathy, including leprosy, syphilis, HIV and diphtheria.

- Diabetes mellitus is perhaps the most common cause of neuropathy in the United States. Both symmetric and asymmetric diabetic neuropathies can occur, as follows:

- Symmetric polyneuropathies: These are the most common and include a sensory/motor polyneuropathy and an autonomic neuropathy.
- Asymmetric neuropathies are less common. Mononeuropathy multiplex results in simultaneous dysfunction of several peripheral nerves, and is due to ischemic infarction of the vasa nervorum. Cranial neuropathies, truncal radiculopathies and diabetic amyotrophy (ischemic infarction of the lumbosacral plexus) are other forms of asymmetric neuropathies. Entrapment neuropathies, including carpal tunnel syndrome, are also commonly seen in diabetics.

Distribution

Nerve damage in peripheral neuropathy may be symmetrical generalized, multifocal or focal.

- **Symmetrical generalized polyneuropathies** produce signs and symptoms in a distal-to-proximal gradient, the so-called "stocking-glove" pattern. The reason for this is that the "offending agent" causing the neuropathy affects protein synthesis in the cell body of the peripheral nerve. Hence, neuronal dysfunction will first occur in the distal portions of the longest axons, and thus produce symptoms of weakness and numbness in the most distal portions of the extremities, i.e. the feet and hands.

- **Multifocal Neuropathies (Mononeuropathy Multiplex):** Patients with these forms of neuropathy develop more-or-less simultaneous dysfunction of several peripheral nerves. The underlying pathologic mechanism is felt to be ischemic infarction of the vasa nervorum due to vasculitis, as can occur with SLE, rheumatoid arthritis, polyarteritis nodosa and diabetes mellitus. These neuropathies are frequently painful and cause profound weakness. Prognosis for recovery is good, assuming that the underlying disease process leading to nerve infarction can be suppressed.

- **Focal Neuropathies (Mononeuropathies):** Traumatic injuries and entrapment of peripheral nerves at the usual sites of compression are the most common causes of focal mononeuropathy. The most frequently seen entrapment neuropathies include:

- Compression of the median nerve across the wrist (carpal tunnel syndrome)
- Compression of the ulnar nerve across the elbow (tardy ulnar palsy)
- Compression of the radial nerve at the spiral groove (Saturday night palsy)
- Compression of the peroneal nerve at the fibular head (peroneal nerve palsy)
- Compression of the distal branches of the tibial nerve at the ankle (tarsal tunnel syndrome)

IV. PATHOPHYSIOLOGY

There are three major pathologic mechanisms causing peripheral neuropathy: distal axonopathy, myelinopathy, and neuronopathy.

Distal Axonopathy:

In this form of neuropathy, a metabolic abnormality causes failure of protein synthesis and axonal transport, resulting in degeneration of distal regions of axons. For this reason, axonal neuropathies characteristically produce a "stocking-glove" distribution of numbness and weakness.

Small-diameter axons are most susceptible to metabolic injury because of their small neuronal size and lack of "reserve". Hence, initial symptoms of an axonal neuropathy typically include autonomic dysfunction and small-fiber sensory modalities, including loss of pain and temperature perception, since these modalities are subserved by small, unmyelinated or thinly myelinated axons.

Myelinopathy:

An immune-mediated attack on peripheral nervous system myelin is the characteristic pathologic change in this group of neuropathies. Guillain-Barre syndrome (GBS) and chronic inflammatory demyelinating polyneuropathy (CIDP) are the two most common forms of demyelinating polyneuropathy.

GBS is a monophasic, immune-mediated demyelinating neuropathy that frequently follows a viral infection and causes an acute and frequently severe progression of weakness and numbness over several weeks. CIDP is a chronic demyelinating polyneuropathy that can have a slowly progressive or a relapsing course. In both of these neuropathies antibodies have been found that cross-react with peripheral nerve myelin. Elevated CSF protein and slowed nerve conduction velocities are characteristic of the demyelinating neuropathies.

In general, demyelinating neuropathies affect large-diameter, myelinated axons at the start of the illness, and hence produce significant motor weakness and large-fiber sensory loss, including loss of vibratory perception and proprioception.

In diphtheritic neuropathy, the bacterium produces a toxin that inhibits Schwann cell synthesis of myelin constituents, producing severe weakness and large-fiber sensory loss.

Neuronopathy:

Selective involvement of the cell bodies of motor, sensory and autonomic nerves is the hallmark of this group of neuropathies.

- **Somatic motor neuronopathies** result from isolated involvement of the anterior horn cells. Amyotrophic lateral sclerosis and the spinal muscular atrophies are two examples of somatic motor neuronopathies.

- **Somatic sensory neuronopathies** result from disruption of the metabolism of sensory nerve cell bodies, followed by degeneration of their processes. Special permeability of the blood vessels in the dorsal root and Gasserian ganglia make these neurons particularly vulnerable to certain toxins. Two common examples of somatic sensory neuronopathies include the paraneoplastic subacute sensory neuropathy seen with oat-cell carcinoma of the lung, and the sensory neuronopathy associated with Sjogren's syndrome.

- **Autonomic Neuronopathy:** This unusual group of neuropathies results from isolated involvement of post-ganglionic autonomic neurons and causes idiopathic orthostatic hypotension.

Modality

Peripheral neuropathies can be sub-classified based on their involvement of motor, sensory or autonomic neurons.

- **Modality-Specific Neuropathies:** The somatic motor, somatic sensory and autonomic neuropathies described above are examples of modality-specific neuropathies. The pathologic lesion in this group of neuropathies is confined to the cell bodies.

- **Mixed-Modality Neuropathies:** The majority of peripheral neuropathies is not modality-specific, and includes various combinations of motor, sensory and autonomic dysfunction. The reason for this finding is that most peripheral nerves include a mixture of motor, sensory and autonomic axons. Hence, axonal neuropathies typically present with mixed symptomatology. Likewise, since most axons are myelinated to a greater or lesser extent, demyelinating neuropathies also produce a mixture of motor, sensory and autonomic symptoms.

V. DIAGNOSIS

1.Clinic

Symptoms

Since the peripheral nervous system consists of motor, sensory and autonomic nerves, symptoms can fall into all three of these categories.

- Sensory symptoms include distal dysesthesias, pain and numbness. A characteristic pattern of numbness is one in which the distal portions of the nerves are first affected, the so-called "stocking-glove" pattern. This pattern occurs because nerve fibers are affected according to length of axon, without regard to root or nerve trunk distribution.

- Motor symptoms include weakness, which once again is distal, and typically involves extensor groups rather than flexor groups of muscles.

- Autonomic dysfunction is common and includes orthostasis, impotence in males and gastroparesis.

Signs

Signs of peripheral neuropathy also include sensory, motor and autonomic components.

- Sensory disturbance is manifest as distal loss of pin, temperature and vibratory perception as well as proprioception. Initial signs are frequently confined to the toes and feet. A positive Romberg sign is frequently present due to proprioceptive loss in the lower extremities.
- Motor signs include distal weakness, primarily in extensor groups, and most prominent in the lower extremities initially. Distal muscles are often atrophic, and one should carefully assess the bulk of the extensor digitorum brevis muscles in the feet and of the intrinsic muscles of the hands. Muscle tone is reduced and often is flaccid.
- Muscle stretch reflexes are frequently lost, and most patients with peripheral neuropathy have absent ankle jerks as one of the first signs of the disorder.
- The most prominent autonomic sign of neuropathy is orthostatic hypotension.

2. Investigation

Laboratory studies play an important role in diagnosing and categorizing the peripheral neuropathies. Electrodiagnostic studies are helpful in quantitating the neuropathy, while blood and urine studies are helpful in identifying an etiology.

Electrodiagnostic Studies

Nerve Conduction Study:

The recording and measurement of the compound nerve and muscle action potential elicited in response to a single supramaximal electrical stimulus, to measure the terminal latency, amplitude and duration of the evoked potential, as well as the conduction velocity.

Nerve conduction studies are helpful in documenting that a neuropathy exists, quantitating the severity, and noting the distribution of the neuropathy, i.e. whether it is distal, proximal or diffuse. In addition, nerve conduction studies can provide information on the modality involved, i.e. motor versus sensory, and can also give clues as to the underlying pathology, whether axonal or demyelinating.

Demyelinating neuropathies (neuropathies due to loss or destruction of myelin) result in slowed conduction velocities and prolonged distal latencies, because conduction velocity is proportional to the velocity of the largest-diameter myelinated fibers. Dispersion of evoked compound action potentials (CAP) can also be seen in demyelinating neuropathies, because all of the action potentials elicited in response to a single electrical stimulus will not reach the recording potential at the same time. Severe demyelinating neuropathies can also produce conduction block, which is a major decrease in amplitude of the muscle CAP upon proximal stimulation of its nerve as compared to distal stimulation.

Axonal neuropathies (neuropathies due to loss of axons or their cell bodies) generally result in a reduced amplitude of the compound motor or sensory nerve action potentials.

Electromyography (EMG):

The recording and study of insertional, spontaneous, and voluntary electrical activity of muscle.

This test allows one to physiologically evaluate the motor unit, including the anterior horn cell, peripheral nerve, and muscle. EMG is helpful when evaluating patients with weakness, in that it can help one determine whether weakness is due to anterior horn cell

disease, nerve root compression, peripheral neuropathy, or an intrinsic disease of muscle itself (myopathy).

EMG can differentiate acute denervation from chronic denervation, and may thus give an indication as to the time course of the lesion causing the neuropathy.

- **Acute Denervation:** Fibrillations and positive waves are present indicating spontaneous discharge of individual muscle fibers.

- **Chronic Denervation:** Voluntary motor unit potentials are of large amplitude and long duration, and are frequently polyphasic, because the motor units are enlarged as a result of re-innervation of adjacent previously denervated muscle fibers. Recruitment of additional motor units in response to increasing the force of muscular contraction is reduced for the same reason.

- **Demyelinating Neuropathy:** A decreased recruitment pattern is seen, since demyelination interferes with conduction of individual action potentials along the course of a peripheral nerve. Because denervation and reinnervation of muscle fibers are not features of demyelinating neuropathies, the configuration of the voluntary motor unit potentials is usually normal, and fibrillation potentials are not seen.

Nerve Biopsy

There are few indications for nerve biopsy when evaluating peripheral neuropathy. In general, a nerve biopsy is performed to evaluate asymmetric, multi-focal neuropathies. The sural nerve is frequently biopsied, since this is a purely sensory nerve that is easily obtained. In the upper extremity, the superficial radial nerve may be biopsied if necessary. The nerve specimen is typically evaluated by means of light and electron microscopy. Semi-thin plastic embedded sections, stained with toluidine blue, are helpful for evaluating the myelin sheaths. Teased nerve fiber preparations are also helpful to look for demyelination and remyelination.

Sural nerve biopsies are particularly helpful when evaluating patients with a clinical picture of mononeuropathy multiplex, the basis of which is still unclear after other laboratory investigations are complete. Vasculitis, amyloidosis, leprosy and sarcoidosis can be accurately diagnosed by means of nerve biopsy.

Performing a nerve biopsy routinely in the evaluation of symmetric, distal polyneuropathies is usually fruitless, in that the pathologic diagnosis most often reveals "chronic neuropathy with mixed axonal-demyelinating features", a non-specific finding of little clinical benefit.

Blood Studies

Routine blood studies should be obtained in all patients with peripheral neuropathy in order to screen for reversible causes. The following blood tests are recommended:

- Complete blood count
- Chemistry profile
- Sedimentation rate
- Thyroid studies
- Vitamin B12 level
- ANA, rheumatoid factor
- Serum protein electrophoresis, serum immuno-electrophoresis
- RPR and HIV (if the clinical situation warrants)

Urine Studies

The following studies are recommended to screen for reversible causes of neuropathy:

- Heavy metal screen (Hg, Pb, Zn, As)
- Urine protein electrophoresis, urine immuno-electrophoresis
- Watson-Schwartz test (qualitative test for porphobilinogen)

Chest x-ray, helpful to screen for asymptomatic lung cancer that can sometimes cause a purely sensory neuropathy.

VI. DIFFERENTIAL DIAGNOSIS

- Radiculopathy
- Neuritis (eg, brachial neuritis)
- Fibromyalgia
- Compartment syndrome
- Peripheral arterial disease
- Vasculitis
- Spinal cord tumor (eg, meningioma, medulloblastoma, metastases, ependymoma)
- Guillain-Barré syndrome
- Sensory ganglionopathy

VII. MANAGEMENT

1. Objective

- Manage the condition causing your neuropathy
- Improve symptoms.

2. Critical sign

Most commonly with sepsis, respiratory failure, multisystem organ failure, or septic inflammatory response syndrome (SIRS)

3. Mortality risk factor

The risks of all-cause and cardiovascular mortality were significantly higher for adults with diabetes and PN compared to those without PN. Adults with diabetes are known to have a 2–4 times higher cardiovascular risk compared to adults without diabetes

4. Outpatient management

- Foot care
- Physical therapy (Gait and balance exercises)
- Rehabilitation Therapy (Ankle supports,..)
- Occupational Therapy

5. First-line treatment

Neuropathy Associated With Systemic Illness

Treatment of the systemic illness frequently results in improvement in neuropathic symptoms. Since nerves regenerate slowly, at a rate of about one mm per day, recovery is often prolonged and may take months to years.

Immune-mediated Neuropathies

The immune-mediated neuropathies include those associated with vasculitis (SLE, rheumatoid arthritis or polyarteritis nodosa), and the immune-mediated demyelinating neuropathies (Guillain-Barre syndrome and CIDP).

Corticosteroids:

By virtue of their immunosuppressive effects, corticosteroids have been found to be effective in treating the vasculitic neuropathies as well as CIDP. Corticosteroids may be given daily or on an every-other-day regimen, depending on the severity and tempo of the disease. The typical initial dose of corticosteroids for the treatment of these neuropathies is 1 mg/kg/day of prednisone (or 2 mg/kg every-other-day).

Corticosteroids have never been proven to be effective in treating Guillain-Barre syndrome, and hence are not recommended for this disorder.

Immunosuppressives:

Azathioprine, cyclophosphamide, cyclosporine, mycophenolate and methotrexate are all used in the treatment of autoimmune diseases. Each of these medications has potential serious toxicity.

Azathioprine is frequently used, in combination with steroids, to treat autoimmune neuropathies because of its steroid-sparing effect. Concurrent use of these two medications allows corticosteroids to be tapered more quickly and more completely once the neuropathy is brought under control.

The usual dosage of azathioprine is 2-3 mg/kg/day given in divided doses. This medication has a delayed beneficial effect and several months are required before an effect may be seen. The erythrocyte mean cell volume (MCV) provides an index of therapeutic effect: a slight elevation in MCV suggests that the dose of azathioprine is appropriate.

Azathioprine is metabolized by xanthine oxidase, and medications that block this enzyme, such as allopurinol, should be avoided since concurrent use can cause azathioprine toxicity.

Plasmapheresis:

In this procedure, blood is removed from the patient, plasma is separated from blood cells and discarded, and blood cells are resuspended in colloid solution and reinfused. Plasmapheresis is effective in treating patients with immune-mediated neuropathies such as those caused by cryoglobulins, SLE, rheumatoid arthritis or polyarteritis nodosa. In addition, individuals with GBS and a large subset of patients with CIDP benefit from this procedure.

The effects of plasma exchange can be summarized as "fast, temporary, and expensive". This procedure does not, by itself, induce permanent remission, and the ideal application is therefore in acute, self-limited disorders such as GBS. Plasma exchange may also have application in chronic diseases, such as CIDP, where rapid therapeutic effects may occasionally be required.

Intravenous Immunoglobulin (IVIg):

IVIg is pooled, human IgG in a form that is safe for intravenous administration. How IVIg affects immunologic function is unknown, but three putative mechanisms have been postulated:

- Administration of IVIg floods the recipient with an enormously diverse array of antibody molecules, some of which are anti-idiotypic antibodies which may neutralize auto-antibodies in the patient, thereby increasing their clearance and perhaps down-regulating their production.
- IVIg inhibits the binding of activated complement to target cells, thus reducing complement-mediated damage to cell membranes.
- IVIg infusion is a potent stimulus that down-regulates immunologic production.

To date, IVIg has been found to be as effective as plasma exchange in treating many immunologic disorders, including Guillain-Barre syndrome and CIDP. IVIg is also reported to be beneficial in treating the polyneuropathy associated with IgG or IgM paraproteins.

Side effects of IVIg include fever, myalgia, headache, rash, and occasionally aseptic meningitis and renal failure. The cost of IVIg is equivalent to that of plasma exchange.

6. Treatment adjustment

Numerous symptomatic treatments for the pain of peripheral neuropathy are available, and all have their relative risks and benefits.

Tricyclic Compounds:

Drugs in this category include amitriptyline, nortriptyline, desipramine, imipramine and duloxetine. These drugs inhibit the re-uptake of the catecholamine neurotransmitters epinephrine and norepinephrine as well as serotonin, and thus may enhance central pathways that suppress pain transmission. The tricyclic compounds are quite useful for treatment of the burning, dysesthetic pains seen with peripheral neuropathies. Most of these drugs are sedating and, when given at bedtime, promote sleep.

Effective dosages of these drugs for treating chronic neuropathic pain are typically lower than the dosages used for treating depression. We recommend starting with a low single dose at bedtime and slowly increasing the dose over several weeks. Patients should be informed that it may take several weeks before full therapeutic effects are realized.

Anticonvulsants:

Drugs in this category include carbamazepine, phenytoin, gabapentin and lamotrigine. These drugs stabilize neuronal membranes and may thus prevent neuronal “short-circuits” that lead to neuropathic pain. The anticonvulsant drugs are quite useful in treating the lancinating pains frequently seen with trigeminal neuralgia and in some other peripheral neuropathies.

Effective dosages for treating neuropathic pain are similar to the dosages used for treating seizures. Blood levels of these drugs may be monitored and dosages adjusted as needed.

Lamotrigine, a new anti-seizure drug effective for primary generalized seizures, can cause a severe and fatal cutaneous hypersensitivity reaction if the starting dosage is high, and therefore must be started at a very low dose with a very gradual dosage escalation.

Topicals:

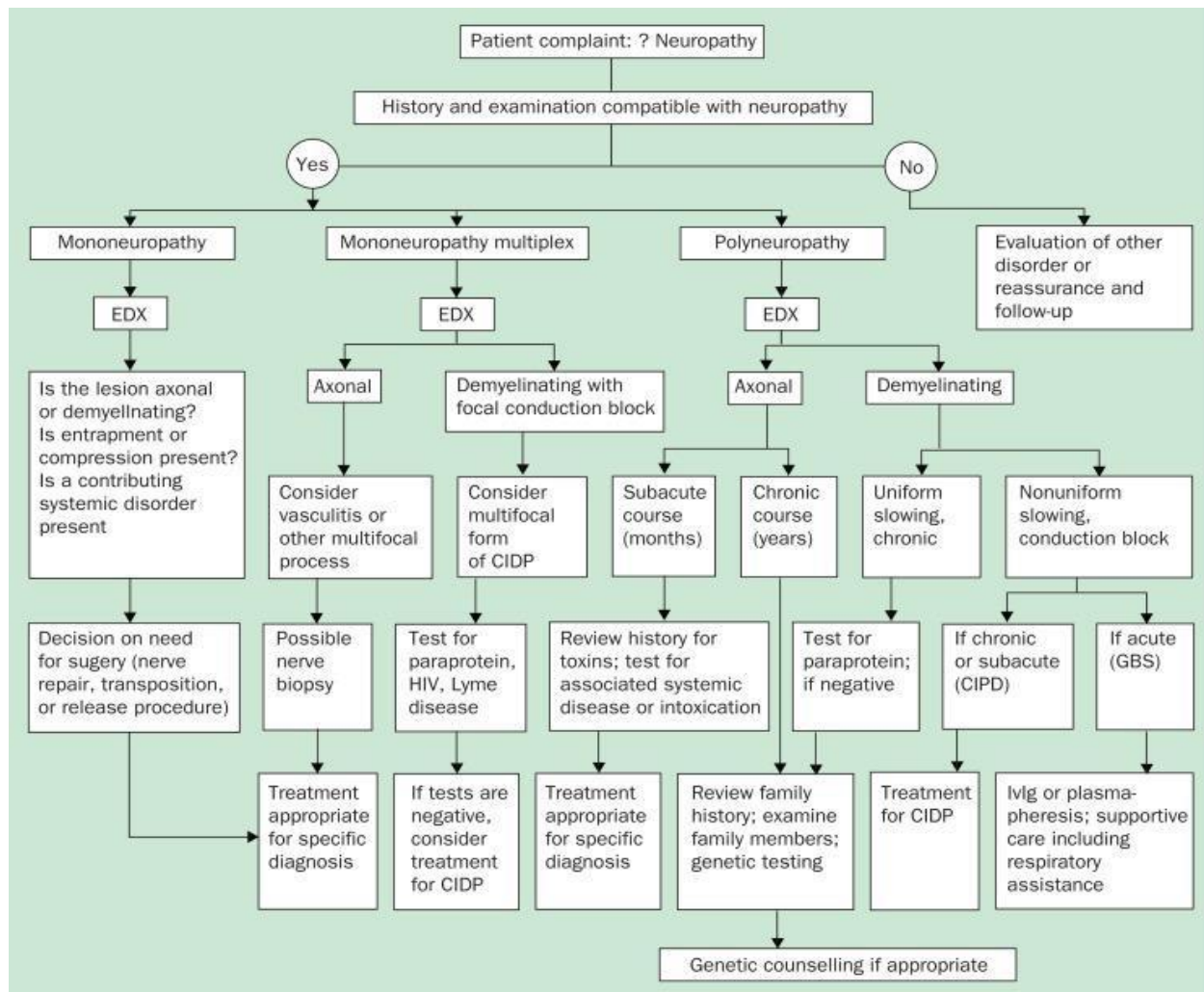
Capsaicin, a drug that impedes pain transmission by depleting substance P from sensory nerve fibers, has been found effective in treating refractory neuralgia in several studies. The cream must be applied several times daily to be effective, and patients frequently experience severe burning following each application for one- or two-weeks following initiation of treatment.

Transdermal lidocaine (Lidoderm), a local anesthetic, also appears to be effective for some patients with neuropathic pain syndromes. This drug is available as an adhesive patch that may be cut to size and replaced every 12 hours.

Surgery

Surgical release of tendons, scar tissue and bony ridges is beneficial in treating many forms of entrapment neuropathy. Nerve transposition may play a role as well in certain clinical circumstances.

7. Algorithm



VIII. SPECIAL CASES

- Sensory nerve dysfunction may produce no symptoms; symptoms of tingling, numbness, or burning; or a sense of “walking on eggshells” or a “funny sensation” under the ball of the foot. When motor nerves are affected, the prominent feature is foot deformities, particularly the claw-toe deformity. When autonomic nerves are affected, sweating is lost, and dry, cracked skin may result.

- In someone complaining of painful, burning feet, differential diagnoses in addition to diabetes would include alcohol toxicity, HIV infection, underlying malignant tumor, or amyloidosis. If carpal tunnel syndrome accompanies peripheral neuropathy, thyroid disease, rheumatoid arthritis, diabetes, and amyloidosis are more likely. Onset of symptoms is another clue to the etiology of different neuropathies, with “entrapment” causes having a more gradual, chronic nature and mononeuritis being sudden and acute.

- In B12 deficiency, neurological symptoms may be present even without anemia and may take up to 18 months to resolve.

IX. EVOLUTION AND PROGNOSIS

The effects of peripheral neuropathy depend on the cause, the nerves it affects, your medical history, treatments you receive and more.

Peripheral neuropathy can be a temporary concern, or it can be permanent. How long it lasts depends on what caused it, the extent of the damage — if any — that it caused, the treatments and more.

Peripheral neuropathy is most likely to be permanent with chronic conditions like type 2 diabetes, autoimmune diseases and genetic conditions.

Peripheral neuropathy is usually not dangerous, but it can have very disruptive effects on your life. These effects are usually not as severe when it only affects one nerve or a limited group of nerves. The more nerves it affects, the greater the potential impact.

The outlook also depends partly on your symptoms. Pain from peripheral neuropathy is usually the most disruptive symptom, but medications or other treatments may help. Autonomic symptoms are among the most serious because they involve your body's vital functions. When those don't work correctly, it can have very severe — and sometimes dangerous — effects.

Motor and sensory symptoms can also greatly disrupt your ability to work and go about your daily activities. They can cause problems — sometimes severe — with mobility, balance and coordination. Sensory symptoms are also disruptive, especially when they involve pain or affect your ability to control what you do with the affected body part(s).

Lastly, treatments can make a big difference in outlook. Some treatments can greatly reduce or even stop symptoms, but this varies.

If you have peripheral neuropathy, it's important to follow your healthcare provider's guidance. That includes seeing them as recommended, taking medications or treatments as prescribed and modifying your life to protect yourself and manage your symptoms. The actions you can take also vary widely depending on many factors, and what helps one person may not be as effective for another.

X. COMPLICATION

Complications of peripheral neuropathy can include:

-Burns, skin injuries and wounds on the feet. You might not feel temperature changes or pain on parts of your body that are numb.

-Infection. Your feet and other areas that lack sensation can become injured without your knowing. Check these areas regularly, wear close-toed, well-fitting shoes and treat minor injuries before they become infected, especially if you have diabetes.

-Falls. Weakness and loss of sensation may be associated with lack of balance and falling. Installing handrails in the bathroom, using canes or walkers when needed, and ensuring that you are walking only in well-lit rooms can decrease fall risk.

XI. PREVENTION AND RECOMMENDATION

Not all cases of peripheral neuropathy can be prevented. However, for people with diabetes, maintaining normal blood sugar levels can help prevent the onset of peripheral neuropathy. Work with your endocrinologist (a doctor who specializes in the conditions affecting the adrenal and pituitary glands, like diabetes) to find the best treatment plan to help prevent this and other potential complications of diabetes.

Manage underlying conditions

The best way to prevent peripheral neuropathy is to manage medical conditions that put you at risk.

Make healthy lifestyle choices

These habits support your nerve health:

Eat a diet rich in fruits, vegetables, whole grains and lean protein to keep nerves healthy. Protect against low levels of vitamin B-12 by eating meats, fish, eggs, low-fat dairy foods and fortified cereals. If you're vegetarian or vegan, fortified cereals are a good source of vitamin B-12, but talk to your health care professional about B-12 supplements.

Exercise regularly. With a health care professional's OK, try to get at least 30 minutes to one hour of exercise at least three times a week.

Avoid factors that may cause nerve damage. These factors can include repetitive motions, exposure to toxic chemicals, smoking and drinking too much alcohol.

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SCIATICA

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I. DEFINITION

Sciatica is a pain starting from the lower lumbar region and radiating to a lower limb (or even both), related to damage to the sciatic nerve (L5 or S1 roots).

II. EPIDEMIOLOGY

- There appears to be no gender predominance.
- Peak incidence occurs in patients in their fourth decade.
- Lifetime incidence is reported to be between 10% to 40%.
- An annual incidence of 1% to 5%.
- No association with body height has been established except in patients aged 50 to 60.
- It rarely occurs before age 20 unless secondary to trauma.
- Some studies do suggest a genetic predisposition.
- Physical activity increases incidence in those with prior sciatic symptoms and decreases in those with no prior symptoms.
- Occupational predisposition has been shown in machine operators, truck drivers, and jobs where workers are subject to physically awkward positions.

III. ETIOLOGY Lumbar disk pathologies (intervertebral disk herniation)

- Lumbarthrosis
- Lumbar spinal stenosis

IV. PATHOPHYSIOLOGY: The pain is caused by any condition that impact or compress The sciatic nerve or its nerve root.

Table 1: Topography of sciatica

		Sensory territory	Deep tendon reflex	Motor deficit
Nerve root	L5	- Posterolateral aspect of thigh - Posterolateral aspect of leg - Ankle bracelet, back of foot, 1st toe		Extensor hallucis longus, extensor digitorum longus
				Anterior tibial
				Fibular
				Gluteus medius
	S1	- Posterior surface of thigh and leg - Heel - Sole, lateral edge of the foot, 5th toe	Achilles	Triceps surae Hamstrings

V. DIAGNOSIS

1. Clinical signs and symptoms

A. History taking

Site and path of pain (or paresthesia if the path of pain is limited):

- Is irradiation accompanied by lower back pain?
- Is pain irradiating to one limb or both limbs?
- Is there a precise path? which could be:
 - **L5**: buttock, posterolateral thigh, lateral knee, lateral or anterolateral leg, lateral malleolus or premalleolar groove, dorsum of foot, hallux or first two or three toes;
 - **S1**: buttock, posterior thigh, popliteal fossa, posterior leg (calf), calcaneal or lateral retro-malleolar tendon, heel, sole or lateral edge of the foot up to the fifth toe.

Type of pain:

- Usually **mechanical** pain (mainly daytime pain, increased pain with activity and improvement with rest).
- Sometimes it can be **inflammatory** pain; meaning nighttime pain, waking up the patient at night, associated with prolonged morning stiffness; inflammatory pain should prompt physician to search for *symptomatic sciatica*.

Onset:

- sudden, after physical effort for common sciatica;
- insidious, progressive pain occurring without a triggering factor is more in favor of a symptomatic origin.

Evolution :

- For common sciatica: usually maximum immediate pain with gradual improvement;
- Progressive worsening of pain should prompt physician to search for symptomatic sciatica.

Previous episodes of low back pain

Thus, the questioning points towards **common sciatica** or **symptomatic sciatica**:

(Note: in french reference, sciatica is further divided into 2 subtypes: "**lombosciatique commune**" and "**lombosciatique symptomatique**". This subdivision is not mentioned in any english references.)

- **Common sciatica**: monoradicular (L5 or S1), mechanical pain, triggered by activity, history of acute lumbar pain;
- **Symptomatic sciatica**: multi-radicular, inflammatory pain, of increasing intensity, without associated low back pain, without triggering by physical effort, occurring in an elderly subject.

B. Physical examination

The physical examination looks for the following:

- An antalgic posture: irreducible lateral inflection due to lumbar contracture, suggestive of a herniated disc;
- **Straight leg raise test (Signe de Lasègue)**: triggering of radicular (and not lumbar) pain when lifting the leg in a straight position; we can also find a **crossed straight**

leg raise test (*Lasègue croisé*) (the radicular pain is reproduced by lifting the limb contralateral to the pain);

- A motor deficit by analytical testing of the muscles of the territory concerned. More generally, walking on heels, which tests the levator muscles of the foot, will detect an L5 deficit. Walking on toes, testing the triceps surae, will identify an S1 deficit. This deficit will be quantified and will make it possible to identify serious sciatica in the event of a motor deficit with a strength score of 3 or less;
- Deep sensory impairment, or superficial sensory impairment to the pinprick test and temperature test (hypoesthesia or dysesthesia), suggestive of posterior cord damage. Sensory impairment is not part of the seriousness of lumboradicular pain;
- Loss of calcaneal tendon reflex, suggestive of S1 damage;

The existence of **cauda equina syndrome**: hypo- or saddle anesthesia on the symptomatic side, loss of tone of the anal sphincter, dysuria or urinary retention, urinary or fecal incontinence.

2. Investigation

In the absence of red flags, typical common sciatica does not initially require any additional testing before initiating drug treatment.

- If symptomatic sciatica is suspected: imaging of the spine and search for an inflammatory syndrome will be prescribed, depending on the etiology sought.
- If invasive procedure planned (infiltration, surgery): spinal imaging is recommended (MRI or, if not available, CT scan) and biological assessment with hemostasis.

Standard x-rays are not mandatory. However, they are useful for detecting subclinical spinal disorders (scoliosis, kypho-scoliosis, spondylolisthesis) or a transitional anomaly, important for the interpretation of cross-sectional images.

It is common to find anatomical anomalies on imaging examinations with no clinical correspondence: a disc overhang or even a herniation without conflict at another level, posterior interapophyseal osteoarthritis, an angioma, etc. These anomalies and their lack of concordance with the clinical must be explained to the patient so that he or she understands why they are not integrated or targeted by the proposed treatment.

When to request them?

- For uncomplicated common sciatica, cross-sectional imaging studies (CT, MRI) are unnecessary unless infiltration or surgical treatment is considered.
- Serious sciatica (paralyzing or with cauda equina syndrome, hyperalgesic) justifies urgent treatment in a medico-surgical environment, during which cross-sectional imaging will be carried out (CT or most often MRI immediately), in order to clarify the mechanism of injury before possible surgery.

Which one(s) to request?

- MRI is more informative, allowing visualization of all structures (bones, discs, roots); it is more and more often favored when it is accessible.
- CT allows you to clearly visualize the bony structures, as well as the herniated disc.

What to expect?

CT: search for concordance with the clinic, specifies the type of hernia (median, paramedian, foraminal and extraforaminal and sub- or extraligamentous, migrated or excluded).

VI. DIFFERENTIAL DIAGNOSIS

- Lumbocruralgia: pain from lesion in L3 and L4 nerve
- Piriformis syndrome: injury or inflammation to the piriformis muscle can produce sciatica symptoms due to its proximity to the sciatic nerve.
- Sciatica due to more severe disease:
 - Cancer metastasis, lymphoma
 - Infectious spondylodiscitis (Pott's disease)
 - Intraspinal tumor: neurinoma, schwannoma
 - Infectious meningoradiculitis (herpes, zona, HIV, Lyme)
 - Spondyloarthritis
 - Epidural abscess
 - Epidural hematoma.

VII. MANAGEMENT

1. Objective Pain relief:

- Screening of patients requiring urgent surgery
- Prevention of recurrences

2. Critical sign that should alert the physician as it may require urgent surgery:

- **Motor deficit**, with muscle strength score of 3 or less;
- **Cauda equina syndrome**: genito-sphincteric dysfunction: urinary urgency or urinary incontinence, fecal incontinence, dysuria, urinary retention, loss of morning erections, saddle anesthesia
- **Severe pain** (resistant to treatment with strong opioids).

3. Mortality risk factor:

- Personal factors:
 - Age (peak 45-64 years)
 - Smoking
 - Mental stress
- Occupational factors:
 - Strenuous physical activity; for example, frequent lifting, especially while bending and twisting
 - Driving, including vibration of whole body.

4. Outpatient management:

Apart from severe sciatica that is associated with critical signs, sciatica patients can be treated symptomatically with medications at home.

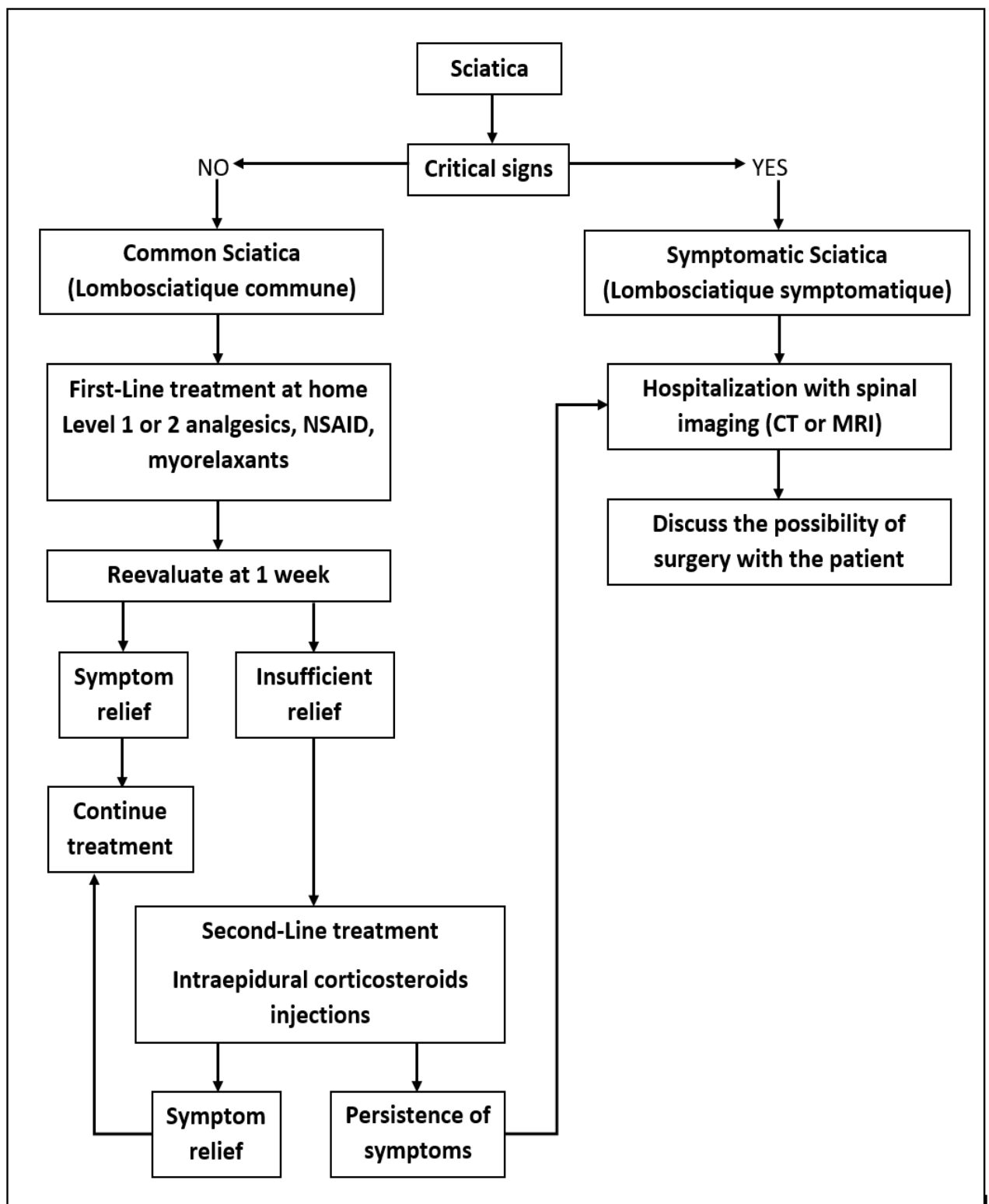
5. First-line treatment Oral analgesics:

Level 1 or 2 analgesics, NSAID, myorelaxants for a few days. Bed rest is not recommended.

6. Treatment adjustment:

In case of symptomatic treatment failure: intra epidural corticosteroids injections under ultrasound can be proposed. Surgical intervention may be discussed as well.

7. Algorithm



VIII. SPECIAL CASES

Symptomatic sciatica

These are pains which are not due to a disc or osteoarthritic pathology but which are the manifestation of more severe conditions or those requiring specific therapeutic care.

The history is the most important, the clinical examination being often poor. Clinical features (Table 2) and history guide the diagnosis.

Table 2: Clinical elements pointing towards symptomatic radiculalgia.

Suspected pathology	History	Physical examination	Additional examinations
Metastasis, lymphoma	– Age > 50 years – History of osteophilic cancer (lung, kidney, breast, prostate, thyroid) – Alteration of general condition – Lower back pain with progressive onset, without triggering factor, worsening pain, mixed or inflammatory pain	– Motor or sensory deficit – Possible cauda equina syndrome – Indurated prostate – Breast tumor – Lymphadenopathy	– MRI of the spine – Search for the primary according to the suspected etiology – Biopsy of the lesion
Infectious spondylodiscitis	– History of procedure on the spine (infiltration, surgery) – Immunosuppressive factors (diabetes, immunosuppressant) – Active infection (dental, endocarditis) – Inflammatory pain	– General signs (fever, chills) – Skin wound or abscess – Major spinal stiffness – Recent heart murmur – Motor or sensory deficit – Possible cauda equina syndrome	– X-ray of the spine (F + P) (normal at the beginning, allows the evolution of bone damage to be assessed) – MRI of the spine – Search for biological inflammatory syndrome – Etiology of infection (blood cultures, ECBU) – Discovertebral biopsy
Intraspinal tumor (neurinoma, schwannoma)	– Inflammatory pain – Nocturnal pain giving in to orthostatism (<i>syndrome du lion en cage</i>)	– Multidirectional stiffness – Lesional or sub-lesional neurological signs	– MRI of the spine – Depending on the suspected etiology: biopsy or immediate excision
Infectious meningoradiculitis (herpes, shingles, HIV, Lyme)	– Absence of triggering traumatic factor, no spinal syndrome – Infectious context, tick bite – Meningeal syndrome	– Multi-radicular irradiation, motor or sensory deficit extended to other territories – Skin lesions	– Microbiological serologies and/or viral PCR – Lumbar puncture

Spondyloarthritis	– Male < 40 years old – Family history of spondyloarthritis – Personal history of tilting buttock pain, talagia, uveitis, psoriasis, diarrhea – Truncated sciatica pain: buttock, posterior irradiation stopping at the knee – Inflammatory pain – Sensitivity to NSAIDs	– Sacroiliac pain – Skin psoriasis	– MRI of the spine and sacroiliacs – HLA-B27
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IX. EVOLUTION AND PROGNOSIS

- The prognosis for sciatica due to disk herniation is generally favorable, with most patients showing substantial improvement over months.
- Two-thirds of patients have spontaneous regression of the herniated portion of the disk over 6 months.
- Surgery is also an important option for patients who have disabling radicular pain despite optimal conservative management over a minimum of 6-8 weeks.

X. COMPLICATIONS

Potential complications of unresolved sciatic nerve compression include:

- Increased pain over time.
- Paresthesia in the affected leg.
- Loss of muscular strength in the affected leg.
- Loss of bowel and/or bladder function.
- Permanent nerve damage.

XI. PREVENTION AND RECOMMENDATION

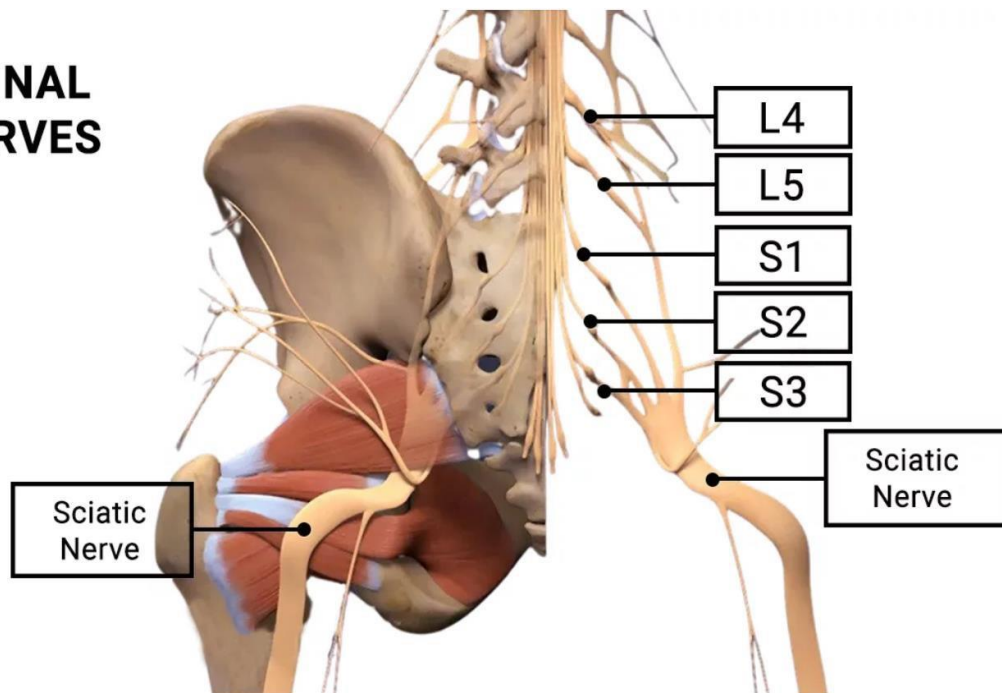
- Adaptation of activities according to pain
- Physical therapy
- Apply heat to reduce pain
- Lumbar belt

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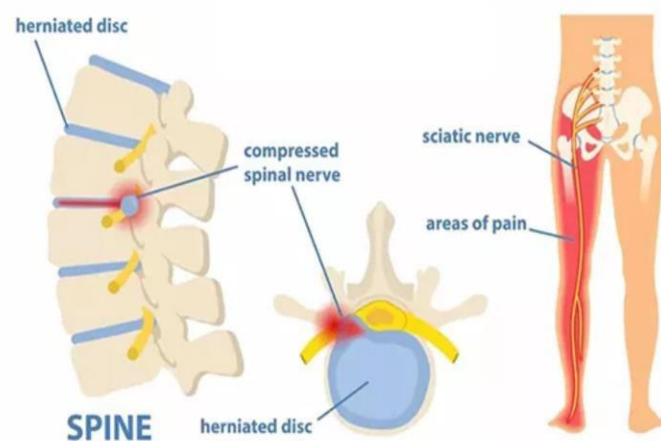
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XII. Appendix

SPINAL NERVES



SCIATICA



It is the thickest and longest nerve in the body and comprises five nerve roots. Sciatica often describes any pain originating from inflammation, irritation, compression, or pinching of the sciatic nerve. People suffering from sciatica experience mild to severe pain that follows the sciatic nerve path. You can experience pain anywhere from the lower back to the buttocks, hips, and down one or both legs.

ACUTE TRANSVERSE MYELITIS

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I. DEFINITION

Acute Transverse Myelitis (ATM) is a pathogenetically heterogeneous focal inflammatory disorder of the spinal cord with varied causes, presenting with sudden or gradual motor weakness, sensory impairment, and autonomic dysfunction. Magnetic resonance imaging (MRI) of the spinal cord typically reveals a focal hyperintense lesion, and cerebrospinal fluid often shows increased white blood cells (pleocytosis).

The etiology of TM is diverse. Partial TM, characterized by asymmetric and short lesions, is commonly associated with multiple sclerosis. In contrast, longitudinally extensive lesions are more indicative of neuromyelitis optica spectrum disorder.

II. EPIDEMIOLOGY

ATM has a conservatively estimated incidence of between 1 and 8 new cases per million per year, or approximately 1,400 new cases each year in the US. Although this disease affects people of all ages, there are peaks between the ages of 10 to 19 years and 30 to 39 years. In addition, approximately 25% of cases are in children. There is no gender or familial association with ATM (9). In 75-90% of cases TM is monophasic, yet a small percentage experience recurrent disease, especially if there is a predisposing underlying illness.

III. ETIOLOGY

There are various causes of transverse myelitis. The most common causes of transverse myelitis (TM) include multiple sclerosis and neuromyelitis optica spectrum disorder. In addition to these, other notable causes can be classified as follows:

- **Para-infectious Causes:** The specific organism responsible is often not identified.
 - **Antecedent Viral Infections:** Various viruses have been implicated, including herpes simplex, varicella zoster, cytomegalovirus, Epstein-Barr virus, enteroviruses, HIV, influenza, human T-cell leukemia virus, West Nile virus, Zika virus, and rabies.
 - **Antecedent Bacterial or Fungal Infections:** Known causative agents include *Mycoplasma pneumoniae*, tuberculosis, syphilis, and Lyme borreliosis.
- **Post-Vaccination:** Cases of TM have been reported following nearly all types of vaccinations.
- **Systemic Autoimmune Causes:** Disorders like systemic lupus erythematosus and Sjogren's syndrome have been linked to TM.
- **Systemic Inflammatory Causes:** Sarcoidosis is one such inflammatory condition associated with TM.
- **Paraneoplastic Syndromes:** TM can also occur in conjunction with paraneoplastic syndromes.

IV. ATHOPHYSIOLOGY

The demyelinating causes of transverse myelitis (TM) primarily include multiple sclerosis (MS) and neuromyelitis optica spectrum disorder (NMOSD). MS is considered an autoimmune disease of the central nervous system (CNS) wherein activated T-cells against an unknown antigen, leading to a cascade of inflammation, demyelination, and loss of axons. NMOSD, on the other hand, is recognized as distinct from MS. Approximately 75% of NMOSD cases are associated with the presence of aquaporin-4 (AQP4)-IgG, an auto-antibody that specifically targets the astrocytic water channel AQP4. NMOSD with AQP4 positivity is characterized as an antibody-mediated astrocytopathy, where complement activation and subsequent complement-mediated injury play crucial roles.

Additionally, antibodies against myelin oligodendrocyte glycoprotein (MOG-IgG) are found in around 25% of patients who are seronegative for AQP4-IgG but present with an NMOSD clinical phenotype. These MOG-IgG antibodies appear to attack myelin rather than astrocytes. Idiopathic complete TM is frequently categorized as para-infectious, leading to the hypothesis that the immune response to a triggering organism may initiate an autoimmune response against myelin or other spinal cord antigens. Potential mechanisms for this include molecular mimicry, where T-cells or antibodies, activated by epitopes on the infectious agent, mistakenly react with similar structures in the CNS, or the triggering of a super-antigen response by the infectious organism. A variety of viruses, bacteria (including certain mycobacteria), fungi, and parasites have been linked to the occurrence of TM.

V. DIAGNOSIS

The diagnosis of transverse myelitis (TM) should follow a systematic approach that includes medical history, physical examination, magnetic resonance imaging (MRI) of both the spinal cord and brain, and analyzing cerebrospinal fluid (CSF). It is crucial to exclude other potential causes of symptoms, such as spinal cord compression and ischemia. Recommended additional tests may include:

- Antibody testing for aquaporin-4 (AQP4) and myelin oligodendrocyte glycoprotein (MOG).
- Specialized CSF stains and cultures to assess for infections.
- Chest X-ray and computed tomography (CT) of the thorax.
- Viral serology testing.
- Testing for auto-antibodies associated with systemic lupus erythematosus (SLE) or Sjogren's syndrome.
- Evaluations such as visual evoked potentials and optical coherence tomography.
- Biopsies, as necessary.

This comprehensive approach ensures appropriate diagnosis and management for patients suspected of having TM.

1. Classification

Acute partial TM:

- Para-infectious, occurring at the time of, and in association with, an acute episode of infection at a site remote from the central nervous system
- Characterized by asymmetric spinal cord lesions typically of 1 or 2 vertebral segments in length

- The clinical syndrome is typically mild to moderate in severity and the motor and sensory symptoms and signs are asymmetric (6)

Longitudinally extensive TM:

- Characterized by asymmetric or symmetric spinal cord lesions that span ≥ 3 contiguous vertebral segments (4)
- The clinical syndrome is usually moderate to very severe, and the motor symptoms and signs are typically bilateral and more symmetric than in acute partial TM

2. Clinical presentation

The typical presentation leading to consideration of TM includes the following:

- Progressive paraparesis or quadriparesis
- Sensory loss and/or paraesthesias
- Autonomic (bowel and bladder) dysfunction
- An evolution of signs over several days.

Table 1: Spinal cord syndromes

Syndrome	Tracts Involved	Clinical Manifestations
Complete transverse myelitis	All	Paresis, sensory loss, and autonomic impairment below the level of the lesion
	Ipsilateral corticospinal;	Ipsilateral paresis and impaired dorsal column
Hemicord (Brown-Sequard)	Ipsilateral dorsal columns; contralateral spinothalamic	sensation; contralateral pain and temperature loss
Dorsal column	Bilateral dorsal columns	Bilateral loss of vibratory and proprioceptive sensation; Lhermitte phenomenon
Subacute Combined Degeneration	Bilateral dorsal columns and corticospinal tracts	Bilateral dorsal column dysfunction; paresis and upper motor neuron signs below the level of the lesion
Central cord	Crossing spinothalamic fibers, and corticospinal tracts	Dissociated sensory loss (diminished pain and temperature with normal dorsal column function) in a shawl-like pattern. Saddle-sparing sensory loss. Upper motor neuron weakness below the level of the lesion.
Conus medullaris	Sacral autonomic fibers	Early and prominent sphincter and sexual impairment; saddle pattern sensory loss; mild weakness

Table 2: Criteria for the Diagnosis of Acute Transverse Myelitis

Inclusion Criteria	Exclusion Criteria
• Development of sensory, motor, or autonomic dysfunction attributable to the spinal cord • Bilateral signs or symptoms (although not necessarily symmetric)	• History of previous radiation to the spine within the last 10 years • Clear arterial distribution clinical deficit consistent with thrombosis of
• Clearly defined sensory level	the anterior spinal artery

- Exclusion of extra-axial compressive etiology by neuroimaging (MRI or myelography; CT of spine not adequate)
- Inflammation within the spinal cord demonstrated by CSF pleocytosis or elevated IgG index or gadolinium enhancement. If no inflammatory criterion is met at symptom onset, repeated MRI and lumbar puncture evaluation between 2 and 7 days after symptom onset meet criteria.
- Progression to nadir between 4 h and 21 days after the onset of symptoms (if patient awakens with symptoms, symptoms must become more pronounced from point of awakening)
- Abnormal flow voids on the surface of the spinal cord consistent with AVM
- Serologic or clinical evidence of connective tissue disease (e.g., sarcoidosis, Behçet disease, Sjögren syndrome, SLE, mixed connective tissue disorder)**
- CNS manifestations of syphilis, Lyme disease, HIV, HTLV-1, mycoplasma, other viral infection (e.g., HSV-1, HSV-2, VZV, EBV, CMV, HHV-6, enteroviruses)**
- Brain MRI abnormalities suggestive of multiple sclerosis**
- History of clinically apparent optic neuritis**

**Do not exclude disease-associated acute transverse myelitis

3. Key diagnostic factors

Common factors:

- Presence of risk factors
- Age 10-19 or 30-39 years
- Motor weakness
- Paraesthesias or sensory loss
- Bladder symptoms: urinary frequency, urgency, incontinence, or retention
- Bowel symptoms: incontinence or constipation
- L'hermitte sign
- McArdle's sign
- Paroxysmal tonic spasms
- Upper motor neuron signs: hyper-reflexia, positive Babinski's sign, limb spasticity
- Sensory loss/sensory level

4. Investigations

Test
MRI spinal cord - Required in all patients to exclude compressive myelopathy. - Detection of intrinsic cord lesion assists in confirmation of myelitis.
MRI brain - Peri-ventricular, brain stem, and juxta-cortical lesions suggest multiple sclerosis (MS) in setting of acute partial TM. - Peri-aqueductal lesions or lesions involving the thalamic and hypothalamic region suggest neuromyelitis optica spectrum disorder (NMOSD). - Normal results suggest NMOSD or idiopathic TM in setting of complete TM.
Serum aquaporin-4 auto-antibodies and myelin oligodendrocyte glycoprotein auto-antibodies - Longitudinally extensive myelitis, defined by a lesion extending over ≥ 3 contiguous vertebral segments, has a greater risk of recurrence if serum aquaporin-4 (AQP4) auto-antibodies are present.

Cerebrospinal fluid cell count, cell differential, protein level, IgG index, oligoclonal bands - Increased cell count (>5 WBC/mm³) supports inflammatory myelitis. - Normal cerebrospinal fluid does not exclude TM but consider repeating lumbar puncture in 7 to 10 days if TM is strongly suspected. - Most causes are associated with 10 to 2000 WBC/mm
Cerebrospinal fluid Gram stain, cultures (bacterial, tubercular, fungal), and India ink smear - Negative in non-infectious TM; helpful when there are constitutional symptoms, fever, and/or cerebrospinal fluid pleocytosis
Cerebrospinal fluid, polymerase chain reaction for herpes simplex virus (HSV)-1, HSV-2, varicella zoster virus (VZV), Borrelia burgdorferi (Lyme disease), cytomegalovirus (CMV), Epstein-Barr virus (EBV), and West Nile virus - Useful for detection of HSV-1, HSV-2, VZV, Borrelia burgdorferi (Lyme disease), CMV, EBV, West Nile virus.
Cerebrospinal fluid Venereal Disease Research Laboratory test - Positive cerebrospinal fluid Venereal Disease Research Laboratory test is virtually 100% specific for syphilis, but a negative result does not exclude the diagnosis
Serum anti-nuclear antibody, double-stranded DNA - Multiple positive auto-antibodies are also often found in association with neuromyelitis optica spectrum disorder.
Extractable nuclear antigen (including SSA and SSB) - Multiple positive auto-antibodies are also often found in association with neuromyelitis optica spectrum disorder.
Serum and cerebrospinal fluid paraneoplastic auto-antibodies - May indicate the presence of an underlying malignancy. The most common are amphiphysin and collapsin response-mediator protein-5.

VI. DIFFERENTIAL DIAGNOSIS

The differential diagnosis of acute transverse myelitis includes the following:

- Compressive myelopathy
- Infectious myelitis (e.g., TB)
- Anterior spinal artery Occlusion
- Posterior spinal artery Occlusion
- Arteriovenous fistula
- Haematomyelia
- Neoplastic myelopathy
- Cerebral lesion
- Guillain-Barre syndrome
- Myasthenia gravis
- Motor neuron disorder

VII. MANAGEMENT

1. Objectives

The goals of treatment are to restore neurological function lost during the acute TM event; to initiate preventive strategies for, and symptomatic relief of, TM complications; and, when applicable, to start therapies aimed at preventing disease relapse.

2. Critical Sign

Acute neurological deficits

- After exclusion of compressive lesions, acute infections, and other non-TM disorders, first-line treatment of the acute neurological deficits related to TM is with intravenous methylprednisolone, typically administered daily for 3 to 5 consecutive days. There are no controlled trials of corticosteroids for TM but the approach is extrapolated from treatment of acute multiple sclerosis (MS) attacks.
- Patients with severe deficits that do not respond to corticosteroid therapy, or whose deficits worsen despite treatment, are candidates for rescue therapy with plasma exchange (plasmapheresis).

3. Mortality risk factors

Red flags arguing against TM in myelopathic patients:

1. Apoplectic onset (reaching the nadir less than 4 hours from onset).
2. Insidious progressive course.
3. Older age of onset.
4. Preceding trauma, pain, and/or vertebral tenderness would suggest traumatic myelopathy.
5. Vascular instrumentation (in particular, aortic and cardiac surgery) or maneuvers that increase intra-abdominal pressure (e.g. weight-lifting or straining) before the acute/subacute appearance of myelopathic features may implicate spinal cord infarction.
6. Prior bariatric surgical procedures, malabsorption syndromes, dietary restrictions, malnutrition, use of zinc supplements, excessive use of zinc-containing denture cream, alcoholism, and/or drug/toxin exposure may implicate a metabolic or toxic etiology.
7. Prior radiation therapy.
8. Immunocompromised state (HIV/AIDS or immunosuppressive therapy).

4. Outpatient Management

- Immediate care includes monitoring and treatment for complications such as respiratory failure (1 week) and sequelae of immobility (deep vein thrombosis; 3 months).
- Follow-up of patients with idiopathic TM should continue until neurological recovery is complete or has reached a plateau, typically for 6 to 12 months, or longer if symptomatic treatment requires re-evaluation.
- Long-term management of symptoms such as pain, depression, or bladder dysfunction may be required.

5. First-line treatment

Please be aware that formulations, routes of administration, and dosages may vary between different drug names and brands, as well as across drug formularies and geographical locations. Treatment recommendations provided are tailored to specific patient populations.

Acute	
Acute neurological deficits	
1st	Corticosteroid Primary options • Methylprednisolone: 1,000 mg intravenously once daily for 3-5 days - There are no controlled trials of corticosteroids for TM, but the approach is extrapolated from treatment of acute multiple sclerosis (MS) attacks. Treatment with 3 to 5 consecutive daily infusions of corticosteroids is associated with more rapid recovery of neurological deficits in MS, and clinical experience suggests similar results in TM. - Some clinicians follow the intravenous therapy with an oral prednisolone taper for 5 to 14 days.
adjunct	Supportive care + acute rehabilitation Treatment recommended for some patients in selected patient group. - In a minority of patients with cervical TM, the lesion extends into the medulla and may cause neurogenic respiratory failure. Close observation of respiratory parameters, including measurement of maximal respiratory pressures and forced vital capacity, and involvement of a skilled critical care team are recommended in cases of ascending cervical myelitis.
2nd	Plasma exchange (plasmapheresis) - Patients with severe deficits that do not respond to corticosteroid therapy, or whose deficits worsen despite treatment, are candidates for rescue therapy with plasma exchange. Intravenous immunoglobulin may be considered but there are no randomized controlled trials that have assessed its efficacy. - Risks of plasma exchange include line infection, sepsis, thrombosis, and bleeding.
adjunct	Supportive care + acute rehabilitation Treatment recommended for some patients in selected patient group. - In a minority of patients with cervical TM, the lesion extends into the medulla and may cause neurogenic respiratory failure. Close observation of respiratory parameters, including measurement of maximal respiratory pressures and forced vital capacity, and involvement of a skilled critical care team are recommended in cases of ascending cervical myelitis.
Ongoing	
Idiopathic TM	
1st	Observation - If the patient is found to have idiopathic TM, no preventive therapy is required.
At risk for multiple sclerosis (typical demyelinating lesions on MRI)	
1st	MS disease-modifying therapies - Acute partial TM is associated with high risk for multiple sclerosis (MS) when the brain MRI shows typical demyelinating lesions. - Disease-modifying therapies (DMT) aim to reduce the risk of MS relapses and disability progression. Selection of initial therapy is guided by characteristics of the disease course, imaging findings, and patient characteristics and preferences.
Aquaporin-4 auto-antibody seropositive	
1st	Immunosuppressive treatment Primary options - Azathioprine: consult specialist for guidance on dose OR

	- Mycophenolate mofetil: consult specialist for guidance on dose OR - Rituximab: consult specialist for guidance on dose
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6. Treatment adjustment

There is no specific treatment adjustment on patients with TM. The treatment is based on the patient's evidence.

7. Algorithm for Approach to Diagnosis of Acute Transverse Myelitis (Jacob, Weinshenker)

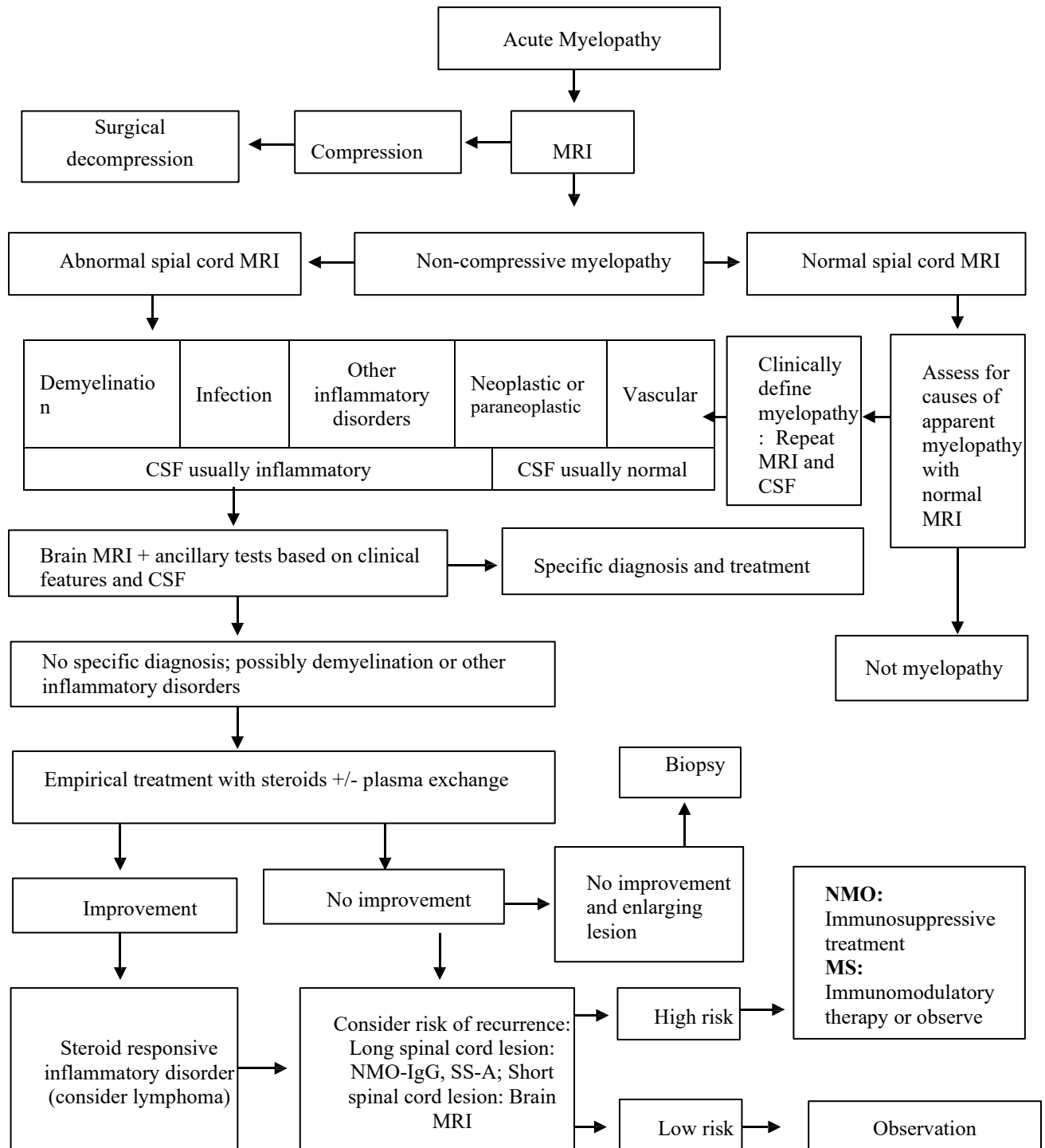


Figure 1: Diagnostic approach to acute myelopathy. MRI, magnetic resonance imaging; CSF, cerebrospinal fluid; NMO, neuromyelitis optica; MS, multiple sclerosis; IgG, immunoglobulin G; SS-A, Sjogren's syndrome antibody (anti-Ro).

VIII. SPECIAL CASE

Case history

A 33-year-old man with no significant medical history presents with a 3-day history of paraesthesias in his left lower extremity, which later progressed to affect the left trunk up to the level of his umbilicus. He also reports experiencing new urinary frequency and that his right leg drags while walking. Upon examination, he has partial Brown-Sequard syndrome, characterized by mild pyramidal weakness and impaired vibration and proprioceptive sensation in his right lower extremity, as well as decreased pain sensitivity throughout the left lower extremity and trunk below the T11 level. The plantar reflex on the right side is extensor.

IX. EVOLUTION AND PROGNOSIS

The course of TM is highly variable. Acute partial TM, such as is seen with multiple sclerosis (MS), has a good prognosis. The neurological deficits typically improve over several weeks, and most patients achieve a complete or near-complete recovery.

Acute partial TM

Acute partial TM typically improves over several weeks to months, with most recovery taking place during the first 3 months after onset, though sometimes slower improvement is noted for up to 1 year after TM onset.

Complete TM

Longitudinally extensive TM has many causes, but most cases are idiopathic monophasic TM or the first event of a neuromyelitis optica spectrum disorder (NMOSD). Maximum deficit usually occurs within a few days, and sometimes within 24 hours. Recovery is highly variable, ranging from return to normal function to complete loss of motor and sensory loss below the level of the lesion. The risk of recurrent TM is unknown in cases associated with systemic processes such as connective tissue diseases; generally, treatment for the underlying disease is recommended (22).

X. COMPLICATIONS

Complications	Timeframe	Likelihood
Multiple sclerosis (with one or more white matter lesion(s))	long term	high
Motor weakness	long term	medium
Spasticity	long term	medium
Neuropathic pain	variable	medium
Bladder dysfunction	variable	medium
Deep venous thrombosis	variable	medium
Pressure ulcers	variable	medium

XI. PREVENTION AND RECOMMENDATION

While transverse myelitis (TM) cannot be prevented, patient education is crucial for understanding the prognosis, disease progression, diagnostic evaluation, and available treatment options. The primary clinical associations with transverse myelitis (TM) include viral infections or vaccinations occurring prior to the onset of the condition. While it is impractical to prevent all future viral infections, it is advisable to refrain from administering new vaccinations during any period of acute disease activity.

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GUILLAIN BARRÉ SYNDROME

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I. DEFINITION

Guillain-Barré Syndrome (GBS) is defined as an acute inflammatory peripheral neuropathy characterized by the following features:

- **Polyradiculoneuropathy:** It is an inflammatory condition affecting multiple nerve roots and peripheral nerves, leading to sensory and motor dysfunction.
- **Demyelinating and Axonal Forms:** GBS can manifest in two main forms:
 - **Demyelinating form:** This involves damage to the myelin sheath surrounding the nerves, leading to impaired nerve conduction.
 - **Acute Motor Axonal Neuropathy (AMAN):** This form is characterized by pure motor impairment due to damage at the axonal level, often associated with infections like *Campylobacter jejuni*.
- **Preceding Infection:** In more than 50% of cases, GBS is preceded by an infectious episode, which can be respiratory or gastrointestinal in nature, often involving viruses such as cytomegalovirus or Epstein-Barr virus, or bacterial infections like *Campylobacter jejuni*.

GBS is an acute, inflammatory, multifocal neuropathy that primarily affects the motor and sensory nerves, often triggered by an infection.

II. EPIDEMIOLOGY

The epidemiology of GBS is a rare neurological disorder with a specific incidence rate, demographic trends, and a strong association with preceding infections, particularly in adults.

- **Incidence:** The estimated incidence of GBS is approximately 1 to 2 cases per 100,000 individuals per year. This indicates that GBS is relatively rare but can occur in various populations worldwide.
- **Age and Gender:** GBS can affect individuals of any age, but it is more commonly seen in adults. There is a slight male predominance, with males being affected more frequently than females.
- **Preceding Infections:** More than 50% of GBS cases are preceded by an infectious event, particularly respiratory or gastrointestinal infections. Common infectious agents include:
 - **Viruses:** Cytomegalovirus, Epstein-Barr virus, and Zika virus.
 - **Bacteria:** *Campylobacter jejuni* is notably associated with the axonal form of GBS (AMAN) and is often linked to gastrointestinal infections.
- **Seasonality:** Some studies suggest that GBS may have seasonal variations, with increased incidence during certain times of the year, potentially correlating with the prevalence of specific infections.
- **Geographic Variation:** The incidence and presentation of GBS can vary by geographic region, influenced by local infectious disease patterns and healthcare access.

III. ETIOLOGY

The etiology of GBS involves several factors, primarily related to immune-mediated responses following infections. Key points include:

- **Infectious Triggers:** A significant proportion of GBS cases (over 50%) are preceded by an infectious illness. Common infectious agents include:
 - **Bacterial Infections:** The most notable is *Campylobacter jejuni*, which is often associated with gastrointestinal infections and is linked to the acute motor axonal neuropathy (AMAN) form of GBS.
 - **Viral Infections:** Various viruses can trigger GBS, including:
 - Cytomegalovirus (CMV)
 - Epstein-Barr virus (EBV)
 - Zika virus
 - Influenza virus
 - Other respiratory viruses.
- **Autoimmune Mechanism:** GBS is considered an autoimmune disorder where the body's immune system mistakenly attacks the peripheral nerves. This is thought to occur due to molecular mimicry, where the immune response to the infectious agent cross-reacts with nerve components, particularly the myelin sheath or axonal structures.
- **Genetic Factors:** While the exact genetic predisposition to GBS is not well-defined, certain genetic factors may influence susceptibility to the syndrome following infections.
- **Environmental Factors:** Some studies suggest that environmental factors, including exposure to certain toxins or vaccines, may also play a role in the development of GBS, although these associations are less clear and require further investigation.

The etiology of GBS is primarily linked to preceding infections that trigger an autoimmune response, leading to inflammation and damage to peripheral nerves.

IV. PATHOPHYSIOLOGY

The GBS involves a complex interplay of immune-mediated mechanisms that lead to nerve damage and include:

- **Immune Response Activation:** GBS is primarily an autoimmune disorder. Following an infection (often viral or bacterial), the immune system is activated, and antibodies are produced. In some cases, these antibodies mistakenly target components of the peripheral nervous system.
- **Molecular Mimicry:** This phenomenon occurs when the immune response to an infectious agent cross-reacts with nerve tissue. For example, the lipooligosaccharides found on the surface of *Campylobacter jejuni* can resemble gangliosides (glycolipids found in nerve cell membranes). This similarity can lead to the production of antibodies that attack the myelin sheath or axonal structures of peripheral nerves.
- **Demyelination and Axonal Damage:**
 - In the demyelinating form of GBS (often referred to as acute inflammatory demyelinating polyneuropathy, AIDP), the immune response leads to inflammation and damage to the myelin sheath surrounding the nerves. This results in slowed or blocked nerve conduction, leading to motor and sensory deficits.
 - In the axonal form of GBS (such as AMAN), the immune attack is directed more towards the axons themselves, leading to more severe motor impairment and a different pattern of nerve damage.
- **Inflammatory Mediators:** The immune response involves the activation of various inflammatory cells, including T cells and macrophages, which infiltrate the nerve

tissues. These cells release pro-inflammatory cytokines that contribute to nerve damage and demyelination.

- **Clinical Manifestations:** The pathophysiological changes result in the clinical features of GBS, including progressive muscle weakness, sensory disturbances, and, in severe cases, respiratory failure due to diaphragm involvement.
- **Recovery Phase:** In many cases, the recovery from GBS occurs as the immune response subsides, and remyelination or axonal regeneration takes place. However, the extent of recovery can vary, and some patients may experience long-term sequelae.

The pathophysiology of GBS is characterized by an autoimmune response triggered by infections, leading to inflammation and damage to peripheral nerves, resulting in the clinical manifestations of the syndrome.

V. DIAGNOSIS

The diagnosis of GBS is primarily based on clinical evaluation and supported by laboratory investigations. Here are the key components involved in the diagnosis:

- **History Taking:**
 - **Preceding Illness:** A history of recent infections (often respiratory or gastrointestinal) is common, particularly infections caused by *Campylobacter jejuni*, cytomegalovirus, or Epstein-Barr virus.
 - **Symptom Onset:** Symptoms typically begin with weakness and may progress rapidly over days to weeks.
- **Clinical Features:**
 - **Motor Weakness:** Symmetrical weakness that usually starts in the legs and may ascend to the upper body and arms. The weakness can progress to paralysis in severe cases.
 - **Sensory Symptoms:** Patients may report sensory disturbances, such as tingling, numbness, or pain, often in a "glove and stocking" distribution.
 - **Areflexia:** The absence of reflexes (areflexia) is a hallmark sign.
 - **Autonomic Dysfunction:** Symptoms may include tachycardia, hypotension and abnormal sweating.
- **Neurological Examination:**
 - A thorough neurological examination is essential to assess the extent of motor and sensory deficits, as well as reflexes.
- **Investigation**
- **Electromyography (EMG) and Nerve Conduction Studies (NCS):**
 - **Demyelinating Form:** EMG may show prolonged latencies, reduced conduction velocities, and conduction blocks, indicating demyelination.
 - **Axonal Form (AMAN):** EMG may show reduced amplitudes of motor potentials without significant demyelination features.
- **Cerebrospinal Fluid (CSF) Analysis:**
 - **Albuminocytologic Dissociation:** This is a classic finding in GBS, where there is an elevated protein level in the CSF with a normal white blood cell count. This indicates a disruption of the blood-CSF barrier due to inflammation.
- **Serological Tests:**
 - While not routinely used for diagnosis, serological tests may be performed to identify specific infections (e.g., *Campylobacter jejuni*, cytomegalovirus) that may have preceded the onset of GBS.

- **Other Investigations:**

- Additional tests may include imaging studies (like MRI) to rule out other causes of weakness, although they are not specific for GBS.

The diagnosis of GBS is primarily clinical, based on the characteristic symptoms and history of preceding infections. Laboratory investigations, particularly EMG/NCS and CSF analysis, provide supportive evidence and help differentiate GBS from other neurological conditions.

VI. DIFFERENTIAL DIAGNOSIS

The differential diagnosis of GBS includes a range of neurological conditions that can present with similar symptoms. A careful clinical evaluation, along with appropriate laboratory and imaging studies, is essential to distinguish GBS from these other conditions. It includes a variety of conditions that can present with similar clinical features, particularly progressive weakness and sensory disturbances to consider:

- **Chronic Inflammatory Demyelinating Polyneuropathy (CIDP):**
 - CIDP is a chronic counterpart to GBS, characterized by progressive or relapsing weakness and sensory loss. Unlike GBS, CIDP has a longer duration of symptoms and may show more prominent sensory involvement.
- **Myasthenia Gravis:**
 - This autoimmune disorder leads to fluctuating muscle weakness and fatigue, particularly with exertion. It can be distinguished by the presence of specific antibodies (e.g., anti-acetylcholine receptor antibodies) and improvement with anticholinesterase medications.
- **Multiple Sclerosis (MS):**
 - MS can present with weakness and sensory changes due to demyelination in the central nervous system. MRI findings and the presence of oligoclonal bands in the CSF can help differentiate it from GBS.
- **Amyotrophic Lateral Sclerosis (ALS):**
 - ALS presents with progressive weakness and muscle atrophy but typically involves upper and lower motor neuron signs. Electromyography (EMG) findings will show both upper and lower motor neuron involvement.
- **Diabetic Neuropathy:**
 - Diabetic neuropathy can cause weakness and sensory loss, particularly in a "glove and stocking" distribution. A history of diabetes and specific nerve conduction study findings can help differentiate it.
- **Infectious Neuropathies:**
 - Conditions such as Lyme disease, HIV-associated neuropathy, or syphilis can present with similar symptoms. Serological tests and clinical history can aid in diagnosis.
- **Toxic Neuropathies:**
 - Exposure to certain toxins (e.g., heavy metals, certain medications) can lead to peripheral neuropathy. A detailed history of exposure is crucial for diagnosis.
- **Vascular Disorders:**
 - Conditions such as vasculitis or ischemic neuropathy can cause acute weakness. Imaging studies may be necessary to identify vascular involvement.
- **Spinal Cord Disorders:**
 - Conditions like transverse myelitis or spinal cord compression can present with weakness and sensory changes. MRI of the spine can help differentiate these conditions.

○ **Peripheral Nerve Injuries:**

- Traumatic injuries to peripheral nerves can cause weakness and sensory loss. A thorough history and physical examination can help identify these injuries.

VII. MANAGEMENT

The management of GBS involves several key components aimed at addressing the condition's acute phase, preventing complications, and facilitating recovery. Here's a detailed overview based on your queries:

1. Objective of GBS Management:

The primary objectives of managing GBS are:

- **Preventing complications:** Such as respiratory failure, deep vein thrombosis (DVT), and pressure sores.
- **Facilitating recovery:** Ensuring the patient regains strength and function as quickly as possible.
- **Symptomatic relief:** Managing pain, autonomic dysfunction, and other symptoms associated with GBS.

2. Critical Signs

Critical signs that require immediate attention in GBS include:

- **Respiratory distress:** Difficulty breathing or decreased respiratory function, which may necessitate mechanical ventilation.
- **Rapidly progressing weakness:** Significant deterioration in muscle strength, particularly if it affects the ability to walk or perform daily activities.
- **Autonomic instability:** Severe fluctuations in heart rate, blood pressure, or other autonomic functions.

3. Mortality Risk Factors

Factors that may increase the risk of mortality in GBS include:

- **Age:** Older age is associated with a higher risk of severe outcomes.
- **Severity of initial symptoms:** More severe initial weakness or respiratory involvement can indicate a worse prognosis.
- **Need for mechanical ventilation:** Patients requiring respiratory support have a higher risk of complications and mortality.
- **Presence of comorbidities:** Underlying health conditions can complicate the clinical course.

4. Outpatient management

Outpatient management may include:

- **Monitoring:** Regular follow-up to assess recovery and manage any ongoing symptoms.
- **Physical therapy:** To aid in rehabilitation and improve strength and mobility.
- **Pain management:** Addressing neuropathic pain with medications as needed.
- **Education:** Informing patients and families about the condition, expected recovery, and signs of complications.

5. First-line treatment

The first-line treatments for GBS include:

- **Intravenous Immunoglobulin (IVIG):** Administered at a dose of 0.4 g/kg per day for 5 days. This treatment is effective in reducing the severity and duration of symptoms.
- **Plasmapheresis:** Plasmapheresis (plasma exchange) can also be used, typically involving four exchanges performed every other day. Both IVIG and plasmapheresis have equivalent efficacy.

6. Treatment adjustment

May be necessary based on:

- **Response to initial therapy:** If there is inadequate improvement, further treatment options may be considered.
- **Complications:** addressing any complications that arise during treatment, such as infections or respiratory issues.
- **Symptomatic Treatment:** Adjustments may be made to manage symptoms such as pain, autonomic dysfunction, and respiratory support. This includes:
 - Pain management with analgesics or neuropathic pain medications.
 - Monitoring and managing respiratory function, which may include mechanical ventilation if necessary.
 - Supportive care to prevent complications such as deep vein thrombosis (DVT) and pressure sores.
- **Rehabilitation:** Early and ongoing physical therapy is crucial to help patients regain strength and mobility. Adjustments in rehabilitation strategies may be made based on the patient's progress and specific needs.
- **Monitoring for Complications:** Continuous assessment for complications such as respiratory failure, autonomic instability, and infections is essential. Treatment plans may need to be adjusted based on the patient's evolving clinical status.
- **Long-term Follow-up:** Patients may require long-term follow-up to address any residual symptoms or complications, and treatment plans may be adjusted accordingly.

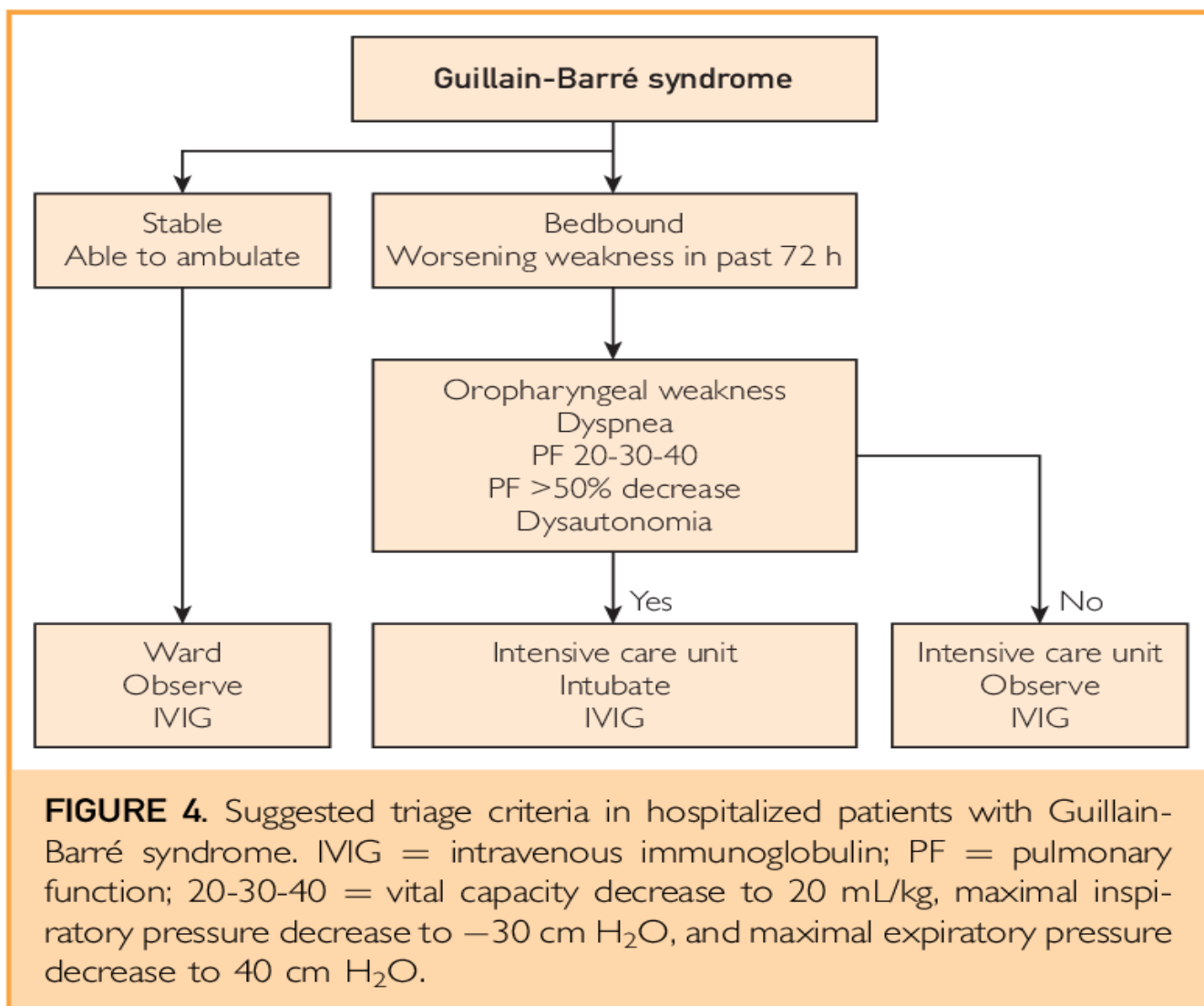
These treatment adjustments are aimed at optimizing recovery and minimizing complications associated with GBS.

7. Algorithm for Management

Algorithms may vary, a general management algorithm for GBS can be outlined as follows:

- **Initial Assessment:**
 - Evaluate clinical symptoms and perform neurological examination.
 - Assess respiratory function and vital signs.
- **Diagnostic Workup:**
 - Conduct EMG/NCS and CSF analysis to confirm diagnosis.
- **Initiate Treatment:**
 - Start IVIG or plasmapheresis based on clinical severity and availability.
- **Monitor for Complications:**
 - Regularly assess respiratory function, autonomic stability, and overall neurological status.
- **Rehabilitation:**
 - Initiate physical therapy and occupational therapy as soon as feasible.
- **Follow-Up:**
 - Schedule regular follow-up visits to monitor recovery and adjust management as needed.
- **Long-Term Management:**
 - Address any residual symptoms and provide support for rehabilitation and reintegration into daily activities.

The management of GBS is multifaceted, focusing on immediate treatment, monitoring for complications, and facilitating recovery through rehabilitation and supportive care. Early intervention with IVIG or plasmapheresis is crucial for improving outcomes.



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VIII. SPECIAL CASES

In the context of GBS, special cases refer to specific presentations or variations of the syndrome that may require tailored management or consideration. Here are some notable special cases of GBS:

- **Miller Fisher Syndrome (MFS)**
 - **Description:** MFS is a variant of GBS characterized by the classic triad of ophthalmoplegia (eye movement paralysis), ataxia (lack of voluntary coordination of muscle movements), and areflexia (absence of reflexes).
 - **Management:** Similar to GBS, MFS is treated with IVIG or plasmapheresis. Most patients have a good prognosis and recover fully.
- **Acute Motor Axonal Neuropathy (AMAN)**
 - **Description:** AMAN is a subtype of GBS that primarily affects motor axons without significant sensory involvement. It is often associated with preceding infections, particularly *Campylobacter jejuni*.
 - **Management:** Treatment options include IVIG or plasmapheresis. Recovery can be variable, and some patients may experience prolonged weakness.

- **Acute Inflammatory Demyelinating Polyneuropathy (AIDP)**
 - **Description:** AIDP is the most common form of GBS, characterized by demyelination of peripheral nerves. It typically presents with progressive weakness and sensory changes.
 - **Management:** Similar to GBS, AIDP is treated with IVIG or plasmapheresis. The recovery process can take weeks to months.
- **GBS in Pregnancy**
 - **Description:** GBS can occur during pregnancy, presenting unique challenges in management due to the potential impact on both the mother and fetus.
 - **Management:** Treatment options remain the same (IVIG or plasmapheresis), but careful monitoring of respiratory function and fetal well-being is essential.
- **Recurrent GBS**
 - **Description:** While GBS typically occurs as a single episode, some patients may experience recurrent episodes. This is rare but can complicate management and prognosis.
 - **Management:** Each episode is treated similarly to the initial presentation, with IVIG or plasmapheresis. Long-term follow-up is necessary to monitor for recurrence.
- **GBS with Autonomic Dysfunction**
 - **Description:** Some patients with GBS may experience significant autonomic dysfunction, leading to symptoms such as severe fluctuations in blood pressure, heart rate abnormalities, and gastrointestinal issues.
 - **Management:** In addition to standard treatments, supportive care for autonomic symptoms is crucial, which may include medications to stabilize blood pressure and heart rate.
- **GBS in the Context of Other Conditions**
 - **Description:** GBS can occur in patients with underlying conditions such as diabetes, HIV, or malignancies, which may complicate the clinical picture and management.
 - **Management:** Treatment should address both GBS and the underlying condition, with careful monitoring for complications.

Special cases of GBS show variability in symptoms and management. Tailored approaches are needed based on unique patient factors and atypical presentations. Early diagnosis and timely treatment are crucial for better outcomes.

IX. EVOLUTION AND PROGNOSIS

The evolution and prognosis of GBS can be characterized by several phases and factors that influence recovery.

1. **Initial Phase (Acute Phase):**
 - **Onset:** GBS typically begins with the onset of weakness, which may start in the legs and ascend to the upper body and arms. This phase can be preceded by an infectious illness, often respiratory or gastrointestinal.
 - **Duration:** This phase usually lasts from days to weeks, with symptoms progressing rapidly.
2. **Plateau Phase:**
 - **Stabilization:** After the initial progression, the symptoms may stabilize, and the patient may reach a plateau where no further deterioration occurs. This phase can last from days to weeks.
 - **Symptoms:** Patients may experience significant weakness, sensory changes, and autonomic dysfunction during this phase.

3. Recovery Phase:

- **Reversal of Symptoms:** Recovery typically occurs in the reverse order of symptom onset. For example, if leg weakness was the first symptom, it may be the last to resolve.
- **Duration:** Recovery can take weeks to months, and in some cases, it may take years. Most patients experience significant improvement, but some may have residual weakness or other complications.

The prognosis of GBS can vary widely based on several factors:

1. Factors Indicating Poor Prognosis:

- **Rapid Progression:** A very rapid progression of symptoms within days is associated with a worse outcome.
- **Age:** Older age (over 60 years) is linked to a higher risk of severe outcomes and prolonged recovery.
- **Initial Severity:** Severe initial weakness, particularly if it leads to respiratory failure, is a negative prognostic indicator.
- **Bilateral Facial Weakness:** The presence of bilateral facial weakness at onset is associated with a poorer prognosis.
- **Inexcitability on Electromyography (EMG):** Lack of nerve excitability on EMG testing indicates a more severe form of the disease and correlates with worse outcomes.

2. Factors Indicating Better Prognosis:

- **Milder Initial Symptoms:** Patients with milder initial symptoms tend to have a better recovery.
- **Younger Age:** Younger patients generally have a more favorable prognosis.
- **Early Treatment:** Prompt initiation of treatment (IVIG or plasmapheresis) within the first two weeks of symptom onset can improve outcomes.

3. Long-term Outcomes:

- **Recovery Rates:** Approximately 70-80% of patients achieve a good recovery, with minimal or no residual symptoms.
- **Residual Symptoms:** About 15% of patients may have long-term residual symptoms, such as weakness or sensory disturbances.
- **Mortality Rate:** The mortality rate is low, around 5%, but can be higher in patients with severe complications.

4. Monitoring and Follow-up

- **Regular Assessments:** Continuous monitoring of respiratory function, motor strength, and autonomic stability is crucial during the acute and recovery phases.
- **Rehabilitation:** Early and ongoing rehabilitation is important to maximize recovery and address any residual deficits.

The evolution of GBS involves distinct phases, from acute onset to recovery, with varying durations. Prognosis is influenced by several clinical factors, and while many patients recover well, some may experience long-term effects. Early recognition and treatment are key to improving outcomes.

X. COMPLICATIONS

GBS can lead to several complications, some of which can be severe and require careful management. Here are the main complications associated with GBS:

1. Respiratory Complications

- **Respiratory Failure:** Weakness of the respiratory muscles can lead to respiratory failure, necessitating mechanical ventilation. This is a critical complication that requires immediate attention.
- **Aspiration Pneumonia:** Due to swallowing difficulties and respiratory muscle weakness, patients are at increased risk of aspiration pneumonia.

2. Autonomic Dysfunction

- **Cardiovascular Issues:** GBS can cause fluctuations in heart rate (tachycardia or bradycardia) and blood pressure (hypotension or hypertension). Severe autonomic dysfunction can lead to life-threatening arrhythmias.
- **Gastrointestinal Problems:** Patients may experience gastrointestinal dysmotility, leading to constipation, ileus, or bowel obstruction.

3. Neurological Complications

- **Persistent Weakness:** Some patients may experience long-term weakness or sensory disturbances, which can affect daily activities and quality of life.
- **Pain:** Neuropathic pain is common in GBS patients and can be debilitating. It may manifest as burning, tingling, or shooting pain.

4. Infections

- **Urinary Tract Infections (UTIs):** Patients may be at risk for UTIs, especially if they require catheterization during hospitalization.
- **Decubitus Ulcers:** Prolonged immobility can lead to pressure sores, particularly in patients who are bedridden.

5. Thromboembolic Events

- **Deep Vein Thrombosis (DVT):** Immobility increases the risk of DVT, which can lead to pulmonary embolism (PE), a serious and potentially fatal complication.

6. Psychological Impact

- **Anxiety and Depression:** The sudden onset of GBS and the potential for long-term disability can lead to psychological issues, including anxiety and depression.

7. Recurrence

- **Recurrent GBS:** Although rare, some patients may experience recurrent episodes of GBS, which can complicate management and recovery.
- **Death:** while the overall mortality rate is low (around 5%), it can occur, particularly in cases with severe complications such as respiratory failure or autonomic instability.

Complications of GBS can significantly impact patient outcomes and quality of life. Early recognition and management of these complications are crucial to improving prognosis and facilitating recovery. Regular monitoring and supportive care are essential components of the management plan for patients with GBS.

XI. PREVENTION AND RECOMMENDATION

Preventing GBS is challenging, as the exact cause is often unknown and it can be triggered by various infections. However, there are some recommendations and preventive measures that can be considered:

1. Prevention Strategies

- **Vaccination:**

- **Influenza Vaccine:** While there has been some concern about a potential association between the influenza vaccine and GBS, the risk is very low. Vaccination is still recommended, especially for high-risk populations, as the benefits of preventing influenza outweigh the risks.
- **Other Vaccines:** Vaccination against other infectious diseases (e.g., COVID-19) is also encouraged, as infections can trigger GBS.
- **Infection Control:**
 - **Hygiene Practices:** Good hygiene practices, such as regular handwashing and avoiding close contact with sick individuals, can help reduce the risk of infections that may trigger GBS.
 - **Prompt Treatment of Infections:** Early diagnosis and treatment of infections, particularly respiratory and gastrointestinal infections, may help reduce the risk of developing GBS.
- **Awareness and Education:**
 - **Public Awareness:** Educating the public and healthcare providers about the signs and symptoms of GBS can lead to earlier recognition and treatment, potentially improving outcomes.
 - **Patient Education:** Individuals who have had GBS should be informed about the possibility of recurrence and the importance of seeking medical attention if they experience new or worsening symptoms.

2. Recommendations for Patients and Caregivers

- **Monitoring for Symptoms:**
 - Patients and caregivers should be vigilant for early signs of GBS, such as sudden weakness, tingling, or numbness, especially following an infection.
- **Rehabilitation:**
 - Early and ongoing rehabilitation is crucial for recovery. Patients should engage in physical therapy to regain strength and mobility and to prevent complications related to immobility.
- **Psychosocial Support:**
 - Providing psychological support and counseling for patients and families can help address the emotional impact of GBS and improve coping strategies.
- **Follow-up Care:**
 - Regular follow-up with healthcare providers is important for monitoring recovery, managing any residual symptoms, and addressing complications.
- **Lifestyle Modifications:**
 - Encouraging a healthy lifestyle, including a balanced diet, regular exercise (as tolerated), and stress management techniques, can support overall health and recovery.

While there is no definitive way to prevent GBS, awareness of potential triggers, good hygiene practices, and vaccination can help reduce the risk of infections that may lead to the syndrome. Education and support for patients and caregivers are essential for managing the condition and improving outcomes. Regular follow-up and rehabilitation are key components of care for individuals recovering from GBS.

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ISCHEMIC STROKE

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I. DEFINITION

Ischemic stroke is a disruption in the flow of blood to the brain, which is typically caused by either a blood clot forming within a blood vessel (thrombosis) or a clot traveling from another part of the body and blocking a blood vessel in the brain (embolism) [1].

Sacco et al Updated Definition of Stroke 2073

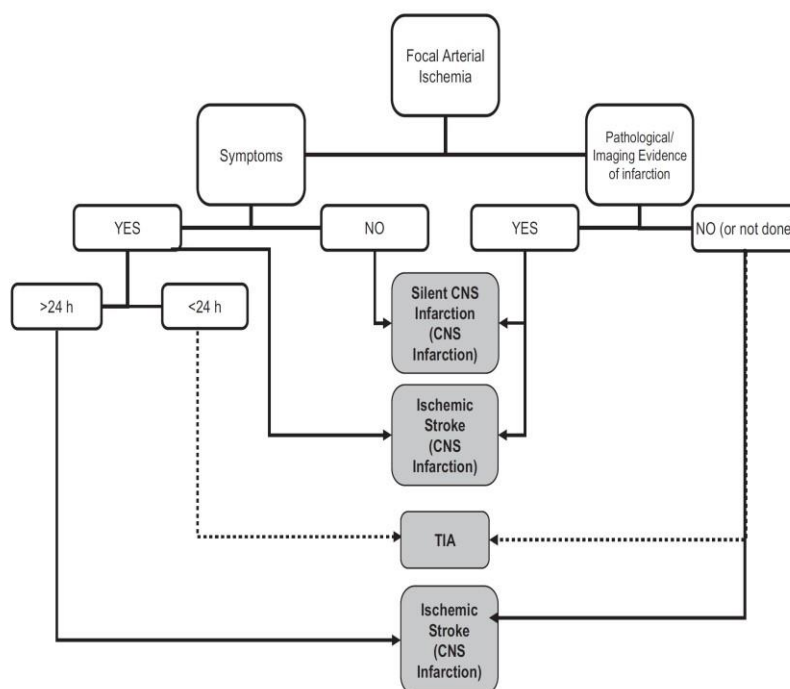


Figure 2. Flow chart depicting a proposed decision tree for determination of a cerebrovascular event definition. Cerebrovascular events are depicted in shaded boxes and are defined by the composite of both features of symptoms on the left and pathological/imaging evidence of infarction on the right. For example, the cerebrovascular event defined as “silent CNS infarction” requires focal arterial ischemia, no symptoms, and pathological or imaging evidence of infarction. CNS indicates central nervous system; and TIA, transient ischemic attack.

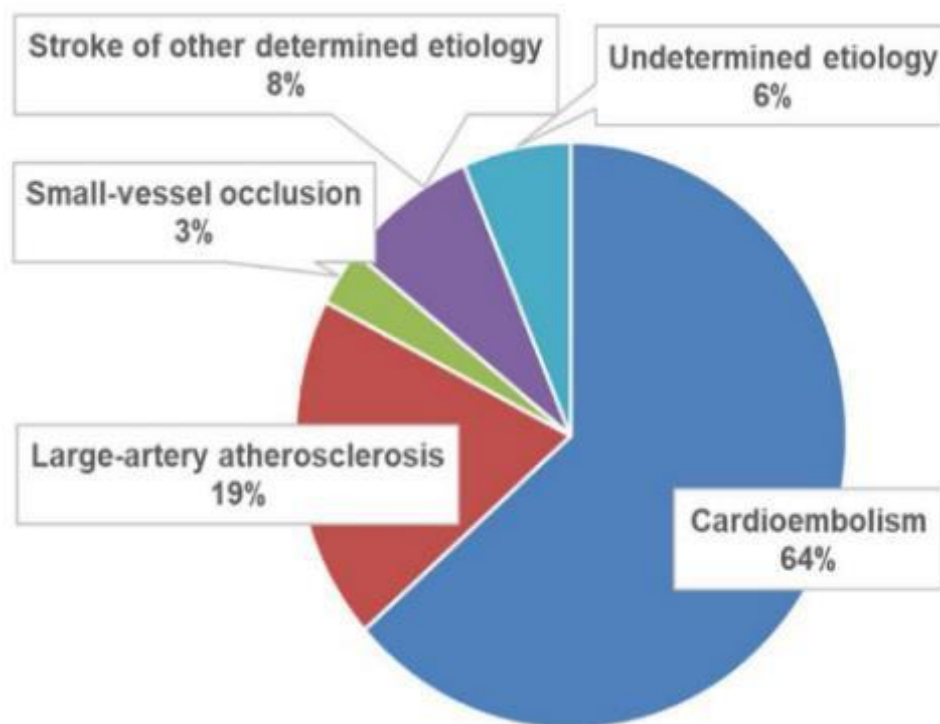
II. EPIDEMIOLOGY

In 2019, the global burden of disease study found that there were 12.2 million stroke incidents, 101 million prevalent strokes, 143 million disability-adjusted life years lost to stroke, and 6.55 million stroke fatalities. Stroke remained the second largest cause of death worldwide and the third major cause of death and disability when combined. Over the span of 1990 to 2019, the number of stroke incidents increased by 70%, prevalent strokes by 85%, stroke fatalities by 43%, and disability-adjusted life years attributed to stroke increased by 32%. Ischemic stroke accounted for 62% of all strokes in 2019. Additionally, the age-standardized incidence of ischemic stroke was significantly higher in the Asian region, according to the Institute for Health Metrics and Evaluation's visualization of global burden of disease data [2]. Regarding to the report by Ministry of Health , Cambodia 2023, the total number of stroke patients was 22,793 (1,176 dead). Annually, 15 million people worldwide suffer a stroke. Of these, 5 million die and another 5 million are left permanently disabled, placing a burden on family and community (WHO , 2023). From 1st January 2023- 30 June 2024 report from Khmer-Soviet Friendship Hospital (KSFH): 1504

Strokes has ischemic stroke 1261 and 208 hemorrhage stroke, prevalence of neurological diseases in Hospitalized Patients at the Neurology Department of KSFH: January 01, 2023 - June 30, 2024.

III. ETIOLOGY

The classification of stroke subtypes was determined by etiology, following the TOAST classification, which included five categories: (1) large-artery atherosclerosis (LAA), (2) small vessel occlusion (SVO), (3) cardio-embolism (CE), (4) stroke of another identified cause, and (5) stroke of unknown cause.



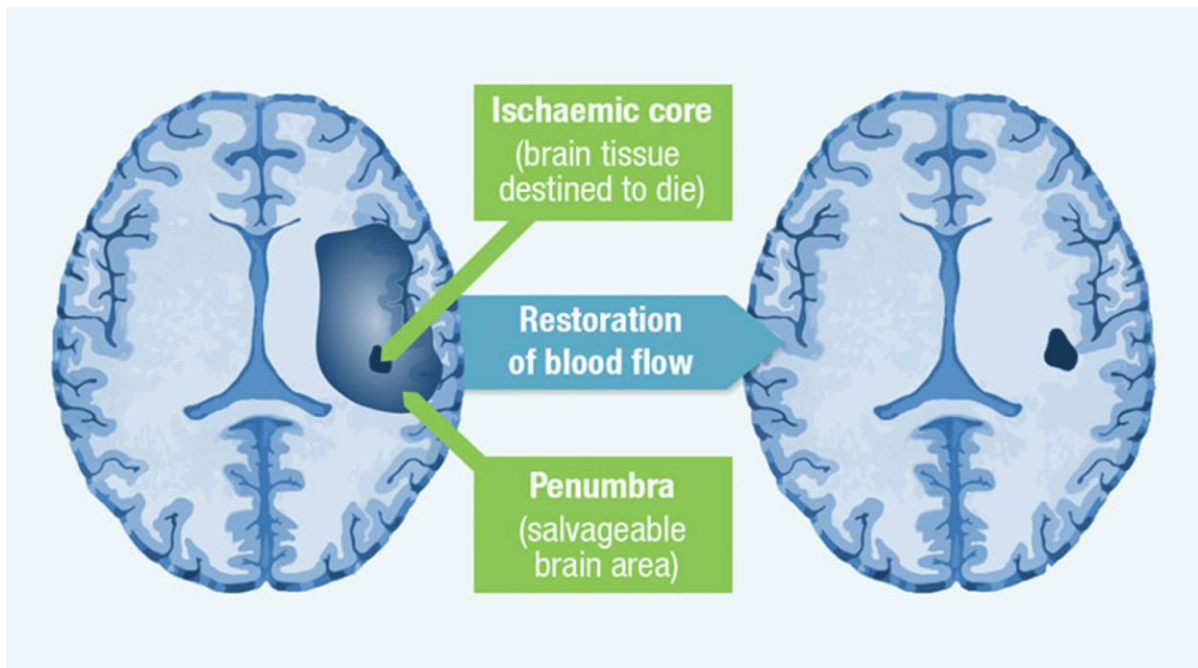
IV. PATHOPHYSIOLOGY

The common pathway of ischemic stroke is lack of sufficient blood flow to perfuse cerebral tissue, due to narrowed or blocked arteries leading to or within the brain. Ischemic strokes can be broadly subdivided into thrombotic and embolic strokes.

Narrowing is commonly the result of atherosclerosis the occurrence of fatty plaques lining the blood vessels. As the plaques grow in size, the blood vessel becomes narrowed and the blood flow to the area beyond is reduced. Damaged areas of an atherosclerotic plaque can cause a blood clot to form, which blocks the blood vessel is a thrombotic stroke. In an embolic stroke, blood clots or debris from elsewhere in the body, typically the heart valves, travel through the circulatory system and block narrower blood vessels. Based on the etiology of ischemic stroke, a more accurate sub-classification is generally used:

- (1) **Large artery disease:** atherosclerosis of large vessels, including the internal carotid artery, vertebral artery, basilar artery, and other major branches of the Circle of Willis.
- (2) **Small vessel disease:** changes due to chronic disease, such as diabetes, hypertension, hyperlipidemia, and smoking, that lead decreased compliance of the arterial walls and/or narrowing and occlusion of the lumen of smaller vessels.

- (3) **Cardio-Embolic stroke:** the most common cause of an embolic stroke is atrial fibrillation.
- (4) **Stroke of determined etiology:** such as inherited diseases, metabolic disorders, and coagulopathies.
- (5) **Stroke of undetermined etiology:** after exclusion of all of the above.



In the core area of a stroke, blood flow is so drastically reduced that cells usually cannot recover and subsequently undergo cellular death. The tissue in the region bordering the infarct core, known as the ischemic penumbra, is less severely affected. This region is rendered functionally silent by reduced blood flow but remains metabolically active. Cells in this area are endangered but not yet irreversibly damaged. They may undergo apoptosis after several hours or days but if blood flow and oxygen delivery is restored shortly after the onset of stroke, they are potentially recoverable.

V. DIAGNOSIS

Diagnosis of ischemic stroke relies on clinical signs and symptoms, along with the use of diagnostic tools such as brain CT, MRI, or laboratory tests. These investigations help in confirming the presence of an ischemic stroke and provide valuable information for further management and treatment.

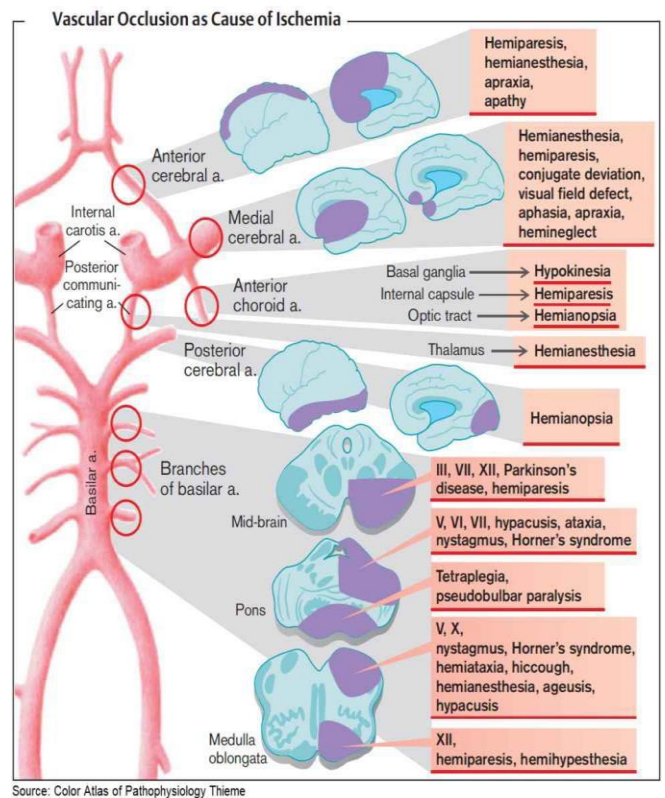
Clinical signs

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ដើម្បីសង្គ្រោះជីវិតជាបន្ទាន់ ទូរស័ព្ទលេខ **119**

- Type of infarctions:
 - (1) **Cerebral hemisphere infarcts:** may be classified by vascular territory:
 - Anterior cerebral artery infarcts:** proximal to the medial striate branch, they cause contralateral hemiplegia; distal to the medial striate branch, they cause contralateral paralysis with sparing of the arm and face.
 - Middle cerebral artery infarcts:** causes contralateral hemiplegia, often sparing the leg, and hemianopia. Aphasia is present if the dominant hemisphere is affected.
 - Posterior cerebral artery infarcts:** causes contralateral hemianopia.
 - Motor signs** are upper motor neurons.
 - (2) **Lacunar strokes:** Lacunar strokes are due to small areas of infarction usually occurring around the basal ganglia, thalamus, and pons. Clinical features include:



- Pure motor or sensory signs
- Mixed motor and sensory picture
- Ataxia
- Dysarthria

NOTE: They may also be asymptomatic and found at post-mortem examination.

- (3) **Brainstem infarcts**: clinical features include:
 - Hemi- or quadriplegia
 - Sensory loss
 - Diplopia
 - Facial weakness and numbness
 - Nystagmus
 - Dysphasia
 - Coma due to damage to the reticular formation
 - Locked-in syndrome from upper brain-stem infarction (the patient is conscious but unable to respond)

The most common brainstem vascular syndrome is lateral medullary syndrome (Wallenberg's syndrome), which is due to thromboembolism of the posterior inferior cerebellar artery. The patient presents with sudden vertigo and vomiting. Ipsilateral signs include 9th and 10th nerve lesions, Horner's syndrome, spinothalamic sensory loss in the face, diplopia, and cerebellar signs in the limbs. Contralateral signs include spinothalamic sensory loss in the body.

5.2. Investigation

The following investigations should be performed in patients who have suffered a stroke:

- Complete blood count (CBC): polycythemia or thrombocytopenia.
- Coagulation studies: bleeding disorders.
- Erythrocyte sedimentation rate (ESR): giant cell arteritis.
- Blood glucose: hyper- or hypoglycemia may lead to impaired consciousness.
- Chest x-ray (CXR): primary tumor; left atrium enlarged with mitral stenosis.
- Syphilis serology.
- Non-contrast computed tomography (CT) scan: this will distinguish between hemorrhagic and ischemic infarction if done within 2 weeks and should be done acutely to rule out hemorrhage whenever thrombolytics are considered; also aids diagnosis of other conditions in cases of uncertainty, such as tumor or subdural hematoma; should always be performed in cerebellar stroke as hematoma requires urgent evacuation.
- MRI: can show ischemia within 15-30 minutes. Different types of MRI can be performed, including MR angiography (MRA), depending on the information needed.
- Carotid duplex scan: stenoses.

VI. DIFFERENTIAL DIAGNOSIS

- Complicated migraines
- Drug toxicity
- Intracranial hemorrhage
- Intracranial tumor
- Intracranial abscess
- Hypoglycemia
- Hyperglycemia
- Hypertensive encephalopathy
- Multiple sclerosis
- Seizure, sepsis
- Syncope
- Wernicke encephalopathy
- Metabolic abnormalities

VII. MANAGEMENT

1. Objectives

Effective management of ischemic stroke involves identifying and managing risk factors such as hypertension, smoking, diabetes mellitus, high cholesterol, and atrial fibrillation. Following a stroke, blood pressure may remain elevated for several days, and the initiation of antihypertensive treatment is typically deferred for about two weeks. Cholesterol reduction is crucial to minimize coronary risks, as myocardial infarction is a significant cause of death in the post-stroke period. Anticoagulation is recommended for patients with atrial fibrillation and thromboembolic stroke. In cases of severe internal carotid artery stenosis (>70%) with recent TIAs, endarterectomy should be considered.

2. Critical Signs

- Sudden weakness or numbness, particularly on one side of the body
- Difficulty speaking or understanding speech
- Vision problems in one or both eyes
- Loss of balance or coordination
- Severe headache with no known cause.

3. Mortality Risk Factors

- Impaired consciousness
- Severe hemiplegia
- Incontinence
- Defect in conjugate gaze
- Advanced age
- Previous history of strokes.

4. Outpatient Management

Post-stroke outpatient care should include:

- Regular follow-up for risk factor management
- Blood pressure monitoring
- Medication adherence (antihypertensives, statins, antiplatelets)

- Support for rehabilitation (physical, occupational, speech therapy)

- **Rehabilitation Interventions for Stroke**

Rehabilitation aims to enable people to reach their optimal physical, cognitive, emotional, communicative, and social activity levels through addressing impairments related to spasticity, upper and lower extremity dysfunction, shoulder and central pain, mobility/gait, dysphagia, vision, and communication. As stroke can cause continuing problems in subsequent years and decades, people living with stroke need lifelong access to rehabilitation services. Furthermore, rehabilitation aims to train and support carers and families by providing appropriate strategies to care for themselves.

Interventions:

- Range of motion exercises
- Positioning for the prevention of contractures
- Balance training
- Gait training
- Constraint induced movement therapy
- Activities of daily living training
- Speech and language interventions
- Fitness training
- Modification of home environment
- Muscle-strengthening exercises
- Physical exercise training
- Provision and training in the use of assistive products for self-care
- Assistive products: Canes/sticks/tetrapod , Rollators , Walking frames/walkers, Orthoses (lower limb) and Wheelchair

- Depression screening and management

5. First-line Treatment

- **Intravenous (IV) Alteplase:** Administered within a 4.5-hour window for eligible patients. The dose is 0.9 mg/kg (max 90 mg), with a 10% bolus given over 1 minute, followed by the remainder over 60 minutes. Inclusion criteria include:
 - Measurable neurological deficit
 - Symptom onset within the treatment window
 - Age 18 or older
- **Mechanical Thrombectomy:** Recommended for patients with large vessel occlusion, can be considered even after fibrinolytic therapy.

6. Treatment Adjustment

Treatment may need to be adjusted based on:

- Blood pressure levels (aim for <180/105 mmHg post-alteplase)
- Blood glucose levels (maintain 140-180 mg/dL)
- Neurological status and response to initial treatment
- Presence of complications (e.g., seizures, edema)

7. Algorithm for Management

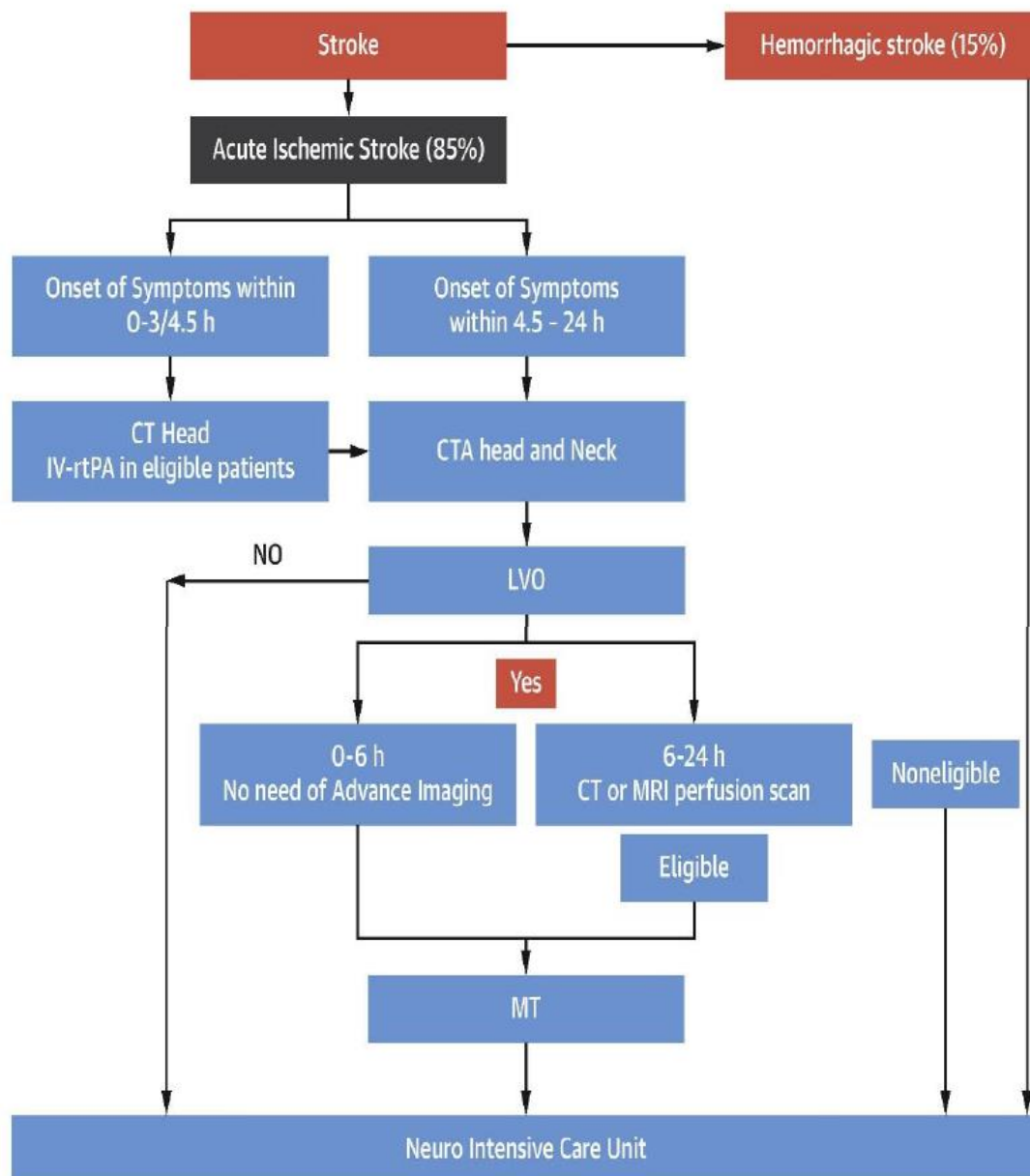
Initial Assessment: Neurological examination, imaging (CT/MRI)

Risk Factor Identification: Evaluate hypertension, diabetes, cholesterol, etc.

- Immediate Treatment:
 - Administer IV alteplase if within criteria.
 - Consider mechanical thrombectomy for eligible patients.
- Post-Treatment Care:

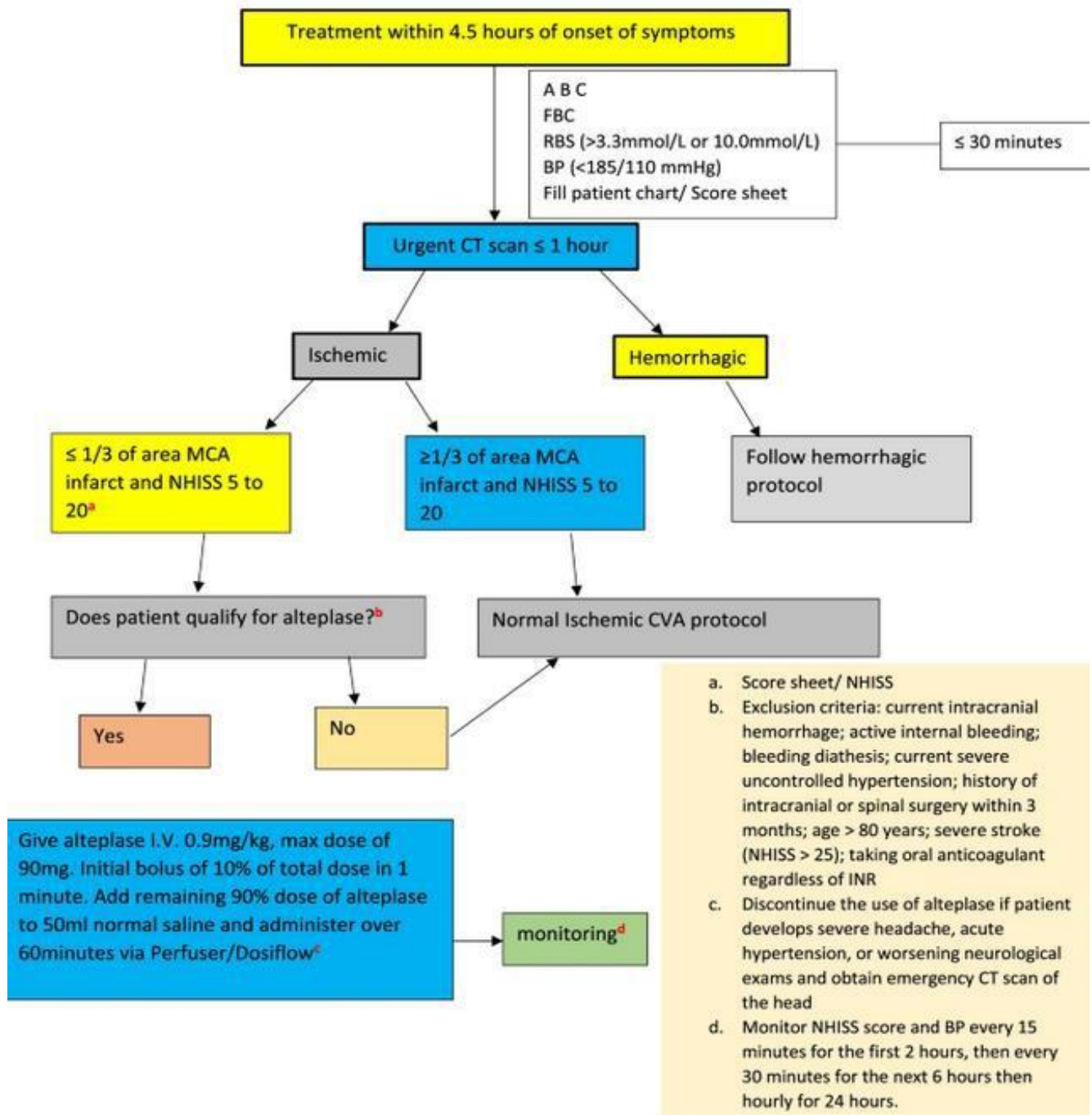
- Monitor vital signs, glucose, temperature, and blood pressure.
- Initiate antiplatelet therapy (aspirin or clopidogrel).
- Address complications (DVT prophylaxis, seizure management).
- Outpatient Follow-up: Establish a rehabilitation plan and regular check-ups.

CENTRAL ILLUSTRATION: Algorithm for Triaging Patients With Acute Ischemic Stroke With Different Imaging Options



Patel, P. et al. J Am Coll Cardiol. 2020;75(15):1844-56.

MANAGEMENT OF CVA USING ALTEPLASE



Drug & Therapeutics Committee, TTH. 2009

MANAGEMENT OF CVA USING ALTEPLASE

VIII. SPECIAL CASES

- **Cerebellar Edema:** Early consultation with neurosurgery for possible interventions like ventriculostomy or decompressive craniectomy.
- **Seizures:** Initiate anti-epileptic medication for recurrent seizures, but routine prophylaxis is not recommended.
- **Cardiac Evaluation:** Monitor for arrhythmias and perform troponin testing.

IX. COMPLICATION

Complications of ischemic stroke can be significant and vary based on the severity of the stroke, the area of the brain affected, and the timeliness of treatment. Common complications include:

- **Neurological Deficits:** These can include weakness or paralysis (hemiparesis), speech and language difficulties (aphasia), and cognitive impairments.
- **Seizures:** Some patients may experience seizures, particularly if the stroke affects certain areas of the brain.
- **Cerebral Edema:** Swelling in the brain can lead to increased intracranial pressure, which may require intervention.
- **Aspiration Pneumonia:** Difficulty swallowing (dysphagia) can increase the risk of food or liquids entering the lungs, leading to pneumonia.
- **Deep Vein Thrombosis (DVT):** Immobility after a stroke can increase the risk of blood clots in the legs.
- **Pressure Sores:** Prolonged immobility can lead to skin breakdown and pressure ulcers.
- **Urinary Incontinence:** Some patients may experience loss of bladder control.
- **Depression and Emotional Changes:** Many strokes' survivors experience depression, anxiety, or emotional lability due to the impact of the stroke.
- **Recurrent Stroke:** Patients who have experienced one stroke are at higher risk for subsequent strokes.
- **Hydrocephalus:** In some cases, fluid buildup in the brain can occur, necessitating further treatment.
- **Cardiac Complications:** Increased risk of arrhythmias, particularly atrial fibrillation, which can lead to additional strokes.

Timely intervention and rehabilitation can help manage these complications and improve outcomes for stroke survivors.

X. PREVENTION AND RECOMMENDATION: FOR ISCHEMIC STROKE

1. Life-style Modifications:

- **Diet:** Adopt a heart-healthy diet, such as the Mediterranean diet, rich in fruits, vegetables, whole grains, lean proteins, and healthy fats. Limit salt, sugar, and saturated fats.
- **Exercise:** Engage in regular physical activity (at least 150 minutes of moderate exercise per week) to help maintain a healthy weight and reduce cardiovascular risk.
- **Smoking Cessation:** Quit smoking and avoid exposure to secondhand smoke, as smoking significantly increases stroke risk.

2. Blood Pressure Management:

- Monitor and control blood pressure regularly, aiming for a target of less than 130/80 mmHg.
- Lifestyle changes and medications may be necessary for those with hypertension.

3. Diabetes Management:

- Keep blood sugar levels within recommended ranges through diet, exercise, and medication adherence for individuals with diabetes.

4. Cholesterol Management:

- Maintain healthy cholesterol levels. Statin therapy may be recommended for those with high LDL cholesterol or existing cardiovascular disease.

5. Atrial Fibrillation Monitoring:

- Regularly check for atrial fibrillation, especially in individuals over 65 or with risk factors. Anticoagulation therapy may be advised if AFib is present.

6. Antiplatelet Therapy:

- For individuals with a history of transient ischemic attacks (TIAs) or minor strokes, low-dose aspirin or clopidogrel may be recommended to reduce the risk of future strokes.

7. Weight Management:

- Achieve and maintain a healthy weight to reduce the risk of stroke and other cardiovascular diseases.

8. Alcohol Consumption:

- Limit alcohol intake to moderate levels (up to one drink per day for women and two for men) to reduce the risk of hypertension and stroke.

9. Regular Health Screenings:

- Routine health check-ups to monitor blood pressure, cholesterol levels, and other risk factors can help identify and manage potential issues early.

10. Education and Awareness:

- Educate individuals about stroke symptoms (e.g., FAST: Face drooping, Arm weakness, Speech difficulties, Time to call emergency services) to ensure prompt treatment.

11. Stress Management:

- Implement stress reduction techniques, such as mindfulness, meditation, or yoga, to help manage overall health and well-being.

Preventive strategies focus on lifestyle modifications, medical management of risk factors, and awareness of stroke symptoms. Regular consultations with healthcare providers can ensure personalized prevention plans and timely interventions.

XI. EVOLUTION AND PROGNOSIS

- Proper hydration, nutrition, and physical therapy are critical.
- Regular monitoring for complications (e.g., pneumonia, depression).
- Approximately 20-25% of patients with thromboembolic infarction and 75% with hemorrhagic strokes may not survive the first month.
- A third of survivors may fully recover, while another third may experience severe disability.

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INTRACEREBRAL HEMORRHAGE STROKE

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I. DEFINITION

Cerebrovascular accident (CVA), otherwise called a stroke, is the third major cause of morbidity and mortality in many developed countries. Stroke can be either ischemic or hemorrhagic.

Hemorrhagic stroke is due to bleeding into the brain by the rupture of a blood vessel. Hemorrhagic stroke may be further subdivided into intra-cerebral hemorrhage (ICH) and subarachnoid hemorrhage (SAH). ICH is bleeding into the brain parenchyma, and SAH is bleeding into the subarachnoid space. Hemorrhagic stroke is associated with severe morbidity and high mortality.

II. EPIDEMIOLOGY

Hemorrhagic stroke contributes to 10% to 20% of strokes annually. The incidence is around 12% to 15% of cases per 1,00,000 per year.

ICH is more common in Asians, advanced age, male sex, and low- and middle-income countries. The incidence is more common in men and increases with age.

The global incidence is increasing, predominantly in African and Asian countries. Japanese data have shown that control of hypertension reduces the incidence of ICH. The case fatality rate is 25% to 30% in high-income countries, while it is 30% to 48% in low- to middle- income countries. The ICH fatality rate depends on the efficacy of critical care.

Based on data from the GBD study, there was a wide range of age- and sex-standardized stroke DALYs lost in Asia, in 2010. The lowest rates are in Japan (706.6/100,000 people) and Singapore (804.2/100,000 people), with low rates also observed in Bangladesh and Bhutan. The highest rates are in Mongolia (4,409.8/100,000 people) and Indonesia (3,382.2/100,000 people), with high rates also observed in Myanmar, Lao PDR, North Korea, and Cambodia.

Based on MoH of Cambodia report 2022 within 5 years (2018-2022) data, total amount of IPD patients with cerebral-cardio vascular disease are 71857 cases. We found that the patients with ischemic stroke are 22793 cases (died 1176 cases), hemorrhagic stroke are 8200 patients (died 328 patients). Particularly, in Khmer-Soviet Friendship Hospital from 2015-2021, our stroke patients are 5620 cases among total IPD patients 250653 cases.

III. ETIOLOGY

Hypertension is the most common cause of hemorrhagic stroke.

Cerebral amyloid angiopathy (CAA) is an important cause of primary lobar intracerebral bleeding in older adults.

Other Important Risk Factors of intracerebral hemorrhage (ICH):

- Cigarette smoking and moderate or heavy alcohol consumption and chronic alcoholism.
- Chronic liver disease with coagulopathy and thrombocytopenia.
- Decreased low-density lipoprotein cholesterol and low triglycerides are also risk factors.

- Dual antiplatelet therapy has an increased risk of ICH than monotherapy.
- Sympathomimetics such as cocaine, heroin, amphetamine, ephedrine, and phenylpropanolamine carry an increased risk of a cerebral hemorrhage.
- Cerebral microbleeds (CMBs) associated with hypertension, diabetes mellitus, and cigarette smoking increase the risk of ICH.
- Old age and male sex. The incidence of ICH increases after 55 years of age. The relative risk after 70 years is 7.
- The tumors which are more prone to bleed are glioblastoma, lymphoma, metastasis, meningioma, pituitary adenoma, and hemangioblastoma.

The usual causes of spontaneous subarachnoid hemorrhage (SAH) are ruptured aneurysm, arteriovenous malformation, vasculitis, cerebral artery dissection, dural sinus thrombosis, and pituitary apoplexy.

IV. PATHOPHYSIOLOGY

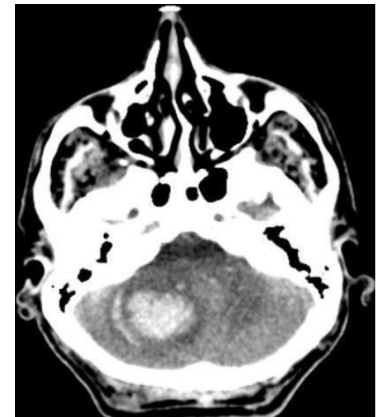
The common sites of the bleed are the basal ganglia (50%), cerebral lobes (10% to 20%), the thalamus (15%), pons and the brain stem (10% to 20%), and the cerebellum (10%).[5]



Lobar hemorrhage



Pontine hemorrhage



Cerebellar hemorrhage

The primary injury is due to the compression of brain tissue by the hematoma and an increase in the intracranial pressure (ICP).

Usually, the hematoma enlarges in 3 hours to 12 hours. The enlargement of hematoma occurs in 3 hours in one-third of cases. The perihematoma edema increases within 24 hours, peaks around 5 to 6 days, and lasts up to 14 days.

V. DIAGNOSIS

Diagnosis of a hemorrhagic stroke is based on a thorough medical history and physical exam, and doctors may strongly suspect stroke based on the patient's symptoms.

Clinical assessment cannot reliably distinguish intracerebral hemorrhage from ischemic stroke; imaging tests include magnetic resonance imaging (MRI) or computed tomography (CT) scans is required.

Electroencephalogram (EEG) or lumbar puncture (LP) may be done in some optional cases to help work up and confirm the diagnosis.

More frequent symptoms of ICH may include:

- Alteration in level of consciousness (present in approximately 50% of patients)
- Nausea and vomiting (approximately 40-50%)
- Sudden, severe headache (approximately 40%)
- Seizures (approximately 6-7%)
- Sudden weakness or paralysis of the face, arm or leg, or numbness, particularly on one side of the body
- Sudden vision changes
- Loss of balance or coordination
- Difficulty understanding, speaking (slurring, confusion), reading, or writing

Investigation:

- Laboratory study (CBC, Coagulation studies, Liver function, Kidney function, blood glucose ...)
- Imaging studies: CT scan / MRI cerebral

VI. DIFFERENTIAL DIAGNOSIS

The differential diagnoses of hemorrhagic stroke are:

- Acute hypertensive crisis
- Pituitary apoplexy
- Cerebral venous thrombosis
- Dural sinus thrombosis
- Cervical artery dissection
- Reversible cerebral vasoconstrictive syndrome (RCVS)
- Hemorrhagic neoplasms
- Arterio-venous malformations
- Meningitis
- Acute subdural hematoma
- Hemorrhagic infarct

Imaging studies such as CT and MRI can rule out these entities.

VII. MANAGEMENT

1. Objectives

- Goal to stop bleeding
- Relieve the symptoms
- Management the cause of ICH
- Prevention the complication

2. Critical sign

- Sudden onset of a severe headache, often described as the "worst headache of my life"
- Weakness or paralysis on one side of the body that develops quickly
- Difficulty speaking or understanding speech
- Seizure
- Loss of consciousness
- Sign of high ICP

3. Mortality risk factors

- Old Age: The risk of ICH increases significantly with age.
- Comorbid conditions: Uncontrolled hypertension, Chronic kidney disease, Diabetes, Coronary artery disease, Atrial fibrillation ...
- Severity of ICH: Larger hemorrhage volume and bleeding in critical areas of the brain.
- Intraventricular hemorrhage: when bleeding extends into the ventricles (fluid- filled cavities) of the brain, it can increase intracranial pressure and worsen the prognosis.
- Level of consciousness: lower Glasgow Coma Scale (GCS) score
- Infection

4. Outpatient management

- Blood pressure control
- Rehabilitation: Physical therapy, Speech therapy, Occupational therapy
- Diet and lifestyle modifications
- Fall prevention
- Education and support

5. Rehabilitation Interventions for Stroke

Rehabilitation aims to enable people to reach their optimal physical, cognitive, emotional, communicative, and social activity levels through addressing impairments related to spasticity, upper and lower extremity dysfunction, shoulder and central pain, mobility/gait, dysphagia, vision, and communication. As stroke can cause continuing problems in subsequent years and decades, people living with stroke need lifelong access to rehabilitation services. Furthermore, rehabilitation aims to train and support carers and families by providing appropriate strategies to care for themselves.

Interventions:

- Range of motion exercises
- Positioning for the prevention of contractures
- Balance training
- Gait training
- Constraint induced movement therapy

- Activities of daily living training
- Speech and language interventions
- Fitness training
- Modification of home environment
- Muscle-strengthening exercises
- Physical exercise training
- Provision and training in the use of assistive products for self-care
- Assistive products: Canes/sticks/tetrapod , Rollators , Walking frames/walkers, Orthoses (lower limb) and Wheelchair
- Depression screening and management

6. First-line management of Intracerebral Hemorrhage

A severity score based on neurological exam findings should be conducted as part of the initial assessment. The National Institute of Health Stroke Score (NIHSS) is preferred for awake or drowsy patients, or a Glasgow Coma Scale (GCS) in patients who are obtunded, semi or fully comatose.

- Patients with declining GCS and/or equal to less than 8 should be rapidly assessed for airway support by endotracheal intubation.
- Patients with reduced level of consciousness, pupillary changes and/or other signs of herniation should have temporizing maneuvers to manage presumed elevation in intracranial pressure (ICP), such as temporary hyperventilation and hyperosmotics (e.g. mannitol or 3% saline)

Patients with suspected ICH should undergo computerized tomography (CT) immediately following stabilization to confirm diagnosis.

Evaluation should include questions about medication history, and antithrombotic therapy, measurement of platelet count, partial thromboplastin time (PTT) and International Normalized Ratio (INR)

Patients should be assessed for clinical signs of increased ICP such as pupil reaction and level of consciousness.

A GCS score and neurovital signs should be conducted at baseline and repeated at least hourly for the first 24 hours, depending on stability of patient.

The use of tranexamic acid has been shown to be safe in a large phase 3 trial (TICH-2) but there was no effect on the primary outcome of functional status at 90 days. Overall, the clinical role of tranexamic acid for spontaneous ICH remains unclear and there is no evidence for its use in the setting of anticoagulant-related ICH.

Obtain immediate neurosurgical consultation for imaging findings indicating need for emergency surgery [8] :

- Cerebellar ICH that is either ≥ 3 cm diameter or causing brainstem compression
- IVH with obstructive hydrocephalus and neurologic deterioration
- Hemispheric ICH with life-threatening brain compression or obstructive hydrocephalus

7. Treatment adjustment:

❖ Blood Pressure Management

Elevated blood pressure is common in patients with acute ICH. Patients may develop elevated blood pressure due to an increase in intracranial pressure (ICP) and pain from the mass effect of the hemorrhage. Uncontrolled elevations in blood pressure and blood pressure variability are risk factors for hemorrhagic expansion and poor outcome.

We manage elevated blood pressure in acute spontaneous ICH as follows, in agreement with guidelines from the American Heart Association:

- In patients with spontaneous ICH requiring acute BP lowering, careful titration to ensure continuous smooth and sustained control of BP, avoiding peaks and large variability in SBP, can be beneficial for improving functional outcomes.
- In patients with spontaneous ICH in whom acute BP lowering is considered, initiating treatment within 2 hours of ICH onset and reaching target within 1 hour can be beneficial to reduce the risk of HE and improve functional outcome.
- In patients with spontaneous ICH of mild to moderate severity presenting with SBP between 150 and 220 mm Hg, acute lowering of SBP to a target of 140 mm Hg with the goal of maintaining in the range of 130 to 150 mm Hg is safe and may be reasonable for improving functional outcomes.
- In patients with spontaneous ICH presenting with large or severe ICH or those requiring surgical decompression, the safety and efficacy of intensive BP lowering are not well established.
- In patients with spontaneous ICH of mild to moderate severity presenting with SBP >150 mm Hg, acute lowering of SBP to <130 mm Hg is potentially harmful.

Parenteral labetalol, hydralazine, nicardapine and/or enalapril (oral or intravenous) may be considered for acute blood pressure reduction.

❖ Management of Anticoagulation

Patients presenting with anticoagulant-related ICH should have their anticoagulation withheld and should be considered for immediate reversal, irrespective of the underlying indication for anticoagulation. Beyond initial investigations, further management should be tailored to the specific antithrombotic agent used. [Figure 1]

- Warfarin should be reversed immediately with prothrombin complex concentrate (PCC) dosed as per local protocols and in conjunction with intravenous Vitamin K 10 mg
- For patients on Direct oral anticoagulants (DOACs), most information about anticoagulation activity would come from establishing time of last dose, creatinine clearance, anti-Factor Xa level if available.
- Factor Xa inhibitors (apixaban, edoxaban, rivaroxaban) should be stopped immediately and PCC administered at a dose of 50 units per Kg with a maximum dose of 3000 u (*no targeted anti-Factor Xa reversal agents available in Cambodia at this time*).
- Dabigatran should be stopped immediately and reversed with idarucizumab; patients should be given a total dose of 5 g, in 2 intravenous bolus doses of 2.5 g

each, given no more than 15 minutes apart (*not available in Cambodia at this time*)

- If the patient has received therapeutic low molecular weight heparin (LMWH) within the past 12 hours, consider administering protamine.
- If the patient is receiving intravenous heparin infusion at the time of ICH, infusion should be immediately discontinued, and consider administering protamine.
- Antiplatelet agents (e.g. acetylsalicylic acid (ASA), clopidogrel, dipyridamole/ASA, ticagrelor) should be stopped immediately.
- Platelet transfusions are not recommended (in the absence of significant thrombocytopenia) and may be harmful.

❖ Seizure Management

People with ICH are at a greater risk of seizures at presentation and should be monitored clinically. Consider continuous EEG for the diagnosis of nonconvulsive status epilepticus in patients with depressed level of consciousness that is out of proportion to the size and location of ICH.

New-onset seizures in patients admitted to hospital with ICH should be treated with antiepileptic medications if they are not self-limiting. A single, self-limiting seizure occurring at the onset, or within 24 hours after an ICH (considered an “immediate” post-stroke seizure) should not be treated with long-term anticonvulsant medications.

Short-term anticonvulsant therapy can be considered in such cases on an individual basis. Prophylactic use of anticonvulsants in patients with ICH is not recommended.

❖ Increased Intracranial Pressure (ICP)

In cases of suspected elevated ICP, conservative methods to decrease ICP (such as elevation of head of bed 30 degrees, methods of neuroprotection (e.g., eutermia, euglycemia), analgesia, and mild sedation) are reasonable.

In the absence of concerns regarding ICP, head of bed positioning does not seem to influence neurological outcomes or serious adverse events in stroke patients, including ICH.

There is insufficient evidence to recommend the routine or prophylactic use of hyperosmotic agents in ICH. Hyperosmotic agents (mannitol and/or 3% normal saline) can be considered as a temporizing measure to decrease ICP in ICH patients with clinical signs of herniation prior to surgical intervention:

- Hypertonic saline 23.4%: 15 to 30 mL IV bolus every 6 hours, or
- Mannitol: 0.25 to 1 g/kg IV bolus every 6 hours

Use of corticosteroids to treat ICP in ICH may cause harm, has no proven benefits, and therefore is not recommended.

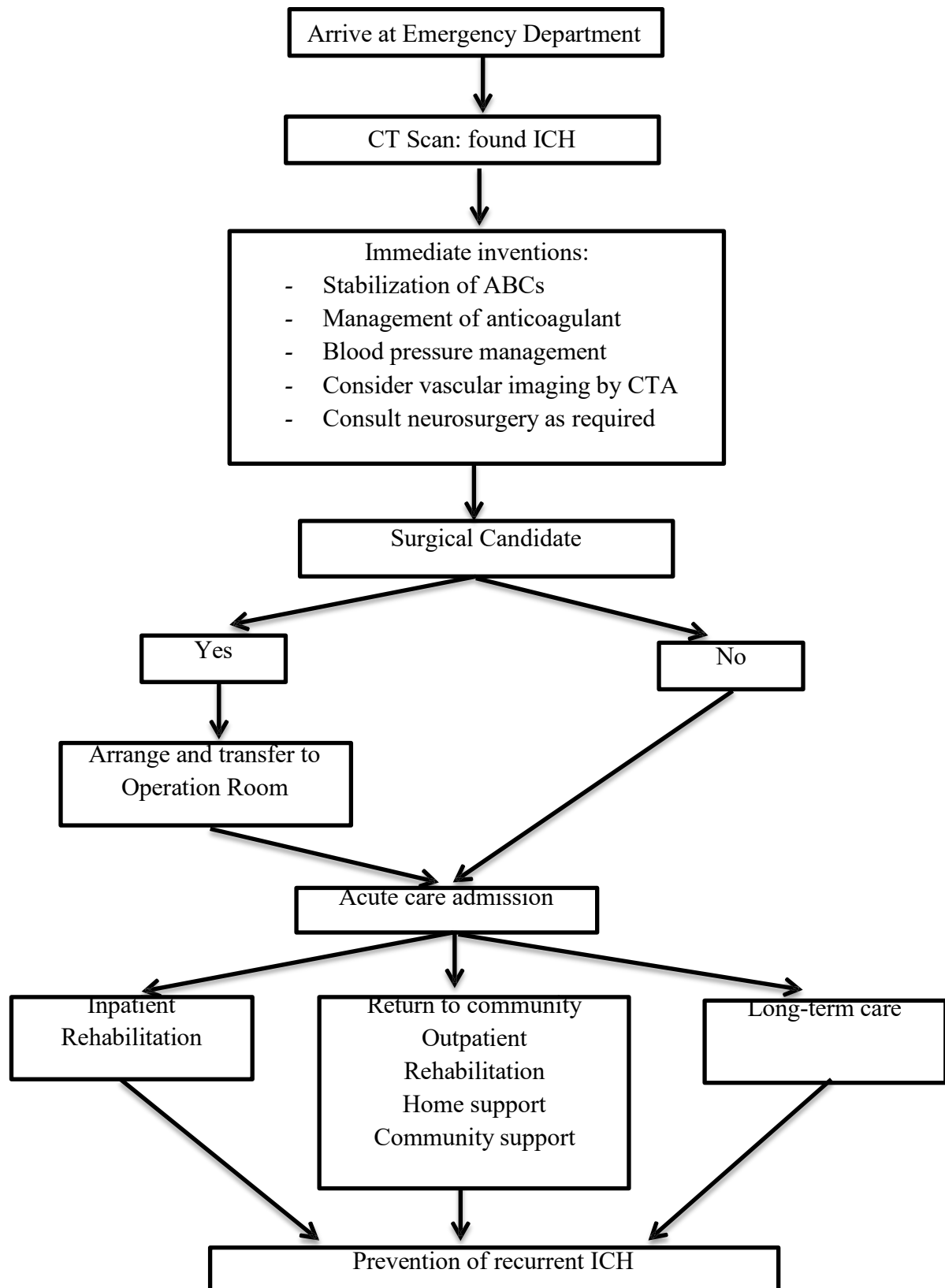
❖ Surgery

The role of surgery in patients with acute ICH varies with the site of the bleed.

- **Cerebrospinal fluid drainage for obstructive hydrocephalus:** Obstructive hydrocephalus may occur when hemorrhage or mass effect obstructs CSF ventricular outflow. This may commonly be seen with ICH in the following locations:

- Thalamic hemorrhage (with third ventricle compression)
- Cerebellar hemorrhage (with fourth ventricle compression) (see 'Cerebellar hemorrhage' below)
- ICH with intraventricular extension
- **Surgical decompression:**
 - **Cerebellar hemorrhage:** The indications for evacuation of a cerebellar hematoma depend on the ICH size and location, the time since onset, and the clinical status of the patient, The example that need urgent surgery :
 - Acute cerebellar hemorrhage greater than 3 cm in diameter (volume of at least 14 cm³) and brainstem compression.
 - A smaller acute cerebellar hematomas may also warrant surgical decompression if they are deteriorating neurologically or have brainstem compression and/or hydrocephalus due to ventricular obstruction.
 - **Supratentorial hemorrhage:** We reserve surgical therapy for patients with life-threatening mass effect from supratentorial ICH, individualizing treatment decisions based on assessments of prognosis with and without surgical therapy. Supratentorial decompression with hematoma evacuation and/or decompressive hemicraniectomy may reduce mortality but not improve functional outcomes. It should only be considered as a life-saving procedure to treat refractory increases in ICP.

7. Algorithm (Intracerebral Hemorrhage Patient Flow Map)



VIII. SPECIAL CASES

- Young patient present with lobar hemorrhage, no risk factor of cerebral vessel disease) suggest to work-up for coagulation profile, toxicology screen and work-up for cerebral vessel imaging (CTA / MRA) to rule out the underlying cause such as Arterio-venous malformation (AVM) or cerebral aneurysm.
- Mid-age female patient with the history of long-term oral contraceptive abuse present with intra-cerebral hemorrhage. Cerebral venous thrombosis is one of the differential diagnoses that should not overlook. Further neuro-imaging such as CTV / MRV should be considered.

IX. EVOLUTION AND PROGNOSIS

The poor prognostic factors are coma, large hematoma with volume greater than 30 ml, intraventricular hemorrhage, posterior fossa hemorrhage, old age greater than 80 years, hyperglycemia, and chronic kidney disease.[5] Early deterioration and death are the major problems with ICH. Coma, at the time of the presentation, indicates a grave prognosis

ICH score – A simple six-point clinical and radiographic grading scale called the ICH score has been devised to predict 30-day mortality after ICH.

Component	ICH score points
GCS score* at presentation	
3 to 4	2
5 to 12	1
13 to 15	0
ICH volume on initial imaging	
≥30 cm ³	1
<30 cm ³	0
Intraventricular extension of ICH	
Present	1
Absent	0
Infratentorial origin of ICH	
Yes	1
No	0
Age (years)	
≥80	1
<80	0
Total	0 to 6

The ICH score results range from 0 to 6, with higher scores being associated with higher predicted mortality risk.

The predicted 30-day mortality:

- ICH score 1 (13 percent)
- ICH score 2 (26 percent)
- ICH score 3 (72 percent)
- ICH score 4 (97 percent)
- ICH score 5 (100 percent)

ICH score to predict 30-day mortality after spontaneous ICH

X. COMPLICATION

Complications of ICH include cerebral edema, increased intracranial pressure, hydrocephalus, seizures, venous thrombotic events, hyperglycemia, increased blood pressure, fever, and infections.

Almost respiratory failure, and respiratory distress. About 4% of patients with ICH suffer one-third of patients with ICH develop pulmonary complications such as pneumonia, aspiration, pulmonary edema, cardiac complications such as

myocardial infarction, atrial fibrillation, ventricular fibrillation, ventricular tachycardia, stress-induced cardiomyopathy, and acute heart failure.

XI. PREVENTION AND RECOMMENDATION

There is a chance of recurrence of ICH. Hypertension and old age are risk factors. BP should be controlled. Lifestyle modifications should be advised, including avoidance of alcohol, tobacco, and illicit drugs. Continued multidisciplinary rehabilitation should be done.

The following are the possible risk factors for a recurrent ICH:

- Lobar location of the initial ICH
- Older age
- Presence and number of microbleeds on gradient echo MRI
- Ongoing anticoagulation

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CEREBRAL VENOUS THROMBOSIS

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I. DEFINITION

Cerebral venous thrombosis (CVT) or cerebral venous sinus thrombosis (CVST): a thrombotic obstruction of the cerebral veins and/or related anatomical structures (dural sinuses) which drain blood from the brain. Common subtypes include transverse sinus thrombosis and superior sagittal sinus thrombosis.

Septic cerebral venous thrombosis: a subtype of CVT of infectious origin (see “Etiology”).

Cavernous sinus thrombosis: a rare subtype of CVT, typically septic in origin, that is associated with cavernous sinus syndrome.

II. EPIDEMIOLOGY

- Sex: F > M, 3:1
- Age: any age group (average age ~ 40 years)
- Incidence: 3–4 cases per million in adults; approx.

III. ETIOLOGY

1. Noninfectious

- Hypercoagulable states
 - Oral contraceptives, pregnancy, and postpartum period
 - Blood clotting disorders (e.g., factor V Leiden, protein C deficiency, protein S deficiency, antiphospholipid syndrome)
 - Hematologic diseases (e.g., polycythemia, sickle-cell anemia)
 - Malignancies
- Head trauma
- Neurosurgical procedures (e.g., lumbar puncture)
- Additional associated systemic conditions
 - Inflammatory bowel disease (e.g., Crohn's disease)
 - Collagen vascular diseases and blood vessel disorders (e.g., lupus erythematosus, granulomatosis with polyangiitis, temporal arteritis)
 - Hyperhomocysteinemia
 - Hematologic conditions (e.g., paroxysmal nocturnal hemoglobinuria, thrombotic thrombocytopenic purpura, sickle cell disease)
 - Nephrotic syndrome
 - Dehydration

2. Infectious

- Rhinogenic (e.g., after sinusitis - paranasal, sphenoid, maxillary, and ethmoid)
- Mid-facial infections (cellulitis or abscess)
- Dental infections
- Otogenic (e.g., after acute otitis media) → generally infection of the lateral sinuses (transverse/sigmoid sinuses) or mastoiditis
- Meningitis ○ Pharyngitis ○ Tonsillitis
- Orbital and periorbital cellulitis
- Covid -19 infection and covid vaccine associated thrombosis.

IV. PATHOPHYSIOLOGY

Thrombogenesis occurs in the cerebral venous system, including the dural sinuses → ↓ cerebral drainage → ↑ intracranial pressure → clinical features; ; (see below).

Additionally, thrombus formation → congestion within the venous system of the brain → blood stasis → ↓ oxygenated blood in brain tissue → cerebral edema and/or infarcts/stroke

V. DIAGNOSTICS

Symptoms and signs

Symptoms and signs of CVT can be grouped into three major syndromes:

- Isolated intracranial hypertension syndrome (headache with or without vomiting, papilledema, and visual problems)
- Focal syndrome (focal deficits, seizures, or both)
- Encephalopathy (multifocal signs, mental status changes, stupor, or coma)

Less common presentations include cavernous sinus syndrome, subarachnoid hemorrhage, and multiple cranial nerve palsies. A case of CVT mimicking a transient ischemic attack has also been reported.

Diagnosis of CVT is based on neuroimaging with venography. Laboratory studies can help identify underlying conditions (e.g., infections) and to assess baseline organ function prior to therapy.

Neuroimaging

- Preferred modalities
 - MRI head with MR venography (MRV): modality of choice (highest sensitivity) ^{[12][13]}
 - CT head with CT venography (CTV): if MRI is unavailable or is contraindicated
 - Findings: Typically, similar in CTV and MRV
 - Absence of flow in venography (Evidenced by filling defect in a vein or sinus)
 - Intraluminal venous thrombus
 - Focal edema secondary to ischemia
 - Intraparenchymal hemorrhage
 - Empty delta sign
 - A radiological finding in contrast-enhanced CT/MRI that resembles the Greek letter delta (Δ)
 - Consists of a central hypodensity (representing the thrombus) with a triangular outline of contrast enhancement
 - May be seen in cerebral venous thrombosis of the superior sagittal sinus.
- Additional modalities
 - Direct cerebral venography: if MRV/CTV are inconclusive or for procedural planning. Findings include filling defects and venous congestion
 - Noncontrast head CT
 - Normal in 70% of patients with CVT ^[9]
 - The thrombus may appear as a hyperdense vein or sinus.
 - Hypodense ischemic area of brain parenchyma with/without a hemorrhagic component

VI. DIFFERENTIAL DIAGNOSIS

- Posterior reversible encephalopathy syndrome (PRES)
- Primary headaches
- Subarachnoid hemorrhage
- Idiopathic intracranial hypertension

VIII. MANAGEMENT

Initial management consists of general stabilization procedures (control of seizures, administration of fluids, treatment of intracranial hypertension) and treatment of the underlying cause.

All patients

require anticoagulation, and those with underlying infection require antimicrobial therapy. Surgical intervention should be considered if there is no improvement with medical therapy.

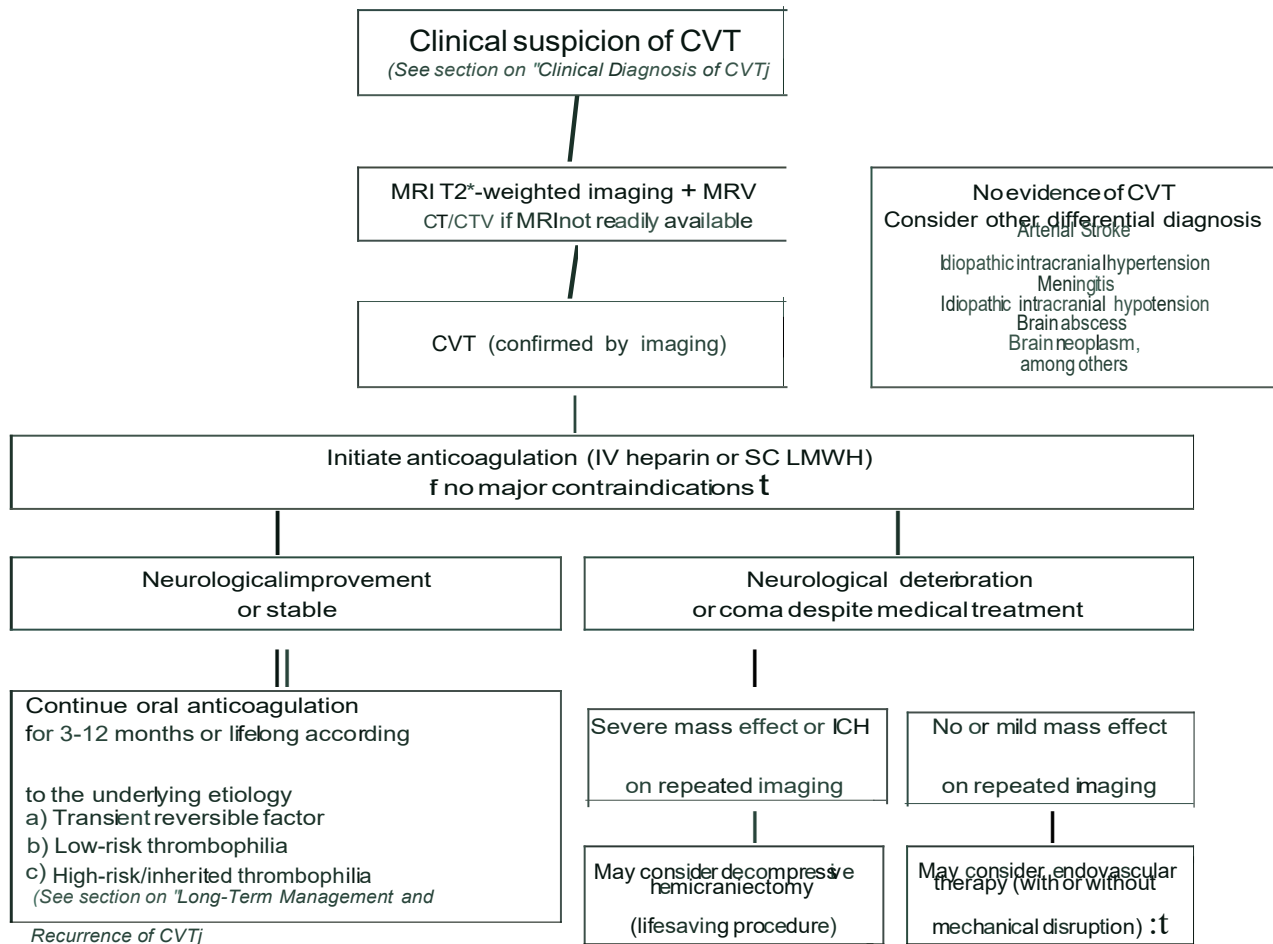
Treatment

- Anticoagulation: indicated for all patients with CVT (intracerebral hemorrhage and underlying infection are not absolute contraindications)
 - Acute phase
 - First line: low molecular weight heparin (LMWH): e.g., Enoxaparin
 - Second-line: unfractionated heparin
 - Long-term anticoagulation: Transition to vitamin K antagonists, e.g., warfarin , while in-hospital.
 - Duration of treatment: 3–12 months
- Empiric antibiotic therapy: only indicated for suspected septic CVT (see also empiric antibiotic therapy for bacterial meningitis).
- Antifungals
 - CVT is rarely caused by fungal infection.
 - Immunosuppression (e.g., neutropenia, diabetes mellitus) is the most important risk factor.
 - Initial treatment option: amphotericin B
- Supportive care
 - Fluid resuscitation
 - Neuroprotective measures
 - ICP management
 - Consider anticonvulsants: Indicated in patients with a seizure and supratentorial lesions on imaging, otherwise not routinely recommended.
 - Consider corticosteroids in the following:
 - Treating other complications, e.g., hypopituitarism
 - Treating underlying conditions: e.g., corticosteroids for or Behcet disease.
 - Highly controversial: reducing vasogenic edema
- Rehabilitation

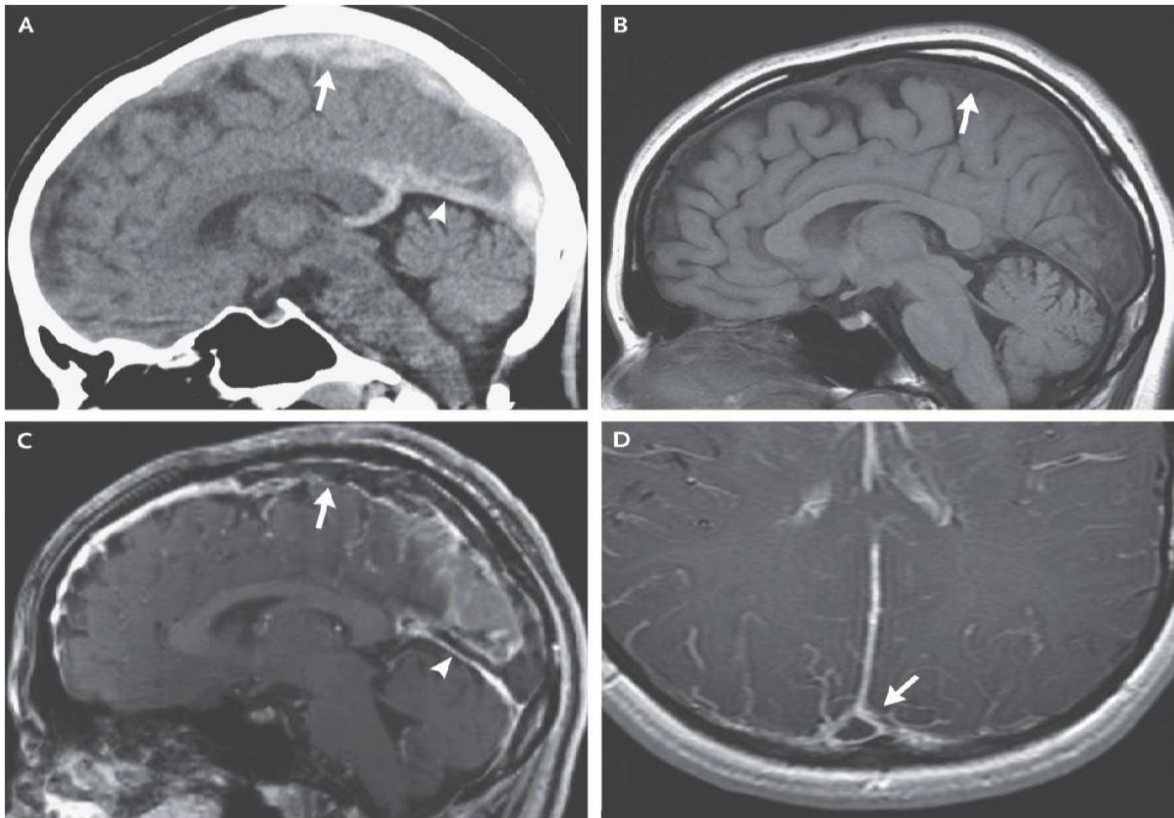
Invasive procedures

- Endovascular interventions
 - Indications
 - Absolute contraindications to anticoagulation
 - Significant symptoms despite anticoagulation
 - Procedures
 - Endovascular thrombolysis
 - Catheter thrombectomy
- Surgery
 - Indications
 - Progressive neurological worsening (despite adequate anticoagulation), e.g., visual loss, signs of ↑ ICP or cerebral herniation syndrome
 - Purulent collections requiring drainage
 - Procedures
 - Decompressive craniectomy
 - Hematoma evacuation
 - Shunt placement
 - Abscess drainage

Proposed Algorithm for the Management of CVT



All patients should receive support for the prevention of complication and symptomatic therapy (eg, management of seizures, intracranial hypertension)



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AMYOTROPHIC LATERAL SCLEROSIS

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I. DEFINITION

Amyotrophic lateral sclerosis (ALS) is a progressive neurodegenerative disease that affects nerve cells in the brain and spinal cord that causes muscle weakness, disability, and eventually death, with a median survival of three to five years.

II. EPIDEMIOLOGY

Amyotrophic lateral sclerosis is classified as either sporadic or familial:

- Sporadic forms account for 90 to 95 percent of ALS cases.
- Familial forms make up 5 to 10 percent of cases.

The incidence of ALS is approximately 1-2.6 cases per 100 000 persons annually, whereas the prevalence is approximately 6 cases per 100 000. The average age of onset of ALS is currently 58-60 years and the male-to-female ratio is approximately 1.3 to 1.5. The average survival from onset to death is 3-4 years.

III. ETIOLOGY

The etiology of ALS is unknown. A number of potential mechanisms have been proposed including abnormal RNA processing, disorders of protein quality control, excitotoxicity, cytoskeletal derangements, mitochondrial dysfunction, viral infections, apoptosis, growth factor abnormalities, inflammatory responses, and others.

IV. PATHOPHYSIOLOGY

Amyotrophic lateral sclerosis is characterized by motor neuron degeneration and death with gliosis replacing lost neurons. Degenerating neurons and glia show intracellular inclusions:

- Cortical motor cells (pyramidal and Betz cells) degenerate, leading to retrograde axonal loss and gliosis in the corticospinal tract.
- The spinal cord becomes atrophic. The ventral roots become thin, and there is a loss of large, myelinated fibers in motor nerves.
- The affected muscles show denervation atrophy with evidence of innervation such as fiber type grouping.

V. DIAGNOSIS

1. Clinic presentation of Amyotrophic lateral sclerosis

Amyotrophic lateral sclerosis is one of multiple degenerative motor neuron diseases that are clinically defined based on the involvement of upper and/or lower motor neurons.

Lower motor neuron signs include the following:

- Weakness
- Fasciculation
- Decreased muscle tone (flaccidity) and reduced or absent reflexes

Upper motor neuron signs include the following:

- Increased tone (spasticity) and increased extremity deep-tendon reflexes
- The presence of any reflexes in muscles that are profoundly weak and wasted
- Pathologic reflexes such as crossed adductors, a jaw jerk, Hoffman sign, or Babinski sign (reflex great toe extension, flexion of the knee, and flexion of the hip)
- The syndrome of pseudobulbar affect, which consists of inappropriate laughing, crying, and/or forced yawning

2. Investigation

The test can help support an amyotrophic lateral sclerosis diagnosis include:

- Electromyography (EMG): evaluates how well nerves and muscles are functioning. This test can include:
 - A nerve conduction study (NCS): measures the electrical activity of nerves and muscles by assessing the nerve's ability to send a signal along the nerve or to the muscle.
 - A needle exam: a recording technique that detects electrical activity in muscle fibers using a needle electrode.
- Magnetic resonance imaging (MRI): uses a magnetic field and radio waves to produce detailed images of the brain and spinal cord. Brain MRI should be performed whenever bulbar disease is present. Cervical and lumbosacral spine MRI can be used to evaluate lower motor neuron findings in the arms and legs.
- Laboratory testing: includes complete blood count with differential, electrolytes including calcium and phosphate, liver function tests, thyroid studies, creatine phosphokinase, erythrocyte sedimentation rate, antinuclear antibody, rheumatoid factor, vitamin B12, anti-GM1 antibody, serum protein electrophoresis with immunofixation, and urine protein electrophoresis with immunofixation.

VI. DIFFERENTIAL DIAGNOSIS

The differential diagnosis for amyotrophic lateral sclerosis includes the following:

- Multifocal motor neuropathy is no involvement of bulbar muscles, conduction blocks.
- Spinal muscular atrophy is only affects LMN, age-related incidence, SMN detection (genetic testing).
- Primary lateral sclerosis is only affects UMN, slower course, survival over 10 years.
- Spinal and bulbar muscular atrophy it develops in middle age, men, fasciculation of the tongue and in the perioral region, gynecomastia, X chromosome, expansion in the androgen receptor gene.
- Hereditary spastic paraplegia are lower limb spasticity, gait with pelvic rotation, minimal upper limb symptoms, familial occurrence, genetic confirmation.
- Myogenic lesions are myopathy syndrome, proximal weakness, laboratory findings, muscle MRI, muscle biopsy.
- Myasthenia gravis is fatigue, localization of impairment (ocular, bulbar, head posture), repetitive stimulation, antibodies.
- Spondylogenic cervical myelopathy, symptoms including sensory disorders, Sphincter disorders, no bulbar symptoms, MRI finding.

- Lumbar spinal stenosis, symptoms of lower extremities only, including sensory or sphincter disorders, fatigue, claudication, MRI findings.

VII. MANAGEMENT

1. Objective:

- Ameliorating the disease
- Slowing down disease progression
- Improving or maintaining quality of life

2. Critical sign:

Breathing (dyspnea): shortness of breath during physical activity and difficulty breathing at night or when lying down that need noninvasive ventilation (NIV).

3. Mortality risk actors:

The only established risk factors to date are older age, male sex, and a family history of ALS. Also, other risk factors include:

- Lifestyle: smoking, dietary factors (vitamin E supplements were associated with a lower risk of ALS)
- Body mass index and physical fitness (Athleticism, chronic traumatic encephalopathy, and physical exercise)
- Occupational and environmental risk factors: Electric occupation, electric shock, and electromagnetic field, Pesticides...

4. Outpatient Management:

- Educating the patients and sharing decision are important, especially since there are many off-label treatment options with potentially serious side effects.
- Prevention exposures and events that drive further disease progression is essential.

5. First line treatment:

*Pharmacologic treatment:

- Riluzole (Rilutek, Exservan, Teglutik). This medicine can increase life expectancy by about 25%. Take this medication by mouth on an empty stomach (at least 1 hour before or 2 hours after a meal) usually twice daily.
- Edaravone (Radicava). This medicine can reduce the speed of decline in daily functioning. It's given through a vein in your arm or by mouth as a liquid. This medicine is given daily for two weeks each month.
- Sodium phenylbutyrate-aurursodiol (Relyvrio). This medicine, approved by the FDA in 2022, can slow the rate of functional decline in people with ALS by about 25%. It also may help people with ALS live about six months longer. This medicine is a powder mixed in room temperature water. The recommended dosage is 1 packet (3 g sodium phenylbutyrate and 1 g aurursodiol) daily for the first 3 weeks, and then 1 packet twice daily thereafter.

*Nonpharmacologic treatment:

When ALS affects ability to breathe, speak and move, therapies and other forms of support can help.

- Breathing care. Most people with ALS eventually have more trouble breathing as muscles weaken (need to use a ventilator with a mask or tracheostomy)
- Physical therapy. A physical therapist can address pain, walking, mobility, bracing and equipment needs that help you stay independent. Practicing low-impact

exercises can help maintain your cardiovascular fitness, muscle strength and range of motion for as long as possible.

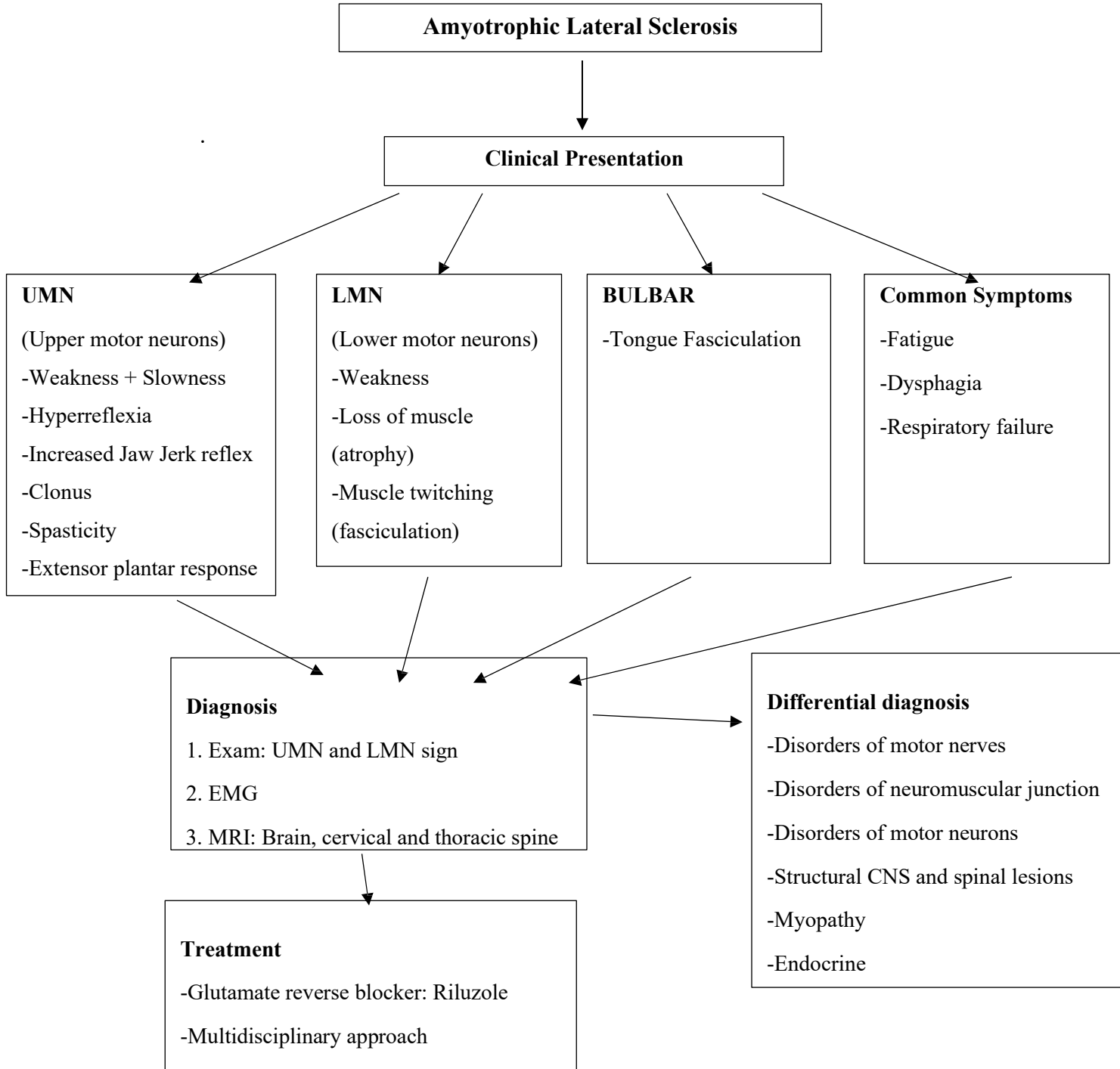
- Occupational therapy. An occupational therapist can help you find ways to remain independent despite hand and arm weakness. Adaptive equipment can help you perform activities such as dressing, grooming, eating and bathing.

- Speech therapy. A speech therapist can teach you adaptive techniques to make your speech more understandable and also can help you find other ways to communicate. These may include using a smart phone app, alphabet board, or pen and paper.

- Nutritional support. Your team typically works with you and your family members to ensure you are eating foods that are easier to swallow and meet your nutritional needs.

- Psychological and social support. Your team might include to help with financial issues, insurance, and getting equipment and paying for devices you need. Psychologists, social workers and others may provide emotional support for you and your family.

6. Algorithm:



VIII. EVOLUTION AND PROGNOSIS

There is no cure for ALS. Over a period of 3 to 5 years, the disease will progress, making voluntary movements of arms and legs impossible. Movement of the diaphragm for breathing is also impaired. You may need a ventilator for breathing. Most people with ALS die from respiratory failure.

IX. COMPLICATION

Complications can arise from the neurological damage of the disease process or from treatments:

- Choking, due to impaired swallowing
- Shortness of breath, due to the atrophy of breathing muscles
- Risk of falling, due to weakness and atrophy of the muscles
- Nutritional deficiencies, due to trouble eating
- Infections, from medical devices like ventilators, feeding tubes, and urinary catheters
- Anxiety and depression, related to an unpredictable and terminal illness
- Inappropriate emotional outbursts, related to damage to that part of the brain
- Cognitive changes and dementia also can occur as ALS progresses.

X. PREVENTION AND RECOMMENDATION

1. Prevention

Amyotrophic lateral sclerosis cannot prevent, some treatments can slow the progression of the disease and lifestyle measures may help with prevention.

2. Recommendation

The care and management of patients with ALS should always be patient focused, with attention to holistic and emotional aspects of well-being. It is the patient who ultimately decides on their treatment; this includes the option of declining interventions.

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MYASTHENIA GRAVIS

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I- DEFINITION

Myasthenia gravis (MG) is an acquired autoimmune neuromuscular disorder characterized by fluctuating motor weakness involving ocular, limb, and/or respiratory muscles.

II- PIDEMIOLOGY

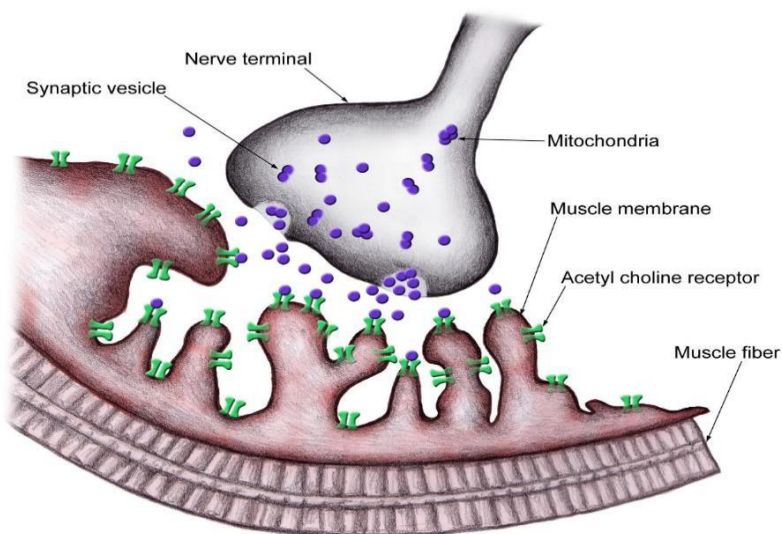
MG is a relatively uncommon disorder with an annual incidence of approximately 7 to 23 new cases per million. The prevalence is approximately 70 to 320 per million. The prevalence of the disease has been increasing since the mid-20th century, likely due to better recognition of the condition, aging of the population, and the longer life span of affected patients.

III- IOLOGY

Myasthenia gravis is caused by an abnormal immune reaction (antibody-mediated autoimmune response) in which the body's immune defenses inappropriately attack certain proteins in muscles that receive nerve impulses. The area of contact between nerve endings and skeletal muscle fibers are known as neuromuscular junctions.

IV- PATHOPHYSIOLOGY

The weakness is due to an antibody-mediated, immunologic attack directed at proteins in the postsynaptic membrane of the neuromuscular junction (acetylcholine receptors or receptor-associated proteins). MG is the most common disorder of neuromuscular transmission.



Normal neuromuscular junction showing a presynaptic terminal with a motor nerve ending in an enlargement (bouton terminale: synaptic cleft and postsynaptic membrane with multiple folds and embedded with several acetylcholine receptors.

V- DIAGNOSIS

Myasthenia gravis is diagnosed based upon a thorough clinical evaluation, detection of characteristic symptoms and physical findings, a detailed patient history, and a variety of specialized tests.

1. **Clinical signs:** Myasthenia gravis should be suspected in patients with fatigable muscle weakness, including those with isolated ptosis and/or diplopia.

2. **Investigation**

- **Serologic Tests:** The anti-AChR Ab test is very specific, and it confirms the diagnosis in ---patients with classical clinical findings. It is present in four-fifths of patients with generalized MG and only in half of the patients with pure ocular MG. The rest of the patients, about 5% to 10%, will demonstrate anti-MuSK antibodies. Only in a few sporadic cases, both anti-AChR and anti-MuSK antibodies are present in the same patient. The 3% to 50% of the remaining patients who are seronegative to either of these antibodies will demonstrate anti-LRP4 antibodies. Anti-striated muscle antibodies are present in 30% of MG patients. They are more useful as a serologic marker for thymoma, especially in younger adult.
- **Electrophysiologic Tests:** These are relevant in patients who are seronegative for antibody testing. Commonly employed tests for MG are the repetitive nerve stimulation (RNS) test and single-fiber electromyography (SFEMG). Both the test assesses for conduction delays in the NMJ. Routine nerve conduction studies are usually performed to determine the functioning of the nerves and muscles before undertaking these tests.
- **Edrophonium (Tensilon) Test:** Edrophonium is a short-acting acetylcholinesterase inhibitor that increases the availability of Ach in the NMJ. This is particularly useful for ocular MG, where electrophysiologic testing cannot be performed. It is administered intravenously, and the patient is observed for improvement in the symptoms of ptosis or diplopia. It has a sensitivity of 71% to 95% for MG diagnosis.
- **Ice-pack Test:** When edrophonium testing is contraindicated, an ice-pack test can be performed. This test requires an ice-pack placed over the eyelid for 2-5 minutes. Then, an assessment for any improvement in ptosis is done. This test cannot be used for evaluation of extraocular muscles.
- **Imaging:** Chest computed tomography(CT) or magnetic resonance imaging (MRI) should be performed in patients diagnosed with MG to assess for thymoma. In cases presenting with pure ocular MG, MRI of orbits and the brain needs to be performed to evaluate for any localized mass lesions.
- **Other Laboratory Tests:** Myasthenia gravis commonly coexists with other autoimmune disorders, and testing for anti-nuclear (ANA antibodies, rheumatoid factor (RF), and baseline thyroid functions are recommended.

VI. Differential Diagnosis

The differential diagnosis for myasthenia gravis includes the following:

- **Lambert-Eaton syndrome** is a fluctuating weakness that improves with exercise, differentiating it from MG. This is usually due to underlying malignancy, commonly small-cell lung cancer. It affects voltage-gated calcium channels in the presynaptic membrane.
- **Cavernous sinus thrombosis** can present with diplopia, photophobia, chemosis, and headaches. It is usually abrupt in onset but can be fulminant, and causes can be septic or aseptic.

- **Brainstem gliomas** are malignant tumors that present with bulbar symptoms, weakness, numbness, balance problems, and seizures depending on its location and structures affected. The symptoms are persistent and usually present with headaches and signs of increased intracranial pressure.
- **Multiple sclerosis** can present with any neurological sign that may fluctuate or persist over hours to days to weeks secondary to demyelination in the central nervous system. This may present with weakness, sensory deficits, cognitive and behavioral issues. Weakness may be unilateral or bilateral and is usually accompanied by upper motor neuron signs like hyperreflexia, spasticity, and a positive Babinski.
- **Botulism** presents with ptosis, double vision, progressive weakness, and pupillary abnormalities accompanied by systemic symptoms. Ingestion of honey or contaminated foods may be elicited in the patient's history.
- **Tick-borne disease** presents with ascending paralysis and respiratory distress and decreased reflexes from a neurotoxin from the saliva of ticks. Minimal constitutional symptoms are seen. It can sometimes present with ophthalmoplegia and bulbar symptoms.
- **Polymyositis and dermatomyositis** cause proximal muscle weakness and is usually associated with pain. The pathology is the inflammation of the muscle itself.
- **Graves ophthalmopathy** presents with eyelid retraction and widened palpebral fissure. These are caused by autoantibodies targeted toward the structures of the eye.

VII- MANAGEMENT

1. Objectives

The mainstay of treatment in myasthenia gravis involves cholinesterase enzyme inhibitors and immunosuppressive agents. Symptoms that are resistant to primary treatment modalities or those requiring rapid resolution of symptoms (myasthenic crisis), plasmapheresis or intravenous immunoglobulins can be used.

Management strategies in myasthenia gravis are based on the following four principles:

2. Critical signs

- Dyspnea
- Severe dysphagia with weak cough and difficulty clearing secretions.
- Signs of respiratory muscle weakness, such as hypophonia, pausing during speech to take a breath, poor respiratory effort, increased respiratory rate with shallow breaths, use of accessory muscles of respiration, and paradoxical abdominal breathing.
- Low baseline vital capacity (VC) that is <30 mL/kg of ideal body weight, even if the patient is breathing without distress.

3. Mortality risk factors

- Sepsis
- Respiratory failure

4. Outpatient Management

We treat the majority of patients in the outpatient setting, giving clear instructions to the patient and their families. In patients demonstrating significant bulbar weakness, our preference is to admit them immediately to the neurology ward and to initiate treatment accordingly including symptomatic treatment with pyridostigmine and where necessary intravenous immunoglobulin (IVIG).

5. First-line treatment

Symptomatic Treatment: Acetylcholinesterase inhibitors increase the level of Acetylcholine at the neuromuscular junction by preventing its enzymatic degradation. Pyridostigmine bromide is preferred over neostigmine because of its longer duration of action. In those with bromide intolerance that leads to gastrointestinal effects, ambenonium chloride can be used. Patients with MuSK MG respond poorly to these drugs and hence require higher dosages.

- First-line treatment: Glucocorticoids (prednisone, prednisolone, and methylprednisolone) and azathioprine.

6. Treatment adjustment

a. Immunosuppressive Treatment: These are indicated in patients who remain symptomatic even after pyridostigmine treatment.

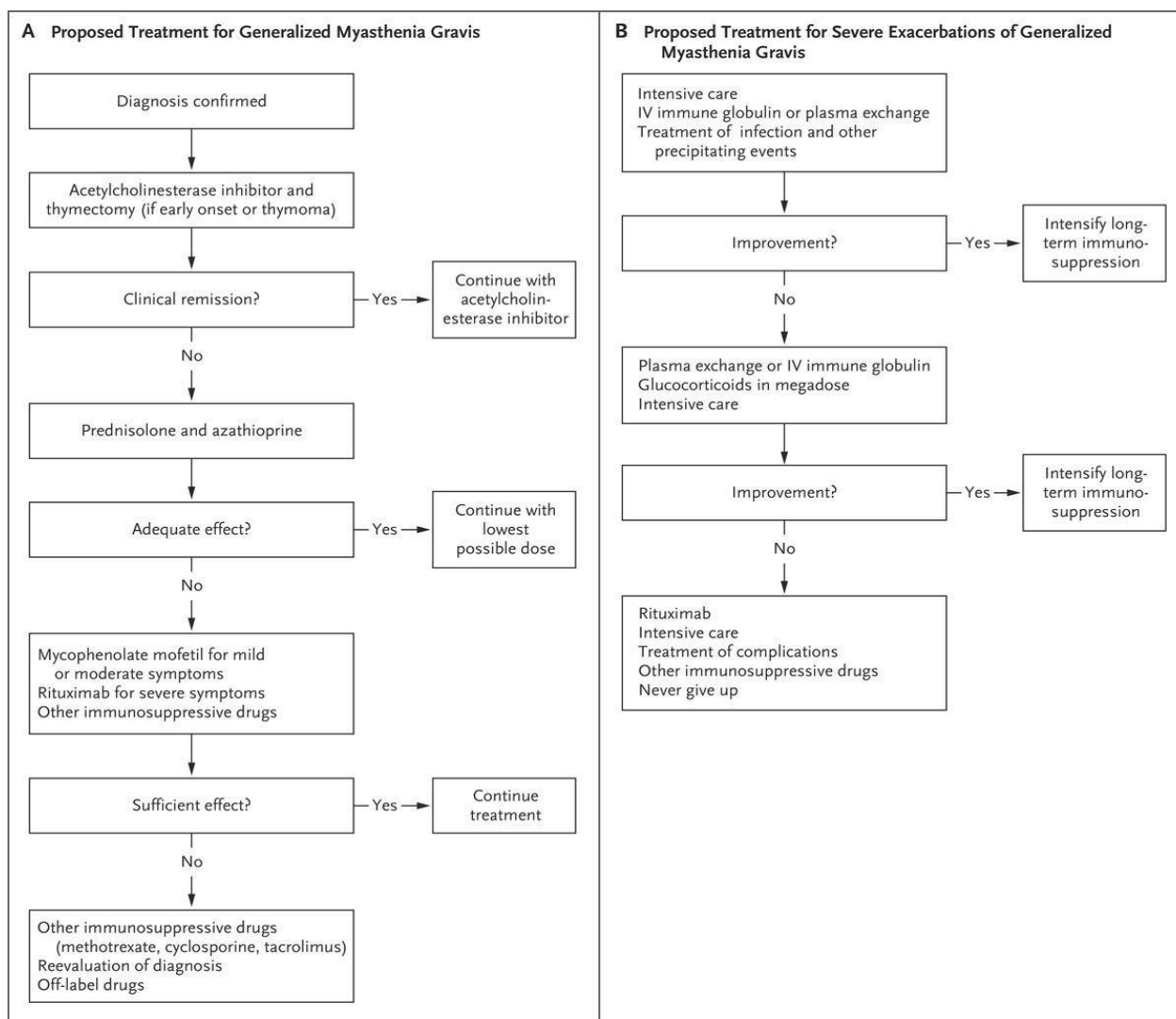
- Second-line treatment: cyclosporine, methotrexate, mycophenolate, cyclophosphamide, and tacrolimus.
- drug-resistant MG: monoclonal antibodies, including rituximab and eculizumab, have been used to treat, but data from clinical trials about their efficacy is yet to be documented.

b. Intravenous immunoglobulins (IVIG) / Plasmapheresis: This is recommended during the perioperative period to stabilize a patient before a procedure. It is also the treatment of choice for the myasthenic crisis due to its rapid onset of action and used in cases that are resistant to immunosuppressive drugs.

c. Thymectomy: This is indicated for the following:

- Any subtypes of myasthenia gravis with evidence of thymoma
- Non-thymomatous n-AChR MG, especially in patients aged 15 to 50 years, performed 1-2 years of disease onset.
- Seronegative non-thymomatous MG

However, it is not recommended for the non-thymomatous MuSK MG since thymic pathology is rare) and non thymomatous ocular MG without secondary generalization.



Proposed treatment algorithms for generalized myasthenia gravis and severe exacerbation generalized myasthenia gravis

VIII. SPECIAL CASES

Isolated ocular myasthenia is rare. While ptosis and diplopia are common presenting symptoms including in patients who will eventually evolve into generalized myasthenia, only 20% of patients will turn out to have pure ocular MG—signifying that these patients will never develop generalized disease). The diagnostic difficulty with this entity is that only 50% have detectable antibodies to the AChR.

Myasthenia and pregnancy: In practice, the majority of MG patients, who are treated adequately before pregnancy, do not experience any complications during pregnancy or in the post-partum phase. However, some report an increased risk of MG relapse during pregnancy that varies between 17% to 41%.

Some patients' MG status improves during pregnancy as one observes with other autoimmune conditions such as multiple sclerosis. In the ideal scenario, the pregnancy is planned to allow optimization of MG status and withdrawal of teratogenic medications where relevant.

Pyridostigmine, corticosteroids, and azathioprine are all safe to be used in pregnancy and should not be discontinued during pregnancy. Mycophenolate mofetil

(MMF) and Methotrexate are teratogenic and should be avoided. Ciclosporin and tacrolimus are not teratogenic but their use can be associated with the development of hypertension and gestational diabetes and, therefore, the patient requires close monitoring. Intravenous immunoglobulins (IVIg) and plasma exchange (PE) are also safe to be used during pregnancy. There are some reports suggesting that MG patients are at risk of preterm rupture of membranes.

MuSK-positive or LRP4-positive MG — Growing clinical evidence suggests that muscle-specific tyrosine kinase (MuSK)-positive MG has important differences with regard to treatment compared with acetylcholine receptor (AChR) antibody-positive and seronegative MG. Most patients with MuSK-positive disease are poorly responsive to anticholinesterase agents as well as thymectomy. Patients with low-density lipoprotein 4 (LRP4)-positive MG may be responsive to immunotherapy, but the role of thymectomy has not been established.

Children — As with adults, the treatment of children with MG should be individualized based on the severity and pace of the disease. [Pyridostigmine](#) is the first line of therapy. If anticholinesterase medications are not sufficient, plasmapheresis or intravenous [immune globulin](#) (IVIg) may be used, but the benefits are transient

IX- EVOLUTION AND PROGNOSIS

Most patient with myasthenia gravis has a near-normal life span with the current treatment modalities. Most people with myasthenia gravis have average life expectancy.

Morbidity results from the intermittent muscle weakness leading to aspiration pneumonia and the adverse effects of medications.

Various clinical and laboratory/imaging findings in myasthenia gravis also have a prognostic significance. Studies have shown the following

- **Risk of secondary generalization:** This is associated with a late age of onset, high titers of anti-acetylcholine receptor (AChR) antibody, and the presence of a thymoma. A recent study predicts this risk by the type of clinical symptoms at the time of presentation. The presence of both ptosis and diplopia at the onset has a higher likelihood of secondary generalization compared to ptosis or diplopia alone. However, early treatment with immunosuppressive drugs such as corticosteroids and azathioprine is associated with reduced risks.
- **Risk of MG relapse:** Age of onset (<40 years), early thymectomy, and administration of prednisolone are found to be associated with reduced risk of relapse. However, patients with anti-K v1.4 antibodies and concomitant autoimmune disease showed a high rate of relapse.

Factors associated with worse prognosis in patients with MG include

- Frequent exacerbation
- Severe bulbar symptoms
- Myasthenic crisis
- Presence of thymoma
- Onset at age > 50 year

X- COMPLICATIONS

Myasthenia gravis causes a significant number of complications:

- Myasthenic crisis,
- Acute respiratory paralysis that requires intensive care
- Adverse events due to long term medication treatment like opportunistic infections and lymphoproliferative malignancies.

XI- PREVENTION AND RECOMMENDATION

- Avoid drugs that may exacerbate MG
- Routine vaccination: seasonal influenza vaccination and pneumococcal vaccine
- Symptomatic therapy with an acetylcholinesterase inhibitor ([pyridostigmine](#))
- Chronic immunotherapies
- Rapid but transient immunomodulatory therapies (plasma exchange and intravenous [immune globulin](#) [IVIG])
- Thymectomy

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MULTIPLES SCLEROSIS

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I. DEFINITION

Multiple Sclerosis (MS) is defined as a chronic and progressive disease of the central nervous system (CNS) characterized by the immune-mediated destruction of myelin, the protective sheath that covers nerve fibers. This demyelination leads to the formation of scar tissue (sclerosis) and disrupts the normal conduction of electrical impulses along the nerves.

The disease typically manifests with a variety of neurological symptoms, which can include fatigue, motor and sensory deficits, visual disturbances, and cognitive changes. MS can present in different forms, including relapsing-remitting MS (the most common form), secondary progressive MS, and primary progressive MS.

II. EPIDEMIOLOGY

Multiple Sclerosis (MS) affects many people in France, with estimates suggesting that between 70,000 and 90,000 individuals live with the disease. Certain areas, especially those with temperate climates, have a higher prevalence of MS. Each year, about 4 to 6 new cases per 100,000 people are reported, indicating a steady, though not extremely high, number of new cases annually.

MS commonly begins between the ages of 20 and 40 and is more frequent in women than in men, with a gender ratio of about 3:2, hinting that hormonal or genetic factors might contribute to its development. Early symptoms of MS are varied and may include fatigue, pain, motor and sensory issues, visual problems, and other neurological symptoms, which can make diagnosing and understanding the disease challenging.

As MS progresses, it often leads to significant disabilities, affecting motor functions, balance, and cognition. It is the primary cause of non-traumatic disability in young adults, severely impacting both personal and professional aspects of their lives.

III. ETIOLOGY

The exact causes of MS are not fully understood, but it is believed to result from a complex interplay of genetic, environmental, and immunological factors. Here are the main contributors thought to be involved in the development of MS:

1. Genetic Factors: There is evidence suggesting a genetic predisposition to MS. Individuals with a family history of the disease are at a higher risk. Certain genetic markers, particularly those related to the immune system, have been associated with an increased susceptibility to MS.

2. Environmental Factors: Various environmental factors may influence the risk of developing MS, including:

- Geographical Location: MS is more prevalent in regions farther from the equator, suggesting that factors such as sunlight exposure and vitamin D levels may play a role.

- Infections: Some viral infections, particularly the Epstein-Barr virus (EBV), have been linked to an increased risk of MS. The hypothesis is that these infections may trigger an autoimmune response in genetically susceptible individuals.

3. Immune System Dysfunction: MS is considered an autoimmune disease, where the immune system mistakenly attacks the body's own myelin, the protective sheath around nerve fibers. This immune-mediated damage leads to inflammation and the formation of scar tissue (sclerosis) in the CNS.

4. Lifestyle Factors: Certain lifestyle factors, such as smoking and obesity, have been associated with an increased risk of developing MS. These factors may influence immune function and overall health.

5. Hormonal Factors: The higher prevalence of MS in women suggests that hormonal differences may contribute to the disease's development. Changes in hormone levels, particularly during puberty, pregnancy, and menopause, may influence the immune response.

while the precise causes of MS remain unclear, it is likely that a combination of genetic susceptibility, environmental influences, and immune system dysregulation contribute to the onset and progression of the disease

IV. PATHOPHYSIOLOGY

The pathophysiology of MS involves a complex interplay of immune, genetic, and environmental factors that lead to the destruction of myelin in the CNS. The key components involved in the pathophysiological process of MS:

1. Autoimmune Response: MS is primarily considered an autoimmune disease. In susceptible individuals, the immune system mistakenly targets the myelin sheath, which insulates nerve fibers. This attack is mediated by activated T lymphocytes (specifically CD4+ T helper cells) that infiltrate the CNS.

2. Inflammation: Once in the CNS, these immune cells release pro-inflammatory cytokines, which promote inflammation and recruit other immune cells, such as B cells and macrophages. This inflammatory response contributes to the damage of oligodendrocytes (the cells responsible for producing myelin) and the myelin itself.

3. Demyelination: The destruction of myelin leads to demyelination, which disrupts the normal conduction of electrical impulses along the affected nerve fibers. This results in the neurological symptoms characteristic of MS, such as motor and sensory deficits, visual disturbances, and cognitive changes.

4. Axonal Damage: In addition to demyelination, there can be direct damage to the axons (the long projections of nerve cells). This axonal injury can lead to irreversible neurological deficits and contributes to the progression of disability in MS patients.

5. Formation of Sclerosis: The term "sclerosis" refers to the scar tissue that forms in place of the damaged myelin. These plaques or lesions can be observed on MRI scans and are indicative of the disease's progression.

6. Chronic Phase: Over time, the inflammatory process may lead to a chronic phase of the disease, characterized by neurodegeneration and progressive disability. In some patients, this can manifest as secondary progressive MS, where there is a gradual worsening of symptoms without distinct relapses.

7. Genetic and Environmental Influences: The pathophysiology of MS is also influenced by genetic predisposition and environmental factors, such as infections and vitamin D levels, which may trigger or exacerbate the autoimmune response.

V. DIAGNOSIS

1. Clinic

The diagnosis of MS is based on a combination of clinical examination findings and laboratory investigations. How each component contributes to the diagnostic process:

- Medical History: A thorough medical history is taken to identify symptoms, their onset, duration, and any previous episodes. Key symptoms may include fatigue, visual disturbances, motor or sensory deficits, balance issues, and cognitive changes.
- Neurological Examination: A comprehensive neurological examination assesses various functions, including:
 - Motor function (strength, coordination)
 - Sensory function (touch, pain, temperature)
 - Reflexes
 - Balance and coordination (e.g., Romberg test)
 - Visual acuity and field testing
- Clinical Criteria : The McDonald criteria are commonly used for diagnosing MS.

These criteria require evidence of :

- Dissémination temporelle (time): Evidence of two or more episodes of neurological dysfunction (pushing) separated in time.
- Dissémination spatiale (space): Evidence of lesions in different parts of the CNS, which can be demonstrated through imaging.

2. Laboratory Investigations

- Magnetic Resonance Imaging (MRI): MRI is a crucial tool in diagnosing MS. It can reveal:
 - Hyperintense lesions on T2-weighted images, indicating areas of demyelination.
 - Gadolinium-enhanced MRI can show active inflammation by highlighting new or active lesions.
- Cerebrospinal Fluid (CSF) Analysis: A lumbar puncture is performed to analyze the CSF for:
 - Oligoclonal bands: The presence of these bands indicates an abnormal immune response in the CNS.
 - Increased immunoglobulin G (IgG): Elevated levels of IgG in the CSF compared to serum can support the diagnosis.
- Evoked Potentials: these tests measure the electrical activity in response to stimuli (visual, auditory, or sensory) and can help identify lesions in the pathways that may not be visible on MRI.

The diagnosis of MS is made by integrating clinical findings with laboratory investigations. The combination of a detailed medical history, neurological examination, MRI findings, CSF analysis, and evoked potentials helps establish the diagnosis based on the **McDonald criteria**. This comprehensive approach ensures that other potential causes of neurological symptoms are ruled out and that the diagnosis of MS is accurate .

VI. DIFFERENTIAL DIAGNOSIS

Differential diagnosis of MS involves distinguishing it from other neurological conditions that can present with similar symptoms. The process typically includes a

thorough clinical evaluation, imaging studies, and laboratory tests to rule out other potential causes of neurological dysfunction. The key conditions to consider in the differential diagnosis of MS:

- **Neuromyelitis Optica (NMO)**
 - Characterized by severe optic neuritis and transverse myelitis.
 - Often associated with the presence of aquaporin-4 antibodies.
 - MRI findings may show longitudinally extensive spinal cord lesions.
- **Acute Disseminated Encephalomyelitis (ADEM)**
 - Typically follows a viral infection or vaccination.
 - Presents with rapid onset of neurological symptoms and multifocal lesions on MRI.
 - More common in children and often has a monophasic course.
- **Chronic Inflammatory Demyelinating Polyneuropathy (CIDP)**
 - Affects peripheral nerves rather than the CNS.
 - Characterized by progressive weakness and sensory loss.
 - Diagnosis is supported by nerve conduction studies.
- **Systemic Lupus Erythematosus (SLE)**
 - Can cause neurological symptoms due to CNS involvement (lupus cerebritis).
 - Associated with other systemic symptoms and laboratory findings (e.g., positive ANA).
- **Sarcoidosis**
 - Can cause neurological symptoms through neurosarcoidosis.
 - MRI may show granulomatous lesions in the CNS.
 - Diagnosis may require biopsy or other specific tests.
- **Vitamin B12 Deficiency**
 - Can lead to subacute combined degeneration of the spinal cord.
 - Symptoms may include weakness, sensory changes, and cognitive impairment.
 - Laboratory tests can confirm deficiency.
- **Thyroid Disorders**
 - Hypothyroidism can cause neurological symptoms, including cognitive changes and peripheral neuropathy.
 - Blood tests can assess thyroid function.
- **Infections**
 - Conditions such as Lyme disease, syphilis, or viral infections (e.g., HIV, herpes) can mimic MS.
 - CSF analysis and serological tests can help identify infectious causes.
- **Tumors**
 - Primary CNS tumors or metastatic lesions can present with focal neurological deficits.
 - MRI can help differentiate tumors from demyelinating lesions.
- **Other Demyelinating Diseases**
 - Conditions like leukodystrophies or other genetic disorders may present with similar symptoms.
 - Genetic testing and specific imaging findings can aid in diagnosis.
- **Diagnostic Approach**
 - Clinical History and Examination: A detailed history of symptoms, their onset, and progression, along with a thorough neurological examination, is essential.

- **Imaging Studies:** MRI is crucial for identifying lesions characteristic of MS and ruling out other conditions.
- **Laboratory Tests:** CSF analysis, blood tests for specific antibodies, and vitamin levels can help exclude other diagnoses.
- **Follow-Up:** In some cases, a follow-up assessment may be necessary to observe the progression of symptoms and the response to treatment.

The differential diagnosis of MS requires a comprehensive approach that includes clinical evaluation, imaging, and laboratory investigations to rule out other conditions that may present with similar neurological symptoms

VII. MANAGEMENT

1.Objective

Effective MS management requires a coordinated, multidisciplinary approach care focused on these objectives to support patients comprehensively.

Primary : *Disease Control and Symptom Management*

- Aim to reduce MS relapses, slow disease progression, and manage symptoms effectively through disease-modifying therapies, symptomatic treatments, and long-term care strategies.

Secondary : *Quality of Life and Support*

- Focus on enhancing the overall quality of life by preventing complications, providing patient and family education, addressing psychosocial needs, and supporting emotional well-being through multidisciplinary

2.Critical sign

- The management of MS is based on several critical factors, including the identification of critical signs, understanding mortality risk factors, outpatient management strategies, first-line treatments, and treatment adjustments. Here's a detailed overview of each aspect:
 - **Management Focus:** Identifying and addressing critical signs such as severe relapses, significant disability progression, or acute exacerbations.
 - **Approach:** Immediate intervention may be required, including high-dose corticosteroids for acute relapses to reduce inflammation and promote recovery. Monitoring for complications and adjusting treatment plans based on the severity of symptoms is essential.

3. Mortality Risk Factors

- **Management Focus:** Understanding factors that may increase mortality risk in MS patients, such as severe disability, respiratory complications, infections, and comorbidities (e.g., cardiovascular disease).
- **Approach:** Regular health assessments to identify and manage comorbid conditions, vaccination against preventable diseases, and proactive measures to prevent infections and complications. Palliative care may be considered for advanced disease stages.

4. Outpatient Management

- **Management Focus:** Providing ongoing care and support in an outpatient setting to monitor disease progression and manage symptoms.
- **Approach:** Regular follow-up appointments for clinical assessments, MRI scans to monitor disease activity, and adjustments to treatment plans as needed. Education on

self-management strategies, lifestyle modifications, and access to rehabilitation services are also crucial.

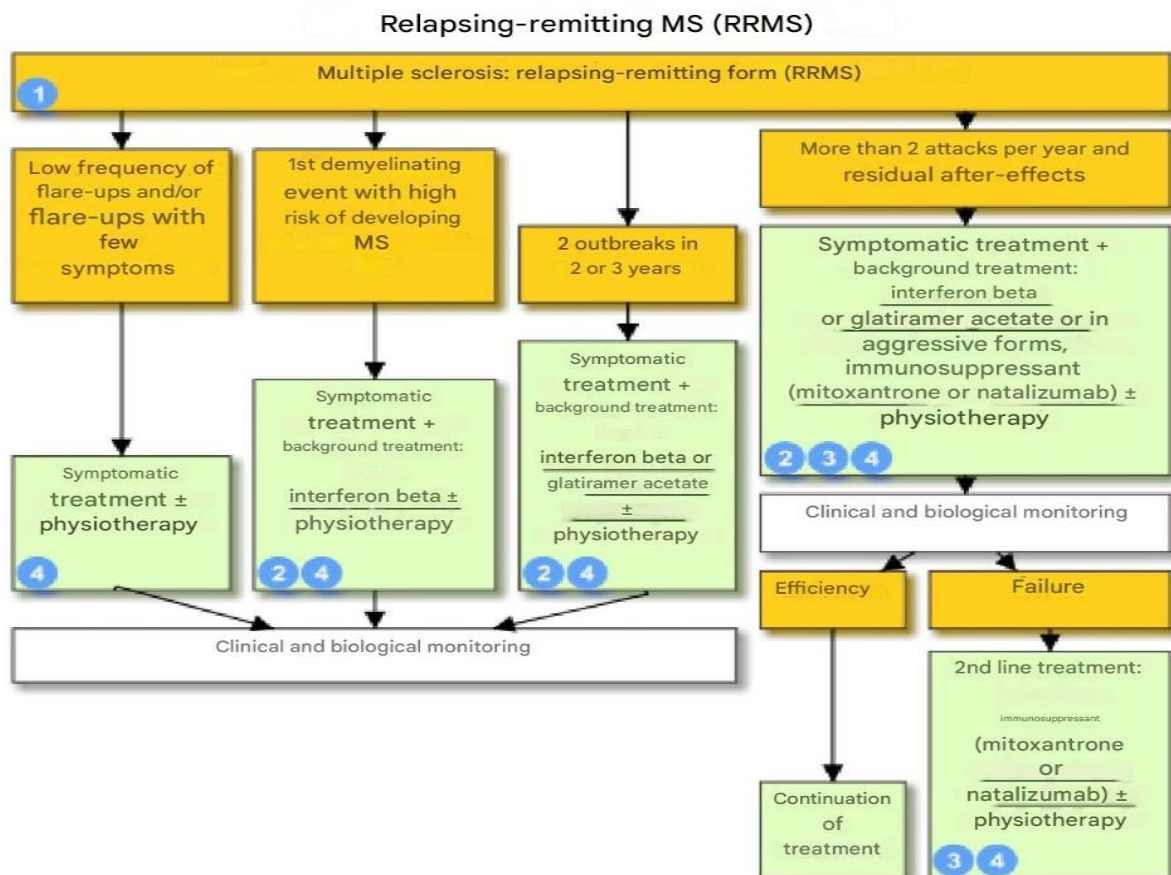
5. First-Line Treatment

- Management Focus: Initiating disease-modifying therapies (DMTs) to reduce relapse rates and slow disease progression.
- Approach: First-line treatments typically include interferon beta preparations (e.g., IFN beta-1a, IFN beta-1b) and glatiramer acetate. These medications are chosen based on the patient's specific clinical profile, preferences, and potential side effects.

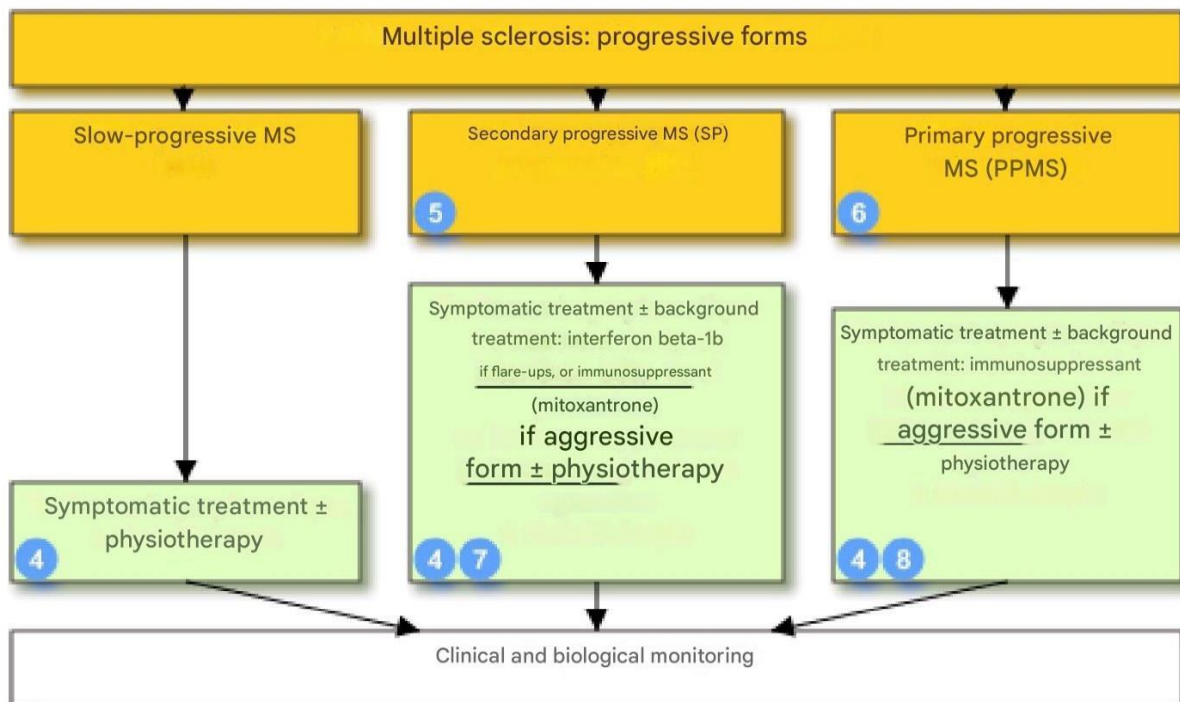
6. Treatment Adjustment

- Management Focus: Regularly evaluating the effectiveness of the current treatment regimen and making necessary adjustments based on clinical response and side effects.
- Approach: If a patient experiences breakthrough disease activity (e.g., new relapses or MRI lesions), a switch to a more effective second-line therapy (e.g., natalizumab, fingolimod, or ocrelizumab) may be warranted. Monitoring for adverse effects and patient adherence to treatment is also essential for successful management.
- The management of MS is a dynamic process that requires careful consideration of critical signs, mortality risk factors, outpatient care strategies, first-line treatments, and ongoing treatment adjustments. A personalized approach, involving regular monitoring and collaboration with a multidisciplinary team, is essential to optimize patient outcomes and enhance quality of life.

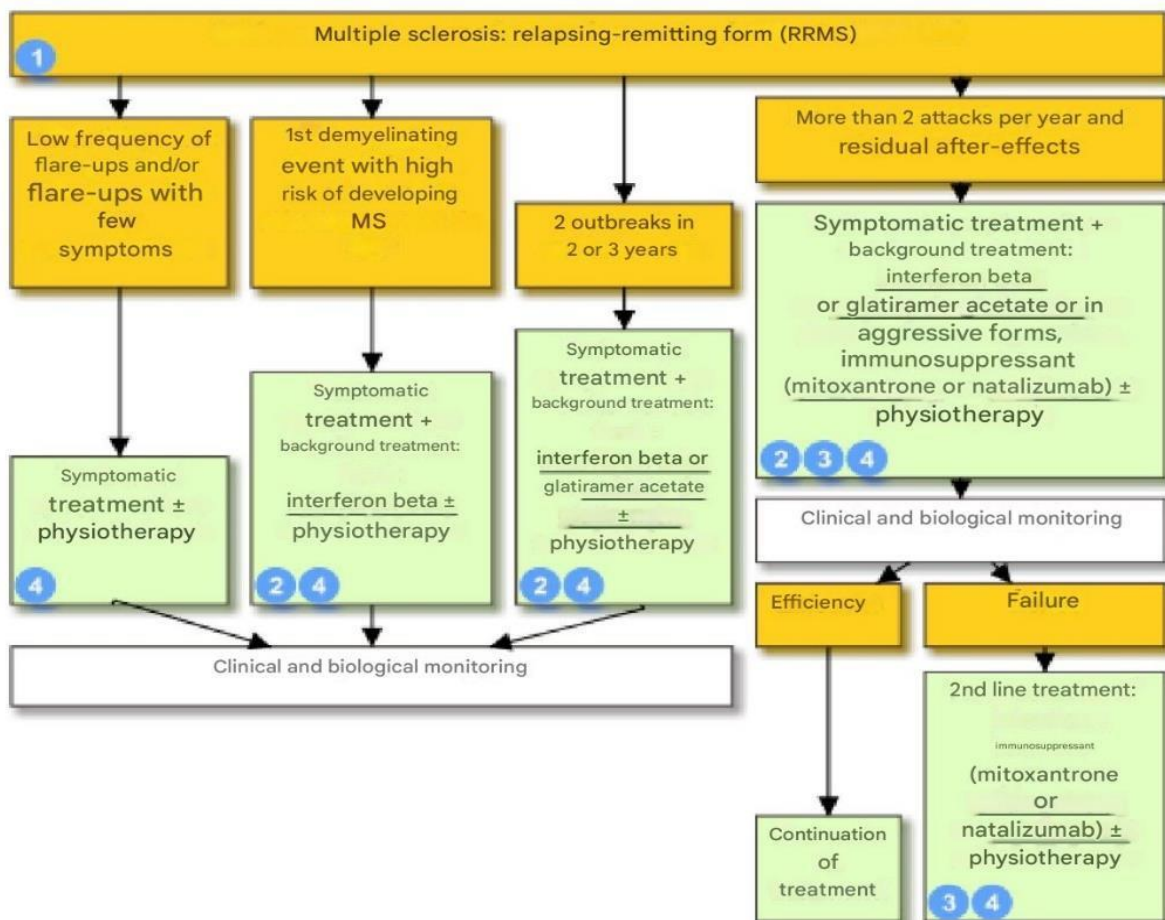
7. Algorithm



Progressive forms of MS



Relapsing-remitting MS (RRMS)



VIII. SPECIAL CASES

A special case of MS refers to specific clinical scenarios or subtypes of the disease that necessitate unique management strategies or considerations. Below are some notable special cases of MS:

- **Primary Progressive Multiple Sclerosis (PPMS)**
 - Description: This form of MS is marked by a gradual increase in disability from the onset, without distinct relapses or remissions, affecting approximately 15% of MS patients.
 - Management: Treatment options are more limited compared to relapsing forms of MS, with Ocrelizumab being the only FDA-approved disease-modifying therapy for PPMS. Management primarily focuses on symptomatic treatment and rehabilitation to preserve function and enhance quality of life.

2. Secondary Progressive Multiple Sclerosis (SPMS)

- Description: Many individuals with relapsing-remitting MS (RRMS) may transition to SPMS after several years, where the disease progresses more steadily, with or without occasional relapses.
- Management: Treatment may involve transitioning to more aggressive therapies if significant disease progression is observed. Symptomatic management and rehabilitation remain critical components of care.

3. Clinically Isolated Syndrome (CIS)

- Description: CIS is characterized by a first episode of neurological symptoms lasting at least 24 hours, suggestive of MS but not yet meeting the criteria for a definitive diagnosis. It may or may not lead to a diagnosis of MS.
- Management: Patients with CIS who are at high risk of developing MS (e.g., those with MRI findings indicative of MS) may be offered early treatment with disease-modifying therapies (DMTs) to potentially delay or prevent the onset of MS.

4. Pregnancy and MS

- Description: Pregnancy can affect the course of MS, with many women experiencing fewer relapses during pregnancy but an increased risk of relapses postpartum.
- Management: Careful planning of treatment around pregnancy is crucial, as some DMTs are contraindicated during this period. A multidisciplinary team approach is essential to address both maternal and fetal health.

5. Pediatric Multiple Sclerosis

- Description: MS can occur in children and adolescents, presenting unique challenges in diagnosis and management.
- Management: Treatment strategies may differ from those used in adults, necessitating age-appropriate therapies and support systems. Consideration of the long-term impact on growth and development is also important.

6. Progressive-Relapsing Multiple Sclerosis (PRMS)

- Description: This rare form of MS is characterized by a progressive course from the onset, accompanied by occasional acute relapses. It is less common than other forms.
- Management: Treatment strategies may include aggressive DMTs and symptomatic management, similar to those employed for SPMS.

- Special cases of MS require tailored management strategies that take into account the unique characteristics of each subtype or scenario. A multidisciplinary approach is vital to address the specific needs of patients, optimize treatment outcomes, and improve quality of life.

IX. EVOLUTION AND PROGNOSIS

The evolution and prognosis of Multiple Sclerosis (MS) can vary significantly among individuals, influenced by factors such as the type of MS, the age of onset, gender, and the presence of specific clinical features.

Prognosis of MS :

➤ Types of MS:

Relapsing-Remitting MS (RRMS): This is the most common form, characterized by clearly defined relapses (exacerbations) followed by periods of remission. Over time, many patients may transition to secondary progressive MS (SPMS).

Secondary Progressive MS (SPMS): Initially starts as RRMS but eventually leads to a progressive worsening of symptoms and disability, with or without occasional relapses.

Primary Progressive MS (PPMS): This form is characterized by a gradual progression of disability from the onset, without distinct relapses or remissions. It typically has a more challenging prognosis.

Progressive-Relapsing MS (PRMS): A rare form that presents with a progressive course from the beginning, with occasional acute relapses.

➤ Clinical Course:

- The clinical course of MS can be unpredictable. Some patients may experience frequent relapses and rapid progression, while others may have long periods of stability. The disease can affect various neurological functions, leading to symptoms such as fatigue, mobility issues, cognitive changes, and sensory disturbances.

➤ Factors Influencing Evolution:

- Age of Onset: Younger age at onset is often associated with a more favorable prognosis.
- Gender: MS is more common in women, and studies suggest that men may experience a more aggressive disease course.
- Initial Symptoms: The type and severity of initial symptoms can influence long-term outcomes. For example, those with motor symptoms may have a different prognosis compared to those with sensory symptoms.

Prognosis of MS

• Long-Term Outcomes:

- The prognosis for MS has improved significantly over the years due to advancements in treatment and management strategies. Many patients can lead active lives with appropriate therapy.
- Approximately 50% of patients with RRMS may transition to SPMS within 10 years, and the rate of progression can vary widely.

• Disability

- The Expanded Disability Status Scale (EDSS) is commonly used to assess disability in MS. Many patients may experience mild to moderate disability over time, while others may develop severe disability.
- The risk of significant disability increases with the duration of the disease, particularly in those who experience frequent relapses.
- **Life Expectancy:**
 - MS is generally not considered a fatal disease, and life expectancy for individuals with MS is close to that of the general population, especially with modern treatments. However, complications related to MS, such as infections or severe disability, can impact overall health and longevity.
- **Psychosocial Factors:**
 - The psychological impact of MS can affect prognosis. Mental health issues, such as depression and anxiety, are common and can influence treatment adherence and quality of life.
 - The evolution and prognosis of MS are complex and individualized. While many patients can manage their symptoms effectively and maintain a good quality of life, others may experience significant challenges.

X. COMPLICATIONS

MS can lead to a variety of complications that affect both physical and mental health. These complications can arise directly from the disease itself or as a result of the disability it causes.

- **Physical Complications**
 - Mobility Issues: Many individuals with MS experience difficulty walking, which can lead to falls and injuries. Muscle weakness, spasticity, and coordination problems contribute to these challenges.
 - Fatigue: A common and debilitating symptom of MS, fatigue can significantly impact daily activities and quality of life.
 - Pain: Patients may experience neuropathic pain (nerve pain) or musculoskeletal pain due to muscle spasms and immobility.
 - Visual Impairments: MS can affect the optic nerve, leading to vision problems such as blurred vision, double vision, or even temporary vision loss (optic neuritis).
 - Bladder and Bowel Dysfunction: Many individuals with MS experience urinary urgency, incontinence, or difficulty emptying the bladder. Bowel issues, such as constipation, are also common.
 - Sexual Dysfunction: MS can lead to sexual problems, including decreased libido, erectile dysfunction in men, and difficulties with arousal and orgasm in women.
- **Cognitive Complications**
 - Cognitive Impairment: Some individuals with MS may experience difficulties with memory, attention, problem-solving, and processing speed. Cognitive changes can affect work and daily functioning.
 - Emotional Changes: MS can lead to mood disorders, including depression and anxiety. The stress of living with a chronic illness can exacerbate these issues.
- **Secondary Complications**

- Infections: Individuals with MS may be at increased risk for infections, particularly urinary tract infections (UTIs) due to bladder dysfunction. Respiratory infections can also occur, especially in those with significant mobility issues.
- Pressure Sores: Reduced mobility can lead to pressure ulcers, particularly in individuals who are wheelchair-bound or have limited movement.
- Musculoskeletal Issues: Prolonged immobility can lead to joint problems, muscle atrophy, and contractures (shortening of muscles or tendons).
- **Social and Economic Complications**
 - Impact on Employment: The physical and cognitive challenges associated with MS can affect a person's ability to work, leading to financial strain and loss of independence.
 - Social Isolation: The symptoms of MS can lead to withdrawal from social activities, contributing to feelings of loneliness and isolation.
- **Long-Term Complications**
 - Disability: Over time, MS can lead to significant disability, affecting the ability to perform daily activities and requiring assistance or adaptive devices.
 - Complications from Disability: Long-term disability can lead to complications such as osteoporosis, fractures, and complications related to immobility.
 - The complications of MS are diverse and can significantly impact the quality of life for individuals living with the disease. A multidisciplinary approach to care, including physical therapy, occupational therapy, psychological support, and medical management, is essential to address these complications and improve overall well-being.

XI. PREVENTION AND RECOMMENDATION

Preventing MS entirely is not currently possible, as the exact cause of the disease is not fully understood. However, certain strategies and recommendations may help reduce the risk of developing MS or manage its progression.

- **Lifestyle Modifications**
 - Healthy Diet: A balanced diet rich in fruits, vegetables, whole grains, and healthy fats may support overall health. Some studies suggest that diets low in saturated fats and high in omega-3 fatty acids may be beneficial.
 - Regular Exercise: Engaging in regular physical activity can help maintain mobility, reduce fatigue, and improve overall well-being. Exercise can also help manage symptoms and prevent secondary complications.
 - Weight Management: Maintaining a healthy weight can reduce the risk of developing other health issues and may help manage MS symptoms.
- **Vitamin D and Sunlight Exposure**
 - Vitamin D: There is evidence suggesting that low levels of vitamin D may be associated with an increased risk of developing MS. Ensuring adequate vitamin D levels through sunlight exposure, diet, or supplements may be beneficial.
 - Sunlight Exposure: Regular exposure to sunlight can help the body produce vitamin D. However, it is essential to balance sun exposure with skin protection to reduce the risk of skin cancer.

- **Avoiding Smoking**
 - Smoking Cessation: Smoking has been linked to an increased risk of developing MS and can worsen the disease's progression. Quitting smoking can have significant health benefits and may reduce the risk of MS.
- **Managing Stress**
 - Stress Reduction Techniques: Chronic stress may negatively impact the immune system. Techniques such as mindfulness, meditation, yoga, and relaxation exercises can help manage stress levels.
- **Regular Medical Check-ups**
 - Monitoring Health: Regular check-ups with healthcare providers can help identify and manage risk factors for MS and other health conditions. Early intervention may improve outcomes.
- **Genetic and Environmental Factors**
 - Awareness of Family History: Individuals with a family history of MS may be at higher risk. Awareness and monitoring of symptoms can lead to early diagnosis and management.
 - Environmental Factors: Some studies suggest that exposure to certain viruses (e.g., Epstein-Barr virus) may be linked to MS. While it is not possible to prevent exposure to all environmental factors, maintaining a healthy immune system may be beneficial.
- **Education and Support**
 - Patient Education: Educating individuals about MS, its symptoms, and management strategies can empower them to seek help early and adhere to treatment plans.
 - Support Networks: Engaging with support groups and networks can provide emotional support and practical advice for managing the disease.
 - While there is no guaranteed way to prevent MS, adopting a healthy lifestyle, managing risk factors, and staying informed can help reduce the risk and improve the quality of life for those affected by the disease. It is essential for individuals to work closely with healthcare providers to develop personalized prevention and management strategies.

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EPILEPSY

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I. DEFINITION

Epilepsy is a disorder of the brain characterized by an enduring predisposition to generate epileptic seizures and by the neurobiological, cognitive, psychological, and social consequences of this condition. The definition of epilepsy requires the occurrence of at least one epileptic seizure.

II. EPIDEMIOLOGY

The prevalence of epileptic disease is stable throughout the world, and is effectively between 0.5 and 0.8% of the population. In Cambodia the prevalence is 5.8/1000. The incidence rate, defined by the occurrence of at least two spontaneous attacks per year, is 50/100,000 inhabitants. Epilepsy occurs more frequently in children and the elderly.

III. ETIOLOGY

The 2017 ILAE Classification of the Epilepsies recognizes six etiologic categories: genetic, structural, metabolic, immune, infectious, and unknown.

- Genetic: A genetic etiology is defined when epilepsy is the direct result of a known or presumed genetic defect and seizures are the core symptom of the disorder.

- Structural: Distinct structural causes may be associated with a substantially increased risk of developing epilepsy. Structural etiologies may be congenital (eg, cortical dysplasia, tuberous sclerosis) or acquired (eg, stroke, trauma, infection, immune-based).

- Metabolic: A metabolic etiology is defined when a patient has a documented metabolic condition that is associated with a substantially increased risk of developing epilepsy. Examples include glucose transporter deficiency, creatine deficiency syndromes, and mitochondrial cytopathies. Many of these conditions are also detected using genetic studies.

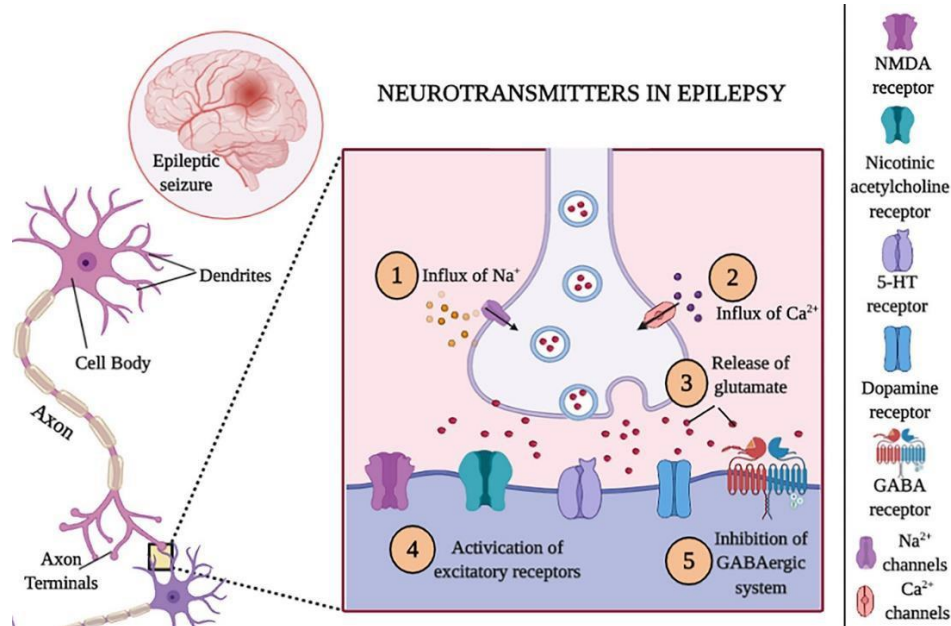
- Immune: Distinct immune mediated etiologies are defined as those in which there is evidence of central nervous system inflammation that results in epilepsy. Examples of immune etiologies include Rasmussen encephalitis, anti-N-methyl-D-aspartate (NMDA) receptor encephalitis.

- Infectious: Central nervous system infection may result in both acute symptomatic seizures as well as epilepsy. Infections are one of the most important causes of epilepsy worldwide. Examples of infectious etiologies of epilepsy include HIV, neurocysticercosis, malaria, and tuberculosis.

- Unknown: the term unknown has replaced the term cryptogenic, and simply means that the nature of the underlying cause is not currently known. All types of epilepsies with normal imaging and no documented genetic, metabolic, immune or infectious etiology are included in this category.

IV. PHYSIOPATHOLOGY

The pathophysiology of epilepsy and seizures is diverse, accounting for the many different types of seizure disorders. However, one commonality across epilepsies is a disrupted balance between excitatory (via glutamatergic signaling) and inhibitory (via GABAergic signaling) drive at the synaptic level that can result in seizure activity.



V. DIAGNOSIS

1. Clinic

The positive diagnosis of epileptic seizure must be evoked in the face of the sudden and unexpected onset of a brief clinical event, stereotyped from one episode to another in the same patient.

The diagnostic process includes 3 successive steps:

- Link the clinical disorder to an epileptic mechanism (ask the patient if at the time of the crisis the consciousness is not altered, otherwise questioning the witnesses is imperative)
- Specify the focal or generalized nature of the crisis (appendix)
- Classify the disease according to the etiology

2. Investigation

For patients with suspected seizures or epilepsy, the evaluation should include a detailed history of the event, electroencephalography (EEG) for select patients to identify epileptiform activity, a neuroimaging study for all patients to detect a structural brain lesion, and laboratory tests to look for metabolic causes.

VI. DIFFERENTIAL DIAGNOSIS

- Syncope
- Transient ischemic attack (particular in older adults)
- Migraine
- Panic attack and anxiety
- Psychogenic non epileptic seizure
- Transient global amnesia

VII. MANAGEMENT

1. Objectives

- Controlling seizures
- Avoiding treatment side effects
- And maintaining or restoring quality of life

2. Critical sign

Status epilepticus is a seizure that lasts longer than 5 minutes, or having more than 1 seizure within a 5 minutes period, without returning to a normal level of consciousness between episodes. It's emergency care.

3. Mortality risk factor

People with epilepsy face higher risk of premature mortality compared to the general population. Several factors contribute to this increased mortality risk including the types of epilepsy, frequency of epilepsy, and external factors, The cause of death can be directly related to seizures or result from other condition or circumstances associated with epilepsy.

4. Outpatient Management

Which patients treated?

Treatment should only be started when the seizures are certain and the epilepsy is sufficiently documented clinically, neurophysiological and neuroradiologically.

5. First-line treatment

- First-line is monotherapy
- Clear explanation of the diagnosis
- Increase progress gradually increase the dose in 7 to 14 day increments to avoid sedative effects linked to an excessive initial dose.
- After a period of 2 to 5 years without seizures, a reduction and then a complete cessation of treatment

6. Treatment adjustment

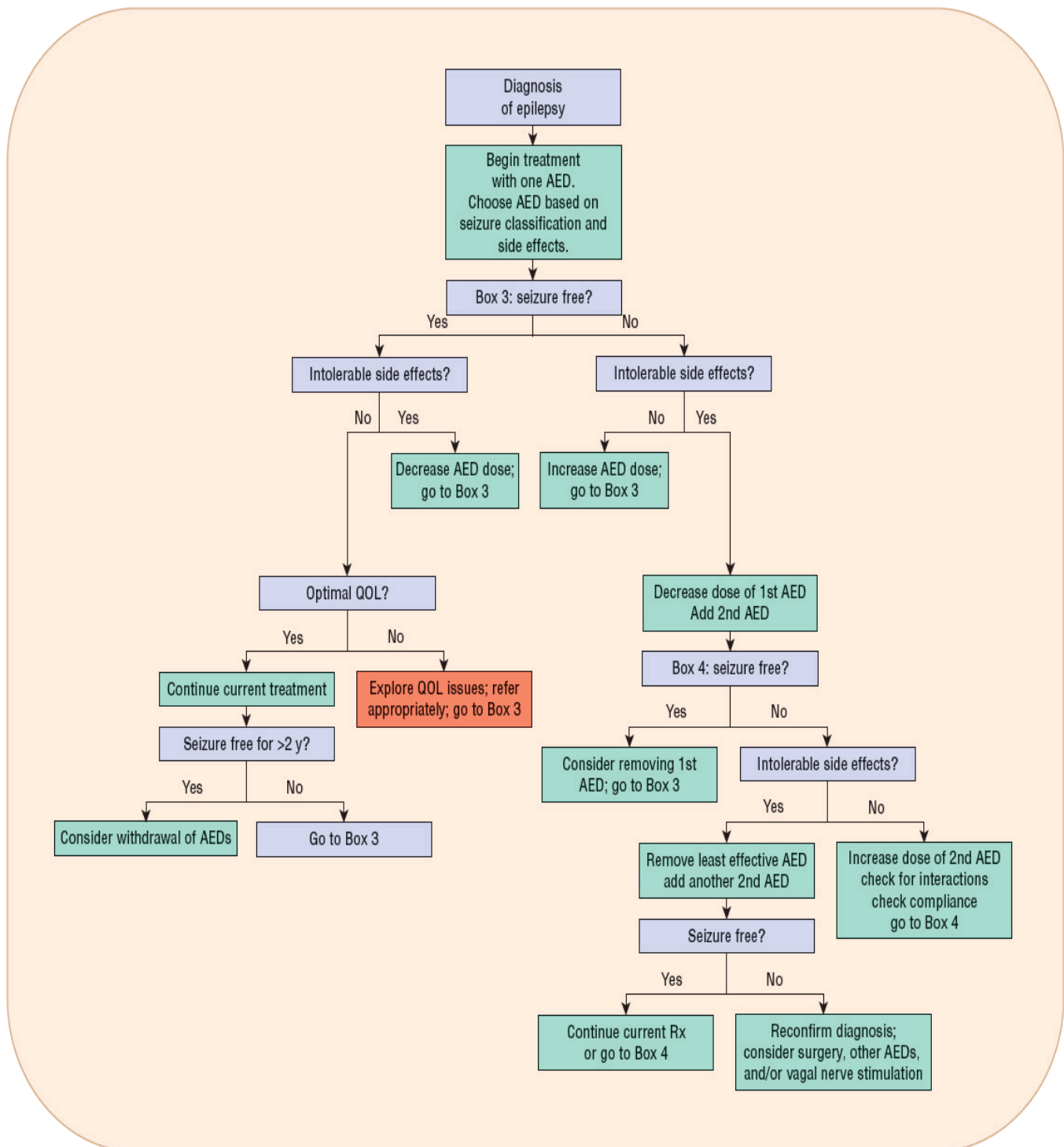
- Initial treatment of focal epilepsy: Lamotrigine, levetiracetam, oxcarbazepine, carbamazepine, lacosamide
- Genetically mediated generalized epilepsies syndrome with mainly generalized tonic clonic seizures: Lamotrigine, levetiracetam, valproate, topiramate, zonisamide
- Absence seizures: Ethosuximide, valproate, lamotrigine
- Genetically mediated generalized epilepsy syndrome with myoclonic seizures: Levetiracetam, valproate, zonisamide
- Female of childbearing age with either genetic generalized epilepsy or focal epilepsy: Lamotrigine, levetiracetam
- Older adult with focal epilepsy: Lamotrigine, levetiracetam, lacosamide
- Comorbid depression with focal epilepsy: Lamotrigine, lacosamide, oxcarbazepine
- Comorbid depression with genetically mediated generalized epilepsy: Lamotrigine, valproate
- Hepatic failure or after organ transplantation: Levetiracetam, gabapentin, lacosamide
- Renal failure on hemodialysis: Lamotrigine, oxcarbazepine, levetiracetam

Antiepileptics cited in the references

INN	Dosage mg/kg/d	Side effects
Phenobarbital	2-3	Severe skin rash Sedations, alterations in cognitive functions, algodystrophies, thickening of facial features, teratogenicity
Phenytoin	3-5	Rash severe, cerebellovestibular syndrome, obtundation, gingival enlargement, cognitive impairment, lupus syndrome, lymphoma, teratogenicity
Carbamazepine	10-20	Rash, lupus syndrome, dizziness, sedation, hyponatremia, teratogenicity
Valproate	15-20	Cytolytic hepatitis, rare pancreatitis, thrombocytopenia, digestive disorders, weight gain, postural tremor, fall hair, teratogenicity
Gabapentin	15- 35	Asthenia, sedation, moderate orexigenic effect
Lamotrigine	3- 7	Severe skin rash, dizziness, abnormal movements (tics)
Oxcarbazepine	10-30	Rash severe, diplopia, dizziness, ataxia, sedation, hyponatremia
Levetiracetam	15- 45	Asthenia, sedation, behavioral disturbances

7. Algorithm

Algorithm for the treatment of epilepsy (AED , antiepileptic drug; QOL, quality of life).



VIII. SPECIAL CASES

For women taking hormonal contraceptives, the use of enzyme-inducing AEDs is associated with an elevated risk of unplanned pregnancies.

- AEDs decrease efficacy of hormonal contraception:
 - Carbamazepine
 - Phenobarbital

- Phenytoin
- Primidone
- AEDs (no interaction with hormonal contraception):
 - Clonazepam
 - Lamotrigine <150mg per day
 - Lacosamide
 - Levetiracetam

Pregnancy: for women on AEDs should be planned. This allows the clinical team and the patient to choose the best treatment regimen for disease control that has an acceptably low structural and neurodevelopmental teratogenic risk profile. Prescription folic acid 5mg/ day at least 4 weeks before conception and continued through pregnancy in lowering the risk for neural tube defect.

IX. EVOLUTION AND PROGNOSIS

Patients with newly diagnosed epilepsy can actually be divided into four fairly quickly identifiable groups.

- Spontaneously benign epilepsies: The seizures are limited to a few episodes, often precipitated by a toxic factor (alcohol, drugs), or metabolic. Epilepsy always remains in spontaneous remission.

- Drug-sensitive epilepsies: also has a good prognosis. Remission, once obtained under appropriate treatment, is permanent and authorizes after a certain period of time for the definitive discontinuation of the drugs.

- Drug-dependent epilepsies: more reserved prognosis in so far as the disease does not seem to heal spontaneously. The various therapeutic adjustments will eventually control the attacks but do not hasten spontaneous remission, which is random. Abandoning treatment favors relapses.

- Drug-resistant epilepsies: is a poor prognosis. Epilepsy is severe because of its chronicity and its relative or absolute resistance to treatment, palliative rather than curative, typically conducted in polytherapy.

X. COMPLICATION

Epilepsy is a chronic disease associated with an increased risk of a variety of psychiatric and medical comorbidities that can adversely impact quality of life as well as life expectancy. Comorbidities can arise due to common underlying predispositions, direct effects of seizures, underlying epilepsy etiologies, and adverse effects of antiepileptics drug (AEDs) and other therapies. Depression and anxiety are particularly common in adults with epilepsy, and screening should be a routine part of long-term follow-up.

XI. PREVENTION AND RECOMMENDATION

Prevention

There are two main approaches to seizure prevention: medication use and trigger avoidance.

Recommendation

- Professional orientation of epileptic patients. Certain professions are prohibited: pilot, truck driver, public transport driver, lifeguard, military, work at height, surgeon, policeman, firefighter, bodyguard.

- Treatment typically involves antiepileptics medications tailored to individual specific type of seizure and medical history. It's essential to take medication as prescribes and attend regular follow-up appointment to monitor effectiveness and adjust treatment as needed.

- Additionally, for individuals with drug-resistant epilepsy, alternative treatments such as neurostimulation therapy or surgery.

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Appendices

A. International classification of epileptic seizures

Generalized crises

- Absences
- Myoclonic seizures
- Clonic seizures
- Tonic seizures
- Tonic-clonic seizures
- Atonic seizures

Partial seizures

- Simple partial seizures
 - With motor signs
 - With somatosensory or sensory signs
 - With vegetative signs
 - With psychic signs
- Complex partial seizures
 - Simple partial onset followed by disturbances of consciousness
 - With impaired consciousness from the onset of the crisis, accompanied or not automation
- Secondly generalized partial seizures
 - Simple partial seizures secondarily generalized
 - Complex partial seizures secondarily generalized
 - Simple partial seizures progressing to a complex partial seizure and then to a secondary generalization

Unclassifiable crises

ALCOHOL WITHDRAWAL SYNDROME

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I. DEFINITION

Alcohol withdrawal syndrome is a sudden cessation of alcohol in the chronic user, the alcohol mediated CNS inhibition is reduced, and the glutamate mediated CNS excitation is left unopposed, resulting in a net CNS excitation.

Alcohol withdrawal occurs in patients who are alcohol dependent and who have stopped or reduced their alcohol intake within hours or days of presentation. Common symptoms include anxiety, nausea or vomiting, autonomic dysfunction, and insomnia. If alcohol withdrawal is not treated or is inadequately treated, 5% of patients will progress to alcohol withdrawal delirium (also known as delirium tremens).

II. EPIDEMIOLOGY

As alcohol use has increased nationally over the last decade, especially during the COVID-19 pandemic, so too has the prevalence of alcohol withdrawal syndrome and its related complications. Those who exhibit unhealthy alcohol use, which includes both risky drinking as well as AUD, are at risk for alcohol withdrawal syndrome when the amount they normally drink is reduced.

Approximately half of those who suddenly stop or reduce their drinking will experience alcohol withdrawal syndrome. However, the severity varies. Not only is alcohol withdrawal syndrome a common cause of hospitalization, but it is also commonly treated in patients hospitalized for another reason. A 2013 sample of over 450,000 veterans' hospital admissions found that 1 in 17 admitted patients were treated for alcohol withdrawal syndrome.

III. ETIOLOGY

Alcohol is a central nervous system (CNS) depressant that works by stimulating GABA, the primary inhibitory neurotransmitter. In acute, episodic intoxication, GABA is produced in excess, resulting in clinical effects such as sedation. Chronic alcohol use changes the balance of inhibitory (GABA) signals and excitatory (glutamate) signals in the CNS, so glutamate is endogenously over-produced to balance the influx of GABA provided by alcohol.

When a patient who chronically drinks alcohol stops drinking or reduces how much they are drinking, the GABA signal is suddenly diminished, resulting in an overrepresentation of glutamate. This excess glutamate causes the autonomic hyperactive state seen in alcohol withdrawal syndrome.

IV. PATHOPHYSIOLOGY

To maintain homeostasis in the CNS, inhibitory signals from the GABAergic system are balanced by excitatory neurotransmitters such as glutamate. Alcohol, a CNS depressant, stimulates the GABAergic system and, in acute intoxication, causes a range of clinical manifestations such as disinhibition, euphoria, and sedation. The effect of alcohol in the acute setting is dose-dependent, with lower doses having a stimulating effect and higher doses having a sedating effect.

Chronic alcohol use results in neuroadaptive changes to the balance of GABA-glutamate by causing an upregulation of glutamate to compensate for alcohol-related increase in GABA. At the same time, endogenous GABA is downregulated. Thus, when alcohol is

withdrawn, a relative deficit of GABA may occur and simultaneous excess in glutamate, resulting in the excitatory symptoms seen in alcohol withdrawal syndrome.

Alcohol withdrawal syndrome exhibits a phenomenon known as kindling or sensitization, meaning successive withdrawal episodes tend to increase in severity, specifically concerning epileptic potential. Repeated episodes of alcohol withdrawal syndrome lower the seizure threshold, making it more likely a person in withdrawal will experience a seizure. For those who have previously had alcohol-withdrawal seizures, their likelihood of experiencing another is quite high.

V. DIAGNOSIS

1. Clinic

Alcohol withdrawal syndrome is a clinical diagnosis that relies heavily on the history and physical, which is also used to gauge disease severity. The signs and symptoms may present within hours of alcohol cessation and peak around 72 hours. In general, these signs and symptoms reflect the heightened excitability that results in the absence of the GABAergic input of alcohol and may include tremulousness, insomnia, agitation, diaphoresis, hypertension, tachycardia, or seizures. When in doubt, clinicians can refer to the DMS-V criteria for diagnosis.

If auditory, visual, or tactile hallucinations in the setting of alcohol withdrawal are present, the patient is likely experiencing alcohol hallucinosis, which affects approximately 2%. The altered mental status that accompanies this presentation is concerning for alcohol withdrawal delirium. When taking the history, it is essential to ask about a history of complicated withdrawal, defined as withdrawal from alcohol that includes either seizures or delirium.

A history of complicated withdrawal increases the likelihood of future complicated withdrawal, and these patients should be treated in a monitored setting.

Making a diagnosis of alcohol withdrawal syndrome

The alcohol withdrawal syndrome is diagnosed when the following two conditions are met

1. A clear evidence of recent cessation or reduction of alcohol after repeated and usually prolonged and/or high-dose use.
2. The patient shows symptoms of alcohol withdrawal that are not accounted for by a medical disorder or by another mental or behavioral disorder.

Table 1: Common signs and symptoms of alcohol withdrawal syndrome

Signs	Symptoms
Elevated blood pressure	Anxiety
Tachycardia	Insomnia
Elevated body temperature	Illusions
Sweating	Hallucinations
Tremulousness of body/increased hand tremor	Paranoid ideas
Dilated pupils	Nausea
Disorientation	Irritability
Hyper arousal	
Grand mal seizure	

The diagnosis requires adequate history of the amount and frequency of alcohol intake, the temporal relation between cessation (or reduction) of alcohol intake and the onset of symptoms

that may resemble a withdrawal state. When the onset of withdrawal-like symptoms or delirium is after 2 weeks of complete cessation of alcohol, the diagnosis of alcohol withdrawal syndrome or DT becomes untenable, regardless of frequent or heavy use of alcohol.

Table 2: Clinical descriptions of alcohol withdrawal syndromes by severity

Minor withdrawal	Starts about 6 h after cessation or decrease in intake, lasts up to 24-48 h: tremors, sweating, tachycardia, GI upset, headache, anxiety and clear sensorium (intact orientation). Usually corresponds to CIWA-Ar <10
Moderate to severe withdrawal	Alcoholic hallucinosis: Hallucinations (visual/tactile/auditory) and illusions in clear sensorium (well oriented) and stable vitals. Lasts 24 h to 6 days. Rarely beyond 1 month but definitely <6 months. Alcohol withdrawal seizure: One or two generalized tonic-clonic seizure. Begins within 6-48 h after the last drink. Multiple seizures (up to 6) can occur, within a span of 6 h. High risk of progression to DT. DT: Tremors, sweating, tachycardia, anxiety, hallucinations/illusions, disorientation, agitation secondary to previous symptoms. Begins 48-72 h after last drink and may last up to 2 weeks. Usually CIWA-Ar >20.

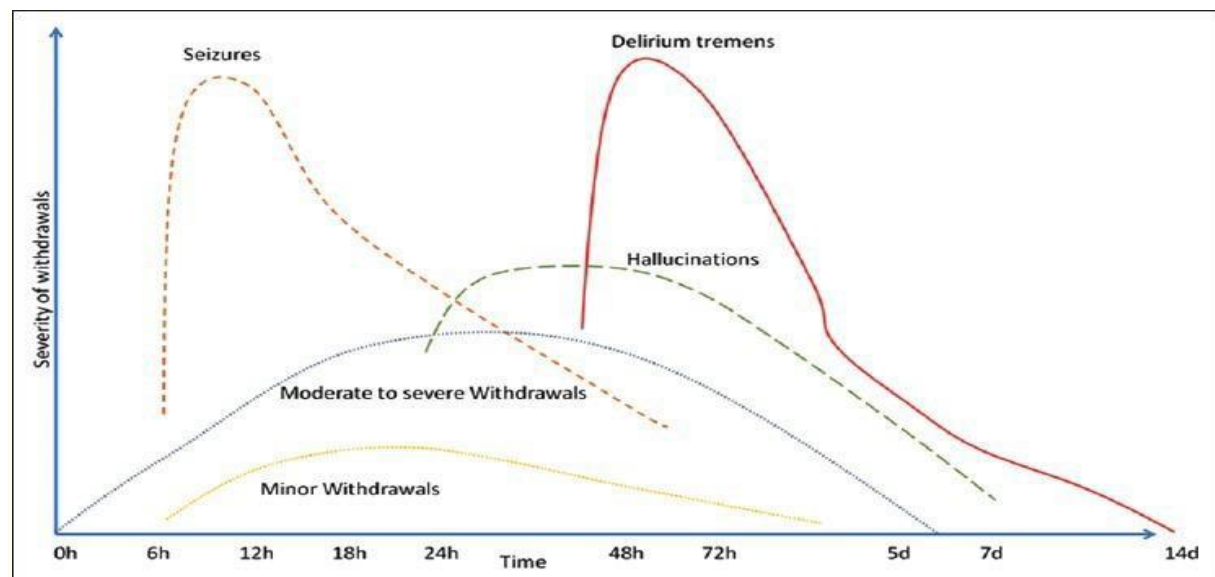


Figure 2: Graph depicting the time course of alcohol withdrawal symptoms

2. Investigation

- A serum glucose or fingerstick glucose test is indicated. Patients with liver disease due to alcoholism have reduced glycogen stores, and ethanol impairs gluconeogenesis. As a consequence, these patients are susceptible to hypoglycemia. Patients in alcohol withdrawal develop anxiety, agitation, tremor, seizure, and diaphoresis, all of which can occur with hypoglycemia.
- Analysis of arterial blood gases may be indicated. Mixed acid-base disorders are common and usually result from alcoholic ketoacidosis (AKA), volume-contraction alkalosis, and respiratory alkalosis. Hypoxia may occur due to aspiration pneumonitis.
- A complete blood cell count (CBC) is recommended. Findings may include the following:
 - Long-term alcohol ingestion leads to myelosuppression with a slight reduction in all cell lines; thrombocytopenia is common.
 - Blood loss from the gastrointestinal (GI) tract and nutritional deficiencies producing anemia are common in alcohol withdrawal.
 - Megaloblastic anemia is observed in patients with alcoholism, due to dietary deficiency of vitamin B-12 and folate; increased mean corpuscular volume suggests this condition
- An article in the Journal of Trauma reported that a mean corpuscular volume (MCV) and aspartate aminotransferase (AST) level in an intoxicated trauma patient admitted to the hospital predicated a low risk for developing delirium tremens. However, this has not been studied in non-trauma patients.
- Urinalysis is indicated, as follows:
 - Routinely check for ketones, as patients may have associated AKA.
 - Ketonuria without glycosuria must be investigated further to exclude AKA and the ingestion of isopropyl alcohol.
 - Myoglobinuria from rhabdomyolysis may first be suspected when hematuria is noted on urinalysis.
- Cardiac markers may be indicated. Findings may include the following:
 - Elevated creatine kinase (CK) and cardiac troponin levels may indicate myocardial infarction resulting from increased demands placed on the heart from hypertension associated with alcohol withdrawal or from hypertension produced by cocaine intoxication prior to the patient's presentation.
- Measurement of prothrombin time (PT) may be indicated, as follows:
- Toxicology screening may be indicated, as follows:
 - Consider measuring serum osmolality and testing for toxic alcohols if the history is suspicious for toxic alcohol ingestion or if the patient has a widened anion gap metabolic acidosis.
 - The ethanol concentration is frequently zero. However, some patients that are habituated to alcohol can be in severe alcohol withdrawal even with a positive serum ethanol concentration.
 - Urine drug immunoassay is of limited utility but may be considered to assess for co-ingestion of other medications (eg, benzodiazepines) and other recreational drugs.

VI. DIFFERENTIAL DIAGNOSIS

When suspecting alcohol withdrawal syndrome, it is important to rule out concurrent withdrawal syndromes, particularly sedative-hypnotics such as benzodiazepines or barbiturates. If the patient is presenting with seizures or delirium, one must consider medical or neurologic causes other than alcohol.

In the workup, it is important to consider clinical entities that exhibit signs and symptoms similar to alcohol withdrawal syndrome, such as thyrotoxicosis, hypoglycemia, diabetic ketoacidosis, stimulant substance use, status epilepticus, CNS infection, withdrawal from sedative-hypnotics, essential tremors, hepatic encephalopathy, intracranial hemorrhage, cerebrovascular accident, and psychosis.

VII. MANAGEMENT

1. Objectives

- Implement evidence-based treatment strategies and monitor for potential complications of alcohol withdrawal syndrome promptly.
- Select the most suitable treatment strategy, such as pharmacotherapy and adjunctive therapies, based on the severity of alcohol withdrawal syndrome.
- Collaborate with interprofessional healthcare team members to optimize patient care and support services for patients recovering from alcohol withdrawal syndrome.

2. Critical Signs

Alcohol withdrawal delirium, formerly known as delirium tremens, is the most severe manifestation of alcohol withdrawal syndrome, characterized by symptoms such as fever, tachycardia, agitation, diaphoresis, hallucinations, disorientation, and hypertension. Patients are at risk of experiencing alcohol withdrawal delirium anywhere from 3 to 8 days following alcohol cessation.

While only approximately 3% to 5% of patients with alcohol withdrawal syndrome will progress to alcohol withdrawal delirium, this condition may prove fatal. Identifying patients at risk for alcohol withdrawal delirium through a validated tool and considering the patient's history of complicated withdrawal is a crucial aspect of managing the syndrome.

3. Mortality risk factor

Objective assessment of the severity of alcohol withdrawal – the Clinical Institutes Withdrawal Assessment-Alcohol Revised (CIWA-Ar) scale

In a patient diagnosed to have alcohol withdrawal syndrome, the CIWA-Ar can be used to measure its severity. The scale is not a diagnostic tool as it has not been found to be useful in differentiating between DT and delirium due to medical illnesses. The scale includes 10 common signs and symptoms of alcohol withdrawal with the notable exceptions of pulse rate and blood pressure, which must be a part of the assessment of alcohol withdrawal states.

Mortality risk factor of alcohol withdrawal's patients
Older age Comorbid medical or surgical illness Past history of DT or alcohol withdrawal seizure Severe withdrawal symptoms at initial assessment, despite having significant blood alcohol levels Presence of dehydration History of having had withdrawal seizure during this current withdrawal state before the assessment Presence of hyponatremia or hypokalemia Elevated AST or GGT levels Low platelet count The presence of structural brain lesions Duration of alcohol use and average daily quantity of alcohol consumed are not consistent predictors of severe alcohol withdrawal

**AST - Aspartate aminotransferase; GGT - Gamma glutamyl transferase;
DT - Delirium tremens**

4. Outpatient management

Alcohol withdrawal syndrome can range in severity from mild to fatal, making it crucial for patients to present to care for evaluation of their symptoms. Patients who have had prior complicated withdrawals should not attempt to decrease their alcohol intake without consultation with their healthcare team. If a patient begins experiencing signs and symptoms of severe withdrawal, including but not limited to seizure, altered mental status, or agitation, they should seek emergency care immediately. When alcohol withdrawal syndrome has resolved, patients ought to be evaluated for AUD and offered treatment, if appropriate, including pharmacotherapy and behavioral treatment.

5. First-line treatment

Symptom-triggered treatment (STT)

The STT was proposed by Saitz *et al.* in 1994 where in chlordiazepoxide was given when CIWA-Ar ratings were eight or more. The STT requires close monitoring as in-patient. Patients who are non-verbal (e.g. stupor due to head injury) may not be suited for this regimen as they may not be able to inform the nursing personnel if they were to experience any withdrawal symptoms. This protocol is not safe in patients with a past history of withdrawal seizures because they can occur even in a patient without overt autonomic arousal or symptoms of alcohol withdrawal. STT decreases the duration of detoxification and dose of benzodiazepine required compared with fixed dose regimen and may be useful in patients who have never had complicated withdrawals.

Rapid loading with close monitoring

This method is recommended only in patients with DT. Frequent boluses of diazepam are given intravenously until the patient is calm and sedated. It is described under management of DT below.

MANAGEMENT OF MINOR ALCOHOL WITHDRAWAL SYNDROME

Minor alcohol withdrawal syndrome may not need pharmacotherapy in all cases. The patient needs supportive care in a calm and quiet environment and observation for a period of up to 36 h, after which he is unlikely to develop withdrawal symptoms.

In the presence of risk factors like an acute medical illness or a past history of severe withdrawals, a single dose of 20 mg of diazepam should be given immediately as a loading dose and the patient be monitored for further signs of alcohol withdrawal, further doses being guided by the appearance of withdrawal symptoms.

Out-patient treatment can be started for patients without these risk factors and is based on the clinical withdrawal signs. Pharmacotherapy is started when the systolic blood pressure exceeds 150 mmHg, diastolic blood pressure exceeds 90 mmHg, body temperature greater than 37.7°C, pulse exceeds 100/min or other withdrawal symptoms such as agitation, insomnia, tremulousness is present without other medical or neurological illness.

Management of moderate to severe alcohol withdrawal syndrome

Without seizures or DT

In these cases, we recommend that patients should be started immediately on a SML dose regimen, while monitoring the withdrawal severity (CIWA-Ar ratings) and clinical signs

of tachycardia and hypertension. A fixed dose regimen can be safely used in such patients in case adequately trained personnel are not available or if outpatient treatment is advised.

Severe alcohol withdrawal with alcohol withdrawal seizures

Regardless of the CIWA-Ar score, the occurrence of seizures during the alcohol withdrawal period is indicative of severe alcohol withdrawal. Seizure prophylaxis with lorazepam 2 mg intravenously must be given to all patients with seizures in the current withdrawal period at presentation and also in those with past history of withdrawal seizure. Lorazepam is more effective than diazepam in preventing seizure recurrence. Unlike diazepam brain tissue levels of lorazepam don't fall rapidly owing to its poor lipid solubility and poor redistribution. Although, a single lorazepam dose given is likely to prevent further seizure recurrences, it may still be required to give SML dose of diazepam of at least 20-60 mg or at times even 80 mg diazepam in such patients. This strategy helps to prevent the development of DT. All patients who presenting with seizures after cessation of alcohol, regardless of previous episodes, must ideally be monitored as inpatients for at least 36-48 h to watch for further seizures or DT.

In patients who present with seizures, a thorough neurological and general medical evaluation is a must to detect alternative cause of seizures. Patients with new onset seizures should preferably undergo brain imaging. Neurological work-up and consultation is essential in patients with more than six seizures during alcohol withdrawal, seizures persisting for more than 6 h or despite adequate dose of benzodiazepines treatment, presence of focal seizures/change in seizure type, history of traumatic brain injury, family history of seizures in relatives who do not use alcohol, status epilepticus, focal neurological deficits, presence of meningeal irritation or patients with worsening sensorium despite at least 8 h of loading dose diazepam (60 to 80 mg).

Severe alcohol withdrawal with DT

Treatment of alcohol withdrawal delirium DT is defined by the goal of achieving a calm, but awake state or light somnolence defined as a sleep from which the patient is easily aroused. This goal is best achieved by the use of intravenous diazepam administered at frequent intervals while closely monitoring the patient during the procedure. Intravenous or intramuscular lorazepam may be used in patients with hepatic disease, pulmonary disease or in the elderly where there is risk of over-sedation and respiratory depression with diazepam.

An initial dose of 10 mg diazepam is given intravenously. Further doses of 10 mg can be repeated every 5-20 min interval. Others recommend increasing the dose to 20 mg per bolus for the subsequent boluses if the first two boluses do not calm the patient down. Once the goal of light somnolence is achieved, the patient is shifted to a SML dose regimen. Though experts' advice the use of rapid loading doses of diazepam for management of DT, no trials of rapid loading with diazepam has been conducted in patients with DT.

There have been trials comparing loading doses of barbiturates (versus diazepam loading), where the drug is given at 2 h intervals and a trial of diazepam (loading dose versus fixed doses) for the management of DT. In practice, loading dose strategy (20 mg diazepam every 2 h) can be safely administered in DT. Vital signs should be used to guide treatment in "DT-plus" condition (DT in presence of medical comorbidities) as these patients have been noted to have failure of loading-dose regimen with falsely high CIWA-Ar scores.

Refractory DT

A review by Hack *et al.* suggests that a high requirement of intravenous diazepam (more than 50 mg in the 1sth, or 200 mg or more within the first 3 h) with poor control of withdrawal symptoms is a marker of non-response of DT to benzodiazepines.

Such patients can be diagnosed to have refractory DT after a review of the clinical condition to rule out medical or neurological causes of delirium. They can be given oral (or intravenous) loading with phenobarbital 100-200 mg/h, which has been shown to be as effective as or better than diazepam front-loading for patients in DT in a more recent retrospective chart review.

The use of barbiturates is justified by the fact that they are also GABA-enhancing drugs that have a different receptor profile than benzodiazepines and have been tested in a double-blind protocol against diazepam for DT. However, it carries the risk of over-sedation, especially in the elderly or in presence of hepatic disease and a risk of respiratory depression in patients with pulmonary disease. There is no antidote to barbiturate toxicity. For these reasons, barbiturates have fallen out of favor.

An alternative adjunctive medication useful in patients with refractory DT is haloperidol given in doses of 0.5-5 mg by intramuscular route every 30-60 min or 2-20 mg/h, while continuing to give diazepam 10-20 mg every 1-2 h. Newer antipsychotics like risperidone (1-5 mg/day) or olanzapine (5-10 mg/day) may have a better safety profile than haloperidol (2, 5-10 mg/day) and are preferred as adjuncts to benzodiazepine treatment.

In patients who do not respond to benzodiazepines and haloperidol, propofol infusion (0.3-1.25 mg/kg/h) in an intensive care setting has been used in a few cases. The risks of propofol infusion include bradycardia, hypotension, metabolic acidosis, acute pancreatitis and lipid abnormalities. Moreover, propofol may not treat the underlying withdrawal syndrome because patients are often noted to exhibit withdrawal symptoms soon after stopping propofol infusion.

Alcoholic hallucinosis

This is a unique form of withdrawal related psychosis which can begin even while the person is continuing to use alcohol or begins after he stops alcohol. Hallucinations occurring in clear sensorium are the hallmark of this disorder. A cluster analysis of alcohol withdrawal symptoms by Driessen *et al.* showed that hallucinosis is a severe form of alcohol withdrawal and is often associated with DT. However, it is one of the conditions that may cause apparent failure of the loading dose regimen and we recommend a fixed dose strategy to cover the period of alcoholic hallucinosis. The patient may be given low doses of antipsychotics like chlorpromazine 100-200 mg/day or risperidone 1-3 mg/day to control severe agitation due to hallucinations. The hallucinations last about a week in most cases, but may last up to 1 month in some patients after which the antipsychotic can be stopped.

Importance of adjunctive supplements (Vitamin B and magnesium)

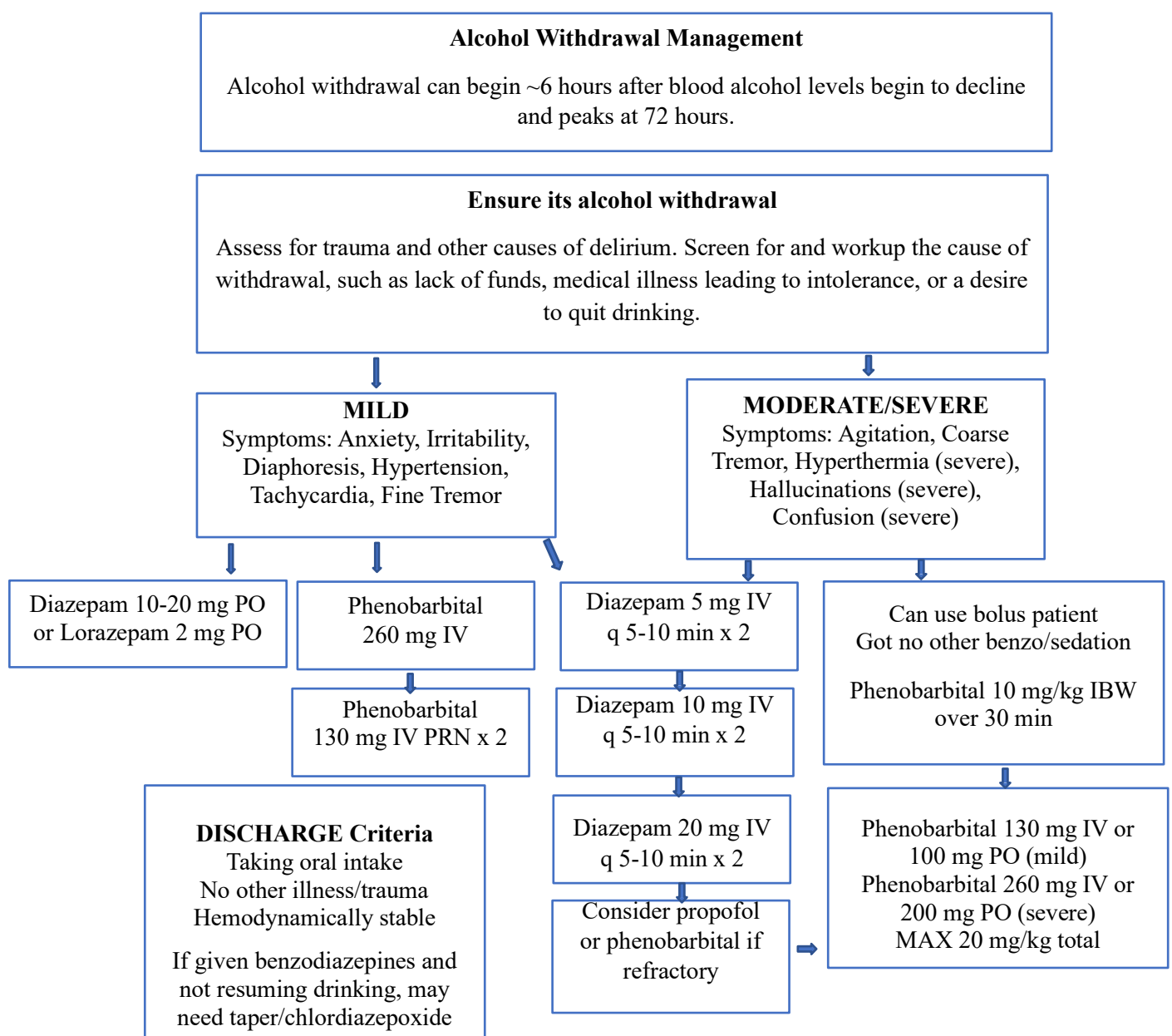
Wernicke's Encephalopathy (WE) results from cell damage due to chronic thiamine deficiency. It rarely presents with the classic triad of confusion, ataxia and ophthalmoplegia and therefore goes undiagnosed in nearly 90% of the cases. It is difficult to diagnose in the presence of alcohol withdrawal symptoms. The presence of small mammillary bodies and thalami on magnetic resonance imaging brain may be helpful in diagnosis, but confirmation is by postmortem examination. WE has an associated mortality of 20%, with 75% developing a permanent severe amnesic syndrome (Korsakoff's encephalopathy).

This can be prevented by administering parenteral thiamine which achieves adequate blood thiamine level much earlier than the oral route. Allergic reactions are rare. All patients in alcohol withdrawal should receive at least 250 mg thiamine by the parenteral route once a day for the first 3-5 days, whereas for those with suspected WE, thiamine 500 mg/day for 3-5 days is advised. If there is clinical improvement the supplementation is continued for total of 2 weeks.

Concurrent administration of parenteral thiamine with glucose is advised traditionally. However, this is only to ensure that thiamine supplementation is not forgotten. Administration of glucose containing fluids before thiamine may not precipitate, WE Due to chronic malnutrition and gastric malabsorption that follows chronic alcohol abuse, many clinicians advise multivitamin supplements (B1 + B2 + B6 + nicotinamide + Vitamin C) in parenteral form for the initial 3-5 days.

Chronic alcohol use is associated with abnormal magnesium metabolism. Those with neuropathy and presenting with severe withdrawal symptoms are more likely to show low serum magnesium level. Oral or parenteral magnesium supplementation may benefit such patients by reducing the severity and duration of alcohol withdrawal. Routine use is not advised.

6. Algorithm



IX. EVOLUTION AND PROGNOSIS

The prognosis of alcohol withdrawal syndrome depends both on the severity at presentation as well as how promptly treatment is initiated. Treatment with benzodiazepines reduces the risk of progression. For patients who experience a withdrawal seizure, they are at risk for recurrent seizures as well as alcohol withdrawal delirium. Historically, the mortality rate of alcohol withdrawal delirium has been as high as 20%, but with advances in critical care, prompt diagnosis, and treatment, the rate is now around 1%.

X. COMPLICATIONS

The following are common complications of alcohol withdrawal syndrome:

- Wernicke encephalopathy: Untreated, severe thiamine deficiency can result in Wernicke encephalopathy, an acute neurological condition that can be fatal in its most severe form. It is characterized by ocular changes, altered mental status, and ataxia.
- Withdrawal seizure: Alcohol-related seizures typically occur between 8 and 48 hours after alcohol cessation. Following a seizure, the patient should be treated with benzodiazepines and monitored closely for the next approximately 24 hours. Alcohol-related seizures can occur in the absence of other signs and symptoms of alcohol withdrawal.
- Alcohol hallucinosis: An alcohol-induced psychotic disorder, alcohol hallucinosis consists primarily of auditory hallucinations and paranoia that resolve within 72 hours.
- Alcohol withdrawal delirium (formerly delirium tremens): Alcohol withdrawal delirium can occur at any point up to 3 to 5 days after alcohol cessation or reduction. Delirium requires treatment and close monitoring, often in an intensive care setting, especially after a patient has received high doses of benzodiazepines or phenobarbital.

XI. PREVENTION AND RECOMMENDATION

Alcohol withdrawal syndrome is a clinical condition that may arise following the cessation or reduction of regular, heavy alcohol consumption. Given its spectrum of manifestations from mild to severe and potentially fatal, all healthcare team members must recognize the signs and symptoms of this condition. Timely assessment and accurate treatment are vital to preventing disease progression. Comprehensive patient care entails acute management and outpatient support in the hospital setting. In the inpatient setting, nurses perform frequent assessments that inform the treatment plan.

Pharmacists are critical in ensuring proper dosing and safety of medications aimed at preventing disease progression. Social workers facilitate the arrangement of outpatient follow-ups with primary care or addiction medicine specialists and provide support in accessing behavioral therapies or rehabilitation programs. Clinicians are responsible for diagnosing and treating sequelae of chronic alcohol use, often necessitating collaboration across multiple specialties. In addition, they conduct screenings for AUD and initiate pharmacotherapy when deemed appropriate. Establishing strong working relationships among various specialties and professions is essential for enhancing the care provided to patients with AUD.

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ACUTE CONFUSION

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I. DEFINITION

Acute confusion, also known as delirium, is defined as a sudden and reversible disturbance in attention and cognition. It is characterized by:

- An abrupt onset and often fluctuating course.
- Global impairment of cognitive functions, including reduced awareness of the environment and difficulty in maintaining attention.
- Disorganized thinking, which may manifest as incoherent or inappropriate speech.
- Disturbances in sleep-wake cycles.

Acute confusion is typically caused by various underlying factors, such as medical illnesses, substance intoxication or withdrawal, infections, or metabolic imbalances. Prompt medical evaluation is essential to identify and treat the underlying cause to facilitate recovery.

II. EPIDEMIOLOGY

The epidemiology of acute confusion (delirium) indicates that it is a common condition, particularly among certain populations.

1. **Prevalence:** Acute confusion is frequently observed in hospitalized patients, with prevalence rates at admission ranging from 11% to 33%. The incidence during hospitalization can vary between 3% and 42%, depending on factors such as patient age, comorbidities, and the specific hospital services involved.
2. **Age Factor:** The condition is especially prevalent in older adults, who are at higher risk due to factors like polypharmacy, pre-existing cognitive impairment, and increased vulnerability to medical illnesses.
3. **Risk Factors:** Various factors contribute to the development of acute confusion, including advanced age, pre-existing cognitive disorders (e.g., dementia), severe illness, and certain medications.
4. **Clinical Settings:** Acute confusion is more commonly seen in intensive care units, post-operative settings, and among patients with significant medical comorbidities.

Understanding the epidemiology of acute confusion is crucial for healthcare providers to identify at-risk populations and implement preventive measures effectively.

Acute confusion, also known as delirium, is a common condition that particularly affects specific populations. The epidemiology of acute confusion reveals its widespread nature, especially among hospitalized patients. Prevalence rates for acute confusion at the time of admission range from 11% to 33%, and the incidence during hospitalization can vary between 3% and 42%. This variation is influenced by factors such as the patient's age, existing medical conditions, and the types of hospital services they receive.

Older adults are particularly vulnerable to acute confusion, as they face increased risks due to factors like polypharmacy, pre-existing cognitive impairments, and heightened susceptibility to medical illnesses. Additionally, several risk factors contribute to the onset of acute confusion, including advanced age, cognitive disorders (such as dementia), severe illnesses, and the use of certain medications.

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Acute confusion is most frequently observed in intensive care units, post-operative settings, and among patients with significant medical comorbidities. Understanding the epidemiology of acute confusion is essential for healthcare providers to identify populations at higher risk and to implement effective preventive measures.

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III. ETIOLOGY

Acute confusion, also known as delirium, is a sudden and often reversible disturbance in attention and cognition, characterized by:

- A rapid onset with fluctuations in severity.
- Widespread cognitive impairment, including reduced environmental awareness and difficulty focusing attention.
- Disorganized thinking, which may present as incoherent or inappropriate speech.
- Disruptions in sleep-wake cycles.

Acute confusion is typically triggered by various underlying factors, including medical conditions, substance intoxication or withdrawal, infections, or metabolic imbalances. Prompt medical evaluation is essential to identify and address the root cause to support recovery.

Main Causes of Acute Confusion

Acute confusion can result from a range of physical and psychological factors. Here are the primary causes:

1. Physical Causes:

- **Intoxications:** Exposure to alcohol, drugs (such as psychotropic drugs, opioids), or other toxic substances.
- **Infections:** Systemic infections, like pneumonia or urinary tract infections, especially in older adults.
- **Metabolic Imbalances:** Electrolyte disturbances (e.g., low sodium or high calcium levels), low blood sugar, or nutritional deficiencies (such as vitamin B1 deficiency).

2. Detailed Physical Causes:

- Poisoning by carbon monoxide (often from faulty heating systems).
- Alcohol dependence.
- Infectious syndromes (e.g., fever, pneumonia).
- Urinary retention or fecal impaction.

- Liver dysfunction.
 - Neurological disorders (e.g., stroke, epilepsy, brain tumors, meningitis).
 - Metabolic issues, including hypoglycemia, dehydration, and electrolyte imbalances.
3. **Medication-Related Causes:**
- Use of psychotropic medications (e.g., benzodiazepines, antipsychotics).
 - Certain antidepressants, pain relievers, anti-Parkinsonian drugs, anticonvulsants, corticosteroids, and some NSAIDs.
4. **Psychological Causes:**
- Psychological stress, often due to environmental changes, such as hospitalization or relocation, particularly affecting the elderly.

Timely identification and treatment of the underlying cause are vital to ensure a quick recovery.

IV. PATHOPHYSIOLOGY

The pathophysiology of acute confusion (delirium) involves a complex interplay of neurobiological mechanisms that disrupt normal brain function. Key aspects include:

- **Neurotransmitter Imbalance:** Acute confusion is often associated with alterations in the balance of neurotransmitters in the brain, particularly acetylcholine and dopamine. A deficiency in acetylcholine and an excess of dopamine are commonly implicated in the development of delirium.
- **Cerebral Hypoperfusion:** Reduced blood flow to the brain can lead to inadequate oxygen and nutrient delivery, contributing to cognitive dysfunction. This is particularly relevant in patients with cardiovascular issues or those undergoing major surgery.
- **Inflammatory Responses:** Systemic inflammation, often triggered by infections or other medical conditions, can affect brain function. Cytokines and other inflammatory mediators can disrupt neurotransmission and contribute to cognitive impairment.
- **Metabolic Disturbances:** Various metabolic imbalances, such as electrolyte disturbances (e.g., hyponatremia, hypercalcemia), hypoglycemia, and dehydration, can adversely affect brain function and lead to acute confusion.
- **Environmental Factors:** Changes in the environment, such as hospitalization, can also trigger acute confusion, especially in vulnerable populations like the elderly. Stressors, sensory deprivation, and sleep disturbances can exacerbate the condition.
- **Neurological Conditions:** Underlying neurological issues, such as strokes, seizures, or infections (e.g., meningitis), can directly impact brain function and contribute to the onset of acute confusion.

V. DIAGNOSIS

1. Clinic:

The clinical diagnosis of acute confusion (delirium) involves a combination of clinical assessment, history-taking, and the application of specific diagnostic criteria.

Clinical Criteria:

According to the DSM-IV-TR (Diagnostic and Statistical Manual of Mental Disorders), the diagnosis of acute confusion includes the following criteria:

- **Attention Deficit:** There is a marked decrease in the ability to maintain attention to external stimuli and to engage appropriately with new stimuli.
- **Disorganized Thinking:** The patient exhibits disorganized thinking, which may manifest as incoherent or inappropriate speech.

- **Additional Symptoms:** At least two of the following symptoms must be present:
 - Altered level of consciousness (obnubilated consciousness).
 - Perceptual disturbances (e.g., illusions or hallucinations).
 - Sleep-wake cycle disturbances.
 - Psychomotor activity changes (either increased or decreased).
 - Disorientation in time and space, or failure to recognize familiar people.
 - Memory disturbances.
- **Duration:** The symptoms must develop over a short period (hours to days) and tend to fluctuate throughout the day.
- **Evidence of an Organic Cause:** There must be evidence of an underlying organic factor contributing to the symptoms, or the symptoms cannot be explained by a primary psychiatric disorder.
- **History and Physical Examination:** A thorough history should be taken, including:
 - Recent medical history, including any infections, medications, or substance use.
 - Changes in environment or significant stressors.
 - A physical examination to assess for signs of infection, dehydration, or other medical conditions.
- **Laboratory and Diagnostic Tests:** Depending on the clinical context, various tests may be performed to identify underlying causes, such as:
 - Blood tests (e.g., complete blood count, electrolytes, liver and kidney function tests).
 - Urinalysis to check for infections.
 - Imaging studies (e.g., CT or MRI) if there are neurological concerns.
 - Toxicology screening if substance use is suspected.
- **Exclusion of Other Conditions:** It is essential to rule out other potential causes of altered mental status, such as dementia, psychiatric disorders, or other neurological conditions.

2. Investigation

Performing a laboratory investigation for acute confusion (delirium) involves a systematic approach to identify potential underlying causes. Here are the key steps and tests typically included in the laboratory investigation:

- **Initial Assessment:**
 - **Clinical History:** Gather a detailed medical history, including recent illnesses, medication use, substance abuse, and any changes in the patient's environment or routine.
 - **Physical Examination:** Conduct a thorough physical examination to identify signs of infection, dehydration, or other medical issues.
- **Basic Laboratory Tests:**
 - **Complete Blood Count (CBC):** To check for signs of infection (e.g., elevated white blood cell count) or anemia.
 - **Electrolytes Panel:** To assess for electrolyte imbalances (e.g., sodium, potassium, calcium) that could contribute to confusion.
 - **Blood Glucose Level:** To rule out hypoglycemia or hyperglycemia as a cause of altered mental status.
 - **Liver Function Tests:** To evaluate liver function and detect any hepatic encephalopathy.
 - **Renal Function Tests:** To assess kidney function, including blood urea nitrogen (BUN) and creatinine levels, which can indicate renal failure.

- **Urinalysis:**
 - A urinalysis can help identify urinary tract infections (UTIs), which are common in elderly patients and can lead to confusion.
- **Toxicology Screening:**
 - If substance use is suspected, a toxicology screen may be performed to detect drugs, alcohol, or other toxic substances in the blood or urine.
- **Additional Tests:**
 - **Thyroid Function Tests:** To rule out thyroid disorders (e.g., hypothyroidism) that can cause confusion.
 - **Vitamin Levels:** Checking for deficiencies, particularly vitamin B1 (thiamine), which can lead to Wernicke's encephalopathy.
 - **Infection Markers:** If there are signs of infection, consider additional tests such as blood cultures or specific tests for infections (e.g., chest X-ray for pneumonia).
- **Imaging Studies:**
 - **CT or MRI of the Brain:** If there are neurological signs or symptoms (e.g., focal deficits, severe headache), imaging studies may be necessary to rule out structural causes such as stroke, hemorrhage, or tumors.
- **Specialized Tests:**
 - **Lumbar Puncture:** If there are signs of meningitis or encephalitis (e.g., fever, neck stiffness), a lumbar puncture may be performed to analyze cerebrospinal fluid (CSF).
- **Follow-Up:**
 - Based on initial findings, further investigations may be warranted to explore specific conditions or complications.
 - The goal of these laboratory investigations is to identify any reversible causes of acute confusion, allowing for appropriate treatment and management of the underlying condition.
 - The diagnosis of acute confusion is primarily clinical, relying on the recognition of characteristic symptoms and the identification of underlying causes to guide treatment and management.

VI. DIFFERENTIAL DIAGNOSIS

Differential diagnosis of acute confusion (delirium) involves distinguishing it from other conditions with similar symptoms. Key steps include:

1. **Clinical History and Examination:**
 - Gather a comprehensive history from the patient, family, or caregivers, covering recent medical events, medication changes, substance use, and life stressors.
 - Perform a thorough physical exam to identify signs of systemic illness or neurological deficits.
2. **Key Conditions to Differentiate:**
 - **Delirium:** Sudden onset with fluctuating symptoms, often from a medical condition.
 - **Dementia:** Gradual, progressive cognitive decline, without the rapid onset or fluctuation typical of delirium.
 - **Psychosis:** May include hallucinations or delusions but lacks delirium's acute fluctuation.
 - **Stroke:** Confusion with possible neurological symptoms; requires rapid imaging.
 - **Seizure Disorders:** Post-seizure confusion, clarified by patient history and EEG.

- **Metabolic Issues:** Disturbances like low blood sugar or high calcium that need lab testing.
 - **Infections:** CNS or systemic infections, particularly in elderly patients.
 - **Substance Intoxication/Withdrawal:** Confusion related to substances like alcohol or benzodiazepines.
 - **Medication Effects:** Common in older adults, especially with anticholinergic or sedative medications.
 - **Severe Depression/Anxiety:** May present with confusion but in a different clinical context.
3. **Diagnostic Testing:**
 - Imaging (CT/MRI) to rule out stroke or brain lesions.
 - Blood tests to detect metabolic imbalances, infections, or organ dysfunction.
 - Lumbar puncture if CNS infection is suspected.
 4. **Symptom Observation:**
 - In complex cases, consulting specialists (neurology, psychiatry) can help guide diagnosis and treatment.

Systematic evaluation of the patient's history, clinical presentation, and diagnostic results enables healthcare providers to distinguish delirium from other conditions and select the appropriate treatment.

VII. MANAGEMENT

1. Objectives

Management of acute confusion (delirium) focuses on two main objectives:

- **Identify and Treat the Cause:** Diagnose and address the underlying issue, such as infection, metabolic imbalance, or medication effects, which often resolves the confusion.
- **Supportive Care:** Provide a safe, calm environment, maintain hydration and nutrition, and use orientation aids. Non-pharmacological approaches are preferred, with limited use of medications for severe agitation only if absolutely necessary.

These strategies help effectively manage delirium and improve patient outcomes.

2. Critical sign

A critical sign of acute confusion (delirium) is **fluctuation in the level of consciousness**. This fluctuation can manifest as periods of confusion interspersed with moments of clarity, and it is a hallmark feature of delirium. Other critical signs include:

- **Inattention:** Difficulty maintaining focus on tasks or conversations.
- **Disorganized thinking:** Thoughts may be incoherent or illogical, leading to inappropriate or nonsensical responses.
- **Altered perception:** This can include hallucinations or illusions, where the patient may see or hear things that are not present.
- **Disorientation:** Patients may be confused about time, place, or identity, failing to recognize familiar surroundings or people.

These signs indicate a significant disturbance in cognitive function and warrant immediate medical evaluation and intervention.

3-Mortality risk factor

The mortality and risk factors associated with acute confusion (delirium) are significant and can impact patient outcomes. Here are the key points regarding mortality and risk factors:

- **Mortality:**
 - **Increased Mortality Risk:** Acute confusion is associated with a higher risk of mortality, particularly in older adults. Studies indicate that delirium can lead to a mortality rate that is significantly higher than that of patients without delirium. The mortality risk can be as high as 10-30% within a year following an episode of delirium, depending on the underlying causes and the patient's overall health status.
 - **Indicator of Poor Prognosis:** The presence of acute confusion is often considered a marker of poor prognosis, indicating underlying severe illness or complications that may lead to increased length of hospital stay, institutionalization, and long-term cognitive decline.
- **Risk Factors:**

Understanding these risk factors is crucial for prevention and early identification of acute confusion, which can help mitigate its impact on patient outcomes. Several factors increase the risk of acute confusion (delirium), especially in older adults:

 - **Age:** Those over 65 are at higher risk due to age-related brain changes and comorbidities.
 - **Pre-existing Cognitive Impairment:** Dementia or mild cognitive impairment makes delirium more likely.
 - **Medical Conditions:** Infections, metabolic imbalances, and organ failure can trigger confusion.
 - **Polypharmacy:** Multiple medications, especially sedatives or anticholinergics, raise the risk.
 - **Environmental Changes:** Hospitalization or sudden environment changes can precipitate delirium.
 - **Substance Use:** Alcohol use, withdrawal, or certain drugs can cause acute confusion.
 - **Surgery and Anesthesia:** Postoperative delirium is common, especially in older adults.

Recognizing these risk factors aids in prevention and early identification to improve outcomes.

4-Outpatient management

The process for managing acute confusion (delirium) in an outpatient setting involves several key steps to ensure proper evaluation, treatment, and follow-up.

- ***Initial Assessment:**
 - **Comprehensive Evaluation:** Conduct a thorough medical history and physical examination to identify potential causes of acute confusion. This includes reviewing medications, recent illnesses, and any changes in the patient's environment or routine.
 - **Cognitive Assessment:** Use standardized tools to assess cognitive function and determine the severity of confusion. This may include tests for attention, orientation, and memory.
- ***Identify and Address Underlying Causes:**
 - **Laboratory Tests:** Order necessary laboratory tests to identify metabolic disturbances, infections, or other medical conditions contributing to confusion (e.g., blood tests, urinalysis).

- **Medication Review:** Evaluate current medications for potential side effects or interactions that could be causing confusion. Adjust or discontinue medications as needed.
- **Management Plan:**
 - **Treat Underlying Conditions:** Initiate treatment for any identified causes, such as antibiotics for infections or electrolyte repletion for metabolic imbalances.
 - **Supportive Care:** Provide recommendations for supportive care, including hydration, nutrition, and a stable environment. Encourage family involvement to help orient the patient and provide reassurance.
- **Education and Counseling:**
 - **Patient and Family Education:** Educate the patient and their family about the nature of acute confusion, its potential causes, and the importance of follow-up care. Discuss strategies to manage confusion at home, such as maintaining a routine and minimizing environmental stressors.
- **Follow-Up Care:**
 - **Schedule Follow-Up Appointments:** Arrange for regular follow-up visits to monitor the patient's progress, reassess cognitive function, and adjust the management plan as necessary.
 - **Monitor for Recurrence:** Keep an eye out for any signs of recurrence or worsening of confusion, and ensure that caregivers are aware of what to look for.
- **Referral to Specialists:**
 - If confusion persists or if there are complex underlying issues, consider referring the patient to specialists such as neurologists, geriatricians, or psychiatrists for further evaluation and management.
- **Preventive Measures:**
 - **Implement Preventive Strategies:** For patients at high risk of developing acute confusion, implement preventive measures such as regular cognitive assessments, medication reviews, and education on managing stressors.
- By following this structured approach, healthcare providers can effectively manage acute confusion in an outpatient setting, ensuring that patients receive appropriate care and support while minimizing the risk of complications.

5-First-line treatment

When prescribing first-line treatment for a patient with acute confusion (delirium), it is essential to focus on addressing the underlying causes and providing supportive care.

Identify and Treat Underlying Causes:

- **Medical Management:** Before considering pharmacological interventions, ensure that any identifiable underlying causes of the acute confusion are addressed. This may include:
 - **Infections:** Prescribe appropriate antibiotics if an infection is diagnosed (e.g., urinary tract infection, pneumonia).
 - **Metabolic Imbalances:** Correct any electrolyte imbalances or dehydration through intravenous fluids or oral rehydration solutions.
 - **Medication Review:** Discontinue or adjust any medications that may contribute to confusion, particularly those with sedative or anticholinergic effects.

- **Supportive Care:**
- **Non-Pharmacological Interventions:** Implement non-pharmacological strategies to help manage symptoms, such as:
 - Ensuring a calm and familiar environment.
 - Providing orientation cues (e.g., clocks, calendars).
 - Encouraging family involvement for reassurance and support.
 - **Pharmacological Treatment** (if necessary):
- **Indications for Medication:** Medications may be considered if the patient is experiencing significant agitation, severe distress, or if non-pharmacological measures are insufficient.
- **First-Line Medications:** The following medications are commonly used as first-line treatments for managing acute confusion:
 - **Haloperidol:** A typical antipsychotic that can help manage agitation and psychotic symptoms.
 - **Dosage:** Start with a low dose (e.g., 0.5 to 1 mg orally or intramuscularly) and titrate based on response and tolerance. The maximum daily dose should not exceed 20 mg.
 - **Risperidone:** An atypical antipsychotic that may have a lower risk of extrapyramidal side effects compared to haloperidol.
 - **Dosage:** Start with 0.5 to 1 mg orally, with adjustments made based on clinical response.
- **Monitoring and Follow-Up:**
 - **Regular Monitoring:** Monitor the patient closely for efficacy and side effects of the prescribed medication. Adjust the dosage as needed based on the patient's response and any adverse effects.
 - **Reevaluation:** Schedule follow-up appointments to reassess the patient's cognitive status and overall health, and to determine if the medication is still necessary.
- **Patient and Family Education:**
 - **Inform the Patient and Family:** Educate the patient and their family about the treatment plan, including the purpose of medications, potential side effects, and the importance of adherence to follow-up appointments.
- **Consideration of Special Populations:**
 - **Geriatric Considerations:** In older adults, start with lower doses and titrate slowly due to increased sensitivity to medications and a higher risk of side effects.

6. Treatment adjustment

Adjusting treatment for acute confusion (delirium) involves ongoing assessment and modification of the management plan based on the patient's response to initial interventions and any changes in their clinical status. Here's a structured approach to adjusting treatment:

- **Regular Monitoring:**
 - **Clinical Assessment:** Continuously monitor the patient's cognitive function, level of consciousness, and overall behavior. Use standardized assessment tools to evaluate changes in confusion severity.
 - **Vital Signs and Laboratory Tests:** Regularly check vital signs and repeat laboratory tests as needed to monitor for any metabolic or infectious complications.

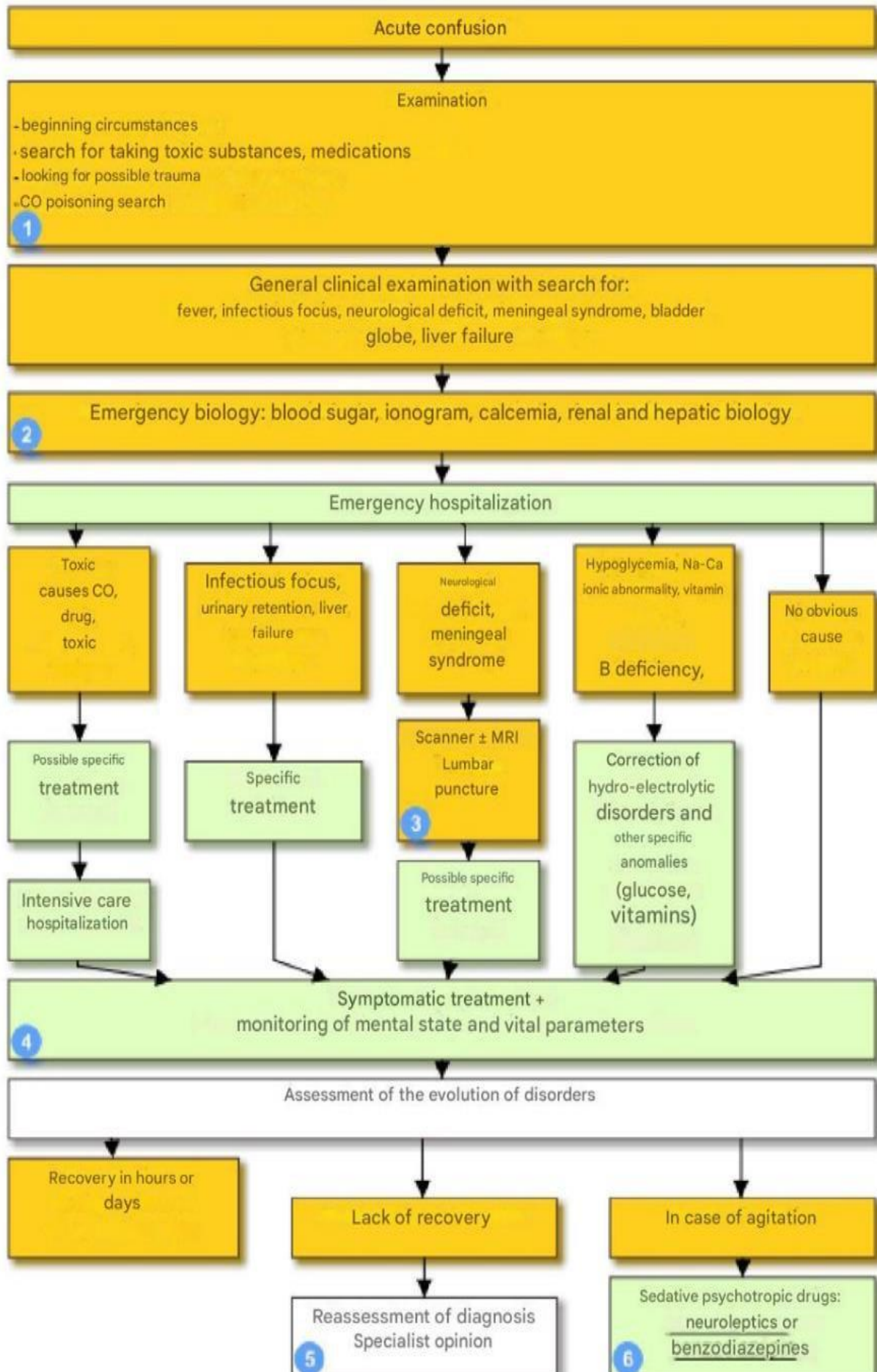
- **Evaluate Response to Treatment:**
 - **Efficacy of Medications:** Assess whether the initial pharmacological treatment (e.g., haloperidol, risperidone) is effectively managing symptoms of confusion and agitation. Look for improvements in attention, orientation, and overall cognitive function.
 - **Side Effects:** Monitor for any adverse effects from medications, such as sedation, extrapyramidal symptoms, or other complications. Adjust the treatment plan if side effects are significant.
- **Adjusting Pharmacological Treatment:**
 - **Titration of Dosage:** If the patient shows inadequate response to the initial medication, consider increasing the dosage gradually while monitoring for side effects. For example:
 - **Haloperidol:** If the initial dose is insufficient, it can be increased by 0.5 to 1 mg every 4 to 6 hours until the desired effect is achieved, without exceeding the maximum daily limit.
 - **Risperidone:** If the initial dose is not effective, it can be increased by 0.5 to 1 mg, with careful monitoring for side effects.
 - **Switching Medications:** If a patient does not respond to one medication or experiences intolerable side effects, consider switching to another agent (e.g., from haloperidol to risperidone or vice versa).
- **Addressing Underlying Causes:**
 - **Reassess Underlying Conditions:** If the patient's confusion persists despite treatment, reevaluate for any underlying causes that may not have been addressed, such as infections, metabolic imbalances, or medication effects.
 - **Modify Supportive Care:** Enhance non-pharmacological interventions based on the patient's needs, such as increasing family involvement, improving environmental cues, or providing additional orientation support.
- **Involve Specialists:**
 - **Consultation:** If confusion persists or worsens, consider consulting specialists such as neurologists, geriatricians, or psychiatrists for further evaluation and management recommendations.
- **Patient and Family Communication:**
 - **Discuss Changes:** Communicate any changes in treatment to the patient and their family, explaining the rationale behind adjustments and the expected outcomes.
 - **Education on Signs of Deterioration:** Educate the family on signs that may indicate worsening confusion or complications, encouraging them to report these promptly.
- **Follow-Up Care:**
 - **Schedule Regular Follow-Ups:** Ensure that follow-up appointments are scheduled to reassess the patient's condition and make further adjustments as necessary.
 - By following this structured approach to adjusting treatment for acute confusion, healthcare providers can optimize patient care, enhance recovery, and minimize the risk of complications .

7.Algorithm

When prescribing first-line treatment for a patient with acute confusion (delirium), it is essential to focus on addressing the underlying causes and providing supportive care.:

- **Identify and Treat Underlying Causes:**
- **Medical Management:** Before considering pharmacological interventions, ensure that any identifiable underlying causes of the acute confusion are addressed. This may include:
 - **Infections:** Prescribe appropriate antibiotics if an infection is diagnosed (e.g., urinary tract infection, pneumonia).
 - **Metabolic Imbalances:** Correct any electrolyte imbalances or dehydration through intravenous fluids or oral rehydration solutions.
 - **Medication Review:** Discontinue or adjust any medications that may contribute to confusion, particularly those with sedative or anticholinergic effects.
 - Supportive Care:
- **Non-Pharmacological Interventions:** Implement non-pharmacological strategies to help manage symptoms, such as:
 - Ensuring a calm and familiar environment.
 - Providing orientation cues (e.g., clocks, calendars).
 - Encouraging family involvement for reassurance and support.
 - Pharmacological Treatment (if necessary):
- **Indications for Medication:** Medications may be considered if the patient is experiencing significant agitation, severe distress, or if non-pharmacological measures are insufficient.
- **First-Line Medications:** The following medications are commonly used as first-line treatments for managing acute confusion:
 - **Haloperidol:** A typical antipsychotic that can help manage agitation and psychotic symptoms.
 - **Dosage:** Start with a low dose (e.g., 0.5 to 1 mg orally or intramuscularly) and titrate based on response and tolerance. The maximum daily dose should not exceed 20 mg.
 - **Risperidone:** An atypical antipsychotic that may have a lower risk of extrapyramidal side effects compared to haloperidol.
 - **Dosage:** Start with 0.5 to 1 mg orally, with adjustments made based on clinical response.
- **Monitoring and Follow-Up:**
 - **Regular Monitoring:** Monitor the patient closely for efficacy and side effects of the prescribed medication. Adjust the dosage as needed based on the patient's response and any adverse effects.
 - **Reevaluation:** Schedule follow-up appointments to reassess the patient's cognitive status and overall health, and to determine if the medication is still necessary.
- **Patient and Family Education:**
 - **Inform the Patient and Family:** Educate the patient and their family about the treatment plan, including the purpose of medications, potential side effects, and the importance of adherence to follow-up appointments.
- **Consideration of Special Populations:**
 - **Geriatric Considerations:** In older adults, start with lower doses and titrate slowly due to increased sensitivity to medications and a higher risk of side effects.

Acute confusion



VIII. COMPLICATIONS:

Acute confusion (delirium) can lead to several complications, which can significantly impact patient outcomes and complications include:

- **Prolonged Hospital Stay:** Patients with acute confusion often experience longer hospitalizations due to the need for additional monitoring, evaluation, and treatment of underlying causes.
- **Increased Mortality:** Acute confusion is associated with a higher risk of mortality, particularly in older adults and those with significant comorbidities. It can be a marker of severe underlying illness and poor prognosis.
- **Functional Decline:** Patients may experience a decline in functional abilities, including difficulties with activities of daily living (ADLs). This decline can persist even after the resolution of acute confusion, leading to long-term dependency.
- **Cognitive Impairment:** There is a risk of developing persistent cognitive deficits following an episode of acute confusion, particularly in older adults. This can progress to more severe forms of cognitive impairment, such as dementia.
- **Increased Risk of Institutionalization:** Patients who experience acute confusion may be more likely to require long-term care or institutionalization after discharge due to the loss of independence and functional abilities.
- **Psychological Effects:** Acute confusion can lead to psychological distress for both patients and their families, including anxiety, depression, and fear related to the sudden changes in mental status.
- **Physical Complications:** The condition can lead to physical complications such as dehydration, malnutrition, and pressure ulcers due to prolonged immobility and altered mental status.

Recognizing and managing acute confusion promptly is crucial to minimize these complications and improve overall patient outcomes.

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PARKINSON DISEASE

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I. DEFINITION

Parkinson's disease (PD) is defined as a progressive neurodegenerative disorder characterized by the degeneration of dopaminergic neurons in the substantia nigra, a part of the brain that plays a crucial role in movement control. The condition is primarily associated with the presence of four cardinal features of parkinsonism:

- **Bradykinesia** - Slowness of movement.
- **Tremor-at-rest** - A resting tremor that typically occurs when the muscles are relaxed.
- **Rigidity** - Increased muscle tone leading to stiffness and resistance to movement.
- **Loss of postural reflexes** - Impairment in the ability to maintain balance and posture.

These features collectively contribute to the motor symptoms of PD, which can significantly impact daily functioning and quality of life. In addition to motor symptoms, PD may also involve non-motor symptoms, including cognitive changes, mood disorders, and autonomic dysfunction.

II. EPIDEMIOLOGY

The epidemiology of Parkinson's disease (PD) encompasses various aspects, including its prevalence, incidence, risk factors, and demographic characteristics. Key points regarding the epidemiology of PD include:

- **Prevalence:** PD affects about 1% of individuals over 60 years and 4% over 80 years, with regional and population variations.
- **Incidence:** Rises with age, from 10–20 cases per 100,000 person-years in individuals under 50 to 100–200 cases per 100,000 in those over 80.
- **Gender Differences:** PD is more common in men (1.5–2:1 male-to-female ratio), potentially due to genetic, hormonal, or environmental factors.
- **Geographic Variation:** Higher rates are reported in North America and Europe compared to Asia.

III. ETIOLOGY

The causes of Parkinson's disease (PD) are multifactorial and can be broadly categorized into genetic, environmental, and age-related factors. Here are the key contributors:

1. Genetic Factors

- **Familial PD:** A small percentage of PD cases (about 10-15%) are familial, meaning they have a genetic basis. Mutations in specific genes have been linked to these cases, including:
 - **SNCA:** This gene encodes alpha-synuclein, a protein that aggregates to form Lewy bodies in PD.
 - **LRRK2:** Mutations in this gene are the most common genetic cause of familial and sporadic PD.
 - **PARK7 (DJ-1), PINK1, and PRKN (parkin):** These genes are associated with autosomal recessive forms of PD.
- **Risk Genes:** Certain genetic variants may increase the risk of developing PD without directly causing it.

2. Environmental Factors

- **Toxin Exposure:** Long-term exposure to environmental toxins, such as pesticides (e.g., paraquat, rotenone) and heavy metals, has been associated with an increased risk of PD.
 - **Occupational Hazards:** Certain occupations that involve exposure to chemicals or solvents may elevate the risk of developing PD.
 - **Head Trauma:** There is evidence suggesting that traumatic brain injury may increase the risk of PD, particularly in individuals with a history of repeated head injuries.
3. **Age**
- Age is the most significant risk factor for PD. The incidence and prevalence of the disease increase with advancing age, with most cases occurring in individuals over 60 years old. The aging process itself may contribute to neurodegenerative changes in the brain.
4. **Sex**
- PD is more common in men than in women, with a male-to-female ratio of approximately 1.5 to 2:1. Hormonal differences may play a role in this disparity.
5. **Other Factors**
- **Lifestyle Factors:** Some studies suggest that lifestyle factors, such as caffeine consumption and physical activity, may have a protective effect against PD.
 - **Comorbidities:** Certain medical conditions, such as diabetes and hypertension, have been associated with an increased risk of PD.

IV. PATHOPHYSIOLOGY

The physiopathology of Parkinson's disease (PD) involves a complex interplay of neurodegenerative processes primarily affecting the basal ganglia, particularly the substantia nigra. Key aspects of the physiopathology of PD include:

- **Neuronal Degeneration:** Progressive loss of dopaminergic neurons in the substantia nigra pars compacta reduces dopamine, impairing movement and coordination.
- **Dopamine Deficiency:** Dopamine depletion in the striatum causes motor symptoms like bradykinesia, rigidity, and tremor.
- **Lewy Bodies:** Abnormal protein aggregates (alpha-synuclein) disrupt cellular processes, contributing to neuronal dysfunction and death.
- **Neuroinflammation:** Activated microglia and inflammatory cytokines exacerbate neurodegeneration.
- **Oxidative Stress:** Imbalance between reactive oxygen species (ROS) and detoxification mechanisms leads to mitochondrial dysfunction and neuronal damage.
- **Genetic Factors:** Mutations in genes like **SNCA**, **LRRK2**, **PINK1**, and **PRKN** influence neurodegenerative processes, particularly in familial PD.
- **Non-Motor Symptoms:** Cognitive decline, mood disorders, and autonomic dysfunction stem from involvement of other neurotransmitter systems (e.g., serotonin, norepinephrine).

V. DIAGNOSIS

1. Clinic

1. Clinical History

- **Symptom Onset and Progression:** Assess the timeline of symptoms, including the onset of motor symptoms (e.g., tremor, rigidity, bradykinesia) and non-motor symptoms (e.g., cognitive changes, mood disorders).
- **Family History:** Inquire about any family history of parkinsonian disorders or related neurodegenerative diseases.

2. Clinical Examination

- **Motor Symptoms:** Evaluate the presence of cardinal features of PD, including:
 - **Bradykinesia:** Slowness of movement.
 - **Resting Tremor:** Typically a "pill-rolling" tremor.
 - **Rigidity:** Increased muscle tone, often described as "cogwheel" rigidity.
 - **Postural Instability:** Difficulty maintaining balance.
- **Non-Motor Symptoms:** Assess for non-motor features such as sleep disturbances, autonomic dysfunction, and cognitive changes.
- **Clinical Diagnostic Criteria:** are commonly used Queen Square Brain Bank (QSBB) and MDS-PD criteria (detail in the below table)

Table 2 Comparison of the Queen Square Brain Bank (QSBB) and the International Parkinson and Movement Disorder Society Clinical Diagnostic (MDS-PD) Criteria.

Criteria	QSBB criteria	MDS-PD criteria
Step 1: Identifying the presence of parkinsonism	Bradykinesia and at least one of the following: (1) Muscular rigidity (2) 4–6 Hz rest tremor (3) Postural instability not cause by primary visual, vestibular, cerebellar or proprioceptive dysfunction	Parkinsonism is defined as bradykinesia, in combination with at least one of rest tremor or rigidity. Diagnosis of clinically established PD requires: (1) Absence of absolute exclusion criteria (2) At least two supportive criteria, and (3) No red flags Diagnosis of clinically probable PD requires: (1) Absence of absolute exclusion criteria (2) Presence of red flags counterbalanced by supportive criteria, with no more than 2 red flags allowed in this category
Step 2: Supportive criteria	(1) Unilateral onset (2) Rest tremor present (3) Progressive disorder (4) Persistent asymmetry affecting the side of onset most (5) Excellent response (70%–100%) to levodopa (6) Severe levodopa-induced chorea (7) Levodopa response for 5 years or more (8) Clinical course of 10 years or more	(1) Clear and dramatic beneficial response to dopaminergic therapy (Marked improvement with dose increases or marked worsening with dose decreases or unequivocal and marked on/off fluctuations) (2) Presence of levodopa-induced dyskinesia (3) Rest tremor of a limb, documented on clinical examination (4) The presence of either olfactory loss or cardiac sympathetic denervation on MIBG scintigraphy
Step 3: Exclusion criteria	(1) History of repeated strokes with stepwise progression (2) History of repeated head injury (3) History of definite encephalitis (4) Oculogyric crises (5) Neuroleptic treatment at onset of symptoms (6) More than one affected relative (7) Sustained remission (8) Strictly unilateral features after 3 years (9) Supranuclear gaze palsy (10) Cerebellar signs (11) Early severe autonomic involvement (12) Early severe dementia with disturbances of memory, language and praxis (13) Babinski's sign (14) Presence of a cerebral tumor or communicating hydrocephalus on CT scan (15) Negative response to large doses of levodopa (16) MPTP exposure	Absolute exclusion criteria (1) Unequivocal cerebellar abnormalities, such as cerebellar gait, limb ataxia, or cerebellar oculomotor abnormalities (2) Downward vertical supranuclear gaze palsy, or selective slowing of downward vertical saccades (3) Diagnosis of probable behavioral variant frontotemporal dementia or primary progressive aphasia (4) Parkinsonian features restricted to the lower limbs for more than 3 years (5) Treatment of dopamine receptor blocker or a dopamine-depleting agent in a dose and time-course consistent with drug-induced parkinsonism (6) Absence of observable response to high-dose levodopa despite at least moderate severity of disease (7) Unequivocal cortical sensory loss, clear limb ideomotor apraxia, or progressive aphasia (8) Normal functional neuroimaging of the presynaptic dopaminergic system (9) Documentation of an alternative condition known to produce parkinsonism and plausibly connected to the patient's symptoms, or, the expert evaluating physician
Step 4: Red flags	Not available	(1) Rapid progression of gait impairment regular use of wheelchair within 5 years (2) A complete absence of progression of motor symptoms or signs over 5 or more years unless stability is related to treatment. (3) Early bulbar dysfunction within first 5 years (4) Inspiratory respiratory dysfunction: inspiratory stridor or inspiratory sighs (5) Severe autonomic failure in the first 5 years (Orthostatic hypotension or severe urinary retention or urinary incontinence) (6) Recurrent (>1/year) falls (7) Disproportionate antecollis or contractures of hand or feet within the first 10 years (8) Absence of any of the common nonmotor features despite 5 years disease duration (9) Otherwise unexplained pyramidal tract signs (10) Bilateral symmetric parkinsonism

3. Ancillary Tests in Parkinson's Disease (PD)

Diagnosis of PD is clinical, with ancillary tests reserved for atypical cases or to rule out other conditions:

- **MRI:** Excludes structural abnormalities; identifies features of atypical parkinsonism (e.g., PSP, MSA).
- **DaTscan:** Detects dopamine deficiency; differentiates PD from essential tremor but not from other parkinsonian syndromes.
- **PET Imaging:** Assesses dopaminergic neuron integrity; useful for distinguishing atypical syndromes but not routinely used.
- **Olfactory Testing:** Supports PD diagnosis; absent in many atypical syndromes.
- **Autonomic Tests:** MIBG scintigraphy detects cardiac denervation, aiding differentiation from atypical parkinsonism.
- **Genetic Testing:** For early-onset or familial cases to identify mutations like **LRRK2** or **SNCA**.
- **Emerging Biomarkers:** Alpha-synuclein assays and advanced imaging show promise but need further validation.

VI. DIFFERENTIAL DIAGNOSIS

Parkinsonism can result from conditions other than idiopathic PD, including neurodegenerative disorders, secondary causes, and functional movement disorders:

- **Neurodegenerative Parkinsonism**
 - **Essential Tremor (ET):** Symmetric action tremor often mistaken for PD; lacks bradykinesia and rigidity but may coexist with PD features.
 - **Dementia with Lewy Bodies (DLB):** Early dementia with hallucinations and fluctuating cognition; parkinsonism occurs alongside or after cognitive symptoms.
 - **Multiple System Atrophy (MSA):** Parkinsonism with dysautonomia, cerebellar signs, and poor levodopa response.
 - **Progressive Supranuclear Palsy (PSP):** Parkinsonism with early falls, vertical gaze palsy, and cognitive changes.
 - **Corticobasal Degeneration (CBD):** Asymmetric rigidity, dystonia, and cognitive dysfunction.
- **Secondary Parkinsonism**
 - **Drug-Induced Parkinsonism:** Caused by antipsychotics or other dopamine-blocking agents; may resolve after withdrawal.
 - **Vascular Parkinsonism:** Lower-body bradykinesia without tremor; associated with cerebrovascular disease.
 - **Toxic Exposure:** Parkinsonism linked to substances like manganese, carbon monoxide, or MPTP.
 - **Trauma or Structural Lesions:** Chronic traumatic encephalopathy, hydrocephalus, or tumors.
- **Functional Parkinsonism**

VII. MANAGEMENT

1. Objectives

- Symptom Control

- Maintenance of Functional Independence
- Minimization of Complication
- Improvement of Quality of Life
- Patient-Centered Care

2. Critical Signs

Identifying critical signs is essential for diagnosis and management. Key signs include:

- Bradykinesia (slowness of movement)
- Resting tremor
- Rigidity (muscle stiffness)
- Postural instability (balance issues)
- Non-motor symptoms such as depression, anxiety, sleep disturbances, and cognitive changes.

3. Mortality Risk Factors

Understanding mortality risk factors in PD is crucial for management. These may include:

- Advanced age
- Presence of comorbidities (e.g., cardiovascular disease, diabetes)
- Severity of motor symptoms
- Cognitive impairment
- History of falls and related complications (e.g., fractures, head injuries).

4. Outpatient Management

Outpatient management involves regular follow-up and monitoring of the patient's condition. Key components include:

- Regular assessments of motor and non-motor symptoms.
- Education and support for patients and caregivers.
- Physical therapy and occupational therapy to improve mobility and daily functioning.
- Nutritional counseling to address dietary needs and manage weight.

5. First-Line Treatment

First-line treatment typically involves pharmacological management. The most common first-line medications include:

- **Levodopa:** Often combined with carbidopa to enhance efficacy and reduce side effects.
- **Dopamine agonists:** Such as pramipexole or ropinirole, which can be used as initial therapy, especially in younger patients.
- **MAO-B inhibitors:** Such as selegiline or rasagiline, which may provide symptomatic relief and have neuroprotective effects. (Detail in Table 2,3)

6. Treatment Adjustment

Treatment adjustments are necessary based on the patient's response and progression of the disease (detail in table 2,3). Considerations include:

- Monitoring for side effects and adjusting dosages accordingly.
- Switching medications if the patient experiences inadequate symptom control or intolerable side effects.
- Considering the addition of other medications (e.g., COMT inhibitors) to enhance the effect of levodopa.
- Regularly reassessing the treatment plan based on changes in symptoms or new health issues.

7. Algorithm

Table 2: Approach to Diagnosing Parkinson Disease

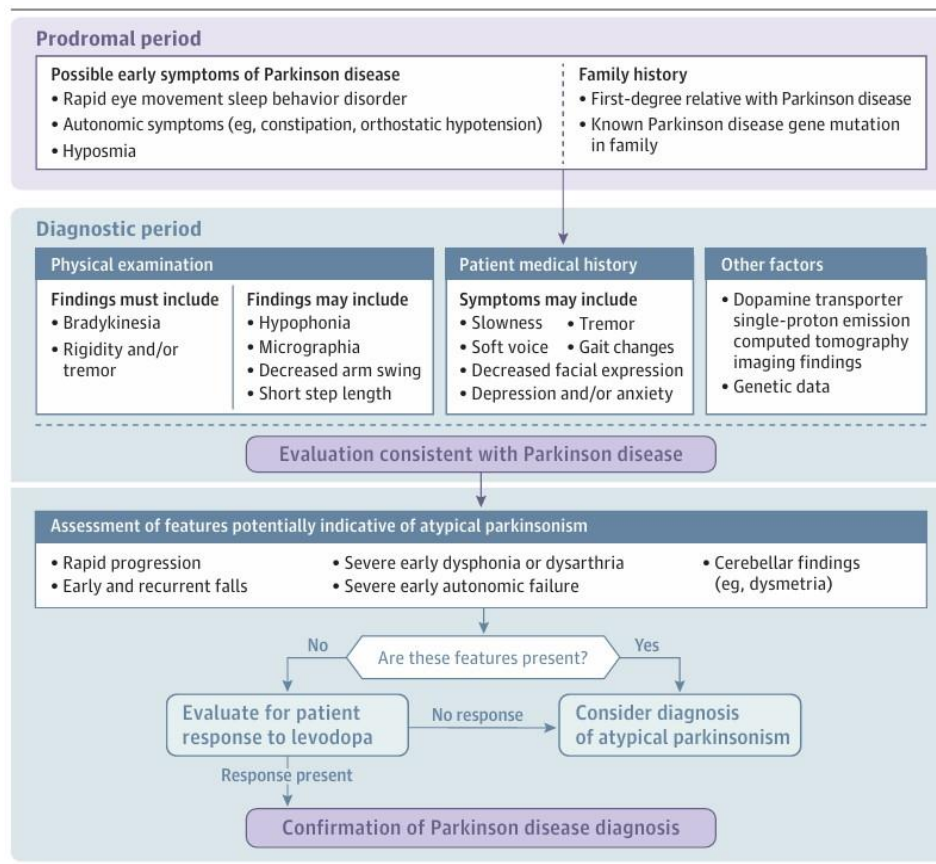


Table 3: Algorithm for approach to treating Parkinson disease

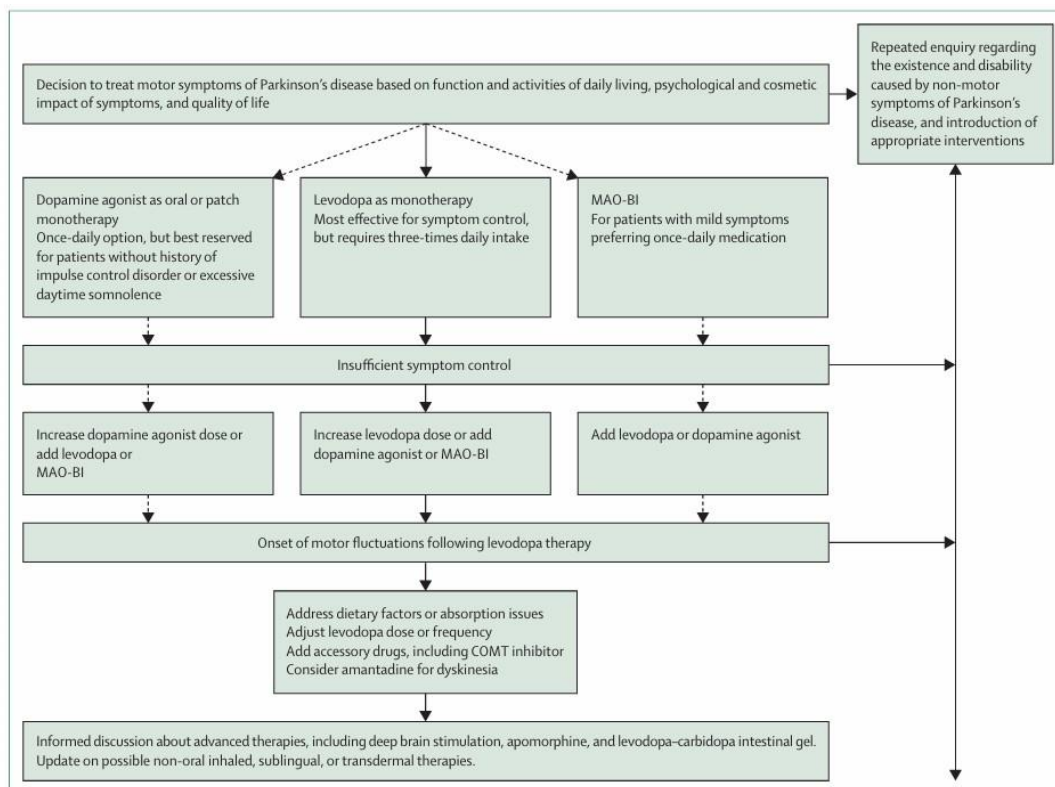


Table 4: Pharmacologic Agents Used for Motor Symptoms in Parkinson Disease

Category	Specific Agents and Typical Starting Dose	Therapeutic Uses				Most Common Adverse Effects Other Than Dyskinesia
		Early Symptomatic	Levodopa Adjunct	Wearing Off	Dyskinesia	
Levodopa preparations	Immediate-release carbidopa-levodopa (25/100 mg, 3 times/d)	●	○	●	●	Nausea
	Controlled-release carbidopa-levodopa (25/100 mg, 3 times/d)	●	○	●	●	Nausea
	Extended-release carbidopa-levodopa (23.75/95 mg, 3 times/d for 3 d; then 36.25/145 mg, 3 times/d for 3 d)	●	○	●	●	Nausea
	Enteral suspension carbidopa-levodopa (clinical titration)		○	●	●	Nausea
	Inhaled levodopa (as needed)		○	● ^b		Nausea, upper respiratory tract infection
Nonergot dopamine agonists ^c	Immediate-release pramipexole (0.125 mg, 3 times/d, increasing weekly) or extended-release pramipexole (0.375 mg, 1 time/d, increasing weekly)	●	●	●	●	Orthostatic hypotension, dizziness, nausea, sleepiness
	Immediate-release ropinirole (0.25 mg, 3 times/d, increasing weekly) or extended-release ropinirole (2 mg, 1 time/d, increasing weekly)	●	●	●	●	Orthostatic hypotension, dizziness, nausea, sleepiness
	Transdermal rotigotine (2 mg/24 h)	●	●	●	●	Site reactions, dizziness, orthostatic hypotension
	Injected apomorphine (as needed)			●		Site reactions, dizziness, orthostatic hypotension
Monoamine oxidase-B inhibitors	Selegiline (5 mg, 2 times/d)	●	●	●	●	Nausea, dizziness, insomnia
	Rasagiline (1 mg every morning)	●	●	●	●	Orthostatic hypotension, nausea
	Safinamide (50 mg/d)	●	●	●	●	Nausea
	Zonisamide (25 to 200 mg/d) ^d			●		Sleepiness, loss of appetite
Catechol-O-methyltransferase inhibitors	Entacapone (200 mg with each levodopa dose)			●	●	Nausea, diarrhea
	Opicapone (50 mg every night) ^e			●	●	Falls, insomnia, orthostatic hypotension
	Tolcapone (100 mg, 3 times/d) ^f			●	●	Gastrointestinal symptoms, orthostatic hypotension, sleep disorders
Other	Anticholinergics (eg, trihexyphenidyl, benztropine; dose varies) ^g	●	●			Dizziness, anxiety
	Amantadine (dose varies by formulation) ^h	●	●		●	Orthostatic hypotension, hallucinations, edema, gastrointestinal symptoms
	Istradefylline (20 mg/d)			●		Nausea, hallucinations
	Clozapine (12.5-25 mg every night) ^f				●	Sleepiness, dizziness, tachycardia, constipation, orthostatic hypotension, sialorrhea

● "Clinically useful" or "possibly useful"ⁱ ● Used in clinical practice outside of evidence base ● Dose reduction or adjustment may reduce dyskinesia ○ Not relevant

VIII. SPECIAL CASE

Some Examples of Special Cases in Parkinson's Disease (PD):

- **Young-Onset PD (YOPD)**
 - Diagnosed before age 50; often linked to genetic mutations like **PARK2**. Symptoms respond well to treatment but complications (e.g., dyskinesia) appear earlier.
- **Atypical Parkinsonism – PSP**
 - Early falls, vertical gaze palsy, and poor response to levodopa suggest Progressive Supranuclear Palsy (PSP), a faster-progressing condition.
- **Drug-Induced Parkinsonism**

- Caused by medications like antipsychotics; resolves after stopping the drug.
- **Parkinsonism-Dementia Complex**
 - Cognitive decline and hallucinations (Parkinson's disease dementia or Lewy body dementia) complicate management.
- **Normal Pressure Hydrocephalus (NPH)**
 - Mimics PD with gait problems, incontinence, and cognitive decline; treatable with shunt placement.
- **Familial PD**
 - Genetic forms linked to mutations like **LRRK2**; may require genetic counseling.
- **Autonomic Dysfunction in PD**
 - Severe symptoms like low blood pressure, constipation, and bladder issues complicate care.
- **Vascular Parkinsonism**
 - Caused by strokes; presents with leg-dominant symptoms and no resting tremor.
- **SWEDD (Scans Without Dopaminergic Deficit)**
 - Tremor without dopamine deficiency; often not true PD and does not progress like PD.

IX. VOLUTION AND PROGNOSIS

- **Progression of Symptoms**
 - **Motor Symptoms:** Initially asymmetric, progressing to bilateral involvement with increasing bradykinesia, rigidity, tremor, and postural instability.
 - **Non-Motor Symptoms:** Cognitive decline, mood disorders, sleep disturbances, and autonomic dysfunction often worsen over time.
- **Stages of PD**
 - Early stage: Mild motor symptoms, good response to dopaminergic therapy.
 - Moderate stage: Motor fluctuations, dyskinesias, and increasing non-motor symptoms.
 - Advanced stage: Significant disability, therapy-resistant symptoms, and potential dementia.
- **Prognosis**
 - PD is a progressive disease.
 - The rate of progression varies widely among individuals.
 - Non-motor symptoms, such as dementia, are key contributors to reduced quality of life and increased mortality.
 - Average survival after diagnosis ranges from 10 to 20 years, depending on age, overall health, and disease progression.
 - The **Movement Disorder Society-Unified Parkinson's Disease Rating Scale (MDS-UPDRS)** is widely used to assess the progression of Parkinson's disease (PD)
- **Impact of Treatment**
 - Medications like levodopa improve symptoms and functional ability, particularly in early stages.
 - Advanced therapies (e.g., deep brain stimulation) can help manage complications in later stages.
 - Multidisciplinary care improves outcomes and delays disability.

X. COMPLICATIONS

Parkinson's disease (PD) leads to a range of motor and non-motor complications that significantly affect quality of life. Key complications include:

- **Motor Complications**
 - Worsening **bradykinesia, rigidity, tremors, postural instability, freezing of gait (FOG), and dystonia.**
- **Falls and Injuries**
 - Increased risk of falls leading to fractures (e.g., hip) and head injuries.
- **Cognitive and Psychiatric Issues**
 - Cognitive decline or **dementia**, depression, anxiety, and psychosis (e.g., hallucinations, delusions).
- **Autonomic Dysfunction**
 - **Orthostatic hypotension**, constipation, dysphagia, and urinary incontinence.
- **Sleep Disorders**
 - **Insomnia**, REM sleep behavior disorder, and daytime sleepiness.
- **Pain and Discomfort**
 - Musculoskeletal pain due to rigidity, immobility, or dystonia.
- **Skin Problems**
 - Issues like sweating (hyperhidrosis) and seborrheic dermatitis.
- **Medication Complications**
 - Long-term dopaminergic therapy may cause **motor fluctuations** and **dyskinesias**.
- **Social and Economic Impact**
 - Social isolation, loss of independence, and financial burdens affect patients and caregivers.

XI. PREVENTION AND RECOMMENDATION

While preventing PD is challenging due to its unclear causes, certain strategies may reduce risk or improve outcomes:

Prevention Strategies:

- **Physical Activity:** Regular exercise (e.g., walking, swimming) may lower risk and improve mobility.
- **Healthy Diet:** A Mediterranean diet rich in antioxidants supports brain health.
- **Caffeine:** Moderate coffee or tea consumption may reduce PD risk (needs more research).
- **Avoid Toxins:** Minimize exposure to pesticides and heavy metals.
- **Healthy Weight:** Prevent obesity to reduce PD-related risks.
- **Mental Engagement:** Activities like reading and socializing promote cognitive health.
- **Stress Management:** Techniques like yoga and meditation protect brain health.
- **Regular Check-ups:** Manage risk factors like hypertension and diabetes.
- **Genetic Counseling:** Beneficial for those with a family history of PD.
- **Awareness:** Early recognition of symptoms allows timely intervention.

Recommendations for PD Patients:

- Optimize medication with healthcare providers.
- Engage in physical, occupational, and speech therapy.
- Join support groups for emotional and practical help.

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ALZHEIMER DISEASE (DEMENCIA SYNDROME)

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I. DEFINITION

Alzheimer's Disease (AD) is defined as a progressive neurodegenerative disorder characterized by the gradual deterioration of cognitive functions, including memory, reasoning, and the ability to perform daily activities. It is the most common cause of dementia, primarily affecting older adults. The disease is marked by specific pathological features in the brain, including the accumulation of amyloid plaques (deposits of beta- amyloid peptide) and neurofibrillary tangles (aggregates of hyperphosphorylated tau protein). These changes lead to neuronal loss and synaptic dysfunction, ultimately resulting in significant impairment in cognitive and functional abilities.

II. EPIDEMIOLOGY

Understanding the epidemiology of Alzheimer's Disease (AD) involves analyzing factors such as prevalence, incidence, age correlation, demographic influences, risk factors, and its impact on healthcare systems. Prevalence measures the total cases in a population, with 4.4% of Europeans over 65 affected and 850,000 cases in France, while incidence tracks new diagnoses, like the 160,000 annual cases in France. Age is a significant risk factor, with prevalence rising from 1.5% in those aged 60-70 to over 15% in those 80 and older. Gender, ethnicity, and socioeconomic status also affect risk, with women generally at higher risk. Modifiable (e.g., lifestyle, cardiovascular health) and non-modifiable (e.g., genetics) factors further shape the epidemiology. The rising prevalence presents challenges for healthcare systems, necessitating enhanced long-term care and support. Research methodologies like cohort and case-control studies help refine understanding and guide interventions.

III. ETIOLOGY

These causes is crucial for developing preventive strategies and therapeutic interventions for Alzheimer's Disease and dementia syndromes.

The causes of AD and dementia syndromes are multifactorial, encompassing genetic, pathological, and environmental factors. Genetic mutations, such as those in the APP, PSEN1, and PSEN2 genes, are linked to familial forms of early-onset AD, while the APOE ε4 allele significantly increases the risk of late-onset AD. Pathological hallmarks include amyloid plaques (beta-amyloid peptide accumulation) and neurofibrillary tangles(hyperphosphorylated tau protein), which disrupt neuronal function and lead to cell death. Neurotransmitter imbalances, particularly reduced acetylcholine levels, impair memory and learning. Chronic neuroinflammation, driven by microglial activation, exacerbates neuronal damage. Vascular factors, such as hypertension, diabetes, and high cholesterol, contribute to cognitive impairment and may coexist with AD. Environmental and lifestyle influences, including poor diet, physical inactivity, and lack of mental engagement, increase susceptibility. Age, the most significant risk factor, sharply raises the likelihood of AD as individuals grow older. Other medical conditions, such as depression, sleep disorders, and cardiovascular diseases, further elevate risk. Understanding these interconnected causes is vital for advancing prevention and treatment strategies for AD and dementia syndromes.

IV. PATHOPHYSIOLOGY

Understanding the pathophysiology of AD and dementia syndromes is essential for developing targeted therapeutic strategies and interventions aimed at slowing disease progression and improving patient outcomes.

The pathophysiology of AD and dementia syndromes involves **complex biological processes** leading to **neuronal degeneration** and **cognitive decline**. Key features include the accumulation of **amyloid plaques** formed by **beta-amyloid peptide aggregation**, disrupting cell communication and triggering **inflammation**, and **neurofibrillary tangles** of **hyperphosphorylated tau protein**, which **destabilize microtubules**, **impair intracellular transport**, and cause **cell death**.

Cholinergic deficits, marked by the **loss of acetylcholine-producing neurons** in memory-related brain regions, further contribute to **cognitive decline**.

Chronic neuroinflammation, driven by microglial activation and pro-inflammatory cytokines, exacerbates neuronal damage, while **oxidative stress** from **reactive oxygen species amplifies neuronal injury**.

Vascular factors, such as impaired cerebral blood flow from conditions like hypertension, also play a role in cognitive impairment.

Genetic predispositions, including the **APOE ε4 allele**, interact with environmental factors, influencing amyloid processing, **tau phosphorylation**, and **inflammation pathways**. Understanding these mechanisms is critical for developing targeted therapies to slow disease progression and improve patient outcomes.

V. DIAGNOSIS

1. Clinic

The clinical diagnosis of AD and dementia syndromes involves a comprehensive assessment that includes clinical evaluation, cognitive testing, and the exclusion of other potential causes of cognitive impairment. Here are the key components of the clinical diagnosis:

- **Clinical History:**
 - A thorough medical history is taken, including information about the patient's cognitive decline, behavioral changes, and functional abilities. It is important to gather information from family members or caregivers to understand the progression of symptoms and their impact on daily life.
- **Cognitive Assessment:**
 - Cognitive function is evaluated using standardized tests. Commonly used tools include:
 - **Mini-Mental State Examination (MMSE):** A brief screening tool that assesses various cognitive domains, including orientation, memory, attention, and language.
 - **Montreal Cognitive Assessment (MoCA):** A more comprehensive assessment that evaluates multiple cognitive domains and is sensitive to mild cognitive impairment.
 - A score below a certain threshold (e.g., MMSE < 24) may indicate cognitive impairment.
- **Functional Assessment:**
 - The impact of cognitive decline on daily functioning is assessed. This may involve evaluating the patient's ability to perform activities of daily living (ADLs) and instrumental activities of daily living (IADLs).
- **Neurological Examination:**

- A thorough neurological examination is conducted to assess motor function, reflexes, coordination, and sensory function. This helps rule out other neurological conditions that may cause similar symptoms.
- **Diagnostic Criteria:**
 - The diagnosis of AD is typically based on established criteria, such as those outlined in the DSM-5 (Diagnostic and Statistical Manual of Mental Disorders) or the NIA-AA (National Institute on Aging and the Alzheimer's Association) criteria. These criteria emphasize the presence of cognitive impairment, functional decline, and the exclusion of other causes .
- **Multidisciplinary Approach:**
 - A multidisciplinary team, including neurologists, geriatricians, psychologists, and social workers, may be involved in the assessment and diagnosis to ensure a comprehensive evaluation and management plan.

The clinical diagnosis of Alzheimer's Disease and dementia syndromes is crucial for guiding treatment decisions, planning for care, and providing support to patients and their families. Early and accurate diagnosis can also help in accessing appropriate resources and interventions.

2. Investigation

Laboratory investigations for AD are vital for confirming the diagnosis, excluding other causes of cognitive impairment, and evaluating the patient's overall health. Key components include:

- **Imaging Studies:** MRI and CT scans assess brain structure, detect hippocampal atrophy, and rule out vascular damage or other abnormalities. PET scans can visualize amyloid plaques and tau pathology.
- **Blood Tests:** Tests include CBC for anemia or infections, thyroid function tests for hormonal imbalances, vitamin B12 and folate levels for deficiencies, and metabolic panels to evaluate liver, kidney, and electrolyte health.
- **CSF Analysis:** Lumbar puncture identifies biomarkers like low beta-amyloid (A β 42), and elevated total tau (t-tau) and phosphorylated tau (p-tau), indicative of AD pathology.
- **Genetic Testing:** Evaluates familial risk, particularly in early-onset AD cases, and assesses the presence of the APOE ϵ 4 allele.
- **Neuropsychological Testing:** Examines cognitive domains like memory, attention, and executive function to differentiate AD from other dementias.
- **Other Investigations:** Tests like urinalysis and cerebral angiography may be conducted to rule out infections, vascular issues, or metabolic problems.

VI. DIFFERENTIAL DIAGNOSIS:

The differential diagnosis of AD involves distinguishing it from other conditions with similar cognitive and behavioral symptoms. Key conditions include:

- **Vascular Dementia:** Caused by reduced brain blood flow, often presenting with sudden or stepwise cognitive decline and neurological signs.
- **Lewy Body Dementia:** Features fluctuating cognition, visual hallucinations, and parkinsonian symptoms, which can overlap with AD.
- **Frontotemporal Dementia (FTD):** Marked by personality, behavior, and language changes rather than memory loss.
- **Depression (Pseudodementia):** Mimics dementia with memory and focus issues **but improves with depression treatment.**

- **Normal Pressure Hydrocephalus (NPH):** Includes gait disturbance, incontinence, and cognitive decline, often identified through imaging.
 - **Parkinson's Disease Dementia:** Cognitive decline follows motor symptoms, with overlapping features of parkinsonism and AD.
 - **Huntington's Disease:** A genetic disorder causing motor dysfunction, cognitive decline, and psychiatric issues.
 - **Thyroid Disorders:** Hypo- or hyperthyroidism can impair cognition; ruled out via blood tests.
 - **Vitamin Deficiencies:** Deficiencies in B12, folate, or thiamine can cause cognitive decline, confirmed through blood tests.
 - **Infections:** Conditions like meningitis or encephalitis can lead to acute cognitive changes, requiring clinical evaluation and possibly lumbar puncture.
 - **Substance Abuse:** Chronic alcohol use or drug abuse may impair cognition and require a substance use history.
 - **Other Neurological Conditions:** Multiple sclerosis, brain injuries, or tumors may also present with cognitive symptoms.
- Accurate diagnosis relies on comprehensive clinical assessments, cognitive testing, neurological exams, and laboratory or imaging studies to guide effective treatment and management.

VII. MANAGEMENT

1.Objectives

The first to alleviate symptoms and improve the quality of life for patients with Alzheimer's Disease through effective management and support and the second to slow the progression of cognitive decline and maintain patient autonomy and independence for as long as possible while supporting.

2. Critical sign

Critical signs of AD typically refer to key symptoms that indicate the presence of cognitive decline and the progression of the disease. These signs can include:

- **Memory Loss:** Particularly difficulty remembering recent events or newly learned information, which is often one of the earliest signs.
 - **Difficulty with Problem-Solving:** Challenges in planning, organizing tasks, or following familiar recipes or instructions.
 - **Disorientation:** Confusion about time or place, including getting lost in familiar environments or forgetting where they are.
 - **Language Problems:** Struggling to find the right words, following or joining conversations, or repeating oneself.
 - **Changes in Mood and Personality:** Noticeable shifts in mood, behavior, or personality, including increased anxiety, confusion, or withdrawal from social activities.
 - **Poor Judgment:** Making uncharacteristic decisions or exhibiting poor judgment, particularly in financial matters or personal care.
 - **Difficulty with Visual and Spatial Relationships:** Problems with reading, judging distances, and determining color or contrast, which can affect driving and navigation.
- These critical signs are essential for early detection and diagnosis of AD, allowing for timely intervention and management .

3. Mortality risk factor

Mortality risk factors associated with AD include a variety of medical, social, and environmental factors that can contribute to increased mortality in affected individuals. Key risk factors include:

- **Advanced Age:** The risk of mortality increases significantly with age, particularly in individuals over 80 years old.
- **Comorbidities:** The presence of other medical conditions, such as cardiovascular disease, diabetes, and respiratory diseases, can complicate the health of individuals with AD and increase mortality risk.
- **Severe Cognitive Impairment:** Greater levels of cognitive decline and functional impairment are associated with higher mortality rates.
- **Behavioral Symptoms:** The presence of severe behavioral symptoms, such as aggression or agitation, can lead to increased caregiver stress and potential neglect, impacting patient health.
- **Infections:** Individuals with AD are at higher risk for infections, particularly pneumonia and urinary tract infections, which can lead to increased mortality.
- **Malnutrition and Dehydration:** Difficulty with eating and drinking can lead to malnutrition and dehydration, significantly affecting health and increasing mortality risk.
- **Falls and Injuries:** Increased risk of falls due to cognitive and physical decline can lead to serious injuries, such as fractures, which can complicate recovery and increase mortality.
- **Social Isolation:** Lack of social support and engagement can lead to depression and neglect, negatively impacting health outcomes.
- **Hospitalization:** frequent hospitalizations for acute medical issues can lead to a decline in overall health and increase the risk of mortality.

Addressing these risk factors through comprehensive care and management strategies can help improve outcomes and potentially reduce mortality in individuals with Alzheimer's Disease .

4-Outpatient management :

AD and dementia syndromes involves a comprehensive approach that includes medical, psychological, and social support. Here are key components of outpatient care:

- **Regular Cognitive Assessments:** Conduct routine cognitive evaluations using standardized tools (e.g., MMSE, MoCA) to monitor cognitive decline and adjust treatment plans accordingly.
- **Medication Management:** Initiate and monitor pharmacological treatments, such as cholinesterase inhibitors or memantine, to manage symptoms. Regularly assess for efficacy and side effects, adjusting dosages as necessary.
- **Behavioral Interventions:** Implement non-pharmacological interventions, such as cognitive stimulation therapy, reminiscence therapy, and behavioral strategies to address challenging behaviors.
- **Multidisciplinary Care Team:** Involve a team of healthcare professionals, including neurologists, geriatricians, psychologists, social workers, and occupational therapists, to provide holistic care tailored to the patient's needs.
- **Caregiver Support and Education:** Offer resources and training for caregivers to help them understand the disease, manage symptoms, and cope with the emotional and physical demands of caregiving.
- **Social Engagement:** Encourage participation in social activities and support groups to reduce isolation and improve quality of life for both patients and caregivers.

- **Nutritional Support:** Assess dietary needs and provide guidance on nutrition to prevent malnutrition and dehydration, which are common in advanced stages of dementia.
 - **Safety Assessments:** Evaluate the home environment for safety risks and recommend modifications to prevent falls and accidents.
 - **Advance Care Planning:** Discuss and document the patient's preferences for future care, including legal and financial planning, to ensure their wishes are respected.
 - **Regular Follow-ups:** Schedule consistent follow-up appointments to monitor disease progression, adjust treatment plans, and provide ongoing support to patients and caregivers.
- By implementing these strategies, outpatient care can significantly enhance the quality of life for individuals with Alzheimer's Disease and dementia syndromes while providing essential support to their caregivers.

5. First line management

The first-line management of AD typically involves a combination of pharmacological and non-pharmacological approaches aimed at alleviating symptoms and improving the quality of life for patients. Here are the key components:

1. Pharmacological Management

- **Cholinesterase Inhibitors:**
 - **Medications:** Donepezil, Rivastigmine, and Galantamine are commonly prescribed.
 - **Indication:** These medications are indicated for patients with mild to moderate Alzheimer's Disease.
 - **Mechanism:** They work by increasing levels of acetylcholine, a neurotransmitter associated with memory and learning, which is often deficient in AD patients.
 - **Dosage:**
 - Donepezil: Initial dose of 5 mg/day, maintenance dose of 10 mg/day.
 - Rivastigmine: Available in oral form (1.5 mg to 6 mg twice daily) and transdermal patches (4.6 mg to 9.5 mg).
 - Galantamine: Initial dose of 8 mg/day, maintenance dose of 16-24 mg/day.
- **Memantine:**
 - **Indication:** This medication is used for moderate to severe Alzheimer's Disease.
 - **Mechanism:** It works as an NMDA receptor antagonist, helping to regulate glutamate activity, which is involved in learning and memory.
 - **Dosage:** Initial dose of 5 mg/day, with a maximum dose of 20 mg/day.

2. Non-Pharmacological Management

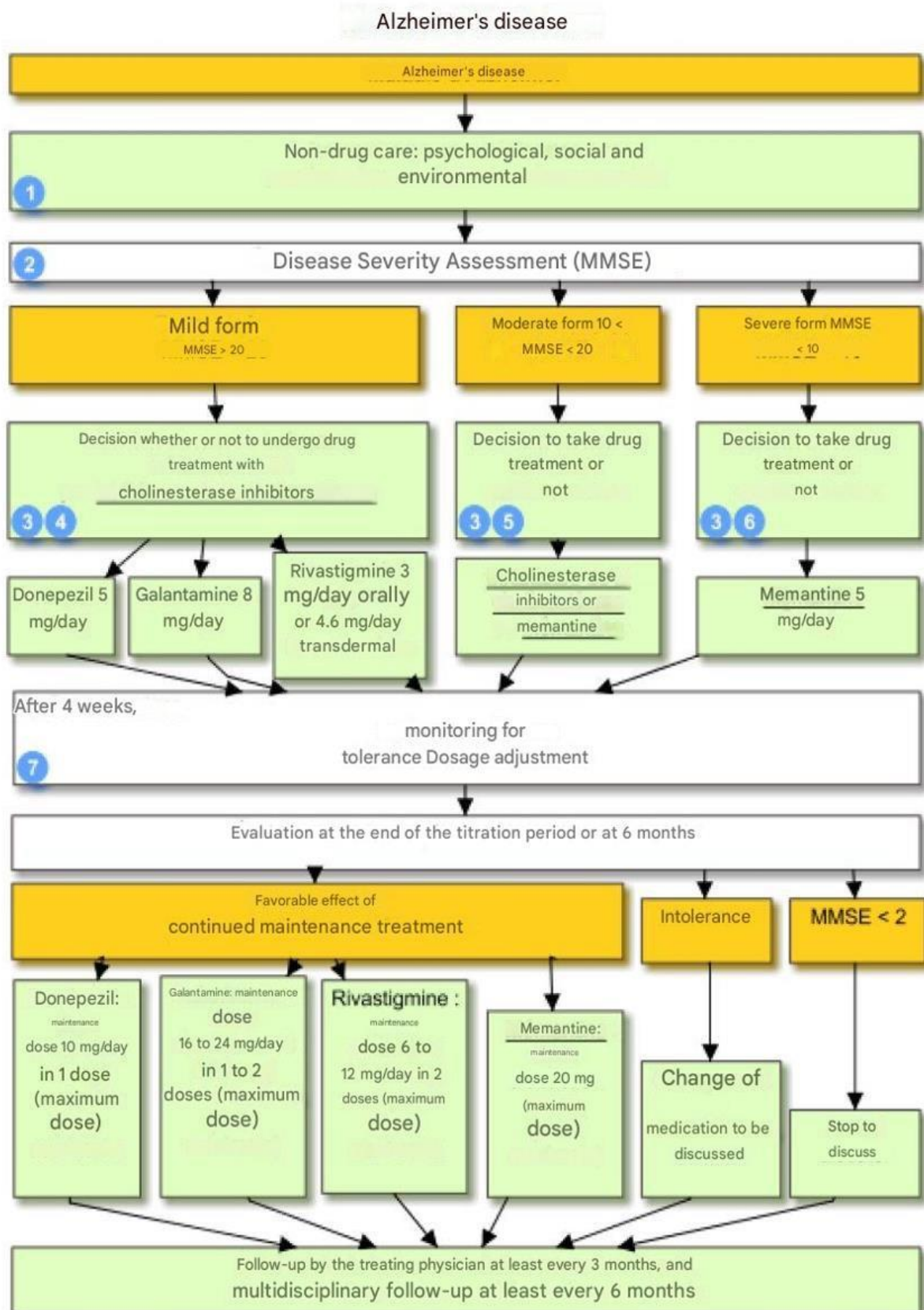
- **Cognitive Stimulation Therapy:** Engaging patients in activities that stimulate thinking and memory can help maintain cognitive function.
- **Behavioral Interventions:** Implementing strategies to manage behavioral symptoms, such as agitation or aggression, through environmental modifications and structured routines.
- **Support for Caregivers:** Providing education and resources to caregivers to help them manage the challenges of caregiving and reduce stress.
- **Lifestyle Modifications:** Encouraging physical activity, social engagement, and a healthy diet to support overall well-being.

- **Regular Monitoring:** Continuous assessment of cognitive function and overall health to adjust treatment plans as necessary.
These first-line management strategies aim to slow the progression of symptoms, enhance the quality of life, and support both patients and their caregivers in navigating the challenges of Alzheimer's Disease.

6. Adjustment management

- **Monitoring and Assessment:**
 - **Cognitive Evaluation:** Periodically assess cognitive function (e.g., MMSE, MoCA) to guide treatment adjustments.
 - **Functional Assessment:** Evaluate daily living skills to determine appropriate support levels.
 - **Medication Management:**
 - **Reassessing Pharmacotherapy:** Review medication effectiveness and side effects, adjusting as needed.
 - **Managing Side Effects:** Monitor for adverse reactions and consider alternative treatments or dosage modifications.
 - **Behavioral Management:**
 - **Tailored Interventions:** Modify behavioral strategies to address specific symptoms (e.g., agitation, aggression, depression).
 - **Environmental Modifications:** Adjust the environment to minimize triggers and enhance safety.
 - **Caregiver Support:**
 - **Education and Training:** Provide ongoing education to empower caregivers.
 - **Respite Care:** Encourage caregivers to utilize respite services to prevent burnout.
 - **Social and Emotional Support:**
 - **Engagement Activities:** Adapt social and cognitive activities to the patient's abilities and interests.
 - **Support Groups:** Facilitate access to support groups for both patients and caregivers.
 - **Nutritional and Physical Health Management:**
 - **Dietary Adjustments:** Monitor nutrition and make dietary changes as needed.
 - **Physical Activity:** Tailor exercise programs to promote mobility and overall health.
 - **Advance Care Planning:**
 - **Future Care Preferences:** Regularly discuss future care wishes, including legal and financial planning.
 - **End-of-Life Care:** Prepare for end-of-life care preferences and ensure they are documented.
 - **Coordination of Care:**
 - **Multidisciplinary Approach:** Collaborate with a team of healthcare professionals to provide comprehensive care.
- By implementing these strategies, healthcare providers can optimize care for individuals with AD, improving their quality of life and well-being.

7. Algorithm of management



VIII. SPECIAL CASES

Understanding these differences is crucial for accurate diagnosis and appropriate management. Frontotemporal Dementia (FTD) is a distinct type of dementia characterized by the progressive degeneration of the frontal and temporal lobes, leading to personality, behavior, and language changes.

Unlike AD, FTD typically occurs at a younger age (40–65) and is associated with the accumulation of proteins like tau, TDP-43, or FUS rather than amyloid plaques and tau tangles. Key symptoms include behavioral changes (apathy, disinhibition), language difficulties (e.g., primary progressive aphasia), and emotional changes (loss of empathy).

Diagnosis relies on clinical evaluation, neuropsychological testing, and imaging to identify specific brain atrophy patterns.

While there is no cure, management focuses on symptom relief, behavioral interventions, and caregiver support. Understanding FTD's unique features is critical for accurate diagnosis and effective care.

IX. EVOLUTION AND PROGNOSIS

The evolution and prognosis AD and dementia syndromes can vary significantly based on several factors, including the type of dementia, the age of onset, the presence of comorbidities, and the individual patient's response to treatment. Here's an overview of the evolution and prognosis for Alzheimer's Disease and dementia syndromes:

Evolution of AD

The progression and prognosis of Alzheimer's Disease (AD) and other dementia syndromes vary based on factors like type, age of onset, comorbidities, and treatment response.

Evolution of AD typically occurs in three stages:

- **Early Stage:** Mild cognitive impairment with subtle memory loss and difficulty in complex tasks.
- **Moderate Stage:** Pronounced cognitive decline, behavioral changes, and challenges in daily activities.
- **Severe Stage:** Loss of communication, dependency on care, and significant cognitive and physical deterioration.

Progression spans 4–20 years, averaging 8–10 years.

Prognosis for all types of dementia is generally poor, with life expectancy post-diagnosis averaging 4–8 years for AD. However, some individuals may live longer, depending on factors like age at diagnosis, overall health, and access to care. For other dementias:

- **Frontotemporal Dementia (FTD)** often progresses faster, with survival ranging from 2 to 10 years..
- **Vascular Dementia** prognosis depends on the extent of vascular damage and comorbidities, with survival typically shorter in severe cases.
- **Dementia with Lewy Bodies** usually results in significant impairment within 5 to 10 years with fluctuating cognition and hallucinations

Supportive care can improve quality of life, but most forms of dementia lead to increasing dependency, cognitive decline, and eventual loss of physical functioning. Early diagnosis and comprehensive management remain essential for optimizing outcomes and comfort.

X. COMPLICATIONS

The complications of AD and other dementia syndromes significantly affect patients' health, quality of life, and care needs. The complications include:

- **Cognitive Decline:** Progressive memory loss and impaired reasoning lead to increased dependency.
- **Behavioral and Psychological Symptoms:** Agitation, depression, anxiety, hallucinations, and delusions complicate care and stress caregivers.
- **Loss of Independence:** Declining abilities necessitate assistance with daily activities like bathing, dressing, and eating.
- **Falls:** Physical and cognitive impairments increase fall risk, often resulting in injuries.
- **Malnutrition and Dehydration:** Difficulty eating and swallowing, especially in advanced stages, may cause these issues.
- **Infections:** Pneumonia and urinary tract infections are common due to immobility and communication challenges.
- **Pressure Ulcers:** Limited mobility can lead to bedsores.
- **End-of-Life Issues:** Advanced stages may require palliative care to manage complex needs.

Specific Dementia Complications:

- **Frontotemporal Dementia (FTD):** Behavioral changes, language difficulties, and emotional blunting heighten caregiver burden.
- **Vascular Dementia:** Recurrent strokes exacerbate cognitive and physical decline.
- **Dementia with Lewy Bodies:** Severe cognitive fluctuations, hallucinations, and parkinsonism complicate care and increase fall risk.

All dementia types share risks like infections, malnutrition, dehydration, and pressure ulcers, emphasizing the need for comprehensive care planning and caregiver support to enhance quality of life.

XI. PREVENTION AND RECOMMENDATION

Preventing Alzheimer's AD and other dementias involves a combination of lifestyle changes, health management, and early interventions. While there is no guaranteed prevention, key strategies can reduce the risk and support cognitive health:

- **Healthy Diet:** Adopt a Mediterranean diet rich in fruits, vegetables, whole grains, fish, nuts, and healthy fats, while limiting processed foods, sugar, and saturated fats.
- **Physical Activity:** Engage in at least 150 minutes of aerobic exercise weekly, which can enhance cognitive function and reduce dementia risk.
- **Cognitive Engagement:** Stay mentally active through reading, puzzles, learning new skills, or musical activities to maintain cognitive health.
- **Social Interaction:** Maintain strong social connections and engage in social activities, which can lower the risk of cognitive decline.
- **Cardiovascular Health:** Control hypertension, diabetes, and high cholesterol with regular check-ups and medication.
- **Avoid Smoking and Limit Alcohol:** Smoking is a major risk factor, and excessive alcohol can harm brain health, so smoking cessation and moderate alcohol consumption are important.
- **Sleep Hygiene:** Ensure 7-9 hours of quality sleep per night, as poor sleep is linked to cognitive decline.
- **Stress Management:** Use techniques like mindfulness, meditation, and deep breathing to reduce stress, which can negatively impact brain health.
- **Regular Health Screenings:** Regular check-ups help manage chronic conditions that may increase dementia risk.

For Caregivers and Families:

- **Education and Training:** Caregivers should learn about dementia to better manage

symptoms.

- Support Networks: Joining support groups helps caregivers share experiences and gain emotional support.
- Create a Safe Environment: Modify living spaces to ensure safety and accessibility.
- Plan for the Future: Discuss and document care preferences early on to ensure the individual's wishes are respected.
- Encourage Routine: Establishing a daily routine can reduce confusion for those with dementia.

While there is no definitive way to prevent dementia, adopting a healthy lifestyle and providing caregiver support can significantly improve quality of life and reduce cognitive decline risks.

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MOVEMENT DISORDERS

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I. DEFINITION

Movement disorders (MDs) are a group of neurological conditions characterized by abnormal or involuntary movements. These disorders can manifest as excessive movements (hyperkinetic disorders) such as dystonia, tremors, tics, and chorea, or as a lack of movement (hypokinetic disorders) such as in Parkinson's disease. MDs often result from dysfunctions in the basal ganglia, cerebellum, or other parts of the nervous system that control movement. They can significantly impact a person's ability to perform daily activities and may be associated with other neurological symptoms or comorbidities.

II. EPIDEMIOLOGY

Movement disorders are common, influenced by age, sex, genetics, and environmental factors like toxins. Coexisting conditions complicate management, requiring better public health strategies and care. The epidemiology of movement disorders highlights their prevalence and impact on various populations.

- **Prevalence of Parkinson's Disease (PD):** The overall prevalence of PD is approximately 1% in individuals aged 65–85 years, increasing to about 4.3% in those over 85 years of age.
- **Essential Tremor:** This is the most common form of tremor, with a prevalence of around 4% in people over 40 years, which rises to 14% in those over 65 years.
- **Tics:** The prevalence of tics in school-age children and adolescents can be as high as 21%.
- **Variability in Presentation:** The clinical presentation of movement disorders is often complex and variable, making accurate diagnosis challenging even for experienced specialists.

III. ETIOLOGY

The etiology of movement disorders (MDs) is diverse and can be classified into several categories, including genetic, environmental, and idiopathic factors. Key points regarding the etiology of MDs include:

- **Genetic Factors:** Some movement disorders have a hereditary component. For example, Huntington's disease is a genetic disorder that leads to progressive movement abnormalities.
- **Neurodegenerative Diseases:** Conditions such as Parkinson's disease and multiple system atrophy are characterized by the degeneration of specific neuronal populations, leading to movement disorders.
- **Metabolic Disorders:** Certain metabolic conditions, such as Wilson's disease, can cause movement disorders due to the accumulation of toxic substances in the body.
- **Structural Brain Changes:** Lesions or structural changes in the brain, such as those seen in stroke or tumors, can lead to movement disorders by affecting the areas responsible for motor control.
- **Environmental Factors:** Exposure to toxins, such as heavy metals or pesticides, has been implicated in the development of some movement disorders, particularly in Parkinson's disease.

- **Psychogenic Factors:** Some movement disorders may have a psychological origin, where stress or psychological trauma can manifest as abnormal movements.

IV. PATHOPHYSIOLOGY

Movement disorders arise from complex interactions between genetic, environmental, and neurobiological factors, leading to abnormal motor function. The pathology of movement disorders involves several key mechanisms:

- **Neurodegeneration:** Progressive loss of specific neurons, such as dopaminergic neurons in Parkinson's disease (PD).
- **Protein Aggregation:** Abnormal accumulation of proteins, like mutant huntingtin in Huntington's disease, leading to neuronal toxicity.
- **Basal Ganglia Dysfunction:** Impairment in the basal ganglia affects motor control, resulting in symptoms like tremors and rigidity.
- **Neurotransmitter Imbalances:** Alterations in neurotransmitters, such as dopamine deficiency in PD and GABA changes in dystonia.
- **Structural Changes:** Atrophy or lesions in brain regions can contribute to movement disorders.
- **Neuroinflammation:** Inflammation may play a role in some disorders, such as multiple system atrophy.

Understanding the physiopathology of movement disorders is crucial for developing targeted therapies and improving patient outcomes, as it provides insights into the underlying mechanisms that drive these complex conditions.

V. DIAGNOSIS

Diagnosing movement disorders (MDs) involves a comprehensive approach that includes clinical evaluation and laboratory investigations. Here's how the diagnosis is typically made:

1. **Clinical Evaluation:** A thorough neurological examination to assess the type and severity of movement abnormalities. The clinical signs of movement disorders can vary widely but generally include:
 - **Tremors:** Involuntary rhythmic shaking, often seen in conditions like Parkinson's disease and essential tremor.
 - **Bradykinesia:** Slowness of movement, commonly observed in Parkinson's disease, affecting daily activities.
 - **Rigidity:** Increased muscle tone leading to stiffness and resistance to movement, typical in Parkinson's disease.
 - **Dystonia:** Sustained muscle contractions causing abnormal postures or twisting movements.
 - **Chorea:** Irregular, non-repetitive, and rapid movements, often seen in Huntington's disease.
 - **Tics:** Sudden, repetitive movements or sounds, characteristic of Tourette syndrome.
 - **Ataxia:** Lack of coordination and balance, often resulting from cerebellar dysfunction.
 - **Myoclonus:** Sudden, brief jerking movements, which can occur in various conditions.
2. **Detailed History:** Gathering a comprehensive clinical history, including symptom onset, duration, progression, associated symptoms, family history, medication use, and any relevant medical or social history.

3. **Diagnostic Tests:** While many movement disorders are diagnosed clinically, additional tests may be used, such as:
 - Neuroimaging: MRI or CT scans to identify structural brain abnormalities.
 - Blood Tests: To check for metabolic or infectious causes.
 - Genetic Testing: For hereditary conditions like Huntington's disease.
 - Electromyography (EMG): To assess muscle activity and identify neuromuscular issues.
 - DaTscan: Visualizes dopamine transporters to differentiate Parkinsonian syndromes.
 - Neuropsychological Testing: Assesses cognitive function.
 - Clinical Rating Scales: Quantifies symptom severity and monitors progression.
4. **Response to Treatment:** Observing how symptoms respond to specific treatments can also provide diagnostic clues, particularly in conditions like Parkinson's disease.
5. **Multidisciplinary Approach:** In some cases, involving specialists such as neurologists, geneticists, and movement disorder experts can enhance diagnostic accuracy.

The diagnosis of movement disorders is often complex and requires a multidisciplinary approach. Clinicians integrate clinical findings with laboratory investigations to arrive at a definitive diagnosis, which is crucial for effective management and treatment planning.

VI. DIFFERENTIAL DIAGNOSIS

The differential diagnosis of movement disorders involves distinguishing between various conditions that present with similar movement abnormalities. Key categories and examples include:

- **Akinetic-Rigid Disorders:**
 - Parkinson's Disease: Characterized by bradykinesia, rigidity, and resting tremor.
 - Multiple System Atrophy: Presents with parkinsonism, autonomic dysfunction, and ataxia.
 - Progressive Supranuclear Palsy: Features vertical gaze palsy and postural instability.
- **Hyperkinetic Disorders:**
 - Essential Tremor: Action tremor that improves with rest.
 - Dystonia: Involuntary muscle contractions causing twisting movements.
 - Chorea: Irregular, non-repetitive movements, as seen in Huntington's disease.
- **Myoclonus:**
 - Epileptic Myoclonus: Associated with seizures.
 - Non-Epileptic Myoclonus: Can be due to metabolic disorders or neurodegenerative diseases.
- **Ataxia:**
 - Cerebellar Ataxia: Can be hereditary or acquired (e.g., due to alcohol or vitamin deficiencies).
 - Vestibular Ataxia: Related to inner ear disorders.
- **Other Conditions:**
 - Wilson's Disease: Can present with tremor, dystonia, and psychiatric symptoms.
 - Drug-Induced Movement Disorders: Such as tardive dyskinesia from antipsychotic medications.
- **Psychogenic Movement Disorders:**

- Movements that are inconsistent with neurological conditions and may be influenced by psychological factors.

The differential diagnosis is guided by the dominant movement disorder, associated symptoms, and clinical history.

VII. MANAGEMENT

1. Objectives

- The primary goal of managing movement disorders (MDs) is to alleviate symptoms and improve functional abilities, enhancing the patient's quality of life and independence.
- Secondary objectives include preventing complications, providing patient education and support, and implementing a tailored, multidisciplinary approach to care.

2. Critical sign

Critical signs of movement disorders vary depending on the specific type of disorder, but some key signs include:

- **Tremor:** Involuntary rhythmic shaking, often seen in conditions like Parkinson's disease and essential tremor. It can be resting (as in Parkinson's) or action-related (as in essential tremor).
- **Bradykinesia:** Slowness of movement, a hallmark of Parkinson's disease, characterized by difficulty initiating movements and reduced amplitude of movements.
- **Rigidity:** Increased muscle tone leading to stiffness and resistance to passive movement, commonly observed in Parkinson's disease.
- **Dystonia:** Sustained muscle contractions causing twisting and abnormal postures, which can occur in various conditions, including primary dystonia and secondary dystonia due to other neurological disorders.
- **Chorea:** Irregular, non-repetitive, and unpredictable movements, often seen in Huntington's disease and other conditions.
- **Myoclonus:** Sudden, brief, shock-like jerks of muscles, which can be seen in various disorders, including epilepsy and certain neurodegenerative diseases.
- **Ataxia:** Lack of voluntary coordination of muscle movements, leading to unsteady gait and difficulty with balance, often associated with cerebellar disorders.
- **Postural Instability:** Difficulty maintaining balance, which can lead to falls, commonly seen in advanced Parkinson's disease and multiple system atrophy.

3. Mortality risk factor

Mortality risk factors in movement disorders include:

- **Age:** Older individuals have a higher mortality risk.
- **Severity of Disease:** Advanced stages with severe symptoms and complications increase mortality.
- **Comorbidities:** Presence of other health issues, like cardiovascular disease, raises risk.
- **Falls and Injuries:** Increased fall risk can lead to serious injuries and death.

- **Dysphagia:** Difficulty swallowing can result in aspiration pneumonia, a common cause of death.
- **Cognitive Impairment:** Dementia and cognitive decline complicate management and increase mortality.
- **Autonomic Dysfunction:** Conditions like multiple system atrophy can lead to serious complications.
- **Medication Side Effects:** Some treatments may have adverse effects that contribute to mortality.

4. Outpatient management

In an outpatient setting, the professional tasks of management of movement disorders includes:

- **Diagnosis and Assessment:** Accurate evaluation and symptom assessment.
- **Medication Management:** Tailored pharmacological treatment with regular monitoring.
- **Physical and Occupational Therapy:** Improving mobility, strength, and daily living activities.
- **Speech Therapy:** Addressing swallowing and communication difficulties.
- **Psychosocial Support:** Managing mental health issues and providing support for patients and families.
- **Education and Counseling:** Informing about the disorder and treatment options.
- **Regular Follow-Up:** Monitoring disease progression and adjusting management plans.
- **Multidisciplinary Approach:** Collaboration among healthcare professionals.
- **Lifestyle Modifications:** Encouraging a healthy diet and regular exercise.

4. First-line treatment

First-line treatments for specific movement disorders include:

- **Parkinson's Disease:**
 - **Levodopa:** Often the first-line treatment, especially for older patients, as it effectively alleviates motor symptoms.
 - **Dopamine Agonists:** Such as pramipexole or ropinirole, may be used as initial therapy in younger patients to delay the onset of levodopa-related complications.
 - **MAO-B Inhibitors:** Rasagiline, Safinamide, and Selegiline offer mild symptom relief.
 - **COMT Inhibitors:** Entacapone, Opicapone, and Tolcapone prolong levodopa effects but require dose adjustment to avoid dyskinesia.
 - **Amantadine:** Helps with motor symptoms and levodopa-induced dyskinesia.
 - **Anticholinergics:** Useful for tremor in younger patients; avoid in older adults or those with cognitive issues.
 - Other advance treatment in PD advance stage such Apomorphine (subcutaneously with injection pen), Deep Brain Stimulation, or Magnetic Resonance (MR)-guided Focused Ultrasound (MRgFUS) is a non-invasive procedure that can treat tremors and other symptoms related to Parkinson's disease.

- **Essential Tremor:**
 - **Propranolol:** A beta-blocker is commonly used as the first-line treatment.
 - **Primidone:** An anticonvulsant that can also be effective.
- **Dystonia:**
 - **Botulinum Toxin Injections:** Considered the first-line treatment for focal dystonia.
 - **Anticholinergics:** Such as trihexyphenidyl, may be used for generalized dystonia.
- **Chorea (e.g., Huntington's Disease):**
 - **Tetrabenazine:** The first-line treatment for chorea, helping to reduce involuntary movements.
- **Myoclonus:**
 - **Clonazepam:** Often the first-line medication for myoclonus.
 - **Valproate:** Another option, particularly for myoclonic seizures.
- **Tics (e.g., Tourette Syndrome):**
 - **Haloperidol:** A traditional first-line treatment for Tourette syndrome.
 - **Pimozide:** Another antipsychotic that can be effective.
- **Restless Legs Syndrome:**
 - **Dopamine Agonists:** Such as pramipexole or ropinirole are often the first-line treatment.
- **Surgical Options:**
 - **Deep Brain Stimulation (DBS):** A surgical treatment for Parkinson's disease and essential tremor that involves implanting a device to regulate abnormal brain activity.
 - **Lesioning Procedures:** In some cases, targeted lesions in specific brain areas may be performed to alleviate symptoms.

Other supportive treatment options :

- **Physical Therapy:** Tailored exercise programs to improve strength, balance, and coordination.
- **Occupational Therapy:** Assistance with daily activities and adaptations to enhance quality of life.
- **Speech Therapy:** For patients with speech or swallowing difficulties, therapy can improve communication and swallowing function
- **Lifestyle Modifications:** Encouraging a healthy diet, regular exercise, and avoiding substances that may worsen symptoms.
- **Psychosocial Support:** Counseling and support groups to help manage the emotional and psychological aspects of living with a movement disorder.

These first-line treatments aim to alleviate symptoms and improve the quality of life for patients with movement disorders. The choice of treatment should be individualized based on the patient's specific condition, symptoms, and overall health.

6. Treatment adjustment

Treatment adjustments for movement disorders involve several key strategies to optimize patient care:

- **Monitoring Symptoms:** Regular assessment of symptom severity and frequency to evaluate treatment effectiveness.
- **Evaluating Side Effects:** Monitoring for adverse effects and adjusting dosages or switching medications if side effects are intolerable.
- **Adjusting Medication Dosages:** Increasing or decreasing dosages based on symptom control and side effects, such as adjusting levodopa in Parkinson's disease.
- **Changing Medications:** Switching to alternative medications if the current one is ineffective or causes significant side effects.
- **Adding Adjunctive Therapies:** Introducing additional medications to manage specific symptoms or side effects.
- **Incorporating Non-Pharmacological Interventions:** Adjusting physical and occupational therapy plans and encouraging lifestyle modifications.
- **Surgical Considerations:** Considering surgical options like Deep Brain Stimulation for patients with refractory symptoms.
- **Regular Follow-Up:** Scheduling follow-up appointments to reassess treatment efficacy and make necessary adjustments.
- **Multidisciplinary Approach:** Collaborating with a healthcare team to provide comprehensive care.

7. Algorithm

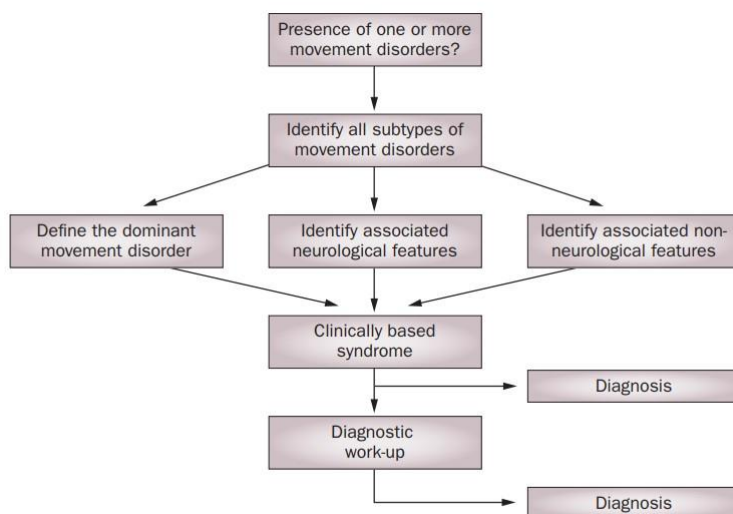


Figure 1 | A systematic approach to diagnosis in patients presenting with movement disorders.

TABLE 1. A classification of movement disorders

I. Hypokinesias	
A. Akinesia/bradykinesia (parkinsonism)	F. Freezing phenomenon
B. Apraxia	G. Hesitant gaits
C. Blocking (holding) tics	H. Hypothyroid slowness
D. Cataplexy and drop attacks	I. Rigidity
E. Catatonia, psychomotor depression, and obsessional slowness	J. Stiff muscles
II. Hyperkinesias	
A. Abdominal dyskinesias	M. Moving toes and fingers
B. Akathitic movements	N. Myoclonus
C. Ataxia/asynergia/dysmetria	O. Myokymia and synkinesia
D. Athetosis	P. Myorhythmia
E. Ballism	Q. Paroxysmal dyskinesias
F. Chorea	R. Periodic movements in sleep
G. Dystonia	S. REM sleep behavior disorder
H. Hemifacial spasm	T. Restless legs
I. Hyperekplexia	U. Stereotypy
J. Hypnogenic dyskinesias	V. Tics
K. Jumping disorders	W. Tremor
L. Jumpy stumps	

This table and the definitions for each item were developed by S. Fahn, C. D. Marsden, and J. Jankovic and published in Table 1.1 in reference 7.

VIII. SPECIAL CASES

- **Case 1: Wilson's Disease with Movement Disorders**
 - 23-year-old woman with progressive tremor, dystonia, and gait difficulty over 6 months.
 - Main Symptoms: Hand tremor (resting and intention), rigidity, dystonia, bradykinesia, dysarthria, and shuffling gait.
 - Key Finding: Kayser-Fleischer rings on eye exam.

- Investigations: Low ceruloplasmin, elevated urinary copper, and MRI showing basal ganglia abnormalities.
 - Diagnosis: Wilson's disease with neurological manifestations.
 - Treatment:
 - Penicillamine (copper chelation).
 - Zinc (reduces copper absorption).
 - Symptomatic management: Levodopa (parkinsonism) and Baclofen (dystonia).
 - Low-copper diet and genetic counseling if available
 - Outcome: Significant improvement in tremor, dystonia, and rigidity after 6 months of treatment.
- **Case 2 : Early-Onset Parkinson's Disease (PD)**
 - 38-year-old man presented with a 1-year history of right-hand resting tremor, stiffness, and slowness of movement.
 - Main Symptoms: Resting tremor in the right hand, bradykinesia, and mild postural instability.
 - History: No family history of PD or neurological disorders. Symptoms interfered with daily activities, including writing and buttoning shirts.
 - Diagnosis: Early-onset Parkinson's disease based on clinical evaluation and responsiveness to levodopa.
 - Management:
 - Initial Treatment: Pramipexole (dopamine agonist) to delay levodopa use and manage symptoms.
 - Lifestyle: Regular exercise and occupational therapy.
 - Education: Counseling on disease progression and genetic testing offered.
 - Outcome: Symptoms improved with treatment, allowing the patient to continue working and maintain quality of life.

IX. EVOLUTION AND PROGNOSIS

The evolution and prognosis of movement disorders vary widely based on the specific type of disorder:

- **Parkinson's Disease (PD):** Progresses gradually with symptoms like tremor and rigidity. Prognosis varies; some patients live for decades with management, but life expectancy may be reduced.
- **Essential Tremor:** Typically starts mild and worsens with age. It is not life-threatening, but it can impair function and quality of life.
- **Dystonia:** Symptoms can fluctuate and vary in severity. Prognosis is variable; some patients improve with treatment, while others may have persistent symptoms.
- **Huntington's Disease (HD):** A progressive disorder with chorea and cognitive decline, leading to significant disability and reduced life expectancy.
- **Tourette Syndrome:** Tics often begin in childhood and may improve with age, though some individuals continue to experience them.
- **Myoclonus:** Depends on the underlying cause; prognosis varies widely, with some forms being treatable.

- **Restless Legs Syndrome (RLS):** Symptoms can worsen over time, impacting quality of life, but treatment can effectively manage symptoms.

X. COMPLICATIONS

Complications of movement disorders can significantly impact patients' quality of life and overall health. Here are some common complications associated with various movement disorders:

- **Functional Impairment:** Many movement disorders lead to difficulties in daily activities, such as walking, writing, and performing self-care tasks, which can result in loss of independence.
- **Falls and Injuries:** Patients with disorders like Parkinson's disease and dystonia are at increased risk of falls due to balance issues, rigidity, or sudden movements, leading to fractures and other injuries.
- **Psychological Issues:** Movement disorders can be associated with depression, anxiety, and social isolation, affecting mental health and emotional well-being.
- **Cognitive Decline:** Some movement disorders, particularly Huntington's disease and Parkinson's disease, can lead to cognitive impairment and dementia over time.
- **Sleep Disturbances:** Conditions like Restless Legs Syndrome and Parkinson's disease often disrupt sleep patterns, leading to insomnia and daytime fatigue.
- **Medication Side Effects:** Treatments for movement disorders can cause side effects, such as dyskinesias in Parkinson's disease, which may complicate management and affect quality of life.
- **Autonomic Dysfunction:** Disorders like multiple system atrophy can lead to autonomic symptoms, including blood pressure fluctuations, urinary incontinence, and gastrointestinal issues.
- **Secondary Conditions:** Patients may develop secondary complications, such as pressure sores from immobility, malnutrition, or aspiration pneumonia due to swallowing difficulties.
- **Social and Economic Impact:** The chronic nature of movement disorders can lead to significant healthcare costs, loss of employment, and strain on caregivers and family members.

XI. PREVENTION AND RECOMMENDATION

To prevent and manage movement disorders, the following strategies are recommended:

- **Prevention Strategies:**
 - **Healthy Lifestyle:** Engage in regular exercise and maintain a balanced diet rich in antioxidants and omega-3 fatty acids to support brain health.
 - **Avoid Risk Factors:** Limit exposure to environmental toxins and manage chronic conditions like hypertension and diabetes.
 - **Genetic Counseling:** For those with a family history of hereditary movement disorders, genetic counseling can provide valuable information.
 - **Mental Health Support:** Addressing mental health issues can improve overall well-being and aid in early detection.
- **Management Recommendations:**

- **Early Diagnosis:** Prompt recognition of symptoms leads to more effective management; regular check-ups with a neurologist are essential.
- **Medication Management:** Adhere to prescribed treatments and have regular follow-ups to adjust medications as needed.
- **Therapies:** Engage in physical and occupational therapy to improve mobility and daily functioning.
- **Support Groups:** Join support groups for emotional support and practical advice.
- **Education:** Educate patients and families about movement disorders for early recognition and care.
- **Regular Monitoring:** Continuously monitor symptoms and treatment side effects for better management.
- **Sleep Hygiene:** Establish good sleep habits to manage sleep disturbances.
- **Nutritional Support:** Consult with a nutritionist for a balanced diet tailored to individual needs.

Overall, while not all movement disorders can be prevented, adopting a healthy lifestyle and comprehensive management strategies can significantly enhance quality of life.

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CHAPTERVIII ENDOCRINOLOGY

1. Dyslipidemia
2. Management of diabetes in hospitalized patient
3. Gestational diabetes mellitus
4. Type 1 Diabetes Mellitus
5. Type 2 Diabetes Mellitus
6. Diabetic nephropathy
7. Management of diabetic foot
8. Management of hyperthyroidism
9. Management of hypothyroidism
10. Management of goiter
11. Adrenal insufficiency in adult
12. Prolong glucocorticoids therapy

DYSLIPIDEMIA

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I. DEFINITION

Dyslipidemia is defined as elevated total cholesterol (TC), low-density lipoprotein cholesterol (LDL-C), or triglycerides (TG); low high-density lipoprotein cholesterol (HDL-C); or a combination of these abnormalities.

II. PIDEMIOLOGY

Based on epidemiological data, in the 2003–2006 National Health and Nutrition Examination Survey (NHANES), 53% (105.3 million) of US adults had lipid abnormalities, of which 27% had high LDL-cholesterol, 23% had low HDL-C, and 30% had high TG. While 21% of US adults were found to have two lipid abnormalities at the same time.

Based on the 2005–2008 NHANES survey, 33.5% (an estimated 71 million) of the US adults aged 20 years had high LDL-C (CDC, 2011). Moreover, according to the 2016 report from the AHA, more than 100 million adults in the US over 20 years of age have total-cholesterol levels 200 mg/dL and almost 31 million have levels 240 mg/dL (Moza arian et al., 2016).

III. IOLOGY

Genetic Disorder of Lipid Metabolism	Secondary Hyperlipidemia
Familial hypercholesterolemia	Diabetes
Elevated plasma Lp (a)	Obesity
Polygenic hypercholesterolemia	Lipodystrophy
Familial combined hyperlipoproteinemia	Hypothyroidism
Familial dysbetalipoproteinemia	Estrogen
Lipoprotein lipase deficiency	Glucocorticoids
Apolipoprotein C-II deficiency	Alcohol
Familial hypertriglyceridemia	Nephrotic syndrome

IV. PATHOPHYSIOLOGY

· In the intestinal lumen, dietary triglycerides and cholesterol esters are hydrolyzed by pancreatic lipase to produce glycerol, FFA, and free cholesterol. Bile acids aid in the formation of micelles. Micelles enable the absorption of glycerol and FFA into the intestinal cells.

· Chylomicrons are produced by small intestine. The remnant particles, after use of lipids by the peripheral tissue, are taken up by the hepatocytes. The liver produces apoB-100 containing VLDL and premature HDL. VLDL is hydrolyzed in circulation and converted into IDL and LDL. Both IDL and LDL can be taken up by the hepatocytes. The discoidal shaped premature

• Dyslipidemia occurs when pathway defects in lipoprotein synthesis, processing, and clearance can lead to accumulation of atherogenic lipids in plasma and endothelium formation. Excess LDL can result in the formation of cholesterol plaques that deposit in arteries (atheroma), skin and tendon (xanthomas), eyelids (xanthelasma), and iris (corneal arcus).

The diagram illustrates the following lipid metabolism pathways:

- Intestine:** The starting point for chylomicron production.
- Chylomicron:** Produced in the intestine, it carries lipids to the liver and peripheral sites.
- Liver:** The central organ for lipid metabolism, receiving lipids from the intestine and peripheral sites, and producing VLDL.
- Peripheral sites:** Sites of lipid storage and release, including adipose tissue.
- Lipid Types and Processes:**
 - Chylomicron:** Carries lipids from the intestine to the liver and peripheral sites.
 - VLDL (Very Low Density Lipoprotein):** Produced by the liver, it carries lipids to the liver and peripheral sites.
 - IDL (Intermediate Density Lipoprotein):** Produced by the liver, it carries lipids to the liver and peripheral sites.
 - LDL (Low Density Lipoprotein):** Produced by the liver, it carries lipids to the liver and peripheral sites.
 - HDL (High Density Lipoprotein):** Produced by the liver, it carries lipids to the liver and peripheral sites.
 - FC (Fatty Cholesterol):** Produced by the liver, it carries lipids to the liver and peripheral sites.
 - FFA (Free Fatty Acid):** Released from adipose tissue, it is taken up by the liver and peripheral sites.
 - LPL (Lipoprotein Lipase):** Enzyme that breaks down lipids in chylomicrons and VLDL into FFA.
 - Bile acids:** Produced by the liver, they are used for fat digestion in the intestine.

V. DIAGNOSIS

- LDL Cholesterol (mg/dl) = total cholesterol – (HDL cholesterol + triglycerides/5)
(This formular becomes inaccurate if serum triglycerides are greater than 400mg/dl)

- Non-HDL cholesterol = total cholesterol – HDL cholesterol
(Non-HDL cholesterol can be determined in a non-fasting state)

VI. IPOPROTEINS AND APOLIPOPROTEINS

Lipoprotein class	Origin	Apolipoproteins	Lipid
Chylomicrons	Intestine	B48, C-II, E	TG (85%), cholesterol (10%)
VLDL	Liver	B100, C-II, E	TG (55%), cholesterol (20%)
IDL	VLDL catabolism	B100, E	TG (25%), cholesterol (35%)
LDL	IDL catabolism	B100	TG (5%), cholesterol (60%)
HDL	Liver, intestine	A-I, E	TG (5%), cholesterol (20%)

VII. RISK FACTORS FOR ATHEROSCLEROTIC CARDIOVASCULAR DISEASE

Risk factors	Definition
Age (years)	Male • 45 years; Female • 55 years
Family history of premature CHD events	Male first degree-relative at < 55 years Female first degree-relative at < 65 years
Hypertension	SBP • 140mmHg DBP • 90mmHg
Low HDL cholesterol	< 40 mg/dl in males < 50mg/dl in females
Cigarette smoking	Current
Risk category	Risk factors/10-year risk
Extreme risk	• Progressive ASCVD, including unstable angina in individuals after achieving an LDL cholesterol <70 mg/dL • Established clinical CVD in individuals with diabetes, stage 3 or 4 CKD, or HeFH • History of premature ASCVD (age <55 years male, <65 years female)
Very high risk	• Established or recent hospitalization for ACS or coronary, carotid, or peripheral vascular disease; 10-year risk >20% • Diabetes or stage 3 or 4 CKD with one or more risk factor • HeFH (Heterozygous Familial Hypercholesterolemia)
High risk	• Two or more risk factors and 10-year risk 10–20% • DM or stage 3 or 4 CKD with no other risk factor
Moderate risk	Two or fewer risk factors and 10-year risk <10%
Low risk	Zero risk factors

VIII. MANAGEMENT

ASCVD risk categories and treatment goals				
Risk Category	Treatment goals (mg/dl)			
	LDL-C	Non-HDL-C	Apo B	Triglyceride
Extreme risk	< 55	< 80	< 70	< 150
Very high risk	< 70	< 100	< 80	< 150
High risk	< 100	< 130	< 90	< 150
Moderate risk	< 100	< 130	< 90	< 150
Low risk	< 130	< 160	NR	< 150

1. HMG CoA Reductase Inhibitors (Statins)

Statins are inhibitors of HMG CoA reductase, the rate limiting enzyme in hepatic cholesterol synthesis. This results in a reduction in intracellular cholesterol which leads to an increase in LDL receptor expression at the surface of the hepatocytes.

Recommended dose of statin therapy

High-Intensity Statin Therapy	Moderate-Intensity Statin Therapy	Low-Intensity Statin Therapy
Atorvastatin 40-80 mg Rosuvastatin 20-40 mg	Atorvastatin 10-20 mg Rosuvastatin 5-10 mg Simvastatin 20-40 mg Pravastatin 40-80 mg Lovastatin 40 mg Fluvastatin 40 mg bid Pitavastatin 2-4 mg	Simvastatin 10 mg Pravastatin 10-20 mg Lovastatin 20 mg Fluvastatin 20-40 mg Pitavastatin 1 mg

2. Cholesterol Absorption Inhibitors

- Cholesterol absorption inhibitors selectively block intestinal absorption of both dietary and biliary cholesterol without altering the absorption of fat-soluble nutrients. This leads to a reduction in hepatic cholesterol delivery.
- It can be used in combination with any dose of statin to further lower LDL-C if targets are not achieved with the maximum tolerated statin dose.
- Ezetimibe-statin combination therapy was found to have CV benefits in individuals with CKD and following vascular surgery and acute coronary syndromes.

3. Proprotein Convertase Subtilisin Kexin type 9 (PCSK9)- Inhibitors

This group of drugs work by inhibiting the binding of PCSK9 to the LDL-receptors. This interaction decreases the degradation of the LDL-receptors, resulting in higher LDL-receptor density at the cell surface. The higher expression of LDL-receptors at the cell surface leads to increased clearance with resulting decrease in LDL-C levels.

4. Small interfering RNA (siRNA) PCSK-9 Inhibitors (Inclisiran)

Small interfering RNA (siRNA) molecules also block the hepatic production of PCSK9. Inclisiran is a long-acting, synthetic siRNA molecule that binds to a RNA-induced silencing complex (RISC), thereby inhibiting the translation of PCSK9 messenger RNA (mRNA) causing a marked reduction in LDL-C levels.

5. Fibric Acid Derivatives (Fibrates)

- Fibrates are Peroxisome Proliferator Activated Receptor (PPAR) - α agonist which have an important role in fatty acid oxidation. They reduce serum TG effectively and increase HDL-C modestly. The initial lipid level has a significant impact on the size of the effect.
- Fibrates reduce the risk of CV and coronary events in primary prevention, according to moderate-quality evidence.
- Its use is limited to the treatment of patients with very high TG levels who do not respond to non-pharmacological measures to prevent pancreatitis.
- Dosage adjustment of fibrates is required in the presence of CKD. Serum alanine aminotransferase should be monitored when starting therapy or when doses are increased.

6. Bempedoic acid

- Bempedoic acid antagonizes ATP-citrate lyase, which is a rate-limiting component of the cholesterol synthesis pathway.
- The combination of bempedoic acid and ezetimibe has been shown to significantly reduce major CV events in statin intolerant patients.
- No dosage adjustment required in patients with mild to moderate kidney or liver impairment.
- The risk of myalgia and myopathy is minimal

7. Bile Acid Sequestrants (Anion exchange resins)

- Bile acid sequestrants bind to bile acids to promote their secretion into the intestines. This leads to bile acid depletion resulting in the liver generating more hepatic cholesterol. This raises the hepatic demand for cholesterol and lowers the level of LDL in the blood.
- Bile acid sequestrants provide a practical add-on option with extra LDL-C-lowering for those who are unable to tolerate a guideline directed statin dose.

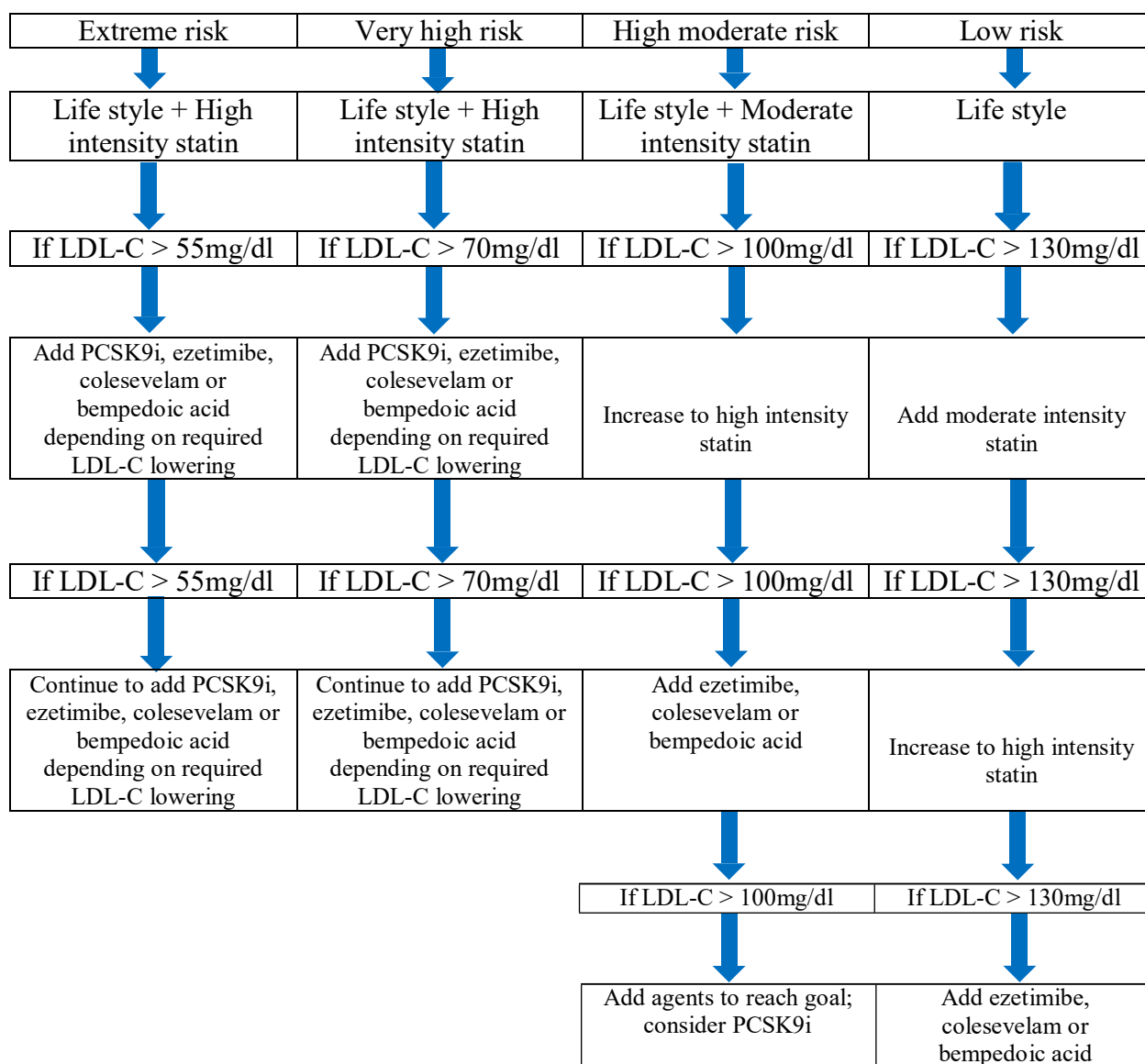
8. Nicotinic Acid (Niacin) and its derivatives

- Nicotinic acid acts in the liver and decreases mobilization of free fatty acids from adipose tissues. It increases HDL-C and lowers TG levels. Its effect on TC reduction is modest.
- It may be considered as an alternative therapy to fibrates in individuals with elevated TG not responsive to other pharmacological options.

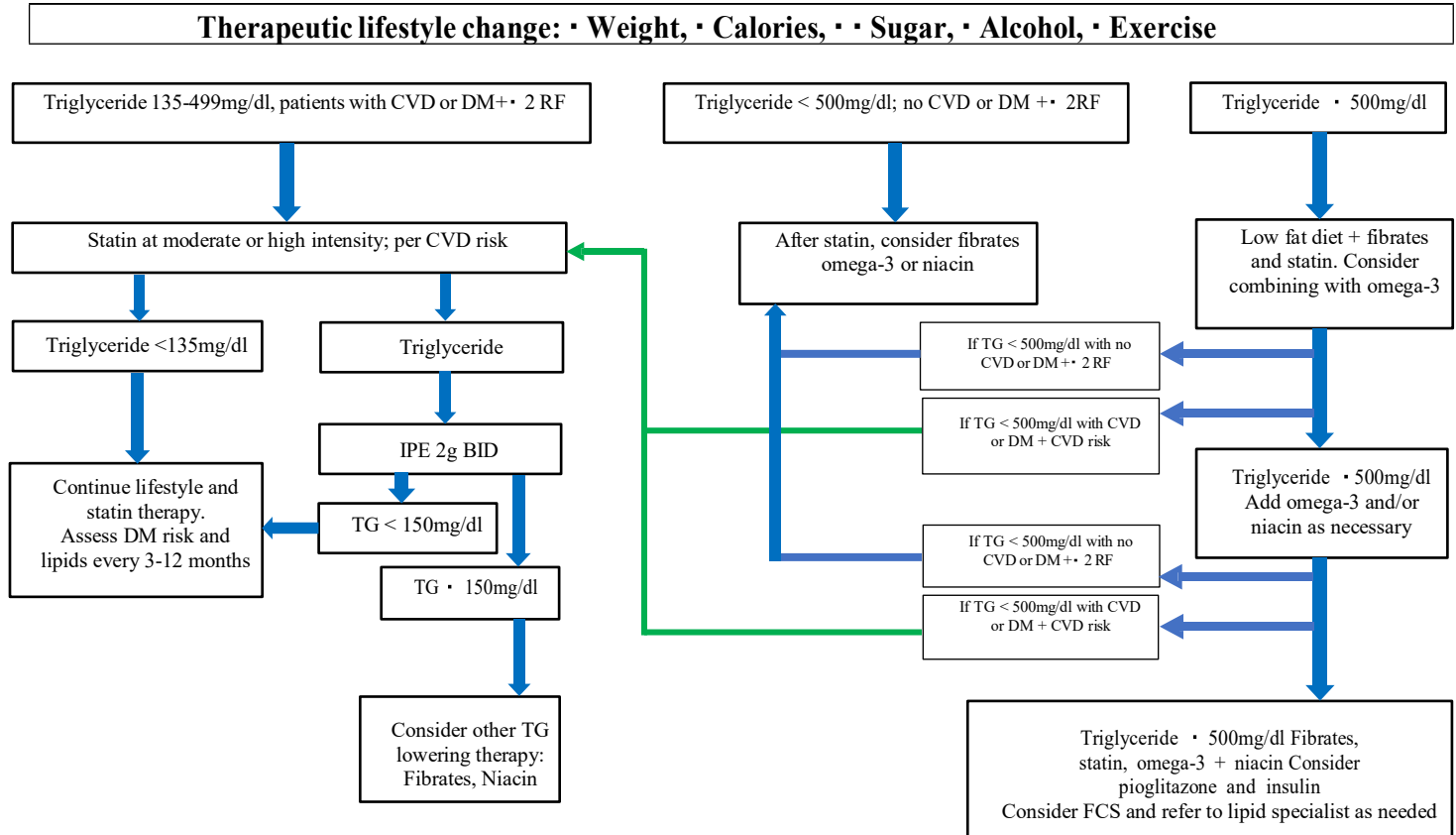
Recommended Dosages for Fibrates

Drugs	Recommended Dosage
Fenofibrate	100mg TDS, 145mg daily (nanoparticles), 160mg daily (micronized), 200mg daily
Gemfibrozil	600-1200mg daily in divided doses 30 minutes before meal (Max:1500mg/day)
Bezafibrate	200mg daily increasing (gradually over 5-7 days) to a maximum dose of 200 mg tds (regular) or 400mg daily (sustained release)
Ciprofibrate	100mg daily

Treating LDL-C to Goal



Management hypertriglyceridemia and the of role of IPE



ASCVD: Atherosclerotic Cardiovascular Disease, CVD: Cardiovascular Disease, BID: Twice Daily, DM: Diabetes Mellitus, FCS: Familial Chylomicronemia Syndrome, IPE: Icosapent Ethyl, RF: Risk Factor, TG: Triglycerides

IX. PROGNOSIS

There is tool that was developed by American College of Cardiology (ACC) / American Heart Association (AHA) to calculate the prognosis of a patient with dyslipidemia in 10-year atherosclerotic cardiovascular disease risk and to assist physicians in implementing shared decision-making before statin therapy initiation. This calculator factors in risk for patients based on lipid levels as well as age, race, tobacco use, diabetes, sex, and hypertension history. This calculation will give a percentage of risk that a cardiovascular event will happen in 10 years.

X. COMPLICATION

The most significant complication for dyslipidemia is cardiovascular disease. Complications of this include sudden cardiac death, acute myocardial infarction, or stroke. Multiple studies have demonstrated that statin and appropriate dyslipidemia treatment has significantly reduced the risk of all-cause mortality, cardiovascular events, and cardiovascular mortality.

XI. REATMENT RECOMMENDATION

1. Familial Hypercholesterolemia

- Persons with FH have a very high lifetime risk of developing ASCVD. FH should be suspected in adults with untreated LDL-C levels ≥ 190 mg/dL or non-HDL-C levels ≥ 220 mg/dL who have a family history of high cholesterol or ASCVD in first-degree relatives.

2. Hypertriglyceridemia

- Elevated TG levels are strongly associated with increased ASCVD risk, but the direct role of TG in ASCVD development is debatable.
- Advise all patients with elevated TG to implement lifestyle interventions shown to reduce TG levels, including 5% -10% reduction in body weight, reduced consumption of sugar and refined carbohydrates, increased physical activity, smoking cessation, and restricted alcohol intake.
- Identify and address secondary causes of hypertriglyceridemia (eg, uncontrolled diabetes and CKD, medications such as protease inhibitors and atypical antipsychotics, glucocorticoides).
- Fasting TG levels >500 mg/dL are associated with pancreatitis and other complications. Dietary fat restriction is essential because it reduces synthesis and entry of additional chylomicrons into the circulation. Lipid-lowering therapies that primarily lower TG (eg, fibrates, omega-3 PUFA, niacin) are recommended. Statins may be reasonable first-line options in patients with an ASCVD risk of $\geq 7.5\%$. Treatment success is defined as reducing TG below 500 mg/dL and preventing pancreatitis.

3. Low HDL Cholesterol

- Low HDL-C is a strong independent risk predictor of coronary heart disease and is defined as <40 mg/dL for men and <50 mg/dL for women. There is no specified goal for HDL-C raising, and the primary target in these patients remains lowering LDL-C.
- Lifestyle modifications (eg, smoking cessation, increasing physical activity) are the preferred treatment approach.
- Niacin can produce the greatest increase in HDL-C compared to other lipid-lowering therapies, but no randomized clinical trial data have shown reduction in ASCVD risk by raising HDL-C.

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MANAGEMENT OF DIABETES IN HOSPITALIZED PATIENT

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I. DEFINITION

Diabetes mellitus is a heterogeneous metabolic disorder characterized by the presence of hyperglycemia due to impairment of insulin secretion, defective insulin action or both. The chronic hyperglycemia of diabetes is associated with relatively specific long-term microvascular complications affecting the eyes, kidneys and nerves, as well as an increased risk for cardiovascular disease (CVD).

Patients with type 1 or type 2 diabetes mellitus are frequently admitted to a hospital, usually for treatment of conditions other than the diabetes. In one study, 25 percent of patients with type 1 diabetes and 30 percent with type 2 diabetes had a hospital admission during one year; patients with higher values for glycated hemoglobin (A1C) were at highest risk for admission. The prevalence of diabetes rises with increasing age, as does the prevalence of other diseases; both factors increase the likelihood that an older person admitted to a hospital will have diabetes.

II. EPIDEMIOLOGY

The IDF Diabetes Atlas (2021) reports that 10.5% of the adult population (20-79 years) has diabetes. By 2045, IDF projections show that 1 in 8 adults, approximately 783 million, will be living with diabetes, an increase of 46%.

In Cambodia:

- STEPS Survey, 2010 (Aged 25-64 years)

Diabetes prevalence is about 3% in rural areas and 6% in city

- STEPS Survey 2016: prevalence 9.6%

Patients with diabetes have a 3-4-fold greater chance of hospitalization compared to those without diabetes

III. ETIOLOGY

- Stress
- Infections
- Perioperative condition
- Uncontrolled diabetes

IV. PATHOPHYSIOLOGY

Major role: Impaired insulin secretion: Absolute or relative insulin deficiency

Impaired insulin action (sensitivity): Insulin resistance

Contributing role: - Decreased incretin effect

- Increased glucagon secretion

- Increased renal glucose reabsorption

- Neurotransmitter dysfunction

The development of hyperglycemia results in an inflammatory state characterized by an elevation of pro-inflammatory cytokines and increased oxidative stress markers. Circulating levels of TNF- α , IL-6, IL-1 β , IL-8, and C-reactive protein are significantly increased two- to fourfold on admission in people with severe hyperglycemia compared with control subjects, and levels returned

to normal levels after insulin treatment and resolution of hyperglycemic crises. Raised concentrations of TNF- α lead to insulin resistance at the level of the insulin receptor and through altered regulation of the insulin-signaling pathway. In addition, preventing insulin-mediated activation of phosphatidylinositol 3- kinase TNF- α reduces insulin-stimulated glucose uptake in peripheral tissues

Counter-regulatory hormones also play an essential role in regulating glucose production and utilization. Glucagon is the most important glycogenolytic hormone, and therefore regulates hepatic glucose production in healthy individuals and in every state of hyperglycemia. During stress, excess concentration of counter-regulatory hormones results in altered carbohydrate metabolism by inducing insulin resistance, increasing hepatic glucose production, and reducing peripheral glucose utilization. In addition, high epinephrine levels stimulate glucagon secretion and inhibit insulin release by pancreatic β -cells

V.DIAGNOSIS

A. Clinic

- Polyuria, Polydipsia, Polyphagia and nocturia
- Weight loss, fatigue, and blurred vision
- Severe asthenia
- Fever
- Dyspnea
- Agitation

B. Investigation

❖ A diagnosis of diabetes can be made from any one of the following four criteria:

1. Fasting Plasma Glucose (FPG) ≥ 126 mg/dL (minimum fasting of 8 hours)
2. 2-hour Plasma Glucose (PG) ≥ 200 mg/dL during the 75-gram oral glucose tolerance test
3. Classic symptoms of hyperglycemia or hyperglycemic crisis associated with a random PG level ≥ 200 mg/dL
4. HbA1c $\geq 6.5\%$ (Note that criteria 1, 2 and 4 should be confirmed by a repeat test)

❖ Other investigations:

- Renal function: plasma creatinine (eGFR), micro-albuminuria
- Lipids: LDL-C, HDL-C, total cholesterol, triglyceride
- Follow up plasma electrolytes
- Consider
 - ECG, cardiovascular screening
 - Thyroid function tests if there is a family history or clinical suspicion
 - Chest X-ray, in case of suspicion for respiratory infections
 - Blood, urine culture

VI.IFFERENCIAL DIAGNOSIS:

- Hyperosmolar hyperglycemic state
- Diabetic ketoacidosis
- Hypoglycemia

VII.MANAGEMENT

1. Objective

- Avoid hypoglycemia
- Avoid severe hyperglycemia, volume depletion, and electrolyte losses

- Assess patient educational needs and address informational deficiencies
- Ensure appropriate glucose management upon discharge until patient can be seen by the clinician managing his or her diabetes as an outpatient

Glycemic Targets

- ❖ **Glycemia targets during hospitalization:**
 - Random blood glucose concentration of more than 140 mg/dL has been regarded as a threshold to consider the diagnosis of inpatient hyperglycaemia. Several target ranges have been investigated in the intensive care unit (ICU) setting.
 - **Noncritically ill** — For most noncritically ill hospitalized patients with diabetes, we suggest a preprandial blood glucose target <140 mg/dL with all random glucose values <180 mg/dL.
 - **Critically ill** — For the majority of critically ill patients, we agree with ADA recommendations for a blood glucose target of 140 to 180 mg/dL. Achieving this goal will usually require an insulin infusion, which should be initiated for persistent hyperglycemia ≥ 180 mg/dL (10.0 mmol/L).

2. Critical signs:

- Severe dehydration
- Shortness of breath
- Confusion
- Coma (ketoacidosis, non- ketotic hyperosmolar)

3. Mortality Risk Factors:

- Diabetic comorbidities such as Cardiovascular-renal disease
- Causal disease such as infections

4. Treatment of hyperglycemia during hospitalization

- Treatment of hyperglycemia in hospitalized patients depends upon the type of diabetes, the patient's current blood glucose concentrations, prior treatment, the clinically assessed severity of illness, and the expected caloric intake during the acute episode.
- **Insulin delivery**
 - **Basal-bolus (or basal-nutritional) insulin regimens** — Although most patients will have type 2 diabetes, many will require at least temporary insulin therapy during inpatient admissions. In such patients, insulin may be given subcutaneously with an intermediate-acting insulin, such as neutral protamine hagedorn (human NPH) or a long-acting (basal) insulin analog, such as glargine, detemir, combined with a pre-meal short acting or rapid insulin (human insulin, lispro, aspart, glulisine) in patients who are eating regular meals (ie, a so-called "basal-bolus" regimen).
 - **Sliding-scale insulin** — In the usual patient with type 2 diabetes managed with oral agents or injectable glucagon-like peptide 1 (GLP-1)-based therapies, and whose glucose management on admission appears at goal, the temporary use of a sliding scale is reasonable for just one to two days as the trajectory of the patient's glycemia becomes apparent. However, after this period of time, a decision should be made about the need for a more physiological glucose management strategy for the remainder of the hospitalization.
 - **Correction insulin** — Varying doses of short acting or rapid-acting insulin can be added to usual pre-meal insulin in patients on basal-bolus regimens to correct pre-meal glucose excursions. In this setting, the additional insulin is referred to as

"correction insulin", which differs from a sliding scale because it is **added** to planned mealtime doses to correct for pre-meal hyperglycemia. Correction dosing typically starts when the pre-meal blood glucose is ≥ 150 mg/dL. The dose of correction insulin should be individualized based upon relevant patient characteristics, such as previous glycemia, previous insulin requirements, and, if possible, the carbohydrate content of meals.

- **Insulin infusion** — Most patients with type 1 or type 2 diabetes admitted to the general medical wards can be treated with subcutaneous insulin. However, some practitioners institute intravenous insulin therapy (human regular insulin in solution) in certain circumstances, such as those with marked hyperglycemia (blood glucose >300 to 350 mg/dL) or in type 1 diabetes, especially those undergoing long and difficult surgery or those who will be expected to have significantly curtailed oral nutritional intake for several days postoperatively. Insulin infusions are typically used in critically ill intensive care unit (ICU) patients, rather than in patients on the general medical wards of the hospital.

For intravenous regular insulin infusion, we recommend starting with approximately half the patient's usual total daily insulin dose, divided into hourly increments until the trend of blood glucose values is known, and then adjusting the dose accordingly. A reasonable regimen usually involves a continuous insulin infusion at a rate of 1 to 5 units of regular insulin per hour; within this range, the dose of insulin is increased or decreased based on frequently measured glucose concentrations, ideally through the use of an approved protocol. In patients who are not eating, concomitant glucose infusion is necessary to provide some calories, reduce protein loss, and decrease the risk of hypoglycemia; separate infusions allow for more flexible management.

When the patient receiving intravenous insulin is more stable and the intercurrent event has passed, the prior insulin regimen can be resumed, assuming that it was effective in achieving glycemic goals. Because of the short half-life of intravenous regular insulin, the first dose of subcutaneous insulin must be given before discontinuation of the intravenous insulin infusion. If intermediate- or long-acting insulin is used, it should be given two to three hours prior to discontinuation, whereas short- or rapid-acting insulin should be given one to two hours prior to stopping the infusion.

- **Patients with type 2 diabetes** — Patients who takes insulin before hospitalization should receive insulin throughout the admission.

For a patient with reasonable glycemic management ($A1C \leq 8$ percent) at home with diet or an non-insulin agent(s), initial correction insulin alone with meals (varying doses of rapid-acting insulin) is a reasonable initial treatment strategy. If the patient is unable to eat normally, oral agents (or injectable GLP-1-based therapies) should be discontinued. In patients who are eating and who do not have contraindications to their oral agent, oral agents (or injectable GLP-1-based therapies) may be cautiously continued if they are on the hospital's formulary. However, if contraindications develop or if blood glucoses are persistently elevated >180 mg/dL, these drugs should be discontinued and a more formal and comprehensive insulin regimen prescribed.

Therapy should be returned to the patient's previous regimen (assuming that it had been effective) as soon as possible after the acute episode, usually as soon as the patient has resumed eating his or her usual diet.

In those with elevated A1C upon admission, the discharge regimen should be modified to improve glycemic management, or at the very least, the patient should be evaluated by the clinician managing his or her diabetes soon after discharge.

- **Diet-treated patients** — Patients with type 2 diabetes treated by diet alone who are to have minor surgery or an imaging procedure, or who have a noncritical acute illness that is expected to be short lived, will typically need no specific antihyperglycemic therapy. Nevertheless, regular blood glucose monitoring is warranted to identify serious hyperglycemia, especially if steroid therapy is administered. Insulin therapy should be instituted if the preprandial blood glucose concentration is persistently ≥ 180 mg/dL, but the patient should be informed that insulin may not be necessary after the episode is over.
- **In patients who are not eating**— correction insulin can be administered every six hours as a correction/sliding scale. Correction insulin with rapid-acting analogs can also be used, but the dosing frequency may need to be every four hours, so the more cost-effective regular insulin is preferred. If substantial doses are required, adding basal insulin will improve glycemia and allow reduced the doses of regular insulin.
- **If the patient is eating**— if fingerstick glucose levels are consistently elevated (≥ 180 mg/dL, a basal-bolus insulin regimen should be initiated. Insulin requirements can be estimated based upon a patient's body weight. Alternatively, requirements can be based upon the total number of units of correction insulin administered over the course of a hospital day. Approximately 50 percent of the total daily dose can be given as basal insulin, and the remaining approximately 50 percent can be given in equally divided doses prior to meals (one-third prior to each meal).

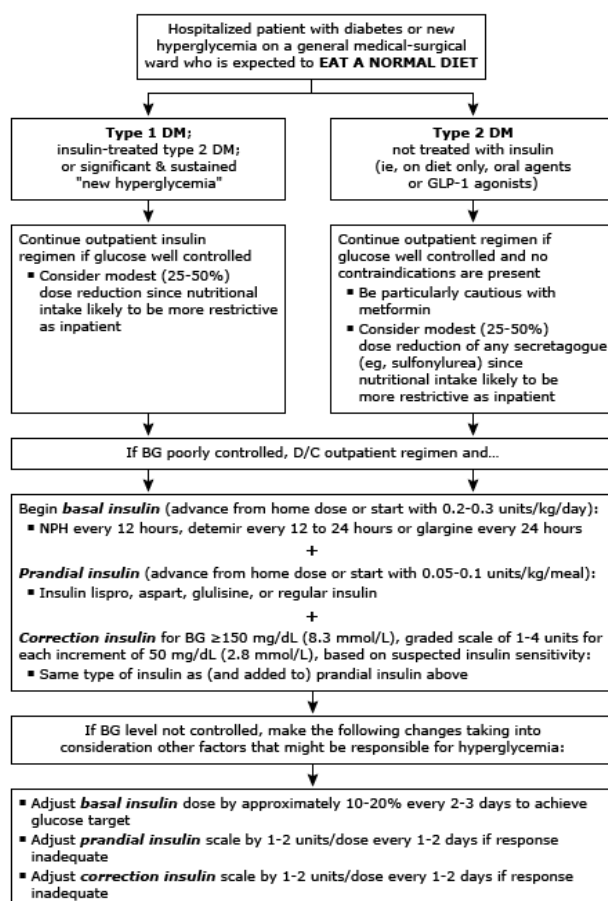


Figure 1: Algorithm for Hospitalized patients who **EAT normal Diet**

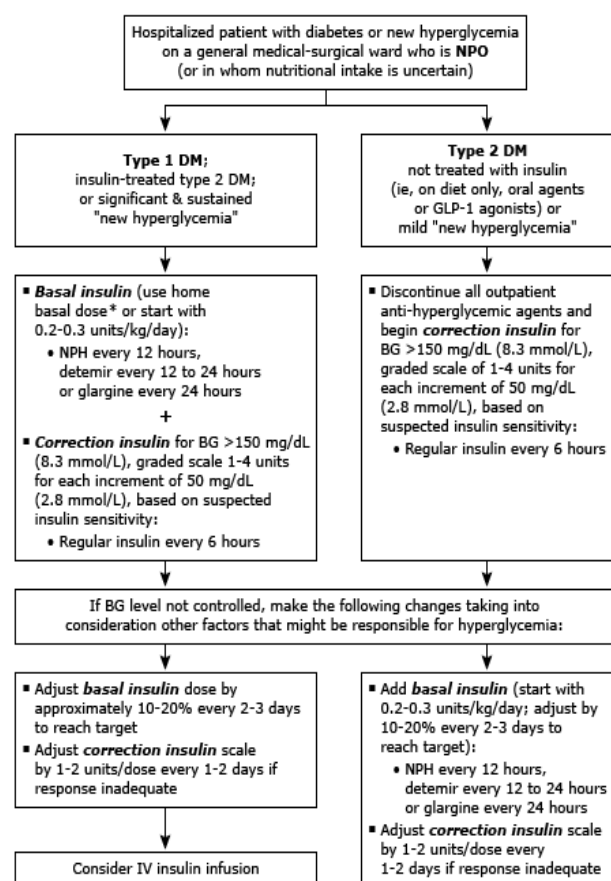


Figure 2: Algorithm for Hospitalized patients who **Do not Eat or NPO**

- **Oral agents**— should generally not be administered to patients who are not eating. In addition, many oral agents have specific contraindications that may emerge in hospitalized patients:
 - **Metformin** – Metformin is contraindicated in situations in which kidney function and/or hemodynamic status is either impaired or threatened, due to the increased risk of lactic acidosis. Examples include patients with acute cardiac or pulmonary decompensation, acute kidney injury, dehydration, sepsis, urinary obstruction, or in those undergoing surgery or radiocontrast studies. Given the typical case mix in most acute care hospitals, metformin should probably be discontinued at least temporarily in most patients.
 - **DPP-4 inhibitors** – Dipeptidyl peptidase 4 (DPP-4) inhibitors (eg, sitagliptin, saxagliptin, linagliptin, alogliptin) exert their effect mainly in the postprandial setting. As a result, they have an uncertain role in patients who are not eating. DPP-4 inhibitors have not been used or studied extensively in the acute care setting. Two studies suggested that they may be reasonably effective in mildly hyperglycemic patients with type 2 diabetes who are eating.
 - **Sulfonylureas** – Sulfonylureas (eg, glyburide, glipizide, glimepiride) are associated with hypoglycemia that can be severe and prolonged. Although they may be continued during the hospitalization in stable patients who are expected to eat regularly, unexpected alterations in meal intake will increase the risk for hypoglycemia. On balance, sulfonylureas should usually be discontinued, at least temporarily, in the hospitalized patient.
 - **GLP-1-based therapies** – GLP-1-based therapies (eg, exenatide, liraglutide, lixisenatide [limited availability], dulaglutide, semaglutide, tirzepatide) may result in nausea and decreased calorie intake and are more effective for postprandial glucose management. As a result, their use should generally be avoided in the acute setting.
 - **SGLT2 inhibitors** – Sodium-glucose cotransporter 2 (SGLT2) inhibitors (eg, dapagliflozin, canagliflozin, empagliflozin) promote the renal excretion of glucose. They increase calorie losses as well as risk of dehydration, volume contraction, and genitourinary tract infections. In addition, euglycemic diabetic ketoacidosis has been reported in patients with both type 1 (during off-label use) and, more rarely, type 2 diabetes who were taking SGLT2 inhibitors. These drugs should therefore generally not be used in the inpatient setting, particularly when patients are acutely ill, although they may be started just prior to discharge if compelling indications are present. For example, these agents may be used by cardiologists and especially heart failure specialists for the benefit of SGLT2 inhibitors in the setting of heart failure.
 - **Thiazolidinediones** – The thiazolidinediones (eg, pioglitazone) are associated with peripheral edema and should be avoided in patients with heart failure. If the question of ventricular dysfunction is raised during a hospitalization, thiazolidinediones should be held until the situation is clarified. The antihyperglycemic effect of this drug class extends for several weeks after discontinuation (as does the fluid-retaining effect), so that temporary interruption of therapy should have little effect on glycemia.
 - **Alpha-glucosidase inhibitors** – Alpha-glucosidase inhibitors (eg, acarbose) are generally used infrequently and are not typically on hospital formularies. Moreover, these inhibitors of intestinal carbohydrate absorption are only effective in patients who are eating and therefore have a limited role in this setting.
 - **Reduced or absent oral intake** – When food intake is diminished or stopped completely in a patient treated with insulin (eg, in preparation for surgery or other procedures), planned pre-meal rapid-acting insulin is discontinued, and the dose of intermediate- or long-acting insulin is reduced. If an episode is clearly short lived (eg, a procedure done in the early morning), subcutaneous insulin (usual dose of short acting and intermediate or long acting) and breakfast can simply be delayed until after the episode.

Once the patient is taking a normal diet, the usual at-home regimen can be restarted, as long as glycemic goals were being met. If altered nutritional intake is present, reduced doses of insulin will be required, starting with doses similar to those administered in the preprocedure setting.

- **Patients with type 1 diabetes** — Patients with type 1 diabetes have an absolute requirement for insulin at **all times**, whether or not they are eating, to prevent ketosis. The doses of insulin needed are usually lower than in patients with type 2 diabetes since most of the former do not have insulin resistance. However, their blood glucose concentrations tend to fluctuate more during the course of the illness or procedure. It is important to avoid hypoglycemia, even if the consequence is a temporary modest rise in the blood glucose concentration.
- If the patient is receiving nothing by mouth, the administration of basal insulin is still required. As examples:
 - **Patients using an insulin pump** – For a patient treated with a continuous insulin infusion via a pump, the basal insulin can be continued at the usual rate, or the overnight basal rate may be reduced by 20 percent to minimize the risk of hypoglycemia the next morning. No boluses would be administered until the patient is able to eat.
 - **Patients treated with multiple daily insulin injections** – For a patient who usually takes 20 units of glargine every evening and rapid-acting insulin before each meal (dose based upon pre-meal blood glucose and carbohydrate content of meal), the usual dose of glargine can be continued. In patients with tightly managed glycemia, an alternative approach is to reduce the dose of glargine by 10 to 20 percent (eg, give 16 to 18 units) to minimize the risk of hypoglycemia that might require oral ingestion of calories and thereby delay the planned procedure. Short-acting insulin should not be given, unless significant hyperglycemia is noted, as described above.

For a patient who usually takes 24 units of NPH and 4 units of regular insulin in the morning, one-half to two-thirds (12 to 16 units) of the usual morning dose of NPH insulin can be given (if the morning blood glucose value is not too low, ie, is at least 120 mg/dL. Short- or rapid-acting insulin should not be given, except if significant hyperglycemia >200 to 250 mg/dL is found in the morning. In such cases, only small doses (1 to 4 units) should be given with the goals of reducing the blood glucose level to <180 mg/dL and avoiding hypoglycemia during the ongoing fast.

Blood glucose should be measured every two to three hours until the first meal is eaten. Small amounts of short- or rapid-acting insulin can be administered every four to six hours to correct hyperglycemia (ie, glucose >150 to 200 mg/dL. Intravenous glucose (at approximately 3.75 to 5 g/hour, eg, 5 percent dextrose at 75 to 100 mL per hour) should be given to limit the metabolic changes of starvation.

5. Monitoring:

- **Blood glucose monitoring** — At the time of admission or before an outpatient procedure or treatment, blood glucose should be measured and the result known. In addition, glucose monitoring should be continued so that appropriate action may be taken.
- **Frequency of monitoring** – The frequency of measurement depends upon the patient's status, the results of earlier measurements, and the steps taken as a result of those measurements. Importantly, in patients with diabetes (or hyperglycemia) who are eating,

the blood glucose monitoring should occur just before the meal. In those who are receiving nothing by mouth, or receiving continuous tube feeds or total parenteral nutrition, the blood glucose monitoring should occur at regular, fixed intervals, usually every six hours.

- **Mode of monitoring** – In hospitalized patients, blood glucose monitoring is typically performed through capillary glucose by glucometer. CGM (Continuous Glucose Monitoring) may be useful in selected inpatients, such as those for whom close contact with inpatient providers should be minimized (eg, COVID-19 or other highly transmissible infection) or possibly, in patients at high risk for hypoglycemia.

VIII.SPECIAL CASES

- **Avoidance of hypoglycemia** — Hypoglycemia should be avoided if at all possible. Measures to reduce the risk of hypoglycemia include:
 - Adequate reduction in antihyperglycemic therapy when caloric intake is stopped or reduced.
 - Avoidance of standard "sliding scale" of insulin without consideration of the patient's specific circumstances.
 - Avoidance of overly aggressive attempts to provide "tight" glycemic management intraoperatively [3] or with intensive insulin therapy in critically ill patients [4]. (See "Glycemic control in critically ill adult and pediatric patients".)
- Although relatively brief and mild hypoglycemia does not usually have clinically significant sequelae, hospitalized patients are particularly vulnerable to severe, prolonged hypoglycemia since they may be unable to sense or respond to the early warning signs and symptoms of low blood glucose.
- Hypoglycemia (ie, serum glucose concentration <70 mg/dL) can be harmful due to the effects of counterregulatory hormones, especially catecholamines, which may possibly induce arrhythmias and other cardiac events. This is especially true in older adults and those with preexisting ischemic heart disease. If the blood glucose falls to or below 54 mg/dL (termed "clinically significant hypoglycemia"), cognitive deficits may also ensue, which can lead to falls or aspiration.

IX.VOLUTION AND PROGNOSIS

- Management on time is a key point for better treatment outcome. Uncontrolled hyperglycemia can lead to acute complications of diabetes.
- **Diabetic Ketoacidosis and Hyperglycemic Hyperosmolar State**— There is considerable variability in the presentation of DKA and HHS, ranging from euglycemia or mild hyperglycemia and acidosis to severe hyperglycemia, dehydration, and coma.
 - **Management goals** include restoration of circulatory volume and tissue perfusion, resolution of ketoacidosis, and correction of electrolyte imbalance and acidosis. It is also essential to treat any correctable underlying cause of DKA, such as sepsis, myocardial infarction, or stroke. In critically ill and mentally obtunded individuals with DKA or HHS, continuous intravenous insulin is the standard of care. Individuals with uncomplicated DKA may sometimes be treated with subcutaneous rapid-acting insulin analogs in the emergency department or step-down units. This approach may be safer and more cost-effective than treatment with intravenous insulin. If subcutaneous insulin administration is used, it is important to provide an adequate fluid replacement, frequent POC blood glucose monitoring, treatment of

any concurrent infections, and appropriate follow-up to avoid recurrent DKA. Several studies have shown that the use of bicarbonate in people with DKA made no difference in the resolution of acidosis or time to discharge, and its use is generally not recommended

X.COMPLICATIONS

- Cardiovascular arrest
- Death
- Severe DKA or HHS
- Severe sepsis

XI.EVENTION AND RECOMMENDATION

General goals – The main goals in patients with diabetes needing hospitalization are to minimize disruption of the metabolic state, prevent adverse glycemic events (especially hypoglycemia), return the patient to a stable glycemic balance as quickly as possible, and ensure a smooth transition to outpatient care.

Glycemic goals – For most noncritically ill hospitalized patients with diabetes, we suggest a preprandial blood glucose target <140 mg/dL with all random glucose values <180 mg/dL.

More stringent goals may be appropriate for stable patients with previous good glycemic management, and more conservative targets should be set for older patients and those with severe comorbidities where the heightened risk of hypoglycemia may outweigh any potential benefit.

•Type 2 diabetes

- Patients treated by diet alone usually need no specific antihyperglycemic therapy. Frequent blood glucose monitoring is warranted to prevent serious hyperglycemia from being unrecognized.
- Patients treated with oral antihyperglycemic drugs or injectable glucagon-like peptide 1 (GLP-1)-based therapies can occasionally continue their usual drug regimen in some circumstances (patient eating, glucose well managed, and no contraindication to oral agents. However, if blood glucose levels are poorly managed with the usual oral agents or if the patient is not eating, drug therapy should be discontinued and insulin initiated.
- Insulin therapy should be continued in all patients already taking it to maintain a reasonably constant basal level of circulating insulin.

•Type 1 diabetes

- Patients with type 1 diabetes require insulin at all times, whether or not they are eating. Insulin can be given either subcutaneously or intravenously. For subcutaneous insulin, sliding scales should never be used as the sole insulin. Optimally, basal insulin (glargine, detemir, or neutral protamine hagedorn [NPH]) should be combined with prandial and correction insulin (short or rapid-acting insulin).
- When intravenous insulin is given, once the blood glucose levels are in goal range, glucose (as dextrose or in total parenteral nutrition [TPN]) should be administered to prevent catabolism. Blood glucose should be measured frequently (every one to two hours in patients receiving an insulin infusion).

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GESTATIONAL DIABETES MELLITUS

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I. DEFINITION

Gestational diabetes mellitus refers to a condition in which glucose intolerance develops during pregnancy.

II. EPIDEMIOLOGY

In 2021, Global prevalence of GDM was 14% according to International Association of Diabetes and Pregnancy Study Group (IADPSG) criteria. In 2019, the prevalence of GDM in South Asia was 11.4%, the highest compared to others region which is 3.6 to 6%. In Asia the gestational diabetes mellitus is increasing in prevalence. It varies from widely from 1.2 to 49.5% based on differences diagnostic criteria. For the time being, there is no formal data for Cambodia.

III. ETIOLOGY

Gestational diabetes mellitus cause by hormonally induced insulin resistance that leads to impaired glucose tolerance.

IV. PATHOPHYSIOLOGY

Gestational diabetes mellitus is thought to be cause by:

- Insulin resistance due to placental secretion of anti-insulin hormones
- Maternal hepatic glucose production increase by 15%-50% to meet fetal demand late in pregnancy
- Pancreatic beta cell dysfunction due to genetic, Autoimmune disorders and chronic insulin resistance.

V. DIAGNOSIS

1. Clinical

GDM present no sign or symptoms. Mostly it can be diagnosis during screening by OGTT 75g. Somehow patient with GDM might have acanthosis nigricans, hydramnios, macrosomia, pregnancy-induced hypertension or symptoms which is compatible with pre-eclampsia or eclampsia.

2. Investigation

OGTT 75g is perform at fasting (patient not eating for more than 8 hours). The blood will be taken at fasting (T0). The patient take 75g of glucose and the blood will be taken at 60minute latter (T60mn) and 120minute latter (T120mn). (see table 1)

Screening for GDM, OGTT 75g is done at 24 to 28 weeks of gestation.

Early screening can be done to screen for pre-existing diabetes in women with diabetes risk factors (see the algorithm).

VI. DIFFERENTIAL DIAGNOSIS

In diabetes and pregnancy, we need to differentiate between pre-existing diabetes (Type 1 or type 2, prior to conception or diagnosis during pregnancy) and gestational diabetes Mellitus by using OGTT.

Table1: OGTT 75g result, one abnormal result the diagnosis can be made

	Normal	GDM	Pre-existing DM
T0	<92mg/dL	92 – 125mg/dL	>125mg/dL
T60mn	<180mg/dL	>180mg/dL	N/A
T120mn	<155mg/dL	155 - 199 mg/dL	>199mg/dL

VII. MANAGEMENT

1.Objective

The objective is to avoid maternal and fetal complication of GDM by achieving and maintaining normoglycemia for pregnancy.

The patient glycemia should be maintain in the target as in table 2.

Table 2: Oral Glucose Tolerance Test (OGTT)

condition	Treatment goal
Pre-prandial glucose	≤95 mg/dL
1 hour post prandial goal	≤140mg/dL
2 hours post prandial goal	≤120mg/dL

The frequency of monitoring is depended on actual plan of treatment. (see table 3)

2.Critical sign

- poor glycemic control
- pre-eclampsia in mother
- Threat for preterm delivery
- fetal complication (hydramnios, Macrosomia)

3. risk factor

- Age > 35 years old
- Obesity (BMI≥23kg/m²)
- Previous history of GDM
- Prior delivery of a large Baby (≥3.5kg)
- History of diabetes in a first-degree relative
- Glycosuria

4.Outpatient Management

GDM patient are treated as outpatient and need to follow up during the pregnancy. It can be face to face consultation or telehealth at a frequency of 4 weeks during 1st and 2nd semester and 2weeks during 3rd semester and can be more frequent if poor glycemic control.

The patient needs to do ultrasound for fetal surveillance 28-36 weeks of gestation:

- Serial growth scan is performed every four weeks to assess fetal growth and amniotic fluid volume.
- The rate of fetal growth which will be use to consider treatment, timing and mode of delivery.

5.First-line treatment

Life style behavior change is an essential component in management of GDM. The Patient Should follow medical nutritional Therapy (Annex 1), physical activity (Annex 2).

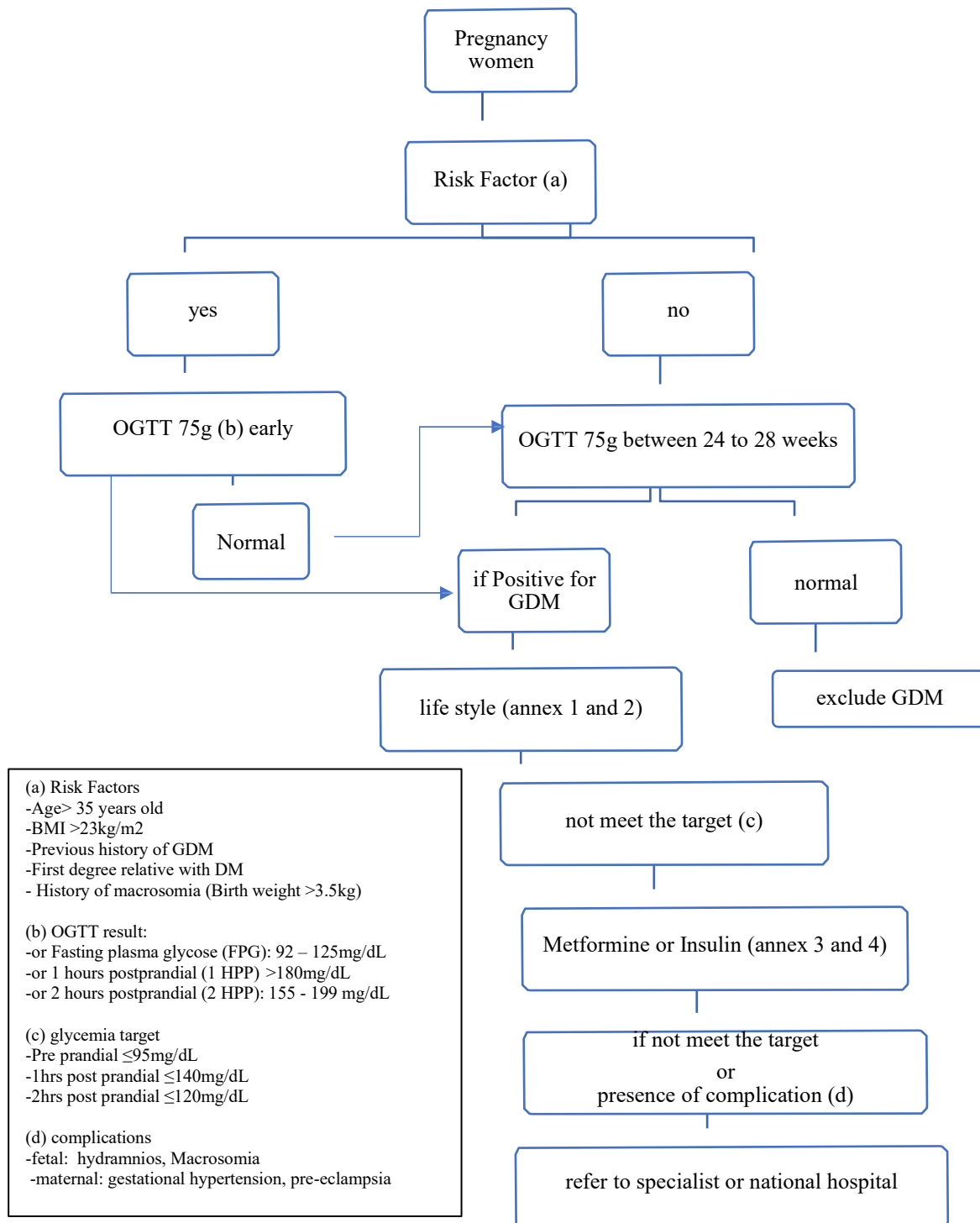
6.Treatment adjustment

Metformine can be given if the blood glucose is not in target (see Table 2) by lifestyle within 1-2 weeks.

Table3: Timing for self-monitoring of blood glucose

Timing for self-monitoring of blood glucose						
	Breakfast		Lunch		Dinner	
	Pre	2h-Post	Pre	2h-Post	Pre	2h-Post
Lifestyle only	√	√		√		√
OAD or Single dose insulin	√	√		√		√
Multiple dose insulin	√	√	√	√	√	√

7.Algorithm



VIII. SPECIAL CASES

All women with GDM are considered as high-risk pregnancy. There need to be managed by a team of multidisciplinary (midwife, nurse, doctor, and if possible, Obstetrician, diabetologist, pediatrician).

IX. EVOLUTION AND PROGNOSIS

- Early management of GDM mostly give good outcome for both mother and baby.
- life style behavior change may suffice.
- Women with GDM have 10 folds increased risk of developing T2DM compared to those without GDM.

X. COMPLICATIONS

Maternal complication:

- Double spontaneous abortion risk
- Hydramnios
- Preterm birth
- Gestational hypertension, preeclampsia
- increased C section delivery
- Increased risk of developing diabetes later in life

Fetal complication:

- Increased risk of still birth
- Increased risk of macrosomia
- neonatal hypoglycemia, hypocalcemia, hyperbilirubinemia
- shoulder dystocia and birth trauma
- Neonatal asphyxia
- Increased risk of developing T2DM later in life

XI. PREVENTION AND RECOMMENDATION

- All pregnancy women shall be screen for GDM at the 24 to 28 weeks of gestation
- Pregnancy women with risk factors shall be screen early.
- Women who was screening positive for GDM should receive nutrition and exercise counseling for life style behavior change.
- life style behavior change is an essential component of management of GDM and may suffice.

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Annex1: Medical nutritional therapy for pregnancy

- Avoid sugars, simple carbohydrates, high processed food, juices and high glucose index fruit.
- Choose healthy low carbohydrate, high fiber source of nutrition, with fresh vegetable
- Eat frequent small meals to reduce risk of postprandial hyperglycemia
- adjust carbohydrate intake based on fasting, premeal and post meal self-blood glucose monitoring.

Annex 2: Physical activities during pregnancy

- Unless contraindicated, physical activity should be included in a pregnant women's daily regimen
- Regular moderate intensity physical activity (ex. Walk, upper body activity that avoid mechanical stress on the abdominal region) can help to reduce glucose level

Annex 3: Insulin type

Name	Type	Onset	Peak Effect	Duration	Recommended dosing interval
Aspart	Rapid acting	15mn	60mn	2hrs	Start of each meal
Regular insulin	Intermediate acting	60mn	2-4 hrs	6hrs	60-90 mn before meal
NPH	Intermediate-acting (basal)	2hrs	4-6hrs	8hrs	Every 8hrs
Detmir	Long-acting (basal)	2hrs	N/A	12hrs	Every 12hrs

Annex 4: Estimation Insulin dosing

Weeks' gestation	Insulin TDD
1-13 weeks	(0,7 x Weight in kg)
14-26 weeks	(0.8 x Weight in kg)
27-37 weeks	(0.9 x Weight in kg)
38 weeks to delivery	(1.0 x Weight in kg)
Postpartum	in GDM, mostly Glycemia return to normal
Use 50% of TDD for basal and 50% for premeal rapid-acting insulin boluses	

TYPE 1 DIABETES MELLITUS: MANAGEMENT AND LONGTERM SURVEILLANCE

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I. DEFINITION

Type 1 diabetes is a chronic illness characterized by the body's inability to produce insulin due to the autoimmune destruction of the beta cells in the pancreas. Although onset frequently occurs in childhood, the disease can also develop in adults.

II. EPIDEMIOLOGY

- Global prevalence of type 1 diabetes is 5.9 per 10,000 people.
- Incidence has risen rapidly over the last 50 years and is currently estimated to be 15 per 100,000 people per year.

III. ETIOLOGY

- Autoimmune destruction of the beta cells of the pancreas
- Genetic predisposition
- Environmental component: viruses (eg, enterovirus, mumps, rubella, and coxsackievirus B4), toxic chemicals, exposure to cow's milk in infancy, and cytotoxins.
- Risk of developing the disease: 2-3% affects to children if mother has type 1 DM, 5-6% if dad has, almost 30% if both have.

IV. PATHOPHYSIOLOGY

Type 1 DM is the culmination of lymphocytic infiltration and destruction of insulin-secreting beta cells of the islets of Langerhans in the pancreas. After 80-90% of the beta cells are destroyed, hyperglycemia develops and diabetes may be diagnosed. Patients need exogenous insulin to reverse this catabolic condition, prevent ketosis, decrease hyperglucagonemia, and normalize lipid and protein metabolism.

Approximately 85% of type 1 DM patients have circulating islet cell antibodies, and the majority also have detectable anti-insulin antibodies before receiving insulin therapy.

V. DIAGNOSIS

1. Clinic

- Signs of uncontrolled hyperglycemia are the same regardless of type of diabetes: polyuria, polydipsia, polyphagia, weight loss, blurred vision.
- 25% of T1DM present is diagnosed by Ketoacidosis onset (DKA): nausea, abdominal pain, severe dehydration, dyspnea, altered mental status.
- Latent autoimmune diabetes in adults (LADA) is a type of diabetes that starts in adulthood and slowly gets worse over time: hyperglycemia will be mild enough as to be asymptomatic.
- Other autoimmune conditions may be present (e.g. hypothyroidism).

2. Investigation

- Criteria for the diagnosis of diabetes:
 - Fasting plasma glucose ≥ 126 mg/dL on more than one occasion

- Random venous plasma glucose ≥ 200 mg/dL in a patient with classic symptoms of hyperglycemia.
- Oral glucose tolerance test (OGTT): Plasma glucose ≥ 200 mg/dL 2 hours after a glucose load of 75 g for adult (1.75 g/kg for children)
- Glycated hemoglobin (A1C) $\geq 6.5\%$ (in children with rapidly evolving type 1 DM, HbA1c may not be significantly elevated)
- Other tests to consider for Type 1 diabetes mellitus:
 - Pancreatic autoantibodies: GAD, others if available (IAA, IA2A, and ZnT8A, while islet cell is not recommended)
 - Urine ketone and Serum β -hydroxybutyrate (BHB)
 - C-peptide level if needed to differentiate from type 2 diabetes because low or no C-peptide indicates insulinopenia
- Screening tests for Associate Disease:
 - White blood cell (WBC) count, blood and urine cultures may be performed to rule out infection.
 - Lipid Profile
 - Thyroid Function test (TSH)

VI. DIFFERENTIAL DIAGNOSIS

- **Other causes of hyperglycemia :** Critically ill patients, medication (glucose, high-dose glucocorticoids, Thiazide diuretics, phenytoin)
- **Other causes of diabetes:** Type 2 diabetes, Monogenic diabetes, Diseases affecting the exocrine pancreas (Cystic fibrosis, hereditary hemochromatosis, and chronic pancreatitis), Endocrine abnormalities in glucose regulation (Cushing syndrome, growth hormone excess, glucagon-secreting tumors, catecholamine excess in pheochromocytoma), Drug-induced diabetes (glucocorticoids, HIV protease inhibitors, cyclosporine, L-asparaginase, tacrolimus and atypical antipsychotic agents)
- **Renal glycosuria without hyperglycemia (Fanconi syndrome)**

VII. MANAGEMENT

1. Objective

Support people with type 1 diabetes to live a long and healthy life:

- Effectively delivering exogenous insulin to maintain glucose levels as close to the individual's target range as is safely possible to prevent the development and progression of diabetes complications while minimizing episodes of hypoglycemia and preventing episodes of DKA.
- Effectively managing cardiovascular risk factors.
- Providing approaches, treatments, and devices that minimize the psychosocial burden of living with type 1 diabetes and, consequently, diabetes-related distress, while promoting psychological well-being.

Table 1: Target for glycemic control in type 1 diabetes

Preprandial glucose	80–130 mg/dL
1–2 h postprandial glucose	<180 mg/dL
Before bedtime	90-150mg/dl
HbA1c	<7.0% without significant hypoglycemia is appropriate

2. Critical Sign : Metabolic acidosis of diabetic ketoacidosis (DKA)
 - Sign: dry mouth, hyperventilation, abdominal pain, nausea, vomiting, and reduced consciousness.
 - Treatment involves rapid hydration, insulin infusion, and correction of electrolyte imbalance. (Cf. DKA, CPG medicine 2013, p.20)
3. Mortality risk factor:
 - conditions related to acute hyperglycemia (DKA), hypoglycemia
 - chronic complications (Kidney and CVD)
4. Outpatient Management
 - Insulin: indicated for everyone with diagnosed T1DM.
 - A clinical decision as to which regimen an individual should follow, and how intensive insulin treatment should be.

Table 2: Types of insulin and their pharmacokinetic profiles

Insulin	Name	Onset (h)	Peak (h)	Duration (h)
Rapid	- Lispro - Aspart	$< \frac{1}{4}$	$\frac{3}{4} - 2\frac{1}{2}$	$3\frac{1}{2} - 4\frac{1}{2}$
Short	Insulin regular	$\frac{1}{2}$	2 – 4	6 – 8
Intermediate	NPH	1 - 3	4 – 12	≥ 18
Long	- Ultralente	4 - 6	8 - 20	≥ 24
	- Glargine	3 - 4	3 - 24	
	- Detemir	1 - 2	3 - 8	
Combination	Premixed rapid- and intermediate-acting	$\frac{1}{2}$	2 – 8	> 8

- Lifestyle modification (Diet modification, weight control, physical activity)
- Self-monitoring blood glucose (SMBG)
- Diabetes self-management education and support (DSME)
- Involvement of different stakeholders like family, school, peers to support T1DM
- Cardiovascular risk factor management: consider low dose aspirin and statin drug therapy for cardiovascular protection in high-risk cardiovascular patients as well as Angiotensin Converting Enzyme Inhibitor or Angiotensin Receptor Blocker therapy for renal protection in patient with hypertension.

5. First line treatment

- Flexible intensive insulin therapy as multiple daily injections (MDI) is the gold standard. It requires 4+ shots per day of basal/bolus insulins.
- To Calculate dose, see algorithm 1. Ex: A patient type 1 diabetes has weight 60kg.
 - $TDD = 60 \times 0.5 \text{ UI} = 30 \text{ UI}$
 - 50% of TDD 15 UI given as basal insulin
 - Other 50% of TDD divided on 3 meals = 5 UI SC given as bolus insulin
- Carbohydrate counting allows a flexible lifestyle with regard to timing and size of meals; meals can even be skipped although not recommended.
- Insulin dose is flexible, encouraging the patient to learn to safely adjust insulin
- Frequent testing of blood glucose guides therapy: test at least before meals

(preprandial goal blood glucose 80-130 mg/dl), occasionally 2 hours after meals (peak postprandial < 180 mg/dl), at bedtime, for symptoms of hyperglycemia or hypoglycemia, and occasionally overnight.

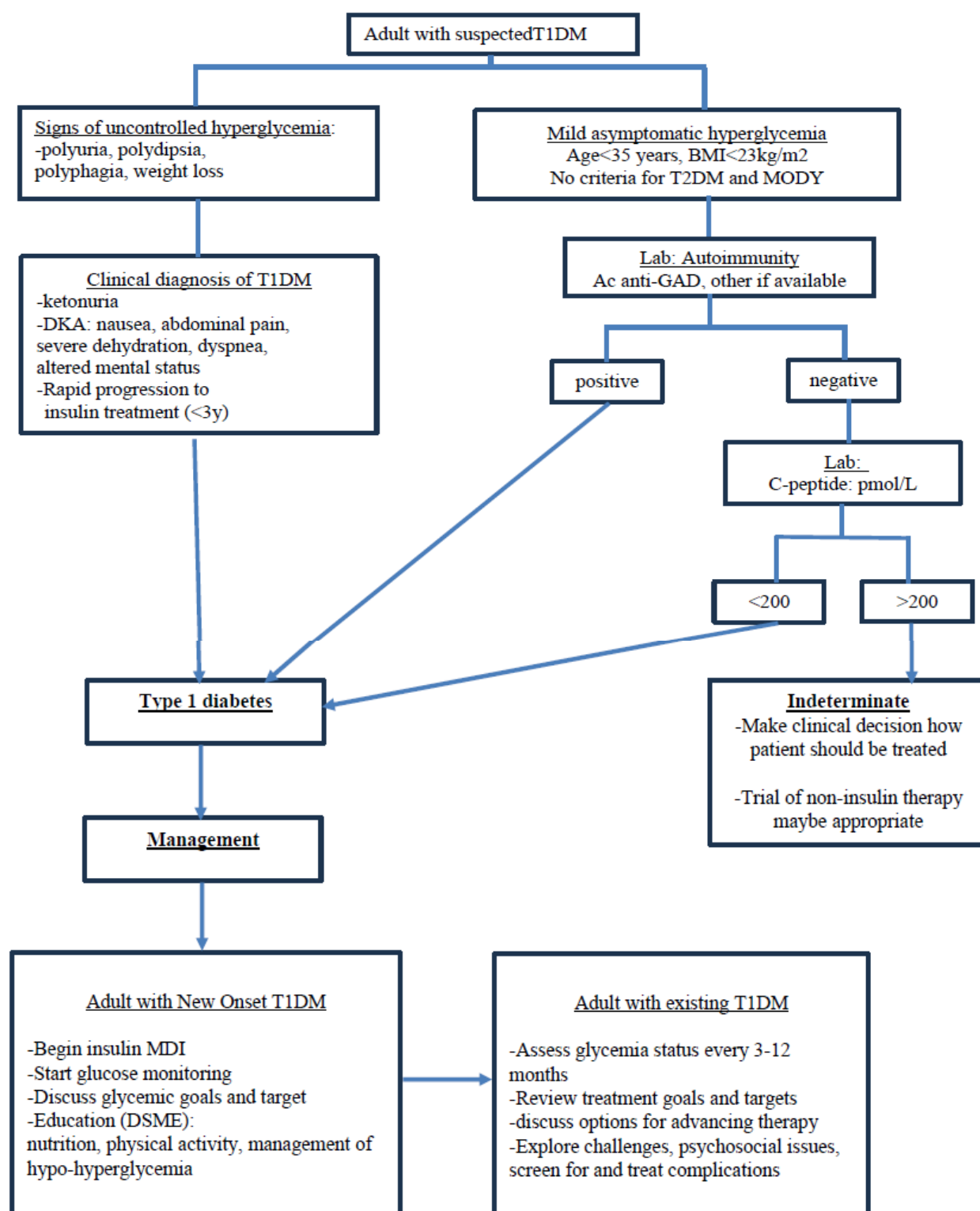
Second line treatment

- Used when patients lack the resources or motivation to adhere to intensive therapy
 - Requires 2 or 3 shots per day Insulin dose is fixed.
 - Diet and exercise must be kept consistent to compensate for fixed insulin doses.
 - Risk of hypoglycemia increased compared to intensive insulin therapy, especially when NPH insulin is peaking.
 - **AM** insulin dose: 2/3 as NPH, 1/3 as Regular (can be mixed in the same syringe or use premixed 70/30 or 75/25 insulin). Breakfast and lunch portions should be matched to this insulin dose. Hyperglycemia at lunch should lead to a smaller lunch. Delayed lunch often causes hypoglycemia.
 - **PM** insulin dose: 2/3 as NPH, 1/3 as Regular or use premixed 70/30 or 75/25 insulin. Dinner and bedtime snack portions should match this insulin dose. Early morning hypoglycemia may result from NPH peak. Move NPH dose to before bed (3 injections/day) to reduce overnight hypoglycemia.
6. Treatment adjustment
- Most insulin dose adjustments should be in 10 to 20% increments depending on the degree of glucose abnormality. Should focus on one particular aspect (exercise, diet, insulin-to-carbohydrate ratios) for a few weeks then switch to another issue.
 - Giving the patient written feedback on diabetes-related management (eg, glycemia chart and/or graphs of the individual's changes in body weight, A1C, etc) is useful.
 - In case of failure under MDI (treatment), consider referring the patient to meet an experienced diabetologist/endocrinologist in T1DM for detailed evaluation and discuss indications for an insulin pump if available.

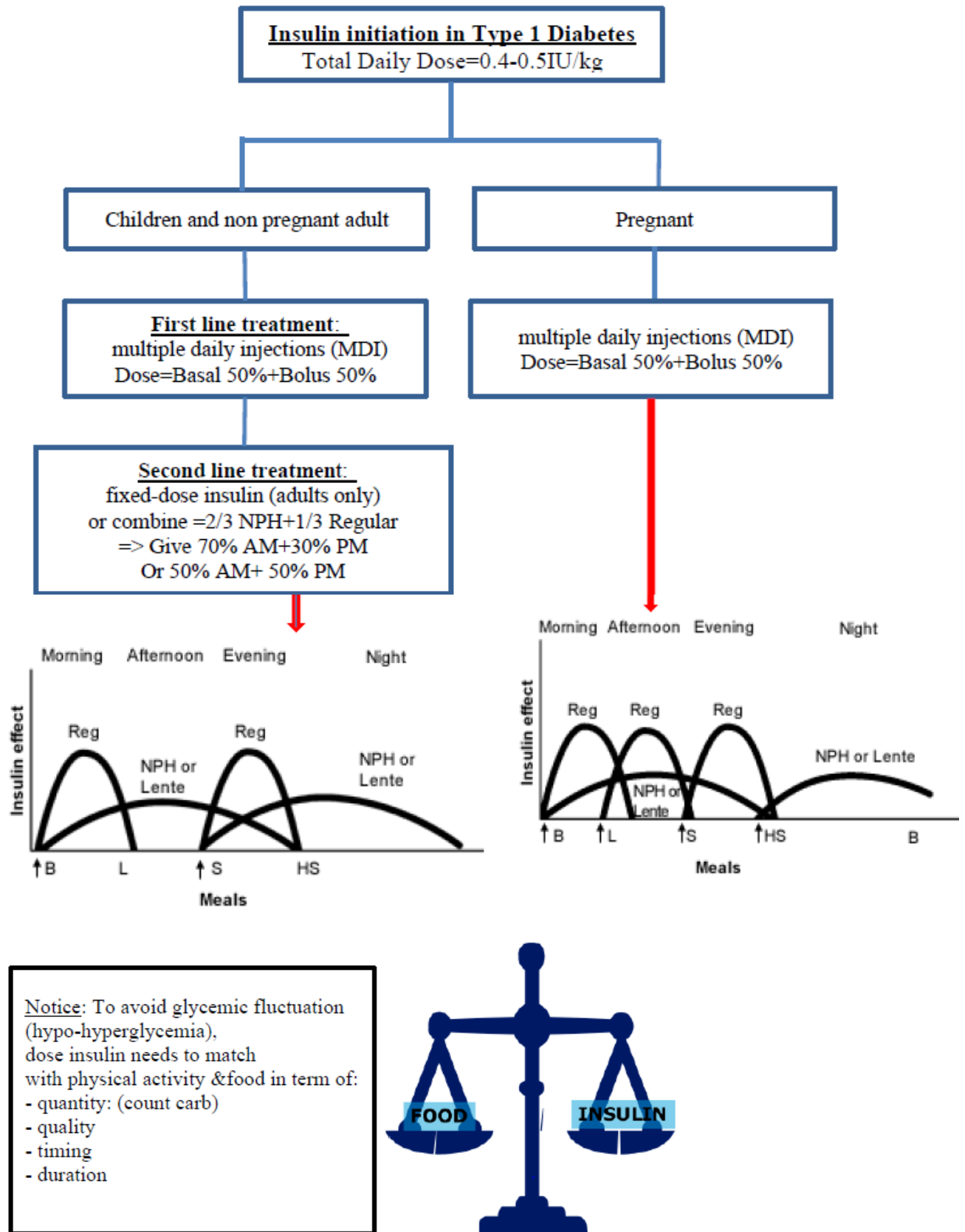
Table 3: Routine care recommendations for patients with type 1 diabetes

	Children/Adolescents (0-19 years)	Adults (≥20 years)
Height	Every 3 months	
Weight	Every 3 months	
Nutritionist	Diagnosis, then annually	
Retinal examination	Begin 5 years after diagnosis Every 1-2 years thereafter	Begin 5 years after diagnosis or Earlier with visual symptoms or if date of T1D onset is unknown Every 1-2 years thereafter
A1C	Every 3 months	
Lipid profile	Annually, once glycemia is stable	Annually or as needed based on treatment
Blood pressure	Every physical examination	
Creatinine/eGFR	At diagnosis, then annually	
Albumin Creatinine Ratio	Begin 5 years after diagnosis, then annually	At diagnosis, then annually

7. Algorithm



Algorithm 2: Insulin initiation in Type 1 Diabetes



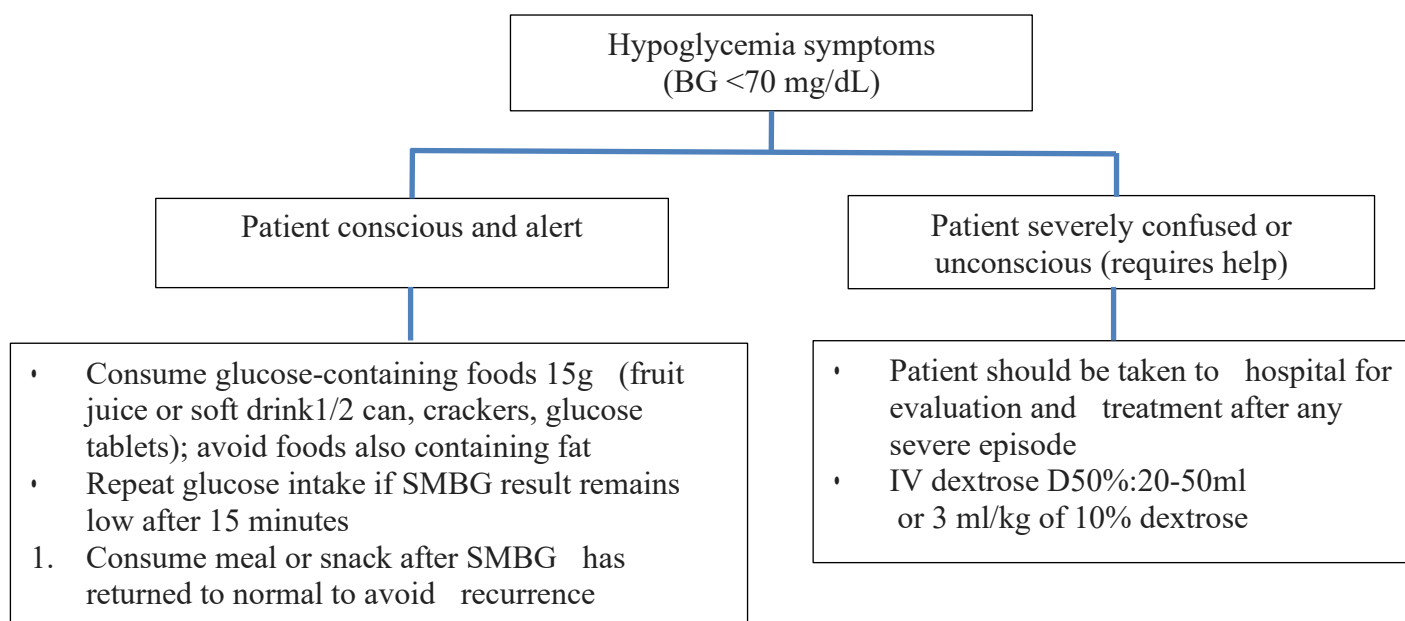
VIII. SPECIAL CASES

1. Hypoglycemia: main limiting factor in the glycemic management of type 1 diabetes.

Table 3: classification and symptom of hypoglycemia

Classification	Glycemia(mg/dL)	Typical Signs and Symptoms
Moderate	<54	Neuroglycopenic: behavioral changes, emotional lability, difficulty thinking, confusion
Severe		Severe confusion, unconsciousness, seizure, coma Requires help from another individual

Algorithm 3: Treatment of hypoglycemia



2. Type 1 diabetes and pregnancy: referral for follow up with multi-disciplinary team, including a dietitian, a nurse, educator, an endocrinologist, and an obstetrician.
3. Diabetes and kidney disease: (cf. Diabetic Nephropathy)
4. Older people with type 1 diabetes: Insulin regimens in older adults should be individualized and patient safety is a key priority. Glycemic targets should be based on functional status and life expectancy, rather than chronological age.
5. Sick-day rules: patient need to do SMBG more frequently during illness (every 2-4h+ night) and check urinary ketones if glycemia >250-300mg/dL for unexplained reason or symptoms as nausea, vomiting, and abdominal pain. Insulin adjustment and hydration are the key to prevent hypoglycemia, significant hyperglycemia, and DKA.
6. Diabetic ketoacidosis (DKA): (cf. CPG medicine 2013, p.20)
7. Lapses in dietary adherence during special occasions:
Ex. restaurant or traveling, religious fasting. Patient should be educated to adjust insulin to prevent marked fluctuations in the plasma glucose.

IX. EVOLUTION AND PROGNOSIS

The evolution and prognosis depend on co-morbidities and complications.

Appropriate management reduce the risk or delay progression of development of the complications.

X. COMPLICATIONS

1. Acute complication: hypoglycemia, Diabetic ketoacidosis (DKA)
2. Chronic complication :
 - Macrovascular complications: Coronary heart disease, Peripheral vascular disease and Cerebrovascular disease
 - Microvascular complications: Nephropathy, Retinopathy and Neuropathy
3. Other complications: Chronic foot ulcer/amputations, Cataract, cutaneous, oral and osteoarticular, increased risk of serious infection.

XI. PREVENTION AND RECOMMENDATION

- No primary prevention for type 1 diabetes. However, acute and chronic complications can be prevented or delayed via effective diabetes health care team with appropriate follow-up.
- The cornerstone of type 1 diabetes therapy is insulin replacement. This is challenging because insulin demands vary widely according to meals, exercise, and many other factors. Furthermore, the insulin doses needed to prevent hyperglycemia are associated with a high risk of hypoglycemia, leaving people with type 1 diabetes walking a tightrope between high and low glucose levels. Insulin management must be supported by adequate monitoring of glucose and education and training to allow the individual with type 1 diabetes to make the most of their treatment regimen.
- Patient education include self-monitoring blood glucose, recognition and treatment of hypoglycemia, DKA, foot care, diet, healthy lifestyle and screening for complications. Education materials include leaflets, books, video, website, digital innovation like mobile application (CarnetDia).
- All patients and their family should be provided with diabetes education and made aware of other available support structures such as peer education or patient associations. It must be emphasized that diabetes is not curable and requires lifelong lifestyle changes and management.

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TYPE 2 DIABETES MELLITUS: MANAGEMENT AND LONG-TERM SURVEILLANCE

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I. DEFINITION

Diabetes mellitus is a heterogeneous metabolic disorder characterized by the presence of hyperglycemia due to impairment of insulin secretion, defective insulin action or both. The chronic hyperglycemia of diabetes is associated with relatively specific long-term microvascular complications affecting the eyes, kidneys and nerves, as well as an increased risk for cardiovascular disease (CVD).

II. EPIDEMIOLOGY

The IDF Diabetes Atlas (2021) reports that 10.5% of the adult population (20-79 years) has diabetes. By 2045, IDF projections show that 1 in 8 adults, approximately 783 million, will be living with diabetes, an increase of 46%.

In Cambodia:

- STEPS Survey, 2010 (Aged 25-64 years)
Diabetes prevalence is about 3% in rural areas and 6% in city
- STEPS Survey 2016: prevalence 9.6%

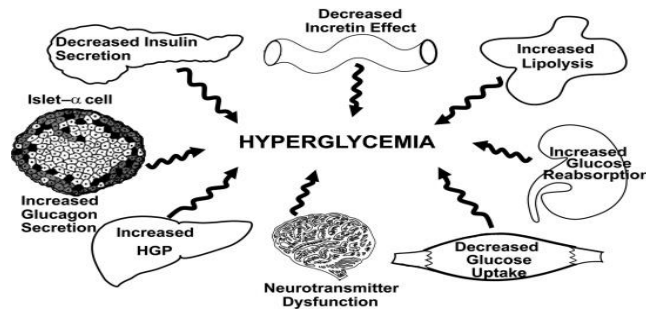
III. ETIOLOGY:

- Genetic
- Environmental factor

Risk factors: Prediabetes, overweight (BMI > 23, waist circumference in men ≥ 85 cm and in women ≥ 80 cm), family history of diabetes, hypertension, dyslipidemia, history of stroke or ischemic heart disease, history of gestational diabetes, history given birth to a large baby (>3500g), age > 35 years old.

IV. PATHOPHYSIOLOGY (summary)

- Impaired insulin secretion: Absolute or relative insulin deficiency
- Impaired insulin action (sensitivity): Insulin resistance



V. DIAGNOSIS

A. Clinic

- Polyuria, Polydipsia, Polyphagia and nocturia
- Weight loss, fatigue, and blurred vision
- Most often asymptomatic in the early stage of diabetes

B. Investigation

- Diagnosis Criteria:

A diagnosis of diabetes can be made from any one of the following four criteria:

1. Fasting Plasma Glucose (FPG) ≥ 126 mg/dL (minimum fasting of 8 hours)
2. 2-hour Plasma Glucose (PG) ≥ 200 mg/dL during the 75-gram oral glucose tolerance test
3. Classic symptoms of hyperglycemia or hyperglycemic crisis associated with a random PG level ≥ 200 mg/dL
4. HbA1c $\geq 6.5\%$ (Note that criteria 1, 2 and 4 should be confirmed by a repeat test)

- Other investigations:

- Renal function: plasma creatinine (eGFR), micro-albuminuria
 - Lipids: LDL-C, HDL-C, total cholesterol, triglyceride
 - Consider
 - ECG, if >50 years old and at least one other cardio vascular risk factor
 - Thyroid function tests if there is a family history or clinical suspicion

V. DIFFERENTIAL DIAGNOSIS

- Type 1 diabetes
- Latent autoimmune diabetes of the adults (LADA)

VI. MANAGEMENT

1. Objective

- Glycemic control
- Cardiovascular risk factor control (blood pressure, dyslipidemia, weight control)
- Prevention of the complications
- Improving quality of life

Glycemic Targets

- HbA1c <7%: long life expectancy, no cardiovascular disease (CVD), short duration of DM, no history of hypoglycemia
- HbA1c <8%: limited life expectancy, advanced micro- or macrovascular complications, extensive comorbidities, history of severe hypoglycemia or long- standing DM where lower goal is difficult to attain.

Glycemic recommendations for many non-pregnant adults with diabetes	
HbA1c	<7% *
Preprandial capillary plasma glucose	80-130 mg/dl
Peak postprandial capillary plasma glucose (1-2 h after the beginning of the meal)*ADA2023	<180 mg/dl

* The target can be modified according to individual with or without complications

2. Critical sign:

- Severe dehydration
- Confusion
- Coma (ketoacidosis, non- ketotic hyperosmolar)

3. Mortality Risk Factors:

- Cardiovascular disease
- Chronic kidney disease

4. Outpatient Management

- Educating patient
- Achieving of targeted glycemic control
- Assess and managing cardiovascular risk factors and co morbidities (control blood pressure, dyslipidemia, smoking cessation, weight reduction...)
- Screening for complications (Diabetic retinopathy, diabetic nephropathy,

neuropathy, coronary heart disease, peripheral artery disease, diabetic foot ulcer)

5. First-line treatment:

- Lifestyle management: Diet modification, weight control, physical activity
- Metformin is the preferred initial pharmacologic agent, if not contraindicated and if tolerated.
- Consider low dose aspirin and statin drug therapy for cardiovascular protection in high risk cardiovascular patients as well as Angiotensin Converting Enzyme Inhibitor or Angiotensin Receptor Blocker therapy for renal protection

6. Treatment adjustment

- Choose initial therapy based on glycemia level and individualized target
- Start with metformin as first line.
- If HbA1c is not at goal in 3 months, add second/third agent. Individualize therapeutic choice based on patient characteristics, amount of glycemic control necessary, availability, adverse effects and cost.
- In patient with established cardiovascular disease, heart failure, or CKD, consider drugs with cardio-renal benefit proven (SGLT2 inhibitor or GLP1 receptor agonist).
- If HbA1c is >9% patient with catabolic symptoms, ketosis, consider adding insulin (starting with basal insulin). If HbA1c not controlled, consider combination injectable therapy with either bolus insulin or switch to premixed insulin.

Medications: Oral and injectable anti hyperglycemic agents

On essential drug list

Drugs	Daily dose	Mode of action	Efficacy	Advantages	Disadvantage
Biguanide*					
Metformine	500-2000mg daily in 2-3 times with/after meals	Reduce hepatic glucose output	Reduce A1c 1% - 2%	No weight gain, may reduce triglycerides	GIT symptoms. Avoid in renal impairment (Cr>1.5 in males or >1.4mg/dL in females or CrCl < 60-70), age>80 years, chronic heart failure and in those with hepatic disease or heavy alcohol intake,
Sulfonylureas (SFUs, second generation)**					
Glibenclamide (glyburide)	2.5 – 20mg 1-2 with meals	Stimulate insulin Release by receptor mediated, glucose independent mechanism	Reduce A1c 1% - 2%	Well tolerated	Hypoglycemia, weight, gain, allergy. Use with precaution in elderly or in patients with liver or renal insufficiency.
Gliclazide	40-320mg 1-2 with meals				
Glimepiride (amaryl)	2-8mg once daily				

Others

Others					
Meglitinides**					
Repaglinide	1.5-16mg 3 times a day	Short acting insulin secretagogue	Reduce A1c 1% - 1.5%	Well tolerated, may have less risk of hypoglycemia compare to sulfonylureas	Hypoglycemia, 3 times a day. More expensive than sulfonylureas
Nateglinide	60-120mg 3 times a day				
Thiazolidinediones (TZDs)*					
Pioglitazone (actos)	15 – 45 mg Once daily	Insulin sensitizer; reduces insulin resistance peripherally	Reduce A1c 0.8% -1.8%	Neutral to Positive effect on CV outcome, carotid IMT	Weight gain, fluid retention, congestive heart failure, anemia; fracture in women; variable lipid effects; rare liver toxicity, increase risk of bladder cancer
Alpha-glucosidase inhibitors (AGIs)					
Acarbose (precise)	50-100mg 3 time a day before meals, start with lower doses	Inhibit gut enzymes that break down carbohydrate	Reduce A1c 0.6% -1.2%	No weight gain	GI side effects, including flatulence, diarrhea, and cramping; rare liver toxicity
Miglitol (Glyset)					

Dipeptidyl peptidase 4 Inhibitor (DPP4-Inhibitors)					
Sitagliptin	50-100mg daily in 1-2 times.	*Enhances glucose dependent insulin secretion. *Suppresses glucagon secretion. *Slows gastric emptying, reducing food intake	Reduce A1c 0.5-1%	Weight neutral, generally well tolerated, minimal risk of hypoglycemia, cardiovascular safety demonstrated	GI side effects including nausea
Vildagliptin	50-100mg daily in 1-2 times				
Linagliptin	2.5mg-5mg once daily				

GLP-1 receptor agonist (GLP1-RA)					
Liraglutide	Injectable S/C Once daily	*Enhances glucose dependent insulin secretion. *Suppresses glucagon secretion.	Reduce A1c 0.6-1.9%	Weight reduction, low risk of hypoglycemia,	GI side effects including nausea, vomiting, diarrhea) Caution: acute pancreatitis, thyroid c- cell tumor
Lixisenatide					
Dulaglutide	Injectable S/C Once weekly	*delay gastric emptying, reducing food intake			
Albiglutide					
Semaglutide (injectable)	Injectable S/C Once weekly				
Semaglutide Oral	2.5mg-5mg once daily				

Sodium-Glucose Transport Protein 2 (SGLT-2 Inhibitors)					
Empagliflozin	10-25mg once daily	*decrease renal glucose reabsorption	Reduce A1c 0.5-1%	Weight loss, no risk of hypoglycemia, benefit on CV mortality reduction, reduce rate of hospitalization of heart failure, slow progression of CKD	Genitourinary tract infections, polyuria, dehydration, risk of DKA
Canagliflozin	100-300mg once daily	*increase glucosuria			
Dapagliflozin	5-10mg once daily	*Slows gastric emptying, reducing food intake			

Insulin:

- Insulin education is necessary before starting insulin therapy. Diabetologist advice in this case is very helpful.
- Generally, starting dose should be ~ 0.1-0.3 UI/kg/day, S/C before breakfast or at bed time.

➤ Indication for insulin injection:

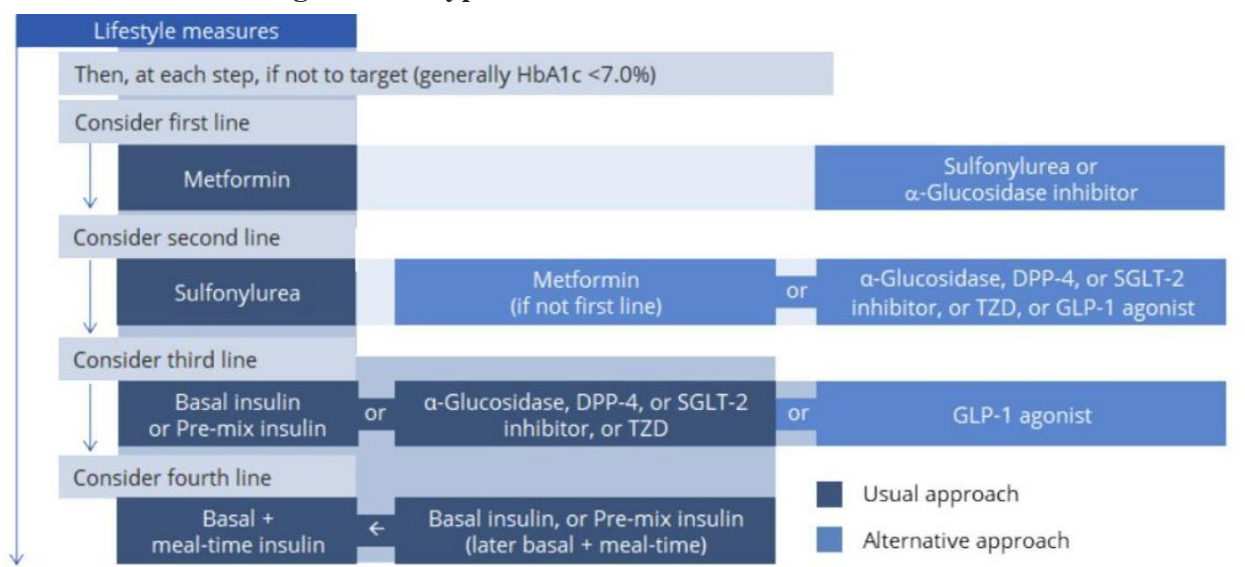
- Type 1 diabetes
- Type 2 diabetes
- Special situation: renal failure, liver disease, use of Rifampicin
- Failure of oral glucose lowering medicines
- T2DM within 5 to 10 years of diagnosis
- Pregnancy or breast-feeding women
- Peri operation conditions
- The early introduction of insulin should be considered if there is evidence of ongoing catabolism (weight loss), if symptoms of hyperglycemia are present, or when A1C levels (>9%) or blood glucose levels > 300mg/dl.

Table 2: Types of insulin and their Pharmacokinetic profiles

Insulin	Name	Onset (h)	Peak (h)	Duration (h)
Rapid	- Lispro - Aspart	< ¼	¾ - 2½	3½ - 4½
Short	Insulin regular	½	2 – 4	6 – 8
Intermediate	NPH	1 - 3	4 – 12	≥ 18
Long	- Ultralente	4 - 6	8 - 20	≥ 24
	- Glargine	3 - 4	3 - 24	
	- Detemir	1 - 2	3 - 8	
Combination	Premixed rapid- and intermediate-acting	½	2 – 8	> 8

7. Treatment Algorithm

Management of type 2 diabetes



DDP-4: Dipeptidyl Peptidase-4; GLP-1: glucagon-like peptid-1, HbA1c: glycated hemoglobin, SGLT-2: Sodium-glucose co-transporter 2; TZD: thiazolidinediones.

VII. SPECIAL CASES

- Type 2 Diabetes and Pregnancy: Specialist advice or referral for follow up with diabetologist is highly recommended.
- Diabetes and Kidney disease. See (cf. Diabetic Nephropathy)
- Diabetes in elderly patients (a more liberal target HbA1C of: 7-8% is recommended for elderly patients). Recognize that sulfonylurea therapy may cause severe hypoglycemia in the elderly.

VIII. EVOLUTION AND PROGNOSIS

The evolution and prognosis depend on co-morbidities and complications.

Optimizing glycemic control reduce the risk or slow progression of development of the complications.

VIII. COMPLICATIONS

- Short term:
 - Hyperglycemia: Diabetic ketoacidosis, Hyperosmolar Hyperglycemic State
 - Hypoglycemia
- Long term:
 - Macrovascular: Coronary heart disease, Peripheral artery disease, Cerebrovascular disease
 - Microvascular: Nephropathy, Retinopathy and Neuropathy

IX. PREVENTION AND RECOMMENDATION

- Patient education include self-monitoring blood glucose, recognition and treatment of hypoglycemia, foot care, diet, healthy lifestyle and screening for complications.
- All patients and their family should be provided with diabetes education and made aware of other available support structures such as peer education or patient associations. It must be emphasized that diabetes is not curable and requires lifelong lifestyle changes and management.
- Early diagnosis and appropriate management are essentials to reduce the risk or delay the progression of the complications.

FOLLOW UP: Patient should be seen every 3 months if stable

- Clinical examination: Glycemia, Height and Weight (BMI), waist circumference, blood pressure, foot exam (every visit)
- Blood test: HbA1c (every 3-6 months), Creatinine (eGFR), liver function test, lipid profile (annually or more frequently depends on previous abnormalities)
- Microalbuminuria or proteinuria (at diagnosis, then annually if normal)
- Fundoscopic exam (annually)

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DIABETIC NEPHROPATHY

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I. DEFINITION

Diabetic Nephropathy is defined by the presence of chronic kidney disease (CKD) characterized by persistently (at least 3 months) elevated urinary albumin excretion (albumin-to-creatinine ratio [ACR] ≥ 30 mg/g) and/or low estimated glomerular filtration rate (eGFR < 60 mL/min/1.73 m²) in a person with diabetes.

II. EPIDEMIOLOGY

Diabetic nephropathy is a common microvascular complication of diabetes and a leading cause of chronic kidney disease (CKD) and end-stage renal disease (ESRD) worldwide. Its prevalence varies based on the population, type of diabetes, and healthcare systems.

Approximately 20-40% of patients with diabetes develop diabetic nephropathy. The burden is increasing due to the global rise in diabetes prevalence, particularly in low and middle-income countries.

III. ETIOLOGY

The exact cause of diabetic nephropathy is unknown, but various postulated mechanisms are hyperglycemia (causing hyperfiltration and renal injury), advanced glycation products, and activation of cytokines.

IV. PATHOPHYSIOLOGY

- **Hyperglycemia** leads to **AGE formation**, which interacts with **RAGEs** on kidney cells, initiating inflammatory and fibrotic pathways.
- **Hyperfiltration and glomerular hypertension** result in **glomerulosclerosis, mesangial expansion, and podocyte injury**.
- **Activation of RAAS** contributes to glomerular injury, fibrosis, and inflammation.
- **Oxidative stress and inflammation** amplify kidney damage, leading to **proteinuria**.
- Progressive fibrosis in the glomeruli and tubulointerstitial space leads to **renal impairment** and eventual **end-stage renal disease (ESRD)**.

V. DIAGNOSIS

Diagnosis is made based on

1. Persistent albuminuria (≥ 30 mg/g ACR) confirmed on at least 2 or 3 tests positive over 3–6 months).
2. Decline in eGFR (< 60 mL/min/1.73 m²).
3. No evidence of other primary renal disease.

VI. DIFFERENTIAL DIAGNOSIS

1. Nondiabetic Glomerular Diseases
2. Hypertensive Nephrosclerosis
3. Acute kidney injury (AKI)
4. Renovascular Disease
5. Primary or secondary Tubulointerstitial
6. Multiple Myeloma and Paraproteinemia-Related Kidney Disease
7. Systemic Diseases Affecting the Kidneys: Systemic Lupus Erythematosus(SLE), Vasculitis, Amyloidosis
8. Obstructive Uropathy
9. Drug-induced Nephrotoxicity

A diagnosis other than diabetic nephropathy should be considered if:

1. Atypical clinical features:
 - Sudden onset or rapid progression of renal dysfunction.
 - Absence of retinopathy (common in DN).
 - Presence of hematuria, especially dysmorphic red blood cells or red blood cell casts.
2. Significant proteinuria without diabetes duration of ≥ 5 years in type 1 diabetes.
3. Presence of systemic symptoms such as fever, rash, or joint pain.
4. Disproportionate kidney size or abnormalities on imaging. eGFR (< 60 mL/min/1.73 m²).
5. No evidence of other primary renal disease.

VII. MANAGEMENT

Effective management involves both preventing progression and controlling associated risk factors.

1. Early Detection

- **Screening for Diabetic Nephropathy:**
 - **Microalbuminuria:** The earliest sign of diabetic nephropathy is the presence of microalbuminuria (albumin excretion 30–300 mg/day), which should be measured annually in patients with type 1 and type 2 diabetes.
 - **Urinary Albumin-to-Creatinine Ratio (ACR):** This test is commonly used for screening.
 - **Serum Creatinine and Glomerular Filtration Rate (GFR):** These help monitor kidney function. An elevated serum creatinine or reduced GFR indicates worsening kidney function.

2. Glycemic Control

- **Tight Glucose Control:** Maintaining HbA1c levels within the target range (usually $< 7\%$) can slow the progression of diabetic nephropathy.
 - **Medications:** Metformin is the first-line oral agent unless contraindicated. SGLT2 inhibitors (e.g., empagliflozin, dapagliflozin) and GLP-1 receptor agonists (e.g., liraglutide, semaglutide) are especially beneficial for patients with diabetic nephropathy as they have renal protective effects.
 - **Insulin:** In patients with type 1 diabetes or uncontrolled type 2 diabetes, insulin therapy may be necessary.

- **Self-monitoring of blood glucose:** Patients should be encouraged to regularly monitor their blood glucose levels.

3. Blood Pressure Control

- **Target BP:** Aim for a target BP of <130/80 mmHg in patients with diabetic nephropathy. Control of hypertension is critical to slowing the progression of renal disease.
- **Antihypertensive Medications:**
 - **Angiotensin-Converting Enzyme (ACE) Inhibitors** (e.g., enalapril, lisinopril) or **Angiotensin II Receptor Blockers (ARBs)** (e.g., losartan, valsartan) are the first-line agents, as they reduce proteinuria and help protect kidney function.
 - **Calcium Channel Blockers** (e.g., amlodipine) and **Diuretics** may be used to manage blood pressure and fluid retention.
 - **Aldosterone Antagonists** (e.g., spironolactone) may also be beneficial in certain cases, particularly in advanced nephropathy with resistant hypertension.

4. Management of Dyslipidemia

- **Statins:** Patients with diabetic nephropathy are at increased cardiovascular risk. Statins (e.g., atorvastatin, rosuvastatin) should be prescribed to manage dyslipidemia and reduce cardiovascular risk.
- **Lipid Targets:** LDL cholesterol should be kept <100 mg/dL, or <70 mg/dL in those at high cardiovascular risk.

5. Dietary Management

- **Protein Restriction:** In patients with significant proteinuria (especially >1 g/day), a moderate reduction in dietary protein intake may help slow the progression of nephropathy. The recommended intake is usually around 0.8 g/kg of body weight per day, but this should be tailored to the individual's needs.
- **Sodium Restriction:** Reducing sodium intake helps control blood pressure and reduce fluid retention.
- **Electrolyte Management:** Monitor potassium and phosphate levels, especially in advanced kidney disease, as imbalances can occur.

6. Renal Protective Therapies

- **SGLT2 Inhibitors:** In addition to their role in glycemic control, these medications (e.g., empagliflozin, dapagliflozin) have shown to reduce the progression of diabetic nephropathy and are now standard care for patients with diabetic kidney disease.
- **Finerenone:** A nonsteroidal mineralocorticoid receptor antagonist, finerenone, has been shown to reduce albuminuria and slow the progression of diabetic kidney disease.

7. Monitoring for Complications

- **Progression to End-Stage Renal Disease (ESRD):** Regular monitoring of GFR and albuminuria to assess progression toward ESRD.
- **Electrolyte Imbalances:** Monitor serum potassium and other electrolytes, especially as kidney function declines.
- **Anemia:** In advanced stages, anemia due to reduced erythropoietin production may occur. Erythropoiesis-stimulating agents (e.g., epoetin alfa) may be needed.
- **Bone Mineral Disorder:** Phosphate binders, vitamin D supplementation, and monitoring for hyperparathyroidism may be necessary in advanced kidney disease.

8. Smoking Cessation

- Smoking accelerates the progression of diabetic nephropathy and increases the risk of cardiovascular disease. All patients should be encouraged to quit smoking.

9. Dialysis and Kidney Transplantation

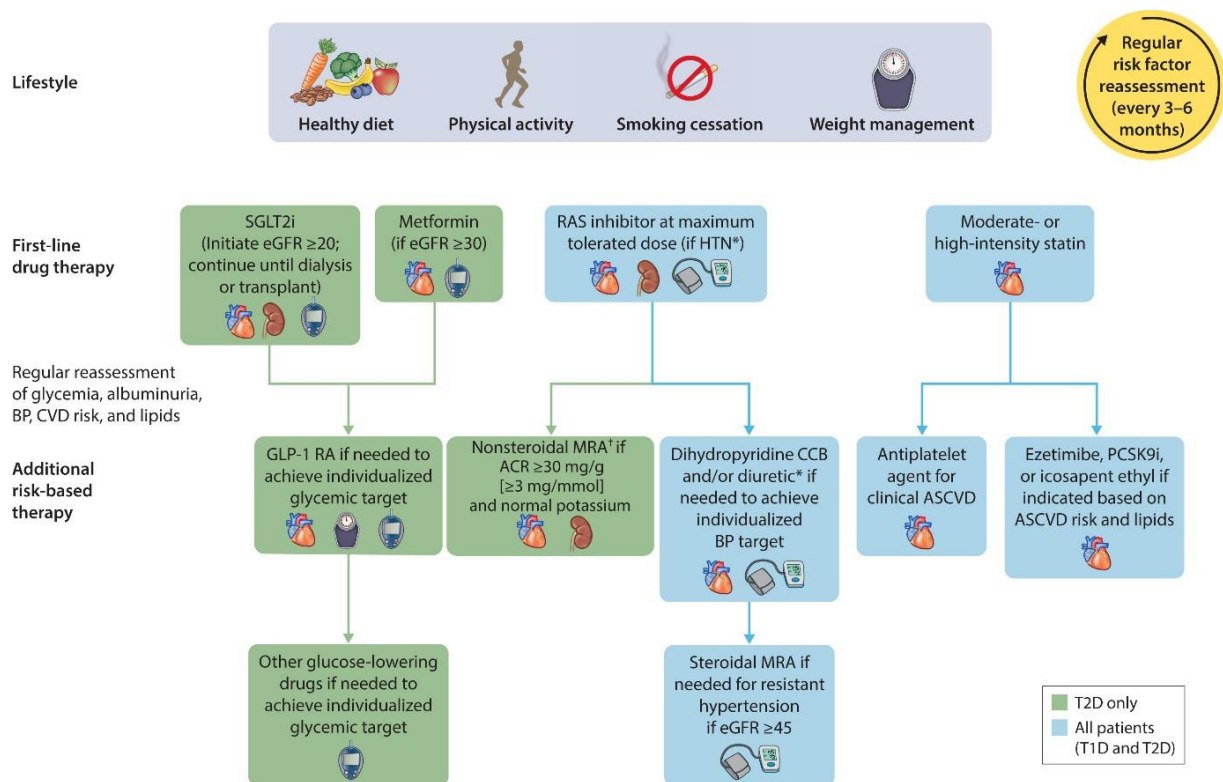
- **Dialysis:** When the patient progresses to ESRD (GFR <15 mL/min/1.73 m²), dialysis may be required. Both hemodialysis and peritoneal dialysis are options.
- **Kidney Transplantation:** If the patient is a suitable candidate, kidney transplantation offers the best option for long-term survival and quality of life.

10. Patient Education and Support

- **Adherence to Medication:** Ensure patients are aware of the importance of taking prescribed medications, especially antihypertensives and renal-protective drugs (e.g., ACE inhibitors, ARBs, SGLT2 inhibitors).
- **Regular Monitoring:** Encourage routine lab tests to monitor kidney function (e.g., serum creatinine, urine albumin/creatinine ratio).
- **Diet and Lifestyle Changes:** Educate on a heart-healthy diet, exercise, weight management, and blood glucose monitoring.
- **Psychosocial Support:** Chronic diseases like diabetic nephropathy can lead to depression or anxiety. Providing counseling or referring to support groups can be helpful.

11. Pharmacotherapy Specific to Diabetic Nephropathy

- **ACE Inhibitors/ARBs:** These reduce proteinuria and help preserve kidney function.
- **SGLT2 Inhibitors:** Reduce blood glucose and have a direct renal protective effect, slowing the progression of kidney damage.
- **Finerenone:** Helps manage diabetic kidney disease, particularly in patients with heart failure and chronic kidney disease.
- **Statins:** For cardiovascular risk reduction.



Algorithm for diabetic nephropathy management

VIII. SPECIAL CASES

- Normoalbuminuric Diabetic Kidney Disease (NADKD):** Some patients with diabetic nephropathy may not exhibit proteinuria. Instead, they present with renal insufficiency and vascular complications¹. This form is characterized by tubulointerstitial injury and fibrosis.
- Diabetic Nephropathy in Patients without a History of Diabetes:** Rarely, diabetic nephropathy can be diagnosed in patients without a history of diabetes. For example, a renal biopsy might reveal diabetic nephropathy in a patient with impaired glucose tolerance (IGT) but no overt diabetes¹.
- Coexistence with Other Renal Diseases:** Diabetic nephropathy can coexist with other renal diseases, such as lupus nephropathy. In such cases, the presence of additional symptoms like accelerated hypertension and an active urinary sediment with red cells and red cell casts can complicate the diagnosis².
- Accelerated Progression in Type 1 Diabetes:** Patients with type 1 diabetes can experience a rapid progression of diabetic nephropathy, often accompanied by nonproliferative retinopathy and hypertension. This can lead to nephrotic syndrome and significant renal insufficiency.

IX. EVOLUTION AND PROGNOSIS

Diabetic nephropathy is a progressive disease that can eventually lead to ESRD if not managed properly. Early detection through regular screening for albuminuria and optimal control of blood glucose and blood pressure are essential in slowing disease progression. Advances in treatment, particularly with SGLT2 inhibitors and renin-angiotensin system blockers, have significantly improved the prognosis of patients with diabetic nephropathy, especially if the disease is caught in the earlier stages.

X. COMPLICATIONS

1. **End-Stage Kidney Disease (ESKD):** This is one of the most severe complications, where the kidneys lose their filtering ability, leading to the need for dialysis or a kidney transplant.
2. **Hypertension:** High blood pressure is both a cause and a complication of diabetic nephropathy. It can further damage the kidneys and other organs.
3. **Cardiovascular Disease:** Patients with diabetic nephropathy are at a higher risk of heart attacks, strokes, and other cardiovascular issues due to the interplay of diabetes, hypertension, and kidney disease.
4. **Nephrotic Syndrome:** This condition is characterized by significant proteinuria (excess protein in the urine), hypoalbuminemia (low levels of albumin in the blood), edema (swelling), and hyperlipidemia (high levels of lipids in the blood).
5. **Anemia:** Reduced kidney function can lead to a decrease in the production of erythropoietin, a hormone that stimulates red blood cell production, resulting in anemia.
6. **Electrolyte Imbalances:** Kidney dysfunction can lead to imbalances in electrolytes such as potassium, calcium, and phosphate, which can cause various symptoms and complications.
7. **Bone Disease:** Chronic kidney disease can affect bone metabolism, leading to conditions such as osteoporosis and bone fractures.

XI. PREVENTION AND RECOMMENDATION

Prevention and management of diabetic nephropathy involve tight control of blood glucose and blood pressure, regular screening for early kidney damage, and lifestyle modifications such as smoking cessation, diet, and exercise. Medications like ACE inhibitors, ARBs, and SGLT2 inhibitors play a critical role in reducing the progression of kidney disease, while managing complications like proteinuria and hyperkalemia is essential for protecting kidney function. Early intervention and regular monitoring are key to preventing ESRD and improving long-term outcomes in diabetic patients.

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MANAGEMENT OF DIABETIC FOOT

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I. DEFINITION

- Diabetic foot syndrome (DFS) refers to the presence of infection, ulceration, or destruction of deep tissues associated with neurological abnormalities and various degrees of peripheral vascular disease in the lower limbs of patients with diabetes.

II. EPIDEMIOLOGY

- Diabetic foot syndrome is a major public health issue, with an estimated global prevalence of 6.3%.
- In France, the prevalence of foot ulcers among diabetic patients is approximately 6%, with a significant proportion leading to hospitalization and, in severe cases, amputation.
- In low- and middle-income countries, the prevalence can be much higher due to limited healthcare infrastructure, lack of proper footwear, and delayed access to preventive care. Studies from countries such as India and parts of Africa suggest that up to 15% of diabetic patients may experience foot ulcers during their lifetime.
- The amputation rate among diabetic patients remains alarmingly high. It is estimated that every 30 seconds, a limb is amputated globally due to diabetes-related complications. The majority of these amputations are preventable with timely intervention, yet up to 85% of amputations are preceded by foot ulcers that could have been managed through conservative treatment if detected early.

III. ETIOLOGY

The primary causes of diabetic foot syndrome include:

- Peripheral neuropathy leading to loss of protective sensation.
- Peripheral arterial disease (PAD) contributing to poor blood supply and delayed wound healing.
- Biomechanical abnormalities, such as foot deformities, increasing the risk of ulcers.

IV. PATHOPHYSIOLOGY

Diabetic foot syndrome typically begins with peripheral neuropathy, causing loss of sensation in the feet. This leads to unrecognized injuries, which, when combined with PAD, result in poor healing and increased susceptibility to infections. The condition may progress to osteomyelitis and gangrene if not adequately managed.

V. DIAGNOSIS

1. Clinical Presentation

The clinical presentation of diabetic foot syndrome varies depending on the underlying pathology:

- Neuropathic ulcers are typically painless and located on pressure points such as the soles of the feet, especially over the metatarsal heads.

- Ischemic ulcers are often painful and found on the margins of the feet, particularly at the tips of the toes.
- Signs of infection include redness, warmth, swelling, purulent discharge, and malodorous wounds. In severe cases, patients may present with fever, indicating systemic infection (e.g., sepsis).

2. Investigations

- **Monofilament Testing:** The use of a 10g monofilament to assess for loss of protective sensation. This is a simple but effective screening tool to identify patients at risk of developing foot ulcers.
- **Ankle-Brachial Index (ABI):** A non-invasive method for evaluating PAD. ABI values of less than 0.9 suggest the presence of PAD, while values below 0.5 indicate critical limb ischemia.
- **Imaging:** X-rays are commonly used to assess bone involvement in infections, such as osteomyelitis. MRI or CT scans may be necessary to evaluate deeper structures, particularly in cases of suspected Charcot foot or soft tissue abscesses.

VI. DIFFERENTIAL DIAGNOSIS

The differential diagnosis of diabetic foot syndrome includes:

- **Venous stasis ulcers,** which are generally located on the medial aspect of the leg and associated with chronic venous insufficiency.
- **Peripheral arterial disease (PAD)** in non-diabetic patients, where ulcers typically develop in areas of reduced perfusion.
- **Osteomyelitis,** where bone involvement must be confirmed with imaging studies.
- **Charcot neuroarthropathy,** characterized by joint destruction and deformity, often mistaken for infection due to the presence of swelling and warmth.

VII. MANAGEMENT

The management of diabetic foot syndrome (DFS) requires a comprehensive, multidisciplinary approach that focuses on early detection, aggressive treatment of ulcers and infections, and the prevention of recurrence. The management strategy must be tailored to the individual patient, considering the severity of the condition, presence of comorbidities, and available healthcare resources.

1. Objectives

The primary objectives of DFS management include:

- **Preventing ulcer formation:** Through regular foot care, proper footwear, and patient education, ulcer formation can be minimized in high-risk individuals.
- **Promoting wound healing:** For patients with established ulcers, the focus shifts to accelerating the healing process and preventing infection.
- **Reducing the risk of amputation:** Prompt management of infections, revascularization when necessary, and wound care are crucial in preserving limb integrity.
- **Improving quality of life:** Ensuring mobility, reducing pain, and preventing complications are key to enhancing the quality of life for patients with DFS.

2. Critical Signs

- **Non-healing ulcers:** Ulcers that do not respond to conservative treatment within 4-6 weeks should be considered for more aggressive interventions.
- **Signs of infection:** These include purulent discharge, erythema extending beyond 2 cm from the wound margin, foul odor, and systemic signs such as fever. Infections

may rapidly progress to deeper tissues, including bones (osteomyelitis), necessitating immediate intervention.

- Gangrene: Tissue necrosis indicates critical ischemia or severe infection and is a medical emergency requiring urgent treatment, often including amputation.

3. Mortality Risk Factors

- Poor glycemic control: Hyperglycemia delays wound healing and increases susceptibility to infections.
- Presence of PAD: Patients with PAD are at increased risk of non-healing ulcers and may require revascularization procedures.
- Delay in treatment: Delayed presentation or management increases the risk of infection and gangrene, leading to higher rates of amputation and mortality.

4. Outpatient Management

The majority of DFS cases can be managed on an outpatient basis, provided the condition is recognized early and aggressive interventions are initiated.

- Foot Care: Regular foot inspections should be part of routine care for all diabetic patients. Appropriate footwear, including custom orthotics, can prevent ulcer formation by reducing pressure on high-risk areas.
- Wound Care: Regular debridement is essential for removing necrotic tissue and promoting a healthy wound bed. Dressing selection should be based on wound characteristics, with moisture balance being a key factor in promoting healing.
- Infection Control: Empirical antibiotic therapy should be initiated based on the clinical presentation, with adjustments made once culture results are available. Superficial infections may be treated with oral antibiotics, while deeper infections may require intravenous antibiotics or hospitalization.

5. First-Line Treatment

The initial management of DFS focuses on conservative treatments aimed at halting the progression of the disease and promoting healing:

- Offloading: Offloading devices, such as total contact casts, removable cast walkers, and specialized therapeutic footwear, reduce pressure on the ulcerated area, promoting healing. Patients should be instructed on the importance of adhering to offloading protocols.
- Glycemic Control: Optimizing blood glucose levels is crucial for enhancing the body's immune response and promoting tissue repair. Insulin therapy or adjustments to oral hypoglycemic agents may be necessary to achieve tight glycemic control.
- Infection Management: Antibiotic therapy should be initiated in cases of suspected or confirmed infection. Treatment should be tailored to the severity of the infection and the results of wound cultures. In cases of osteomyelitis, prolonged antibiotic therapy (6 weeks or more) may be required.

6. Treatment Adjustments

In cases where conservative treatment fails, or the ulcer progresses, more aggressive interventions may be necessary:

- Revascularization: Patients with significant PAD and non-healing ulcers may benefit from revascularization procedures such as angioplasty or bypass surgery to restore adequate blood flow to the lower extremities. Vascular surgeons should be involved early in the management of these patients.

- **Surgical Debridement:** For patients with necrotic tissue, surgical debridement may be necessary to remove infected or dead tissue and promote the formation of healthy granulation tissue.
- **Amputation:** Amputation may be required in cases where limb salvage is no longer possible due to severe infection, gangrene, or extensive tissue necrosis. The goal of amputation is to control infection and preserve as much of the limb as possible to maintain mobility.
- **Advanced Wound Care:** For ulcers that fail to heal with standard treatments, advanced therapies such as negative pressure wound therapy (NPWT), skin substitutes, growth factors, or hyperbaric oxygen therapy (HBOT) may be considered. These therapies are typically used in specialized wound care centers.

VIII. SPECIAL CASES

DFS presents with unique challenges in special cases, such as in patients with Charcot foot or those with severe, deep infections like osteomyelitis. Each case must be carefully evaluated and managed with a specialized approach.

Charcot Foot

Charcot neuroarthropathy is a rare but serious complication of diabetic neuropathy, characterized by joint destruction and significant deformity of the foot. Patients with Charcot foot often present with swelling, warmth, and redness, mimicking an infection. Early diagnosis is critical, as delayed recognition can result in severe deformities that are difficult to manage.

Management of Charcot foot involves:

- **Immobilization:** The affected foot should be immobilized in a total contact cast or a removable walker boot to offload the area and prevent further damage.
- **Temperature Monitoring:** Frequent monitoring of skin temperature can help assess the activity of the Charcot process and guide the duration of offloading.
- **Surgical Stabilization:** In cases where the deformity is severe, or the foot becomes unstable, surgical stabilization with internal fixation may be necessary.

Osteomyelitis

Osteomyelitis, or infection of the bone, is a common complication of diabetic foot ulcers, particularly those that are long-standing or associated with deep tissue involvement.

Diagnosis of osteomyelitis requires a combination of clinical suspicion, imaging (X-ray, MRI, or bone scan), and microbiological testing from deep tissue or bone biopsies.

Management of osteomyelitis includes:

- **Long-term Antibiotic Therapy:** Treatment typically involves 6 weeks of intravenous or oral antibiotics tailored to culture results.
- **Surgical Debridement:** In cases of extensive bone involvement, surgical debridement may be required to remove infected bone and prevent further spread of the infection. In some cases, partial or total amputation may be necessary to control the infection.

IX. EVOLUTION AND PROGNOSIS

The prognosis of diabetic foot ulcers depends on several factors, including the size and depth of the ulcer, the presence of infection, and the adequacy of blood supply. Patients with early-stage ulcers that are managed promptly and effectively have a good chance of healing without complications. However, patients with deep ulcers, infection, or critical ischemia are at higher risk of poor outcomes.

Long-Term Outcomes

Patients with a history of DFS are at increased risk of recurrence. Studies show that up to

70% of diabetic foot ulcers will recur within 5 years, emphasizing the importance of ongoing preventive care and patient education. Regular follow-up with a multidisciplinary team, including podiatrists, endocrinologists, and vascular surgeons, is essential to monitor for signs of recurrence and to implement early interventions when necessary.

Amputation Risk

The risk of amputation is significantly increased in patients with delayed treatment, infection, and ischemia. Following major amputation, mortality rates are high, with up to 50% of patients dying within 5 years. This underscores the critical need for early intervention and aggressive management of DFS.

X. COMPLICATIONS

- Infection: Can progress to osteomyelitis or sepsis.
- Amputation: High risk in cases of severe infection or gangrene.
- Charcot Foot: Leads to significant foot deformity and instability if not treated promptly.

XI. PREVENTION AND RECOMMENDATION

- Patient Education: Emphasize the importance of foot care, regular inspections, and early reporting of foot injuries.
- Regular Screening: Annual foot exams for all diabetic patients to assess for neuropathy and PAD.
- Multidisciplinary Approach: Involving endocrinologists, podiatrists, vascular surgeons, and wound care specialists in the management of diabetic foot patients.

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Figure 1: Diabetic foot management



Figure 2: Types of off-loading shoe



MANAGEMENT OF HYPERTHYROIDISM

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I. DEFINITION

Hyperthyroidism is a set of disorders that involve excess synthesis and secretion of thyroid hormones by the thyroid gland, which leads to the hypermetabolic condition of thyrotoxicosis. The most common forms of hyperthyroidism include diffuse toxic goiter (Grave's disease), toxic multinodular goiter and toxic adenoma.

II. EPIDEMIOLOGY

Thyrotoxicosis is a common condition: the estimated global prevalence of overt hyperthyroidism (ie, elevated T₄ or T₃, or both) in iodine-sufficient populations is 0.2–1.3%. In Europe, the prevalence of overt hyperthyroidism is 0.75%, and the annual incidence is 51 cases per 100 000 people per year.¹ The prevalence and incidence of hyperthyroidism are higher in historically iodine-deficient areas than in iodine-sufficient area.

Graves' disease is the most common form of hyperthyroidism, causing approximately 60-80% of cases of thyrotoxicosis. The annual incidence of Graves' disease was found to be 0.5 cases per 1000 population during a 20-year period, with the peak occurrence in people aged 20-40 years.^[19]

Toxic multinodular goiter (15-20% of thyrotoxicosis) occurs more frequently in regions of iodine deficiency. Toxic adenoma is the cause of 3-5% of cases of thyrotoxicosis

III. ETIOLOGY

1 Hyperthyroidism due to increased thyroid hormone synthesis

A- Autoimmune thyroid disease

- **Grave's disease**

Very common cause of hyperthyroidism. The most important autoantibody is thyroid-stimulating antibody (TSab; also called TSI, LATs, or TRab), which is directed toward epitopes of the TSH receptor and acts as a TSH-receptor agonist. Like TSH, TSab binds to the TSH receptor on the thyroid follicular cells to activate thyroid hormone synthesis and release and thyroid gland growth (hypertrophy).

- **Hashitoxicosis**

Hashitoxicosis is a variant of Hashimoto's thyroiditis that presents with high thyrotropin receptor antibodies and hyperthyroidism (high radioiodine uptake) with transition to overt hypothyroidism over 3 to 12 months.

B- Autonomous thyroid tissue

- **Toxic multinodular goiter**

Toxic multinodular goiter (Plummer disease) accounts for 15-20% of thyrotoxicosis cases (see the image below). It occurs more commonly in elderly individuals, especially those with a long-standing goiter. Thyroid hormone excess develops very slowly over time and often is only mildly elevated at the time of diagnosis.

- **Toxic adenoma**

Toxic adenoma is caused by a single hyperfunctioning follicular thyroid adenoma. This disorder accounts for approximately 3-5% of thyrotoxicosis cases. The excess secretion of thyroid hormone occurs from a benign monoclonal tumor that usually is larger than 2.5 cm in diameter. The excess thyroid hormone suppresses TSH levels. RAIU usually is normal, and

the radioactive iodine scan shows only the hot nodule, with the remainder of the normal thyroid gland suppressed because the TSH level is low

C- TSH-mediated hyperthyroidism

- TSH-producing pituitary adenoma
- Non-neoplastic TSH-mediated hyperthyroidism

D- Human chorionic gonadotropin-mediated hyperthyroidism

- Hyperemesis gravidarum
- Trophoblastic disease (choriocarcinoma, molar pregnancy)

2 Hyperthyroidism due to release of preformed thyroid hormone

Thyroiditis

- Subacute granulomatous (de Quervain's) thyroiditis
- Painless thyroiditis (silent thyroiditis, lymphocytic thyroiditis)
- Postpartum thyroiditis
- Amiodarone (also may cause iodine-induced hyperthyroidism)
- Radiation thyroiditis
- Palpation thyroiditis

3 Hyperthyroidism due to exogenous thyroid hormone intake

- Excessive replacement therapy
- Intentional suppressive therapy
- Factitious hyperthyroidism

4 Ectopic hyperthyroidism

- Struma ovarii (ectopic thyroid tissue in an ovarian teratoma)
- Metastatic follicular thyroid cancer

IV. PATHOPHYSIOLOGY

Binding of TSH to receptors on the thyroid gland leads to the release of thyroid hormones primarily T₄ and to a lesser extent T₃. In turn, elevated levels of these hormones act on the hypothalamus to decrease TRH secretion and thus the synthesis of TSH. Any process that causes an increase in the peripheral circulation of unbound thyroid hormone can cause thyrotoxicosis.

Disturbances of the normal homeostatic mechanism can occur at the level of the pituitary gland, the thyroid gland, or in the periphery. Regardless of etiology, the result is an increase in transcription in cellular proteins, causing an increase in the basal metabolic rate. In many ways, signs and symptoms of hyperthyroidism resemble a state of catecholamine excess, and adrenergic blockade can improve these symptoms.

In Graves disease, a circulating autoantibody against the thyrotropin receptor provides continuous stimulation of the thyroid gland. This stimulatory immunoglobulin is diagnostic for Graves disease and has been called thyroid-stimulating immunoglobulin (TSI), thyroid-stimulating antibody (TSab), and TSH-receptor antibody (TRab). These antibodies stimulate the production and release of thyroid hormones and thyroglobulin; they also stimulate iodine uptake, protein synthesis, and thyroid gland growth.

OPHTHALMOPATHY IN GRAVE'S DISEASE

These immune processes lead to an active phase of inflammation, with lymphocyte infiltration of the orbital tissue and release of cytokines that stimulate orbital fibroblasts to multiply and produce mucopolysaccharides (glycosaminoglycans), which absorb water. In consequence, the extraocular muscles thicken and the adipose and connective tissue of the retro-orbit increase in volume.

Cigarette smoking and a high TSH-receptor autoantibody level are significant risk factors for ophthalmopathy

V.DIAGNOSIS

1.Clinical signs and symptoms

Classic symptoms

- Anxiety, emotional lability,
- Weakness
- Tremor
- Palpitations,
- Heat intolerance,
- Increased perspiration
- Weight loss despite a normal or increased appetite
- Other symptoms that may be present include hyperdefecation (not diarrhea), urinary frequency, oligomenorrhea or amenorrhea in women, and gynecomastia and erectile dysfunction in men

Specific organ systems

- **Skin:** sweating, heat intolerance, onycholysis, hyperpigmentation, pruritus and hives, Alopecia, thinning of the hair
- **Eyes:** Stare and lid lag occur in all patients with hyperthyroidism
- **Cardiovascular:** Tachycardia, Atrial fibrillation, congestive heart failure
- **Metabolic and endocrine:**
 - **Bone:** osteoporosis
 - **Serum lipids:** low serum total and high-density lipoprotein (HDL) cholesterol concentrations and a low total cholesterol/HDL cholesterol ratio
 - **Hyperglycemia:** Impair glucose tolerance tendency
 - **Adrenal function:** Low serum cortisol finding in hyperthyroidism
 - **Gonadal function:** oligomenorrhea, Amenorrhea and anovulatory (infertility), can occur in women with severe hyperthyroidism.
 - **Respiratory:** Dyspnea and dyspnea on exertion, respiratory muscles weakness, compression by volume goiter
 - **Gastrointestinal:** Weight loss, hyperdefecation, rare vomiting abdominal pain, dysphagia due to goiter. Abnormalities in liver function tests: increase PAL, ASAT, ALAT
 - **Hematologic:** normochromic, normocytic anemia. Serum ferritin concentrations may be high
 - **Genitourinary:** Urinary frequency and nocturia
 - **Neuropsychiatric:** anxiety, restlessness, irritability, emotional lability and insomnia, cognitive impairments and impaired concentration
 - **Thyrotoxic periodic paralysis:** a rare presentation of thyrotoxicosis from any cause is periodic paralysis, marked muscle weakness associated with hypokalemia,

frequently precipitated by exercise, fasting, or a high carbohydrate meal, and seen most frequently in East Asian males

2. Investigation

2.1 Thyroid function tests

In patients in whom there is a clinical suspicion of hyperthyroidism, the best initial test is serum TSH with reflex free T4 and T3 measured automatically if the TSH is low. If reflex ordering is not available, and clinical suspicion is high, order TSH, T3, and free T4 as initial tests. In addition, if hyperthyroidism is strongly suspected despite a normal or elevated serum TSH, serum free T4 and T3 should be measured.

Serum TSH	Serum free T4	Serum T3	Assessment
Normal	Normal	Normal	Euthyroid
Normal hypothalamic-pituitary function			
Low	High (or normal)	High	Hyperthyroidism
	Normal	High	T3 toxicosis
	High	Normal	T4 toxicosis
Low	Normal	Normal	Subclinical hyperthyroidism*
Abnormal hypothalamic-pituitary function			
Normal or high	High	High	TSH-mediated hyperthyroidism
Low [¶]	Normal	Low or normal	Euthyroid with nonthyroidal illness or hyperthyroidism in a critically ill patient. Consider hyperthyroidism in a critically ill patient when TSH is lower [¶] and free T4 and T3 are higher than expected in nonthyroidal illness*.

2.2. Thyroid Autoantibodies

- we measure TRAb to determine the etiology of hyperthyroidism. Grave's disease is caused by autoantibodies to the TSH (thyrotropin) receptor that activate the receptor, thereby stimulating thyroid hormone synthesis and secretion as well as thyroid growth (causing a diffuse goiter).
- TRab positive confirms diagnosis of Grave's disease

2.3. Radioiodine uptake

- Hyperthyroidism with a **high** (or normal) radioiodine uptake indicates de novo synthesis of hormone: Grave's disease, GMNT, toxic Adenoma
- Hyperthyroidism with a **low** (nearly absent) radioiodine uptake indicates either inflammation and destruction of thyroid tissue (Thyroiditis, factitious hyperthyroidism...)

2.4. Thyroid ultrasonography

- Thyroid ultrasonography to detect thyroid nodule and other multiple nodules

- Quantitative thyroid blood flow by ultrasonography: Doppler flow is elevated in Graves' disease and depressed in a destructive thyroiditis

VI. DIFFERENTIAL DIAGNOSIS

Euthyroid hyperthyroxinemic

The presence of hyperthyroidism may be incorrectly suspected in patients who have one of several abnormalities in serum thyroid hormone-binding proteins that result in high serum total (and sometimes free) T4 concentrations and normal (or slightly high) T3 concentrations. These patients have normal TSH concentrations and are euthyroid (euthyroid hyperthyroxinemic)

Low serum TSH without hyperthyroidism

- Central hypothyroidism
- Nonthyroidal illness
- Recovery from hyperthyroidism
- The "physiologic" lowering of serum TSH in pregnancy

Assay interference with biotin ingestion

VII. MANAGEMENT

1. Objective

The primary objective in managing hyperthyroidism is to restore normal thyroid hormone levels and reduce symptoms. Individualizing curative treatment based on patient characteristics, preferences and comorbidities.

2. Critical Signs

- Stroke
- Shortness of breath
- Fever/hyperthermia
- Fast heartbeat
- Nausea, vomiting, diarrhea
- Dehydration
- Confusion
- Delirium

3. Mortality risk factors

- untreated or late management of hyperthyroidism can lead to complications, including
- Cardiovascular disease: Development of heart failure, cardiac arrhythmia, Increased risk of heart disease, hypertension
- Thyroid storm: Life-threatening condition caused by severe hyperthyroidism.
- Mental Health Issues: Increased risk of delirium and cognitive impairment.
- Infertility: Potential reproductive issues in women.

4. Management of Grave's disease

4.1-The first-line treatment

4.1.1 Medical treatment Attacking phase (4-6 weeks)

- Antithyroid drug
 - Propylthiouracil PTU dose 600-800mg/day
 - Carbimazole CBZ dose 20-60mg/day
 - Methimazole MMI dose 10-30mg/day
- Beta-blockers
 - Propranolol dose 40mg 1 tab 2-3 times per day

- Atenolol dose: 50-100mg/day
- Bisoprolol 5-10mg/day

4.1.2 Treatment adjustment

- Maintenance dose: Block and replace strategy has no superiority over degrading to minimal efficacy. Monitoring Free T3 and Free T4 level and dose adjustment should be done in 4 weeks. TSH level should be measured in 2-4 months because of the long blockade of TSH serum in hyperthyroidism is noted.
- Dose adjust of ATD will be done according to thyroid function evolution, PTU 50-100mg/day, CBZ dose 5-20mg/d and MMI dose 2.5-10mg/day
- Beta blocker will be reduced in function of heart rate and normalization of thyroid function test.

Medical treatment recommended for duration 12-18 months, if relapse after treatment, we indicate the radical treatment by iodine or thyroidectomy. In some case, long term with small dose of ATD is also considered.

Low probability of remission

- Trab titer very high
- Big volume goiter
- Severe hyperthyroidism
- Complication related to ATD

4.2 Iodine treatment (iodine 131)

If radical treatment by iodine indicated, the contra-indication of iodine 131 must be noticed

- Suspicion of thyroid malignancy
- Compressive goiter
- Active Grave's eyes disease
- Wish to be pregnant in short term

4.3 Surgical treatment

The thyroidectomy is not the first choice recommended for treatment of Grave's disease.

thyroidectomy indicate in case of:

- Suspicion of thyroid malignancy
- Compressive goiter
- Active Grave's eyes disease
- Wish to pregnant in short term

5-Management of Toxic Multinodular Goiter and Toxic adenoma

Iodine treatment is the privilege indication of the treatment of toxic multinodular goiter and toxic adenoma seen its non-invasive procedure.

Surgical treatment is privilege in particular clinical context as below:

- Suspicion of thyroid malignancy
- Volume goiter with or without compression signs
- Patient inability to respect radioprotection protocol
- Patient's preferences
- Pregnancy plan less than 4-6 months

The privilege long term of small dose ATD (methimazole, carbimazole):

- Frailty, comorbidities and short life expectancy
- Contra-indication of iodine and surgical treatment

- Patient's preferences despite detail explanation of modality treatment by long term ATD (Agranulocytosis, Hepatitis induced by ATD)

6-Outpatient Management involves

- Regular monitoring of free T3 and free T4 in first 1 and 2 months, TSH in 2nd month
- Control Trab at the end of medical treatment of Grave's disease (12 or 18 months)
- Educating patients about symptoms and the importance of adherence to therapy.
- Assessing for comorbidities that may impact treatment (e.g., cardiovascular disease).
- Remission rate of medical treatment 50-60%
- Benefit and risk of iodine 131 and surgical indication
- Education side effect of therapy, in case of fever, check CBC to rule out agranulocytosis

VIII.SPECIAL CASES

- Thyroid storm
- Grave's eyes disease
- Grave's disease and pregnancy:
- Breastfeeding Considerations:
- Hyperthyroidism post iodine treatment:
- Hyperthyroidism post-surgical treatment
- Subacute thyroiditis:
- Subclinical hyperthyroidism

IX. EVOLUTION AND PROGNOSIS

- Hyperthyroidism is a curative condition that usually requires differences therapeutic options.
- Without treatment, it can lead to severe complications, life threatening especially cardiac complications
- Early diagnosis and treatment can improve outcomes.

X. COMPLICATIONS

- Congestive heart failure, Pulmonary edema
- Atrial fibrillation
- Embolic stroke secondary from atrial fibrillation
- thyrotoxic periodic paralysis
- Osteoporosis
- Infertility

XI. PREVENTION AND RECOMMENDATION

- Screening: in some cases, hyperthyroidism present without or less symptoms disease, we recommend to screen thyroid function test
- Healthy lifestyle: Balanced diet, regular exercise, and stress management
- Regular medical check-ups: For early detection and treatment

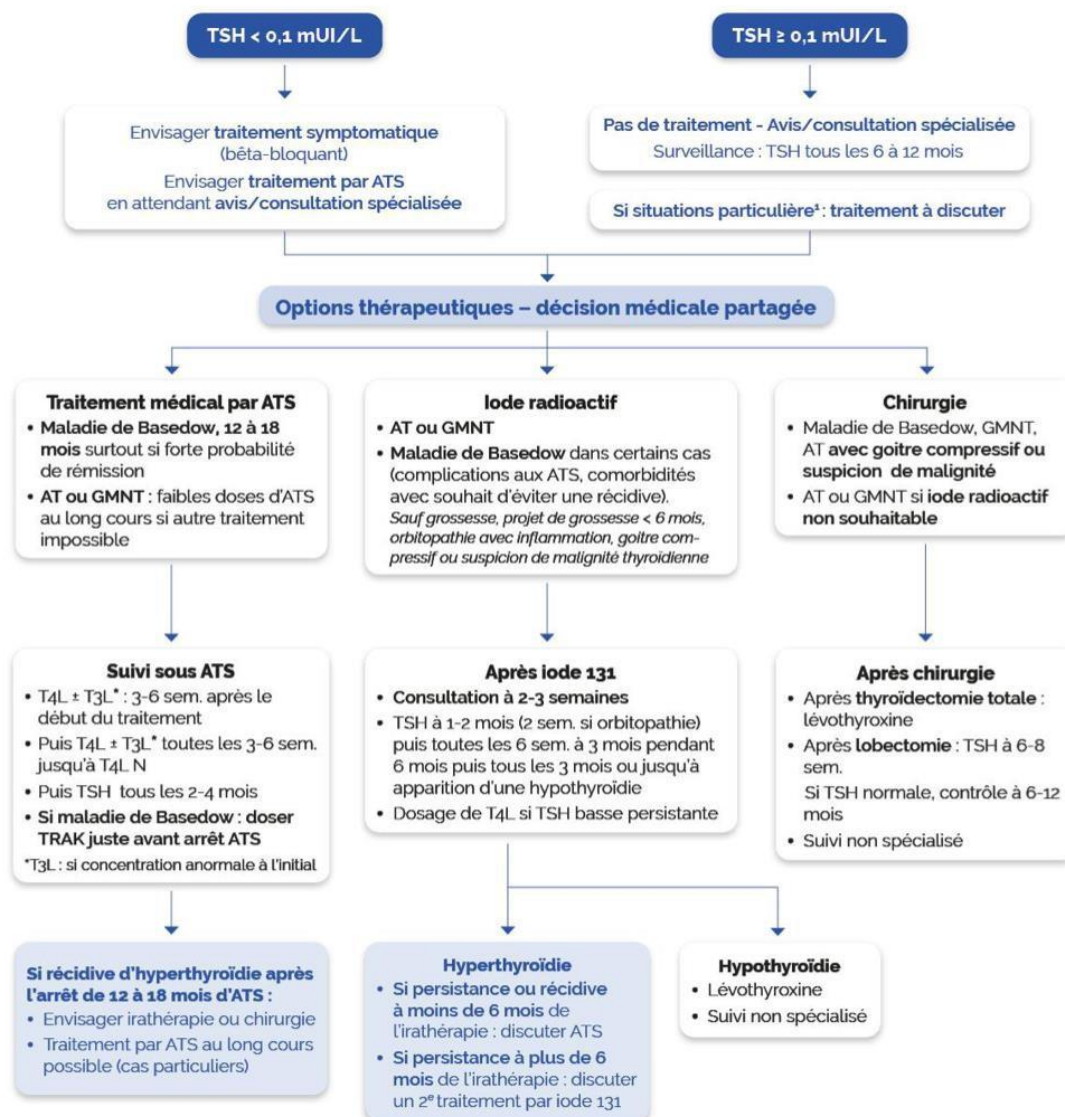
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Annexe 2. Prise en charge thérapeutique de l'hyperthyroïdie primaire chez l'adulte de moins de 65 ans



1. Situations particulières : pathologies cardiaques ou facteurs de risque cardiovasculaires, symptômes par excès d'hormones thyroïdienne, ostéoporose, femme en péri-ménopause ou ménopause sans oestrogènes, ni bisphosphonates.

ATS : anti-thyroïdien de synthèse ; AT : adénome toxique ; GMNT : goitre multinodulaire toxique ; Sem : semaine ; TRAK : anticorps anti-récepteur de la TSH ; TSH N (normale) / T4L N (normale) : résultat dans l'intervalle de référence du laboratoire.

MANAGEMENT OF HYPOTHYROIDISM

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I. DEFINITION

Hypothyroidism is a condition characterized by an insufficient production of thyroid hormones by the thyroid gland. These hormones, thyroxine (T4) and triiodothyronine (T3), regulate various metabolic processes in the body.

II. EPIDEMIOLOGY

The prevalence of hypothyroidism varies from 3 to 10% depending on the studies and, especially, the populations studied, with a female predominance, especially during pregnancy and post-partum, with higher rates in iodine-deficient regions, increases with age. The prevalence of subclinical hypothyroidism is higher in the elderly or in cases with a history of thyroid disorder or under treatments of amiodarone, lithium, interferon. However, Cambodian epidemiology is still to be established.

III. ETIOLOGY

We classified hypothyroidism into 3 main categories:

III.1 Primary Hypothyroidism: most common, caused by thyroid gland dysfunction.

- Autoimmune thyroiditis (Hashimoto's disease)
- Postpartum thyroiditis is usually seen within one year after giving birth, and usually goes away within 18 months. Women who develop this form of thyroiditis are thought to have an underlying autoimmune disease that causes the inflammation. Symptoms occur in two phases, starting with hyperthyroid symptoms and moving to hypothyroid symptoms.
- Sub-acute thyroiditis: mostly due to viral infection, called subacute granulomatous thyroiditis, and called subacute lymphocytic thyroiditis, also occurs during the postpartum period. Hyperthyroid symptoms develop earlier (usually within three months after giving birth), and hypothyroid symptoms can last for several months after.
- Post-thyroid surgery, radio iodine therapy (Iodine 131) and radiation therapy
- Medications (e.g., lithium, amiodarone, interferon, etc.)
- Congenital hypothyroidism
- Iodine deficiency (presented with goiter).

2. Secondary Hypothyroidism: Due to pituitary gland dysfunction, leading to insufficient thyroid-stimulating hormone (TSH) production.

3. Tertiary Hypothyroidism: Results from hypothalamic dysfunction, affecting TSH release.

IV.PATHOPHYSIOLOGY

1. Autoimmune mechanisms like Hashimoto's Thyroiditis is the most common cause of hypothyroidism in around 50% of cases, characterized by the presence of autoantibodies (anti-thyroid peroxidase and anti-thyroglobulin) that attack thyroid tissue, leading to inflammation and decreased hormone production.
2. Thyroid gland damage by surgical removal like thyroidectomy can lead to hypothyroidism due to reduced hormone production.
3. Radiation therapy in such treatment for head and neck cancers may damage thyroid cells, impairing hormone synthesis.
4. Post Radioactive iodine (RAI) hypothyroidism occurs typically 3 to 6 months after RAI therapy is administered. Radioactive iodine (RAI, also called iodine-131 or I-131) can be used to treat some types of thyroid cancer. Patients are told to expect life-long thyroid hormone replacement therapy to be required within this timeframe and many providers expect this post-RAI hypothyroidism to be complete and permanent.
5. Medications like lithium amiodarone and interferon can disrupt thyroid function.
6. Congenital factors which lead to congenital hypothyroidism include agenesis or dysgenesis of the thyroid gland, often due to genetic factors.
7. Central causes by pituitary disorders induced insufficient TSH production due to pituitary damage can lead to secondary hypothyroidism. Hypothalamic dysfunction led to disruption in TRH secretion which affects TSH and thyroid hormone levels.
8. Metabolic factors like severe illness or starvation may impair the conversion of T4 to T3, causing symptoms of hypothyroidism despite normal TSH levels.
9. Iodine deficiency, essential for thyroid hormone synthesis, inadequate iodine intake leads to reduced production of T3 and T4, stimulating increased TSH levels, which can result in goiter and eventual hypothyroidism.

V.DIAGNOSIS

The diagnosis of hypothyroidism involves a systematic approach that includes clinical evaluation, laboratory testing, and sometimes imaging studies.

Hypothyroidism is classified as clinical (or symptomatic) when the clinical signs are pronounced, with elevated TSH level and lowered free T4 (or lower free T3). This form is typically associated with clear symptoms such as fatigue, weight gain constipation and cold intolerance etc.

On the other hand, it is classified as subclinical (or asymptomatic) when the signs are minimal or absent, characterized by elevated TSH confirmed by a second measurement after one month, while free T4 and free T3 levels remain normal. Subclinical hypothyroidism can often go unnoticed and may not require immediate treatment, though monitoring is essential.

1.Clinical signs and symptoms

When thyroid hormone levels are too low, the body's cells can't get enough thyroid hormone and the body's processes start slowing down. As the body slows, you may notice that you feel colder, you tire more easily, your skin is getting drier, you're becoming forgetful and depressed, and you've started getting constipated. Other symptoms included muscle weakness, hoarseness, menstrual irregularities, cognitive impairment, joint pain.

Physical Examination: Signs may include bradycardia (slow heart rate), dry skin, and a goiter (enlarged thyroid gland).

2. Investigation:

2.1. Laboratory Testing: The first test should be serum thyroid-stimulating immunoglobulin (TSH). TSH is usually elevated (>4.0 mIU/L) due to decreased thyroid hormone levels. Free T4, measurement of free T4 (thyroxine) is crucial. In primary hypothyroidism, free T4 is typically low.

2.2. Thyroid Autoantibodies: If the etiology is unclear, testing for anti-thyroid peroxidase (anti-TPO) antibodies can help diagnose autoimmune thyroiditis (e.g., Hashimoto's thyroiditis).

2.3. Nonspecific laboratory test: decreased glucogenesis, increased total and LDL Cholesterol, increase lipoprotein(a), anemia (micro-normo-macrocytic), coagulopathy.

2.4. Imaging Studies: Thyroid ultrasound may help visualize small nodules that are not palpable or structural changes in the thyroid gland during a clinical examination, while not routinely used for diagnosis, and is therefore included as part of a comprehensive evaluation.

VI. DIFFERENTIAL DIAGNOSIS:

- Hyperthyroidism: Low TSH level, Elevated T3 and T4 levels.
- Non-thyroidal illness syndrome: Low T3 levels due to illness
- Primary hypothyroidism is usually caused by dysfunction of the thyroid gland itself (e.g., autoimmune disease, iodine deficiency, surgery, radiotherapy and after radioiodine therapy).
- Secondary hypothyroidism: due to inadequate stimulation by TSH from the pituitary gland or TRH from the hypothalamus.

VII. MANAGEMENT

1. **Objective:** the primary objective in managing hypothyroidism is to restore normal thyroid hormone levels and reduce symptoms. This involves individualizing treatment based on patient characteristics, preferences, and comorbidities.

2. **The first-line treatment** is thyroid hormonal supplement by levothyroxine (synthetic T4). The initial dosage varies based on weight and severity of symptoms, but a target dose of 1.6 mcg/kg per day can be given upfront, typically in doses of 50, 75, or 100 mcg per day.

In cases of mild hypothyroidism, a lower starting dose of 25 mcg may be appropriate. Adjustments should be made based on TSH levels and clinical response.

Administration: Taken on an empty stomach, preferably in the morning, to enhance absorption.

Target Dosage: for replacement therapy, the target dosage should normalize TSH levels. There is inter-individual variability in responses. The average dosage typically ranges between 100 and 150 mcg per day, but it can be lower than 100 mcg per day, especially in elderly patients or those with mild hypothyroidism.

Note: After subtotal or total thyroidectomy, the target dose can be administered right away by 1.6mcg per kilogram per day.

3. Treatment adjustment:

Monitoring: TSH levels should be checked every 6-8 weeks after initiating therapy or changing the dose, until stable. An earlier measurement (before TSH stabilization) could lead to unnecessary increases in dosage.

Adjustment: Once stable, monitoring can be extended to every 3-6 months. Final dosage adjustments are rarely needed and are typically made in increments of 12.5 mcg to 25mcg until TSH levels normalize and no further adjustments are usually necessary.

Normalized TSH: patients should be informed about the gradual regression of symptoms. If symptoms persist despite normalized TSH, consider that they may not be solely attributed to hypothyroidism. Treatment is usually continued for life. If your dosage is too high and you get too many hormones, you could have these side effects: appetite increases, can't sleep, heart palpitations, shakiness and weight loss.

4. **Critical Signs:** in case of severe hypothyroidism which leads to myxedema or crisis, patients are often presented with fatigue, confusion state, hypothermia, cold intolerance, drowsiness, depression, quick changes in blood pressure, swelling in the abdomen caused by chronic constipation, slow heart rate, weight gained, dry skin, breathing issues, breathing issues, movement problems, nerve issues.
5. **Mortality risk factors:** untreated or poorly managed hypothyroidism can lead to complications, including:
 - Cardiovascular Disease: Increased risk of heart disease, hypertension and arrhythmia.
 - Myxedema Coma: A rare but life-threatening condition caused by severe hypothyroidism.
 - Mental Health Issues: Increased risk of depression and cognitive impairment.
 - Infertility: Potential reproductive issues in women.
6. **Outpatient Management involves:**
 - Regular monitoring of TSH and free T4 levels.
 - Educating patients about symptoms and the importance of adherence to therapy.
 - Assessing for comorbidities that may impact treatment (e.g., cardiovascular disease).
7. **Drug interactions and dietary precautions:**
 - a. Calcium carbonate, ferrous fumarate, and sevelamer bind to levothyroxine and reduce the absorption of levothyroxine. Administer levothyroxine 4 hours apart from the medications mentioned above.
 - b. Bile acid sequestrants (e.g., colestyramine, cholestyramine, colestipol) and ion exchange resins, e.g., polystyrene sulfonate, reduce levothyroxine absorption. Administer levothyroxine at least 4 hours before these drugs and monitor TSH levels. Monitor patients treated concomitantly with levothyroxine and orlistat for changes in thyroid function due to reduced absorption.
 - c. Gastric acidity is a prerequisite for adequate absorption of levothyroxine. Sucralfate, antacids, and proton pump inhibitors can cause hypochlorhydria and decrease levothyroxine absorption.
 - d. Furosemide inhibits the protein binding of T4 to thyroid binding globulin (TBG), leading to an increase in free T4; it also competes for T4-binding sites on TBG, so a single high dose of furosemide can acutely lower the total T4 level.
8. The absorption of levothyroxine can be impaired by soybean, papaya, and grapefruit.
9. Biotin (vitamin B7) can interfere with thyroid function tests, leading to inaccurate results and potentially incorrect diagnoses: Falsely elevated levels: Biotin can cause falsely elevated levels of T3 and T4, falsely low levels: Biotin can cause falsely low levels of TSH.

10. Patients treated with immune checkpoint inhibitor therapy (ICPi- pembrolizumab, nivolumab, ipilimumab) may develop hypothyroidism due to immune-related adverse events (IRAE), which may require discontinuation of ICPi.

VIII.SPECIAL CASES

1. **Myxedema Coma or crisis:** initiated treatment by Levothyroxine 200 to 400 mcg initial IV loading dose is followed by a daily dose of 1.2 mcg/kg/day, considering lower doses in patients with a history of cardiac disease, arrhythmia, or older patients more than 65 years old. Switch to oral therapy (8 mcg/kg/day) when symptoms are resolved. The equivalence between intravenous and oral levothyroxine is 0.75 to 1 (for example, 200 mcg IV of levothyroxine equals 266 mcg of oral levothyroxine).
2. **Subclinical hypothyroidism:** may require treatment if symptoms are present or if there is an increased risk of complications, is well defined by a TSH level greater than 5 mUI/L, confirmed by a second measurement after one month, with normal free thyroxine (free T4) levels. Progression to overt hypothyroidism is not inevitable and occurs in only about one-third of cases. The clinical repercussions are inconsistent, nonspecific, and generally negligible for TSH levels below 10 mUI/L.
3. **Special Populations:** in elderly patients or those with cardiovascular conditions carefully, often starting at lower doses, initial dose of 25 mcg/day with a dose adjustment of 12.5 to 25 mcg every 4 to 6 weeks as needed. Targeted screening is recommended in the following at-risk situations:
 - Women over 65 years old with a history of thyroid issues.
 - Presence of anti-thyroid antibodies.
 - History of thyroid or cervical surgery or radiation.
 - Treatments that may affect thyroid function (e.g., amiodarone, lithium, interferon)

A follow up ECG is mandatory, among older persons treated with levothyroxine at doses more than 75 mcg per day is associated with an increased risk of atrial fibrillation.

4. **Hypothyroidism and Pregnancy:** the need for thyroid hormones increases by up to 50% during pregnancy. Due to the risk of complications during pregnancy and potential developmental delays for the child, clinical and hormonal monitoring (TSH and free T4) should be performed at least every 2 months, with dose adjustments as needed. Mild hypothyroidism may be associated with an increased risk of retroplacental hematoma, preterm birth, and neonatal respiratory distress.

Pregnant patients with newly diagnosed hypothyroidism should receive initial treatment at 1.8 mcg/kg/day. Adjust the dose every four weeks as needed. If a patient is diagnosed with hypothyroidism before pregnancy, adjust the dose of levothyroxine as needed. After pregnancy, levothyroxine should decrease to 1.6 mcg/kg/day. After delivery, levothyroxine should be reduced to the patient's preconception dose. Further thyroid function testing should be performed at approximately six weeks postpartum.
5. **Breastfeeding Considerations:** Levothyroxine (T4) is a component of human milk. Adequate levothyroxine during lactation may normalize milk production in hypothyroid lactating mothers. The health benefits of breastfeeding should be evaluated, along with the mother's clinical requirement for levothyroxine and possible adverse effects on the breastfed infant from levothyroxine. Exogenous replacement of levothyroxine during breastfeeding has no adverse effects in infants.

6. **Postpartum thyroiditis:** transient condition that usually resolves on its own; may require temporary treatment or lead to overt hypothyroidism in some cases which need thyroid hormonal supplement as well.
7. **Congenital hypothyroidism:** Requires immediate treatment to prevent intellectual disability. (see in pediatric modules)
8. **Screening in special pregnant patients:** Targeted screening (TSH, free T4, and anti-TPO antibodies) is recommended for any pregnant woman exhibiting suggestive signs: personal or family history of thyroid disease, history of thyroid dysfunction, thyroid surgery, elevated thyroid antibodies, or autoimmune disease. A TSH level greater than 3 mUI/L should lead to enhanced thyroid monitoring: TSH should be checked in 1 month, along with measurement of anti-TPO antibodies. Treatment is justified as soon as TSH exceeds 4 mUI/L, with a therapeutic target of TSH <2.5 mUI/L.

IX.VOLUTION AND PROGNOSIS

- Hypothyroidism is a chronic condition that usually requires lifelong treatment.
- Without treatment, it can lead to severe complications which are life-threatening like Myxedema coma, often presented with confusion state, hypothermia, drowsiness, breathing issues, movement problems, nerve issues, quick changes in blood pressure, swelling in the abdomen, and slow heart rate.
- Early diagnosis and treatment can improve outcomes.

X.COMPLICATIONS

- Myxedema coma is a rare but life-threatening condition caused by severe hypothyroidism.
- Cardiovascular disease resulted in an increased risk of heart failure, arrhythmias, and atherosclerosis due to elevated cholesterol levels.
- Osteoporosis resulted in an increased risk of fractures.
- Neurocognitive disorders: increased risk of depression, impaired memory, concentration, and cognitive impairment.
- Infertility and miscarriage are potential reproductive issues in women.
- Neonatal hypothyroidism in newborns, if not treated correctly during pregnancy.

XI. REVENTION AND RECOMMENDATION

- Screening: For thyroid disease, especially in pregnant women and at-risk populations
- Healthy lifestyle: Balanced diet, regular exercise, and stress management
- Regular medical check-ups: For early detection and treatment
- Iodine supplementation: In iodine-deficient regions

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MANAGEMENT OF GOITER

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I. DEFINITION

Goiter refers to abnormal growth of the thyroid gland. Adults may have goiters that are diffuse or nodular, and the goiters may be associated with normal, decreased, or increased thyroid hormone production.

II. EPIDEMIOLOGY

Simple goiter is present in about 5% of the general population and 10% for multinodular goiter. Unfortunately, the current prevalence of goiters is not known in Cambodia.

III. ETIOLOGY

There are many factors in etiology for the development of thyroid nodules. Most common cause of goiter are classified as iodine dependent and non-iodine dependent factors basically.

1. Hyperthyroidism goiters

- a) Graves' disease is an autoimmune disorder called Graves' disease occurs when the immune system produces a protein that mimics TSH. This rogue protein prompts the thyroid to overproduce hormones (hyperthyroidism) and can result in thyroid growth.
- b) Thyroid nodules. A nodule is the irregular growth of thyroid cells that form a lump. A person may have one nodule or several nodules (multinodular goiter). The cause of nodules is not clear, but there may be multiple factors: genetics, diet, lifestyle and environment (goitrogens vegetables). Most thyroid nodules are noncancerous (benign).

2. Hypothyroidism Goiter

- a) Iodine deficiency, iodine is essential to produce thyroid hormones. If a person does not get enough dietary iodine, hormone production drops and the pituitary gland signal the thyroid to make more. This increased signal results in thyroid growth. In the United States, this cause is uncommon because of iodine added to table salt.
- b) Hashimoto's disease is an autoimmune disorder, an illness caused by the immune system attacking healthy tissues. The damaged and inflamed tissues of the thyroid don't produce enough hormones (hypothyroidism). When the pituitary gland detects the decline and prompts the thyroid to create more hormones, the thyroid can become enlarged.

Other thyroid dysfunction goiter: such as Pregnancy, a hormone produced during pregnancy, human chorionic gonadotropin (HCG), may cause the thyroid gland to be overactive and enlarge slightly.

Thyroiditis is inflammation of the thyroid gland caused by an autoimmune disorder, bacterial or viral infection, or medication. Inflammation may cause hyperthyroidism or hypothyroidism. Thyroid cancer is less common than other cancers and generally treatable. About 5% of people with thyroid nodules are found to have cancer.

IV.PATHOPHYSIOLOGY

In patients with iodine deficiency or chronic autoimmune (Hashimoto's) thyroiditis, an increase in thyroid stimulating hormone (TSH) secretion is the predominant cause of goiter. In contrast, most patients with sporadic nontoxic multinodular goiters have normal serum TSH concentrations. In them, the thyroid enlargement is probably caused by several growth factors (including TSH) that act over time on thyroid follicular cells that have different synthetic and growth potentials.

In patients with Graves' disease, thyroid stimulating immunoglobulin stimulates the TSH receptor to cause thyroid growth and excessive hormonal secretion.

There is often a family history of goiter, suggesting that genetic factors may also play a role. The result is diffuse and later multinodular thyroid enlargement. Some nodules eventually become autonomous due to activating mutations in the TSH receptor or G proteins within the thyroid follicular cells. The following observations support this sequence of events:

- Thyroid volume is larger in older patients.
- The longer the patient has a goiter, the larger the size of the goiter.
- The larger the size of the goiter, the lower the serum TSH concentration.

V.DIAGNOSIS

The diagnosis of goiter is usually done by perform a physical examination and feel that you have an enlarged thyroid. However, the presence of a goiter indicates that there's an issue with your thyroid gland.

1.Clinical signs and symptoms

The main symptoms of goiter include: a lump in the front of your neck, just below your Adam's apple, a feeling of tightness in your throat area, hoarseness (scratchy voice), neck vein swelling, dizziness when you raise your arms above your head.

Other, less common symptoms include: difficulty breathing (shortness of breath), coughing, wheezing (due to squeezing of your windpipe), difficulty swallowing (due to squeezing of your esophagus).

Some people who have a goiter may also have hyperthyroidism (overactive thyroid). Symptoms of hyperthyroidism include: rapid heart rate (tachycardia), unexplained weight loss diarrhea, sweating without exercise or increased room temperature, shaking, agitation. Some patients present with Thyroid eyes disease symptoms which is bulging or protruding eyeball or eyeballs.

Some people with goiter may also have hypothyroidism (underactive thyroid). Symptoms of hypothyroidism include: fatigue (feeling tired), constipation, dry skin, unexplained weight gain and abnormal menstruation (periods).

2.Investigation

- a) Thyroid function tests: Serum TSH should be measured in all patients with goiter. Patients with goiter may have normal thyroid function, hyperthyroidism (subclinical or overt), or hypothyroidism (subclinical or overt). If the TSH is below normal, serum free T4 and total T3 should also be measured. In patients with overt or subclinical hyperthyroidism and goiter, multinodular goiter with autonomy or Graves' disease is the most likely diagnosis. If serum TSH is above normal, free T4 should be measured.
- b) Thyroid ultrasound: should be obtained in all patients with nontoxic goiter unrelated to Hashimoto's thyroiditis or iodine deficiency. In the latter two circumstances, ultrasound should be reserved for patients with thyroid asymmetry or palpable nodules. Thyroid ultrasound in patients with Hashimoto's thyroiditis should be interpreted with caution due to the diffuse heterogeneity and the presence of pseudo-nodules related to ongoing

inflammation. Ultrasound may identify distinctive, nonpalpable thyroid nodules within a nodular or diffuse goiter. Nodules with indeterminate or suspicious ultrasound features should be considered for biopsy because the prevalence of cancer in an individual nodule in a goiter is independent of the number of sonographically identified nodules. Ultrasound may be useful in evaluating a region of a goiter with unusual characteristics on palpation, such as hard consistency or tenderness, findings suggestive of possible coexisting cancer or lymphoma.

- c) CT scan or MRI (magnetic resonance imaging) of thyroid gland: If the goiter is very large or spreads into your chest, a CT scan or MRI is used to measure the size and spread of the goiter.

VI. DIFFERENTIAL DIAGNOSIS

Thyromegaly can be diagnosed if the lateral lobe of the thyroid is larger than the terminal phalanx of the child's thumb.

VII. MANAGEMENT

1.Objectives

All goiters need to be assessed, at least for the thyroid function (TSH and/or T4) and a thyroid ultrasound.

2.Critical sign

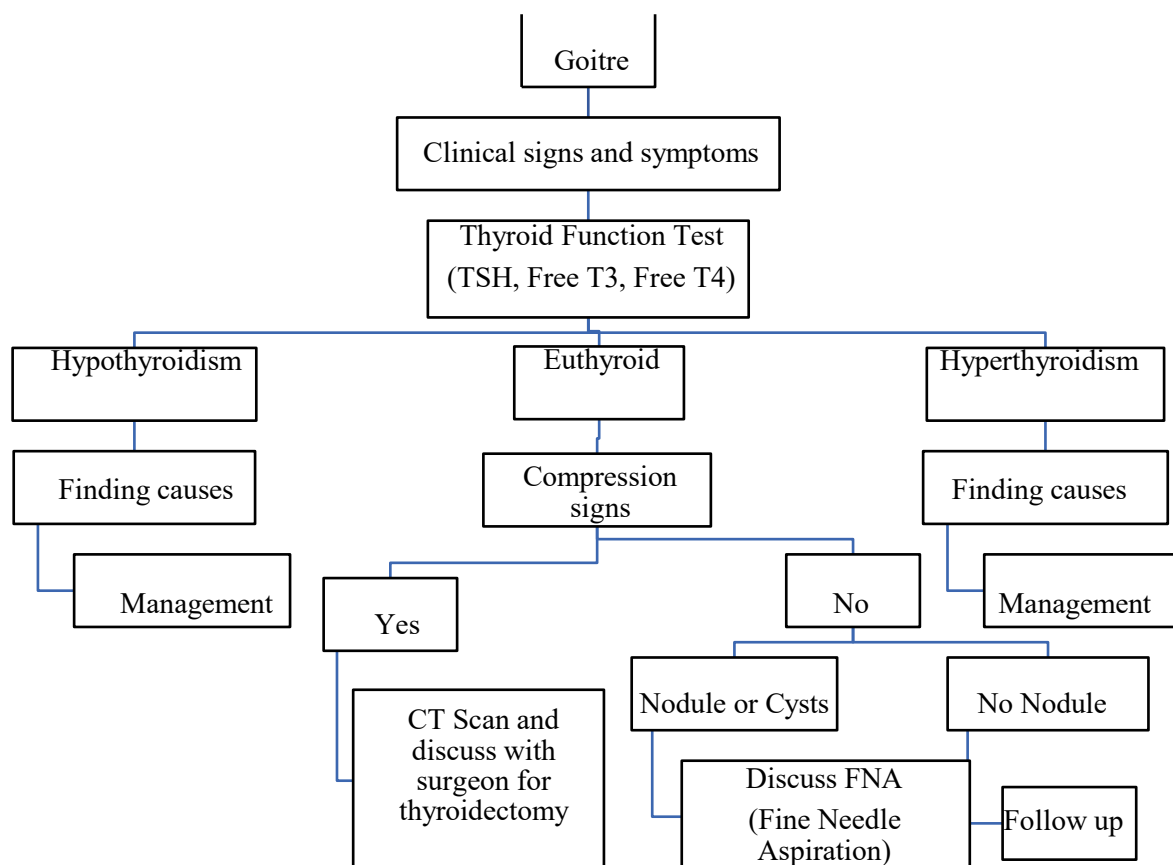
Most goiters are painless, only if you have thyroiditis or that can be painful around the thyroid gland or lead to thyroid storm symptoms (Critical sign of Hyperthyroidism).

3. Mortality risk factor: there is no data elsewhere for mortality rate of goiter.

4.Outpatient Management

Treatment of goiter depends on its etiology. (See Hyperthyroidism and hypothyroidism module).

5.Algorithm



VIII. SPECIAL CASES

Rarest of all is the plunging goiter (goitre plongeant) or substernal goiter also said cervico-mediastinic goiter, initially described and so named by Malard, is a thyroid formation with cervical departure that goes beyond, with stretched neck, the superior thoracic strait for at least 3 cm and that preserves, generally, the parenchymal or fibrous connections between the cervical and thoracic portion. In this

Compressing sign of goiter is indicated for CT Scan and refer to see surgeon for next step management if surgery needed.

IX. EVOLUTION AND PROGNOSIS

A simple goiter may happen for only a short time and may go away on its own without treatment. Many goiters, such as multinodular goiter, are associated with normal levels of thyroid hormone. These goiters usually don't require any specific treatment after your healthcare provider has diagnosed it. However, you may be at risk of developing hypothyroidism or hyperthyroidism in the future.

The prognosis for goiter depends on its type and what caused it. Simple goiter has a good prognosis. If your thyroid continues to enlarge, it may compress the surrounding structures and may cause difficulty in breathing and swallowing and hoarseness. If the goiter is a sign of another thyroid disease like Graves' disease or Hashimoto's disease, the prognosis depends on the underlying cause of thyroid enlargement.

X. COMPLICATIONS

Most goiters grow very slowly over many decades. Therefore, the majority of patients with goiter are asymptomatic. Patients with long-standing goiters may also develop symptoms of obstruction, due to progressive compression of the trachea or sudden enlargement (usually accompanied by pain), secondary to hemorrhage into a nodule.

Rapid thyroid growth also occurs in patients with infectious (suppurative) or sub-acute thyroiditis. However, the growth is typically accompanied by pain and fever.

In patients with infection, the thyroid abnormalities are predominantly unilateral, whereas most patients with sub-acute thyroiditis have bilateral thyroid enlargement and pain.

Thyroid cancers (differentiated) grow slowly and may be indistinguishable clinically from a uni-nodular or multinodular benign goiter. The rare anaplastic thyroid cancers or primary lymphomas of the thyroid often present with a rapidly enlarging neck mass that may cause obstructive symptoms, such as dyspnea, wheezing and coughing.

XI. PREVENTION AND RECOMMENDATION

Goiter caused by iodine deficiency (simple goiter) is generally the only type of goiter you can prevent. Consuming a diet that includes fish, dairy and a healthy amount of iodized table salt prevents these types of goiters.

Regardless of the cause, it's important to see your healthcare provider regularly (at least annually) if you've been diagnosed with goiter so they can monitor it. If you develop new symptoms, talk to your healthcare provider.

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ADRENAL INSUFFICIENCY IN ADULT

Asst. Prof. EM Savoeun, Dr. THEL Sophea

I. DEFINITION

Adrenal insufficiency is the decreased production of adrenocortical hormones

- glucocorticoids,
- mineralocorticoids, and
- adrenal androgens

Classified as primary, secondary, or tertiary

- Primary adrenal insufficiency (**Addison disease**) is caused by a *disorder* of the adrenal glands.
 - ✓ The most frequent cause of primary adrenal insufficiency in the USA is autoimmune adrenalitis
- Secondary adrenal insufficiency is the result of *decreased* production of **ACTH** (adrenocorticotrophic hormone)
- Tertiary adrenal insufficiency is the result of **decreased** production of **CRH** (corticotropin-releasing hormone) by the **hypothalamus**
- Serum cortisol levels that remain low even after the administration of exogenous ACTH (ACTH stimulation test) confirm the diagnosis of primary adrenal insufficiency.
- Glucocorticoid replacement therapy with hydrocortisone is required for all forms of adrenal insufficiency.

II. ETIOLOGY

A. Primary adrenal insufficiency (Addison disease)

a. Autoimmune adrenalitis

- Most common cause in the US (~ 80–90% of all cases of primary adrenal insufficiency)
- Associated with other autoimmune endocrinopathies

b. Infectious adrenalitis

- Tuberculosis: most common cause worldwide, but rare in the USA
- CMV disease in immunosuppressed states (especially AIDS)
- Histoplasmosis

c. Adrenal hemorrhage ^{[1][2]}

- Sepsis: especially meningococcal sepsis (endotoxic shock) → hemorrhagic necrosis (Waterhouse-Friderichsen syndrome)
- Disseminated intravascular coagulation (DIC)
- Anticoagulation: especially heparin (heparin-induced thrombocytopenia)
- Venous thromboembolism, especially in antiphospholipid syndrome (APS)
- Adrenal tumor (most commonly pheochromocytoma) → intra-tumoral bleeding
- Short-term steroid usage
- Trauma (mostly blunt trauma, can also occur postoperatively)

d. Infiltration of the adrenal glands

- Tumors (adrenocortical tumors, lymphomas, metastatic carcinoma)
- Amyloidosis
- Hemochromatosis
- e. Adrenalectomy
- f. Impaired activity of enzymes that are responsible for cortisol synthesis
 - Cortisol synthesis inhibitors (e.g., rifampin, fluconazole, phenytoin, ketoconazole): drug-induced adrenal insufficiency
 - 21 β -hydroxylase deficiency
- g. Vitamin B5 deficiency ^[3]

B. Secondary adrenal insufficiency

Secondary adrenal insufficiency is caused by conditions that decrease ACTH production (impaired hypothalamic-pituitary-adrenal axis).

- Sudden discontinuation of chronic glucocorticoid therapy or stress (e.g., infection, trauma, surgery) during prolonged glucocorticoid therapy
 - Prolonged iatrogenic suppression of the hypothalamic-pituitary-adrenal axis
 - Impaired endogenous cortisol production in addition to discontinuation of steroid treatment, decrease in dosage, or increase in requirement (e.g., stress) → acute glucocorticoid deficiency
- Hypopituitarism: ↓ ACTH → ↓ endogenous cortisol

C. Tertiary adrenal insufficiency

Tertiary adrenal insufficiency is caused by conditions that decrease CRH production.

- The most common cause is sudden discontinuation of chronic glucocorticoid therapy.
- Rarer causes include hypothalamic dysfunction (e.g., due to trauma, mass, hemorrhage, or anorexia): ↓ CRH → ↓ ACTH → ↓ cortisol release

III. PATHOPHYSIOLOGY

A. Primary adrenal insufficiency (Addison disease)

Damage to the adrenal gland leads to the deficiency in all three hormones produced by the adrenal cortex: androgen, cortisol, and aldosterone.

- Hypoandrogenism
 - Loss of libido
 - Impaired spermatogenesis (in men)
- Hypocortisolism leads to:
 - ↑ ACTH → ↑ production of Proopiomelanocortin [POMC] (in order to increase ACTH production) → ↑ melanocyte-stimulating hormone (MSH) → hyperpigmentation of the skin (bronze skin)
 - ↑ ADH level → retention of free water → dilutional hyponatremia
 - ↓ Expression of enzymes involved in gluconeogenesis → ↓ rate of gluconeogenesis → hypoglycemia
 - Lack of potentiation of catecholamines action → hypotension

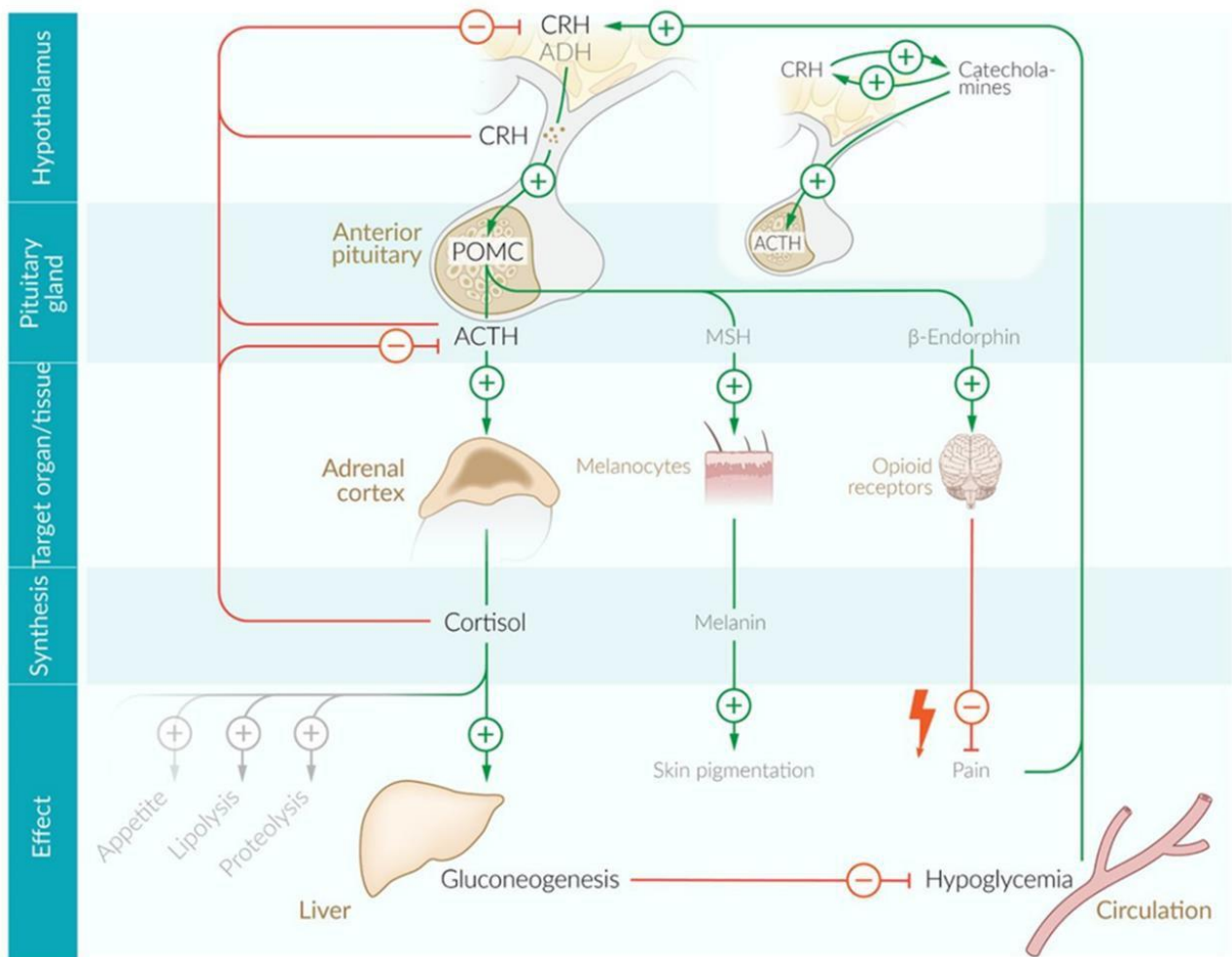
- c. Hypoaldosteronism → hypotension (hypotonic hyponatremia and volume contraction), hyperkalemia, metabolic acidosis

B. Secondary adrenal insufficiency

- ↓ ACTH → hypoandrogenism and hypocortisolism
- Aldosterone synthesis is *not affected* (mineralocorticoid production is controlled by RAAS and angiotensin II, not by ACTH).

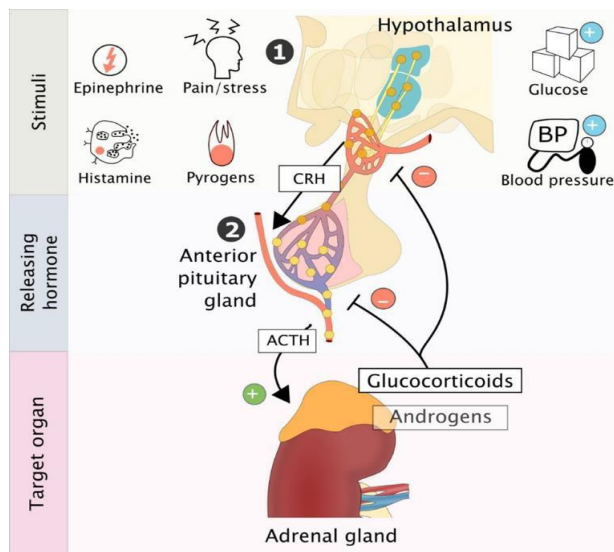
C. Tertiary adrenal insufficiency

- ↓ CRH → ↓ ACTH → hypoandrogenism and hypocortisolism
- Aldosterone synthesis is *not affected*.



Regulation of the hypothalamic-pituitary-adrenal axis

Melanocyte-stimulating hormone (MSH)



Hormonal feedback control of the adrenal hormones ^[4]

- (1) Stimuli such as pain, stress (physical/psychological), pyrogens, epinephrine, histamine, hypoglycemia, and hypotonia cause the release of corticotropin-releasing hormone (CRH) in the hypothalamus.
- (2) In the anterior pituitary gland, CRH stimulates the release of adrenocorticotropic hormone (ACTH), which stimulates the secretion of glucocorticoids and androgens in the adrenal cortex. Glucocorticoids inhibit the secretion of CRH and ACTH via a negative feedback loop.

IV. Clinical features and Diagnosis

Hormonal changes	Clinical features	Laboratory findings	Primary adrenal insufficiency	Secondary adrenal insufficiency	Tertiary adrenal insufficiency
Hypoaldosteronism	Hypotension Salt craving	Hyponatremia Hyperkalemia Normal anion gap metabolic acidosis	✓	Absent	Absent
Hypocortisolism	Weight loss, anorexia Fatigue, lethargy , depression Muscle aches Weakness Gastrointestinal complaints (e.g., nausea, vomiting, diarrhea) Sugar cravings (Orthostatic) hypotension	Hypoglycemia Hyponatremia	✓	✓	✓
Hypoandrogenism	Loss of libido Loss of axillary and pubic <u>hair</u>	↓ DHEA-S	✓	✓	✓
Elevated ACTH	Hyperpigmentation of areas that are not normally exposed to sunlight palmar creases, mucous membrane of the <u>oral cavity</u>)	↑ MSH	✓	Absent	Absent

Melanocyte-stimulating hormone (MSH), DHEA-S: dehydroepiandrosterone sulfate

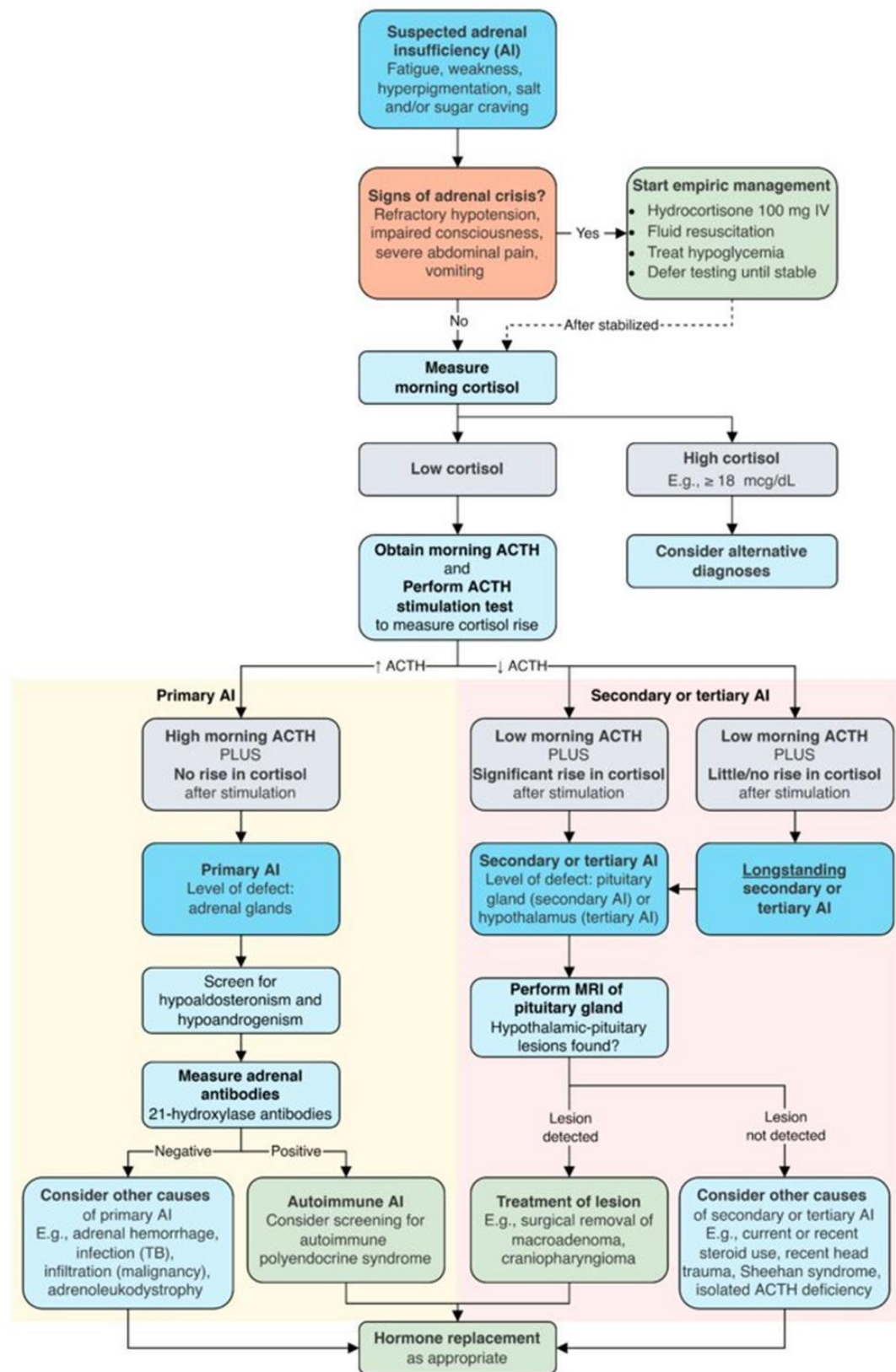
- Most cases of adrenal insufficiency are subclinical and only become apparent during periods of stress (e.g., surgery, trauma, infections), when the cortisol requirement is higher
- Primary adrenal insufficiency *Pigments* the skin
- Secondary adrenal insufficiency *Spare*s the skin.
- Tertiary adrenal insufficiency is due to *Treatment* (cortisol).



Addison hyperpigmentation ^[5]

Black pigmentation is visible on the left buccal mucosa and hyperpigmented spots are visible on the front area of the tongue
Hyperpigmentation of mucous membranes of the oral cavity is a characteristic feature of Addison disease.

Endocrine testing for adrenal insufficiency ^[6]			
	Morning cortisol	Morning <u>ACTH</u>	ACTH stimulation test
Primary adrenal insufficiency [<i>Level of defect: adrenal glands</i>]	↓	↑	No increase in serum <u>cortisol</u> after stimulation
Secondary/tertiary adrenal insufficiency [<i>Level of defect: pituitary gland/ hypothalamus</i>]		↓	Increase in serum <u>cortisol</u> after stimulation
Longstanding secondary/tertiary adrenal insufficiency [<i>Level of defect: pituitary gland/hypothalamus and subsequent atrophy of adrenal glands</i>]		↓	No (or very little) increase in serum <u>cortisol</u> after stimulation



Diagnosis of adrenal insufficiency

V. LABORATORY TEST

1. Routine laboratory studies

- a. Primary adrenal insufficiency
 - Basic Metabolic Panel (BMP)
 - Na^+ ↓, K^+ ↑, Ca^{2+} ↑
 - Normal anion gap metabolic acidosis
 - Hypoglycemia
 - Creatinine ↑, BUN ↑
 - CBC: mild anemia, lymphocytosis, eosinophilia
 - Thyroid function abnormalities
- b. Secondary adrenal insufficiency: potentially only ↓ Na^+ or hypoglycemia

2. Endocrine studies

Endocrine testing is typically performed sequentially

- 1) Morning cortisol level: *initial test* ^[6]
 - Levels ≥ 18 mcg/dL have a high negative predictive value for ruling out adrenal insufficiency
 - Levels < 3 mcg/dL strongly suggest hypocortisolism
 - Random cortisol levels are of limited value, as cortisol secretion varies *diurnally* and with physiological stress
 - Cortisol levels are influenced by cortisol-binding globulin (CBG) and albumin levels
- 2) Morning ACTH level: obtain if morning cortisol is low ^[6]
 - **Primary** adrenal insufficiency: elevated ACTH levels > 100 pg/mL
 - **Secondary/tertiary** adrenal insufficiency: ACTH levels low to normal
 - ACTH secretion is subject to *diurnal* variation, which is why a morning sample is desirable
 - Exogenous glucocorticoids (via any route) can suppress ACTH secretion through negative feedback
- 3) Standard-dose ACTH stimulation test (cosyntropin test): **gold standard** to confirm the diagnosis of primary adrenal insufficiency ^[6]
 - Method
 - Administration of 250 mcg exogenous ACTH to stimulate cortisol secretion
 - Measurement of cortisol levels before and 30 and/or 60 minutes after injection
 - Physiological response: exogenous ACTH → ↑ cortisol
 - If a patient is on prednisone, prednisolone, or dexamethasone, temporarily switch them to hydrocortisone and hold hydrocortisone 24 hours prior to testing
 - Interpretation
 - In primary adrenal insufficiency: peak cortisol level < 18–20 µg/dL (< 500–550 nmol/L)
 - In secondary/tertiary adrenal insufficiency: usually a rise in cortisol > 18–20 µg/dL (> 500–550 nmol/L) ^[7]

3. Screening for hypoaldosteronism and hypoandrogenism

The following parameters should be measured in patients with primary adrenal insufficiency.

- Hypoaldosteronism
 - ↑ Plasma renin activity (PRA) or ↑ plasma renin concentration (PRC)
 - AND inappropriately normal or ↓ serum aldosterone
- Hypoandrogenism
 - ↓ DHEA-S
 - Only relevant in patients without testes, for whom the adrenal glands are one of the main sources of androgens

4. Other dynamic endocrine studies

These can be performed in consultation with an endocrinologist, e.g., if previous tests have been inconclusive

- Insulin tolerance test (also insulin hypoglycemia test) [6]
 - Physiological response: administration of insulin → hypoglycemia (strong stimulator for ACTH and cortisol secretion) → stress-induced ↑ in plasma ACTH → ↑ in cortisol levels
 - Secondary adrenal insufficiency: no incremental rise in ACTH and cortisol levels
 - Unsafe in the elderly and in patients with known seizures and cardiac disease
- Overnight metyrapone stimulation test [8]
 - Physiological response: metyrapone inhibits 11 β hydroxylase → impaired conversion of 11-deoxycortisol to cortisol (last step of cortisol synthesis) → ↓ serum cortisol → ↑ in CRH and plasma ACTH (negative feedback) → ↑ in adrenal steroidogenesis → ↑ in 11-deoxycortisol level
 - In primary adrenal insufficiency: metyrapone → ↓ cortisol synthesis → ↑ in CRH/ACTH → no increase in adrenal steroid production → no increase in 11-deoxycortisol or cortisol levels
 - In secondary/tertiary adrenal insufficiency: metyrapone → ↓ cortisol → no increase in CRH/ACTH → no increase in adrenal steroid production → no increase in 11-deoxycortisol or cortisol levels

5. CRH stimulation test: used to distinguish between secondary and tertiary adrenal insufficiency [8]

- Physiological response: CRH → ↑ ACTH → ↑ cortisol
- Secondary adrenal insufficiency: CRH → no increase in ACTH → no increase in cortisol
- Tertiary adrenal insufficiency: CRH → ↑ ACTH → ↑ cortisol

VI. TEST TO INVESTIGATE UNDERLYING CAUSES

a) Primary adrenal insufficiency

The most common cause of primary adrenal insufficiency in high-income countries is autoimmune adrenalitis

- Autoimmune screen
 - Antibodies against 21-hydroxylase

- Consider further screening for autoimmune polyendocrine syndromes if antibodies against 21-hydroxylase are present:
 - Hypoparathyroidism: calcium, PTH
 - Diabetes: anti-GAD antibodies, islet cell antigen antibodies
 - Pernicious anemia: antiparietal cell antibodies, anti-IF antibodies
 - Thyroid disease: TSH-receptor antibodies, TPO antibodies
- Imaging
 - CXR: Screen for tuberculosis if an infective cause is suspected
 - CT or MRI adrenal glands: Screen for adrenal hemorrhage and malignant or infiltrative disease
- b) Secondary/tertiary adrenal insufficiency

The most common cause of tertiary adrenal insufficiency is exogenous glucocorticoid administration

 - Imaging: MRI head to investigate for pituitary/hypothalamic lesion
 - Advanced: Screen for congenital disorders, e.g., DNA methylation analysis for Prader-Willi syndrome

VII. TREATMENT APPROACH

1. Primary adrenal insufficiency: replacement for hypocortisolism, hypoaldosteronism, and hypoandrogenism
 - Glucocorticoids
 - Mineralocorticoids
 - Androgens (as needed)
2. Secondary/tertiary adrenal insufficiency: replacement for hypocortisolism and hypoandrogenism
 - Glucocorticoids
 - Androgens (as needed)
 - Replacement of other hormones if there is concomitant hypopituitarism
3. Underlying causes: Identify and treat reversible conditions in all patients
4. Prevention of adrenal crisis: “Stress-dose steroids.”

A. Steroid replacement

Glucocorticoids ^[9]: Agents

- Hydrocortisone DOSAGE Hydrocortisone 15–25 mg/day PO in 2–3 divided doses; e.g., 10 mg in the morning, 5 mg at lunchtime, and 5 mg in the evening
- Cortisone acetate DOSAGE Cortisone acetate 20–35 mg/day PO in 2–3 divided doses; e.g., 15 mg in the morning, 10 mg at lunchtime, and 10 mg in the evening
- Prednisolone DOSAGE Prednisolone 3–5 mg/day PO in 1–2 divided doses; e.g., 3 mg in the morning, 2 mg in the afternoon

NB: Stress-dose steroids ^[10, 11]

- ✓ Steroid doses should be increased to prevent adrenal crisis in at-risk patients, e.g., in acute illness, surgery, or trauma
- ✓ Inpatient steroid doses are adjusted according to the level of stress

- Febrile illness: Double the usual oral dose until recovery; consider IV route if the patient is vomiting.
- Critically ill patients: IV hydrocortisone DOSAGE Hydrocortisone 100 mg IV once, then 200 mg IV over 24 hours; either as 50 mg IV every 6 hours OR 8.33 mg/hour IV infusion
- Perioperative patients
 - Minor and moderate procedures: usual daily steroid dose PLUS stress-dose hydrocortisone DOSAGE Hydrocortisone 50 mg IV once pre-incision, then 25 mg IV every 8 hours for 24 hours followed by a return to the usual steroid dose
 - Major procedure: usual daily steroid dose PLUS stress dose hydrocortisone DOSAGE Hydrocortisone 100 mg IV once pre-incision, then 200 mg IV over 24 hours; either as 50 mg IV every 6 hours OR 8.33 mg/hour IV infusion followed by steroid taper as needed

B. Mineralocorticoids

Agent: fludrocortisone DOSAGE Fludrocortisone 50–100 mcg PO daily (a synthetic mineralocorticoid with mostly mineralocorticoid and limited glucocorticoid effects)

C. Androgens

Consider treatment in anatomically female patients with low libido, depressive symptoms, and low energy level

Agent: DHEA DOSAGE DHEA 25–50 mg PO daily

D. Treatment of the underlying cause and associated conditions

- Drug-induced adrenal insufficiency: Consider medication adjustment
- Hypopituitarism: Substitute other pituitary hormones
- Infectious adrenalitis: “Tuberculosis,” “HIV,” and “Cytomegalovirus infection.”
- Adrenal hemorrhage: e.g., Waterhouse-Friderichsen syndrome
- Infiltration of the adrenal glands: Treat malignancy, e.g., via tumor resection
- Autoimmune adrenalitis: Treat other associated autoimmune endocrinopathies

VIII. OUTPATIENT SICK DAY RULES

The following rules should be taught to patients with primary adrenal insufficiency so that they can self-administer glucocorticoids in times of illness/physiological stress:

- ✓ Sick day *rule one*: circumstances requiring double the routine oral glucocorticoid dose
 - Illness with any of the following: Fever, need for bed rest, need for antibiotics
 - Minor outpatient procedures (e.g., dental intervention)
- ✓ Sick day *rule two*: circumstances requiring IM/IV glucocorticoids or suppositories
 - Intractable vomiting or diarrhea
 - Severe illness
 - Trauma
 - Fasting required for a procedure (e.g., colonoscopy)
 - Perioperative period

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PROLONG GLUCOCORTICOID THERAPY

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I. DEFINITION

In general, patients taking more than 5 mg of prednisolone or the equivalent for longer than a few weeks are at risk for secondary adrenal insufficiency.

II. EPIDEMIOLOGY

- Systemic glucocorticoids are used by 1-3% of the general population. Prednisone, methylprednisolone, and dexamethasone are used in chronic therapy because of a stronger anti-inflammatory activity and lesser sodium retention effect than hydrocortisone. Focusing on the long-term use of oral glucocorticoids, the prevalence is 0.5-1.8%.
- Oral glucocorticoids are more commonly used by women and older people; chronic use is reported in 1.4% of patients older than 55 years and in 3% of women over the age of 80. A systematic review and meta-analysis found that the absolute risk of adrenal insufficiency induced by oral glucocorticoids was 48.7%, with the highest risk in people with hematologic malignancies (60%), followed by patients with a history of renal transplant (56.2%), inflammatory bowel disease (52.2%), and rheumatologic disorders (39.4%).
- The use of inhaled glucocorticoids has also increased over time in both children and adults, and the overall risk of GI-AI in these patients is 7.8%.
- Adult patients treated with topical or nasal glucocorticoids were reported to have a lower risk of adrenal insufficiency of 4.2-4.7%.
- Patients treated with intra-articular glucocorticoids had biochemical evidence of adrenal insufficiency in 52.2% of cases; however, this estimate.

III. ETIOLOGY

The etiology of conditions necessitating chronic glucocorticoid therapy is diverse, encompassing a range of autoimmune, inflammatory, and endocrine disorders. These include:

- **Autoimmune Diseases:** rheumatoid arthritis, systemic lupus erythematosus, Purpura thrombopenia immunologic (PTI), Graves' ophthalmopathy and multiple sclerosis often requires prolonged glucocorticoid therapy to control systemic inflammation and prevent organ damage.
- **Chronic Inflammatory Diseases:** Chronic obstructive pulmonary disease (COPD), inflammatory bowel disease (IBD), Chronic infections like tuberculosis, asthma are common indications for long-term glucocorticoid use. In these conditions, glucocorticoids reduce inflammation in the airways or intestines, thereby preventing flare-ups and maintaining disease remission.
- **Endocrine Disorders:** In adrenal insufficiency, such as Addison's disease, glucocorticoids are used as hormone replacement therapy. These patients lack

sufficient endogenous cortisol due to adrenal gland dysfunction and require exogenous glucocorticoids to maintain normal physiological functions.

- **Post-Transplantation Management:** Glucocorticoids are a key component of immunosuppressive regimens in organ transplantation. They help prevent the immune system from attacking the transplanted organ, thereby reducing the risk of rejection.
- **Oncological Conditions:** Certain cancers, particularly hematologic malignancies like lymphomas and leukemias, are treated with glucocorticoids as part of combination chemotherapy regimens. In these cases, glucocorticoids help reduce inflammation and manage symptoms associated with the disease or its treatment.
- **Local Administration:**
Glucocorticoid administration can be via several non-systemic routes, including intra-articular joint injections for joint inflammation, inhalational for asthma, topical for dermatological problems, ocular drops for eye conditions, and intra-nasal for seasonal rhinitis.

IV. PATHOPHYSIOLOGY

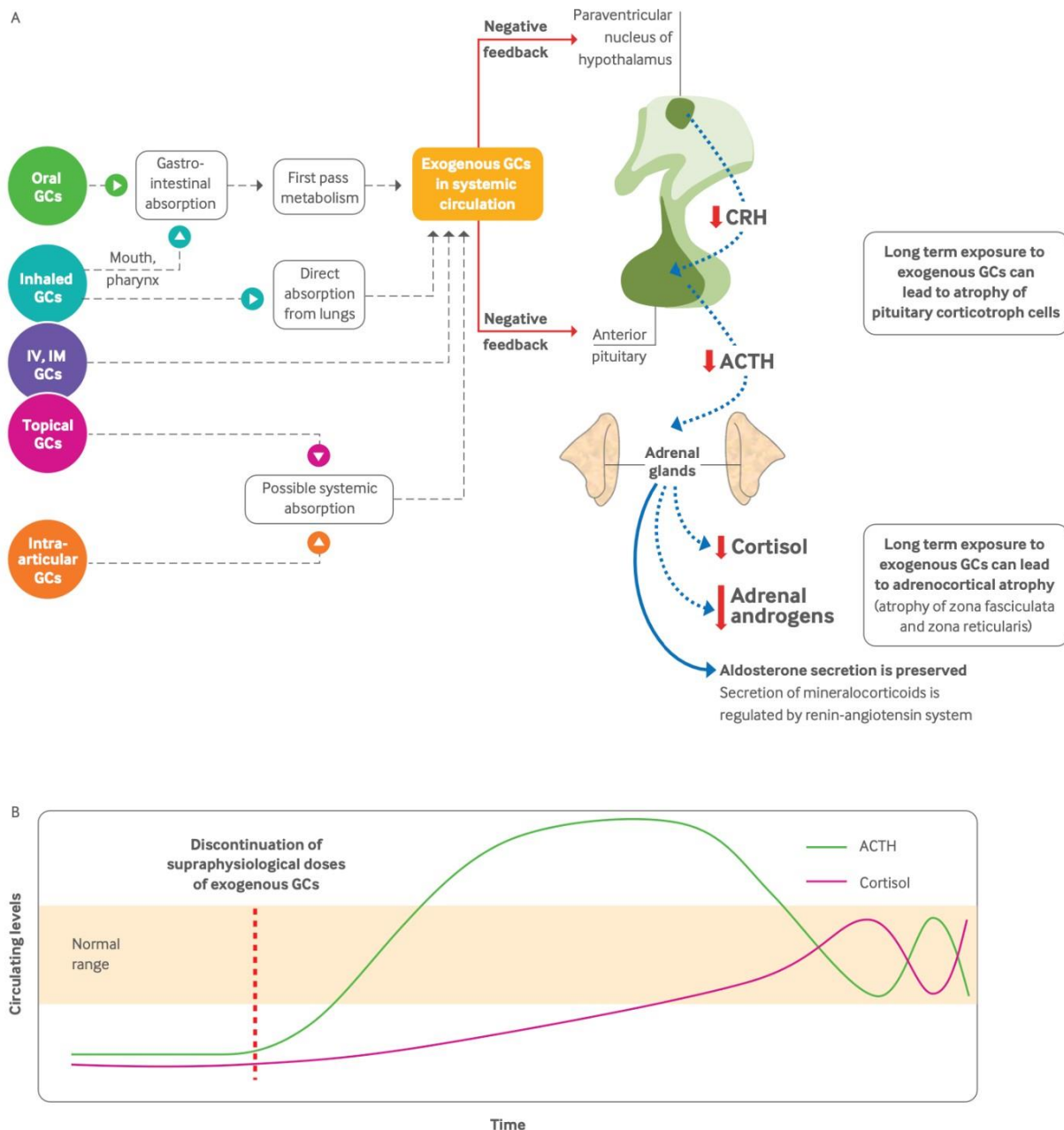
1- Mechanism of Action:

Glucocorticoids that directly enter the systemic circulation and those surviving the first pass metabolism after gastrointestinal absorption exert negative feedback on the corticotropin releasing hormone producing neurons and pituitary corticotrope cells. This leads to reduced adrenal cortisol production and, after prolonged exposure, adrenal cortical hypoplasia and atrophy. After the effect of glucocorticoid therapy on the hypothalamic-pituitary-adrenal axis (HPAA) has worn off, production of ACTH and corticotropin releasing hormone typically recovers first, followed by that of cortisol, which can remain suppressed in the long term if adrenal atrophy has ensued.

2- Pharmacologic use of glucocorticoids:

Glucocorticoids	Equivalent Physiological Dose (mg/day)	Antiinflammatory Activity relative to hydrocortisone	Mineralocorticoid Activity Relative to Hydrocortisone	Duration of Action (Hours)
Hydrocortisone	20	1	1	8-12
Prednisone	5	4	0.8	12-36
Methylprednisolone	4	5	0.5	12-36
Triamcinolone	4	5	0	12-36
Dexamethasone	0.75	30	0	36-72
Betamethasone	0.6	30	0	36-72

Figure 1 : Mechanism of Action



V. DIAGNOSIS

In general, patients taking more than 5 mg of prednisolone or the equivalent for longer than a few weeks are at risk for secondary adrenal insufficiency. Adverse effects of corticosteroids are both dose and time-dependent. Several other factors may influence the adverse effects of glucocorticoids. Older age, comorbid conditions (such as diabetes mellitus), concomitant use of other immunosuppressive agents, severity and nature of the underlying disease, and poor nutritional status can all influence the occurrence and

magnitude of side effects. Not only oral but also inhalation, topical, nasal, intra-articular and intravenous administration of glucocorticoids may lead to adrenal suppression.

VI. MANAGEMENT

A- Withdrawal of Glucocorticoid Therapy

In general, patients who are given acute corticosteroid therapy for less than 14 to 21 days do not develop hypothalamic-pituitary-adrenal axis (HPA) suppression, and treatment can stop with no need for any tapering regime in them. If the therapy has been ongoing for greater than three weeks, tapering is needed.

glucocorticoid therapy may be reduced according to the following scheme:

- 5 to 10 mg/day every one to two weeks from an initial dose >40 mg/day of prednisone (or equivalent)
- 5 mg/day every one to two weeks at prednisone doses ranging from 40 to 20 mg/day
- 2.5 mg/day every two to three weeks at prednisone doses ranging from 20 to 10 mg/day
- 1 mg/day every two to four weeks at prednisone doses ranging from 10 to 5 mg/day
- 0.5 mg/day every two to four weeks at prednisone doses <5 mg/day.

B- Dynamic test

- Performing short stimulation test with synthetic ACTH (SST) can also be used to assess the HPA axis recovery after the diagnosis of glucocorticoids induced adrenal insufficiency. Cortisol level is usually assessed at baseline, 30 and 60 min after IV administration of 250 µg (standard dose test-SST). However, a peak cortisol level higher than 18–20 µg/dL practically rule out glucocorticoids induced adrenal insufficiency.
- Repeating SST annually is suggested if the patient shows a response to SST with delta cortisol <3.6 µg/dL (where delta cortisol means a cortisol value at 30-min minus basal cortisol value) or 30-min cortisol <12.7 µg/dL. SST should be performed every 6 months if delta cortisol is >3.6 µg/dL or 30-min cortisol >12.7 µg/dL. If the response in SST is adequate, glucocorticoids replacement can be safely withdrawn.

VII. SPECIAL CASES

- **Pregnancy and lactation:** glucocorticoids including betamethasone and dexamethasone are used when an effect on the fetus is desired (eg, to speed fetal lung maturity in when anticipating premature delivery) because these cross the placenta. By contrast, the placenta inactivates prednisolone and prednisone and therefore little active drug reaches the fetus. Glucocorticoids are excreted in small amounts in human milk and adjustment to breastfeeding may be indicated for certain doses. A more detailed discussion of the effects of glucocorticoids in pregnancy and lactation is available elsewhere.
- **End-stage kidney disease:** Patients with end-stage kidney disease (ESKD) typically do not require dose adjustments for synthetic, glucocorticoids although the pharmacokinetics may be altered. Among patients treated with hemodialysis, the clearance of total prednisolone is dose dependent, while the clearance of

unbound prednisolone is constant . Hemodialysis also removes significant amounts of methylprednisolone, thereby resulting in a 32 percent reduction in plasma half-life relative to those without ESKD. The removal rate of unbound cortisol in patients treated with peritoneal dialysis is similar to that in patients without ESKD.

- **Severe liver disease** — In the presence of severe liver disease, we commonly use prednisolone rather than prednisone or methylprednisolone for these patients. This is because the activation of prednisone via metabolism to 6-beta-hydroxyl compounds may be impaired, potentially affecting the efficacy of glucocorticoid therapy.

VIII. COMPLICATION

There are numerous potential adverse effects from glucocorticoids, which are most notable for patients receiving high doses and or those with a high cumulative steroid burden over time . However, adverse effects can happen even at lower doses, underscoring the importance of prescribing glucocorticoids. Adverse effects and recommended monitoring are described in more detail elsewhere:

1. **Hypothalamic–Pituitary–Adrenal Axis Suppression**
2. **Bone and Muscle** : Osteoporosis, Osteonecrosis, Myopathy
3. **Metabolic** : Hyperglycemia and Diabetes, Weight Gain , Fat Redistribution, dyslipidemia, Hypokalemia
4. **Ocular Complications** : Cataracts, Glaucoma
5. **Cardiovascular Effects** : Hypertension, Premature atherosclerotic (Ischemic heart disease, stroke) heart failure , Arrhythmias, Venous thromboembolism
6. **Immune system** : increase risk of Infection, decreased response to vaccinations
7. **Gastrointestinal Complications** : increases the risk of peptic ulcer disease, gastritis and upper gastrointestinal bleeding
8. **Dermatologic** : Skin thinning, ecchymoses, Acne, Hirsutism, facial erythema
9. **Neuropsychiatric Complications** : mild anxiety and sleep disturbance to severe mood disorders, psychotic depression, and delirium.

IX. PREVENTION AND RECOMMENDATION

1- Patient Education:

- Educate patients about the potential side effects of glucocorticoid therapy and the importance of adhering to their treatment plan.
- Education on symptoms and management of adrenal insufficiency

2- Assessment of Bone Health

- All adults receiving prednisone 2.5 mg or more daily for more than three months shall be encouraged to optimize calcium and vitamin D intake.
- Bone mineral density (BMD) measurement via DEXA scan shall be performed ideally before or within six months after the initiation of glucocorticoid therapy in all adults 40 years of age or more, and in adults younger than 40 years of age if there is a history of osteoporotic fractures or other risk factors for osteoporosis.
- Bisphosphonates, teriparatide, or denosumab shall be recommended in patients less than 40 years of age but in the moderate or high fracture risk category.

- Bisphosphonates, teriparatide, denosumab, or raloxifene shall be recommended in patients 40 years of age or more in the moderate or high fracture risk category.
 - Oral bisphosphonates are preferred in all these patients.
 - Lateral spine X-ray shall be considered in adults 65 years of age or older to evaluate for vertebral fractures.
- 3- **Assessment of Dyslipidemia and Cardiovascular Risk (Adults)**
- Lipid profile shall be monitored one month after glucocorticoid initiation and then every 6 to 12 months.
 - Glycemic control requires assessment via screening for classic symptoms at every visit: polyuria, polydipsia, weight loss. Monitor glucose parameters for at least 48 hours after glucocorticoids initiation, then every 3 to 6 months for the first year and annually afterward.
- 4- **Assessment of Ophthalmological Complications**
- An annual ophthalmological examination shall be considered, especially for those with symptoms of cataracts, and early referral for intraocular pressure assessment should occur if there is a personal or family history of open-angle glaucoma, diabetes mellitus, or high myopia.
- 5- **Vaccination:** Ensure that patients are up-to-date with vaccinations before starting long-term glucocorticoid therapy. Vaccines such as influenza, pneumococcal, and varicella are recommended to reduce the risk of infections.
- 6- Patients who also require concomitant treatment with non-steroidal anti-inflammatory drugs (NSAIDs) or anticoagulants shall receive therapy with proton pump inhibitors (PPI). There is no consensus on PPI treatment of patients on glucocorticoids alone without NSAIDs and no other risk factors for peptic complications.

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CHAPTER IX INFECTIOLOGY

1. Acute fever
2. Melioidosis
3. Meningitis
4. Cholera
5. Tetanus
6. Leptospirosis
7. Rickettsial infection
8. Enteric Fever
9. Soft Tissue Infection
10. Osteomyelitis
11. Principal of antibiotherapy

ACUTE FEVER

Koy Sophany, Yung Youda, Meng SreyHuoch, Chea Sereywatana, Vann Pisey, Sotharith Bory

I. DEFINITION

Fever is an elevation in core body temperature above the daily range for an individual. However, in practice, a general threshold of temperature $> 38^{\circ}\text{C}$ is often used. Acute fever in adults can result from various infectious or non-infectious causes.

II. EPIDEMIOLOGY

Epidemiology of acute fever in adults involves studying the distribution, determinant, and pattern of fever-related illnesses in populations. Fever can result from a wide range of infectious and non-infectious causes, and its prevalence varies by geographic location, season, and underlying health condition.

1. Infectious Causes

A. Respiratory Infection

- **Pneumonia:** Leading cause of fever in adults worldwide. *Streptococcus pneumoniae*, *Haemophilus influenzae*, *Klebsiella pneumoniae*, and respiratory virus are the common pathogens.

- **Upper Respiratory Infection (URTI):** Common cause of fever, Virus like *rhinovirus*, *influenza*, and *coronaviruses* are predominant.

B. Gastrointestinal Infection

- **Gastroenteritis:** Often caused by *rotavirus*, *norovirus*, and enteric bacteria like *Salmonella* and *Campylobacter*

- **Typhoid Fever:** Caused by *Salmonella enterica*, more prevalent in developing country like Cambodia. Annual incidence estimated at 11-12 millions cases globally.

C. Genitourinary Infections

- **Urinary Tract Infections (UTIs):** Common cause of fever, especially in women. UTI is most common in young women and elderly individuals.

D. Skin and Soft Tissue Infections:

- **Cellulitis/Erysipelas/Abscesses:** Caused by *Streptococcus* and *Staphylococcus* species. It is common in adults with diabetes, immunocompromised states, or peripheral vascular diseases.

E. Vector-Borne Diseases

- **Dengue:** Endemic in tropical and subtropical area. Around 390 million infections annually worldwide.

- **Malaria:** Common in sub-Saharan Africa, Southeast Asia, and part of Latin America. An estimated 229 million cases and 409,000 deaths globally in 2019.

- **Leptospirosis:** Contaminated water exposure.

2. Non-Infectious Causes

- A. **Drug Fever:** Can occur with many medications. The frequency of drug-induced fever is difficult to determine but may account for a significant portion of fever cases in hospitalized adults.

- B. **Autoimmune Diseases:** Condition like systemic lupus erythematosus (SLE), rheumatoid arthritis, and vasculitis may present with acute fever. More common in young to middle-aged women.

- C. **Malignancies:** Some cancers, especially hematologic malignancies (leukemia, lymphoma), present with fever as a primary symptom. Increased incidence in older adults.

D. Heat-Related Illnesses: Fever due to heatstroke can occur in hot climate, especially in the dry season.

III. DIAGNOSIS

1. Clinical Assessment

A. History

1. Onset and duration: Sudden or gradual? Continuous, intermittent, or remittent?

2. Associated symptoms

- Reparatory: Cough, sore throat, shortness of breath
- Gastrointestinal: Nausea, vomiting, diarrhea, abdominal pain
- Skin and soft tissue: Rash, swelling, redness
- Neurological: Headache, altered mental status, seizures
- Musculoskeletal: Joint pain, myalgia

3. Exposure history: Travel, animal bites, mosquito/tick exposure, food or waterborne source, contact with sick individuals (human, animals).

4. Past medical history: Immunosuppression, chronic diseases (diabetes, cancer, autoimmune diseases).

5. Medical history: Recent antibiotic or drug use (drug fever).

B. Physical examination:

1. Vital signs: Temperature, heart rate, blood pressure, respiratory rate, oxygen saturation

2. General inspection: Rash, jaundice, dehydration, lymphadenopathy

3. Focused examination based on symptoms:

- **Respiratory:** Lung auscultation for pneumonia
- **Cardio:** Auscultation for murmur (endocarditis)
- **Gastrointestinal:** Abdominal tenderness, hepatosplenomegaly
- **Neurological:** Neck stiffness, photophobia
- **Skin:** Petechiae, abscesses, erythema

2. Laboratory

A. Basic Laboratory Tests

1. Complete blood cell count (CBC):

- Leukocytosis (Neutrophils) → Bacterial infection
- Leukopenia → Viral infection, Severe sepsis
- Thrombopenia → Dengue, Sepsis

2. C-Reactive Protein (CRP) or Procalcitonin: Elevated in bacterial infection

3. Liver and Renal function tests: To assess organ involvement

4. Lactate: Tissue hypoxia, or sepsis

B. Specific Investigations

1. Urinalysis or Urine cultures: If UTI suspected (better before any Antibiotics treatment)

2. Blood culture: If sepsis suspected or before starting any Antibiotics Treatment, or before changing the Antibiotics Regimen. Two blood bottle cultures from two different sides each time is recommended.

3. Autoantibody: If suspect of Autoimmune diseases

4. Lumbar puncture (LP): If meningitis suspected

5. Dengue NS1 antigen, IgM/IgG
6. Malaria smear or rapid diagnostic test
7. COVID-19 or Influenza PCR
8. Serology of Leptospirosis, Rickettsiosis

C. Laboratory Red Flags

Table 1: Laboratory Red Flags

Test	Critical Value	Possible Condition
WBC	< 4,000 or > 15,000/mm³	Sepsis, leukemia, severe infection
Platelets	< 50,000/mm³	Dengue hemorrhagic fever, severe sepsis, dissemination intravascular coagulation (DIC)
Lactate	> 2mmol/L	Tissue hypoxia, sepsis
CRP/Procalcitonin	Markedly elevated	Sepsis, severe bacterial infection
Liver Enzymes (AST/ALT)	> 100 U/L	Severe sepsis, dengue shock, severe hepatitis
Serum Creatinine	> 2x baseline	Sepsis, Acute kidney injury
Blood Glucose	> 250mg/dL or < 60mg/dL	Hypoglycemia in sepsis, diabetic ketoacidosis

3. Imaging

- a. **CXR:** If the patient come with the symptoms of pneumonia or broncho-pneumonia
- b. **Abdominal ultrasound:** Find the infection foci, deep collection, or abscesses intra-abdominal.
- c. **Heart ultrasound:** If endocarditis suspected
- d. **Brain CT:** if patient present with localized neurological signs (brain abscesses)

IV. MANAGEMENT

A. General Red Flag Signs: Patient need to be hospitalized

- a. **Vital Signs Abnormalities:** **Hight Fever >40c** (suggest for severe infection or sepsis), **Tachycardia >100/mn** (may indicate dehydration, sepsis, systemic inflammation, or pre-shock syndrome), **Hypotension SBP < 90mmHg or MAP < 65mmHg** (suggest shock: septic shock, dengue shock, etc...), **Tachypnea RR > 22/mn** (indicated respiratory distress, pneumonia, etc...), **SaO2 <92%** (suggested severe pneumonia, COVID-19, or ARDS).
- b. **Neurological Signs:** **Altered mental status** (suggested meningitis, encephalitis, sepsis, or metabolic dysfunction), **Focal neurological deficits** (possible stroke or brain abscess).
- c. **Circulatory Signs:** **Weak or absent peripheral pulse** (severe hypotension or cardiovascular collapse), **Cold, clammy skin, mottling, prolonged capillary refill > 3s** (suggested shock).
- d. **Respiratory Distress:** **Intercostal retractions** (impending respiratory failure), **Severe dyspnea and/or cyanosis** (severe pneumonia, ARDS, or sepsis).
- e. **Hemorrhagic or Shock Features:** **Petechiae, purpura, spontaneous bleeding** (dengue hemorrhagic fever, meningococcemia, or DIC), **Severe thrombopenia < 50Giga/L**

B. Severity Classification & Management

Table 2: Severity Classification & Management

Severity Level	Criteria	Management
Mild	No red flags, stable vitals, mild symptoms	Outpatient management with more hydration, symptomatic treatment
Moderate	Persistent fever (> 5days), mild dehydration, mild organ dysfunction	Consider hospitalization for observation and IV fluids
Severe	Any red flag sign (shock, respiratory distress, altered mental status, severe bleeding, organ failure)	Immediate admission, aggressive resuscitation

C. Treatment**❖ Supportive Care (All patients)**

- **Hydration:** Oral or IV fluids if needed
- **Antipyretics:**
 - **Paracetamol:** 500 – 1000mg every 6 to 8 hours
 - **Avoid NSAIDs** in suspected dengue to prevent bleeding risk
- **Monitoring:** Reassess vital signs, urine output, and mental status

❖ Empiric Antibiotic Therapy (If Bacterial Infection Suspected)

- **Community-acquired pneumonia (CAP):**
 - **Mild:** **Amoxicillin** 1g three times per day or **Azithromycin** 500mg daily
 - **Severe:** **Ceftriaxone + Azithromycin**
- **UTI:**
 - **Ceftriaxone** 2g daily or **Ciprofloxacin** 500mg twice daily
- **Meningitis:**
 - **Ceftriaxone + Vancomycin + Dexamethasone**
- **Sepsis (Unknown Source):**
 - **Pip/Tazo or Meropenem** (if suspected of resistant bacterial)
 - **+ Vancomycin** in case suspected of *Staphylococcus* Infection.

V. COMPLICATIONS

Complications (depended on the cause of infection) include sepsis, renal failure (may be chronic), acute respiratory distress syndrome (ARDS), massive hemoptysis due to alveolar hemorrhage (pulmonary hemorrhage syndrome), neurologic complications (meningitis, transverse myelitis, Guillain-Barre syndrome)

VI. PREVENTION AND PUBLIC HEALTH MEASURES

- **Vaccination:** Influenza, COVID-19, pneumococcal vaccine, typhoid vaccine
- **Vector control:** Mosquito nets, repellents (for dengue/malaria prevention)
- **Hand hygiene & sanitation:** Reduce risk of gastroenteritis and respiratory infections
- **Avoid self-medication:** Unnecessary antibiotics increase resistance.

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MELIOIDOSIS

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I. DEFINITION

Melioidosis is an infection caused by a soil and water bacterium called *Burkholderia pseudomallei*. Infection occurs through skin inoculation, ingestion or inhalation.

II. EPIDEMIOLOGY

Melioidosis has most often been described in Southeast Asia and northern Australia however there are increasing reports of infection from South Asia and other parts of the tropical belt. In January 2016, experts on melioidosis predicted that *B.pseudomallei* is ubiquitous through the tropics with the highest risk zones in Asia(southeast and south), tropical Australia, Western sub-Saharan Africa and South America. High rainfall, temperature and soils (both clay and sandy) are strongly associated with the presence of the bacteria. Soil studies in Siem Reap in 2006 confirmed the presence of the bacteria in the soil but no further research (soil or water) has yet been carried out in Cambodia

In Cambodia, the first recorded cases were in 1928 when a Russian visiting from Bangkok, became ill and died soon after arrival. He almost certainly became infected in Thailand however his diagnosis was made for the first time in Cambodia. No further human cases were recorded until October 2005 at Angkor Hospital for Children in Siem Reap.

1- Risk Factors:

All age groups can develop melioidosis but the most common age group affected in between 40 and 60 years. Children account for about 5-15% of all cases. The majority of infections (75-80%) occur during the wet season when there is more contact with soil and water through wind and rainfall. The majority of patients have risk factors which can be divided into medical and occupational/ lifestyle risk factors. Adults are more likely to have medical risk factors than children.

Medical risk factor

- Diabetes mellitus (may not be diagnosed until admission with melioidosis)
- Chronic lung disease
- Chronic renal disease
- Hazardous alcohol excess(including binge drinking)
- Chronic liver disease
- Immunosuppressive therapy e.g.corticosteroids
- Haemoglobinopathies e.g. Thalassemia
- Other: Malignancy, Iron overload, Rheumatic heart disease, Cardiac failure.

At least 20% of patients have no known risk factors. Children are more likely to have no risk factors than adults.

Occupational and Lifestyle risk factors

As the bacteria reside in soil and water, those in close contact with soil and/or water are more likely to develop the infections. These include:

- Rice farmers (when ploughing fields, transplanting rice, skin is often unprotected)
- Non-rice farmer (in contact with water and soil) e.g. rubber plantations, vegetable growing, fruit orchards
- Freshwater fishing, lily collectors, boat workers
- Construction workers (soil disturbance and in contact with soil/water)
- Gardening and recreational activities such as playing football on a wet, muddy field
- Drinking untreated or unboiled water
- Other lifestyle and occupational activities where one is in contact with soil and water.

2. Transmission

Infections are thought to mostly occur through

- 1) skin inoculation
- 2) inhalation
- 3) ingestion un-chlorinated water

III. ETIOLOGY

Gram negative bacteria: *Burkholderia pseudomallei*.

IV. PHYSIOPATHOLOGY

B.pseudomallei is an opportunistic, facultative intracellular, motile saprophyte (an organism that obtains its energy from decaying organic matter) that possesses a remarkable intrinsic array of virulence factors and broad antimicrobial drug resistance. *B.pseudomallei* is highly adaptable, a property that enables it to generate a variety of clinical manifestations, depending on the infected tissue, and to maintain a survival advantage in infected hosts and the environment.

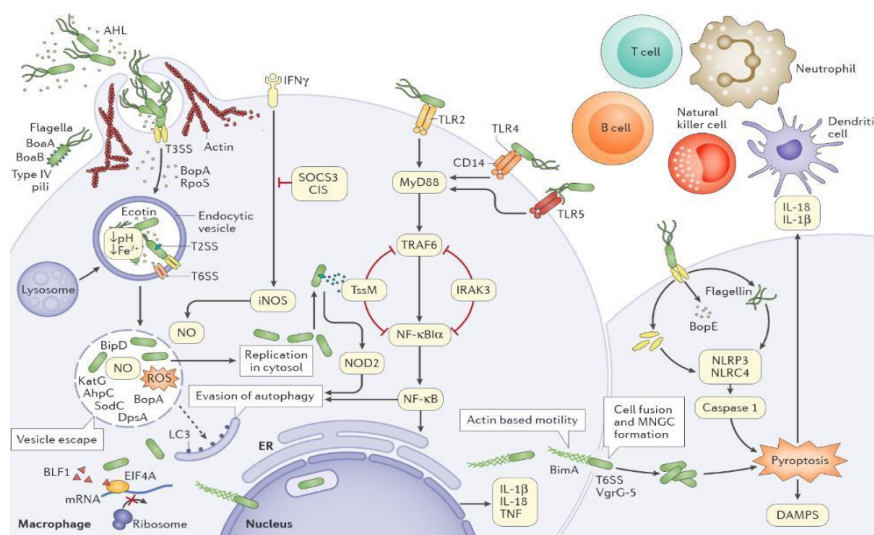


Figure 1: Schematic model of host-pathogen interactions and pathophysiology of melioidosis. Wiersinga WJ, Virk HS, Torres AG, Currie BJ, Peacock SJ, Dance DAB, et al. *Melioidosis*. *Nat Rev Dis Primers*. 2018 Feb 1;4(1):17107.

V. DIAGNOSTIC

A. Clinic

- Asymptomatic infection is common and due to exposure to the bacterium from the environment.
- Symptomatic infection includes acute presentations (85% of cases), chronic infection (those with symptoms for more than 2 months) which accounts for 11% of infections-or latent infection (<4%).
- Clinical presentation is varied and includes acute life-threatening pneumonia and/or sepsis to deep visceral abscess (liver, spleen,kidney,prostate), head and neck abscesses, skin infections(ulcers, abscesses,cellulitis),joint and bone infections and bacteraemia with no focus of infections. Melioidosis can affect any organ of the body and is a great mimicker of other diseases particularly TB.
- More than half of all patients are blood culture positive on admission to hospital, so blood cultures are important to collect on all suspect melioidosis patients.
- Chronic infection can present with chronic respiratory symptoms, cervical lymphadenopathy, chronic skin ulcers/abscesses or deep visceral abscesses. These presentations can often be confused with TB or other infections.

B. Para clinic

- Diagnosis can only be confirmed by culture of *B.pseudomallei* from blood,urine,sputum, pus, bodily fluids, bone, throat or rectal swabs.
- All patients suspected to have an infection should have **blood, urine and throat swabs** collected. Additional specimens should also be collected when present e.g. sputum,pus.
It is recommended that laboratories use selective media (Asdown's broth and agar) for all non-sterile specimens to increase the chance of isolation and identification.
- Doctors should always include clinical presentation, risk factors and the suspicion of melioidosis on the microbiology request form so laboratory staff can process specimens appropriately.
- All suspect and culture confirmed cases should have chest X-ray and abdominal ultrasound performed to rule out additional sites of infection.

VI. DIFFERENTIAL DIAGNOSIS

- Acute lung symptoms- other causes of community acquired pneumonia
- Chronic lung symptoms- TB (especially those with AFB smear negative sputum or those with previously treated TB), other infections, malignancy including metastases)
- Sepsis-other common causes of sepsis (E. coli, Staphylococcal infection, Salmonella Typhi/Paratyphi etc)
- Deep abscesses- Staphylococcus aureus, Klebsiella pneumoniae, TB, amoebic abscess

- Lymphadenopathy-TB (especially those who may have previously been unsuccessfully treated with anti-tuberculous treatment)
- Skin abscesses-Staphylococcus aureus, other infections
- Chronic skin ulcers/wounds unresponsive to commonly prescribed antibiotics
- Septic arthritis/osteomyelitis-Staphylococcus aureus, TB.

VII. TREATMENT

- There are 2 phases of treatment-intensive and eradication phases.

Intensive phase:

Ceftazidime 2g IV every 6-8 hours (q6-8h) for at least 14 days (maximum 8g per day) (Don't change treatment if patient's fever continues: median time for fever clearance is 9 days)

Meropenem 1g IV 8h can be given for those with septic shock (and de-escalate to ceftazidime after improvement).

Consider adding co-trimoxazole (TMP-SMX) to intensive phase if

- Deep tissue infection or
- Patient's condition worsening despite drainage of pus (and isolate shows susceptibility to ceftazidime) or
- Wanting to ensure no side effects to co-trimoxazole prior to discharge on oral medications.

Eradication Phase: After at least 14 days IV therapy and clinical improvement, change to oral

1st choice: co-trimoxazole for at least 3 months (12 weeks). Osteomyelitis and CNS infection required 6 months treatment. Folic acid is also recommended when prescribing co-trimoxazole.

Table 1 Dosing of co-trimoxazole and folic acid by weight (14,39)

Tablet and strength	Children 6/30mg/kg	Adults		
		<40 kg (6/30mg/kg)	40-60 kg	>60 kg
80 mg TMP-400mg SMX (single strength)	Maximum 3 po q12h	Maximum 3 po q12h	3 po q12h	4 po q12h
160mg TMP-800mg SMX (double strength)	Maximum 1.5 po q12h	Maximum 1.5 po q12h	1.5 po q12	2po q12h
Folic acid maximum 5 mg	0.1 mg/kg	0.1 mg/kg	5mg	5mg

2nd choice : The most effective drug for eradication therapy is co-trimoxazole however where this is contraindicated (neonate, first trimester of pregnancy) or not tolerated, co-amoxiclav is second choice providing the correct ratio of amoxicillin to clavulanic

acid (4:1) is used e.g. amoxicillin 500mg: clavulanic acid 125mg. Many preparations in Cambodia have a different ratio so ensure the correct dose is available.

Table 2 dosing of co-amoxiclav by weight (35)

Tablet and strength	<60 kg 20/5mg/kg	≥60kg
Co-amoxiclav (amoxicillin/clavulanate) 500 mg/125 mg tablet 8 hourly (must have ratio 4:1 of amox:clav)	Maximum 2 tabs po q8h	Maximum 3 tabs po q8h

3rd choice

Doxycycline has higher relapse rates than both co-trimoxazole and co-amoxiclav but where co-trimoxazole is unable to be given due to adverse effects or co-amoxiclav is not available in the correct ratio, doxycycline 100mg q12h should be given (avoid use in children under 8 years or pregnant women).

VIII. PATIENT CONSULTATION AND PREVENTION

- Prevention should be targeted at the most common methods of transmission- skin inoculation, ingestion and inhalation particularly in those with risk factors. This includes protection of skin when working with water and soil, drinking appropriately treated water and avoiding rainstorms.
- Person to person transmission in hospitals is very unlikely, ward staff do not need to take special precautions and patients do not have to be isolated.

IX. ANNEXES

Table 3 Australian Northern Territory guidelines for duration of antibiotics by site of infection (14)

Focus of infection	Intensive phase duration (weeks)*	Eradication Phase duration (months)
Skin abscess	2	3
Bacteraemia (no focus) - (always ensure repeated and careful examination, check for abscesses with ultrasound or CT, CXR, urine and throat swabs)	2	3
Pneumonia – no intubation/inotropes; no mediastinal lymphadenopathy	2	3
Pneumonia – severe requiring intubation/inotropes or mediastinal lymphadenopathy (>10 mm diameter)	4	3
Parotitis; other head and neck abscesses	2	3
Deep seated abscess (liver, spleen, prostate, psoas etc)	4**	3
Septic arthritis	4	3
Osteomyelitis	6	6
Central nervous system	8	6
Vasculitis	8	May be lifelong

Table 4 Treatment dosing in renal impaired adults (14)

	Dose adjustment by Creatinine Clearance (ml/min)*		
	31-50	15-30	<15
Ceftazidime	<60 kg 1g q8h	< 60 kg 1g q12h	<60kg 1 g q24h
	>60 kg 2g q8h	>60 kg 2g q12h	> 60kg 2g q24h
Meropenem	1g q12hrs	1g q12h	1g q24h
TMP + SMX**	<60 kg 240+1200mg q12h	<60 kg 240+1200mg q24h	<60 kg 240+1200mg q24h
	>60 kg 320+1600mg q12h	>60 kg 320+1600mg q24h	>60 kg 320+1600mg q24h
Co-amoxiclav	Follow dosage adjustment recommendations provided in paediatric renal dosing table below.		

* Creatinine clearance is calculated by Cockcroft-Gault method:

$$\text{CrCl (mL/min)} = (140 - \text{age (yrs)}) \times \text{ideal body weight} \times 0.85 (\text{female}) / (\text{serum creatinine (mg/dL)} \times 72)$$

**Trimethoprim + sulfamethoxazole. Folic acid 5mg daily is added for the duration of therapy

Figure 2 Melioidosis - hospital WITH microbiology laboratory available

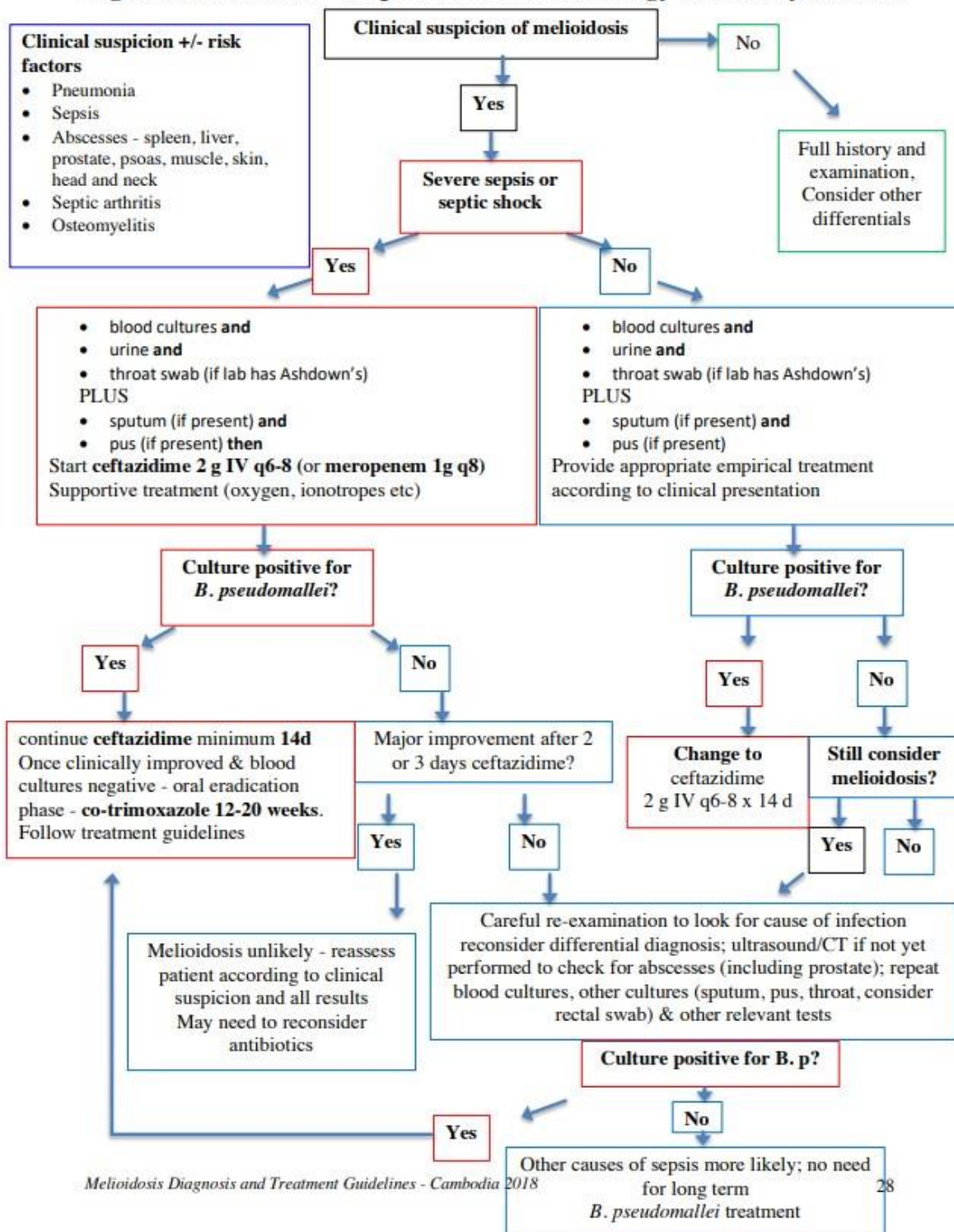
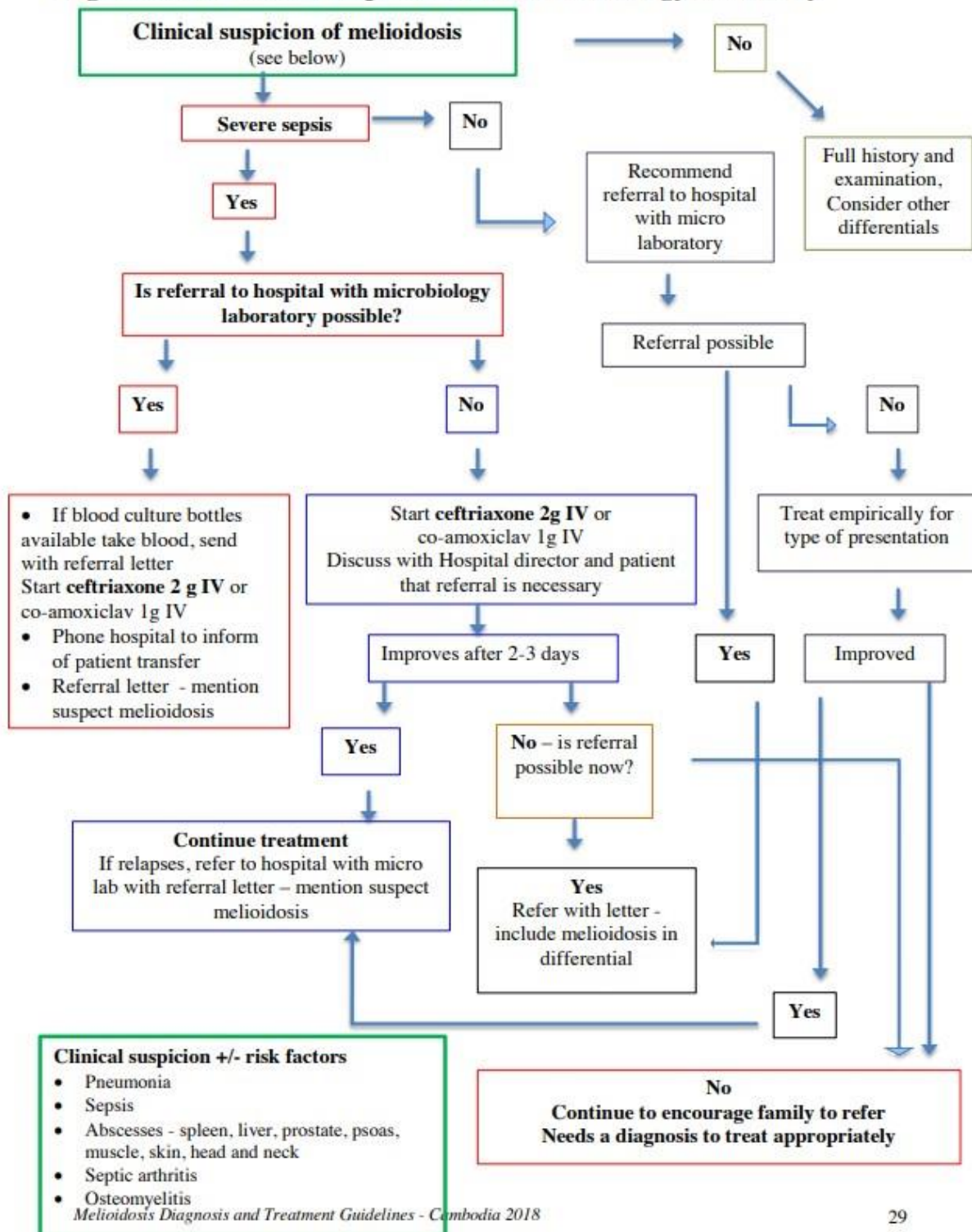


Figure 3 Melioidosis - hospital with NO microbiology laboratory



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MENINGITIS

Meng Sreyhuoch, Koy Sophany, Young Youda, Chea Sereywatana, Lim Krui, Vann Pisey, Choun Kimcheng, Phe Thong, Bory Sotharith

I. DEFINITION

Meningitis is acute inflammatory disease of leptomeninges, tissue surrounding brain and spinal cord and is characterized by an abnormal number of immune cells (mostly Neutrophils and lymphocytes) in the cerebrospinal fluid (CSF) associated with exudate protein. The meninges consist of three parts: the pia, arachnoid, and dura mater. The meningitis reflects infection of the arachnoid mater and the CSF in both the subarachnoid space and the cerebral ventricles. The main causes are infection, cancer and auto-immune disease.

II. EPIDEMIOLOGY

Meningitis is considered as a serious health problem that needs rapid diagnostic, effective management and prevention strategies, especially in endemic countries and periods.

Incidence of meningitis is annually estimated at 20 cases per 100 000 people, and it can lead to hundreds of thousands of deaths worldwide.

III. ETIOLOGY

1. Bacterial meningitis

The most frequent causes of acute bacteria are *Haemophilus influenzae*, *Streptococcus pneumoniae*, *Neisseria meningitidis* and *Listeria monocytogenes*.

- *Streptococcus pneumoniae* is the most frequent etiology of meningitis in adult age over 20 years old. Normally associated with ENT infection like sinusitis, otitis media, splenectomy, diabetes, alcoholism. The mortality rate is around 20% despite antibiotic therapy.
- *Neisseria meningitidis* is frequently cause of meningitis in 15-25 years of age but now it has decreased with routine immunization of 11-18 years old with the quadrivalent (Serogroup A, C, W-135 and Y).
- *Haemophilus influenza* type B is usually seen in neonatal meningitis and has only 5% of cases in all age patients.
- *Listeria monocytogenes* is common cause of meningitis in neonates (<1 month of age), pregnancy women, patients > 60 years, and immunocompromised individuals of all ages. Infection is acquired by ingesting foods contaminated by *Listeria* (coleslaw, milk, soft cheeses, and several types of "ready-to-eat" foods, including delicatessen meat and uncooked hotdogs).
- *Staphylococcus aureus* and *coagulase negative* are the main etiology of meningitis post neurosurgery procedure.

- Gram-negative bacilli are rare but can cause of meningitis in patient with chronic condition like diabetes, alcoholism, cirrhose, chronic urinary infection and complicated from neurosurgery.
- *Mycobacterium tuberculosis* (MBT) is the one of important etiology of meningitis in the location with high prevalence of MBT in population and immunocompromised patient.
- *Streptococcus suis* is zoonotic causes of bacterial meningitis and most prevalent in Southeast Asia (like Vietnam...), is associated with exposure to pigs and alcoholism.

2. Viral meningitis

- Herpes simplex viral 2 (HSV-2): more commonly seen in young females with 50 percent of them having a history of genital herpes and 30 percent having a history of previous viral meningitis.
- Viral infection can associate meningitis (Varicella-zona virus, HIV, mumps...)
- Enteroviruses and parechoviruses : the most commun cause of viral meningitis at summer and fall.

3. Fungal meningitis

Fungal meningitis is rare but normally present in immunocompromised patient like HIV, on immunosuppressor...

- Cryptococcal infection: commonly present in with severe deficiencies in cell-mediated immunity (HIV with CD4 lower than 100/mm³, immunosuppressor medication...).
- Candida species and others fungal infection

4. Parasitic meningitis

Parasitic meningitis is called eosinophilic meningitis (EM) and less common cause than bacteria and viral.

Risk factor vary but can include age, animal contact and geographic location. There are 3 main parasites that cause EM.

- *Angiostrongylus cantonensis* (neurologic angiostrongyliasis): most of these infections are diagnosed in Southeast Asia and the pacific Islands included in Hawaii by eating undercooked or exposed to infected animal (snail or slugs).
- *Baylisascaris procyonis*
- *Gnathostoma spinigerum*: most present in Thailand and Japan (contaminated by eating freshwater fish or eels, frogs, poultry and snaked)
- Others parasite: *toxocariasis* (visceral larva migrans), *trichinellosis*, *echinococcosis*, *neurocysticercosis*, *fascioliasis*, *paragonimiasis*.

IV. PATHOGENESIS

Micro-organism reach the central nervous system by 3 main mechanisms

- Hematogenous spread: bacteria crossing the blood-brain barrier (most notably *Streptococcus pneumonia*, *Listeria monocytogenes*, *N.meningitidis*...) and some virus.
- Direct extension of infection from ENT spread: otitis media, sinusitis and mastoiditis (*Streptococcus pneumonia*)
- Direct inoculation: the result of the manipulation of the meninges during neurosurgical produces (Nosocomial infection) or traumatism.

V. DIAGNOSIS

1. Risk factor

- Chronic medical disorder (Diabetes, Chronic kidney disease, cystic fibrosis)
- Extremes age
- Immunosuppressed states
- Travel to endemic areas
- Bacteria endocarditis
- IV drug use
- Sickle cell anemia
- Splenectomy
- Contiguous infection (Sinusitis, mastoiditis, otitis media and pneumonia)
- Cranial anatomical defects and ventriculoperitoneal shunt

2. Clinical manifestation

Presenting manifestation:

The classic clinical trial of meningitis: consists of fever, nuchal rigidity, and a change in mental status with headache

- Infection syndrome: Fever with temperature greater than 38 C ° presents 77% of cases. Sometime it can be miss because of use antipyretic or corticosteroids.
- Severe headache: diffuse and permanent, associated with photophobia and phonophobia.
- Nuchal rigidity: present around 77% of cases.
- Mental status change is also report in meningitis infection.

Examination for meningeal irritation

- The classic Brudzinski sign refers to spontaneous flexion of the hips during attempted passive flexion of the neck.
- The Kernig sign refers to the inability or reluctance to allow full extension of the knee when the hip is flexed 90 degrees. The Kernig test is usually performed in the supine position, but it can be tested in the seated patient.

Specific clinical features: A rapidly spreading of petechial rash are found on the trunk and lower extremities, in the mucous membranes and conjunctiva, and occasionally on the palms and soles known as purpura fulminans would suggest a Meningococcal infection.

Sign of gravity

- Unstable hemodynamic
- Purpura fulminans
- Brain suffering

3. Laboratory

➤ General blood test finding:

- Serology HIV should do all cases suspected meningitis
- CBC: increased of WBC, low red blood cell and low plaquette in some. Neutropenia can be found in some severe infection
- Coagulation test should perform
- C-reactive protein: is likely to high in present of bacterial infection
- Electrolytes: low sodium may point toward tuberculous meningitis
- Elevated of lactate ($>4\text{mmol/L}$) is normally in sepsis or septic shock

➤ Diagnostic testing:

- Blood culture: should be immediately obtained and empirical antimicrobial and adjunctive dexamethasone therapy initiated without delay
- Lumbar puncture: analyze of cerebrospinal fluid (CSF) for gram stain, culture, complete cell count, glucose, protein and Gen X-pert, BAAR in case suspect tuberculosis meningitis and others rapid test, polymerase chain reaction (PCR) depends on clinical suspicion.
 - Contre indication to LP
 - Skin infection in the place of LP
 - Unstable hemodynamic or unstable respiratory
 - Problem of coagulation as known (Coagulopathy hemophilia, thrombopenia plaquette less than 50 G/L, on anticoagulant dose treatment, spontaneous hemorrhage result from CIVD).
 - **Indications for CT scan before LP (one or more following risk factors:** 1. Immunocompromised state (HIV, solid organ or hematopoietic cell transplantation), 2. History of central nervous system diseases (mass lesion, stroke or focal infection), 3. Papilledema, 4. Abnormal level of consciousness, 5. Focal neurologic deficit.
 - Analyze LP
 - **Normal findings:** Normal CSF is clear and colorless, WBC less than 5/mm³ (No neutrophile and no hematic), glycorrhachia more than 2/3 of glycemia, protein concentration less than 0.4g/L and culture negative.

- **Meningitis with purulent CSF:** Cloudy, WBC more than 100/mm³ (Predominant PMN), decreased glycorrhachia less than 0.4 of glycemia, increased protein concentration 1g/L, culture positive in 60-80% of cases (if no antibiotic before LP), CSF Gram's stain demonstrates organisms in >60%
- **Meningitis with clear CSF:** WBC 5-100/mm³ (lymphocyte >50%), Glycorrhachia 2/3 of glycemia (viral meningitis), Glycorrhachia less 0.4 of glycemia (BK or listeriosis), protein concentration less than 1g (viral) and 1-2g/l (bacteria)
- Cryptococcus can detect in direct analyze of CSF, find antigen cryptococcus in CSF.
- PCR of virus herpes simplex (can do if suspect meningitis by herpes simplex).
- Some examen and imagery should be done to find out the primary foyer of meningitis (ENT examination, Chest X-ray, Brain scanner or cerebral IRM in case
- *Choice of methods of confirmation diagnostic will depend on availability at local laboratories and physicians should consult local testing protocol.*

VI. DIFFERENTIAL DIAGNOSTIC

- Subarachnoid hemorrhage (SAH)
- Acute brain abscess
- Acute encephalitis
- Non infection meningitis (tumoral, auto-immune, chimiotherapy)

VII. MANAGEMENT

1. Supportive care

- Hospitalization at emergency or conventional department depends on their gravity
- Fluid and electrolyze management
- Apyretic management
- Reduce intracranial pressure (ICP) if need
- Treatment of comitial crise and prevention recidivism

2. Acute bacterial meningitis

- Bacterial meningitis is medical emergency treatment. The goal is to begin antibiotic therapy within 60 min of a patient's arrival in the emergency room.
- The antimicrobial therapy needs to against *S. pneumoniae* and *N. meningitidis* that are most common etiologic organisms of community-acquired bacterial meningitis. The antibiotic the choice should depend on the local epidemiology of antibiotic resistance.

- Empirical therapy:
 - Antibiotic: Third generation cephalosporin (Ceftriaxone 2 g every 12 hours or Cefotaxime 300mg/kg/day divided in 4-6 perfusion) plus
 - Vancomycin
 - Loading dose 20 to 35 mg/kg (based on actual body weight, rounded to the nearest 250 mg increment; not to exceed 3000 mg) with infusion rate 10-15mg/min.
 - Maintenance or typically dose 15 to 20 mg/kg every 8 to 12 hours for most patients (based on actual body weight, rounded to the nearest 250 mg increment) with infusion rate 10-15mg/min.
 - Corticoid: Dexamethasone 0.4mg/kg/12h or 0.15mg/kg/6h duration 4days (Dexamethasone should 15-20 min before administration or not later than concurrent with the first antibiotic)
 - Adaptation after antibiogram
- Pneumococcal Meningitis:
 - Antibiotic:
 - a. In case penicillin G MIC < 0.1ug/ml: Penicillin G 4million units IV q4h or Ampicillin 2gm IV q4h
 - b. Penicillin G MIC > 0.1ug/ml and ceftriaxone < 0.5ug/ml: Ceftriaxone 2g in every 12h or Cefotaxime 200mg/kg/day divided in 4-6 perfusion)
 - c. Penicillin G MIC > 0.1ug/ml and ceftriaxone > 0.5ug/ml: Ceftriaxone 2g in every 12h or Cefotaxime 300mg/kg/day divided in 4-6 perfusion plus Vancomycin 15-20mg/kg/8-12h with adaptation kidney function.
 - Dexamethasone 0.4mg/kg/12h [14]or 0.15mg/kg/6h intravenous
 - Duration 10-14days
- Listeria Meningitis:
 - Antibiotic:
 - a. Ampicillin 2gm IV q4h associated with (gentamycin 5mg/kg/day q24h perfusion 30 min only first 5 days)
 - b. Alternative regimes: The combination of trimethoprim (10–20 mg/kg per day) and sulfamethoxazole (50–100 mg/kg per day) given every 6h may provide an alternative in penicillin-allergic patients.
 - Duration 3 weeks totally.
- Meningococcus meningitis:
 - Antibiotic: Ceftriaxone 2g in every 12h or Cefotaxime 200mg/kg/day divided in 4-6 perfusion
 - Dexamethasone 0.4mg/kg/12h or 0.15mg/kg/6h intravenous
 - Duration: 4-7days

- Droplet precautions should be used for 24 hours after starting effective antimicrobial therapy in patients with suspected or confirmed *N. meningitidis* infection
- Chemoprophylaxis with Ciprofloxacin/azithromycin/ceftriaxone for all closed contact person (Expose to oropharyngeal secretion of the patient).
- *Staphylococcus aureus*
 - Antibiotic:
 - a. Methicillin-susceptible *S. aureus* (MSSA) : Nafcillin (2 g IV every four hours) or oxacillin (2 g IV every four hours)
 - b. Methicillin-resistance *S. aureus* (MRSA):
 - Loading dose 20 to 35 mg/kg (based on actual body weight, rounded to the nearest 250 mg increment; not to exceed 3000 mg) with infusion rate 10-15mg/min.
 - Maintenance or typically dose 15 to 20 mg/kg every 8 to 12 hours for most patients (based on actual body weight, rounded to the nearest 250 mg increment) with infusion rate 10-15mg/min.
 - c. Adaptation dose depends on kidney fonction
 - Duration: 2 weeks and can be extend up to 4-6weeks (in the setting of endocarditis, discitis, or other associated metastatic infections).
- *Haemophilus influenzae*:
 - Antibiotic: Ceftriaxone 2g in every 12h or Cefotaxime 200mg/kg/day divided in 4-6 perfusion.
 - Dexamethasone 0.4mg/kg/12h or 0.15mg/kg/6h intravenous
 - Duration: 7days.
- Gram negative bacilli
 - Enterobacteria (Mostly *E. coli*): Ceftriaxone 2g in every 12h or Cefotaxime 200mg/kg/day divided in 4-6 perfusion plus (gentamycin 5mg/kg/24h in the first 2 days)
 - Non enterobacteria (*Pseudomonas aeruginosa*): Ceftazidime or cefepime or meropenem.
 - Duration: 21 days.
- Tuberculosis meningitis: Management follow by TB national guideline

3. Viral meningitis

- Viral meningitis: primarily symptomatic and includes use of analgesics, antipyretics, and antiemetics. Fluid and electrolyte status should be monitored.
- In case herpes meningitis: Acyclovir 10mg/kg/q8h infusion over 1h each dose (to reduce nephrotoxicity with larger doses of acyclovir) until clinical improved then Valacyclovir 1000mg q8h for duration total 10-14days

- In case herpes encephalitis: Acyclovir 10mg/kg/q8h infusion over 1h each dose (to reduce nephrotoxicity with larger doses of acyclovir) for duration 14-21days.

4. Fungal meningitis

- Cryptococcosis meningitis in non-HIV patient
 - o Induction phase:
 - Amphotericin B 0.7-1mg/kg q24h in perfusion plus Flucytosine 25mg/kg oral q6h.
 - Duration: minimum 4 weeks (treat until patient is afebrile and culture are negative)
 - o Consolidation therapy: Fluconazole (400 to 800 mg/day) for 8 weeks.
 - o Maintenance therapy: Fluconazole (200 mg/day) for 6 to 12 months.
- Cryptococcosis meningitis in transplantation patient
 - o Induction phase:
 - Amphotericin B 0.7-1mg/kg q24h in perfusion plus Flucytosine 25mg/kg oral q6h.
 - Duration: minimum 2 weeks (treat until patient is afebrile and culture are negative)
 - o Consolidation therapy: Fluconazole (400 to 800 mg/day) for 8 weeks.
 - o Maintenance therapy: Fluconazole (200 mg/day) for 6 to 12 months in transplantation patient.
- Cryptococcosis meningitis in HIV patient
 - o Induction phase:
 - Amphotericin B 0.7-1mg/kg q24h in perfusion plus Flucytosine 25mg/kg oral q6h.
 - Duration: minimum 2 weeks (treat until patient is afebrile and culture are negative)
 - o Consolidation therapy: Fluconazole (400 mg/day) for 10 weeks
 - o Secondary prophylaxis: Fluconazole (200 mg/day).
 - o Discontinuation of antifungal therapy can be considered among patient who have completed induction therapy and received consolidation for at least 12months and successful antiretroviral therapy (viral load <50/ml) and CD4>, = 100/mm3 for 3months.
- LP repeat every week of the treatment (in order to measure pressure and re-evaluation of the treatment)
- No corticoid uses in fungal meningitis

VIII. PRONOSIS

- Outcomes depend on patient characteristics such as age and immune status but also vary depending on the etiologic organism.
- Mortality rate is 3–7% for meningitis caused by *H. influenzae*, *N. meningitidis*, or group B streptococci; 15% for that due to *L. monocytogenes*; and 20% for *S. pneumoniae*.

IX. COMPLICATION

- Acute complication:
 - Encephalitis
 - Collection intra-dural or epidurals (Empyema)
 - Cerebral Thrombophlebitis
 - Brain abscesses
- Complication neurologic:
 - Hearing loss
 - Trouble behavioral and cognitive difficulties
 - Motor deficit
 - Seizure disorder
 - Visual impairment
- Hydrocephalus
- Cerebrovascular complications

X. PREVENTION

- All healthcare workers (nurses, physicians, and pharmacists) should educate patients and parents in regards to vaccine-preventable meningitis (*H. influenzae* type B, *S. pneumoniae*, *N. meningitidis*, Measles, and Varicella).
- Consult with doctor before you travel to endemic region

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CHOLERA

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I. DEFINITION

Cholera is an acute and life-threatening diarrheal infection caused by consuming food or water contaminated with the bacterium *Vibrio cholerae* leading to significant public health concerns, especially in areas with inadequate sanitation and unsafe drinking water. Cholera is a preventable and treatable disease where awareness and prompt treatment are essential to reduce mortality rates associated with this disease.

II. PHYSIOPATHOLOGY

1. Transmission

Vibrio cholerae is transmitted via the fecal-oral route which includes:

- Transmission within households or between close contacts.
- Outbreaks related to contaminated food and water.
- The infectious dose of *V. cholerae* required to cause cholera is thought to be relatively high; an inoculum of 10^8 *V. cholerae* or higher resulted in severe infection among healthy volunteers.

2. *Vibrio cholerae* strains

Vibrio cholerae is the bacterium responsible for cholera, and it is classified into several strains based on serogroups and biotypes. Only two serogroups, O1 and O139, cause cholera outbreaks. *V. cholerae* O1 has caused all recent outbreaks. *V. cholerae* O139 caused outbreaks in Asia in the past but has recently only been identified in sporadic cases. There is no difference in the illness caused by the two serogroups.

- Serogroups:
 - O1: This is the most common serogroup associated with cholera epidemics. It is further divided into two biotypes:
 - O139: This serogroup emerged in the early 1990s and caused significant outbreaks, particularly in Asia. It is genetically distinct from O1 and has been linked to severe cholera cases.
- Virulence Factors:
 - Pathogenic strains of *V. cholerae* possess virulence factors such as cholera toxin and toxin co-regulated pilus (TCP), which are crucial for their ability to cause disease. Cholera toxin is responsible for the severe diarrhea seen in infected individuals.

3. Mechanism of infection

There are certain conditions for infection to occur:

- *V. cholerae* must survive the acidic environment of the stomach, then attach to the small intestine and produce cholera toxin, which is the main virulence factor of pathogenic *V. cholerae* strains.
- It activates an intracellular pathway resulting in massive fluid secretion into the small intestine, leading to large volumes of fluid loss with high concentrations of sodium, chloride, bicarbonate, and potassium.
- The use of proton pump inhibitors and antihistamines can increase the risk of infection and make the patient susceptible to more severe symptoms.

- The fluid losses typically occur from the duodenum, whereas the colon is insensitive to the toxin. Because the enterotoxin has a local effect and is not invasive, in most cases no neutrophils are observed in fecal specimens.

III. EPIDEMIOLOGY

Each year, there are an estimated 3 million cases of diarrheal illness and approximately 100,000 deaths worldwide caused by *Vibrio cholerae*. Cholera occurs primarily in settings where there is inadequate access to clean water and sanitation. The number of cholera cases reported to WHO has continued to rise in recent years. In 2023, 535 321 cases and 4007 deaths were reported to WHO from 45 countries, up from 472 697 in 2022.

1. Cambodia: Disease Outbreak Reported

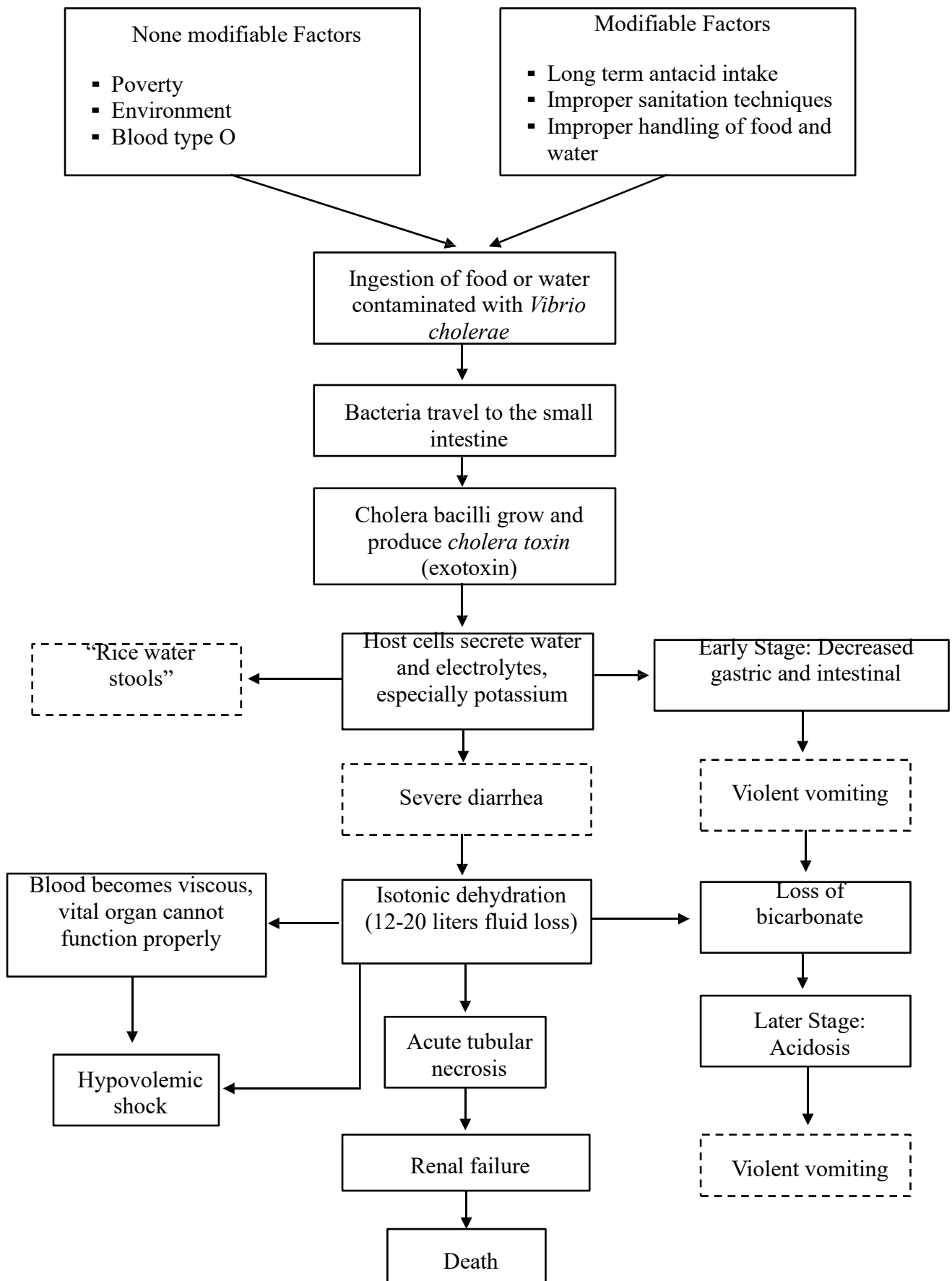
- In 21 May 1999, the Ministry of Health of Cambodia reported a cholera outbreak, started on 16 April to 16 May. in Rotanakiri province.
- A total of 874 cases with 56 deaths was reported.
- Lack of good hygiene and sanitation facilities, as well as difficulty in sending supplies because of poor road conditions in the area contributed to the spread of the disease.
- The Ministry of Health is providing oral rehydration salts and antibiotics and is also organizing health education campaigns for the affected and surrounding villages.

2. Risk factors

Cholera infection was found to have been associated with the following conditions:

- People whose incomes are under poverty thresholds and lack of access to safe food, water, and adequate sanitation.
- Large cholera epidemics often occur in populations impacted by natural disaster or human conflict.
- For patients with hypochlorhydria or partial gastrectomy, a lower infectious dose is needed to cause infection.
- The organism is acquired via the fecal-oral route and a large dose is required to develop infectivity.
- Use of proton-pump inhibitors (PPIs) and antihistamines
- Having type O blood
- Overcrowding
- Prior vagotomy
- *Helicobacter pylori* infection

Pathophysiology



IV. DIAGNOSIS

Patients with suspected cholera should undergo careful assessment for signs and symptoms of volume depletion.

1. Incubation period

The incubation period is one to two days; however, it can range from several hours to as long as three to five days. Factors influencing the incubation period include:

- Host susceptibility
- Inoculum size

Vibrio cholerae infection is associated with a spectrum of manifestations ranging from asymptomatic intestinal colonization to severe diarrhea.

2. Gastrointestinal symptoms

- Vomiting may begin before or after onset of diarrhea. Patients may have abdominal cramping but usually do not have frank abdominal pain typically associated with dysentery.
- In early illness, cholera stools may contain fecal matter and bile.
- Fever is uncommon; presence of fever should prompt evaluation for concurrent infection.

Severe cholera ("cholera gravis") is characterized by passage of profuse "rice-water" stool, a watery stool with flecks of mucous. It typically has a fishy odor. The diarrhea is usually painless, without tenesmus.

- Approximately 5 to 10 percent of patients develop severe hypovolemia.
- In children with severe cholera, stool output is typically between 10 and 20 mL/kg per hour.
- In adults with severe cholera, stool output can be as high as 1000 mL per hour. The total volume loss over the course of illness may be up to 100 percent of body weight.

3. Manifestations of hypovolemia and electrolyte loss

Severe cholera is associated with rapid development of hypovolemic shock due to profound fluid and electrolyte losses, often within hours of the symptom onset.

- Patients with severe hypovolemia may have sunken eyes, dry mouth, cold clammy skin, decreased skin turgor, and wrinkled hands and feet ("washer woman's hands").
- Patients are frequently apathetic and lethargic.
- Kussmaul breathing (deep respirations reflecting compensatory hyperventilation) may occur due to acidosis from loss of stool bicarbonate as well as lactic acidosis from poor perfusion.
- Initially the pulse is rapid and thready; as the blood pressure drops, the pulse may become difficult to palpate.
- Muscle cramping and weakness due to loss of potassium and calcium are common.

Laboratory testing may demonstrate hypokalemia, hyponatremia or hypernatremia (although cholera is most often associated with isonatremic dehydration), hypocalcemia, hypoglycemia, and acidosis. Renal failure with acute tubular necrosis may occur.

Severity of cholera can be classified clinically as:

- Grade I: Subclinical.
- Grade II: Clinical signs and symptoms with stool volume of less than 5 liters.
- Grade III: Clinical signs and symptoms with stool volume of more than 5 liters.

Table 1: Assessment of Dehydration		
No dehydration (<5%)	Some dehydration (5 to 10%)	Severe dehydration (>10%)
	Two or more of the following signs:	Two or more of the following signs:
▪ Well, alert	▪ Restlessness, irritability	▪ Lethargy or unconsciousness
▪ Normal eyes	▪ Sunken eyes	▪ Sunken eyes
▪ Drinks normally, not thirsty	▪ Thirsty, drinks eagerly	▪ Unable to drink or drinks poorly
▪ Skin pinch goes back quickly	▪ Skin pinch goes back slowly	▪ Skin pinch goes back very slowly (≥ 2 seconds)
▪ Estimated fluid deficit: <50 mL/kg	▪ Estimated fluid deficit: 50 to 100 mL/kg	▪ Estimated fluid deficit: >100 mL/kg

4. Diagnosis

In most cases, cholera is diagnosed presumptively in patients with severe acute watery diarrhea. However, there are no signs or symptoms that can unequivocally distinguish cholera from other infectious causes.

- A presumptive bacteriologic diagnosis can be made by phase contrast of dark field microscopy of hanging-drop preparation of fresh liquid stool by detecting the characteristic shooting star motion.
- The diagnosis of cholera can be confirmed by isolation of *V. cholerae* from stool cultures.
- In patients with ileus, confusion, seizure, or no urine output in response to fluid replacement, serum electrolyte, and glucose testing may be useful to guide further management.

5. Differential Diagnosis¹

Differential diagnosis is essential for accurate identification, appropriate treatment and understanding the severity of the disease. The following conditions should be assessed along with cholera symptoms:

- Iatrogenic diarrhea
- Poisoning Food
- HVI+ diarrhea,
- Diabetes diarrhea
- Escherichia coli infection
- Salmonellosis
- Shigellosis
- Typhoid fever
- Rotavirus infection

6. Complications

Cholera can lead to several serious complications, primarily due to the rapid loss of fluids and electrolytes. Here are some of the key complications associated with cholera:

- Hypovolemic shock and electrolyte loss
- Metabolic acidosis
- Acute tubular necrosis
- Renal failure
- Pulmonary oedema due to excessive IV fluid given, especially when metabolic

¹ Source: <https://www.ncbi.nlm.nih.gov/books/NBK470232/>

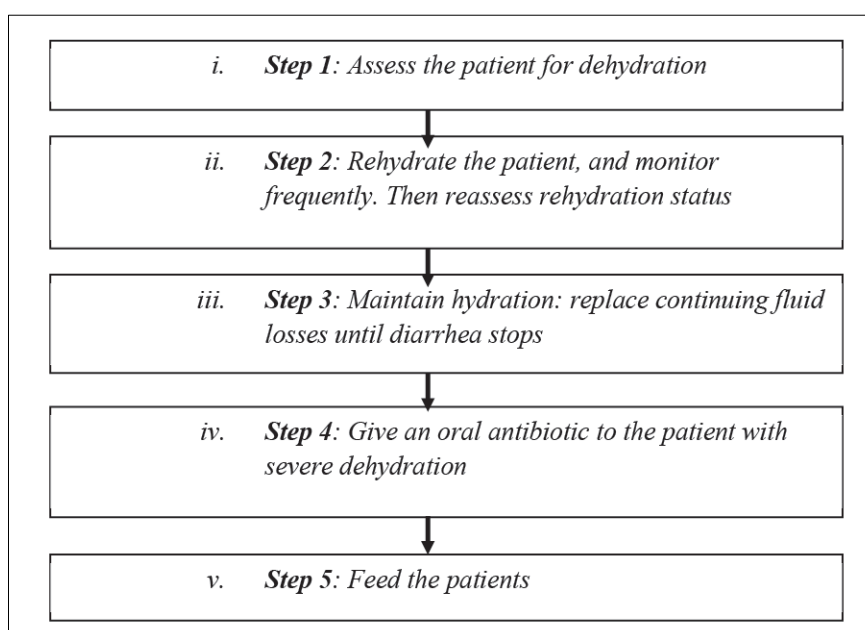
- acidosis has not been corrected.
- Pneumonia (potentially from aspiration in the setting of vomiting)
- Pregnant patients with cholera are at increased risk of miscarriage, premature delivery, and stillbirth.
- "Cholera sicca" is an unusual form of the cholera in which fluid accumulates in the intestinal lumen; circulatory collapse and death can occur in the absence of diarrhea.
- Bacteremia due to *V. cholera* is rare.
- Death

V. OBJECTIVES OF MANAGEMENT

- Replace lost fluids and electrolytes properly by oral rehydration solution (ORS) and intravenous hydration (in severe dehydrated cases).

VI. MANAGEMENT

The management of cholera follows the following steps:



Source: (Hak et al., 2013)

1. Assessment of fluid loss

Volume status may be determined by assessment of the mental status, eyes, mouth, and skin.

- Based on physical findings, the degree of volume depletion can be categorized by World Health Organization (WHO) criteria as no dehydration (<5 percent of body weight), some dehydration (5 to 10 percent of body weight), or severe dehydration (>10 percent of body weight).
- Cholera cots are useful for estimating stool volume losses; if these are not available, ongoing losses may be estimated at 10 to 20 mL/kg of body weight for each episode of diarrhea or vomiting.
- Fluid resuscitation — The type and quantity of fluids to administer is determined by the level of volume depletion and an assessment of ongoing fluid losses.

The most common error in caring for patients with cholera is underestimating the amount of fluid needed to correct volume depletion and replace ongoing losses. Patients with cholera typically present with more severe volume depletion, have more rapid fluid losses (typically 10 to 20 mL/kg/hour), and have proportionally greater stool electrolyte losses than patients with non-cholera gastroenteritis.

a. No dehydration (<5 percent) or some dehydration (5 to 10 percent):

- Use of oral rehydration solution (ORS).
- For patients with no dehydration, ORS should be administered to replace ongoing losses.
- or patients with some dehydration, ORS should be administered to correct the fluid deficit, then continued for replacement of ongoing losses.

Indications for intravenous hydration include:

- Patients with profound vomiting (more than three times in one hour) or ongoing stool losses; in such cases, administration of intravenous therapy in conjunction with ORS may be warranted to prevent progression to severe dehydration.
- Patients who are unable to drink because of vomiting or altered mental status.

b. Severe dehydration (>10 percent)

For patients with severe dehydration, intravenous fluids should be administered urgently:

→ Fluid quantity:

- fluid volume of 100 mL/kg should be administered over three hours; an initial volume of 30 mL/kg should be given over the first 30 minutes.
- In the first 24 hours of treatment, patients with severe cholera typically require an average of 200 mL/kg of isotonic oral or intravenous fluids and may require more than 350 mL/kg.

→ Fluid formulation:

Intravenous Ringer's lactate is the best commercially available intravenous solution for rehydration of patients with severe dehydration due to cholera, since it includes potassium and sodium bicarbonate — both of which are lost in cholera stools.

2. Continued monitoring:

Volume status should be assessed via physical exam on an ongoing basis, and the rate of fluid repletion should be adjusted if needed. In patients treated with proper rehydration, diarrhea is most severe during the first two days and resolves after four to six days.

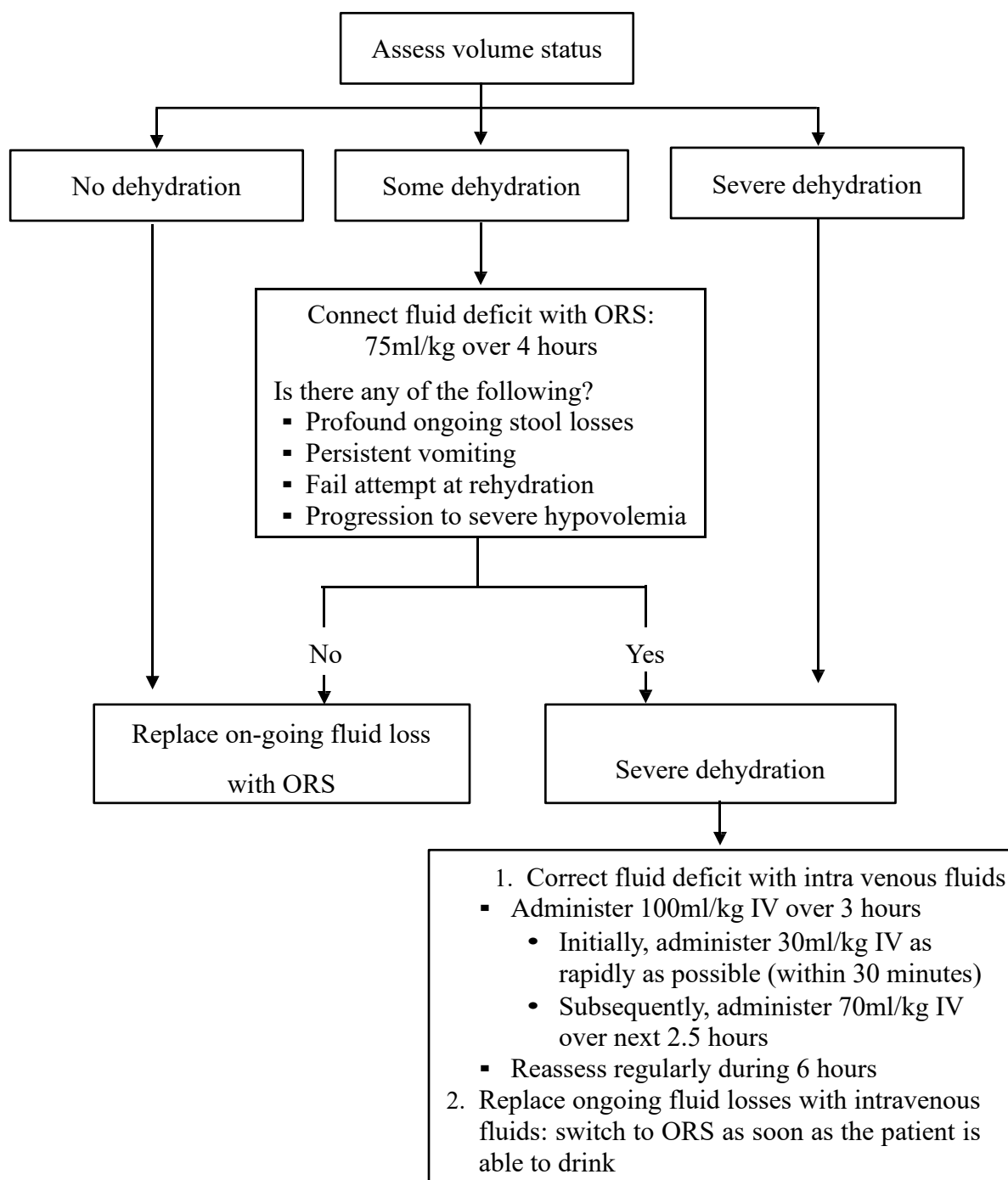
3. Antibiotic therapy:

for patients with cholera and moderate to severe dehydration. The options include macrolides, fluoroquinolones, and tetracyclines.

Class	Antibiotic	Adult dose	Comments
Tetracyclines	Doxycycline	300 mg (single dose)	• Antibiotic resistance to all tetracyclines is common • Empiric use is appropriate in epidemics caused by documented susceptible isolates • Not recommended for pregnant patients and children <8 years of age if other agents are appropriate and available.
	Tetracycline	500 mg 4 times per day for 3 days	
Macrolides	Azithromycin	1 g (single dose)	• Single-dose azithromycin is preferred therapy. • Rare reports of macrolide resistance • Azithromycin is associated with less gastrointestinal distress (e.g., vomiting) than erythromycin
	Erythromycin	500 mg 4 times per day for 3 days	

Fluoroquinolones	Ciprofloxacin	500 mg twice daily for 3 days	- Reduced susceptibility to fluoroquinolones has been reported in Asia and Africa. - Not recommended for pregnant patients and children <8 years of age if other agents are appropriate and available
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Approaches to fluid management of patients with cholera



Oral Rehydration Solution (ORS) is a critical treatment for dehydration, particularly in cases caused by diarrhea and vomiting.

VII. RISK FACTOR OF MORTALITY

Mortality in untreated patients may reach 50 to 70 percent in severe cholera.

Administration of appropriate rehydration therapy can reduce the mortality of severe cholera to <0.5 percent.

- In endemic areas, the risk of mortality risk is highest in children (10 times greater than in adults).
- Early studies suggested a high risk of fetal death associated with cholera during pregnancy (up to 50 percent during the third trimester); subsequent studies have reported a risk of approximately 8 percent.

Other risk factors:

- Old age
- Comorbidity
- Severe dehydration

VIII. PREVENTION

1. Prevention measures

Preventing cholera involves a combination of:

- improving water, sanitation and hygiene,
- increasing risk communication and community engagement,
- improving access to quality treatment and implementing oral cholera vaccine campaigns.

2. Cholera vaccines

parenteral vaccine is not recommended because of short lasting immunity (6 months) significant side effects and lack of efficacy preventing spread.

Currently, three WHO pre-qualified oral cholera vaccines (OCV) are available: Dukoral®, Euvichol-Plus®, and Euvichol-S®. All require two doses to fully protect an adult. One dose of Euvichol-Plus® or Euvichol-S® provides good short-term protection. Euvichol-S®, pre-qualified by WHO in 2024, is a simplified version of Euvichol-Plus®. The Shancol vaccine is no longer produced. Euvichol-Plus® and Euvichol-S® do not require a buffer solution and can be given to anyone over one year old. OCV is safe during pregnancy.

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TETANUS

Vann Pisey, Yung Youda, Koy Sophany, Meng Sreyhuoch, Chea Sereywatana, Sotharith Bory

I. DEFINITION

Tetanus is a serious disease affecting the nervous system, characterized by muscle spasms due to a neurotoxin produced by the bacterium *Clostridium tetani*. This bacterium is commonly found in soil and can enter the body through wounds, leading to the proliferation of the bacteria and subsequent toxin release.

II. EPIDEMIOLOGY

It is more common in rural and agricultural regions; areas where contact with soil or animal excreta is likely. Because the spores exist in the environment, tetanus cannot be eradicated.

- In 2020, over 11,750 tetanus cases across the globe were reported to the World Health Organization / United Nations Children's Fund, of which 2,230 occurred in neonates. Most tetanus cases were reported from countries in Africa and Southeast Asia. Tetanus can affect any age group¹.
- Mortality rates from tetanus vary greatly across the world, depending on access to healthcare, and approach 100% in the absence of medical treatment

III. ETIOLOGY

bacterium *Clostridium tetani*, which is an anaerobic, gram-positive bacillus that forms spores.

IV. PHYSIOPATHOLOGY

The pathophysiology of tetanus involves several key processes that lead to the characteristic symptoms of the disease.

Infection and Toxin Production

- **Entry of Spores:** *C. tetani* spores are ubiquitous in the environment, commonly found in soil, dust, and animal feces. They can enter the body through wounds, especially deep puncture wounds that create anaerobic conditions conducive to germination.
- **Germination:** Once inside the body, the spores germinate in anaerobic conditions, leading to the production of two main toxins:
 - **Tetanolysin:** A hemolysin with no known pathological activity.
 - **Tetanospasmin:** The primary toxin responsible for the clinical manifestations of tetanus, which is extremely potent with a minimum lethal dose of approximately 2.5 ng/kg body weight.

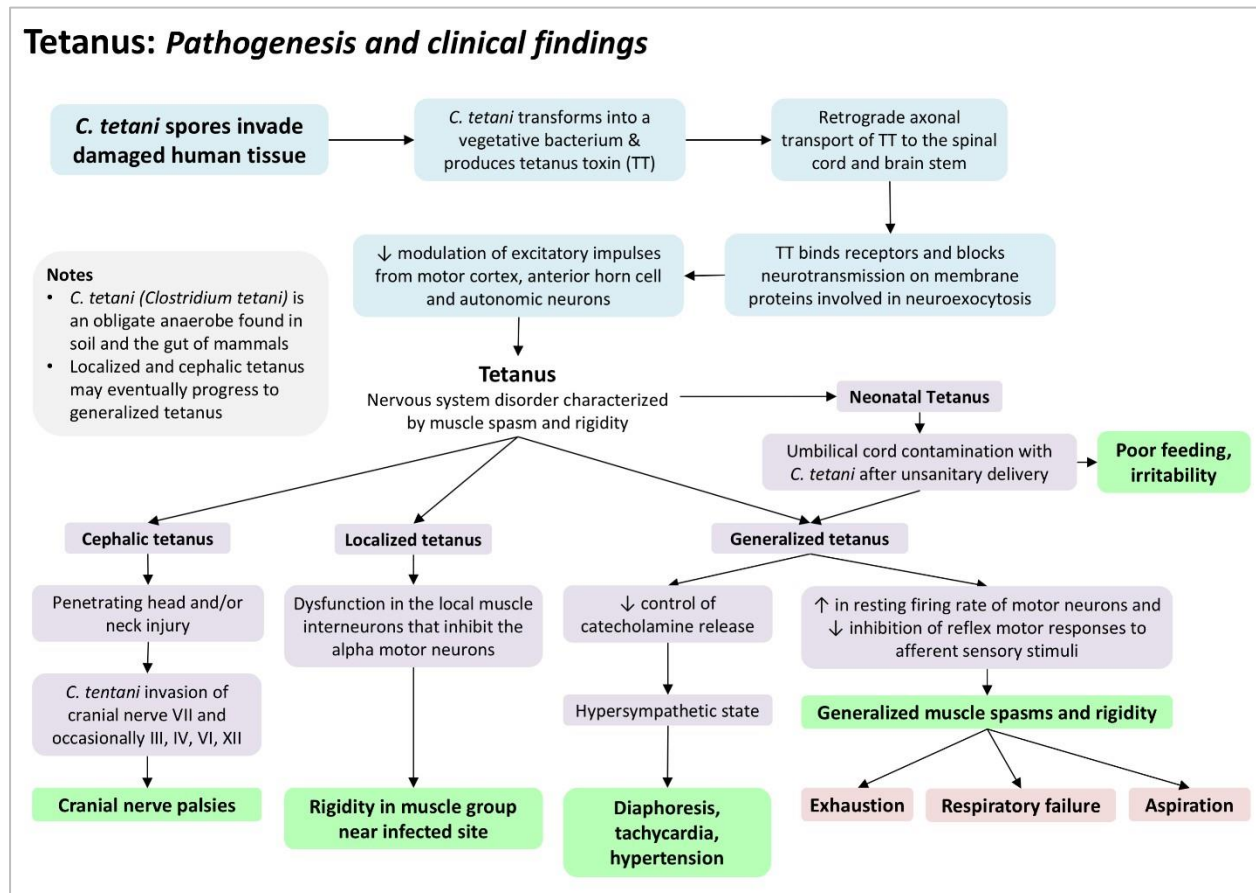
Mechanism of Action of Tetanospasmin:

- **Neuronal Uptake:** Tetanospasmin is taken up by the terminals of lower motor neurons and transported retrogradely (backwards) along the axons to the spinal cord and brainstem.
- **Trans-synaptic Movement:** Once in the central nervous system, the toxin moves trans-synaptically into inhibitory nerve terminals. Here, it interferes with the release

¹ Source: <https://wwwnc.cdc.gov/travel/yellowbook/2024/infections-diseases/tetanus>

of inhibitory neurotransmitters, specifically gamma-aminobutyric acid (GABA) and glycine.

- **Inhibition of Neurotransmitter Release:** Tetanospasmin acts as a zinc-dependent protease that cleaves synaptobrevin, a protein essential for the docking and release of neurotransmitter vesicles. This blockage prevents the release of GABA and glycine, which are crucial for inhibiting motor neuron activity.



Source: (Chloe, 2017)

1. Complications

Nervous system abnormalities, as well as a variety of complications related to severe spasms and prolonged hospitalization, can accompany generalized tetanus. Complications of tetanus can also include (CDC.GOV, 2024):

- Aspiration pneumonia
- Fractures
- Hypertension
- Laryngospasms
- Nosocomial infections
- Pulmonary embolism
- Death

V. DIAGNOSIS

1. Clinic

Diagnosis is based on clinical findings; no confirmatory laboratory tests are available or even rule out a tetanus diagnosis. Tetanus is a nationally notifiable disease.

- **Loss of Inhibition:** The disruption of inhibitory neurotransmitter release leads to disinhibition of lower motor neurons, resulting in uncontrolled muscle contractions and spasms. This is characterized by:
 - **Trismus (Lockjaw):** Inability to open the mouth due to masseter muscle spasm.
 - **Opisthotonos:** A spasm causing hyperextension of the back and neck.
 - **Generalized Muscle Rigidity:** Intermittent spasms triggered by stimuli such as noise or touch.
- **Autonomic Dysregulation:** The loss of inhibitory control also affects autonomic neurons, leading to symptoms such as autonomic hyperactivity, which can manifest as sweating, tachycardia, and hypertension.
- **Pain and Sensory Integrity:** Despite the severe muscle spasms, patients typically maintain an intact sensorium and experience significant pain during spasms.

2. Clinical presentations

Based on location, there are three forms of tetanus:

- Generalized
- Localized
- Cephalic

2.1. Generalized

- Generalized tetanus is the most common form of the disease. It accounts for more than 4 in 5 cases.
- Neonatal tetanus is a form of generalized tetanus. It occurs in newborns who lack the passive protection derived from maternal antibodies.

2.2. Localized

- Localized tetanus is an unusual form of the disease. It's possible for it to progress to generalized tetanus.
- It typically occurs in people with partial tetanus immunity.

2.3. Cephalic

- Cephalic tetanus is the rarest form. It's associated with lesions of the head or face. It may be associated with otitis media. It's possible for it to progress to generalized tetanus.

3. Common signs and symptoms

3.1. Generalized tetanus

Painful muscular contractions are characteristic of tetanus:

- Difficulty in swallowing or breathing
- Generalized spasms (frequently induced by sensory stimuli)
- Rigidity
- Seizures
- Trismus, or lockjaw painful muscle contractions

3.2. Localized tetanus

Like generalized tetanus, localized tetanus is characterized by painful muscle spasms. However, they're in a confined area close to the injury site.

3.3. Cephalic tetanus

Unlike the other two forms, cephalic tetanus results in flaccid cranial nerve palsies rather than spasm.

Common first signs in older children and adults

- Abdominal rigidity is a common first sign suggestive of tetanus in older children and adults. However, it's sometimes confined to the region of injury.
- Generalized spasms occur, frequently induced by sensory stimuli.

History of an injury or apparent portal of entry may be lacking. In addition, healthcare providers rarely recover tetanus bacteria from the site of infection.

Table 1: Types of tetanus

Classification	Description
Generalized	Begins with trismus and risus sardonicus (spasm of the facial muscles), then proceeds to generalized spasms and opisthotonos.
Localized	Muscle rigidity limited to the site of spore inoculation
Cephalic	Form of localized tetanus affecting cranial nerves, often following a head injury.
Neonatal	Generalized tetanus in newborns resulting from infection of the umbilical stump.

Based on severity, there are 4 grades:

Table 2: Severity of tetanus

No	Severity	Description
1	Mild	Mild trismus, general spasticity, no respiratory compromise, no spasms, no dysphagia
2	Moderate	Moderate trismus, rigidity, short spasms, mild dysphagia, moderate respiratory involvement, respiratory rate >30 breaths/min.
3	Severe	Severe trismus, generalized rigidity, prolonged spasms, severe dysphagia, apneic spells, pulse > 120 beats/min, respiratory rate >40 breaths/min.
4	Very Severe	Grade 3 with automatic dysfunction

Source: (Afshar et al., 2011)

VI. DIFFERENTIAL DIAGNOSIS

- Drug-induced dystonias such as those due to phenothiazines
- Trismus due to dental infection
- Malignant neuroleptic syndrome
- Stiff-person syndrome

VII. REATMENT

Objectives

- Halting the toxin production
- Neutralization of the unbound toxin
- Airway management
- Control of muscle spasms
- Management of dysautonomia
- General supportive management

1. Halting toxin production

Wound management: wound debridement to eradicate spores and necrotic tissue, which could lead to conditions ideal for germination.

1.1. Antimicrobial therapy

- Metronidazole (500 mg intravenously [IV] every six to eight hours) is the preferred treatment for tetanus,
- Or penicillin G (2 to 4 million units IV every four to six hours) is a safe and effective alternative,
- The duration of treatment is 7 to 10 days.
- If a mixed infection is suspected, a first-, second-, or third-generation cephalosporin such as cefazolin (1 to 2 g IV every 8hours), cefuroxime (2 g IV every 6 hours), or ceftriaxone (1 to 2 g IV every 24 hours) can be used.

1.2. Neutralization of unbound toxin

- Intramuscular antitoxin – Human tetanus immune globulin (HTIG) is the antitoxin and is used 500 units intramuscularly (IM). If HTIG is not available, human immunoglobulin is reasonable alternative.
- Active immunization — Since tetanus is one of the few bacterial diseases that does not confer immunity following recovery from acute illness, **all** patients with tetanus should receive active immunization with a full series (eg, three doses in adults and children >7 years old) of tetanus and diphtheria toxoid-containing vaccines, commencing immediately upon diagnosis.

2. Control of muscle spasms

- Placement of the patient and control of light or noise in the room
- Benzodiazepines
 - Benzodiazepines have been used traditionally and are generally effective in controlling the rigidity and spasms associated with tetanus [14]. They also provide a sedative effect. Diazepam has been used most frequently, but other benzodiazepines are as effective as diazepam.
 - For tetanus, the usual starting dose of diazepam for an adult is 10 to 30 mg IV and is repeated as needed every 1 to 4 hours. Patients with tetanus often show tolerance to the sedating effects of benzodiazepines and may remain awake and alert after receiving doses that would sedate or cause anesthesia in other patients [30].

- Since benzodiazepines may be required for a prolonged period of time (often weeks), they should be tapered gradually to avoid withdrawal reactions and ventilator weaning.
- **Other sedatives** – Infusion of the anesthetic propofol may also control spasms and rigidity. Its prolonged use has been associated with lactic acidosis, hypertriglyceridemia, and pancreatic dysfunction.
- **Neuromuscular blocking agents:** are used when sedation alone is inadequate. Options depend on availability. Vecuronium or other cardiovascularly inert neuromuscular blockers are preferred. Pancuronium, a long-acting agent, has been traditionally used, but it may worsen autonomic instability because it is an inhibitor of catecholamine reuptake. Neuromuscular blocking agents are generally given as continuous infusions.

2.1. Management of autonomic dysfunction

Several drugs have been used to produce adrenergic blockade and suppress autonomic hyperactivity; only treatment with magnesium sulfate has been studied in a randomized clinical trial in tetanus [54].

- **Magnesium sulfate** — Magnesium sulfate acts as a presynaptic neuromuscular blocker, blocks catecholamine release from nerves, and reduces receptor responsiveness to catecholamines [55]. In patients with tetanus, it is used for the management of autonomic dysfunction and as adjunctive treatment for controlling spasms [54,56-59].
- **Beta blockade** — (0.25 to 1 mg/min) has frequently been administered because of its dual alpha- and beta-blocking properties. Beta blockade alone with propranolol, for example, should be avoided because of reports of sudden death [62]. Morphine sulfate (0.5 to 1 mg/kg per hour by continuous intravenous infusion) is commonly used to control autonomic dysfunction as well as to induce sedation.
- **Other drugs** — Other drugs for the treatment of various autonomic events, which have been reported to be useful, are dexmedetomidine, atropine, clonidine, and epidural bupivacaine.

2.2. Airway management and other supportive measures (Hassel, 2013)

Since tetanus toxin cannot be displaced from the nervous system once bound to neurons, supportive care is the main treatment for tetanus. In patients with severe tetanus, prolonged immobility in the intensive care unit is common, much of which is on mechanical ventilation and may last for weeks. Such patients are predisposed to nosocomial infections, decubitus ulcers, tracheal stenosis, gastrointestinal hemorrhage, and thromboembolic disease.

- Endotracheal intubation is justified initially, but early tracheostomy is frequently indicated because of the likelihood of prolonged mechanical ventilation.
- Energy demands in tetanus may be extremely high, so early nutritional support is mandatory. Enteral feeding is preferred. Prophylactic treatment with sucralfate or acid blockers may be used to prevent gastroesophageal hemorrhage from stress ulceration.
- Prophylaxis of thromboembolism with heparin, low molecular weight heparin, or other anticoagulants should be administered early.

- Physical therapy should be started as soon as spasms have ceased, since tetanus patients often are left with disability from prolonged muscle wasting and contractures.
- In the absence of an ICU, ideally a separate ward or room should be designated for patients with tetanus, and sensory stimuli should be kept to a minimum since loud noises, physical contact, and light can trigger tetanic spasms. Other options include eye shades and ear plugs to reduce stimuli. However, benzodiazepines and baclofen can be used in such situations if doses are carefully titrated to avoid respiratory depression. Magnesium sulfate may be used to manage autonomic dysfunction and as an adjunctive for muscle spasm.
- Tetanus antitoxin is given once intramuscularly; doses of 500 IU, 3000 IU. The antitoxin is given to inactivate any free tetanus toxin. The toxin that has been taken up into nerve terminals is probably not available to the antitoxin. Therefore, muscle symptoms may develop further, although the clostridia have been eradicated and antitoxin has been given, because tetanus toxin continues to be transported axonally and trans-synaptically and to cleave VAMP. Intrathecal administration of antitoxin, e.g. via lumbar puncture, could inactivate tetanus toxin during its trans-synaptic transport.
- Because immunity may not develop after an episode of tetanus, vaccination is included in the treatment.

VIII. REVENTION

Prophylaxis against tetanus consists of:

- immunization with formaldehyde-inactivated tetanus toxin (toxoid) and
- measures to achieve good hygiene.
- For instance, contamination of the umbilical stump of the newborn is a primary cause of neonatal tetanus. These issues are interrelated: a good immunization status in pregnant women leads to reduction in the prevalence of neonatal tetanus (Lambo et al., 2011), because maternal anti-tetanus toxin antibodies are transferred across the placenta to the child in utero (Van Den Berg et al., 2010).

The schedule of tetanus vaccine:

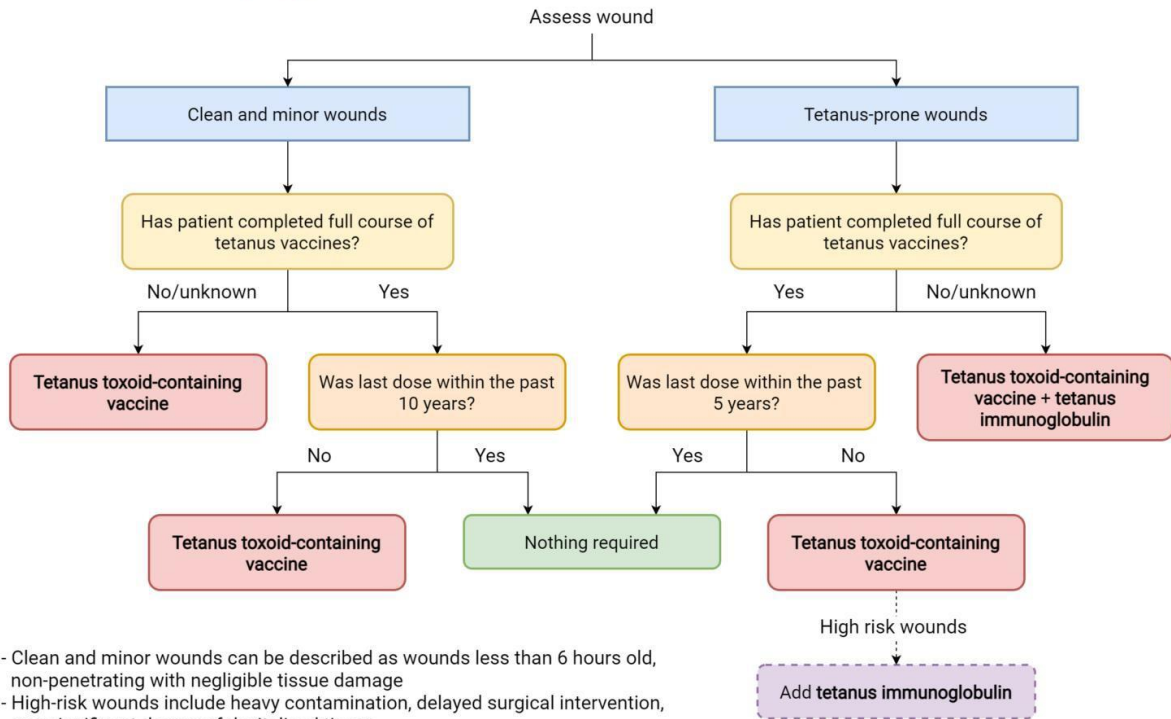
- Infants: 3 doses of pentavalent vaccine (Diphtheria, Tetanus, Pertussis, Hib, Hep B) at 6, 10, and 14 weeks, with a booster at 18 months.
- Pregnant Women: 2 doses of Td vaccine during pregnancy, with a third dose in subsequent pregnancies.
- Adolescents and Adults: Primary series of 3 doses if unvaccinated, with boosters every 10 years.

In case, there is wound:

- Wound management: wound debridement to eradicate spores and necrotic tissue
- Tetanus vaccine and tetanus immunoglobulin



Tetanus Prophylaxis in Wound Management



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LEPTOSPIROSIS

Yung Youda, Koy Sophany, Chea Sereywatana, Lim Kruy, Vann Pisey, Choun Kimcheng, Phe Thong, Sotharith Bory

I. DEFINITION

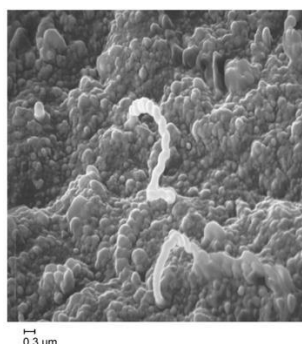
Leptospirosis is a widespread zoonotic infection caused by spirochetes of the *Leptospira* genus. Transmission occurs via direct contact with urine, blood, or tissue of an infected animal or via contact with contaminated freshwater.

II. EPIDEMIOLOGY

Leptospirosis is thought to be the most widespread zoonosis in the world. It is an under-reported that more than one million human cases occur worldwide annually, including almost 60,000 deaths.

III. ETIOLOGY

Leptospira spp., are obligate aerobic, gram-negative spirochete bacteria. Organism characteristics include: 0.15-0.3 micrometers in diameter and 6-20 micrometers in length, slow growing (culture up to 3 months) and endoflagella allow for strong mobility even through viscous materials. 20 species are described (9 pathogenic, 6 saprophytes and 5 intermediate strains).



Transmission electron microscopical image of *Leptospira interrogans* invading equine conjunctival tissue. (Image kindly provided by Dr. JE Nally, National Animal Disease Center, U.S. Department of Agriculture, Ames, IA. This image appears on the homepage of the European Leptospirosis Society website [<http://eurolepto.ucd.ie/>].)

IV. PATHOGENESIS

- The main risk factor for leptospirosis is direct or indirect contact with urine of infected animals (rats, bats, seals, dogs, cattle, swine, horses etc).
- Transmission to humans occurs through either direct contact with infected animals or by indirect contact with contaminated urine, blood, tissue, freshwater, or soil via: breaches in the skin, mucosal membranes (such as the conjunctiva or oral mucosa), intact skin (after prolonged contact with contaminated water), ingestion of contaminated water.
- After entry, the organisms proliferate, cross tissue barriers, and disseminate hematogenously to all organs (leptospiemic phase). During this initial incubation period, leptospirae can be isolated from the blood stream. Moreover, pathogenic leptospirae resist ingestion and killing by neutrophils, monocytes, and macrophages.
- Infection results in biphasic illness:
 - Acute/initial (leptospiemic phase or septicemic) phase: high fevers, malaise, headache, myalgia, and abdominal pain, followed 5-7 days

- Immune phase, associated with antibody production and excretion of the organism in urine. During the immune phase, patients can have severe systemic manifestations, including renal failure, hepatic failure, and pulmonary hemorrhages, which can be fatal

V. DIAGNOSIS

1. Clinical manifestation

Most infections (approximately 90%) are subclinical. Patients who develop symptoms usually experience an acute, self-limited febrile illness; however, 5% to 10% develop a more severe illness, which can lead to multi-organ failure and death.

Syndromes caused by leptospirosis are often divided into two categories: *anicteric leptospirosis* and *icteric leptospirosis*.

Anicteric leptospirosis:

- Acute septicemic phase (1st week of illness): high fevers, chills/rigors, headache, myalgia (the calves: *pathognomonic*), abdominal pain, diarrhea, nausea and vomiting, asthenia, anorexia, photophobia and bilateral conjunctival suffusion.
- Immune phase (after 5-7 days): fever resolves, severe eye pain, headache, photophobia, pulmonary symptoms, palpitations, bilateral conjunctival suffusion (*pathognomonic*), muscle tenderness, mental status changes (delirium, coma), and focal neurological deficit.



Subconjunctival hemorrhage in leptospirosis
Scleral icterus and subconjunctival hemorrhage in both eyes (panels A and B).
View In Context: [Leptospirosis](#)
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Icteric leptospirosis (Weil's disease):

Severe disease is associated with the immune phase and may manifest with renal failure, hepatic failure, and/or pulmonary hemorrhages. Other presentations during this phase include myocarditis with arrhythmias, aseptic meningitis, and pancreatitis.

Patients who are immunocompromised may manifest with cardiovascular complications such as complete heart block, although this is a very rare presentation.

2. Laboratory

A. General findings on blood tests may include:

- Normal to moderately elevated WBC count, thrombocytopenia
- Elevated hepatic transaminases (mild elevation or normal), alkaline phosphatase, and direct bilirubin
- Elevated erythrocyte sedimentation rate (ESR)
- Elevated creatinine and urea if renal is involved
- Hypokalemia (due to potassium-wasting nephropathy)
- Coagulopathy (prolonged prothrombin time)

B. Diagnostic Testing

- Polymerase chain reaction (PCR): blood (1st week illness), urine and CFS (after 1st week of illness)
- Serologic tests: specific immunoglobulin (Ig)M and IgG antibodies

against *Leptospira* organisms (sample should be obtained after 1st week)

- Other tests: microscopic agglutination test (MAT), direct examination (by dark field microscopy), Blood cultures (by Fletcher's media) and tissues biopsy.

*Choice of methods of confirmation will depend on availability at local laboratories, and physicians should consult local testing protocols.

3. Investigation of Complication

- Cerebrospinal Fluid (CSF) Analysis: **aseptic meningitis**
 - Normal or slightly elevated CSF pressure
 - Pleocytosis of about 500/mm³, lymphocyte predominate
 - Elevated total protein (50-100 g/L)
 - Normal glucose
 - Xanthochromia may occur
- Chest Xray or CT scan for pulmonary hemorrhage that is a serious complication of Weil's disease.

4. Imaging

CXR may show bilateral small nodular densities and patchy alveolar densities in patients with pulmonary involvement during the immune phase. Cases in which chest radiographs detect air-space nodules have been associated with severe leptospirosis.

Abdominal ultrasound may show acalculous cholecystitis, hepatomegaly, and splenomegaly.

VI. DIFFERENTIAL DIAGNOSIS

The differential diagnosis of leptospirosis is broad, reflecting the diverse clinical presentations of the disease: Influenza, dengue and chikungunya, rickettsial diseases, acute HIV infection, hantavirus infections, malaria, typhoid fever, ehrlichiosis, viral hepatitis.

VII. MANAGEMENT

A. Objectives

Effective management of leptospirosis involves a combination of antibiotic therapy and appropriate supportive therapy. It is generally accepted that antibiotic therapy should be initiated as soon as possible to prevent the development of major organ system failure or lessen its severity.

B. Warning Signs

- Increased age (>40)
- Pregnancy (can cause fetal death or abortion)
- Signs of *Weil's syndrome*: jaundice, renal failure, and hemorrhagic diathesis (hemoptysis, melena, hematuria, bleeding from venipuncture sites)
- Altered mental status, signs of meningitis
- Hypotension or septic shock
- Hyponatremia, hypokalemia and arrhythmia

C. Treatment

- ❖ Antibiotic

- **Mild leptospirosis:** *Doxycycline* 100 mg orally twice daily or *Azithromycin* 500 mg orally once daily or *Amoxicillin* (500 mg PO tid).
- **Moderate/severe leptospirosis:** *Doxycycline* (loading dose of 200 mg IV, then 100 mg IV q12h), or *Ceftriaxone* 1g IV every 24 hours or *Penicillin* 1.5 million units IV every 6 hours, or *Ampicillin* 0.5-1 g IV every 6 hours or *Cefotaxime* 1 g IV every 6 hours.
- **Duration:** 7 days for mild, moderate and severe leptospirosis.
- ❖ Supportive: IV hydration to prevent renal failure, potassium supplementation for patients with hypokalemia, considered dialysis for acute kidney injury, mechanical ventilation for patients with acute respiratory distress syndrome (ARDS) or pulmonary hemorrhage. Intravenous methylprednisolone has been used successfully in patients with pulmonary leptospirosis, but one systematic review found limited evidence; further trials are needed to determine whether corticosteroids should routinely be given to patients with severe leptospirosis and pulmonary involvement.

VIII. PREGNANCY

Leptospirosis in pregnancy may be misdiagnosed as it can mimic other pregnancy-associated conditions, including other infections, pregnancy-associated hypertension, acute fatty liver, and HELLP (hemolysis, elevated liver enzymes, and low platelet count syndrome).

We favor treatment with either azithromycin or amoxicillin. Azithromycin is preferred over amoxicillin if the differential diagnosis includes rickettsial infection.

IX. PROGNOSIS

- Most patients recover fully.
- Severe disease develops in about 10% of cases.
- Mortality: 5%-15% mortality is reported in patients with severe disease.
- Severe leptospirosis with pulmonary haemorrhage has a mortality rate of >50%.

X. COMPLICATIONS

Complications include: sepsis, renal failure (may be chronic), acute respiratory distress syndrome (ARDS), massive hemoptysis due to alveolar hemorrhage (pulmonary hemorrhage syndrome), neurologic complications (meningitis, transverse myelitis, Guillain-Barre syndrome).

XI. PATIENT ADVICE AND PREVENTION

❖ General measures include

- The first line of leptospirosis prevention is to avoid exposure.
- Avoid wading, swimming, bathing, swallowing, or submersing head in potentially contaminated freshwater (rivers, streams) especially after periods of heavy rainfall or flooding.
- Avoid contact with floodwater, and do not eat food contaminated with floodwater.
- If exposure cannot be avoided, wear appropriate personal protective equipment (rubber boots, waterproof coveralls/ clothing, gloves). Cover open wounds with waterproof dressings.
- Treat unsafe or potentially contaminated drinking water by boiling or chemically treating.
- Keep rodent populations (rats and mice) or other animal pests under control. Do not

eat food that may have been exposed to rodents and possibly contaminated with their urine.

✦ **Chemoprophylaxis:**

Options for postexposure prophylaxis include: Doxycycline (200 mg orally in a single dose or one a week) or azithromycin (250 mg PO once or twice a week).

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RICKETTSIAL INFECTION

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Choun Kimcheng, Phe Thong, Bory Sotharith

I. DEFINITION

Rickettsia is the one common diseases of group vector-borne organisms and obligate intracellular bacteria within the *Rickettsia* and *Orientia* genera, and *Rickettsiaceae* family that cause acute non-malarial febrile illnesses in Africa and Asia. Although rickettsial infections have a similar clinical appearance, the causative species and epidemiology might differ geographically.

II. EPIDEMIOLOGY.

There are 3 types of *Rickettsia* depend on geographically and clinical presentation

Typhus group rickettsioses

- **Louseborne typhus** or endemic typhus that caused by *R. prowazekii* transmitted by human body louse (*Pediculus humanus corporis*). This is rare now but found in cold session. The endemic of louseborne typhus bacterium is presented in Central Africa, South America, Asia.
- **Murine typhus** is flea borne typhus caused by *R. typhi* which can infect to human by travelling to zone endemic and when expose to cat or dog exposed to flea-infested cats, dogs, and peridomestic animals, or enter or sleep in areas infested with rodents. Murine typhus has been reported among travelers returning from Africa, Asia, and the Mediterranean Basin. Most cases acquired in the United States are reported from California, Hawaii, and Texas.

Scrub typhus group rickettsioses: is endemic in East-Asia (China, northern Japan), Southeast Asia (India, Indonesia, Sri Lanka), the Pacific (eastern Australia), and several parts of south-central Russia. Globally, more than 1 million cases of scrub typhus are reported annually. In one meta-analysis reported incidence in Asia ranged from 1.22/100000 in China to 17.7/100000 in South Korea.

Spotted fever group rickettsioses: There are three important species at risk globally *Rickettsia felis*, *R. conorii* and *R. africae*.

- African tick-bites fever
- Cat-flea rickettsiosis
- Mediterranean Spotted Fever

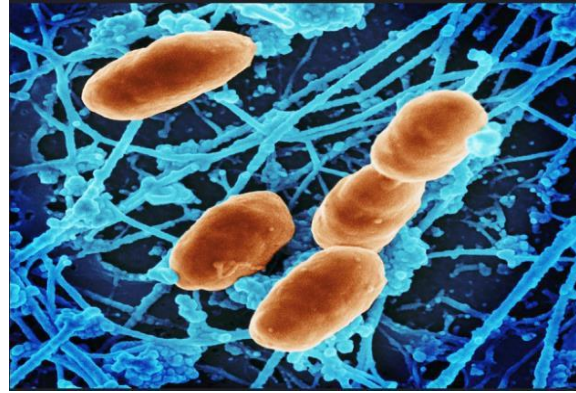
III. ETIOLOGY

Rickettsiosis is aarthropod-borne, gram-negative and obligate intracellular bacteria. Organism characteristic include small size (ie, 0.3 by 1.0 micrometers) and the cell wall contain peptidoglycan and liposaccharide.

There are 2 common features, they grow inside a host animal eukaryotic cell and live in the cells of both vertebrate (Human and rats) and invertebrate animals (lice, fleas, mites and ticks) which is the host and the vector for transmission of the rickettsia to human and vertebrate animals.



This photograph depicts a dorsal view of a female Gulf Coast tick, *Amblyomma maculatum* from PHIL

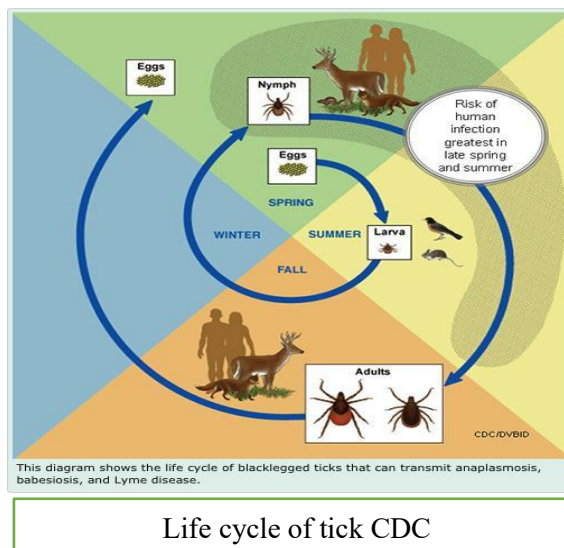


Scanning electron micrograph of *Rickettsia typhi* bacteria. Image captured at NIAID's Rocky Mountain Laboratories (RML) in Hamilton, Montana. Credit: NIAID

IV. PATHOGENESIS

Transmission of rickettsia is usually transmitted by tick-bite. After the tick has been attached to the host for 6 to 10 hours, *rickettsiae* are released from the salivary glands of the ticks and spread via the bloodstream to infect the endothelium and sometimes the vascular smooth muscle cells. In addition, humans may become infected by contact with tick tissues or fluids during the process of tick removal.

The CD68+ (Macrophage and dendritic) is the important target cell of this organism. After entry, it will spread via the lymphatic vessels to region lymph nodes give lymphangitis, spread hematogenously through body mainly infected to epithelium cell and others organs like skin, liver, brain, kidneys, lung, gastrointestinal track, heart, etc.



V. DIAGNOSIS

1. Risk factor

- Exposure to human body lice
- Exposure to rodents (rats and mice) and their fleas
- After tick and mites' bites
- Travel to the tropical area

2. Clinical manifestation

- Rickettsia diseases are difficult to diagnostic. The clinical manifestations are irritable depend on the type of rickettsia and present mostly fever, rash and eschar.
- The incubation period of rickettsia is usually 5-10days after exposure to the Rickettsia species via a flea or tick bite.
- Symptoms classically include:
 - Fever always presents to all patients who are infected by rickettsia
 - Rash is common in all rickettsia infections and normally present after a few days to week of infection. It can be macular or popular but also vesicular in some cases (E.g. Queensland tick typhus due to *Rickettsia australis* and *Rickettsia pox*) or petechial (Epidemic typhus due to *R. prowazekii*)
 - Eschar: The presence or absent of an eschar depends on the specific strain of Rickettsia that is involved as well as patient to patient and geographical location. Normally we can find hair, under scrotum, natal cleft, under breasts and groin.
 - Lymphadenopathy is not common but it can associate with the eschar.
- Unspecific symptom
 - Headache, myalgia, fatigue, nausea or vomiting and abdominal pain usually presented in Rickettsial infection CDC.
 - Hepatosplenomegaly may have in only some case



A. characteristic spotted rash of Rocky Mountain spotted fever (PHIL)



B. Eschar of Rocky Mountain spotted fever

3.

Laboratory

- General blood test finding:
 - CBC: thrombocytopenia mild to severe and usually present in acute phase of rickettsial disease. Lymphopenia is usually mild and associated with abnormal morphology. WBC is mostly in the normal range.
 - Elevation hepatic transaminase (being < 200 in the former and > 500 U/L in the latter).
 - Serum electrolyte levels: Hyponatremia is often seen at the severe of infection.
 - CRP: Raised > 100mg/L in rickettsial infection (differential from viral infection <50mg/L)
 - Elevated creatinine and urea if renal is involved
- Diagnostic Testing
 - PCR (Polymerase Chain Reaction) : best done on a swab or biopsy of eschar if present, from blood in case no eschar (But it can be false negative during the first several days of the illness).
 - Serology test: specific immunoglobulin IgM and IgG antibodies against *Rickettsial* organism (can be negative in the 1st week of infection so if negative, can repeat in 7-10days after)
 - Others test: Immunohistochemistry (Detection of special anti-rickettsia antisera), blood culture or biopsy culture (growth only intracell).
- Choice of methods of confirmation diagnostic will depend on availability at local laboratories and physicians should consult local testing protocol.
- Imagery
 - Chest X ray: find out pneumonia
 - Abdominal ultra-sound: Find out hepatosplenomegaly.
- Trial of doxycycline therapy: Rapid clinical improved after give doxycycline 48h.

VI. DIFFERENTIAL DIAGNOSIS

The differential diagnostic is all eruption fever. The elements of clinical and epidemiology are very important of diagnostic. Malaria infection, dengue fever, chikungunya viral infection, lyme disease, meningococcal disease, leptospirosis and others eruption fever should be eliminated.

VII. MANAGEMENT

Administration of appropriated empirical antibiotic for rickettsial infection as soon as possible when suspected without waiting for the result of tests to confirm the diagnostic.

- Determining disease severity

- **Mild to moderate disease:** Patients with mild to moderate disease typically present with fever, myalgia, and headache as well as a rash (with or without eschar) and/or cough
- **Severe:** Patients with severe disease typically present with the findings above in combination with clinical or laboratory manifestations that indicate end-organ damage. These include hyperbilirubinemia, renal failure, cardiovascular collapse (hypotension/shock), acute respiratory distress syndrome (ARDS), and meningoencephalitis.
- Antibiotic
 - Doxycycline 100mg orally twice daily or Azithromycin 500mg orally once daily in case patient contraindication of Doxycycline.
 - Duration 5-7days or 2-3days after apyretic
 - Commonly used antibiotic (Penicillin and cephalosporins) do not work against rickettsiae because of the unusual chemical nature of their bacterial cell wall and sulfonamides have been associated with increased severity and death.
- Supportive care: antipyretic, IV hydration and others supportive care should add if need.
- In pregnancy people: Azithromycin is choice of treatment for 7 days of duration.

VIII. PROGNOSIS

Rickettsial infections vary in severity based on the virulence of the *Rickettsia* species and host factors, such as the degree of immunocompetence. *R. rickettsii* and *R. prowazekii* are considered the most virulent species and are associated with significant mortality if the infected individual is not diagnosed and treated early. For example, patients with Rocky Mountain spotted fever have mortality rates as high as 30% without prompt diagnosis and antibiotic treatment. Appropriate antibiotics started in the first week of the patient's illness are very effective and are associated with excellent outcomes.

IX. COMPLICATION

- Sepsis and septic shock
- Acute respiratory distress syndrome
- Other organism failure due to metastatic of infection (brain, kidneys, liver, heart...)

X. PREVENTION

- No vaccine is available for preventing rickettsial infections. Antibiotic prophylaxis is not recommended for rickettsial diseases, and antimicrobial agents should not be given to asymptomatic people.
- Prevention relies on avoidance of exposure to tick, lice, mite, and flea bites, particularly when residing or traveling to endemic areas.
- Patients should be educated on careful inspection for insects after outdoor activities, the importance of wearing long-sleeved shirts, pants, socks, and closed-toe shoes, and the use of insect repellent to minimize the risk of rickettsial infection.

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ENTERIC FEVER

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I. DEFINITION

Enteric fever (also known as Typhoid and Paratyphoid Fever) is a potentially fatal multisystemic infection produced primarily by *Salmonella enterica* serotype typhi and to a lesser extent *Salmonella enterica* serotypes (paratyphi A, B, and C).

II. EPIDEMIOLOGY

The highest burden of enteric fever can be found in Africa, Latin America, and *Asia* in areas with poor sanitation and limited access to safe food and water. An estimated 11-21 million cases of typhoid fever and an estimated 5 million cases of paratyphoid fever occur annually worldwide.

Risk factors for enteric fever include:

- Close contact with persons with typhoid fever
- Ingestion of contaminated water or food
- Poor sanitation and hygiene
- Lack of vaccination against typhoid fever
- Low socioeconomic level
- Travel to or residence in an endemic area
- Sexual contact, in particular for men who have sex with men (rare).

III. ETIOLOGY

Salmonella enterica is a gram-negative, rod-shaped, anaerobic, motile pathogen and member of the Enterobacteriaceae family. The serotypes causative for enteric fever are typhoidal:

- *S. enterica* serovar Typhi causes typhoid fever.
- *S. enterica* serovar Paratyphi (A, B and C) causes paratyphoid fever.

Humans are the only known host and reservoir of infection for typhoidal serotypes. Nontyphoidal serotypes may infect humans and animals and typically cause self-limiting gastroenteritis and diarrhea.

IV. PATHOGENESIS

- Transmission of infection is through person-to-person via the *fecal oral route*.
- Upon ingestion, some bacteria survive the gastric acid barrier to reach the small intestine:
 - *Salmonella enterica* cross the intestinal epithelium to invade M cells and access Peyer patches.
 - Within Peyer patches, bacteria are phagocytosed by antigen-presenting cells (APCs).
 - Infected APCs form discrete foci that become pathological lesions.
 - Bacteria multiply within *Salmonella*-containing vacuoles within phagocytes.
 - When phagocytes undergo apoptosis, bacteria are released and can infect additional epithelial or phagocytic cells.
 - With growing infection, inflammatory lesions in the intestinal wall worsen. Lymphoid hyperplasia may be followed by ulceration and necrosis leading to abdominal pain and possible intestinal perforation.

- The bacteria burst from Peyer patches and disseminate via the lymphatic or hematogenous route to liver (hepatomegaly), spleen (splenomegaly), bone marrow and gallbladder.
- Bacteria can persist in lymph nodes, bone marrow, and gallbladder for life in chronic carriers.
- Inflammatory response:
 - Early Th1-type and macrophage (recruitment of neutrophils and mononuclear cells) immune response prevents or delays bacterial spread.
 - A later shift to a Th2-type response is characterized by interleukin 4 (IL-4), IL-10, IL-13, and transforming growth factor (TGF)-beta production which protect infected tissues from aggressive and destructive Type 1 response.
 - B cells produce IgM, IgG, and secretory IgA.
 - Dimeric IgA is secreted into the intestinal lumen to provide mucosal defense.

V. DIAGNOSIS

Enteric fever should be suspected in a febrile patient living in, traveling from, or visiting from an endemic area, particularly if the duration of fever is more than three days or if the patient also has gastrointestinal symptoms. The diagnosis of enteric fever is made by isolating *S. Typhi* or *Paratyphi* from a culture specimen in the setting of a compatible clinical illness.

1. Clinical manifestation and Laboratory

	1 st week of illness	2 nd week of illness	3 rd week of illness
Clinical feature	• Fever rising ("stepwise") • Frontal headache, asthenia, insomnia, anorexia and epistaxis • The pulse is dissociated and gurgling right iliac fossa	• Fever rises to a plateau (39-40°C) • Digestive: abdominal pain, diarrhea, hepatosplenomegaly • Neurological: Confusion, delirium, or stupor • Rash (Rose spots), white coated tongue • Pulse is dissociated	• Fever with complication: - intestinal bleeding, or perforation - peritonitis signs - septic shock or an altered level of consciousness - hepatosplenomegaly
General laboratory	• Leukopenia, sometimes thrombocytopenia; • Moderately elevated transaminases. • Nonspecific, serum CRP is often elevated	• The same as 1st week of illness • Anemia and either leukopenia shift to leukocytosis (suspicion of intestinal perforation or hemorrhage)	• Anemia and either leukopenia shift to leukocytosis (suspicion of intestinal perforation or hemorrhage)
Confirmed diagnosis	• Blood culture: increases the sensitivity to ≈80% • Stool culture: not usually positive (less diagnostic sensitivity)	• Blood culture and stool culture	• Stool culture • Blood cultures: Sensitivity declines (prior antibiotic use)

VI. DIFFERENTIAL DIAGNOSIS

The differential diagnosis of enteric fever: Malaria, Dengue, Chikungunya virus infection, Leptospirosis, Rickettsia illnesses, Brucellosis, Amebiasis, Meningococcal disease, Bacteremia with gram-negative bacilli and Hepatitis A virus (HAV) infection

VII. MANAGEMENT

A. Objectives

Antibiotic therapy shortens the clinical course of enteric fever and reduces the risk for death. Antibiotic susceptibility testing should guide the choice of antibiotic therapy. If antibiotic susceptibility testing results are unavailable, empiric antibiotic therapy should be initiated and based on local patterns of antibiotic resistance for *Salmonella enterica* serovar Typhi and Paratyphi (or the pattern of resistance for the location of recent travel)

B. Warning Signs

- Two or more of the following the Quick Sepsis-related Organ Failure Assessment (qSOFA) criteria on initial examination:
 - Altered mental status
 - Respiratory rate ≥ 22 breaths/min
 - Systolic blood pressure ≤ 100 mm Hg
- Worsening abdominal pain and discomfort (suspicion of intestinal perforation or hemorrhage)

C. Treatment

† Antibiotics

i) Empiric Antibiotic Therapy

- (1) Uncomplicated disease: no evidence of systemic toxicity and tolerate oral therapy
 - **Azithromycin:** 1 g orally on day 1 followed by 500 mg orally once daily for the next 6 days
- (2) Severe or Complicated disease: systemic toxicity, depressed consciousness, prolonged fever, organ system dysfunction, or other feature that prompts hospitalization
 - **Ceftriaxone:** 2 g IV once or twice daily (50 to 100 mg/kg IV in one or two divided doses, maximum 4 g per day) for 10 to 14 days
 - **Meropenem** 1 to 2 g IV every eight hours for 10 to 14 days
 - ⇒ Patient who recent travel to Pakistan or Iraq, or
 - ⇒ Intestinal perforation or hemorrhage
 - ⇒ Encephalopathy with hemorrhagic shock

ii) Directed therapy

- The enteric fever should be based on results of susceptibility testing especially for fluoroquinolone in Cambodia.
- Antibiotics:

- Azithromycin: 1 g orally on day 1 followed by 500 mg orally once daily for the next 6 days
- Cefixime 200 mg orally twice daily for 10 to 14 days
- Ciprofloxacin 500 mg orally or IV 400 mg twice daily for 7 to 10 days
- Ofloxacin 400 mg orally or IV twice daily for 7 to 10 days
- TMP-SMX: 1 double-strength tablet (160/800 mg) orally twice daily 10 to 14 days

† **Adjunctive corticosteroids**

For patients with suspected or known enteric fever and severe systemic illness (delirium, obtundation, stupor, coma, or shock), we suggest adjunctive dexamethasone (Randomized control) (3 mg/kg followed by 1 mg/kg every 6 hours for a total of 48 hours)

In a randomized, prospective, double-blind study performed in Indonesia in the early 1980s among 38 adults and children with severe enteric fever (shock or obtundation), the addition of high-dose dexamethasone to chloramphenicol treatment reduced mortality compared with chloramphenicol alone (10 versus 55 percent)

† **Surgery and Procedures**

Surgery may be necessary in patients with complications such as intestinal perforation. Blood transfusion may be required for patients with severe gastrointestinal bleeding

VIII. SPECIAL CASE

- **Unresponsive to treatment:** Patients treated with antimicrobial agents *can continue to have fever for 3–5 days*, but the maximum temperature generally decreases each day. If fever in a person with typhoid or paratyphoid infection does not subside within 5 days of initiating antibiotic therapy, however, consider treatment with alternative antibiotics or begin looking for a persistent focus of infection (e.g., an abscess, or an infection in a bone, joint, or other extraintestinal site).
- **Chronic Carriage:** An estimated 1%–4% of treated patients become asymptomatic chronic carriers (defined as people who excrete the organism in stool for ≥ 12 months); a prolonged antibiotic course is usually required to eradicate the organism.

IX. PROGNOSIS

An estimated 135,000-230,000 deaths due to enteric fever occur annually worldwide. Mortality in patients with enteric fever correlates with a lack of prompt administration of appropriate antibiotics and local prevalence of antimicrobial resistance. Mortality is estimated to be:

- < 1% in patients who receive prompt therapy with appropriate antibiotics
- 10%-20% in patients treated with inappropriate antibiotics
- 10%-30% in patients who are untreated

Mortality may be higher in

- Children < 4 years old compared to older children
- Patients with complications of enteric fever such as intestinal perforation or encephalopathy with hemodynamic shock.

X. COMPLICATIONS

The most complications of enteric fever include: gastrointestinal bleeding, intestinal perforation of the ileum or colon, and encephalopathy with hemodynamic shock. The less

common complication is meningitis (rare), myocarditis and bradycardia, septic arthritis and osteomyelitis.

XI. PATIENT ADVICE AND PREVENTION

- Food & Water Precautions: Prevention of enteric fever relies on safe food and water precautions and frequent handwashing (in particular prior to meals)
- **Immunization:** Vaccination is recommended for travelers to high-risk areas and those in close contact with chronic carriers or laboratory workers who work with *S. enterica* serovar Typhi. No vaccines are active against *S. enterica* serovar Paratyphi

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SOFT TISSUE INFECTION

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I. DEFINITION

A soft tissue infection refers to an infection that effects of the skin, subcutaneous tissues, muscles, tendons, and fascia, which are collectively termed “soft tissue”. This infection can occur due to the bacterial, viral, fungal, or parasitic pathogens and may involve localized or widespread areas of the body.

II. EPIDEMIOLOGY

1. **In cidence and Prevalence:** STIs account for a significant portion of emergency

department visits and hospital admission, with cellulitis being one of the most common reasons for emergency care.

- **Cellulitis**: is estimated to account for about 7 – 10% of all hospital admissions for infections.
- **Necrotizing fasciitis**: has an incidence approximately 0.4 – 0.6 per 100,000 annually.

2. **R isk factors:**

- Chronic diseases: Diabetes, Obesity, Peripheral vascular disease, chronic kidney disease
- Immunosuppression: HIV/AIDS, on immunosuppressive drugs, on going chemotherapy
- Trauma/Physical Injury
- Intravenous Drug Use

III. ETIOLOGY

The primary cause if soft tissue infection is the introduction of pathogens through breaks in the skin (cuts, trauma, surgical wound, etc...), direct inoculation (bites, injections, burns etc...), underlying condition (diabetes, vascular insufficiency, etc..).

1. **Microorganisms res pon sible for Soft Tissu e Inf ection**

1. **Bacterial Causes:**

- i. **Gram-Positive Bacteria:**

- ***Staphylococcus aureus***: leading cause of cellulitis, abscesses, and surgical site infection.
- ***Streptococcus pyogenes***: responsible for cellulitis, erysipelas, and necrotizing fasciitis.
- ***Streptococcus agalactiae***: causes infection in neonates and adults with diabetes or chronic diseases.
- ***Enterococcus spp***: cause in mixed infection, especially in immunocompromised patients.

- ii. **Gram-Negative Bacteria:**

- ***Escherichia coli, Klebsiella spp, Proteus spp***: seems in polymicrobial infection such as diabetes foot
- ***Pseudomonas aeruginosa***: seems burn wounds, water related-injury and in immunocompromised patients.
- ***Aeromonas hydrophila***: associated with freshwater related trauma

- ***Vibrio vulnificus***: seems in wound infections related with the seawater or raw seafood.
 - iii. **Anaerobic Bacteria**:
 - ***Clostridium perfringens***: responsible in gas gangrene (a severe and rapid progressive infection with tissue necrosis).
 - ***Bacterioides spp.*, *Peptostreptococcus spp.***: seems in polymicrobial infections and especially in necrotizing fasciitis.
- 2. **Viral Causes**:
 - i. ***Herpes Simplex Virus (HPV)***: responsible in localized infections such as herpetic whitlow.
 - ii. ***Varicella-Zoster Virus (VZV)***: leads to shingles with secondary bacterial soft tissue infection.
- 3. **Fungal Causes**:
 - i. ***Candida spp.***: infection in moist areas or in immunocompromised patients.
 - ii. ***Aspergillus spp.* and *Zygomycetes***: cause severe infections especially in immunocompromised patients
- 4. **Parasitic Causes**:
 - i. ***Leishmania spp.***: causes cutaneous leishmaniasis.
 - ii. ***Dracunculus medinensis*** (Guinea worm): may lead to the secondary bacterial infection in affected tissue.
- 2. **Specific Types of Soft Tissue Infection and Etiology**
 - a. **Cellulitis**: commonly caused by *Staphylococcus aureus* and *Streptococcus pyogenes*.
 - b. **Abscesses**: Caused most frequently by *Staphylococcus aureus* (including MRSA)
 - c. **Necrotizing Fasciitis**:
 - i. **Monomicrobial**: often caused by *Streptococcus pyogenes* or *Vibrio vulnificus*
 - ii. **Polymicrobial**: Gram-positive cocci, Gram-negative bacilli and anaerobes.
 - d. **Burn Wound Infections**: *Pseudomonas aeruginosa* and other waterborne organisms are common
 - e. **Diabetic Foot Infection**: Typically polymicrobial, involving *S. aureus*, *E.coli*, *Pseudomonas aeruginosa*, and anaerobes.

IV. PATHOGENESIS

Soft Tissue Infections (STIs) developed through the interplay of host factors, pathogen, virulence, and environment exposure. The process involves the introduction of microorganisms into the skin or underlying soft tissue, followed by a sequence of microbial proliferation, tissue invasion, immune responses, and, in some cases, systemic involvement.

1. **Cellulitis**: typically caused by *S. pyogenes* or *S. aureus*. Pathogens enter through a skin breach, adhere to tissue, and spread along the dermis and subcutaneous layers. Inflammatory mediators cause swelling, redness, warmth, and pain.

2. **Abscesses:** caused by *S. aureus* (often MRSA). Bacteria from a localized pocket of infection surrounded by a wall of inflammatory cells and necrotic debris. Abscesses protect bacteria from systemic immune responses, often requiring drainage for resolution.

V. DIAGNOSIS

1. Clinical manifestation

- a) **Erysipelas:** Well-demarcated area of redness, warmth, swelling and pain accompanying the systemic symptoms (it is a subset of cellulitis but differs in its more superficial involvement).
- b) **Cellulitis:** Clinical diagnosis based on localized of redness, swelling, warmth, and tenderness. Cultures rarely required unless systemic symptoms are present.
- c) **Abscess:** Fluctuant mass with localized pain.
- d) **Necrotizing fasciitis:** Suspected in a rapidly progressing with severe pain, skin discoloration, and systemic toxicity.

2. Laboratory

a. Blood Tests:

- **Complete Blood Count (CBC):** Elevated white blood cell (WBC) count with neutrophilia. Leucopenia may indicate severe sepsis or immunosuppression.
- **Inflammatory markers:**
 - **CRP and ESR:** Are elevation in acute infections.
 - **Procalcitonin:** May be elevated in bacterial infections, particularly in systemic involvement.
- **Lactate:** Elevated level may indicate tissue hypoperfusion and sepsis
- **Glucose level:** Critical in diabetic patients to detect poor glycemic control, which exacerbate infection.
- **Renal and Liver Function Tests:** Useful to assess organ involvement in severe infection and help for the adaptation of Antibiotic dosage.

b. Microbiological Testing:

- **Blood cultures:** Essential in case of systemic symptoms, especially for detecting bacteremia (e.g., *Streptococcus pyogenes*, *Staphylococcus aureus*) and their antibiotics susceptibilities.
- **Wound cultures:** Swab, aspirates, or tissue biopsies from infected sites help identify causative organisms.
 - **Gram Stain:** Provide rapid preliminary identification of Gram-positive or Gram-negative organisms
- **Anaerobic Cultures:** particularly important in polymicrobial infection like diabetic foot ulcer or necrotizing fasciitis.
- **Molecular Diagnostic:** PCR test for specific pathogens such as *Streptococcus pyogenes*, MRSA, etc...

3. Imaging Studies:

- a. **X-Ray:** To detect gas in the tissue (indicative of gas gangrene or necrotizing fasciitis) or underlying bone involvement (osteomyelitis)
- b. **Ultrasound:** Useful for evaluating abscess formation, cellulitis with fluid collections, guidance for aspiration or drainage.

c. **CT-Scan:** Provides detailed imaging of deep tissue involvement, fascial thickening, gas formation, or fluid collections in necrotizing fasciitis.

d. **MRI:** Gold standard for identifying the extent of soft tissue involvement. It could help to early detect of necrotizing infection and differentiating cellulitis from deeper infection.

VI. DIFFERENTIAL DIAGNOSIS

1. **Inflammatory or Autoimmune Conditions:** Acute contact dermatitis, Psoriasis, Gout or Pseudogout, Lupus etc..

2. **Vascular Condition:** Deep vein thrombosis, Stasis dermatitis, Superficial thrombophlebitis.

3. **Malignant Conditions:** Inflammatory breast cancer, Angiosarcoma

4. **Other rare conditions:** Sweet syndrome, Erythema migrans, Erythema nodosum (Lyme disease)

VII. MANAGEMENT

1. General Principles of Management:

- **Early Recognition:** Prompt diagnosis is critical to prevent progression to severe infection or systemic complication.

- **Risk assessment:** Access for signs of severe infection such as systemic toxicity, rapid progression, necrosis.

- **Empirical Treatment:** Initiate Antibiotics based on the likely causative organisms, site of infection, and local antimicrobial resistance patterns.

- **Targeted Therapy:**

2. Classification - Based Treatment:

a. **Mild Infections:** (Uncomplicated cellulitis, erysipelas, minor abscesses)
→ Oral Antibiotics targeting common pathogens (*Streptococcus pyogenes*, *Staphylococcus aureus*)

- **Penicilline VK, Amoxicilline, Clindamycine, or Cephalixin**

- If MRSA is suspected: **Trimethoprim-sulfamethoxazole or Doxycycline**

→ Supportive care: Elevation of the affected limb, Pain management

b. **Moderate Infection:** Large or deeper infection, abscesses requiring drainage, systemic signs (fever, tachycardia).

→ Oral or Intravenous (IV) antibiotics depending on severity:

- IV options: **Cefazolin, Clindamycin, or Vancomycin for MRSA**

- For polymicrobial infection: **Ampicillin-Sulbactam or Piperacillin-Tazobactam**

→ **Drainage:** Surgical incision and drainage for abscesses

c. **Severe Infections:** Necrotizing fasciitis, gas gangrene, infection with sepsis or organ dysfunction.

→ Immediate surgical intervention: surgical debridement is critical for necrotizing infections to remove devitalized tissue.

→ Empirical broad-spectrum Antibiotics: Combination therapy

- **Vancomycin** (for MRSA) + **Piperacillin-Tazobactam** or **Meropenem** (for polymicrobial infections)

- Add **Clindamycin** to inhibit toxin production in *Streptococcus pyogenes* or *Clostridium* infection.
- Supportive care:
 - Aggressive fluid resuscitation
 - Hemodynamic support in case of septic shock.
- d. **Bite wounds**: *Pasterella*, *Capnocytophaga* (animal bites), *Eikenella corrodens* (human bites)
 - Empirical therapy: **Amoxicillin-clavulanate** (covers aerobes and anaerobes)
 - Tetanus prophylaxis and rabies post-exposure prophylaxis as needed
- e. **Water-associated infection**: *Aeromonas species* (fresh water), *Vibrio vulnificus* (seawater)
 - Empirical coverage with **Doxycycline** + **Third-generation cephalosporin** (**Ceftriaxone** or **Ceftazidime**).
- 3. **Duration of Therapy**:
 - a. **Mild to Moderate Infection**: 5-10 days, depending on clinical response.
 - b. **Severe Infection**: 2-4 weeks, depending on the extend of tissue involvement and microbiological response.
- 4. **Monitoring and Follow - Up**:
 - a. **Clinical response**: Monitoring for resolution of erythema, swelling, and systemic symptoms.
 - b. **Imaging**: Repeat imaging (e.g., ultrasound, MRI) if clinical improvement is not observed or complications are suspected.

VIII. PROGNOSIS

The prognosis of soft tissue infection (STIs) varies depending on factors such as the type and severity of infection, the causative organisms, the patient's overall health, and timeliness of intervention. Early detection might lead to the good outcome while delaying management or severe infection might result in complications and increased mortality.

1. **Erysipelas and uncomplicated cellulitis**: Excellent prognosis with early Antibiotic therapy. Mortality is rare and typically associated with untreated or immunocompromised patients.
2. **Abscesses**: Full recovery is expected with proper drainage and Antibiotics. Recurrent abscesses may occur in patients which conditions like hidradenitis suppurative or diabetes.
3. **Necrotizing Fasciitis**: High mortality rate (20-40%), especially without timely surgical intervention. Prognosis improves with early debridement, appropriate Antibiotics, and supportive care.
4. **Gas Gangrene**: Prognosis is poor without immediate surgical intervention and hyperbaric oxygen therapy. Mortality rate can reach 30-50% in severe cases, particularly with sepsis or liver disease.
5. **Diabetic Foot Infection**: Prognosis depends on the extend of infection and glycemic control. Severe infections may lead to chronic wound, osteomyelitis, or amputations if not manage promptly.

IX. COMPLICATION

1. **Localized Complications:** Abscess formation, Chronic wound or ulcers, Lymphatic damage (chronic lymphedema).
2. **Systemic Complications:** Sepsis or Septic shock, Multiorgan failure, post-streptococcal glomerulonephritis (rare with Group A Streptococcus infection).
3. **Long-term Consequences:** Recurrent infections in predisposed individuals (those with venous insufficiency or diabetes). Functional impairment or amputation (diabetic foot infections).

X. PREVENTION

Prevention soft tissue infections require a combination of strategies to reduce the risk of infection, manage predisposing conditions, and educate patients on proper wound care and hygiene.

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OSTEOMYELITIS

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I. DEFINITION

Osteomyelitis is an infection of the bone, typically caused by bacteria, although fungi or other microorganisms can also be responsible. The infection may reach the bone through the bloodstream, from nearby infected tissue, or directly following an injury, surgery, or the presence of a foreign foreign device like a metal implant.

II. EPIDEMIOLOGY

A-Incidence and Prevalence:

- The overall incidence of osteomyelitis is approximately 2 to 5 cases per 10,000 people annually.

A- e Distribution:

- More common in children under the age of 5, typically affecting long bones such as the femur or tibia.
- In adults, it often occurs in the vertebrae or bones of the feet, especially in diabetics.

C-Geographical Distribution:

- Higher rates are observed in regions with limited access to healthcare and sanitation, where open wounds and untreated infections are more prevalent.
- In developed countries, most cases are associated with diabetes, surgical interventions, or intravenous drug use.

III. ETIOLOGY

1. Routes of Infection

- Hematogenous Spread, contiguous Spread, direct Inoculation.

2. Causative Organisms

Bacterial Infections (Most Common):

- Staphylococcus aureus, Streptococcal species, Gram-negative bacteria: Escherichia coli, Pseudomonas aeruginosa.
- Polymicrobial Infections
- Tuberculous Osteomyelitis
- Fungal Infections.

3. Patient-Specific Risk Factors

- Diabetes Mellitus
- Peripheral Vascular Disease (PVD)
- Trauma or Surgery
- Immunosuppression
- Prosthetic Devices
- Pediatric Factors: High vascularity in metaphyseal bones makes children more susceptible to hematogenous osteomyelitis.

IV. PATHOPHYSIOLOGY

1. Initiation of Infection

- Microbial Entry: via hematogenous spread, direct inoculation (trauma or surgery), or contiguous spread from adjacent tissues.
- Common sites vary by age: In children, metaphyses of long bones are typically affected. In adults, vertebrae are commonly involved in hematogenous osteomyelitis.

2. Localized Inflammatory Response

- Recruitment of Immune Cells: Release of pro-inflammatory cytokines (e.g., IL-1, IL-

6, TNF- α) and reactive oxygen species damages bone tissue.

- Edema Formation: Increased pressure can reduce blood flow, leading to ischemia and hypoxia.

3. Bone Necrosis

- ischemic Damage: Compromised blood flow results in bone necrosis and the formation of sequestra (dead bone segments).
- The necrotic bone acts as a foreign body, perpetuating infection and inflammation.
- Biofilm Formation: Biofilms protect bacteria from the immune system and antibiotics, promoting chronic infection.

4. Chronic Inflammatory Cycle

- Formation of Involucrum: the periosteum produces new bone, called involucrum, which may surround the sequestra.
- Chronic Infection: the infection becomes chronic, with alternating periods of activity and remission. Sinus tracts may develop, allowing purulent material to drain to the skin surface.

5. Spread of Infection

- Adjacent Structures: Infection can spread to surrounding soft tissues, joints, and muscles, leading to cellulitis, septic arthritis, or abscess formation.
- Systemic Spread: pathogens can disseminate via the bloodstream, causing sepsis or secondary infections in other organs.

6. Differences in Acute vs. Chronic Osteomyelitis

- Acute Osteomyelitis: Rapid onset of infection with intense inflammatory response. Necrosis and sequestrum formation may be minimal if treated early.
- Chronic Osteomyelitis: Persistent infection with significant sequestrum and involucrum formation. Biofilms play a major role in resistance to treatment.

V. DIAGNOSIS

1. Clinical Evaluation

A-History and Symptoms:

- Pain, Fever and Chills, Recent Trauma or Surgery.
- Underlying Conditions: diabetes, immunosuppression, or vascular insufficiency increase the likelihood of osteomyelitis.

B-Physical Examination:

- Localized Redness, Swelling, and Warmth.
- Bone Deformity: In chronic osteomyelitis, deformities and draining sinus tracts may be present.

2. Laboratory Tests

- Blood Cultures
- C-reactive protein (CRP) and erythrocyte sedimentation rate (ESR) are elevated in osteomyelitis.
- Leukocytosis

3. Imaging Studies

- X-rays: Initial Imaging Tool. Bone changes may not be visible in the first 1-2 weeks after infection.
- Magnetic Resonance Imaging (MRI): Gold Standard for Diagnosis especially in early stages and for spinal or soft tissue involvement.
- Computed Tomography (CT) Scan: CT is helpful when MRI is not available or in evaluating bony deformities and the extent of infection, particularly in the pelvis, spine, or in the presence of foreign objects (e.g., implants).
- Ultrasound: Guided Aspiration, helping to confirm the diagnosis and identify microorganisms.

4. Microbiological Testing

- Aspiration or Biopsy: When blood cultures are negative, aspiration or bone biopsy is the gold standard for identifying the microorganism and determining the appropriate antibiotic therapy.

VI. DIFFERENTIAL DIAGNOSIS

- Septic Arthritis
- Soft Tissue Abscess
- Bone Tumors (Benign or Malignant)
- Fractures or Bone Trauma
- Gout
- Rheumatic Fever (with Migratory Arthritis)
- Abscess or Infection in Adjacent Tissues (e.g., Soft Tissue Infection)
- Paget's Disease of Bone
- Osteoarthritis
- Osteonecrosis (Avascular Necrosis).

V. MANAGEMENT

1. Objectives

- Eliminate the infection using targeted antibiotic therapy.
- Prevent complications like chronic infection, systemic spread, or bone deformities.
- Restore bone and joint function through surgical interventions and rehabilitation.
- Manage pain to ensure comfort and improve quality of life.
- Optimize overall health and address underlying risk factors (e.g., diabetes, immunosuppression).
- Ensure patient adherence to treatment and follow-up care to prevent recurrence.

2. Critical sign

- Severe Localized Pain: it may signal worsening infection or abscess formation.
- Fever and Chills: suggests that the infection has reached a significant level (bacteremia).
- Pus Drainage (Sinus Tract Formation): indicated a more severe or long-standing infection which could lead
- to widespread tissue destruction.
- Systemic Symptoms of Sepsis: Critical signs include, Fever or hypothermia (low body temperature), Tachycardia (heart rate > 100 bpm), Hypotension (blood pressure < 90/60 mmHg), Altered mental status or confusion (due to inadequate oxygen supply to the brain), Shortness of breath and respiratory distress.
- Extensive Bone Tenderness or Palpable Sequestrum: Is an indication of necrosis (dead bone tissue) and active infection, which requires surgical intervention to remove the infected and necrotic bone.
- Signs of Necrosis or Gangrene: suggests that the infection has compromised blood flow, often requiring urgent surgical debridement or even amputation in extreme cases.
- Worsening or Non-Resolving Symptoms Despite Antibiotics: indicate the presence of an abscess, sequestrum, or an antibiotic-resistant organism (e.g., MRSA), and further intervention (e.g., surgical debridement or a change in antibiotics) may be necessary.
- Persistent or Escalating Inflammation After Trauma or Surgery: suggest that infection may be developing in the bone, particularly if there is a history of surgery or trauma, requiring early intervention to prevent osteomyelitis.

3. Mortality risk factor

- Age: Older Adult, Neonates/Infants.
- Immunocompromised State: such as HIV/AIDS, diabetes mellitus, chronic corticosteroid use, immunosuppressive therapy and cancer.

- Chronic Comorbidities: Diabetes Mellitus, Peripheral Artery Disease (PAD), Renal Failure.
- Delayed Diagnosis and Treatment
- Type and Severity of Infection: Methicillin-Resistant *Staphylococcus aureus* (MRSA), Fungal or Multidrug-Resistant Infections, Co-Infections.
- Presence of Sepsis: indicating that the infection is systemic and the patient is at high risk of death.
- Advanced Bone Involvement: chronic Osteomyelitis, recurrent or Persistent Infection
- Poor Nutritional Status: Malnutrition, Weight Loss and Anemia
- Bone Involvement in High-Risk Locations: Spinal Osteomyelitis, Pelvic Osteomyelitis.

4. Outpatient management

A. Antibiotic Therapy

- Oral Antibiotics: Once the causative organism has been identified and the patient's condition is stable, oral antibiotics can be used.
- Duration: Typically, antibiotics are continued for 4–6 weeks depending on the severity and location.
- Common Antibiotics:
 - For *Staphylococcus aureus* (including MRSA): Oral options include clindamycin, trimethoprim-sulfamethoxazole (Bactrim), or doxycycline.
 - For non-resistant organisms: Cephalexin may be used.
 - For Gram-negative organisms: Options like ciprofloxacin or amoxicillin-clavulanate may be appropriate.

B. Monitoring and Follow-up Care

- Regular Follow-ups: to monitor the patient's progress, assess clinical response to antibiotics, and check for signs of complications are critical.
- Blood Tests: Periodic blood tests to assess the level of inflammation and monitor the resolution of infection.
- Imaging: Follow-up imaging, such as X-rays, MRI, or CT scans, to assess bone healing or any possible complications.

C. Wound Care (if applicable)

- Dressings, drain Care, Hygiene.

D. Pain Management

- Analgesics, Opioids and Physical Therapy.

E. Nutrition and General Health Support

- Proper Nutrition: Protein-rich foods and vitamin D supplementation may be recommended.
- If the patient is malnourished or has difficulty eating, nutritional supplements or referral to a dietitian may be needed.
- Optimizing Comorbidities: controlling the underlying health issues is essential for ensuring proper healing and preventing further complications.

F. Patient Education

- Adherence to Antibiotic Regimen: understands the importance of completing the full course of antibiotics.
- Warning Signs of Complications: such as increased pain, redness, swelling, fever, or drainage, which should prompt immediate medical attention.
- Wound Care Instructions: cleaning techniques and when to change dressings.

G. Avoiding Risk Factors

- Avoiding Weight-Bearing Activities, Smoking Cessation, Alcohol Moderation.

5. First-line treatment

A-Empiric Antibiotic Therapy

- Initiated immediately after blood cultures or biopsy samples are taken, targeting the most likely pathogens.
- Common empiric choices include:
 - Vancomycin: Covers *Staphylococcus aureus* (including MRSA).
 - Plus, a broad-spectrum β -lactam (e.g., ceftriaxone, cefepime, or piperacillin-tazobactam) to cover Gram-negative organisms, especially in polymicrobial infections or immunocompromised patients.

B-Targeted Therapy

- Adjusted based on culture and sensitivity results.
- Typical antibiotics for common pathogens:
 - MSSA: Nafcillin, oxacillin, or cefazolin.
 - MRSA: Vancomycin or daptomycin.
 - Gram-negative organisms: Ciprofloxacin, ceftriaxone, or piperacillin-tazobactam.

C-Duration of Therapy

- Acute osteomyelitis: Typically, 4–6 weeks of antibiotics.
- Chronic osteomyelitis: May require extended therapy, often with surgical intervention.

D- Adjunct Measures

- Surgical Debridement: Removal of necrotic bone or infected hardware, especially in chronic or refractory cases.
- Drainage and Stabilization: For abscess formation or severe infections compromising bone integrity.

6. Treatment adjustment

A. Adjusting Based on Culture and Sensitivity Results

- Empiric Therapy to Targeted Therapy:
 - Empiric broad-spectrum antibiotics (e.g., vancomycin + ceftriaxone) are adjusted once microbiological results are available.
- Specific adjustments include:
 - Methicillin-sensitive *Staphylococcus aureus* (MSSA): Switch to oxacillin, or cefazolin.
 - Methicillin-resistant *Staphylococcus aureus* (MRSA): Continue vancomycin or consider alternatives like linezolid.
 - Gram-negative organisms (e.g., *E. coli*, *Pseudomonas*): Use ceftazidime, or ciprofloxacin based on sensitivity.
 - Anaerobes: Add metronidazole or clindamycin if anaerobic pathogens are isolated (e.g., in polymicrobial infections).

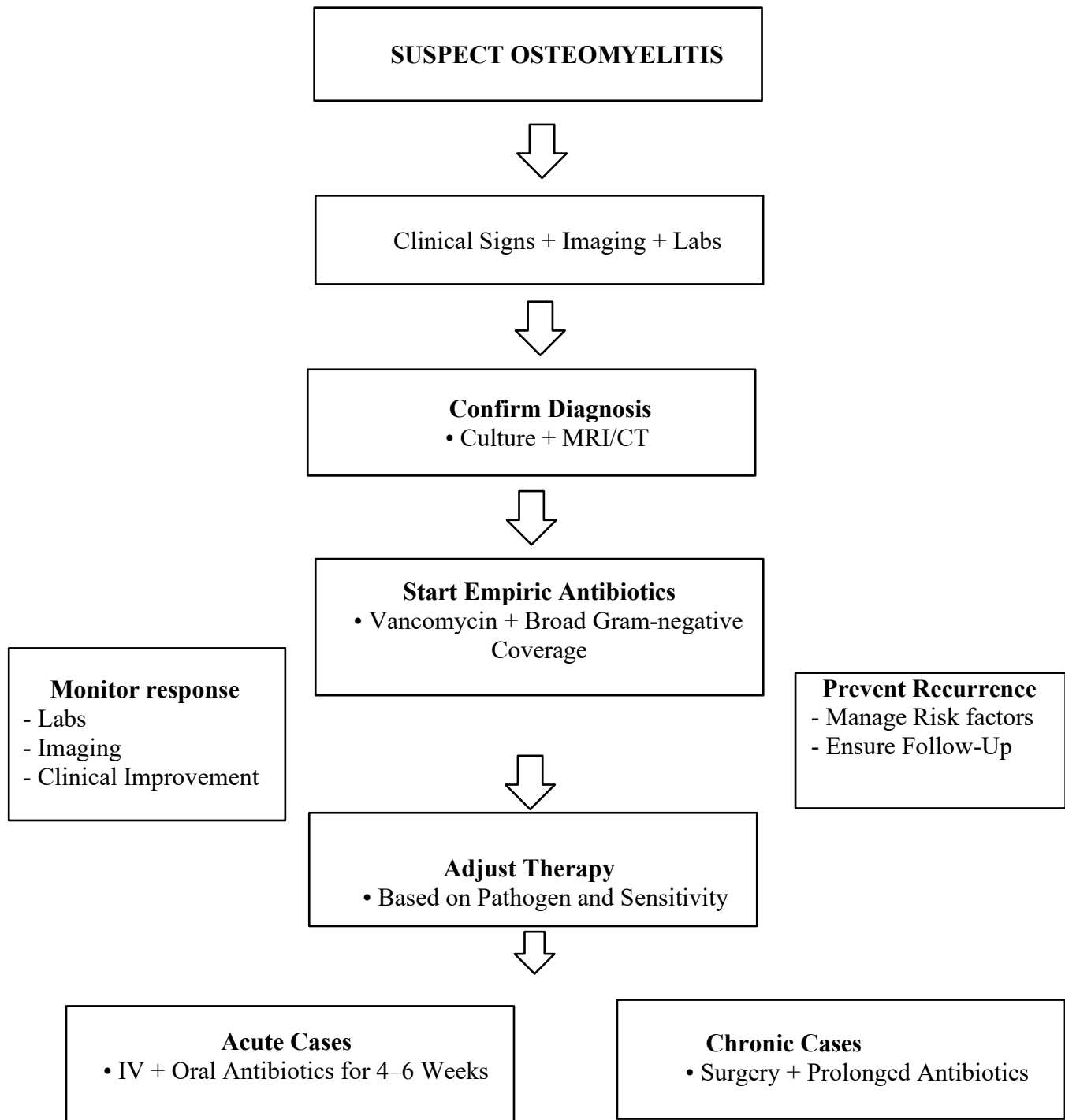
B. Adjustments in Special Populations

- Diabetic Foot Osteomyelitis: Broaden coverage to include polymicrobial organisms (aerobes and anaerobes) plus glycemic control.
- Prosthetic or Hardware-Associated Infections: Consider hardware removal if infection persists despite antibiotics.
- Immunocompromised Patients: Extend coverage to include fungal or atypical pathogens. Longer antibiotic courses or combination therapy may be necessary.

C. Addressing Complications

- Recurrent Infections:
 - Evaluate for sequester (dead bone) or abscesses that require surgical removal.
 - Adjust antibiotics based on new culture results.
- Multidrug-Resistant Organisms: Refer to specialized center.

7. Algorithm



VI. PECIAL CASE

1. Osteomyelitis in Diabetic Patients (Diabetic Foot Osteomyelitis)

- It typically occurs as a result of diabetic neuropathy and peripheral vascular disease, leading to foot ulcers that become infected and spread to the underlying bone.
- Antibiotics: Empiric therapy should cover Gram-positive cocci (e.g., *Staphylococcus aureus*) and Gram-negative organisms (e.g., *Pseudomonas aeruginosa*), with adjustments based on culture results.

- Blood Sugar Control: Strict glycemic control is essential to optimize immune function and promote healing.
- Surgical Debridement: In cases of extensive bone involvement or necrosis, surgical removal of infected or necrotic tissue may be necessary.
- Amputation: In severe, refractory cases with widespread infection or non-healing ulcers, amputation may be required

2. Spinal Osteomyelitis

- It is a serious condition that can result in severe complications, including spinal cord compression, nerve damage, and sepsis.
- Infection in the spine can cause compression of the spinal cord or nerves, leading to permanent neurologic deficits if not treated promptly.
- Early identification via imaging (MRI is preferred) and blood cultures is critical.

3. Chronic Osteomyelitis

- It often involves sequestrum (dead bone), biofilm formation, and a low-grade infection that is difficult to treat.
- The bacteria in chronic osteomyelitis often form biofilms, which protect them from the immune system and antibiotics.
- Removal of infected and necrotic tissue is often necessary, including the sequestrum.

4. Osteomyelitis in Immunocompromised Patients

- These patients are more likely to develop severe, aggressive infections, including those caused by opportunistic pathogens such as fungi and atypical bacteria.
- Standard antibiotic regimens may not be effective, and treatment may require long courses or higher doses of medication.
- Empiric therapy should cover a wide range of potential pathogens, including resistant organisms (e.g., MRSA, multidrug-resistant organisms) and fungi (e.g., Candida, Aspergillus).
- These patients require close monitoring for side effects of long-term antibiotic use and potential complications such as sepsis or organ dysfunction.

5. Osteomyelitis in Prosthetic Joints (Periprosthetic Osteomyelitis)

- Infection around prosthetic joints can lead to periprosthetic osteomyelitis. This type of infection can be more challenging to treat due to the presence of the foreign material.
- The bacteria can form biofilms on the prosthetic joint, making them harder to treat with antibiotics alone.
- The prosthetic joint may need to be removed or replaced to fully resolve the infection.

VI. EVOLUTION AND PROGNOSIS

- Good Prognosis: With early diagnosis, appropriate antibiotic therapy, and surgical intervention (when needed), acute osteomyelitis can be effectively treated with a high likelihood of full recovery.
- Chronic or Recurrent Osteomyelitis: Chronic or recurrent osteomyelitis may lead to prolonged treatment courses, functional impairment, and sometimes permanent bone damage, though patients can often manage these conditions with careful follow-up and intervention.
- Severe Cases: In more severe cases, especially when osteomyelitis leads to complications such as sepsis, spinal cord compression, or limb loss, the prognosis may be poor, with a higher risk of long-term disability or death. may be poor, with a higher risk of long-term disability or death.

VII. COMPLICATION

- Sepsis
- Chronic Osteomyelitis
- Bone and Joint Destruction
- Soft Tissue Involvement
- Neurologic Complications (Especially in Spinal Osteomyelitis)
- Amputation
- Spread of Infection to Other Organs
- Osteonecrosis (Bone Death)
- Sepsis-Induced Organ Dysfunction
- Persistent Sinus Tract and Wound Drainage

VIII. PREVENTION AND RECOMMENDATION

- Preventing osteomyelitis revolves around early identification of infections, prompt treatment, and addressing the underlying risk factors such as chronic conditions (e.g., diabetes, immunocompromised states), poor hygiene, or poor wound care.
- Preventive measures in surgical settings, appropriate wound management, and lifestyle changes (e.g., smoking cessation, fall prevention) all play critical roles in reducing the incidence of osteomyelitis.

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PRINCIPAL OF ANTIBIOTHERAPY

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I. DEFINITION

Antibiotherapy refers to the use of antimicrobial agents to treat infections caused by bacteria, viruses, fungi, or parasites. It involves selecting appropriate antibiotics to eliminate pathogens while minimizing harm to the host and preventing resistance.

II. EPIDEMIOLOGY

- Antibiotics are among the most commonly prescribed medications worldwide.
- Inappropriate use, such as over prescription and misuse, has led to a global rise in antimicrobial resistance (AMR).
- AMR causes approximately 1.27 million deaths annually and is projected to worsen without intervention.

III. ETIOLOGY

A. Pathogens:

- Bacterial: Gram-positive (e.g., *Staphylococcus aureus*), Gram-negative (e.g., *Escherichia coli*), and atypical organisms (e.g., *Mycoplasma pneumoniae*).
- Viral: Influenza, herpesviruses.
- Fungal: *Candida* species.
- Parasitic: *Plasmodium* (malaria).

B. Contributing factors:

- Inappropriate prescribing.
- Poor infection control practices.
- Lack of patient adherence.

IV. PATHOPHYSIOLOGY

- Infection establishment: Pathogens invade and multiply, overcoming the host's immune defenses.
- Host response: Inflammation, fever, and tissue damage occur as the body attempts to control the infection.
- Antibiotic action:
 - + Bactericidal agents kill bacteria (e.g., beta-lactams).
 - + Bacteriostatic agents inhibit bacterial growth (e.g., tetracyclines).
 - + Narrow-spectrum antibiotics target specific pathogens; broad-spectrum agents cover a wider range.

V. DIAGNOSIS

1. Clinical evaluation:

- Signs and symptoms: Fever, localized pain, swelling, or discharge.
- History of recent infections or exposure.

2. Laboratory tests:

- Blood tests: Complete blood count (CBC), inflammatory markers (CRP, ESR).
- Microbiological analysis: Gram stain, culture, and sensitivity testing.

3. Imaging:

- Radiographs, CT, or MRI for localized infections (e.g., osteomyelitis).
- Ultrasound for fluid collection (e.g., abscess).

VI. DIFFERENTIAL DIAGNOSIS

- Viral infections: Differentiated by lack of bacterial growth in cultures.
- Autoimmune diseases: Mimic infections with elevated inflammatory markers.
- Non-infectious causes: Drug fever, malignancies, or post-surgical inflammation.

VII. MANAGEMENT

1. Objectives

- Eradicate the pathogen.
- Prevent complications and resistance.
- Minimize adverse effects.

2. Critical sign

- Signs of sepsis: Fever, tachycardia, hypotension.
- Multi-organ dysfunction.
- Rapidly progressing local infections (e.g., necrotizing fasciitis).

3. Mortality risk factor

- Delay in appropriate treatment.
- Multidrug-resistant pathogens.
- Immunosuppression or comorbidities (e.g., diabetes, malignancies).
- Severe infection (e.g., endocarditis, meningitis).

4. Outpatient management

- Suitable for mild infections in stable patients.
- Oral antibiotics with close follow-up.
- Examples: Uncomplicated urinary tract infections or cellulitis.

5. First-line treatment

A-Respiratory Tract Infections Community-Acquired Pneumonia (CAP):

- Amoxicillin: 1 g PO every 8 hours for 5–7 days.
- Azithromycin: 500 mg PO for 5-7 days.

B-Sinusitis & Pharyngitis

- Amoxicillin-Clavulanate: 500 mg/125 mg PO every 8 hours for 7–10 days.

C-Urinary Tract Infections (UTIs):

- Uncomplicated UTI:
 - + Nitrofurantoin: 100 mg PO every 12 hours for 5 days.
 - + Trimethoprim-Sulfamethoxazole (TMP-SMX): 160/800 mg PO every 12 hours for 3 days.
- Complicated UTI/Pyelonephritis:
 - + Ciprofloxacin: 500 mg PO every 12 hours for 7 days.
 - + Ceftriaxone: 1-2 g IV once daily for 7–14 days.

D-Skin and Soft Tissue Infections

- Cellulitis:
 - + Amoxicillin-Clavulanate: 500 mg/125 mg PO every 8 hours for 7–10 days.
 - + Clindamycin: 300 mg PO every 8 hours for 7–10 days.
- MRSA Suspected:
 - + Doxycycline: 100 mg PO every 12 hours for 10 days.
 - + TMP-SMX: 160/800 mg PO every 12 hours for 10 days.

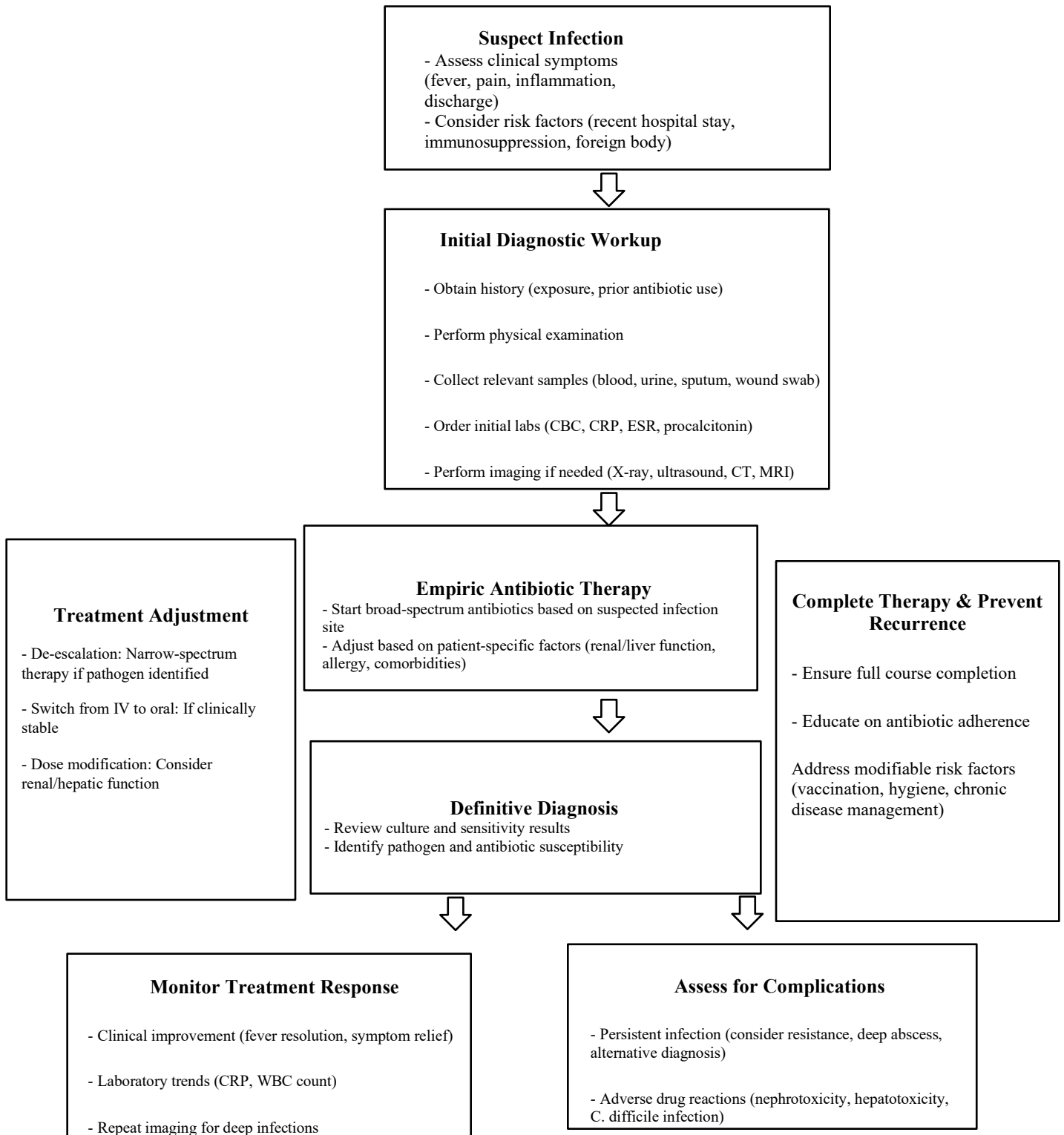
E-Gastrointestinal Infections

- Clostridioides difficile infection :
 - + Oral Vancomycin: 125 mg PO every 6 hours for 10 days.
 - + Oral metronidazole 500mg PO every 6 hours for 10 days.
- Traveler's Diarrhea:
 - + Azithromycin: 1 g PO single dose or 500 mg PO daily for 3 days.
 - + Ciprofloxacin: 500 mg PO every 12 hours for 1–3 days.

6. Treatment adjustment

- De-escalation:
 - + Adjust based on culture and sensitivity results.
 - + Shift from broad-spectrum to narrow-spectrum agents when appropriate.
- Transition from IV to oral:
 - + Consider after 48–72 hours if the patient is stable.
 - + Example: IV ceftriaxone → oral cefuroxime or amoxicillin/clavulanate acid for pneumonia.
- Renal and hepatic impairment:
 - + Adjust dosages of aminoglycosides, vancomycin, and fluoroquinolones.
- Adverse effects monitoring:
 - + Monitor renal function (aminoglycosides, vancomycin), QT prolongation (fluoroquinolones, macrolides).

7. Algorithm



VIII. SPECIAL CASE

1. Immunocompromised Patients (Neutropenia, HIV, Transplant)

- Empiric Broad-Spectrum Therapy:
 - + Piperacillin-tazobactam: 4.5g IV every 6 hours
 - + Meropenem: 1g IV every 8 hours
 - + Vancomycin: 15–20 mg/kg IV every 8–12 hours (adjust based on renal function)

- Fungal Prophylaxis (Neutropenia or Transplant)

- + Fluconazole: 400mg IV/PO once daily

2. Pregnancy & Lactation (Safe Antibiotics)

- Urinary Tract Infection (UTI) in Pregnancy:
 - + Amoxicillin-clavulanate: 500/125 mg PO every 8 hours for 7 days
 - + Nitrofurantoin (avoid in 1st trimester): 100 mg PO every 12 hours for 5 days
- Bacterial Vaginosis:
 - + Metronidazole: 500 mg PO every 12 hours for 7 days.
- Pneumonia:
 - + Amoxicillin-Clavulanate: 500 mg/125 mg PO every 8 hours for 7–10 days.
 - + Azithromycin: 500 mg PO on day for 5 days.

3. Geriatric Patients (Renal Dose Adjustments Needed)

- UTI (Complicated or Catheter-Associated):
 - + Ceftriaxone: 1-2g IV every 24 hours.
- Pneumonia:
 - + Levofloxacin: 750 mg IV/PO once daily (reduce to 500 mg if CrCl <50 mL/min)
- Skin & Soft Tissue Infection (MRSA Coverage Needed):
 - + Vancomycin: 15 mg/kg IV every 12 hours (adjust per renal function)

4. Severe Drug Allergy (Beta-Lactam Alternative Therapy)

- Sepsis in Penicillin Allergy (suspected Gram positive):
 - + Vancomycin: 15 mg/kg IV every 12 hours
- Pneumonia (Beta-Lactam Allergy):
 - + Levofloxacin: 750 mg IV/PO once daily

5. Antibiotic-Resistant Infections (MDR, ESBL, MRSA, VRE)

- Refer patient to specialist center

6. Prosthetic-Associated Infections (Biofilm Involvement)

- Prosthetic Joint Infection (Staphylococcus aureus)
 - + Levofloxacin: 750 mg PO once daily for 6 weeks

7. Osteomyelitis & Endocarditis (Long-Term Therapy Required)

- Osteomyelitis (Staphylococcus aureus)
 - + Vancomycin: 15 mg/kg IV every 12 hours for 4–6 weeks
 - + Ceftriaxone: 2g IV once daily
- Endocarditis (Empiric Therapy)
 - + Ceftriaxone: 2g IV once daily
 - + Gentamicin: 5 mg/kg IV once daily for synergy

8. Atypical Pathogens (leptospirosis, Rickettsia)

- Leptospirosis:
 - + Mild Cases: Doxycycline: 100 mg PO every 12 hours for 7 days
 - + Severe Cases: Ceftriaxone 1-2g IV once daily or Cefotaxime 1g IV every 6 hours for 7 days

- Rickettsiosis:
 - + First-Line Treatment (Adults & Children >45 kg): Doxycycline: 100 mg PO every 12 hours for 7–14 days

IX. EVOLUTION AND PROGNOSIS

- Favorable outcomes with timely and appropriate treatment.
- Delayed or inappropriate therapy can lead to complications, including chronic infections, sepsis, or death.
- Prognosis depends on infection severity, pathogen resistance, and patient comorbidities.

X. COMPLICATION

1. Allergic Reactions

- Mild Reactions: Rash, itching, and hives.
- Severe Reactions: Anaphylaxis, which can be life-threatening and may present with difficulty breathing, swelling, and cardiovascular collapse.

2. Gastrointestinal Disturbances

- Diarrhea: Often due to alterations in gut microbiota or overgrowth of pathogens like *Clostridium difficile*, leading to antibiotic-associated diarrhea.
- Nausea and Vomiting: Can occur with certain antibiotics, leading to poor adherence to treatment.

3. Antibiotic-Associated Colitis

- Caused primarily by *C. difficile* after the use of broad-spectrum antibiotics, resulting in severe diarrhea, abdominal pain, and potentially life-threatening complications.

4. Hepatotoxicity

- Some antibiotics can lead to liver enzyme elevations and, in severe cases, acute liver injury or failure (e.g., isoniazid, amoxicillin-clavulanate).

5. Nephrotoxicity

- Certain antibiotics (e.g., aminoglycosides, vancomycin) can cause kidney damage, leading to acute kidney injury.

6. Hematological Effects

- Agranulocytosis: Severe reduction in white blood cells, leading to increased infection risk.
- Hemolytic Anemia: Destruction of red blood cells, possibly due to an immune reaction.
- Thrombocytopenia: Low platelet count can increase bleeding risk.

7. Superinfection

- Overgrowth of non-susceptible organisms (e.g., fungi or antibiotic-resistant bacteria) that can occur when normal microbial flora is disrupted.

8. Development of Antibiotic Resistance

- The inappropriate or excessive use of antibiotics can lead to the emergence of resistant strains of bacteria, complicating future treatment options.

9. Drug Interactions

- Antibiotics can interact with other medications, altering their effectiveness or increasing potential side effects. For example, some antibiotics can reduce the efficacy of oral contraceptives.

10. Photosensitivity

- Certain antibiotics (e.g., doxycycline, ciprofloxacin) can cause heightened sensitivity to sunlight, increasing the risk of sunburn or skin reactions.

11. Tendinopathy

- Fluoroquinolones may increase the risk of tendon rupture, particularly in older patients or those taking corticosteroids.

XI. PREVENTION AND RECOMMENDATION

- Antibiotic adherence: Complete prescribed course even if symptoms resolve.
- Avoid misuse: Do not demand antibiotics for viral infections.
- Infection prevention: Vaccination, Hand hygiene and proper wound care.
- Follow-up care: Monitor for treatment response and complications.
- Awareness campaigns: Educate on antimicrobial resistance and its impact.

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CHAPTER X HEMATO-ONCOLOGY

1. Work-up for abnormality of complete blood count (Red Blood Cells, White Blood Cells)
2. Work-up for abnormality of complete blood count (Thrombocytosis)
3. Work-up for abnormality of complete blood count (Thrombocytopenia)
4. Immune thrombocytopenia
5. Aplastic anemia
6. Hodgkin Lymphoma
7. Non-Hodgkin lymphoma
8. Thalassemia
9. Indication of transfusion and its management

WORK-UP

FOR ABNORMALITY OF COMPLETE BLOOD COUNT

(Red Blood Cells, White Blood Cells)

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I. INTRODUCTION

Complete blood count (CBC) is one of the most common blood tests requested by clinicians and evaluates the total numbers and characteristics of cell components in the blood, such as white blood cells (WBCs), red blood cells (RBCs), and platelets. CBC includes: WBC total and differential count, erythrogram (RBC count, determination of hemoglobin (Hb) and hematocrit, and indices calculation (mean corpuscular volume (MCV), mean corpuscular hemoglobin (MCH), mean corpuscular hemoglobin concentration (MCHC), and red cell distribution width (RDW), and platelet count indices calculation (mean platelet volume (MPV).

1.1. WBC Count

One of the most important indicators in interpreting CBC results is WBC count. Several types of circulating WBCs can be classified according to differentiation process, size, and shape, including neutrophils, lymphocytes, monocytes, eosinophils, and basophils. WBCs are an essential part of the immune system, and the different types of WBCs have distinct roles in the human body. For example, an increase in WBCs can be a sign of infection in any part of the body and can indicate the presence of a tumor. A decrease in WBCs can suggest a problem with bone marrow production or be a side effect of certain drugs, including anticancer drugs. In addition, the WBC component of CBC is clinically useful in follow-up to determine the therapeutic effects of treatments for infection or inflammation

1.2. RBC Parameters

RBC count is another crucial CBC result. RBCs are created in the bone marrow and released into the peripheral blood circulation after maturation. RBCs are usually uniform in size and shape, but their appearance can be affected by various conditions

1.3. Platelet Parameters

Platelets are anucleate cells with a short lifespan of 7–10 days and are eliminated in the spleen or liver after circulating in the body. The primary function of the platelet is hemostasis. Platelets stick to damaged blood vessel walls or clump together, causing blood clotting and stopping blood loss. In addition, platelets play an essential role in systemic inflammation, and an activated platelet contributes to vascular inflammation and damage, atherogenesis, and thrombosis. Several components related to platelets in CBC are platelet count, MPV, and platelet distribution width (PDW), and each factor reflects the condition of various diseases. Table 1.

Table 1. Complete blood count

Variable	Male	Female
Hemoglobin (g/dL)	130-170	120-160
RBCs ($\times 10^{12}/L$)	4.50-5.90	4.00-5.20
Mean corpuscular volume (fL)	80-95	80-95
Platelets ($\times 10^9/L$)	150-450	150-450
WBCs ($\times 10^9/L$)	4.0-9.0	4.0-9.0
Neutrophils ($\times 10^9/L$)	1,5-7,5	1,5-7,5
Lymphocytes ($\times 10^9/L$)	1.0-4.0	1.0-4.0
Monocytes ($\times 10^9/L$)	0.1-1.0	0.1-1.0
Eosinophils ($\times 10^9/L$)	0.2-0.7	0.2-0.7
Basophils ($\times 10^9/L$)	0.0-0.2	0.0-0.2

II. ANEMIA

Anemia is defined as a low number of red blood cells.

- Men > 13g/L
- Women >12g/L
- Pregnancy >10,5g/L (> 4month).

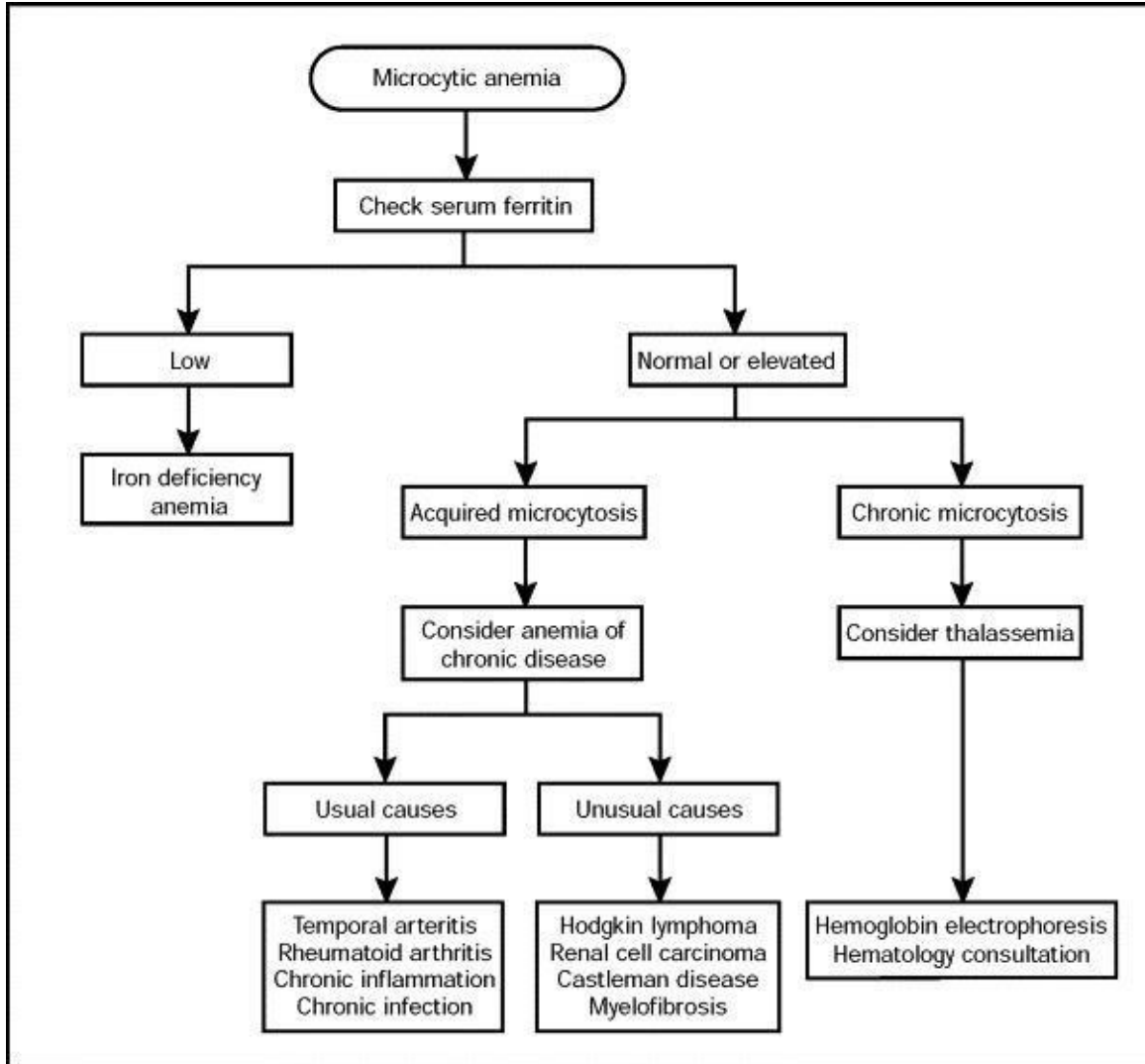
2.1. Microcytic anemia

The 3 major diagnostic possibilities for microcytic anemia are iron deficiency anemia (IDA), thalassemia, and anemia of chronic disease (ACD). A fourth possibility, sideroblastic anemia presenting with microcytosis, is not prevalent enough for routine consideration. Clues from the CBC and PBS for the differential diagnosis of microcytic anemia are outlined. Since the most common of the microcytic anemias is IDA, we recommend determination of the serum ferritin level as the initial step for all patients with microcytic anemia. A low serum ferritin level is diagnostic of IDA.

Table 2. Clues from CBC and PBS in the Differential Diagnosis of Anemias

Clues from CBC and PBS in the Differential Diagnosis of Anemias			
Category of anemia	Differential diagnosis	CBC clues	PBS clues
Microcytic	Iron deficiency anemia	Increased RDW	Anisocytosis
		Thrombocytosis	Poikilocytosis
			Elliptocytosis
	Thalassemia	Normal / elevated RBC count	Polychromasia
			Target cells
	Anemia of chronic disease	Normal / elevated RDW	Basophilic stippling
		Normal RDW	Unremarkable (typically)
			Rouleaux formation (CD)
			Myelophthisis (MMM)

*Diagnostic algorithm for microcytic anemia

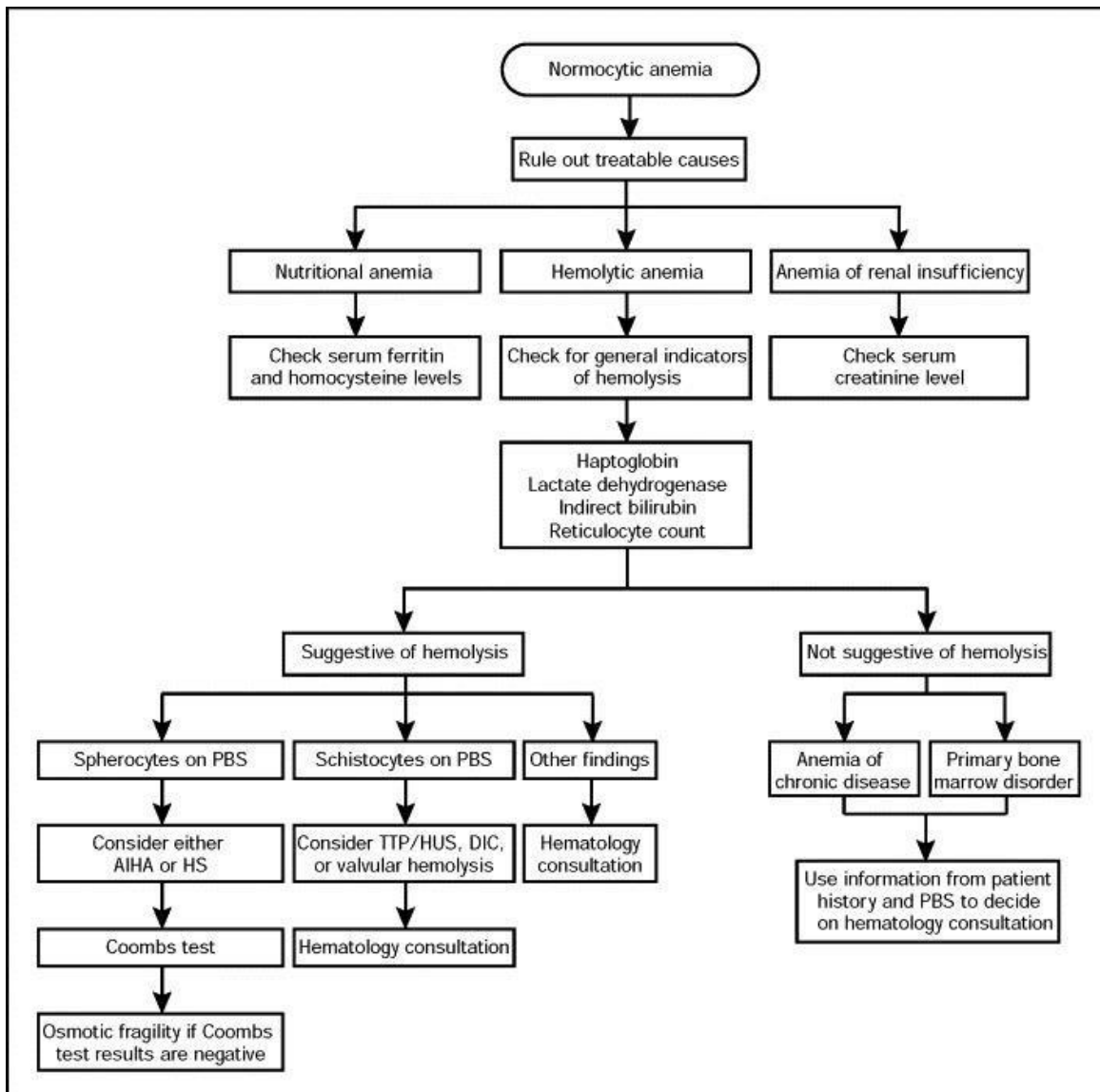


2.2. Normocytic Anemia

The first step in approaching normocytic anemia is to exclude potentially treatable causes from the standpoint of anemia, including bleeding, nutritional anemia, anemia of renal insufficiency, and hemolysis. Clues from the CBC and PBS for each of these categories are listed in. Patient history is key in implicating bleeding as a cause of anemia, and a fecal occult blood test can be ordered if indicated. Regarding nutritional anemia, it should be noted that both iron and vitamin B₁₂/folate deficiencies are possible causes of “normocytic” anemia, despite their usual association with microcytic and macrocytic anemia, respectively. Anemia of renal insufficiency is addressed easily by checking the serum creatinine level. Hemolytic anemia is usually normocytic but can be macrocytic if accompanied by marked reticulocytosis. Table 3.

Table 3. Clues from CBC and PBS in the Differential Diagnosis of Anemias

Clues from CBC and PBS in the Differential Diagnosis of Anemias			
Normocytic	Bleeding	Usually unremarkable	Polychromasia
	Nutritional anemia	Increased RDW	Anisocytosis
			Dimorphic RBCs
	Anemia of renal insufficiency	Normal RDW	Usually unremarkable
	Hemolysis	Normal/elevated RDW	Polychromasia
		Thrombocytosis	Spherocytes
			Spherocytes
			Bite cells
	Anemia of chronic disease	Normal RDW	Unremarkable
	A primary bone marrow disorder	Increased RDW	Dimorphic RBCs (MDS)
		Other cytopenia	Pseudo Pelger-Huët anomaly (MDS)
		Monocytosis	Oval macrocytes (MDS)
		Leukocytosis	Myelophthisis (MMM) [‡]
		Thrombocytosis	Rouleaux (myeloma)
		Abnormal differential	Blasts (acute leukemia)
			Presence of abnormal cells



*Diagnostic algorithm for normocytic anemia. AIHA = autoimmune hemolytic anemia; DIC = disseminated intravascular coagulation; HS = hereditary spherocytosis; PBS = peripheral blood smear; TTP/HUS = thrombotic thrombocytopenic purpura/ hemolytic uremic syndrome.

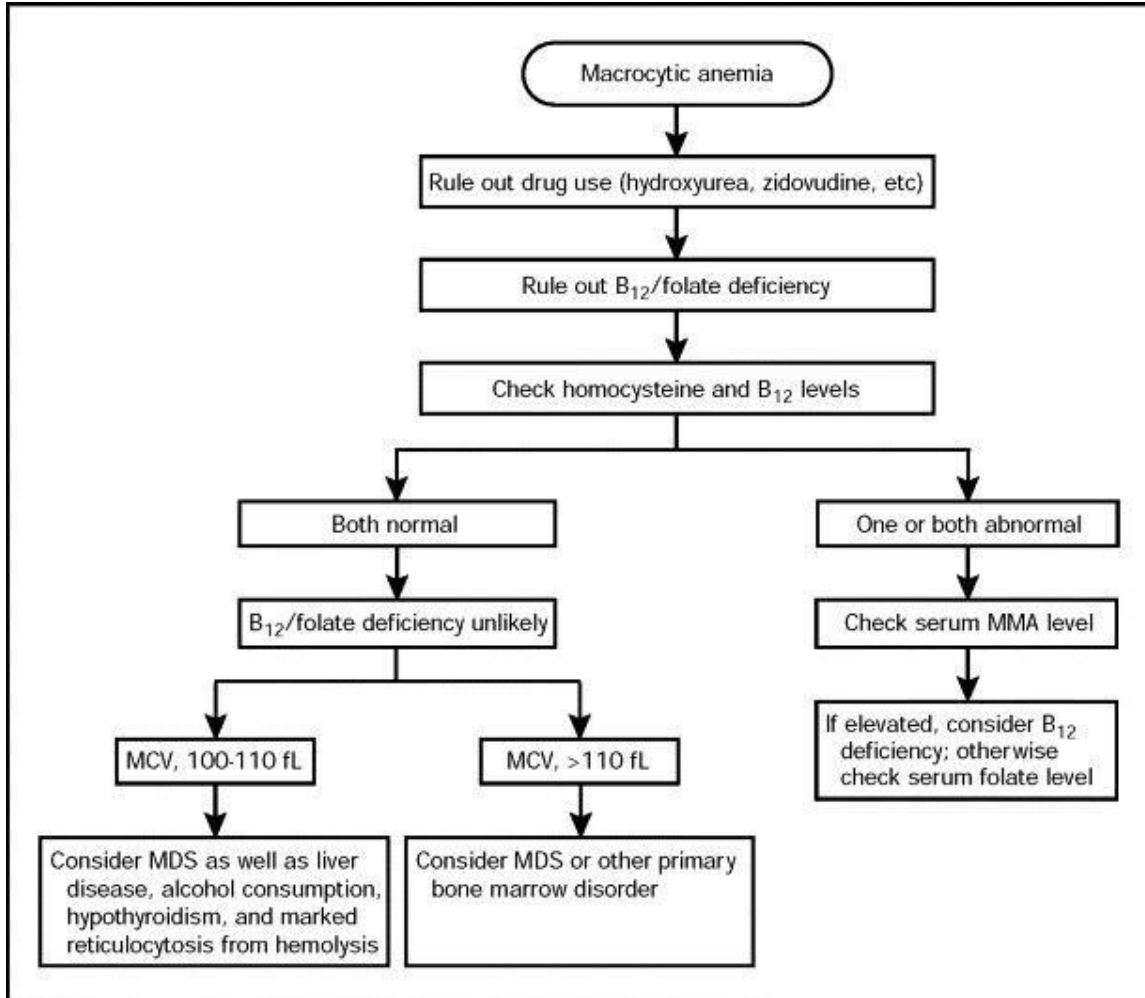
2.3. Macrocytic anemia

Use of certain drugs (eg, hydroxyurea, zidovudine) and alcohol consumption are notoriously associated with macrocytosis and should be first considerations during evaluation of macrocytic anemia. The next step is to rule out nutritional causes (B_{12} or folate deficiency); we prefer to use serum homocysteine for initial screening because of its higher test sensitivity. However, we advocate concomitant determination of the serum B_{12} level to safeguard against laboratory error in view of the dire clinical consequences associated with vitamin B_{12} deficiency. If 1 of the 2 tests has abnormal results, the serum methylmalonic acid level should be checked; an increased level strongly suggests B_{12} deficiency. Table 4.

Table 4. Clues from CBC and PBS in the Differential Diagnosis of Anemias

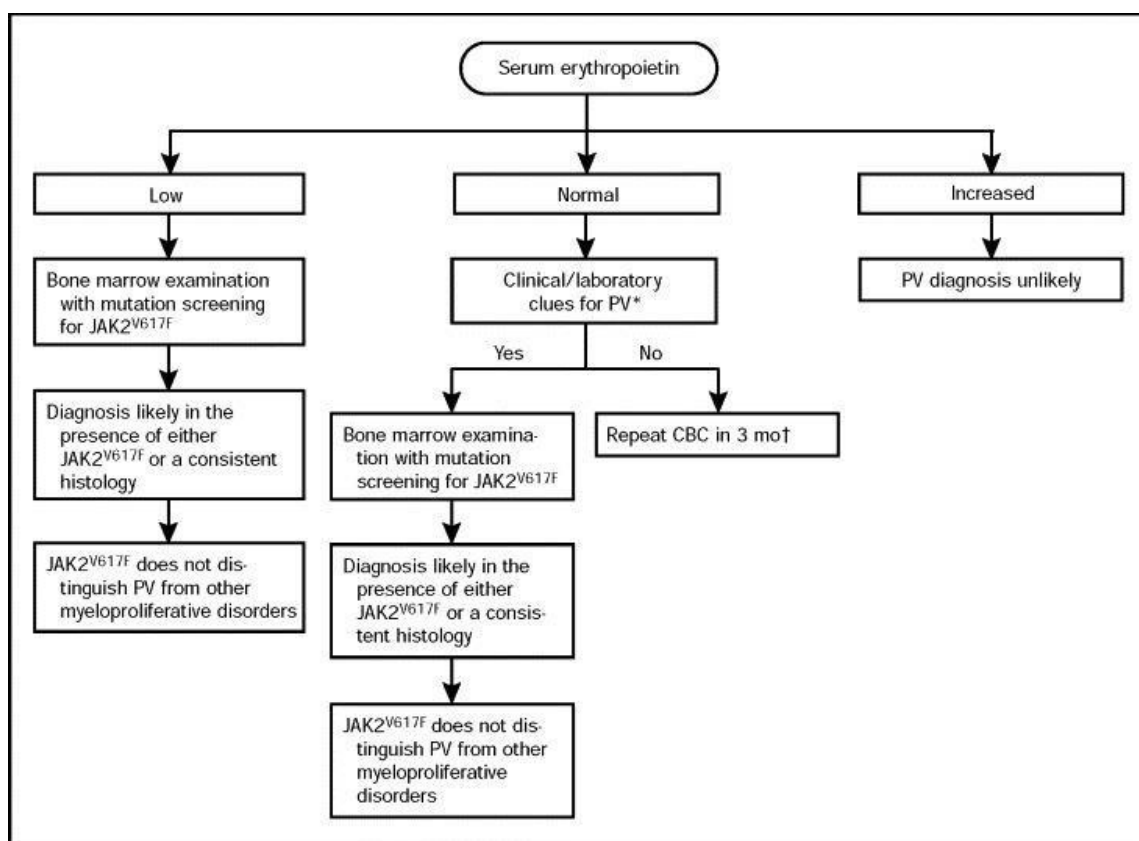
Clues from CBC and PBS in the Differential Diagnosis of Anemias			
Macrocytic	Drug-induced	Increased RDW	Oval macrocytes
		Marked or mild macrocytosis	
	Nutritional	Increased RDW	Oval macrocytes
		Marked or mild macrocytosis	Hyper segmented neutrophils
	MDS or other bone marrow disorder	Increased RDW	Dimorphic RBCs
			Pseudo Pelger-Huët anomaly cells
			Oval macrocytes
	Liver disease, alcohol use	Normal RDW	Round macrocytes
		Thrombocytopenia	Target cells
	Hypothyroidism	Normal RDW	Round macrocytes
	Hemolysis	Normal or elevated RDW	Polychromasia

*Diagnostic algorithm for normocytic anemia



III. POLYCYTHEMIA

An “increased” Hgb always raises the possibility of polycythemia vera (PV). However, many other conditions are associated with increased Hgb that indicate either a real increase in RBC mass (RCM) (true polycythemia) or a spurious perception of an increase in RCM (apparent polycythemia). True polycythemia is caused by either PV, which is a clonal myeloproliferative disorder, or a nonclonal increase in RCM that is often, but not always, driven by erythropoietin (secondary polycythemia). Therefore, PV must be distinguished from both apparent and secondary polycythemia. shows a way to accomplish this distinction without measuring RCM



IV. EUCOPENIA

4.1. utropenia

The absolute neutrophil count (ANC) is either derived by multiplying the total leukocyte count by the percentage of band neutrophils and segmented neutrophils or obtained directly from an electronic cell counter. Neutropenia is clinically most relevant when it is severe (ANC, $<0.5 \times 10^9/\text{L}$) because of the associated risk of infection. Severe neutropenia is classified into congenital and acquired categories. The most frequent cause of acquired neutropenia is drug therapy; the most commonly implicated agents are listed in. However, any drug should be assumed to be a potential offender until proved otherwise. Infection is another common cause of neutropenia, and the major culprits are viruses and sepsis.

Table 5. Drugs Frequently Implicated in Neutropenia

Drugs Frequently Implicated in Neutropenia	
Anticonvulsants	Carbamazepine, valproic acid, diphenylhydantoin
Thyroid inhibitors	Carbimazole, methimazole, propylthiouracil
Antibiotics	Penicillins, cephalosporins, sulfonamides, chloramphenicol, vancomycin, trimethoprim-sulfamethoxazole
Antipsychotics	Antipsychotics
Antiarrhythmics	Procainamide
Antirheumatics	Gold salts, hydroxychloroquine, penicillamine
Aminosalicylates	

Nonsteroidal anti-inflammatory drugs	
--------------------------------------	--

4.2. lymphopenia

The possibility of recent therapy with immunosuppressive drugs, including corticosteroids and antilymphocyte monoclonal antibodies, must be considered first in treating the patient with lymphopenia. Other causes of acquired lymphopenia, which should be familiar to the primary care physician, include viral infections such as acquired immunodeficiency syndrome and severe acute respiratory syndrome, critical illness including sepsis, autoimmune and connective tissue diseases including lupus and rheumatoid arthritis, sarcoidosis, chronic renal failure, excess alcohol use, older age, thymoma, and tuberculosis and other bacterial infections.

V. LEUKOCYTOSIS

The first step in evaluating an increased WBC count (leukocytosis) is to examine the WBC differential to determine which WBC type is in excess. The differential usually is reported along with the WBC count at no extra charge. The increase in WBCs may be secondary to either immature precursors or blasts (acute leukemia) or expansion of the aforementioned mature leukocyte types (granulocytes, lymphocytes, monocytes). Therefore, a PBS is recommended to exclude the possibility of acute leukemia and to classify the process as granulocytosis, monocytosis, or lymphocytosis. Each of these can be reactive or neoplastic (clonal).

Abbreviation: *ACD*, anemia of chronic disease; *ANC*, absolute neutrophil count; *CBC*, complete blood cell count; *CML*, chronic myeloid leukemia; *ET*, essential thrombocythemia; *FISH*, fluorescence in situ hybridization; *Hct*, hematocrit; *HES*, hypereosinophilic syndrome; *Hgb*, hemoglobin; *HIV*, human immunodeficiency virus; *IDA*, iron deficiency anemia; *ITP*, idiopathic thrombocytopenic purpura; *LDH*, lactate dehydrogenase; *LGL*, large granular lymphocyte; *MCV*, mean corpuscular volume; *MDS*, myelodysplastic syndrome; *PA*, pernicious anemia; *PBS*, peripheral blood smear; *PT*, primary thrombocytosis; *PV*, polycythemia vera; *RBC*, red blood cell; *RCM*, RBC mass; *RT*, reactive thrombocytosis; *TCR*, T-cell receptor; *TTP/HUS*, thrombotic thrombocytopenic purpura/hemolytic uremic syndrome; *WBC*, white blood cell

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WORK-UP FOR ABNORMALITY OF COMPLETE BLOOD COUNT Thrombocytosis

Dr. KAO Vannarith, Dr. SUONG Meng Hour, Dr. NOU Bona, Dr. OU Kakada, Dr. CHIN Roza, Dr. TOUCH Khannara, Dr. RATH Beata, Dr. HENG Viroath, Dr. TOUCH Socheat, Prof KUOY Samnang

I. OVERVIEWS

Thrombocytosis defined as a platelet count $\geq 450 \times 10^9/L$. The differential diagnosis for thrombocytosis is broad and the diagnostic process can be challenging. Thrombocytosis can be attributed to a reactive process or due to clonal disorder. This distinction is important as it carries implications for evaluation, prognosis, and treatment

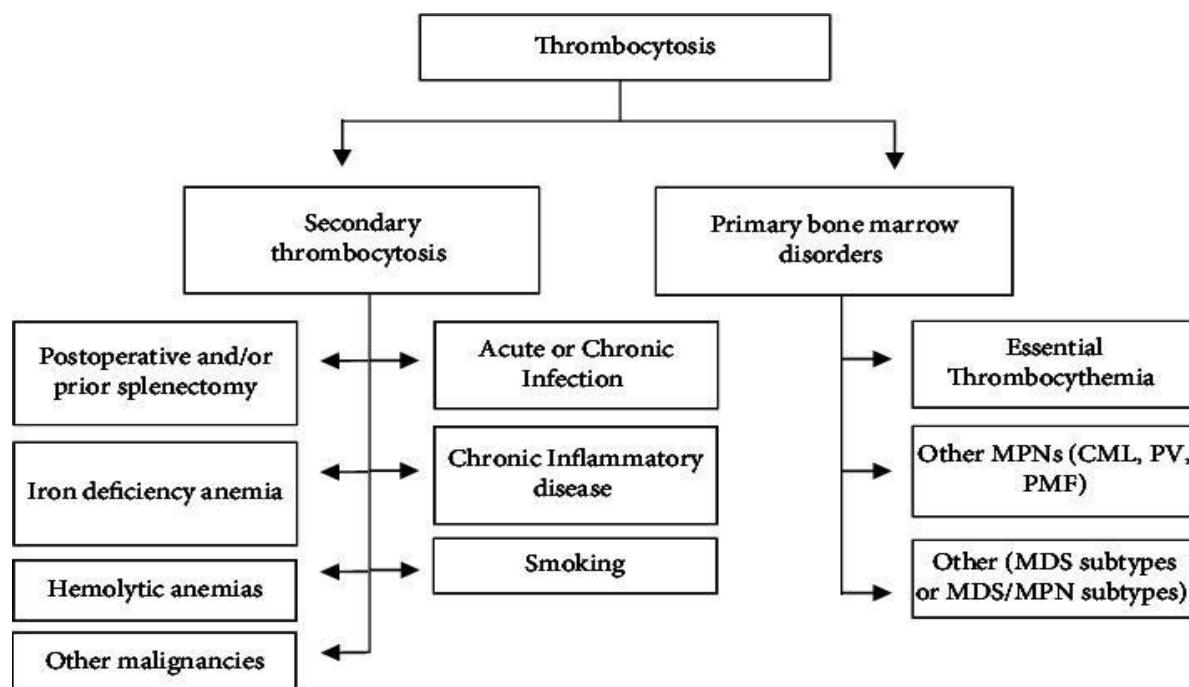
II. ETIOLOGY

1. Primary or Clonal Thrombocytosis

Once a reactive thrombocytosis is excluded and thrombocytosis is persistent, the diagnostic evaluation should turn to distinguishing between the various causes of clonal thrombocytosis. The myeloproliferative neoplasms (MPNs), comprised of essential thrombocythemia, chronic myeloid leukemia, polycythemia-vera, and primary myelofibrosis are the common clonal processes associated with thrombocytosis.

2. Secondary or Reactive Thrombocytosis

Once the diagnosis of thrombocytosis is confirmed by peripheral blood smear review, the diagnostic evaluation turns to determining whether the process is reactive or clonal in nature. An important initial step in this determination is familiarity with the underlying causes of reactive thrombocytosis. In adults, infection (typically acute), tissue damage, chronic inflammatory disorder, and malignancy, iron deficiency anemia are the most common causes of reactive thrombocytosis. In children, the most common causes of reactive thrombo-cytosis are similar, although hemolytic anemias, especially thalassemia, were a relatively common etiology.

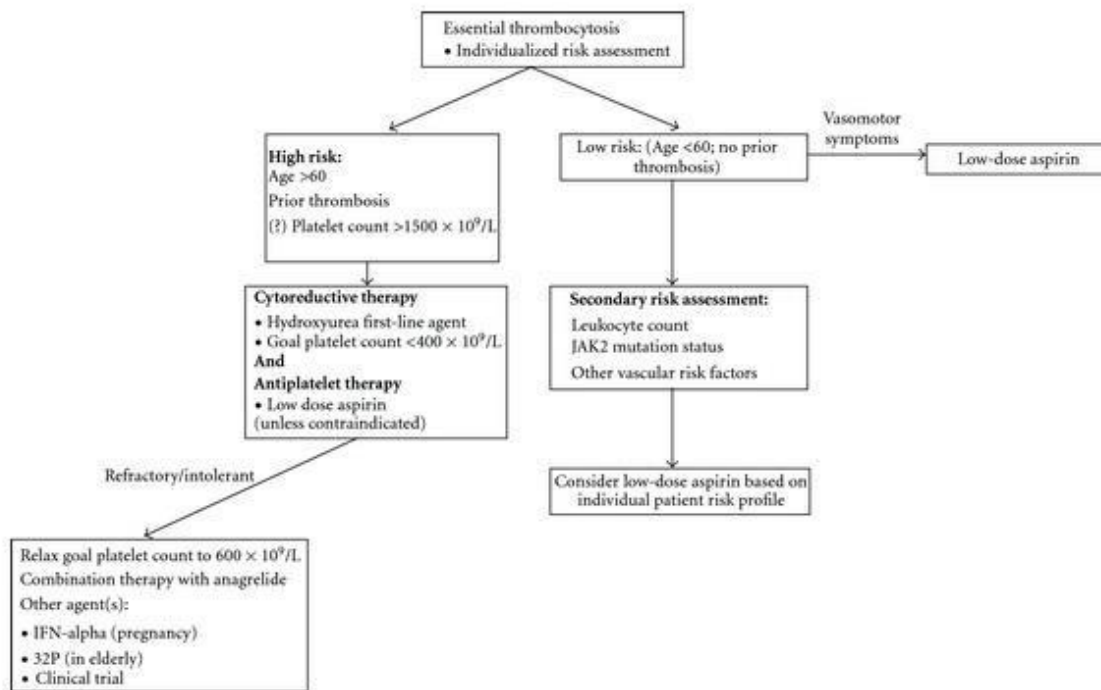


Differential diagnosis of thrombocytosis consisting of the most common causes of secondary thrombocytosis and primary bone marrow disorders. MPNs, myeloproliferative neoplasms; CML, chronic myeloid leukemia; PV, polycythemia vera; PMF, primary myelofibrosis; and MDS, myelodysplastic neoplasms.

III. MANAGEMENT

Secondary thrombocytosis is generally felt thought to be a self-limited process which resolves with resolution of the underlying disorder when possible. In primary thrombocytosis, especially in essential thrombocythemia, thrombotic complication are a major cause of morbidity and mortality and the primary factors in the determining treatment strategy.

Treatments to this end fall into essentially two categories: cytoreductive therapy with an aim to decrease circulating platelet count (as well as hematocrit in PV) and antiplatelet therapy, usually in the form of aspirin. Given these treatment approaches, much work has been done in an effort to risk-stratify patients to determine which patients will benefit from either or both of the above classes of therapy. Risk assessment and subsequent therapy based on risk of thrombosis has led to a variety of treatment strategies based on risk



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WORK-UP FOR ABNORMALITY OF COMPLETE BLOOD COUNT

Thrombocytopenia

Dr. KAO Vannarith, Dr. SUONG Meng Hour, Dr. NOU Bona, Dr. OU Kakada, Dr. CHIN Roza, Dr. TOUCH Khannara, Dr. RATH Beata, Dr. HENG Viroath, Dr. TOUCH Socheat, Prof KUOY Samnang

I. OVERVIEW

Thrombocytopenia is defined as platelet count less than 150×10^3 per μl . This defines mild thrombocytopenia as a platelet count from 100 to 150×10^3 per μl , moderate as 50 to 99×10^3 per μl , and severe when platelet counts are less than 50×10^3 per μl .

II. ETIOLOGY

1. Decreased platelet production
 - Bone marrow failure (e.g., aplastic anemia, paroxysmal nocturnal hemoglobinuria)
 - Bone marrow suppression (e.g., from medication, chemotherapy or irradiation)
 - Chronic alcohol abuse
 - Infection (e.g., cytomegalovirus, Epstein-Barr virus, Hepatitis C virus, mump, parvovirus 19, rubella, varicella-zoster virus)
 - Myelodysplastic syndrome
 - Neoplastic marrow infiltration
 - Nutritional deficiencies (vitamin B12, folate)
2. Increased platelet consumption
 - Alloimmune destruction (e.g., post transfusion, post transplantation)
 - Autoimmune syndrome (e.g., antiphospholipid syndrome, systemic lupus erythematosus, sarcoidosis)
 - Disseminated intravascular coagulations/severe sepsis
 - Drug-induced thrombocytopenia (e.g., heparin, cimetidine, vancomycin, piperacillin)
 - Immune thrombocytopenic purpura
 - Mechanical destruction (e.g., mechanical valve, extracorporeal bypass)
 - Preeclampsia/HELLP syndrome
 - Thrombotic thrombocytopenic purpura/hemolytic uremic syndrome
3. Sequestration/other
 - Dilutional thrombocytopenia (e.g., hemorrhage)
 - Gestational thrombocytopenia
 - Hypersplenism
 - Liver disease (e.g., cirrhosis, fibrosis, portal hypertension).

III. PHYSICAL EXAMINATION

The physical examination should assess for petechiae, purpura, ecchymosis, and evidence of mild bleeding (e.g., epistaxis, mucosal). All patients should be evaluated for lymphadenopathy, hepatosplenomegaly, venous thromboembolism, and vascular

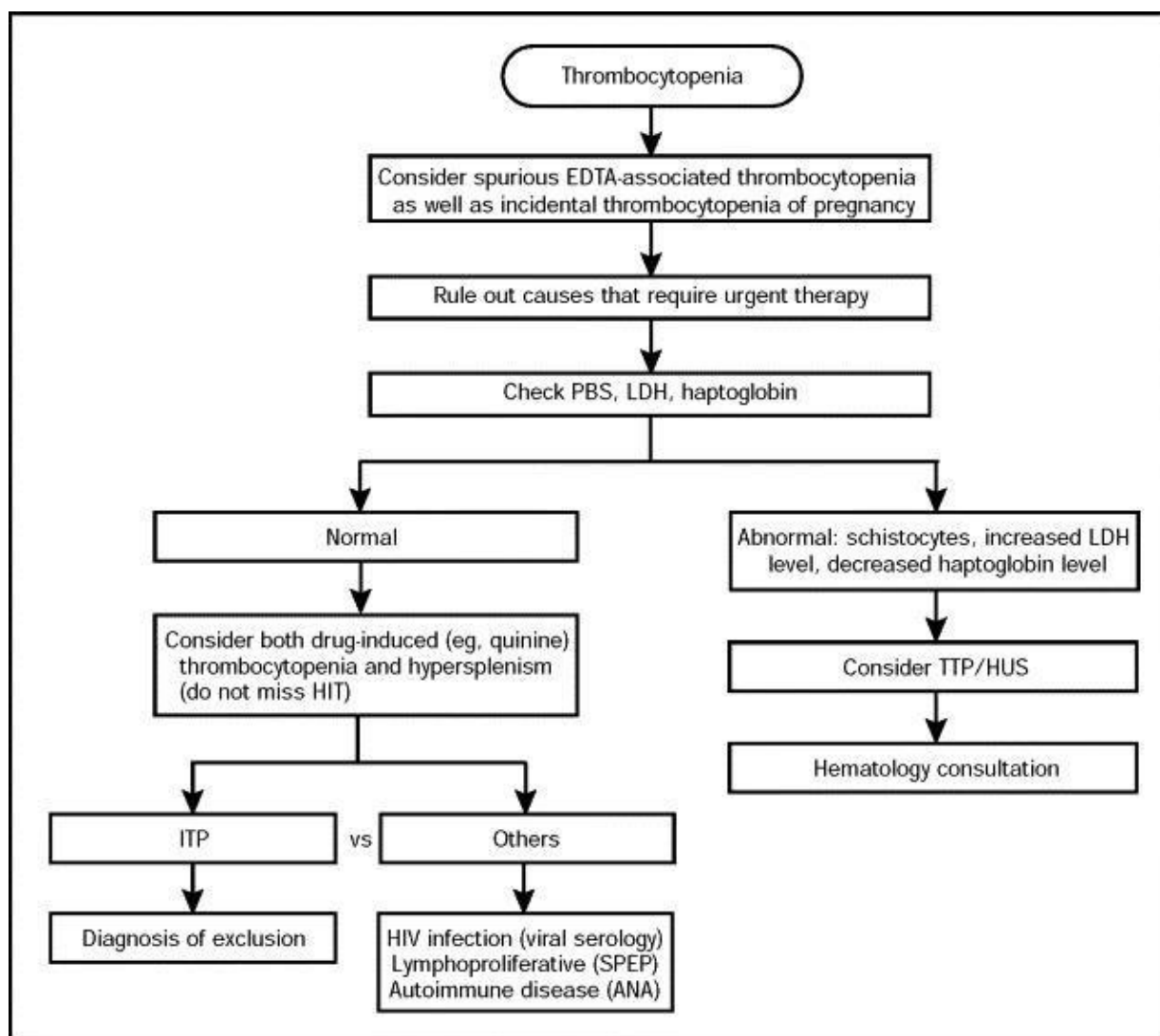
necrosis, which may suggest a systemic condition (e.g., thrombotic microangiopathies, lymphoproliferative disorder).

IV. LABORATORY TESTING

A peripheral blood smear is recommended for all patients with thrombocytopenia to evaluate the presence of fragmented red cells (schistocytes), which suggests microangiopathic hemolytic anemia.

In patients who are asymptomatic, a reasonable approach includes a disseminated intravascular coagulation panel (e.g., fibrinogen, D-dimer, coagulation factors); hemolysis screen (i.e., direct antiglobulin test, haptoglobin, and lactate dehydrogenase); renal and hepatic function tests; selected viral serologies (e.g., hepatitis, HIV); *Helicobacter pylori* stool antigen testing, fecal occult blood test, vitamin B₁₂, and folate levels. Additional testing for autoimmune syndromes, antiplatelet antibodies, and thyroid function testing should be considered when clinically indicated.

V. DIAGNOSIS AND MANAGEMENT



Diagnostic approach to thrombocytopenia. ANA = antinuclear antibody; DIC = disseminated intravascular coagulation; HIT = heparin-induced thrombocytopenia; HIV = human immunodeficiency virus; ITP = idiopathic thrombocytopenic purpura; LDH = lactate dehydrogenase; PBS = peripheral blood smear; SPEP = serum protein electrophoresis; TTP/HUS = thrombotic thrombocytopenic purpura/hemolytic uremic syndrome.

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IMMUNE THROMBOCYTOPENIA

Dr. KAO Vannarith, Dr. RATH Beauta, Dr. OU Kakada, Dr. TOUCH Khannara

I. DEFINITION

An autoimmune disorder with isolated thrombocytopenia: platelet count $< 100 \times 10^9/L$ and absence of other causes or disorders that may be associated with thrombocytopenia. It increased risk of bleeding (very heterogeneous)

II. EPIDEMIOLOGY

- Female 3F/1M. Young age (25-35 y-o)
- USA: Incidence: 2-5/100,000 person a years. Prevalence: 9.5 to 23.6 per 100,000 persons
- France: 4000 new adult cases vs 400 new child cases a year
- South Korea: incidence 5.3/100,000 person per year. The overall incidence rate ratios of children under 15years old to adults and females to males were 3.8 (95% CI: 3.7-3.9) and 1.3 (95% CI: 1.2-1.4), respectively.

III. ETIOLOGY AND DIFFERENTIAL DIAGNOSIS

- Can be primary: 80%, secondary: 20%
- Primary by immunology reaction
- Secondary of further etiologies (to eliminate)
 - o Bleeding by certain pathology
 - o Drug induce: vaccination, Heparin, NSAIDs, Aspirin, Antibiotic (Meropenem, Bactrim, chloramphenicol...), PPIs, Anti-convulsion drug...
 - o Dysthyroidism
 - o Infection: Sepsis, Hepatitis, HIV, EBV, CMV, Parvovirus, various viral infection, H-pylori, Malaria...
 - o Hypersplenism (cirrhosis, splenomegaly)
 - o Other Immunological Diseases: Systemic lupus erythematosus, Antiphospholipid antibody syndrome, Evans syndrome, Microangiopathy Thrombotic, Thrombotic Thrombocytopenic Purpura, Disseminated intravascular coagulation (DIC), TTP, HELLP syndrome,
 - o Bone marrow problem: Folate/B12 deficiency, Myelodysplasia syndrome, Leukemia, Lymphoma, Multiple myeloma, Cancer invading to bone marrow

IV. ATHOPHYSIOLOGY (SUMMARY)

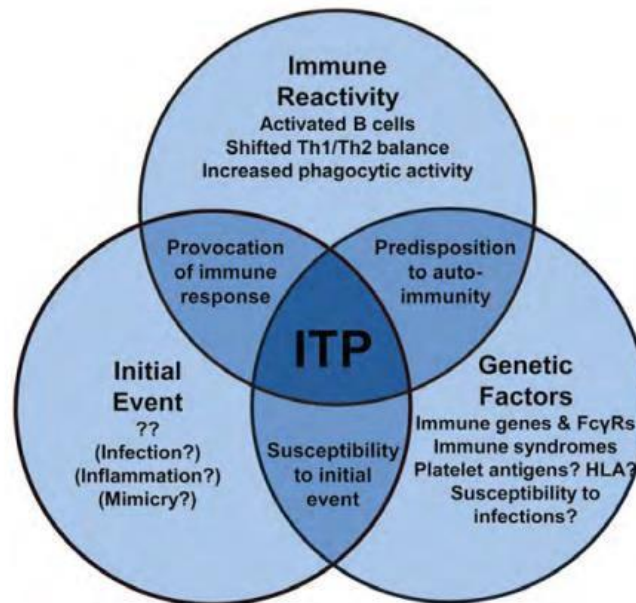


Figure 1. Model of relationship of contributing factors in ITP.

(Johnsen J. Pathogenesis in immune thrombocytopenia: new insights. Hematology Am Soc Hematol Educ Program. 2012;2012:306-12).

V. DIAGNOSIS

1. Clinic

- Sudden or insidious onset. Advanced age and prior bleeding are established independent risk factors
- Fatigue (very common)
- No spontaneous bleeding manifestation, if Platelet > 50k/mm³. Normal size spleen +++
- Skin: Petechial purpura, ecchymosis purpura (does not fade with pressure), need to eliminate: fulminant purpura (Meningococcal), vascular purpura (vasculitis)
- Vibice: linear streak, mostly at bending fold area
- Mucous membrane bleeding: gum bleeding, oral blood blisters, oral cavity bleeding, epistaxis, menorrhagia,
- Organs bleeding: hematuria, melena, Intracranial hemorrhage (ICH) :most concern when platelet < 20k/mm³ (RR: 3.2)

2. Investigation

- Thrombocytopenia = platelet < normal range
- Fake thrombocytopenia? = Platelet clumping
 - ⇒ Repeat platelet count +/- EDTA citrate tube +/- blood smear
- Standard blood test for ITP:

⇒ Blood group, CBC, blood smear,

- ⇒ PT, ACT, fibrinogen,
- ⇒ ALT, AST, BUN, Creatinine, CRP, VS
- ⇒ HIV, serology of Hepatitis B and C,
- ⇒ ANA, RAI, Coombs test
- ⇒ Abdominal ultrasound
- Bone marrow analysis (Myelogram): Optional but Essential when:
 - ⇒ Age > 60 years old
 - ⇒ Abnormal of WBC, Normocytic normochromic anemia, abnormal blood smear
 - ⇒ Splenomegaly, lymph nodes,
 - ⇒ Lack of clear response to corticosteroid or to IVIg ttt
 - ⇒ Before planning to do splenectomy
- If thrombocytopenia + various clinical syndrome
 - ⇒ Have to think about others differential disease (see disease in paragraph III above)
 - ⇒ Required laboratory test accordingly
- ITP = elimination diagnosis = various etiologies provoking thrombocytopenia

Annexe 3a. Score hémorragique utilisable chez l'adulte pour guider la prescription d'immunoglobulines intraveineuses*

(d'après Khellaf et al, Haematologica 2005; 90 : 829-32).

Age		Saignement gastrointestinal	
Age > 65 ans	2	Saignement digestif sans anémie	4
Age > 75 ans	5	Saignement digestif avec anémie (perte de plus de 2 g d'hémoglobine) et/ou choc	15
Saignement cutané		Saignement urinaire	
Purpura pétiéchal localisé (membres)	1	Hématurie macroscopique sans anémie	4
Purpura ecchymotique	2	Hématurie macroscopique avec anémie aiguë	10
Purpura pétiéchal avec localisations multiples	3	Saignement du système nerveux central (SNC)	
Purpura pétiéchal généralisé	3	Saignement du SNC ou saignement avec mise en jeu du pronostic vital	15
Purpura ecchymotique généralisé	4		
Saignements muqueux			
Epistaxis unilatérale	2		
Epistaxis bilatérale	3		
Bulles hémorragiques spontanées ou gingivorragies spontanées	5		

* Pour chaque rubrique, seul le score le plus élevé est pris en compte. Un traitement par IgIV associées aux corticoïdes est proposé pour les patients ayant un score hémorragique supérieur à 8. En l'absence de contre-indication, les corticoïdes sont proposés en monothérapie en première intention en cas de score hémorragique ≤ 8.

VI. ANAGEMENT

1. Objectives

- Do not treat platelet rate, but based on patient history, comorbidities, disease stage, treatment side effects, age, concomitant medications, and patient preference.
- The goal of treatment is to achieve normal hemostasis, not to reach a normal platelet count. Most of the cases, the safety platelet rate $> 50\text{k/mm}^3$

2. When to treat? IN or OUT?

- Platelet $> 30\text{k/mm}^3$: “Watch-and-Wait strategy” and educate the patient:
 - Bleeding sign,
 - Control CBC monthly for 3 months then every 6 months for 1 year,
 - Control CBC, if have symptom
- Platelet $< 30\text{k/mm}^3$, Start treatment
 - $20 > \text{platelet} < 30\text{k/mm}^3$ +/- minor symptom: Start treatment as Outpatient
 - Platelet $< 20\text{k/mm}^3$ without symptom: start treatment as outpatient
 - Platelet $< 20\text{k/mm}^3$ with symptom: start treatment as inpatient
- No/Mild bleeding + Platelets $> 30\text{k/mm}^3$:
- Unsupportable fatigue and assessment of health-related quality of life
- Medical procedure planning: Start at least 1 week prior the procedure
 - Invasive procedure (endoscopy, tooth treatment, biopsy...) $\Rightarrow$ Platelet $> 50\text{k/mm}^3$
 - Small to moderate surgery $\Rightarrow$ Platelet $> 80\text{k/mm}^3$
 - Large surgery or life threatening surgery (brain surgery, poly trauma, heavy bleeding surgery...) $\Rightarrow$ Platelet $> 120\text{k/mm}^3$. (MDT advice must be considered)

3. First Line therapy

- **Score bleeding $< 8 \Rightarrow$ Steroid**
Corticosteroids:
 - $\Rightarrow$ Adults:
 - Prednisone 0.5 to 2.0 mg/kg/day, QD
 - Or Dexamethasone 40 mg/day IV, QD for 4 days
 - $\Rightarrow$ Children:
 - Prednisone 2 to 4 mg/kg, QD; maximum, 120 mg QD, for 5-7 days or
 - Dexamethasone 0.6 mg/kg, QD; maximum 40 mg, QD, for 4 days
- **Score bleeding $> 8 \Rightarrow$ MDT for IVIg decision is a “MUST”**
Intravenous Immunoglobulin (IVIg): Platelet recovery within 24-48h
 - 0.8-1.0g/kg IV, QD for 1-2 days
 - Or up to 2g/kg (max 100g/dose)

4. Second Line therapy

- TPO-RAs: 25-50mg/Day for 3-6 months
- Rituximab: 375mg/m², weekly for 4 weeks
- Splenectomy

5. Treatment of emergency bleeding in ITP

The life-threatening bleeding condition requires emergency treatment.

MDT advice is a “MUST” for the following management below:

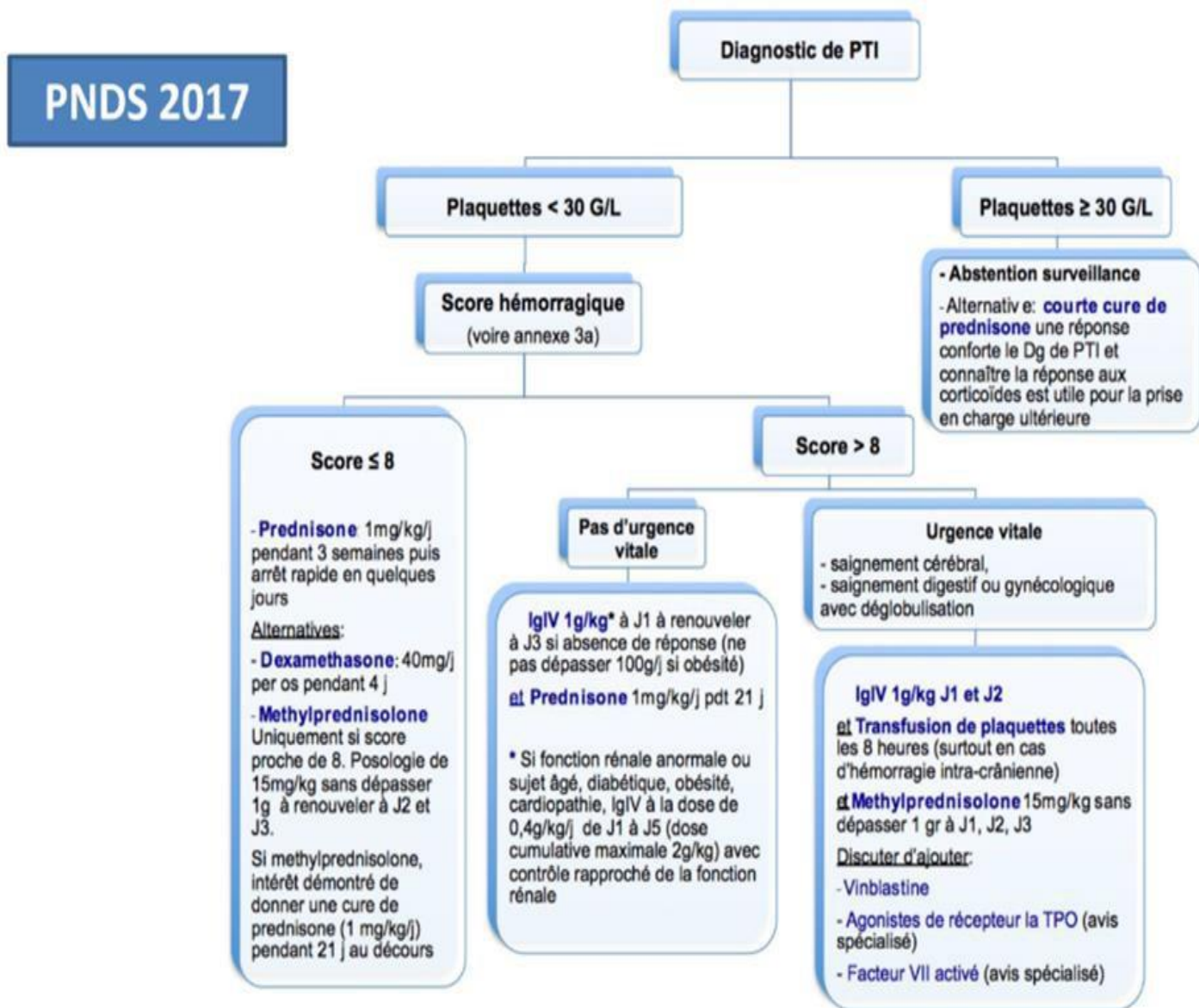
Platelet transfusions to achieve hemostasis,

IVIg 1–2g/kg, QD, for 1-2 days

And Steroids, Methylprednisolone 15mg/kg (max 1000mg) IV, QD, for 1–3 days

Or High dose Dexamethasone 40 mg, QD, IV/PO, for 4 days

6. Algorithm



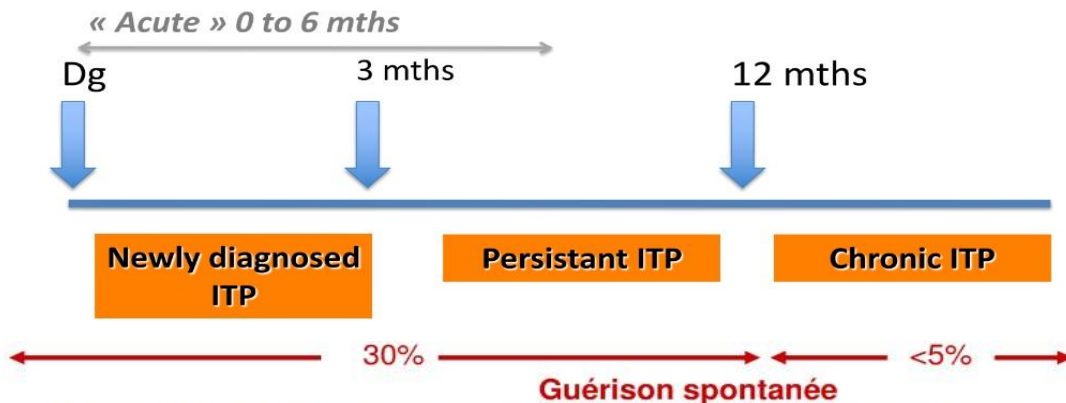
VII. VOLUTION AND PROGNOSIS

Standardization of terminology, definitions and outcome criteria in immune thrombocytopenic purpura of adults and children: report from an international working group

F Rodeghiero et al

BLOOD, 12 MARCH 2009; 113, 2386-93

ITP duration



- PTI (ou ITP) = Thrombopénie Immune ou Immune ThrombocytoPenia
Thrombopénie < 100G/L
Thrombopénie Immune **primaire** = isolée
Thrombopénie Immune **secondaire** = ex PTI-LLC
- PTI **réfractaire** = échec de splénectomie
- **Rémission complète** : Plaquettes > 100G/L, **partielle** > 30G/L + 2 x base

Rodeghiero F et al, Blood 2009;113:2386-93.

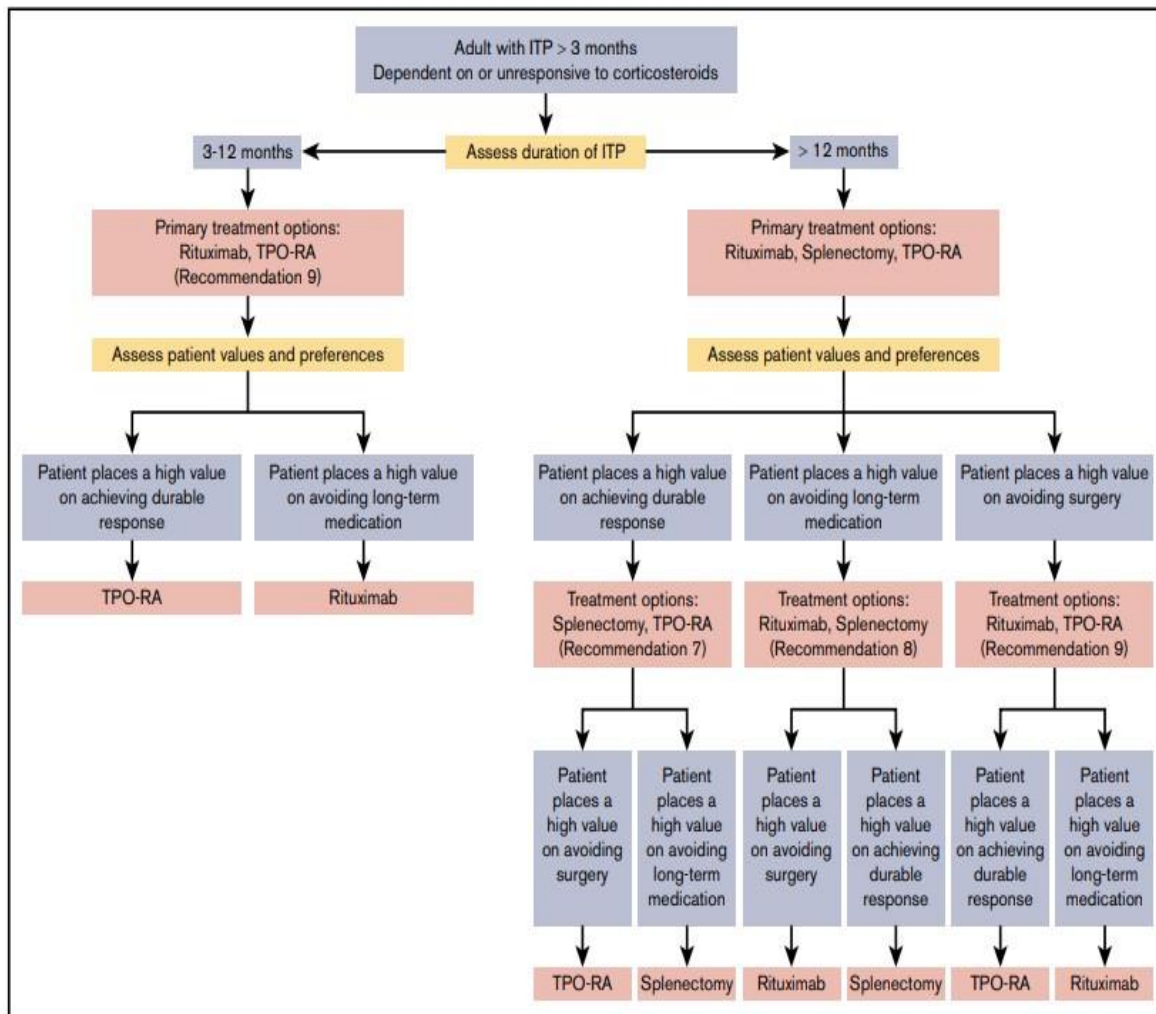


Figure 1. Algorithm for the selection of second-line therapy in adults with ITP. Selection of second-line therapy in adults with ITP should be individualized based on duration of disease and patient values and preferences. Other factors that may influence treatment decisions include frequency of bleeding sufficient to require hospitalization or rescue medication, comorbidities, compliance, medical and social support networks, cost, and availability of treatments. Patient education and shared decision-making is encouraged. Patient characteristics are shown in blue boxes, actions in yellow boxes, and treatment options in red boxes. Numbered recommendations corresponding to each treatment option are provided.

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APLASTIC ANEMIA

Dr. OU Kakada, Dr. RATH Beauta, Dr. KAO Vannarith, Dr. TOUCH Khannara

I. DEFINITION

Aplastic Anemia (AA) is a bone marrow failure (BMF) disorder characterized by pancytopenia with bone marrow hypoplasia/aplasia due to loss of hematopoietic stem cells (HSCs).

II. EPIDERMIOLOGY

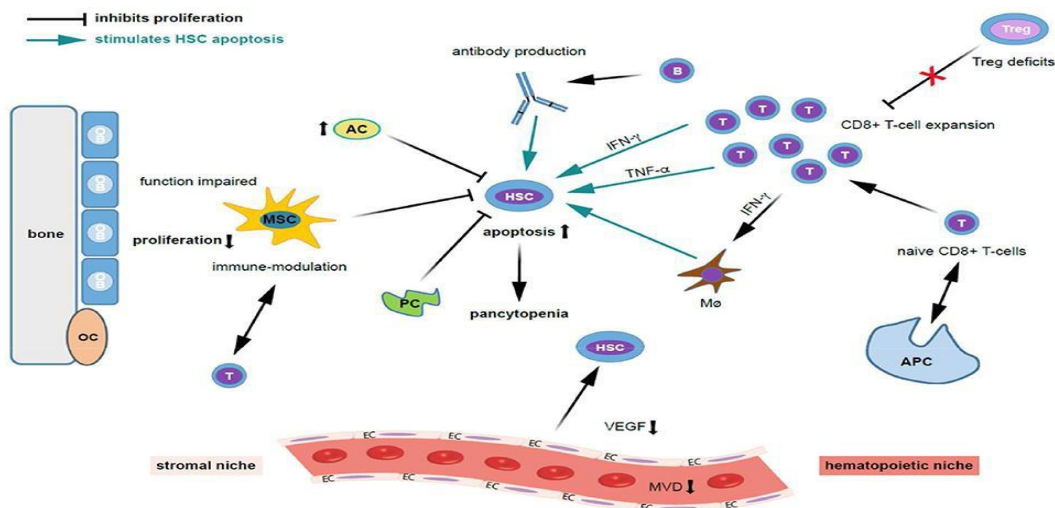
- Ratio 3 Female/1 Male. Young age (15-29 y-o) and above 60 y-o
- USA and Europe: Incidence: 2-3 per million person per years, its incidence relatively high elsewhere, particularly in Asia. The incidence rate was reported to be 3-5 per million in Thailand.
- South Korea: incidence 5.3 per million person per year.

III. ETIOLOGY

- Can be idiopathic: 90%, acquire/secondary: 10%
- Idiopathic by autoimmune destruction of hematopoietic stem cells (HSCs).
- Secondary of further etiologies (to eliminate)
 - o Toxins (Benzene, Solvents, Glue vapor ...)
 - o Drug: (Chemotherapy, carbamazepine, phenytoin...)
 - o Viral infection (EBV, Hepatitis, HIV, Other herpes virus...)
 - o Ionizing radiation (dose dependent bone marrow suppressive effect).

IV. PATHOPHYSIOLOGY (SUMMARY)

In most cases, aplastic anemia behaves as an immune-mediated disease. Cellular and molecular pathways have been mapped in some detail for both effector (T lymphocyte) and target (hematopoietic stem and progenitor) cells.



V. DIAGNOSIS

- Clinic:

- Recurrent infection (Low WBC)
- Anemia/fatigue (Low Hb)
- Bleeding tendency (Gum bleeding, petechiae, ecchymosis, GI and CNS bleeding)

- Investigation:

- Blood test:
 - ⇒ Blood group, CBC, reticulocyte, blood smear
 - ⇒ Coagulation panel (PT, aPTT, fibrinogen...)
 - ⇒ LFT (ALT, AST), Creatinine, BUN, LDH, Bilirubin
 - ⇒ Iron profiles (Ferritin)
 - ⇒ Serology (HIV, EBV, HBV, HCV ...)
 - ⇒ ANA, RAI, Coombs test
 - ⇒ Abdominal ultrasound
- Bone marrow examination: Bone marrow aspiration and biopsy is required to establish the diagnosis of AA and exclude other causes of pancytopenia.
Diagnostic findings from bone marrow examination include:
 - ⇒ Hypocellular with a decrease in all elements and composed mostly of fat cells.
 - ⇒ Hematopoietic cells are morphologically normal.
 - ⇒ Infiltration of the bone marrow with malignant cells or fibrosis is not present.
- Definition of aplastic anemia with severity criteria— AA is defined as pancytopenia with a hypocellular bone marrow in the absence of an abnormal infiltrate or marrow fibrosis.

Classification	Criteria
Severe	BM cellularity < 25% (or < 50% if < 30% of BM is hematopoietic cells) AND ≥ 2 of the following: • Peripheral blood neutrophil count < $0.5 \times 10^9/L$ • Peripheral blood platelet count < $20 \times 10^9/L$ • Peripheral blood reticulocyte count < $20 \times 10^9/L$
Very severe	As above, but peripheral blood neutrophil count must be < $0.2 \times 10^9/L$
Nonsevere	Hypocellular BM with peripheral blood values not meeting criteria for severe aplastic anemia

VI. MANAGEMENT

1. Objectives

- Decrease mortality while patients with severe complication.
- The goal of treatment is to achieve enough function of bone marrow (Prevent infection, tolerate anemia, No bleeding)

2. Critical Signs

Emergencies associated with pancytopenia

- Neutropenia (new diagnosis or associated with fever/infection)
- Symptomatic anemia (eg, cardiac ischemia, hemodynamic instability, worsening congestive heart failure)
- Thrombocytopenia (platelets <10,000/microL, or <50,000/microL associated with bleeding)
- Disseminated intravascular coagulation
- Abnormal peripheral blood smear (eg, microangiopathy, blasts)
- Severe aplastic anemia
- Hemophagocytic lymphohistiocytosis
- Metabolic emergencies (eg, symptomatic hypercalcemia, hyperkalemia, tumor lysis syndrome)

3. Mortality risk factor:

- Large Surgery
- Sepsis/DIC
- Ischemic heart disease
- Polytrauma

4. Outpatient management

- Oral immunosuppressive therapy
- Follow up blood result, if necessary for transfusion, antibiotic therapy

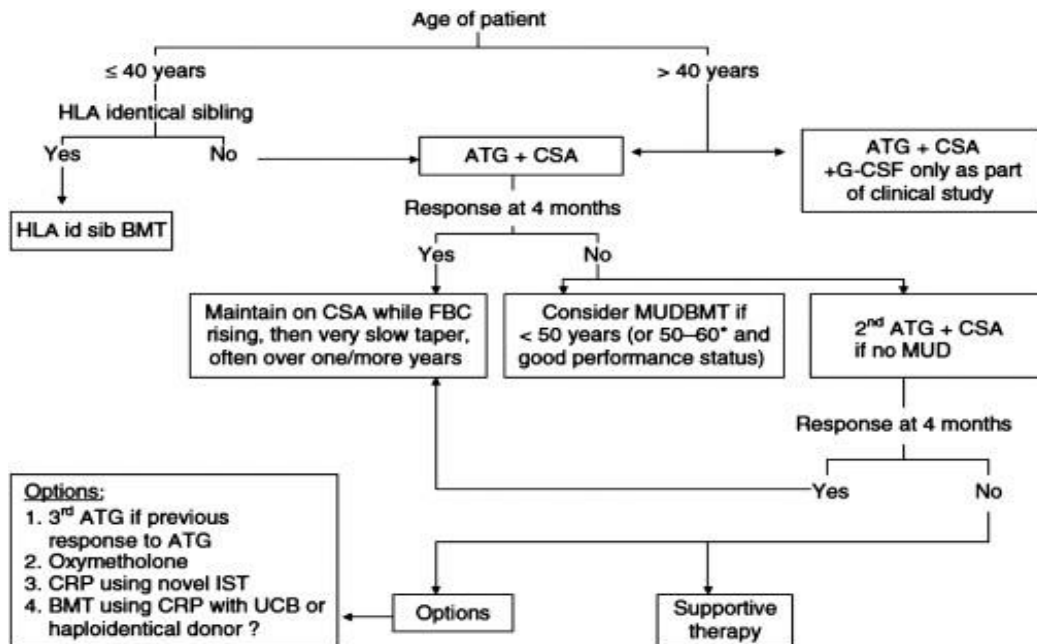
5. Treatment First line

- Double immunosuppressive therapy- CsA, EPAG
- Lower-intensity treatments-EPAG alone
- Triple Immunosuppressive therapy – hATG, CsA, EPAG (if hATG available)
- Hematopoietic Stem Cells Transplantation (if available)
- Transfusion supportive care

6. Treatment Second line

- Androgen
- Chemotherapy
- GCF-S
- EPO

7. Algorithm



VIII. SPECIAL CASES

- Patient with very severe aplastic anemia associate sepsis or DIC should be receive large spectrum of antibiotic first.
- Patient with pancytopenia associated with hemolysis, considered as life threatening condition must transferred to referral hospital with hematology unit available.

IX. VOLUTION AND PROGNOSIS

- The overall prognosis of patients with severe AA (SAA) has improved with advances in supportive care, immunosuppressive therapy (IST) and triple IST, and hematopoietic cell transplantation (HCT). Current 5- or 10-year survival rates are as high as 80 to 90 percent, compared with 10 to 20 percent in the 1960s.
- Evolution, about 5–15% of cases with acquired AA may be susceptible to myelodysplastic syndrome (MDS) and leukemia.

X. COMPLICATION

Most common complications:

- Systemic infection
- Multiple transfusion increase ferritin level (organs toxic, liver, thyroid gland, heart..)
- GI and CNS Bleeding

XI. PREVENTION AND RECOMMENDATION

- There's no prevention for most cases of aplastic anemia.
 - Avoiding exposure to insecticides, herbicides, organic solvents, paint removers and other toxic chemicals might lower your risk of the disease.
 - Annual checkup CBC if fatigue, easy bruising, recurrent infection.

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HODGKIN LYMPHOMA

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I. DEFINITION

Hodgkin lymphoma (HL) is a malignancy which in the vast majority of cases is derived from germinal centre B-lymphocytes, being of T-cell origin.

II. EPIDEMIOLOGY

In 2021, approximately 8830 patients are expected to be diagnosed with HL in the United States, and 960 patients are expected to die from this malignancy. The disease has a bimodal age distribution with one peak in the early 20s and the second in the mid-70s. there is a slight male predominance (male: female incidence ratio of 1:3).

Nodular sclerosis (NS) classic Hodgkin lymphoma: Males and females are affected in equal proportion, and, at diagnosis, most patients are between the ages of 15 and 35 years.

Mixed-cellularity cHL is the second most common subtype, representing 20% of cHL. There is a male predominance.

III. ETIOLOGY

The risk factor of Hodgkin lymphoma:

- Epstein-Barr virus infection
- Primary and secondary immune deficiencies
- Autoimmune and allergic disorder
- Environnemental expose : ultraviolet light, Tobacco, alcohol.

IV. PATHOPHYSIOLOGY

HL is monoclonal lymphoid neoplasm derived from B cells in most cases and is divided into 2 distinct entities:

- Classical Hodgkin lymphoma (cHL): 95% of cases
- Nodular lymphocyte-predominant Hodgkin lymphoma (NLPHL).

The malignant cell in cHL, the Hodgkin Reed-Sternberg (HRS) cell, is a large, bilobed cell with 2 or more nuclei with eosinophilic nucleoli. HRS cells are derived from germinal center B lymphocytes but lack a B-cell receptor and several B-cell associated genes and proteins. In cHL, the HRS cells express CD30 and CD15.

Within cHL, there are 4 histologic subtypes:

- Nodular sclerosis (NS) Hodgkin lymphoma
- Mixed cellularity (MC) Hodgkin lymphoma
- Lymphocytic rich (LR) Hodgkin lymphoma
- Lymphocytic depleted (LD) Hodgkin lymphoma.

Nodular lymphocyte-predominant Hodgkin lymphoma (NLPHL) is morphologically and immunophenotypically distinct from cHL. NLPHL is derived from antigen-selected B cells and express typical germinal center B-marker including BCL-6. The cells are typically CD30- and CD15-negative, and positive for CD20+, CD79a+, PAX-5 and OCT-2.

V. DIAGNOSIS

1. Clinic

Patients with cHL often present with non-tender lymphadenopathy. The neck is the most commonly involved site of disease.

- B symptoms, defined as fever $> 38.0^{\circ}\text{C}$, drenching night sweats, and involuntary weight loss of $> 10\%$ of body weight in the preceding 6 months
- Pruritus is seen in 10% to 15% of patients
- Although it occurs rarely ($<5\%$ of cases), patients may experience intense pain in the sites of disease upon alcohol ingestion.
- Mediastinal involvement, which may be bulky, is more common in NS cHL, and patients may present with respiratory symptoms.
- LD cHL is the least common subtype, median age of onset is in the 30s, and male are more often affected. Extranodal and intra-abdominal disease, advanced-stage disease, and systemic symptoms are common.

2. Investigation

- Biopsy lymph node

Adequate tissue biopsy $\Leftrightarrow$ Is critical



Fine needle aspirate is not adequate $\Leftrightarrow$ incisional or excisional biopsy is preferred

- ☐ Cervical-thorax and abdominopelvic CT scan.
- ☐ [^{18}F] fluorodeoxyglucose (FDG-18) positron emission tomography (PET), computerized tomography scanning (CT), in the Central (or tertiary) hospital.
- ☐ Bone marrow biopsies done by the specialized haematologist.
- Biological test
 - Initial laboratory assessment should include:
 - ☐ Complete blood count (CBC)
 - ☐ Renal and hepatic function, including albumin,
 - ☐ Erythrocyte sedimentation rate (ESR): elevated, prognostic in early-stage disease.
 - ☐ Viral serology: Hepatitis B, C, HIV.
 - ☐ Serology of syphilis
 - ☐ Lactate dehydrogenase (elevated in extensive stage, advance-stage disease)
 - ☐ Protein electrophoresis (EPP)
 - ☐ Thyroid function: T3, T4 and TSH.
- Other assessment/pre-therapeutic assessment
 - ☐ Pulmonary function testing
 - ☐ Assessment of cardiac function
 - ☐ Dental panorama
 - ☐ Pregnancy test

3. Staging and workup

Staging should be performed with CT scan. Ann Arbor staging classification should be used for disease localization.

Table 1. Staging of Hodgkin lymphoma/ Ann-Arbor classification

Staging of Hodgkin lymphoma/ Ann-Arbor classification	
Stage I	Involvement of one lymph node region.
Stage II	Involvement of 2 or more lymph node regions – or lymph node structures on the same side of the diaphragm.
Stage III	Involvement of lymph node regions or lymphoid structures on both sides of the diaphragm.
Stage IV	Diffuse or disseminated involvement of one or more extranodal organs or Tissue beyond that designated E, with or without associated lymph node involvement.
All cases are subclassified to indicate the absence (A) or presence (B) of B symptoms	
Designation “E”	Extranodal contiguous extension (ie, proximal or contiguous extranodal disease) that can be encompassed within an irradiation field appropriate for nodal disease of the same anatomic extent. More extensive extranodal disease is designated stage IV.
Subscript “X”	Is used if bulky disease is present. A mediastinal mass with a maximum width that is equal to or greater than 1/3 of the internal transverse diameter of thorax, and > 10 cm maximum dimension of a nodal mass.

4. Risk factor

It has got many organizations use different criteria in clinical trials to risk stratify patients. The German Hodgkin Study Group (GHSG) scale includes 5 risk factors:

- Bulky mediastinal disease: defined by more than 1/3 of the maximal intrathoracic cavity
- ESR of 30 in the presence of B symptoms or
- ESR of more than 50 without B symptoms
- Extra-nodal extension of disease
- 3 or more lymph node sites of involvement.

GHSG criteria + Ann-Arbor classification ➔ Risk factor for Hodgkin lymphoma



- Early favourable disease
- Early unfavourable disease
 - Intermediate disease
 - Advanced disease

	IA, IB, IIA	IIB	IIIA/B	IVA/B
Absence of risk factors	Early favourable			
≥ 3 lymph node sites	Early Intermediate		Advanced stage	
ESR > 50	unfavourable			
Bulky mediastinal	Early			
disease	unfavourable			
Extra-nodal extension				

II. DIFFERENTIAL DIAGNOSIS

Other cause of lymphadenopathy Non-Hodgkin lymphoma

III. MANAGEMENT

1. Objectives

- Approach the complete tumour response and gain the best quality of life
- Palliative care in case of relapse or advance stage.

2. Critical sign

- Tumor lysis syndrome (TLS): read in Non-Hodgkin Lymphoma (2.1)
- Superior Vena Cava syndrome: read in Non-Hodgkin Lymphoma (2.2)

3. Principle of treatment

- Multi Disciplinary Team (MDT): haematologist, radiation oncologist, nuclear medicine oncologist, Radiologist – Pathologist

Chemotherapy + Radiation therapy



Chemotherapy administered under tight supervision by professionals in the cancer care unit

4. Regimen of chemotherapy

- Standard regimen: ABVD (Doxorubicine, Bleomycine, Vinblastine, Dacarbazine)
- Others regimen need to be confirmed by MDT decision

5. Radiation therapy

Involved site radiation therapy (ISRT): 20 to 30 Gy.

IV. COMPLICATIONS

Complication related to radiation therapy:

- Cervical irradiation => Hypothyroidism: pic 3-5 years

- Mediastinal irradiation => Coronaropathy (increased risk in 5-10 years), Breast cancer, lung cancer, pulmonary fibrosis
- Abdominal irradiation => early menopause

V. PREVENTION AND RECOMMENDATION

There is not clear about the cause enhanced Hodgkin lymphoma. That why the way to prevent this disease is not published. By the way, the prevention of all the above risk factors, could minimize the risk of Hodgkin lymphoma. EBV infection is the most common risk factor.

Secondary prevention for the already diagnosis patients:

- Lifestyle modification: smoking cessation, no alcohol consumption
- Healthy diet
- Exercise

All the management of Hodgkin lymphoma need the Multidisciplinary Team discussion.

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NON-HODGKIN LYMPHOMA

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I. DEFINITION

Non-Hodgkin Lymphoma (NHL) are the malignant monoclonal proliferations of lymphoid cells of variable degree of maturation, with a peripheral point of origin.

The entire lymphoid system can be affected simultaneously or secondary: lymph nodes, spleen, tonsils, digestive tract, lung, skin ...etc.

II. EPIDEMIOLOGY

Unfortunately, there is limited data available specifically on the epidemiology of Non-Hodgkin Lymphoma in Cambodia.

Globocan 2022 estimates that in 2020 was listed in the 8th range of all other cancer (after the liver, lung, breast, colorectum, cervix uteri, thyroid and leukaemia), were approximately 315 new cases of NHL in men (the 5th after the liver, the lung, colorectum and leukaemia) and 293 in women (the 9th after the breast, cervix uteri, liver, colorectum, lung, thyroid, ovary and leukaemia) with a combined total of 608 cases.

The 9th most case of mortality related to cancer in both sexes in Cambodia after the liver, lung, colorectum, breast cervix uteri and leukaemia. The man's death rate was recorded as 199 cases per year, ranking seventh behind liver, lung, colon, leukaemia, oesophagus, and stomach. The 176 cases of deaths annually for the woman fell into the tenth range.

Based on global trends, the most common subtypes of NHL are likely to be diffuse large B-cell lymphoma (DLBCL), follicular lymphoma (FL) and marginal zone lymphoma (MZL).

III. ETIOLOGY

As with most cancers, the exact causes of NHL are not fully understood, but the doctors believe that a virus or the activation of abnormal genes could be the cause of some cases of NHL condition.

However, certain risk factors have been identified, including:

- Age: mostly the case of NHL increases in older than 60 years old.
- Medication: immunosuppressor after the transplantation, phenytoin, digoxin and TNF antagonist, chemotherapy, radiation exposure.
- Infection : VIH, Epstein-Barr Virus, Helicobacter pylori bacteria
- Certain chemicals and Pesticides – insecticides - herbicides
- Family history: people with a family history of NHL are an increased risk of developing the disease.

IV. PATHOPHYSIOLOGY

Most (80 to 85%) non-Hodgkin lymphomas arise from B lymphocytes; the remainder arise from T lymphocytes or natural killer cells. The stage of lymphocyte differentiation at which the oncogenic event occurs determines the disease presentation and outcome.

Most lymphomas are nodal with variable involvement of the bone marrow and peripheral blood. A leukemia-like picture with peripheral lymphocytosis and bone marrow involvement may be present in up to 50% of children and about 20% of adults with some types of non-Hodgkin lymphoma.

Hypogammaglobulinemia caused by a progressive decrease in immunoglobulin production is present in 15% of patients at diagnosis. Hypogammaglobulinemia increases the risk of serious

bacterial infection, and patients may require IV immune globulin to replace deficient immunoglobulins.

Non-Hodgkin lymphoma arises due to chromosomal translocation or mutation/deletion. Proto-oncogenes are activated by chromosomal translocation, and tumor suppressor genes are inactivated by chromosomal deletion or mutation.

- The t (14;18) translocation is the most common chromosomal abnormality in NHL (especially follicular lymphoma).
- The t (11;14) translocation is associated with mantle cell lymphoma, results in the overexpression of cyclin D1
- The t (8;14) translocation of c-myc and heavy chain Ig is associated with Burkitt lymphoma.
- Alteration in BCL-2 and BCL-6 is associated with diffuse large B-cell lymphoma.

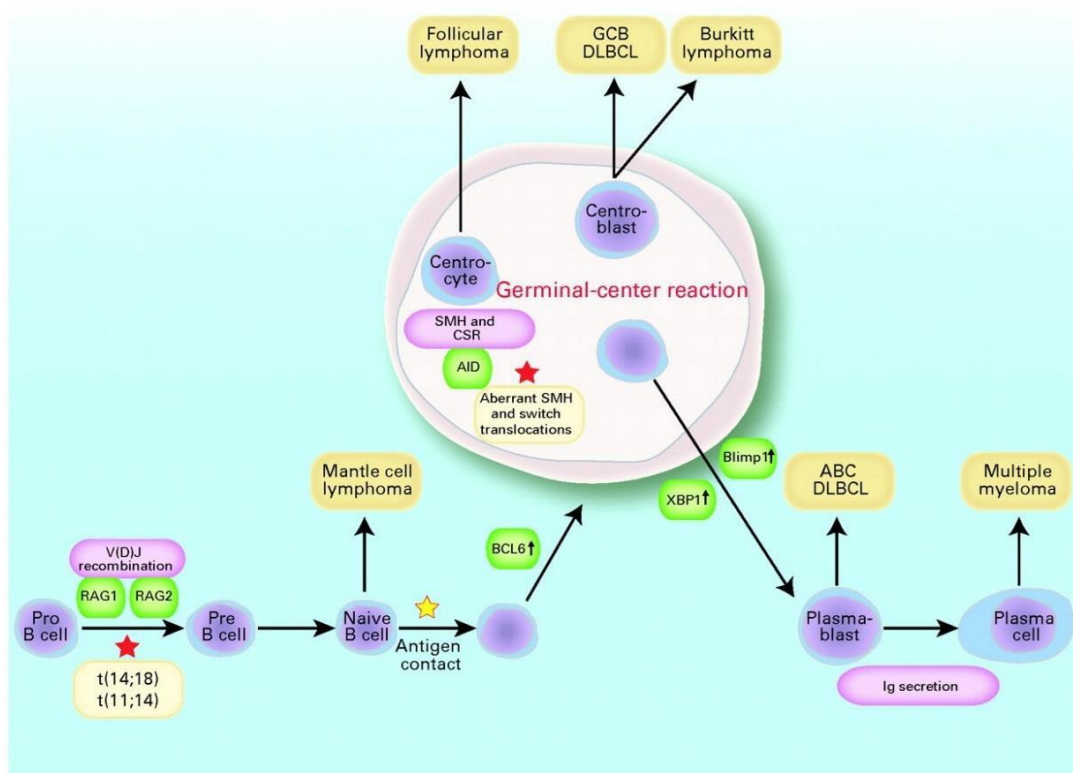


Fig: Pathophysiology, lymphomas arise at different stages of B cell differentiation

V. DIAGNOSIS

1. Clinic

1.1- General symptoms:

- Fever without any infectious causes, unexplainable, chronic more than 7 days.
- Weight loss more than 10% of actual weight
- Sweat +++ (at night)
- Inflammatory syndrome unexplainable

1.2- Lymph nodes presentation:

- Superficial lymphadenopathy: 80% of cases
- Swollen lymph nodes in the groin, armpit and the neck
- One or more lymph nodes, asymmetric lymphadenopathy, painless,

firm, mobile, non-inflammatory and non-compressible.

- Deep lymphadenopathy:
 - Mediastinal area:
 - cough, dyspnea, hoarseness, dysphagia, pleural/cardiac effusion
 - Superior Vena cava Syndrome
 - Abdominal-pelvic area:
 - Spleen enlargement on palpation

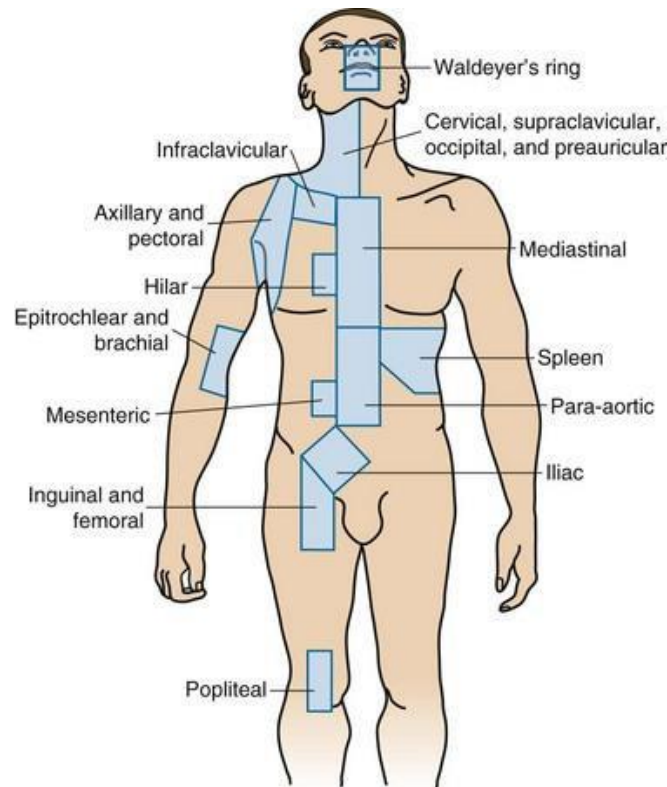


Fig: Lymph node localisation

1.3- Extra-lymph nodes presentation: around 1/4 of patients

- ENT: hypertrophy of Waldeyer ring
- Digestif: gastric symptoms, colorectum symptoms,
- Skin: hematodermie
- Breast: nodular
- Central Nervous System or epidural: intracerebral lymphoma, medullar compression.
- Gonads:
- Bone: associate or not with leukemic forms
- Bone marrow: invasion signs => cytopenia.

Emergency signs and symptoms:

- **Superior Vena Cava Syndrome:** Facial oedema, Arm oedema, distended neck-chest veins, facial plethora, visual symptoms, respiratory signs (dyspnea, cough, hoarseness, stridor), neurologic signs (syncope, headaches, dizziness, confusion)
- **Spinal cord compression:**
- **Tumor Lysis Syndrome:** hyperuricemia, hyperkalemia, hyperphosphatemia, hypocalcemia, kidney failure.

2. Investigation

2.1. Biopsy of lymph nodes or operated pieces:

- Send to the reference hospital: Calmette Hospital, Khmer-Soviet Hospital...
- The histological analysis of a biopsy is the only basis for the diagnosis.

QUIT BELIEVING BIOPSY PROVIDES RAPID METASTASIS

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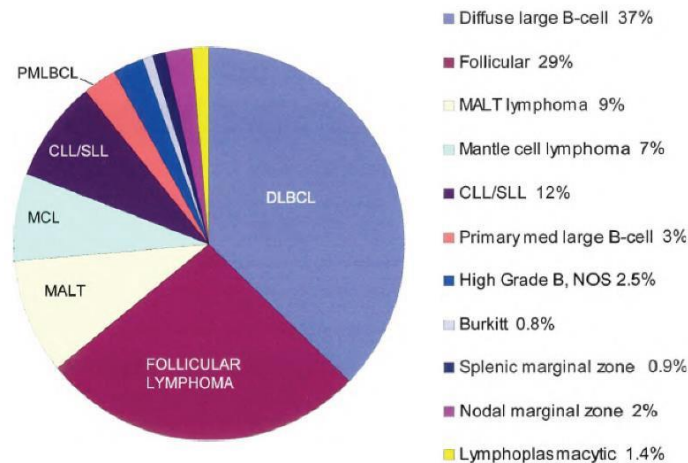


Fig: Distribution of non-Hodgkin lymphoma

2.2. Biology

- Complete blood count (CBC)
- VS (sedimentation test) prolonged
- Lacticodehydrogenase (LDH), Beta 2 microglobuline (B2M), Protein electrophoresis
- Tumor lysis syndrome: uricemia, calcemia, phosphoremia
- Viral test: HIV, HBV, HCV
- Hepatic and renal function

2.3. Radiology – Extension assessment – pretherapeutic assessment

- Chest X-ray: to evaluation of mediastinal enlargement
- CT scan cervical-thorax and abdominopelvic with injection:
 - Located the lymph node invasion and extra lymph node's location.
 - Classification of lymph node stage belong to Ann Arbor staging
 - Well located the location of biopsy in case of deep lymph nodes invasion
 - Define the complication of organ compression due to lymph node.
- Heart ultrasound, ECG

Ann Arbor Classification	
Stade I	Involvement of single lymph node region
Stade II	Involvement of ≥ 2 lymph node regions on the same side of diaphragm
Stade III	Involvement of lymph node regions on both sides of diaphragm
Stade IV	Diffuse or disseminated involvement of ≥ 1 extralymphatic organ or tissues $\pm$ associated lymph node enlargement
A or B designations - A: no systemic symptoms - B: Unexplained loss of more than 10% of body weight, unexplained fever, drenching night sweats E designations for $\Rightarrow$ extralymphatic organ involvement S designations for $\Rightarrow$ involvement the spleen	

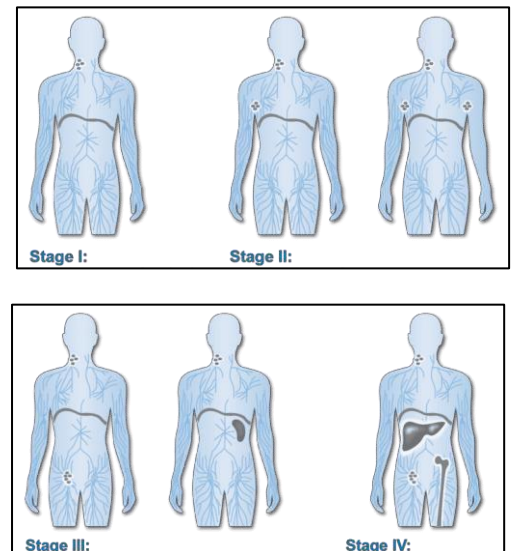


Fig: Non-Hodgkin Lymphoma's classification

VI. IFFERENTIAL DIAGNOSIS

- Other cause of lymphadenopathy: infection, systemic disease and other solid cancers.
- Histologic based result

VII. ANAGEMENT

1. Objectives

- Depend on the severity criteria and histology
- Approach the complete remission or to gain the best quality of life
- Palliative treatment in case of advance stage or palliative stage

2. Critical sign

2.1. Tumor Lysis Syndrome

Classification of Cairo and Bishop	Prevention is better than cure
Biologic: - Uricemia $> 476 \mu\text{mol/l}$ (8mg/l) or increase $> 25\%$ - Kaliemia $> 6 \text{ mmol/l}$ or increase $> 25\%$ - Phosphor $> 1.45 \text{ mmol/l}$ (4.5 mg/dl) or increase $> 25\%$ - Calcemia $< 1.75 \text{ mmol/l}$ (7 mg/dl) or decrease $> 25\%$	- Hydration: crystalloids $2\text{-}3\text{L/m}^2$ (with heart tolerance + follow up the urine output 70-100 ml/m²) - Hypouricemic therapy: - Allopurinol: ASAP – 100mg/m² every 8h (max 800mg per day) - Rasburicase: 0.2 mg/kg once daily for 5-7 days. - Electrolyte abnormality $\rightarrow$ - Monitoring serum K⁺/ Phosphor/ Ca²⁺ every 4-6H - Avoid exogenous potassium /phosphate intake - Administration IV insulin-glucose, inhaled beta-agonists - Hemodialysis
Clinic: - Creatininemia $> 1.5\text{N}$ - Seizure - Arrhythmia	

2.2. Superior Vena Cava Syndrome

- Discussion about transfer patient to the Onco-haematology unit
- Management:

Biopsy is required immediately, and shouldn't be significant delay a specific treatment	
Treatment of Obstruction	Treatment the causes
- Elevation of the head - Corticotherapy (1mg/kg) - Diuretic – Anticoagulation (may discuss cas by cas)	- Chemotherapy - Radiation therapy is possible
<i>The efficacy of any of these treatments has not been established. None of these therapies stood out as being better than the others.</i>	

3. Mortality risk factor

- Bulky (tumor mass > 7 centimetres or mediastinal mass located more than 1/3 of transthoracic diameter)
- Histology sub-type – aggressive lymphoma
- Diffuse or disseminated lymphadenopathy
- Sensitive to treatment by corticotherapy

4. Outpatient management

- Mostly of case => need to admit in hospital.

5. First-line treatment

Chemotherapy



Chemotherapy administered under tight supervision by professionals in the cancer care unit

Front of the bulky and mediastinal mass (with compression sign)

- Corticotherapy: Solumedrol 48 mg/m² or Prednisolone 60 mg/m²

The chemotherapy regimen for the first line:

- COP (Cyclophosphamide – Oncovin – Prednisolone)
- CHOP (Cyclophosphamide – Anthracycline – Oncovin – Prednisolone)

The immunotherapy anti-CD20 cluster should be add to improve the response rate of non-Hodgkin lymphoma.

6. Follow up:

The relapse or refractory risk are essential increased in the first 2 years.

- **Physical examination:** every 3 months for the first year, then every 6 months for 2 years, then once a year. Annual monitoring can be entrusted to the attending physician.
- **Biology assessment:** 3th, 6th, 12th, 24th months then on demand.
- **CT scan with injection** on 12th, and 24th months.

VIII. SPECIAL CASES

1. Non-hodgkin lymphoma with HIV or hepatitis seropositive:

- Multi-disciplinary team discussion for the management of patient
- Antiretroviral recommended in the period of the chemotherapy

2. Transformation of indolent lymphoma to aggressive lymphoma

- Chemotherapy intensive

3. Leukemic form of lymphoma (invade the bone marrow)

- Treatment as acute leukemia
- Chemotherapy intensive adapted beyond the MDT discussion

4. Non-Hodgkin lymphoma of testis

- Chemotherapy and systemic neuro-meningeal prophylaxis.
- Testis radiation therapy + option in discuss about pan-encephalic prophylaxis radiation therapy.

5. Non-Hodgkin lymphoma of the breast

- Chemotherapy and neuro-meningeal prophylaxis
- Breast radiation therapy + pan-encephalic prophylaxis radiation.

6. Non-Hodgkin lymphoma of the mediastinal and ENT:

- Chemotherapy + radiation therapy

IX. VOLUTION AND PROGNOSIS

The prognosis of non-Hodgkin Lymphoma mainly depends on histopathology, the extent of involvement, and patient factor.

International Prognostic Index (IPI) and its variant are used as the main prognostic tools in patients with non-Hodgkin lymphoma.

Congenital or acquired immunodeficiency states are associated with an increased risk of lymphoma, and response to therapy is poor. IPI has been modified for most of the NHL for better assessment of prognosis, e.g., FLIPI for follicular lymphoma, MIPI for mantle cell lymphoma, and so on.

Patient with aggressive T- or NK cell lymphoma usually have a worse prognosis.

X. COMPLICATIONS

Life-threatening emergent complications on NHL should be considered during the initial workup and evaluation. The complications include:

- Febrile neutropenia
- Hyperuricemia and tumor lysis syndrome
- Spinal cord or brain compression
- Focal compression depending on the location: airway obstruction, intestinal obstruction and intussusception, ureteral obstruction.
- Superior or inferior vena cava obstruction
- Hyperleukocytosis
- Venous thromboembolic disease

XI. PREVENTION AND RECOMMENDATION

a. Prevention

There is no sure way to prevent non-Hodgkin lymphoma.

But there are some things to limit the risk of certain infections such as HIV, HTLV-1, *Helicobacter pylori* (*H. pylori*) infection.

Secondary prevention for the patient:

- Lifestyle modification: smoking cessation, no alcohol consumption
- Healthy diet
- Exercise

b. Recommendation

- The management of NHL patient is Multidisciplinary team (oncohematologist, oncologist, radiation oncologists, nurse, and primary care physician (PCP)).
- The biopsy is the core-diagnosis for non-Hodgkin lymphoma. The surgery should be done in case of complication, and refer than to the cancer care unit for next treatment step.

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THALASSEMIA

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I. DEFENITION

Thalassemia refers to a group of blood diseases characterized by decreased or absent synthesis of one or more of the normal globin chains. According to the chain whose synthesis is impaired, the thalassemia is called α , β , γ , δ , $\delta\beta$ or $\epsilon\gamma\delta\beta$ thalassemia. Most thalassemia is inherited as recessive traits.

II. EPIDEMIOLOGY

Thalassemia is estimated to affect 1:2000 conceptions globally, and predominantly affect people whose family origins are from South Asia, Southeast Asia, the Mediterranean or the Middle East, although people who are of African or European origin may also be affected.

In Cambodia, based on certain studies, the affected Hb gene were within 23 –51% of general population in which 1% are diseases, which are similar to data in neighboring countries such as Thailand (30-40%), Vietnam (30%).

III. PHYSIOLOGY

1. Beta thalassemia

The basic defect in β thalassemia is a reduced or absent production of β globin chains with relative excess of α chains which accumulate and precipitate in the erythroid precursors forming inclusion bodies that, bound to the membrane skeleton, cause oxidative membrane damage and extensive premature destruction by apoptosis of the red blood cell precursors in the bone marrow (ineffective erythropoiesis).

Hemolysis plays a secondary role and is less prominent in thalassemia major than in thalassemia intermedia.

2. Alpha thalassemia

Reduction of α globin synthesis results in decreased production of Hb A ($\alpha_2\beta_2$) and reduced hemoglobin synthesis. In addition, excess unpaired β globin chains can form tetramers (β_4) that are not physically stable and precipitate, attaching to the red cell surface membrane and causing oxidative damage, thus shortening of red cell survival.

IV. DIAGNOSIS

1. Clinic

- Fatigue/weakness/shortness of breath
- Pale or yellowish skin/anemia
- Facial bone deformity
- Slow growth/failure to thrive
- Abdominal swelling
- Hepatosplenomegaly.

2. Investigation

- Complete blood count and peripheral blood film
- Hb typing
- Serum ferritin
- Liver function tests
- Infective screen for HIV, Hepatitis B and C, Syphilis
- Red cell phenotyping before transfusion (if available)
- Genotyping (DNA analysis) is an optional test which can be done to confirm unusual Hb variant and carrier status for prenatal investigation

V. DIFFERENTIAL DIAGNOSIS

- Sickle cell anemia
- Iron deficiency anemia
- Hemolytic anemia
- Sideroblastic anemia
- Anemia of chronic disease
- Erythropoietin deficiency
- Vitamin B12 or folate deficiency
- Acute blood loss
- Bone marrow failure

VI. MANAGEMENT

1. Blood transfusion

a. Objectives

The aim of blood transfusion in thalassemia is to deliver a safe and effective transfusion regimen whilst minimizing the burden of transfusion therapy on everyday life.

An effective transfusion regimen will result in:

- Good growth and development
- Good energy levels
- Sufficient suppression of intra and extra-medullary hematopoiesis

b. Criteria for initiating transfusion therapy

For deciding whom to transfuse, the following should be included in the investigations:

- Confirmed diagnosis of thalassemia
- Laboratory criteria:
 - * Hemoglobin level (Hb) <70 g/l on 2 occasions, > 2 weeks apart (excluding all other contributory causes such as infections) AND/OR
 - * Clinical criteria irrespective of hemoglobin level: Hemoglobin >70 g/l with any of the following significant symptoms of anemia:
 - Poor growth / failure to thrive
 - Complications from excessive intramedullary hematopoiesis such as pathological fractures and facial change
 - * Clinically significant extra-medullary hematopoiesis

2. Transfusion programs

- The recommended treatment for thalassemia major involves lifelong regular blood transfusions, usually administered every two to five weeks.
- 15–20 mL/kg of optimally screened (ideally NAT tested), leukodepleted packed red cells, to maintain a pretransfusion hemoglobin of 9.5–10.5 g/dL.
- Pre-transfusion hemoglobin level of 110–120 g/l may be appropriate for patients with heart disease, clinically significant extramedullary hematopoiesis or other medical conditions, and for those patients who do not achieve adequate suppression of bone marrow activity at the lower hemoglobin level.

3. Iron chelation

Indication for iron chelation therapy

Iron chelation must commence once serum ferritin exceeds 1,000 µg/L, which is usually reached after 10–20 transfusions.

Table 1. Iron chelation

Chelator	Deferasirox	Deferoxamine	Deferiprone
Route	Oral (dispersible tablet/ film-coated tablet)	Subcutaneous/IV if intensive chelation warranted	Oral
Dose	DT: 20–40 mg/kg/d FCT: 14–28 mg/kg/d	30–40 mg/kg/d, 5–7 days/week	75–100 mg/kg/d
Frequency	Once daily	12-hour infusions	Three divided doses
Side effect	Gastrointestinal (GI) disturbances, transaminitis, azotemia, Microalbuminuria	Local reactions, ototoxicity, and retinopathy	Agranulocytosis and arthralgias
Monitoring	LFT/serum creatinine, urine monthly to begin with then 3 monthly	Annual ophthalmic and auditory review	CBC every 2 weeks

4. Splenectomy

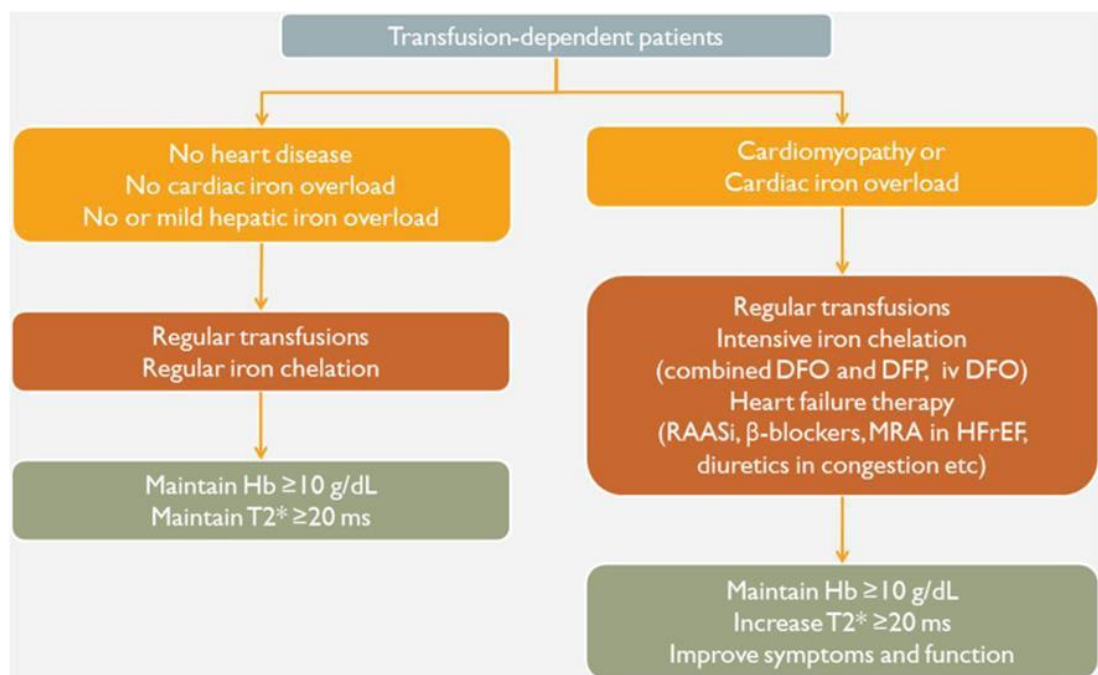
- a. Indication for splenectomy:
 - Increased blood requirement, but not due to allo/auto-antibodies or blood loss (more than 200 -275 ml/kg/year red cells)
 - Hypersplenism
 - Symptomatic splenomegaly
- b. Immunoprophylaxis before splenectomy
 - *Streptococcus pneumoniae*: 4-6 weeks or at least 2 weeks pre-splenectomy, repeat 5 yearly thereafter
 - *Haemophilus influenzae* B: 4-6 weeks/or at least 2 weeks pre-splenectomy.

- Meningococcal C: 4-6 weeks or at least 2 weeks pre-splenectomy. Repeat 5 yearly thereafter.
- Influenza virus vaccination: Yearly
- c. Chemoprophylaxis after splenectomy
 - Penicilline chemoprophylaxis for splenectomised patient is recommended up to the age of 16 years or for 5 years in children and for 2 years for adult post splenectomy.
 - < 2 years old: oral penicillin 125 mg BID
 - 2-5 years old: oral penicillin 250 mg BID
 - Adult: oral penicillin 250 mg BID

5. Supportive treatment

- Folic acid: 1-5mg/day
- Vitamin C: 2-3 mg/Kg/day, recommended only in conjunction with deferoxamine infusion.
- Calcium and vitamin D: 2000 iu per day, check vitamin D level every 6 months
- Drinking tea with meals may be of benefit for patients with non-transfusion dependent thalassemia to reduce iron absorption.

6. Algorithm



A basic therapeutic algorithm for thalassemia patients on regular blood transfusions.

DFO: deferoxamine; DFP: deferiprone; Hb: hemoglobin concentration; HFrEF: heart failure with reduced left ventricular ejection fraction; MRA: mineralocorticoid receptor antagonists; RAASi: renin-angiotensin-aldosterone system inhibitors.

VII. SPECIAL CASES

Pregnancy

- Increase frequency of blood transfusion to maintain pre-transfusion hemoglobin above 100 g/l
- Patients on oral chelators (Deferasirox or Deferiprone) should be advised to switch to Deferoxamine, prior to induction of ovulation/spermatogenesis
- Deferoxamine is not required during pregnancy in patients who are not iron overloaded and have adequate cardiac function prior to pregnancy
- If cardiac function deteriorates during pregnancy, deferoxamine may be used with caution after the first trimester
- Check cardiac, liver and thyroid function once each trimester
- Screen for gestational diabetes
- Serial ultrasound scans to monitor fetal growth
- Resume Deferoxamine after delivery.

VIII. EVOLUTION AND PROGNOSIS

If untreated, patients of thalassemia major die by the age of 3 to 4 years due to severe anemia. Transfusion therapy alone, patients with thalassemia major children died due to cardiac complications (related to iron overload) as early as 10 to 12 years of age

IX. COMPLICATION

- Jaundice and pigment gallstones
- Hypersplenism
- Complications of extramedullary hematopoiesis
- Skeletal changes
- Iron overload
- Growth impairment
- Endocrine and metabolic abnormalities
- Heart failure and arrhythmias
- Pulmonary abnormalities

X. PREVENTION AND RECOMMENDATION

Screening for thalassemia in couple reproductive life to allow primary prevention including primary screening to determine red cell indices, Hb typing in subject with reduced MCV and/or MCH.

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INDICATION OF TRANFUSION AND ITS MANAGEMENT

(Before, during and after blood transfusion)

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I. INTRODUCTION

Blood transfusions are a relatively common medical procedure, and while typically safe, there are multiple complications that practitioners need to be able to recognize and treat. The indications for blood transfusion, including for special patient populations, the pre-transfusion preparation, and the potential complications of blood transfusions.

II. PRETRANSFUSION ASSESSMENT

Before any first transfusion of any labile blood product:

- ⇒ 2 determinations of the ABO-RH1 blood group and the RH2, 3, 4, 5 and KEL1 phenotype must be carried out.

For RBC, depending on the level of urgency, the search for irregular agglutinins (RAI)¹ dating back less than 72 hours is mandatory (deadline extended to 21 days in the absence of a history of pregnancy, transfusion or transplantation within months).

Regardless of the level of urgency specified on the prescription, the group and the RAI must be collected but the transfusion can be carried out before the results in the event of a vital emergency according to the following table:

Table 1. Pretransfusion assessment

	Relative emergency	Vital emergency	Immediate vital emergency
Delivery time	2-3H	< 30 min	Without delay
2 determinations of ABO-RH1 blood group and RH-KEL1 phenotype (Delay 30 min)	Isogroup or compatible transfusion		Collected but transfused without waiting for the result: O, RH: 1, KEL: -1 Except for women until the end of the reproductive period: O, RH: -1, KEL: -1 (within the limits of availability)
RAI (Delay 45 min)		Mandatory result before transfusion. <u>If RAI +:</u> direct compatibility test in the laboratory	Collected but transfused without waiting for results

¹Attention, following a transfusion episode, an RAI must be performed within 1 to 3 months. It is also necessary to give their group card to any transfused patient.

*** 3 causes of false positive RAI:**

- auto-antibodies with positive direct Coombs test.
- treatment with antibody monoclonals such as daratumumab (anti-CD38 indicated in myeloma because it fixes CD38 on red blood cells);
- treatment with anti-D (Rhophylac®): positive RAI for at least 3 weeks, check for negativation at a distance

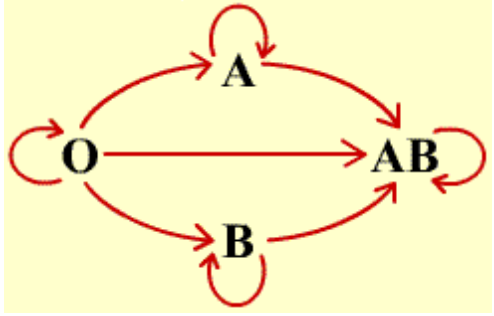
III. BLOOD COMPONENTS AND INDICATION

3.1- Red Blood Cell Concentrates (RBC)

One unit of Red Blood Cell Concentrates

- Contain on average 40 g of Hb
- Hematocrit between 50% and 70%
- Volume 280 ml
- Storage: 42 days at 4°C
- Are leukoreduced (< 1.106)

3.1.1- when to transfuse?

(The first criterial is tolerance to anemia) ¹	• 7 g/dL in patients > 80 years old without cardiovascular disease or poor clinical tolerance². • 8 g/dL in the general hematology patient. • 10 g/dL in the case of associated cardiovascular disease. • No threshold for hemolytic anemia³ and hemoglobinopathies (clinical tolerance +++).
Dosage (Adult) and administration	• One unit of RBC increases Hb by 1 g/dL or raise HCT ~ 3% (on average in 280 ml)⁴. • Rate of transfusion: - Transfuse slowly for first 15 minutes - Complete transfusion within 4 hours
ABO Compatibility Rules	 The diagram illustrates the ABO compatibility rules. It shows four blood types: A, B, AB, and O. Arrows indicate the direction of compatibility: from donor to recipient. - Self-compatibility: A to A, B to B, AB to AB, and O to O (indicated by curved arrows). - Forward compatibility: A to AB and B to AB (indicated by straight arrows). - Reverse compatibility: O to A, O to B, and O to AB (indicated by curved arrows). - Incompatibility: No arrows point from AB to A or B, and no arrows point from A to B.

¹ Well-tolerated deficiencies: iron and B9 (IV if PO route ineffective), B12 (PO or deep IM/SC if neurological signs).

² Monitor FR and saturation, if possible, only 1 RBC, no systematic preventive

diuretics.

³ In case of cold agglutinins, remember to warm the pellet!

⁴ According to ANAES, the quantity of RBC to be transfused is the minimum quantity to eliminate the observed signs of poor tolerance and/or to raise the Hb concentration to an acceptable level.

Transformation (Complex situations should be anticipated due to additional delay)	
Deplasmatised (3-4 hours more)	• IgA deficiency with presence of anti-IgA antibodies in the recipient's plasma. • History of severe (shock, Quincke's edema) or repeated allergic transfusion reactions.
Cryopreserved (Extra hours)	• Patients with a rare blood group. • Patients who have developed an association of several antibodies directed against blood group antigens of balanced frequencies.
Qualification	
Extended phenotype (FY/JK/MNS)	• Patients who have developed one or more anti-erythrocyte alloantibodies presenting a transfusion risk against an antigen in blood group systems other than RH and KEL1 (in this case it is desirable to also respect the RH-KEL1 phenotype as a preventive measure).
Compatible ¹	• Sickle cell patients regardless of the RAI result. • Patient with RAI or a history of positive RAI.

¹In some cases, when the antibody is directed against a high-frequency antigen (present in 99.9% of the population), it is not possible to find compatible RBCs. In this case, the head of the transfusion center may decide to issue RBCs that have had the weakest reactions during the compatibility test.

4. Platelets

Platelet concentrates have a very low residual RBC content. There is therefore no risk of hemolytic accident. On the other hand, this small quantity of RBCs may be sufficient to trigger alloimmunizations against the erythrocyte antigens received.

Platelet antigens belong to several systems: ABO (not expressed by platelets but passively transported by platelets from a release of the donor endothelium, a variable phenomenon in the general population), HLA class I (HLA A, HLA B and HLA C) and HPA system antigens (HPA1, HPA2, HPA3, HPA4) which are specific to platelets.

Platelet incompatibilities can lead to immunization of the recipient (rarer than erythrocyte alloimmunizations). Here again, two causes can lead to this alloimmunization: transfusion of platelet concentrates or pregnancy.

There is bioequivalence between apheresis platelet concentrates (APC) and whole blood platelets (WBP), so the prescription must be made indifferently, except for HLA compatible where APCs will only be used.

The first criterion is the presence of a hemorrhagic syndrome)². The hemorrhagic phenotype and the kinetics of decline are elements to be taken into account)	• 10 G/L if no risk factor • 20 G/L if fever ≥ 38.5 °C, infection, arterial hypertension, grade ≥ 2 mucositis, lesion with hemorrhagic potential, rapid decrease kinetics of NP in 72 hours, or even MDS (<i>thrombopathies</i> +++). • 50 G/L if fibrinolysis, anticoagulant, DIC, invasive procedure (Lumbar puncture...), to be modulated if peripheral thrombocytopenia². • 80 G/L if epidural. • 100 G/L if neurosurgery.
Dosage (adult)	One unit / 7-10 kg (1-600mL)
ABO Compatibility Rules	Préférable de respecter la compatibilité ABO, mais transfusion possible dans toutes les combinaisons de groupes sanguins sous réserve de taux anti-A et/ou anti-B faibles dans la poche et chez le receveur.

¹Apheresis platelet concentrates (APC from a single donor) or Whole blood platelets (WBP from 4-6 donors of the same ABO group).

²Peripheral thrombocytopenia (ITP, heparins, etc.) is not an indication for transfusion, except in the case of life-threatening hemorrhage or essential invasive procedures.

In the case of Disseminated intravascular coagulation (DIC) during acute leukemia (in particular AML3), it is recommended to perform platelet transfusions to maintain a threshold of 50 Giga/L.

WARNING: RH1 compatibility (= Rh D) must be respected for the transfusion of packed red blood cells, platelets and leukocytes (not taken into account for plasma) within the limit of availability in women with obstetric future other than allogeneic transplantation.

If impossible: administration of at least 100 µg of anti-D immunoglobulins (Rhophylac®) within 72 hours. Beyond 3 weeks, a search for residual anti-D allows to know if there is still protection.

5. Fresh Frozen Plasma (FFP)

Plasmas provide all plasma proteins, in particular coagulation factors and complement fractions. Plasmas are the only products capable of providing factor V, protein S, plasminogen and von Willebrand factor-cleaving metalloprotease (ADAMTS 13), as well as certain complement fractions. These factors are not available in stable purified form among blood-derived drugs.

FFP transfusion can sometimes exceed its indications due to the urgency and unavailability of specific treatments:

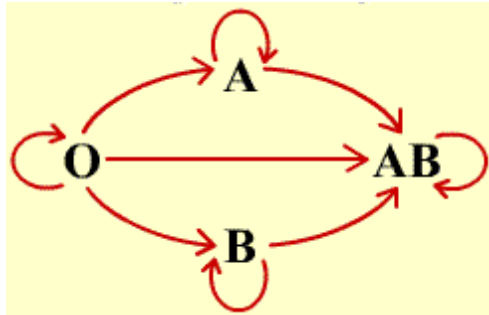
- **Acute attacks of hereditary angioneurotic edema (laryngeal+++)**
The use of therapeutic plasma in this indication can only be considered in the absence of immediate availability of specific treatments C1-INH IV or Icatibant SC.
- **Thrombotic microangiopathies (TTP)**

In thrombotic microangiopathies, plasma has a recognized therapeutic effect at large volume, between 40 and 60 mL/kg, or 1 to 1.5 plasma mass. This volume ensures the supply of a significant quantity of the ADAMTS 13 molecule in TTP and the supply of complement proteins in HUS. This is why plasma administration is preferentially done by plasma exchange.

- **In case of severe overdose of AVK, in two rare situations**

Absence of prothrombin complex concentrates (PCC).

Absence of PCC not containing heparin in case of history of HIT.

Threshold (no threshold if hemorrhagic shock) ¹	• TP <40% (<50-60% in neurosurgery). • Fibrinogen < 1g/L post plasma exchange.
Dosage (Adulte) ²	• 10 to 15 mL/kg (DIC or FV/Protein S/plasminogen³ deficiency, associated with active/potential hemorrhage). • 10 to 30 mL/kg (after plasma exchange or, non-serious hemorrhage).
ABO Compatibility Rules	

¹ with FFP: RBC ratio between 1:2 and 1:1

² 1 FFP = 200 to 300ml

³ others if no purified factor available

IV. COMPLICATION RELATED TO TRANFUSION AND MANAGEMENT

Transfusion is a medical act and the doctor is responsible for prescribing it. A doctor must be available in case of complications. Whenever possible, transfusions should be performed during the day rather than at night. It is important to provide therapeutic education to the patient on the warning signs. The spectrum of transfusion reactions is very varied and it is difficult to predict their severity because they often start with the same non-specific symptoms. And to complicate matters, the patient's condition before the transfusion does not make things easier!

4.1- Acute (<24h):

	Clinical	Paraclinical examinations	Treatment
Graves ¹			
Transfusion-associated circulatory overload (TACO): Pulmonary edema overload	Acute respiratory distress, mainly	Bilateral infiltrate on Chest X-Ray,	• Transfuse 1 pellet at a time and slow down the transfusion rate over 3-4 hours in
Risk factors: History of cardiovascular disease, Hypertensive surge during the first bag	hypertension, possible associated signs of ICD	increased NTproBNP, if pleural puncture = transudate	patients at risk (cardio history, hypertensive surge during the 1st pellet)
Anaphylaxis (APCs>WBP>RBC)	Often rapid reaction (minutes) Hypotension arterial, even anaphylactic shock, urticaria/angioedema, bronchospasm, digestive disorders, rare fever	Serum tryptase and histamine	• Furosemide curative • Stop transfusion immediately and definitively • IM or even IV adrenaline + antihistamine treatment Discuss NIV and bronchodilators Search for anti-IgA Ac and quantitative IgA dosage Next PSL: deplasmated
TRALI (all)	Acute respiratory distress rather arterial hypotension, possible fever: ARDS picture	Bilateral infiltrate on chest x-ray, leukopenia, positive anti-PNN/HLA antibodies in the donor, pleural exudate	• Immediate and definitive transfusion stop • Transfer to intensive care (poor prognosis)
Acute hemolytic accident (RBC and plasma) ^{2,3}	Fever, chills, low blood pressure or even vasoplegic shock, acute lower back pain, oliguria/anuria possible	Coombs + (low VPN), increased LDH and BiliC, collapsed haptoglobinuria (BU+), often IRA and CIVD	• Stop transfusion immediately and definitively • Redo RAI test • Perfusion serum saline for hydration • Next transfusions with compatible RBCs if RAI +
Transfusion-transmitted bacterial infection (all)	Fever, chills, hypotension or even septic shock Possibility of a predominantly	Bacteremia, hyperleukocytosis, possible disseminated intravascular coagulation (DIC)	• Stop immediate and definitive transfusion • Patient and bag blood cultures then ATB anti-BGN

	digestive picture with profuse diarrhea, abdominal pain, vomiting		
Hypocalcemia (in case of massive transfusion) ⁴	Paresthesia, malaise, rigidity, rhythm disturbances Hypothermia		• Calcium IV or PO
Post-transfusion purpura (all) ⁵	Mucocutaneous hemorrhagic syndrome over several hours, occurs mainly in female recipients	Acute deep thrombocytopenia	• Anti-HLA/HPA antibody search • Discuss IVIG 1g/kg/day • contraindication relative to platelet transfusion. • If hemorrhagic syndrome: transfusion of compatible platelets SB
Pulmonary leukostasis (Acute leukemia with hyperleukocytosis)	Pulmonary leukostasis syndrome	-	• Stop immediate and definitive transfusion • IV saline for hydration
Benin: monitor the evolution +++			
Urticaria (especially with RBCs and CPA)	Isolated urticarial	-	• Curative: Polaramine® 5mg (to be repeated once SB) or even corticosteroids • If repeated: deplasmated PSL
Chills-hyperthermia syndromes	Fever, myalgia	-	• Slow down flow • Paracetamol

¹ Do not forget the other potential more general complications such as gas embolism!

² Rare non-immunological cause of acute hemolysis: thermal shock.

³ ABO, irregular immune antibodies of the Rh, Kell, Duffy, Kidd, MNS systems, antibodies directed against high frequency antigens (HFA) and anti-Wra (Diego system).

⁴ During hemorrhagic shock, in addition to iatrogenic dilution, a very early coagulopathy specific to polytrauma and the "lethal triad" consisting of the association of coagulopathy, hypothermia and acidosis are incriminated. The supply of platelets must then be systematic in the following ratio: 1 unit of platelet (APCs or WBP) for 4 to 6 RBC/FFP.

⁵ Up to 10 days after the transfusion!

Practical management

- Common signs: fever, chills, sometimes pruritus.
- Signs that should alert: dyspnea/desaturation, hemoglobinuria, impaired consciousness, high blood pressure/low blood pressure, acute lower back pain, hemorrhagic syndrome, oliguria.
- a. For all: pause the transfusion, Perfusion IV of saline NaCl 0.9%, identity check on blood bag with patient, clinical examination and oxygen therapy if needed.
- b. If there is any doubt, scope the patient, redo Beth Vincent and contact the laboratory of hospital or blood bank then send an assessment with: CBC, ionogramme, urea, creatinine, Coombs +/- according to clinic, TP, TCA, D-dimers, NTproBNP, chest x-ray, blood gas test, IgA Ac anti-IgA, tryptase, histamine, Calcium, Phosphors.
- c. Return the blood product to the Laboratory or blood bank after calling the Laboratory (who will fill out the adverse event form).
- d. Reassess the need for a new transfusion.

4.2- Delayed

4.2.1- Infectious risks (all labels of blood products)

Despite all the precautions taken in securing labile blood products in France, there is still a residual risk of viral transmission (HIV, HCV, HBV, HTLV1, etc. of the order of 1/10,000,000), as well as parasitic (P. Falciparum... etc.).

4.2.2- Transfusion-associated graft-versus-host disease (GVHD):

Occurs within 4 to 30 days, GVHD can be cutaneous, digestive, hepatic and especially medullary (expression of HLA class II) in patients with cellular immunodepression (except HIV) and partial HLA matching. The diagnosis is difficult and there are often many possible differentials. Mortality is high (over 90%) and treatment is mainly based on IS treatments, hence the importance of prevention by irradiation!

4.2.3- Platelet transfusion inefficiency (platelets)

Inefficiency is determined after a 2nd transfusion of a number of platelets unit adapted to the patient's weight, ABO compatible¹, and stored for less than 72 hours, the CCI is less than 7. This is frequently associated with fever during transfusion.

$$CCI = \frac{[NP \text{ après transfusion} - NP \text{ avant transfusion}] \times \text{surface corporelle (m}^2\text{)} \times 100}{\text{Nombre de plaquettes transfusées (x } 10^{11}\text{)}^*}$$

Platelet count in Giga/L; PN performed 18-24 hours after transfusion

Other causes may be responsible for a refractory state:

- Fever, with or without documented infection.
- Disseminated intravascular coagulation.
- Splenomegaly.
- Complications of a hematopoietic stem cell transplant (MVO, CMV, GVH, TMA, etc.).
- Drug interference (amphotericin B, vancomycin, etc.).
- Quality of the transfused product: dose, ABO¹ compatibility and shelf life.

In case of HLA/HPA immunization, it is recommended to look for phenotyped or even compatible APCs. Prophylactic transfusion is not recommended and if necessary, fractionated transfusions in the nychthemeral are recommended. One in case of ABO incompatible transfusion, the transfusion yield may be reduced, especially in recipients with a high serum anti A and/or B level.

4.2.4- Delayed hemolysis (potentially all)

This is a hemolytic anemia, often asymptomatic¹ and often extravascular occurring up to 4 weeks post-transfusion. It is necessary to monitor clinical signs +/- renal function. It is systematically recommended to perform an RAI +/- CBC at 1-3 months post-transfusion.

¹ delayed hyperhemolysis crisis in multi-transfused hemoglobinopathies with severe CVO clinical, discuss treatment with corticosteroid therapy and/or IV Ig or even Eculizumab.

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CHAPTER XI DERMATOLOGY

1. Scabies infestation
2. Dermatophytes infections (Tinea)
3. Varicella
4. Herpes simplex viruses
5. Herpes zoster
6. Atopic Dermatitis
7. Acne vulgaris
8. Gonorrhea
9. Seborrheic dermatitis
10. Psoriasis

SCABIES INFESTATION

*Dr SEANG Sokheng, Dr BORN Bori, Dr SRON Bulong, Dr CHUKMOL Sirithida,
Prof MEY Sithach*

I. DEFINITION

Scabies is an infectious disease caused by the mite *Sarcoptes scabiei var. hominis*, which burrows into the upper layer of the skin—the stratum corneum. The eggs and mite proteins produce an allergic reaction, responsible for the characteristic itching and rash. The patient's main complaint is nocturnal itching.

II. PIDEMIOLOGY

According to WHO, Scabies is a common parasitic infestation and can affect people of all ages, races, and socioeconomic statuses, with an estimated 200 million cases at any given time globally. The prevalence of scabies can vary significantly across different regions and populations. Scabies outbreaks are frequently reported in crowded living conditions, such as in schools, dormitories, refugee camps, pagodas, prison...

III. IOLOGY AND PATHOGENESIS

The infestation of *Sarcoptes scabiei var. hominis* occurs through skin-to-skin contact with another individual who has scabies or by contact with infested fomites (e.g., clothing and towels). It is sometimes transmitted via care providers or bedding. The disease can be sexually transmitted. The incubation period for those without previous exposure to scabies is 2 to 6 weeks. Individuals who have been previously infested with scabies develop symptoms within 1 to 5 days of re-exposure. In classical scabies, there are under 5-15 mites per host. Crusted scabies is characterized by a much higher burden of mites in the infested individual.

IV. DIAGNOSIS

Clinical manifestation:

- **Classical Scabies** The symptoms include itching, which usually develops within 2-6 weeks after infestation. The itching is usually generalized, intense, and intractable, worsening at night. And For physical examination ,look for small erythematous papulovesicular lesions predominantly over anterior axillary folds, nipple area, periumbilical skin, elbows, the volar surface of the wrists, interdigital web spaces, belt line, thighs, buttocks, penis, scrotum, ankles, and typically sparing the head, face, and neck in adults. Scabies burrows are most easily found on the hands, especially finger webs, and on the wrists; other sites of predilection are the feet, axillae, umbilicus, male genitalia, and breast areolas in women. For infants and young children, unlike adults, lesions are common on the face, scalp, neck, palms, and soles. Burrows are common on the palms, soles, and sides of the feet. The reddish-brown nodules of scabies are seen in axillary and inguinal regions, wrists, and male genitalia, and these may persist for several months. Some lesions may be altered by excoriations, eczematization, and secondary bacterial infection.
- **Crusted Scabies** It is an uncommon form of scabies, highly contagious with a large number of mites. It is more common in immunocompromised and neurologically impaired individuals but may also occur in otherwise healthy subjects. Pruritus may be minimal or absent, or the host may be physically incapable of scratching. Characterized by diffuse hyperkeratosis associated with variable degrees of underlying erythroderma. Hyperkeratosis and crusting are particularly severe on the

hands, including the palms and soles, under the fingernails, on the ears, trunk, and extremities.

Investigation:

The diagnosis of scabies is based on a general assessment of clinical symptoms, results of microscopy, dermoscopy, and other methods for detecting mites, and the epidemiological status of the outbreak, including information about the chance of contact with scabies patients. A definitive diagnosis is supported by a positive microscopic examination of skin scrapings, which identifies mites, eggs, or fecal pellets (scybala). Dermoscopy and digital photography are non-invasive and effective methods for identifying the presence of scabies mites; the 'delta' sign at the end of the burrow represents the anterior body of the adult female mite.

V. DIFFERENTIAL DIAGNOSIS

We need to differentiate scabies with common skin problems below:

1. Atopic dermatitis is a chronic relapsing, pruritic condition that is often associated with allergic rhinitis and/or asthma. Look for scaly erythematous scaly papules and plaques involving the flexural surface, particularly antecubital fossae and popliteal fossae, face neck, and extremities in general. For chronic cases, hypo/hyper pigmentation or lichenification might be seen.
2. Id reaction: An Autoeczematization reactions that occur in relation to a dermatophyte infection. Its pathogenesis may involve an immunologic reaction to fungal antigen (delayed-type hypersensitivity).
3. Acropustulosis of infancy: A benign recurrent vesiculopustular eruption characterized by pruritic vesiculopustules on extremities which are palms and soles.
4. Arthropod bites: An inflammatory reaction after bite or sting and might manifested by pruritus, redness, swelling. Look for small central punctum with surrounding erythema and swelling
5. Bullous Pemphigoid in elderly: An autoimmune blistering disorder with a prodromic phase of pruritic, eczematous, popular or urticaria-like skin lesions preceding the development of classic tense blister or bullae.
6. Dermatitis herpetiformis: an autoimmune blistering disorder associated with gluten sensitivity classic clinical manifestation include intensively itching with inflammatory papules and vesicles on predilection areas such as forearms kness scalp and buttocks.

VI. MANAGEMENT

General Measures: Patients must receive detailed information about scabies infestation and therapeutic options, including the amount of drug to be used and proper administration. Patients with scabies and their close physical contacts, even without symptoms, should receive treatment simultaneously. Prescriptions must be provided for all household members and sexual partners. Topical treatment must be applied to the entire skin surface from the jawline downwards, including all body folds, groin, navel, and external genitalia, as well as the skin under the nails (especially in crusted scabies). In adults with classical scabies, treating the face is controversial, but in babies, the face must be treated because transmission may occur from breastfeeding. Medications should be re-applied if washed off during treatment, such as after hand washing. After treatment completion, patients should use fresh clean bedding and clothing. If possible, potentially contaminated clothes and bedding should be washed at high temperatures (>50°C) or kept in a plastic bag for up to 72 hours since mites separated from the human host will die within that period.

Drug Therapy

Treatment	Treatment Regime	Contraindication / Caution	Side Effects	Comments
Permethrin 5% Cream / Lotion	Applied head to toes; rinse off after 8 to 12 hours & repeat 1 week later		Itching & burning/stinging sensation on application	
Benzyl Benzoate 10-25% Lotion	Applied head to toes; rinse off after 8 to 12 hours & repeat continuously for 3 consecutive nights (with baths taken between each application).	Pregnant & breastfeeding women and infants less than 2 years	Skin irritation and burning sensation. May cause conjunctivitis if exposed to the eyes. May worsen/cause post-scabetic eczematous reaction.	
Precipitated Sulphur 6-10%	Applied head to toes; rinse off after 8 to 12 hours & repeat continuously for 3 consecutive nights (with baths taken between each application).	Low toxicity	Messy, malodorous, stained clothing; causes irritant contact dermatitis	Safe for infants, pregnant, and breastfeeding women
Crotamiton 10% Ointment	Applied head to toes; rinse off after 8 to 12 hours & repeat continuously for 3 consecutive nights (with baths taken between each application). Nodular scabies: Apply to the nodules 3 times a day for 7-14 days.	Avoid massive & prolonged use in pregnant women and infants	Irritant contact dermatitis	
Lindane 1% Lotion / Cream	Applied head to toes; rinse off after 8 to 12 hours (single application)	Pregnant & breastfeeding women, infants, children <10 years, patients with seizure disorders. Avoid applying after a hot bath to prevent/reduce percutaneous absorption	Neurotoxicity; cramps, dizziness, seizures in children. Hypoplastic anemia and cancer	

Ivermectin	Oral drug 200 ug/kg single dose and repeat after 2 weeks	Not for children below 5 years old or less than 15kg. Avoid in pregnant and lactating women.	Use with other drugs which reinforce GABA activity can lead to augmented activity (valproate, barbiturates, benzodiazepines)	
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Treatment in Specific Considerations

Clinical condition Classical scabies	Recommended therapy	Alternative therapy	Additional measures
Infants < 2 months	Sulphur 6% in petroleum in ointment base for 3 days	--	Treat whole body including the face (avoid eyes and mouth)
Children >2months- 2 years	Two applications of Permethrin 5% for 8-12 hours at one week apart	Sulphur 6% in petroleum for 3 days	Treat the whole body including the face (avoid eyes and mouth)
Children >2- 12 years	Two applications of permethrin 5% for 8-12 hours at one week apart	Benzyl Benzoate 12.5% Whole body neck and below for 3 consecutive days	--
Pregnancy and breast-feeding	Two applications of permethrin 5% for 8-12 hours at one week apart.	--	--
Crusted scabies (better refer to specialist center)	Permethrin 5% cream or benzyl benzoate lotion 25% repeated daily for 7 days then 2 times weekly until cure Oral Ivermectin 200 mcg/kg on day 1, 2 and 8.	Oral ivermectin alone or in combination with permethrin is very useful OR Several applications of Benzyl Benzoate .	Apply keratolytic agents (salicylic acid ointment) to hyperkeratotic areas. Keep nails short and apply medication to subungual areas.
Nodular scabies	- Topical anti-inflammatory agents; e.g.		

	topical corticosteroids of mid potent to potent for a short duration of 2 weeks. - Crotamiton cream twice daily for 7 to 14 days.		
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Treatment of Infection in Scabies

Secondary bacterial infection is a common complication in scabies due to disruption of the epidermal barrier caused by excoriations. The timing of antibiotic commencement depends on the condition of the patient; it can be started concurrently with scabicides or delayed for 48 hours to allow partial healing of the erosions. Treat with systemic antibiotics covering gram-positive organisms for a minimum of 7 days. Use antiseptic soaks/baths, e.g., KMnO₄ (1:10,000) for oozing lesions. Clean excoriated/crusted lesions with antibacterial soap or antiseptic soaks before applying the prescribed drug. Apply topical antibiotics (Fusidic acid cream/Mupirocin ointment) to localized infected lesions. For extensive bacterial infections, use systemic antibiotics.

Treatment of Itching in Scabies

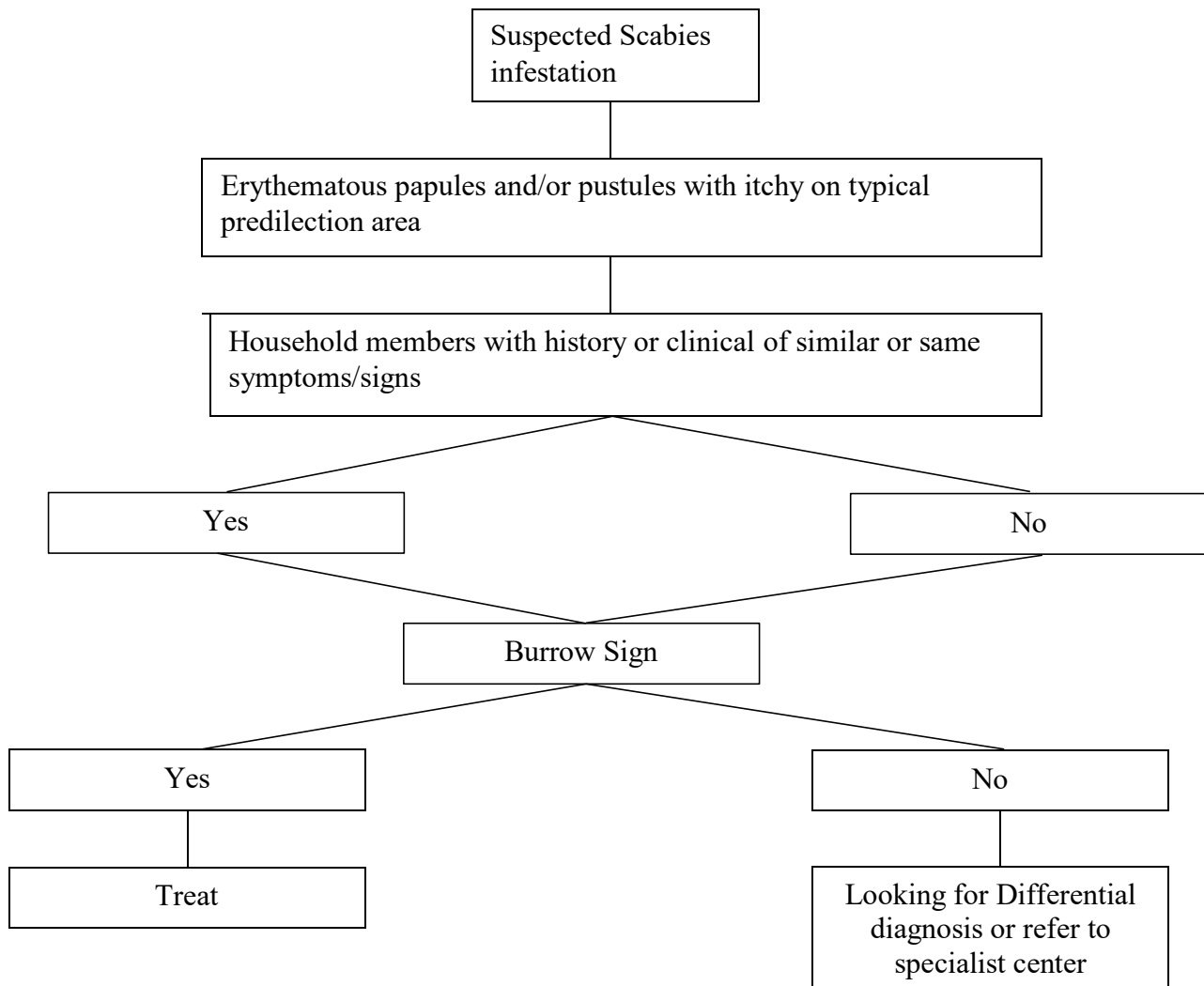
Itching usually persists for about one to two weeks after successful treatment, but it should be evaluated if prolonged. The persistent itching could be caused by irritation, allergic contact dermatitis to medicaments, post-scabetic eczema, or possible treatment failure. The post-treatment itch includes:

- Antihistamines: non-sedative antihistamine for day time and sedative antihistamines for night time, but sedative antihistamine must be used with caution in children less than 2 years old.
- Corticosteroids: topical or short course of oral steroids (0.5 mg/kg depending on the severity).
- Emollients: regular application of emollients for dry and eczematous skin.

Follow-up

Repeat visits 2 weeks and 4 weeks after initial treatment are recommended. Patients should be reviewed again at the end of one month to ensure they are cured of scabies. This is the length of time taken for lesions to heal and in cases where there is inadequate treatment for residual eggs and mites to reach maturity, causing symptoms to reappear. Patients can be re-treated if necessary.

Algorithm for diagnosis and treating scabies



VII. EVOLUTION, PROGNOSIS AND COMPLICATION

- Scabies typically evolves from initial skin infestation to symptomatic disease over a period of 2 to 6 weeks in individuals without previous exposure. Those previously infested may develop symptoms within 1 to 4 days due to sensitization.
- Prognosis is favorable with proper treatment, but severe cases can lead to secondary infections
- Complications include secondary bacterial infections such as impetigo, cellulitis, and eczema. In severe cases, especially in immunocompromised individuals, crusted scabies can occur, leading to more extensive skin involvement and higher risk of bacterial superinfection.

VIII. PREVENTION/HEALTH PROMOTION

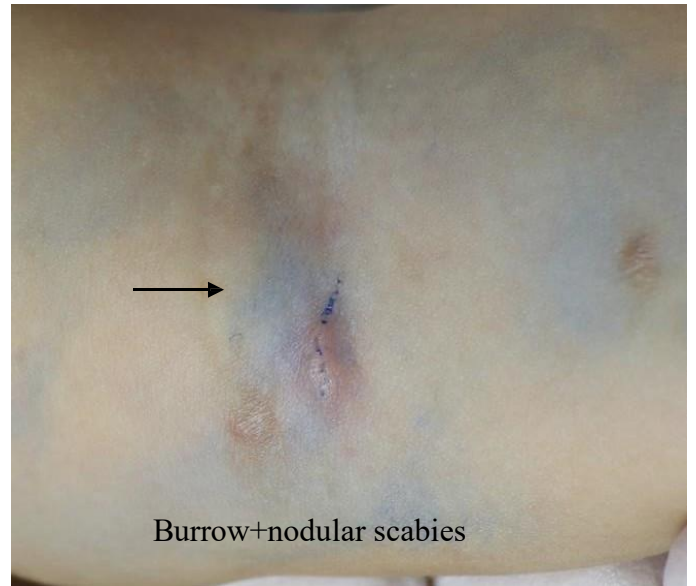
The risk of scabies can be reduced by limiting the number of sexual partners and observing strict personal hygiene when living in crowded spaces (e.g., no sharing of clothes, bedding, and towels, and avoidance of skin-to-skin contact).

IX. INDICATIONS FOR REFERRAL TO SPECIALIST CARE

- Diagnostic uncertainty/failure to respond to adequate treatment of the patient and contacts
- Crusted scabies
- Patients with complications such as severe infections



Burrow



Burrow+nodular scabies



Nodular Scabies



Crusted scabies

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DERMATOPHYTES INFECTIONS (Tinea)

Dr SRON Bunlong, Dr BORN Bori, Dr CHUKMOL Sirithida, Prof MEY Sithach

I. DEFINITION

Dermatophytes are fungi that invade and multiply within keratinized tissues (skin, hair and nails). These infections are common worldwide with prevalence of 10-20% of the world population. Risk factors are poor hygiene, stray pet, share clothing or close contact with infected persons or animals.

II. EPIDEMIOLOGY

Tinea infections, caused primarily by dermatophytes such as *Trichophyton* and *Microsporum* species, are a significant global health concern. Approximately 20-25% of the world's population is affected by superficial fungal infections. The incidence of tinea infections has been rising, compounded by increasing antifungal resistance. Notably, *Trichophyton indotineae* has shown a high resistance to terbinafine, a commonly used antifungal treatment

III. ETIOLOGY

Dermatophytes are filamentous fungi in the genera *Trichophyton*, *Microsporum*, and *Epidermophyton*.

IV. ATHOGENESIS

Transmission: Transmission of dermatophytes to humans occurs via three major sources: human to human, animal to human, and soil to human.

Adherence and Invasion: The initial step involves contact between the infectious elements of the fungus and the skin. The fungi then produce keratinase, an enzyme that allows them to adhere to the host and invade the stratum corneum of the skin.

Severity of Infection: The severity of clinical symptoms is influenced by the extent of invasion and the host's immune response.

1. Clinical subtypes

The major clinical subtypes of dermatophyte infections include infections of the hair, epidermis, and nails:

Localization of the infection

- **Hair:**
 - **Tinea capitis:** Infection of scalp hair
 - **Tinea barbae:** Infection of beard hair
 - **Severe forms of tinea capitis:** **Kerion** and **Favus**
- **Tinea faciei:** Infection of the face
- **Tinea manuum:** Infection of the hand
- **Tinea corporis:** Infection of body area
- **Tinea cruris:** Infection of the groin
- **Tinea pedis:** Infection of the foot (also known as athlete's foot)
- **Onychomycosis:** Infection of the nail

V. DIAGNOSIS

Cutaneous dermatophyte infection maybe strongly suspected based upon clinical finding. However, these findings can overlap with other cutaneous disorders, we typically confirm the diagnosis with testing:

5.1. Physical examination:

A careful physical examination of the affected areas allows to identify of findings consistent with or inconsistent with the suspected dermatophyte infection.

- **Tinea capitis:** scaly patches with alopecia or patches of alopecia with visible black dots.
- **Kerion:** boggy plaque that is often contains pustules. Purulent drainage may be present.
- **Tinea corporis:** pruritic, circular or oval, erythematous scaling patch or plaque that spreads centrifugally. Central clearing, active and raised border.
- **Tinea cruris:** erythematous patch on the proximal medial thigh, slightly elevated, sharply demarcated border.
- **Tinea pedis:** pruritic erosions or scales between the toes, and some hyperkeratotic lesion on the plantar area.
- **Onychomycosis:** whitish, yellowish, or brownish discoloration of a distal corner of a nail. The great toe is often the initial site of infection.

5.2. Diagnostic tests – The approach to diagnostic testing differs based upon the suspected type of infection:

- 1- **Potassium hydroxide (KOH):** fungal hyphae will appear as thin, filamentous tree branches with direct microscopic examination.
- 2- **Culture:** fungal culture (on Sabouraud's medium) allows for both identification and speciation of fungal infection. But the result takes a few weeks.
- 3- **Biopsy:** Using PAS reveal fungal elements.

VI. IFFERENTIAL DIAGNOSIS

7.1 Subacute cutaneous lupus erythematosus (SCLE) – SCLE is a subtype of cutaneous lupus erythematosus that can manifest as annular or polycyclic, scaly plaques on sun-exposed skin SCLE can be idiopathic or occur in association with systemic lupus erythematosus or drug exposure.

7.2 Granuloma annulare – Granuloma annulare is a benign, inflammatory condition that classically presents with one or more erythematous or violaceous, annular plaques on the extremities. Unlike tinea corporis, scale is absent.

7.3 Erythema annulare centrifugum – Erythema annulare centrifugum, an inflammatory skin disorder of unknown etiology, exhibits annular plaques. A trailing rim of scale is often evident in the superficial variant of this disorder.

VIII. TREATMENT PRINCIPLES

a. Topical antifungals

- **Azole (Imidazoles):** Clotrimazole, Ketoconazole, Econazole, Miconazole, Sertaconazole, Luliconazole, Eberconazole: once to twice daily.
- **Allylamines:** Terbinafine, Naftifne: Can also be use once to twice dialy.

b. Oral antifungals

i. Adults

Griseofulvin:

- Griseofulvin microsize 500 to 1000 mg/day for two to four weeks

- Griseofulvin ultramicrosize 375 to 500 mg per day for two to four weeks

Terbinafine

Terbinafine 250 mg per day for one to two weeks

Azole derivative

Itraconazole 200 mg per day for one week (longer courses may be necessary in some patients).

Fluconazole 150 to 200 mg once weekly for two to four weeks

Use of oral ketoconazole is no longer recommended because of risk for severe liver injury, adrenal insufficiency, and drug interactions most importantly with HIV medications.

ii. **Children** – Children are treated for similar durations.

-Terbinafine tablets:

10 to 20 kg – 62.5 mg per day

20 to 40 kg – 125 mg per day

Above 40 kg – 250 mg per day

-Itraconazole: 3 to 5 mg/kg per day (up to 200 mg per day)

-Fluconazole: 6 mg/kg once weekly for two to four weeks

-Griseofulvin:

Griseofulvin microsize 10 to 20 mg/kg per day for two to four weeks

Griseofulvin ultramicrosize 5 to 15 mg/kg per day for two to four weeks

• **Onychomycosis**

Terbinafine is the first line treatment:

• **Fingernail onychomycosis** – 250 mg per day for 6 weeks

• **Toenail onychomycosis** – 250 mg per day for 12 weeks

9. Special cases in Pregnancy and lactation

Absorbent antifungal dusting powders (clotrimazole) may be used once the active infection is taken care of. Onychomycosis treatment may be attempted after lactation is stopped as it mostly requires oral antifungals for prolonged duration.

10. Evolution and prognosis

Evolution: The evolution of the infection can vary based on the site and extent of involvement, as well as the host's immune response. Initially, symptoms might be mild and localized but can spread and intensify without proper treatment.

Prognosis: The prognosis for dermatophyte infections is generally good with appropriate treatment. Most infections respond well to topical antifungals, though more extensive or resistant cases may require oral antifungal therapy.

IX. COMPLICATION

Potential complications of dermatophyte infection include secondary bacterial infection, tinea incognito, Majocchi's granuloma, Id reactions, and severe tinea capitis can lead to scarring alopecia.

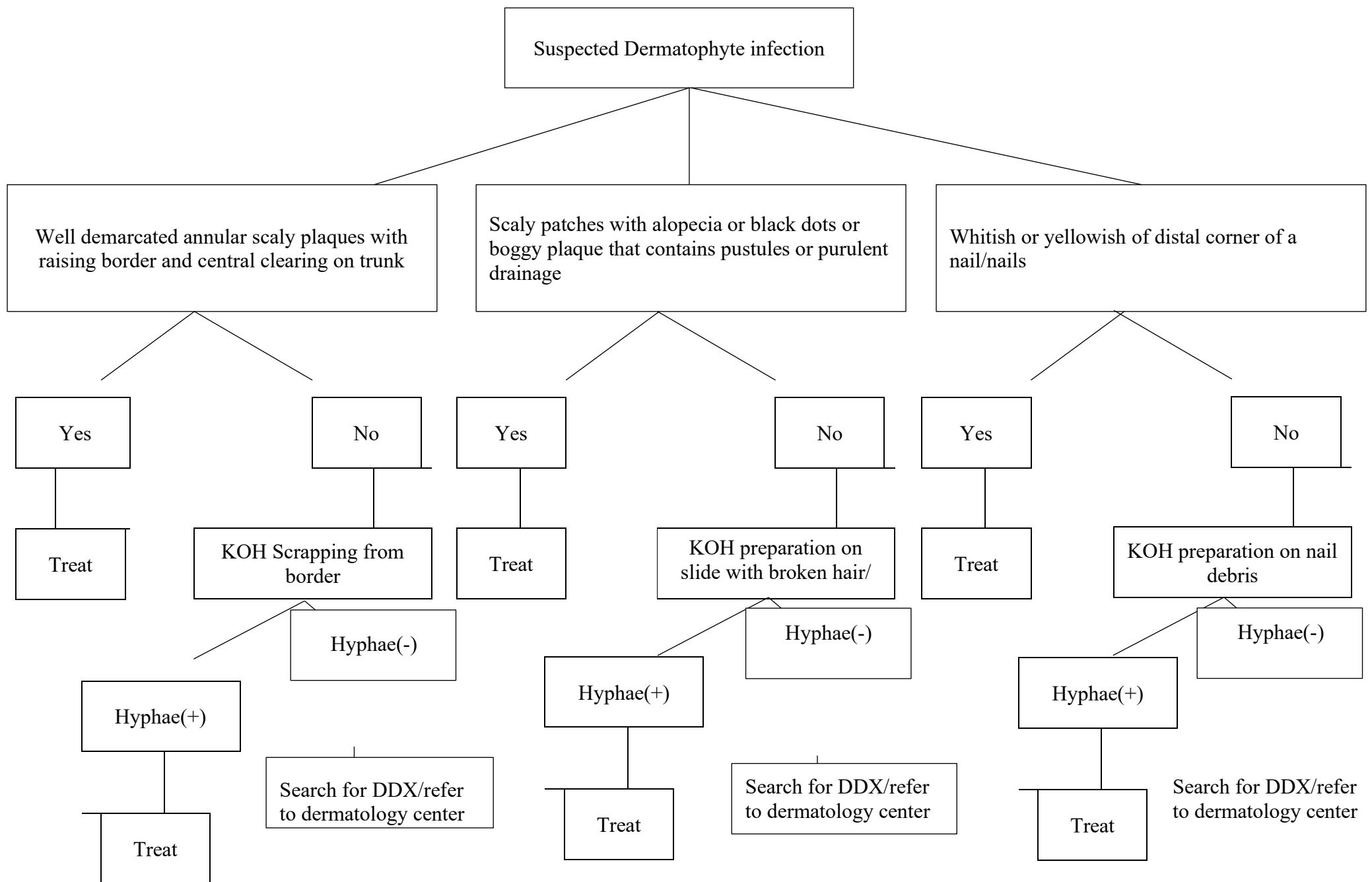
- Secondary infection:** can occur in association with dermatophyte infections. Patients who exhibit significant erosions, ulceration, pain, or malodor in the affected area should have a Gram stain and culture to evaluate for secondary bacterial infection.
- Tinea incognito:** may result after the use of topical corticosteroids for treatment. Alterations to the clinical presentation may include diminished erythema and scale or loss of a well-defined border.

- c. **Majocchi's granuloma** characterized by the spread of epidermal infection into deep portions of the hair follicle and dermis with erythematous, perifollicular papules or small pustules.
- d. **Id reactions** also known as Autoeczematization reactions are secondary, dermatitic eruptions that occur in association with primary, often inflammatory, skin disorders.

X. Prevention and recommendation

- Keep your skin clean and dry.
- Wear shoes that allow air to circulate freely around your feet.
- Don't walk barefoot in areas like locker rooms or public showers.
- Clip your fingernails and toenails short and keep them clean.
- Change your socks and underwear at least once a day.
- Don't share clothing, towels, sheets, or other personal items with someone who has ringworm.
- Wash your hands with soap and running water after playing with pets. If you suspect that your pet has ringworm, take it to see a veterinarian. If your pet has ringworm, follow the steps below to prevent spreading the infection.
- If you're an athlete involved in close contact sports, shower immediately after your practice session or match, and keep all of your sports gear and uniform clean. Don't share sports gear (helmet, etc.) with other players
- Refer to specialist center if uncertain diagnosis or treatment failure





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VARICELLA

Uk Vannady, CHUKMOL Sirithidai, SRON Bunlong, BORN Bori, MEY Sithach

I. DEFINITION

Varicella, commonly known as chickenpox, is a highly contagious viral infection that causes an acute fever and blistered rash, mainly in children. This condition is the primary infection with the *Varicella-zoster virus* (VZV).

II. EPIDEMIOLOGY

Chickenpox is a common disease, with most cases occurring in the pediatric population. Varicella has neither a racial nor a sexual predilection. In temperate countries, it is a common disease of children, with most cases occurring during the winter and spring.

III. ETIOLOGY AND MICROBIOLOGY

Varicella-zoster virus also known as human herpesvirus 3 (HHV-3) of herpes virus subfamily Alphaherpesvirinae. Airborne droplets are the usual route of transmission of varicella, although direct contact with vesicular fluid is another mode of spread, and the incubation period is 11–20 days.

Varicella is extremely contagious, and 80–90% of susceptible household contacts develop a clinically evident infection. The affected individual is infectious from 1–2 days before skin lesions appear until all the vesicles have crusted.

IV. PHYSIOPATHOLOGY

Entry of VZV is through the mucosa of the upper respiratory tract and oropharynx, where VZV infects tonsillar T cells that disseminate virus via the blood and lymphatics (the primary viremia). Infected T cells carry virus to the reticuloendothelial system, the major site of virus replication during the remainder of the incubation period, and to the skin. Viremia occurs early in the incubation period, but VZV replication and rash formation are delayed by innate immune responses, for example, by interferon and natural killer cells, and by developing VZV-specific immune responses.

Virus replication eventually overcomes these host defenses, so that about 10 to 14 days after infection a much larger (secondary) viremia occurs, resulting in systemic symptoms and skin lesions.

In addition, immunopathology contributes to rash development. Skin lesions appear in successive crops, reflecting a cyclic viremia that in the normal host is terminated after about 3 to 5 days by VZV-specific immune responses, primarily VZV-specific T cell-mediated immune responses. Virus circulates in mononuclear leukocytes, primarily lymphocytes.

The immunologic memory that develops during varicella persists for life and is essential to protect against second episodes of varicella and against herpes zoster.

V. DIAGNOSIS

Chickenpox is usually diagnosed clinically based on the characteristic rash and successive crops of lesions. Lesions may be found in all stages of development and healing in affected sites.

1. Clinical presentations

- **History:** Exposure to an infected contact within the incubation period of 10-21 days.

- **Prodromal phase:** 5 days before eruptions and most common in adults. Symptoms included aching muscles, nausea, decreased appetite, and headache followed by a rash, oral sores, malaise, and a low-grade fever. In children, the illness may not be preceded by prodromal symptoms, and the initial sign could be a rash or oral cavity lesion.
- **Skin lesion phase:** The rash begins as small red dots on the face, scalp, torso, upper arms and legs. Over the next ten to 12 hours, it progresses to small bumps, blisters, and pustules; and eventually umbilication and scab formation in different stages of development. New vesicle formation generally stops within four days, and most lesions have fully crusted by day six in normal hosts. The vesicular rash of varicella, which is usually pruritic or might be painful, appears in successive crops over several days.
- **Recovery phase:** Crusts tend to fall off within about one to two weeks. The blisters clear up might leave a temporary area of hypo/hyperpigmentation or a few scars in the skin.

2. Investigations

Laboratory tests are undertaken to confirm the diagnosis if needed but usually unnecessary.

- Tzanck smear from a vesicle will usually show characteristic multinucleate giant cells can confirm the infection and type the virus.
- PCR to detect VZV DNA from the base of a deroofed vesicular lesion is the most reliable and sensitive method, and is now considered the gold standard.
- Viral culture of VZV is possible but is insensitive, time-consuming, and expensive because it requires special media.

VI. DIFFERENTIAL DIAGNOSIS

- **Arthropod bites:** An inflammatory reaction after bite or sting and might manifested by pruritus, redness, swelling. Look for small central punctum with surrounding erythema and swelling.
- **Impetigo:** An infectious disease presents as erythematous pustules and papules with minimal or no surrounding erythema, ruptures releasing serous contents which leaving typical honey-colored crust.
- **Drug eruptions:** History of taking suspected drugs around 1 week to 2 months prior to skin presentation which appear as mild itchy erythematous maculo-papules distributed on trunk, face, lower and upper extremities.
- **Dermatitis herpetiformis:** An autoimmune blistering disorder associated with gluten sensitivity. Classic clinical manifestation includes intensely itching with inflammatory papules and vesicles within predilected areas such as forearms, knees, scalp and buttocks.

VII. MANAGEMENT

1. Objective of therapy

Treatment mainly aims to relieve symptoms, reduce the risk of disseminated disease, and prevent complications.

2. Critical signs

- High Fever > 39 °C
- Severe Rash especially if the lesions are hemorrhagic (bleeding) or rapidly spreading, can be a sign of a more severe form of varicella, such as disseminated varicella.
- Neurological symptoms such as severe headache, vomiting, confusion, seizures, or signs of meningitis/encephalitis.

- Respiratory Distress: difficulty breathing, persistent cough, chest pain, or signs of pneumonia.
- Thrombocytopenia or bleeding tendencies: the presence of petechiae (small red or purple spots on the skin) or other signs of bleeding, such as easy bruising or mucosal bleeding.
- Hepatic dysfunction: jaundice, abdominal pain, or signs of liver failure could indicate varicella hepatitis, which is rare but severe.

3. Mortality risk factors

- Elderly or Patients with weakened immune systems, such as those with HIV/AIDS, cancer, organ transplants, or those on immunosuppressive therapies (e.g., corticosteroids), are at a significantly higher risk of severe complications and mortality from varicella.
- Neonates particularly those born to mothers who develop varicella shortly before or after delivery, are at high risk for severe disease and mortality.
- Pregnant women, particularly in the first and second trimesters, are at risk for severe complications such as pneumonia, which can be fatal.
- Complications such as varicella pneumonia, encephalitis, and bacterial superinfections significantly increase the risk of mortality.

4. Outpatient management

Varicella in immunocompetent children can be treated symptomatically with antipyretics acetaminophen but must avoid aspirin), antihistamines, calamine lotion, and lukewarm baths (29C-35C). If started within 24–72 hours after the onset of the cutaneous eruption, acyclovir has been shown to decrease the duration and severity of varicella. Oral acyclovir and valacyclovir are FDA-approved for the treatment of varicella in children (2–17 years of age) while acyclovir is approved for adults.

These antiviral agents are recommended for varicella in healthy adolescents and adults as well as in children with chronic cutaneous or pulmonary disorders and in those receiving chronic salicylate therapy, inhaled corticosteroids, or intermittent oral corticosteroids. However, routine antiviral therapy is not recommended for otherwise healthy children with varicella due to the self-limited disease course and modest benefits of treatment.

5. Algorithm and special cases

The decision to initiate antiviral therapy depends upon the patient's age, the presence or absence of comorbid conditions, and the patient's clinical presentation. Although varicella is usually a self-limited disease, if complications develop, they can be life-threatening.

Figure 1: Approach to antiviral therapy in immunocompetent individual without complications

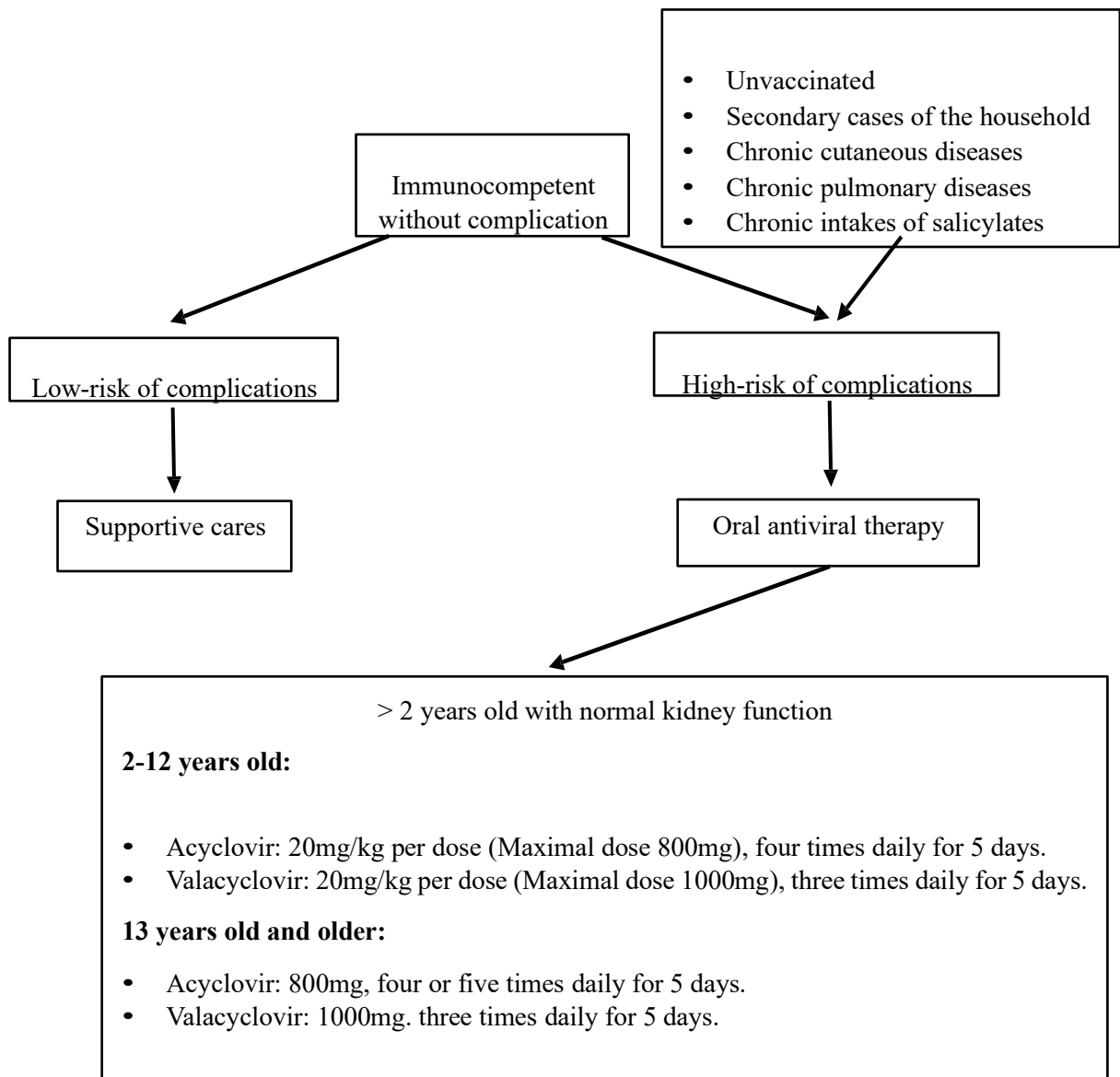


Figure 2: Approach to antiviral therapy in immunocompromised individual without complications

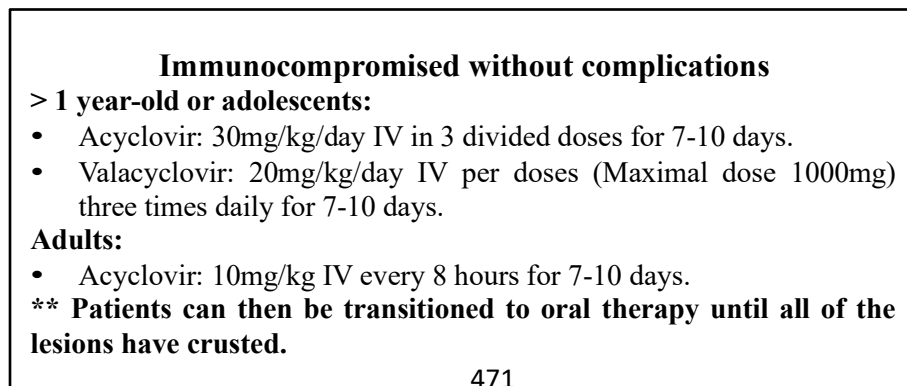


Figure 3: Approach to antiviral therapy in pregnancy individual

Pregnancy without complications • Oral acyclovir, 800mg five times daily for 7 days. • Oral valacyclovir, 1000mg three times daily for 7 days. Pregnancy with complications • IV acyclovir: 10-15mg/kg every 8 hours for 5-10 days.

Figure 4: Approach to antiviral therapy in individual with complications

Individual with complications > 1 year-old or adolescents: • Acyclovir: 30mg/kg/day IV in 3 divided doses for 7-10 days. • Valacyclovir: 20mg/kg IV per doses (Maximal dose 1000mg) three times daily for 7-10 days. Adults: • Acyclovir: 10mg/kg IV every 8 hours for 7-10 days. ** Patients can then be transitioned to oral therapy until all of the lesions have crusted.
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VIII. EVOLUTION AND PROGNOSIS

Chicken-pox is mostly a self-limited disease with preferable prognosis. The blister clears up within one to three weeks, but may leave a few scars. These are most often depressed (anetoderma), but they may be thickened (hypertrophic scars). Scarring is prominent when the lesions get infected with bacteria.

IX. COMPLICATIONS

- Neurological complications: Encephalitis and, mostly in the past, Reye syndrome, are the most serious complications of VZV infection, although they are uncommonly seen.
- Pneumonia: Rare but accounts for the majority of morbidity and mortality seen in adults with varicella.
- Varicella Hepatitis
- Secondary Bacterial infection
- Congenital Varicella Syndrome: If a woman contracts varicella during the first or second trimester, the fetus is at risk which can cause limb hypoplasia, eye defects, and neurological abnormalities.
- Hemorrhagic Complications

X. PREVENTION AND RECOMMENDATION

Postexposure varicella vaccination (within 72–120 hours) may prevent or modify disease in nonimmune individuals who are ≥ 12 months of age, immunocompetent, and eligible to receive this live attenuated vaccine.

For VZV vaccine which is approved by the FDA in 1995, the live attenuated VZV vaccine is highly efficacious, with seroprotection rates of $\sim 85\%$ after one dose and $\geq 99\%$ after two doses in healthy children. Two doses of the vaccine, routinely given at ages 12–15 months and 4–6 years, are recommended to improve protection and counteract waning vaccine-induced immunity.

However, in a report based on surveillance of 350 000 subjects who received one vaccine dose, loss of vaccine-induced immunity over time was observed (within 1 year of vaccination, there were 1.6 cases of varicella per 1000 person-years versus 58.2 per 1000 person-years at 9 years postvaccination).

Administration of VZV Vaccine is contraindicated during pregnancy and in individuals with immunosuppression due to hematologic malignancies, HIV infection ($CD4^+ < 15\%$ or < 200 cells/mcl), other T-cell immunodeficiencies, or systemic immunosuppressive therapy such as TNF inhibitors or high dose of prednisone ($> 2\text{mg/kg/bw}$).

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Annex: Courtesy of varicella patients at the dermatology department of Cambodia-China Friendship Preah Kossamak Hospital.





Generalized eruption of varicella in adult patient

HERPES SIMPLEX VIRUSES

HUM Davina, UK Vannady, SRON Bunlong, CHUKMOL Sirithida, BORN Bori MEY Sithach

I. DEFINITION

Herpes is a common infection caused by herpes simplex viruses. It can cause painful blisters that come and go, usually around the mouth and genitalia.

II. EPIDEMIOLOGY

Herpes simplex viruses have a worldwide distribution. It is estimated that approximately one-third of the world's population has experienced a symptomatic HSV infection.

In children less than 10 years of age, most herpetic infections are caused by HSV-1 (>80–90%). Most HSV-1 infections occurred during the first five years of life in South-East Asia. Prevalence increased with age, and was high across all regions, but highest in Africa (87% overall prevalence) and lowest in the Americas (40–50%) and in South-East Asia (58–59%).

III. ETIOLOGY and Microbiology

HSV belongs to the alpha herpesvirus group. It is an enveloped virus of linear with double-stranded DNA genome. There are two types of herpes simplex virus: HSV-1 and HSV-2.

HSV-1 is mainly transmitted via contact with the virus in sores, saliva or surfaces in or around the mouth. Less commonly, HSV-1 can be transmitted to the genital area through oral-genital contact to cause genital herpes. It can be transmitted from oral or skin surfaces that appear normal; however, the greatest risk of transmission is when there are active sores. People who already have HSV-1 are not at risk of reinfection, but they are still at risk of acquiring HSV-2.

HSV-2 is mainly transmitted during sex through contact with genital or anal surfaces, skin, sores or fluids of someone infected with the virus. HSV-2 can be transmitted even if the skin looks normal and is often transmitted in the absence of symptoms.

In rare circumstances, herpes (HSV-1 and HSV-2) can be transmitted from mother to child during delivery, causing neonatal herpes.

IV. PATHOPHYSIOLOGY

Transmission of HSV can occur during asymptomatic periods of viral shedding. HSV-1 is spread primarily through direct contact with contaminated saliva or other infected secretions, while HSV-2 is spread primarily by sexual contact. Virus replicates at the mucocutaneous site of infection and then travels by retrograde axonal flow to the dorsal root ganglia, where it establishes latency until reactivation. Latency enables the virus to exist in a relatively noninfectious state for varying periods of time in its host.

HSV can be reactivated either spontaneously or by a trigger such as emotional stress, ultraviolet light, fever, menstruation, immunosuppression, surgical or dental procedures, and other local tissue damage.

V. DIAGNOSIS

1. Clinic

- **Herpes labialis:** this is the most common manifestation of recurrent HSV-1 infection, referred to by many as “cold sores”. A prodrome of pain, burning, and tingling often occurs at the affected site, commonly the face, around the lips, followed by the development of erythematous papules that rapidly develop into tiny, thin-walled, intraepidermal vesicles that become pustular and ulcerate. Maximum viral shedding is in

the first 24 hours of the acute illness but may last 5 days.

- **Acute herpetic gingivostomatitis:** often present as gingivostomatitis in children or as pharyngitis and a mononucleosis-like syndrome in young adults. The mouth and lips are the most common sites of involvement, with oral lesions typically appearing on the buccal mucosa and gingivae. Edema and painful oropharyngeal ulcerations can lead to dysphagia and drooling.
- **Herpetic whitlow:** typically present with a painful swollen digit, often the thumb or index finger (distal phalanx) with one or more (often clustered) 1–3 mm fluid-filled vesicles with surrounding erythema. Vesicular fluid is usually clear but can progress to become turbid, seropurulent, or hemorrhagic. Vesicles can coalesce into larger bullae and can spread proximally. Itching, burning or altered sensation on affected finger or malaise prodrome and fever.
- **Herpes gladiatorum** it presents initially with a vesiculopustular rash on an erythematous base that progresses to ulceration and may be misdiagnosed as folliculitis or impetigo. Outbreaks of HSV-1 skin infections have occurred among wrestlers or boxers in the context of close contact to skin, likely exposure to saliva, and superficial skin abrasions, which can occur during matches.
- **Eczema herpeticum:** an explosive development of a vesiculopustular eruption caused by herpes simplex virus in an individual with an atopic condition. The clustering and often umbilication of the vesicles are characteristic with sites of the dermatitis most commonly affected.
- **Genital herpes:** painful, grouped vesicles appear on an erythematous base and may become umbilicated, followed by progression to pustules, erosions, and/or ulcerations with a characteristic scalloped border. In males, lesions commonly occur on the glans penis or the penile shaft; in females, lesions may involve the vulva, perineum, buttocks, vagina, or cervix. There is accompanying pain, itching, dysuria, vaginal and urethral discharge, and tender inguinal lymphadenopathy

2. Investigation

In many instances the history and clinical findings may be sufficient but the social, emotional, and therapeutic implications of a diagnosis dictate that it be confirmed by laboratory testing when possible

- **PCR:** is more sensitive than viral isolation and has become the preferred method for diagnosis. PCR has been extensively used for the diagnosis of CNS infections and neonatal herpes.
- **Viral Culture** In culture, HSV causes typical cytopathic effects, and most specimens will prove positive within 48 to 96 hours after inoculation.
- **Direct Fluorescent Antibody (DFA) Test** Direct fluorescent antibody staining of lesion scrapings also can be used, but its sensitivity is lower than that of viral culture.
- **Tzanck Smear:** scrapings from early lesions, in particular the base/edges of a freshly unroofed vesicle, reveals multinucleated epithelial giant cells in ~60–75% of HSV outbreaks.
- **Serology test:** Serologic detection of IgG antibodies to HSV can be helpful in certain settings, but the results are often misinterpreted. The main function of serologic testing is to differentiate a primary episode from a recurrent infection.

VI. DIFFERENTIAL DIAGNOSIS

Differential diagnosis of Herpes labialis:

- **Oral aphthae** typically lesion(s) on the buccal mucosa without associated vesicles. Aphthous ulcers almost always involve only the nonkeratinized mucosa, while recrudescing HSV almost always involves the keratinized mucosa in healthy hosts.
- **Erythema multiforme:** classical target lesions are well-defined, circular, erythematous macules or papules that are less than 3 cm in diameter and have 3 distinct color zones, with a central zone that has a bulla or crust. EM lesions first appear in a symmetrical distribution on acral sites and progress in a centripetal fashion.
- **Steven-Johnson syndrome:** associated with severe cutaneous involvement characterized by extensive necrosis and detachment of the epidermis. Mucous membranes are affected in more than 90 percent of patients, usually at two or more distinct sites. The initial cutaneous presentation of SJS may mimic an exanthematous rash which evolve over the course of one to two days to develop dusky erythema, purpuric spots, atypical targets, and flaccid bullae. Lesions start on the face and thorax before spreading to other areas and are symmetrically distributed.
- **Oral candidiasis:** a fungal infection of the mouth caused by *Candida albicans*, may arise suddenly as an acute infection or persist for long periods as a chronic infection and characterize as white cottony patches on gums, tongue and inside the mouth that can be peeled off easily leaving a raw area.
- **Hand-Foot-Mouth disease:** a common, self-limiting viral infection that causes blisters on the hands, feet, and inside or around the mouth. It mainly affects children under the age of 5 years.

Differential diagnosis of Herpes genitalis:

- **Primary syphilis:** a sexually transmitted infection is characterized by chancre(s) which typically, is a solitary, small firm red painless papule on the genital area that quickly becomes a painless ulcer with a well-defined margin and an indurated base. Non-tender lymphadenopathy might occur in the region of ulceration (eg, inguinal lymphadenopathy in ulcers on the genitals and cervical lymphadenopathy with an oral ulcer).
- **Chancroid:** a sexually transmitted infection which is initially developed one or more red erythematous papules which quickly evolve to pustules and become larger until they break down into an ulcer. A typical ulcer is deep, has shaggy undermined borders and is very painful. The base of the ulcer has a purulent exudate and bleeds easily.
- **Lymphogranuloma venereum:** a sexually transmitted infection which is appeared as small painless genital papule, pustules, or shallow ulcers appear on the skin. These initial lesions are transient, meaning they heal quickly and disappear. They often go unnoticed during primary infection and developed into secondary infection with painful and inflamed swollen lymph glands develop in the groin area which can rupture and drain pus.

VII. MANAGEMENT

1. Objective of management

The management of HSV-1 infection in the immunocompetent host depends upon a variety of considerations, including:

- Whether the patient has primary HSV-1 infection or reactivation disease
- The severity of symptoms
- The site of infection (eg, mucosal versus disseminated disease)
- The frequency of recurrences

2. Critical signs and Mortality risk factors

In immunocompetent individuals, both primary and recurrent HSV infections are most commonly asymptomatic and, when manifesting clinically, they are usually transient with a moderate to mild severity, respectively. In contrast, in immunocompromised hosts, such as cancer patients undergoing chemo/radiotherapy, HIV positive individuals and solid organ and bone marrow transplant recipients both primary and secondary HSV-1 and HSV-2 orogenital infections are more aggressive, extensive, and persistent as well as on neonates.

Look for signs and symptoms of meningoencephalitis which include fever, headache, nausea, vomiting, and nuchal rigidity, including abnormalities in brain function, altered mental status, personality changes, behavioral changes, sensory deficits, motor deficits, speech disorders, hemiparesis, flaccid paralysis, and paresthesia.

For Herpes simplex keratitis looking for herpetic lesions on the eyelids or periorbital skin, the conjunctiva will be injected, and a lymph node may be palpable on the involved side. Corneal sensation is often reduced due to damage to the trigeminal nerve.

3. First line treatment

Most patients with a first episode of oral/genital HSV can be treated with oral therapy, IV therapy is typically reserved for the management of complicated infection (eg, immunocompromised, central nervous system, end-organ diseases including hepatitis and pneumonitis and disseminated disease).

• Uncomplicated diseases

- Acyclovir: 400 mg orally 3times/day for 7–10 days OR
- Valacyclovir: 1000 mg orally 2times/day for 7–10 days

• Complicated diseases

- Acyclovir PIV : 5-10mg/kg every 8 hours several days until clinical improvement) followed by Valacyclovir 1000 mg orally 3times/day for Complete 10-21 days of total treatment

• Recurrent diseases

- Episodic therapy
 - + Acyclovir 800 mg orally 2times/day for 7–10 days OR
 - + Acyclovir 800 mg orally 3times/day for 2 days OR
 - + Valacyclovir 500 mg orally 3times/day for 3 days OR
 - + Valacyclovir 1000 mg orally once daily for 5 days
- Suppressive therapy (usually for patients with ≥ 6 episodes/year)

- + Acyclovir 400 mg orally 2times/day OR
- + Valacyclovir 1000 mg orally once daily

VIII. SPECIAL CASES

1. Patients with HIV

- Episodic therapy
 - + Acyclovir 400 mg orally 3times/day for 5–10 days OR
 - + Valacyclovir 1000mg orally 2 times/day for 5–10 days
- Suppressive therapy (usually for patients with ≥ 6 episodes/year)
 - + Acyclovir 400–800 mg orally 2-3times/day OR
 - + Valacyclovir 500 mg orally 2 times/day

2. Pregnant women and breastfeeding

- Acyclovir can be administered orally to pregnant women with first-episode genital herpes or recurrent herpes and should be administered IV to pregnant women with severe HSV
- Acyclovir is also believed to be safe during breastfeeding with recommended dose for uncomplicated immunocompetent above
- Suppressive therapy (usually for patients with ≥ 6 episodes/year)
 - + Acyclovir 400 mg orally 3 times/day OR
 - + Valacyclovir 500 mg orally 2 times/day

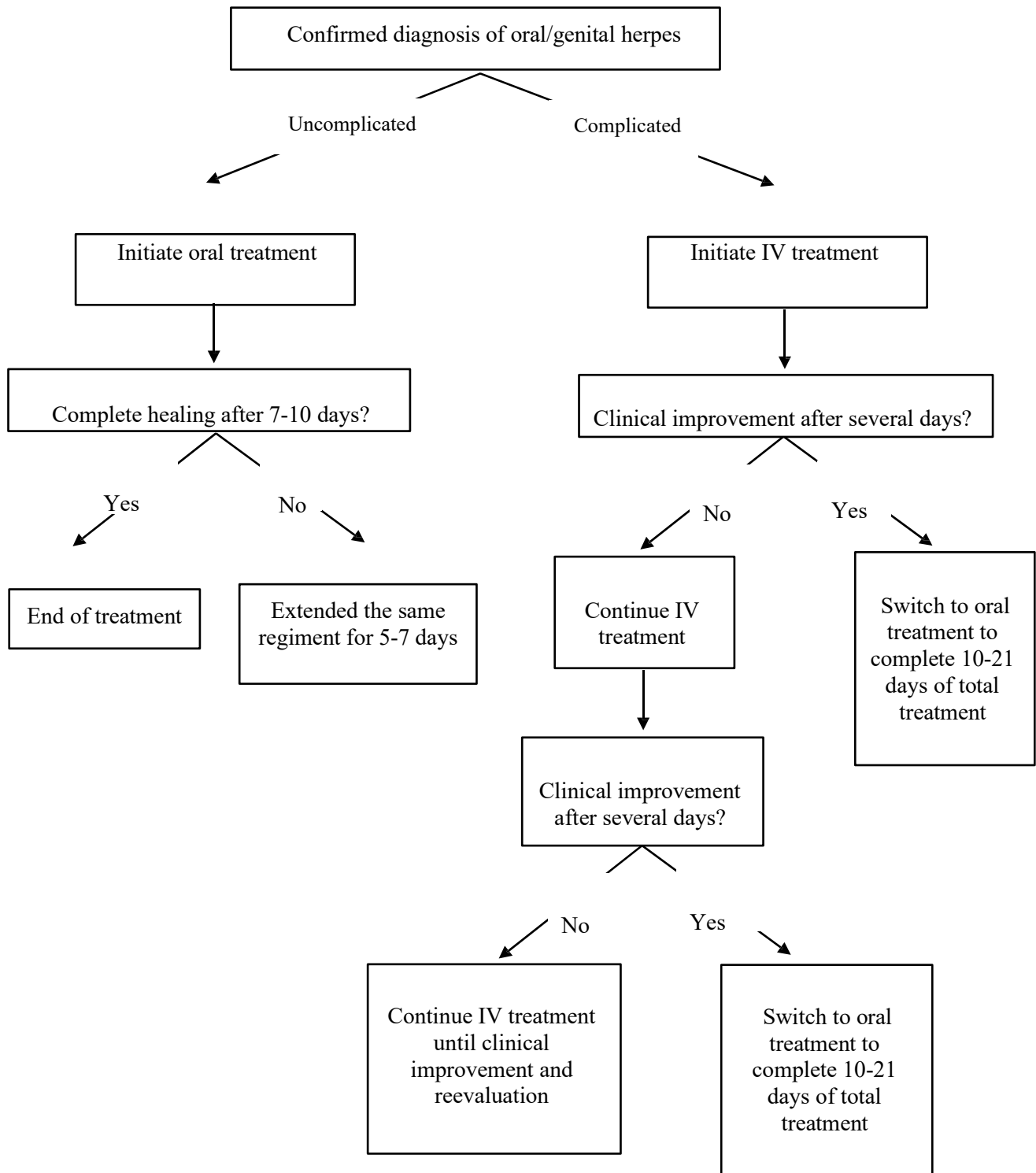
** Treatment recommended starting at 36 weeks' gestation.

3. Neonates

- Acyclovir 20mg/kg body weight IV every 8 hours for 14 days if disease is limited to the skin and mucous membranes or for 21 days for disseminated disease and Central nervous system involvement.

4. Algorithm

Figure 1: Approach to antiviral therapy in oral/genital herpes



IX. EVOLUTION AND PROGNOSIS

For acute herpetic gingivostomatitis lasts 5-7 days, and the symptoms subside in 2 weeks and in men and women, the ulcerative lesions of primary herpes genitalis persist from 4-15 days until encrusting and reepithelialization occur and around 12 days for recurrent herpes genitalis.

X. COMPLICATIONS

- Secondary bacterial infection
- Keratitis or ocular infection
- Meningitis/Encephalitis
- Respiratory tracts infection
- Hepatitis
- Increased the risk of HIV infection for patients with genital herpes

XI. PREVENTIONS AND RECOMMENDATIONS

1. Transmission prevention

- People with symptoms of oral herpes should avoid oral contact with others (including oral sex) and sharing objects that touched saliva.
Individuals with symptoms of genital herpes should abstain from sexual activity while experiencing symptoms. For sexually active people, consistent and correct use of condoms is the best way to prevent genital herpes and other STIs. Condoms reduce the risk; however, HSV infection can still occur through contact with genital or anal areas not covered by the condom.
- Both HSV-1 and HSV-2 are most contagious when sores are present but can also be transmitted when no symptoms are felt or visible.
- Healthcare workers: must use gloves during physical examination of a patient with active HSV lesions and for examination of mucus membranes can reduce the risk of transmission. Providers involved with procedures that may involve exposure to infected secretions, such as suctioning or bronchoscopy, should also wear eye and mouth protection.

2. Prophylaxis for recurrent HSV with identified triggers

- **Sunlight exposure:** Sun protection using high protection factor sunscreens and protective clothing are important.
- **Surgical procedures:** HSV reactivation can be seen in the setting of surgical procedures, such as trigeminal nerve root decompression, facial dermabrasion, and ablative laser resurfacing. Given the high risk of HSV reactivation associated with these procedures, we typically administer antiviral prophylaxis around the time of surgery, even to those without a known history of HSV.

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Annex 1: Iconography of herpes simplex patients at the dermatology department of Cambodia-China Friendship Preah Kossamak Hospital.



Figure 2: Herpes simplex on buttocks (left), Herpes simplex on shoulder (right)

Annex 2: Iconography of herpes simplex



Figure 3: Herpes labialis gingivostomatitis



Figure 4: Herpetic



Figure 5: Herpetic Whitlow



Figure 6: Eczema Herpeticum



Figure 7: Male genital herpes

HERPES ZOSTER

HUM Davina, UK Vannady, SRON Bunlong, CHUKMOL Sirithida, BORN Bori, MEY Sithach

I. DEFINITION

Herpes zoster, also known as shingles, is a viral disease caused by reactivation of latent VZV (*varicella-zoster virus*) that gained access to sensory ganglia during varicella (chicken pox).

II. EPIDEMIOLOGY

The incidence of herpes zoster ranges from 1.2 to 3.4 per 1000 persons per year among younger healthy individuals while incidence is 3.9 to 11.8 per 1000 persons per year among patients older than 65 years. There is no seasonal variation seen with herpes zoster. The recurrences of the disease are uncommon unless patients' immune systems are weakened.

III. Etiology

The disease is caused by the reactivation and replication of the varicella zoster virus that involve the sensory ganglion nerves.

Triggers for herpes zoster include:

- Emotional stress
- Use of medications (immunosuppressants)
- Acute or chronic illness
- Presence of a malignancy

IV. PHYSIOPATHOLOGY

Upon reactivation, the virus replicates in neuronal cell bodies, and virions shed from the cells which are carried down the nerve to the area of skin innervated by that ganglion. In the skin, the virus causes local inflammation and blistering and painful sensation.

The dermatological involvement is centripetal and follows a dermatomal distribution. In most cases, it is the lumbar and cervical roots that are involved, whereas motor involvement is rare. The infection is contagious to individuals who have no prior immunity to varicella-zoster virus however, the rates of transmission are low. The virus can be transmitted either via direct skin contact or by inhaling infected droplets.

V. DIAGNOSIS

Herpes Zoster is usually diagnosed clinically on the basis of the prodromes, characteristic rash and associated signs.

1. Clinical presentations

Clinically, lesions start as closely grouped erythematous papules which, rapidly become vesicles on an erythematous and edematous base and may occur in continuous or interrupted bands in one, two, or more contiguous dermatomes unilaterally. Dermatomes commonly involved are thoracic (53%), cervical (20%), and trigeminal (15%) including ophthalmic and lumbosacral (11%).

The three phases of the infection include:

- Pre-eruptive stage: Presents with abnormal skin sensations or pain within the dermatome affected. this phase appears at least 48 hours prior to any obvious lesions. At the same time, the individual may experience headaches, general malaise, and photophobia.

- **Acute eruptive phase:** Presents of the vesicles and the symptoms seen in the pre-eruptive phase. The lesions initially start as macules and quickly transform into painful grouped vesicles. Those vesicles often ruptured, ulcerated and eventually crusted. Patients are most infectious in this stage until the lesion dry out. Pain is severe during this phase and often unresponsive to traditional pain medications. The phase may last 2-4 weeks but the pain may last longer.
- **Chronic infection phase:** Characterized by recurrent pain that lasts more than 4 weeks. Besides the pain, patients experience paresthesia, shock-like sensations, and dysesthesias. The pain is disabling and may last 12 months or longer.

Special manifestations:

- **Ocular manifestations (either of the following):**
 - Herpes zoster ophthalmicus, suggested by:
 - + Tearing, ocular pain, blurred vision, conjunctival injection, and/or swelling/erythema of the lid and periorbital structures
 - + Uveitis, episcleritis, and/or keratitis
 - Acute retinal necrosis, suggested by:
 - + Blurred vision
 - + Acute iridocyclitis (may have "ciliary flush" - injection that gives the appearance of a red ring around the iris).
- **Ramsay Hunt syndrome:**
 - Triad of ipsilateral facial paralysis, ear pain, and vesicles in the auditory canal and/or auricle. Other symptoms may include tinnitus or nystagmus.

2. Investigations

Laboratory tests are undertaken to confirm the diagnosis if needed but usually unnecessary.

- The Tzanck smear of vesicular fluid shows multinucleated giant cells.
- Varicella-zoster virus-specific IgM antibody in blood is detected during the active infection of chickenpox or shingles but not when the virus is dormant/inactive.
- Direct fluorescent antibody testing of vesicular fluid or corneal fluid can be done when there is eye involvement.
- PCR testing of vesicular fluid, a corneal lesion, or blood in a case with eye involvement or disseminated infection.

VI. DIFFERENTIAL DIAGNOSIS

Cutaneous lesions should be differentiated with:

- **Herpes simplex:** A common infection caused by herpes simplex viruses. It can cause painful blisters that come and go, usually around the mouth and genitals.
- **Dermatitis herpetiformis:** An autoimmune blistering disorder associated with gluten sensitivity. Classic clinical manifestation includes intensely itching with inflammatory papules and vesicles on predilection areas such as forearms, knees, scalp and buttocks.
- **Impetigo:** An infectious disease presents as erythematous pustules and papules with minimal or no surrounding erythema, ruptures releasing serous contents which leaving typical honey-colored crust.
- **Contact dermatitis:** A group of skin disorders in which the skin reaction is due to direct contact with the causative agent. The appearance of contact dermatitis is highly variable. It may affect any area of the body and may be any shape (linear, round, polygonal, irregular). The rash has occurred in the areas contact with causative agents.

- **Steven-Johnson syndrome/Toxic epidermal necrolysis (SJS/TEN):** A severe cutaneous adverse reactions characterized by extensive necrosis and detachment of the epidermis. Mucous membranes are affected in more than 90 percent of patients, usually at two or more distinct sites. The initial cutaneous presentation of SJS/TEN may mimic an exanthematous rashes which evolve over the course of one to two days to develop dusky erythema, purpuric spots, atypical targets, and flaccid bullae. Lesions start on the face and thorax before spreading to other areas and are symmetrically distributed.
- **Arthropod bites:** An inflammatory reaction after bite or sting and might manifested by pruritus, redness, swelling. Look for small central punctum with surrounding erythema and swelling.
- **Cellulitis/Erysipelas:** A common bacterial skin infection that mostly affects the lower superficial dermis, dermis and subcutaneous tissue. It results in a localized area of red, painful, swollen skin, and systemic symptoms. Left untreated, cellulitis can be life-threatening.
- **Lichen striatus:** Lichen striatus is an uncommon self-limiting skin rash that occurs mainly in children. It presents as pink raised spots that join together to form one or more dull red slightly scaly linear bands.

VII. MANAGEMENT

1. Objectives

Goals of therapy is to hasten healing of cutaneous lesions, decrease the duration and severity of acute neuritis, and prevent new lesion formation.

2. Critical signs

- Herpes zoster ophthalmicus
- Disseminated diseases involving more than 2 organs
- More than 2 dermatomes involved
- Bilateral involvement of the same dermatome
- Severe, long-lasting rashes
- Herpes zoster in patients with no history of chicken pox

3. Mortality risk factors

- Older age > 70-years-old, especially female
- Immunocompromised host
- Organ transplant recipients
- Neonates

4. First-line treatment

Treatments of herpes zoster mostly includes antiviral therapy, anti-inflammatory and analgesia.

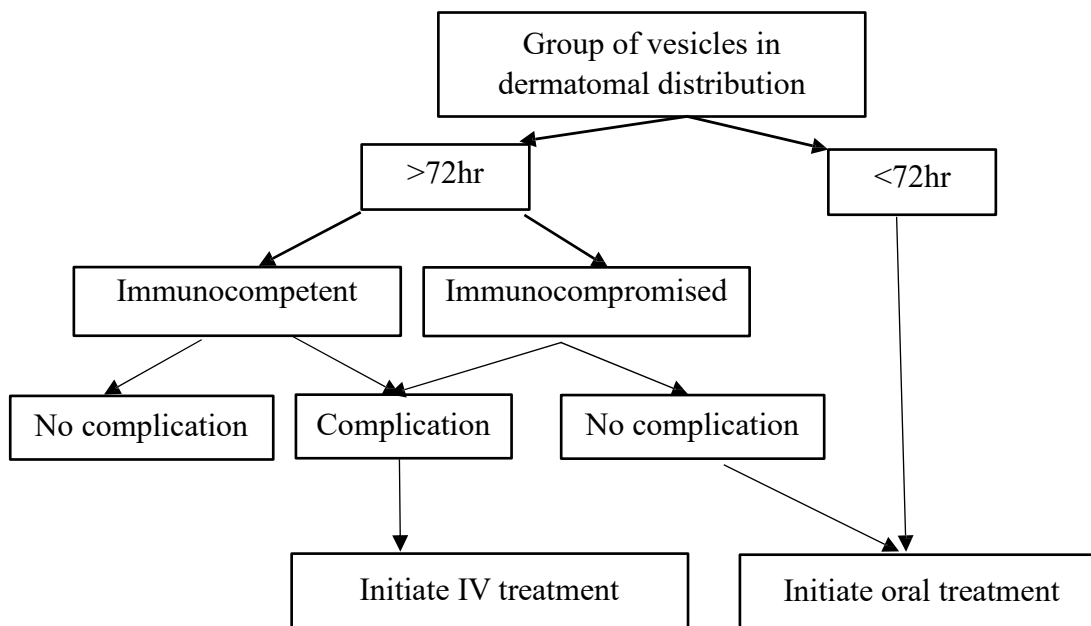
- Dosing of antiviral agents for treatment of adult herpes zoster:
 - Valacyclovir 1000mg PO thrice daily (7-10 days)
 - Acyclovir 800mg PO five times daily (7-10 days)
 - Acyclovir 10mg/kg IV every 8hr (7-14 days)
- Additional symptomatic treatment includes:
 - Topical antibiotic creams like fusidic or mupirocin help to prevent secondary bacterial infection.
 - Analgesics help to relieve the pain. Occasionally, severe pain may require an opioid or gabapentinoid medications.
 - Topical lidocaine and nerve blocks may also reduce pain

5. Algorithm

The indications for treatment depend upon uncomplicated and complicated natures of the diseases. Complicated herpes zoster includes:

- Disseminated herpes zoster
- Ocular manifestations
- Neurologic complications
- Ramsay Hunt syndrome

Figure 1: Approach to antiviral therapy in adults with uncomplicated herpes zoster (9,10)



Are there circumstances in which treatment could still be beneficial

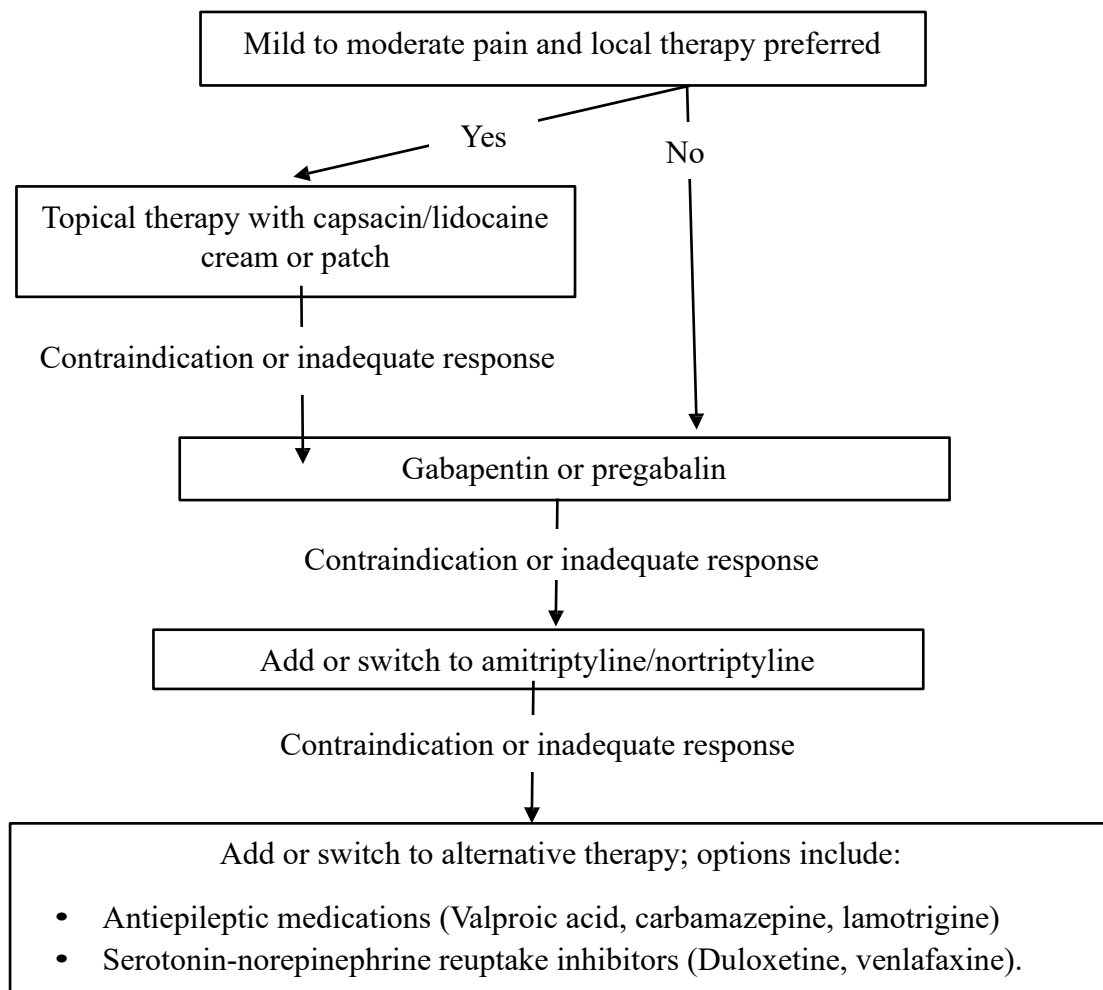
- Age > 65-years-old
- Complicated locations (eye, ear)
- Cutaneous with visceral or neurological involvement
- Organ transplant recipient
- Cancer and leukopenia patients
- In patient with renal insufficiency, the dose of antiviral should be adapted according to EGFR.

Figure 2: Approach to antiviral therapy in adults with complicated herpes zoster

Disseminated herpes zoster (14) • IV acyclovir 10mg/kg thrice daily (10-14 days) • PO valacyclovir 1g thrice daily (10-14 days) Neurological complications • IV acyclovir 10mg/kg thrice daily (10-14 days) ** Duration of treatment may need to be extended in those with ongoing symptoms.

Ramsay Hunt syndrome (15) • PO valacyclovir 1g thrice daily (7-10 days) + prednisone 1mg/kg (5 days) ** In severe cases (eg, vertigo, tinnitus, or hearing loss), IV therapy can be initiated, and the patient can then be transitioned to an oral antiviral agent when the lesions begin to crust.
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Figure 3: Approach to post-herpetic neuralgia in adults



VIII. SPECIAL CASES

- **For pregnant women:** acyclovir is preferred since there is the most experience with this medication in pregnancy. Treatment should continue until all lesions have crusted (typically 7 days for immunocompetent patients; 7 to 14 days for immunocompromised patients).

IX. EVOLUTIONS AND PROGNOSIS

Herpes zoster usually clears in 2 to 3 weeks and rarely returns. If the virus affects the nerves that control movement (the motor nerves), you may have temporary or permanent weakness or paralysis. Sometimes the pain in the area where the shingles occurred may last from months to years. Less common, but often serious, ophthalmic locations and related complications can have permanent detrimental effects.

X. COMPLICATIONS

- Disseminated zoster (any of the following):
 - Extensive cutaneous involvement (e.g. >2 contiguous dermatomes, bilateral involvement lesions crossing the midline, involvement of non-contiguous dermatomes)
 - Visceral involvement

- Post-herpetic neuralgia is the persistence of pain after a month of onset of herpes zoster. It is the commonest side effect seen in elderly patients with involvement of the ophthalmic division of trigeminal nerve.
- Complications like cranial neuropathies, polyneuritis, myelitis, aseptic meningitis, or partial facial paralysis occur due to the involvement of the nervous system.
- During pregnancy, varicella may lead to infection in the fetus and complications in the newborn, but chronic infection or reactivation, in other words, herpes zoster, is not associated with fetal infection.

XI. PREVENTION AND RECOMMENDATION

- **Vaccination:** Prevention of herpes zoster and its complications by vaccination would improve the life of older people and also reduce the societal impact of herpes zoster. A live-attenuated VZV vaccine (Zostavax) against herpes zoster, which has been licensed in a number of countries worldwide for administration to adults aged over 50 years, has been shown to reduce the incidence of herpes zoster, herpetic neuralgia and other complications in immunocompetent adults. Herpes zoster vaccination is contraindicated in immunosuppressed patients due to the risk of it causing disseminated herpes zoster infection.

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Annex: Annex: Courtesy of Herpes zoster patients at the dermatology department of Cambodia-China Friendship Preah Kossamak Hospital.



Figure 1: Herpes zoster affected the C6 and C7 dermatomes.



Figure 2: Herpes zoster affected T1 to T4 dermatomes.



Figure 3: Herpes zoster ophthalmicus



Figure 4: Herpes zoster ophthalmicus

ATOPIC DERMATITIS

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I. DEFINITION

Atopic dermatitis (AD) or Atopic Eczema is a chronic, pruritic, inflammatory skin disease that occurs most frequently in children but also in adults. It follows a relapsing course. AD is often associated with elevated serum immunoglobulin E (IgE) levels and a personal or family history of type I allergies, allergic rhinitis, and asthma.

II. EPIDEMIOLOGY

- Affects persons of all races.
- Sex: male-to-female ratio is 1:1.4
- Age: 85% of cases, atopic dermatitis occurs in the first year of life; 95% of cases it occurs before age 5 years.
- Incidence of atopic dermatitis is highest in early infancy and childhood. The disease may have periods of complete remission, particularly in adolescence, and may then recur in early adult life.
- In the adult population, the rate of atopic dermatitis frequency is 3% or higher, but onset may be delayed until adulthood.

III. ETIOLOGY

- Changes (mutations) in genes
- Problems with the immune system
- Exposure to certain things in the environment

IV. PATHOPHYSIOLOGY

Pathogenesis driven by skin barrier defects (most importantly in the filaggrin (FLG) gene), environmental effects and alterations in immunologic responses in T cell, antigen processing, inflammatory cytokines, host defense proteins, allergen sensitivity, and infection.

V. DIAGNOSIS

The diagnosis of atopic dermatitis

- Based upon history
- Morphology, and distribution of skin lesions
- Associated clinical sign
- Criteria for diagnosis by Hanifin & Rajka

1. Clinical findings:

Atopic dermatitis symptoms can appear anywhere on the body and vary widely from person to person. you may notice one or more of the following: dry skin, cracked skin, itchy, rash on swollen skin that varies in color depending on skin color, small, raised bumps on brown or black skin, oozing and crusting, thickened skin, darkening of the skin around the eyes and raw skin from scratching.

It is divided into infantile, childhood, and adolescent/adult stages.

Infants < 2 years of age

- Signs of inflammation usually develop during the 3rd month of life
- Lesions usually found on the facial cheeks and/ or chin

Children 2-12 years of age

- Inflammation in the flexural areas
- Lesions usually found on the neck, wrists, ankles, popliteal fossae, antecubital fossae

13 years old – adults

- Resurgence of inflammation that recurs near puberty onset
- Eruptions more likely develop on the upper body including the face, neck, chest and back
- In severe cases, rare erythroderma may occur due to extension of eruptions all over the body
- Pattern of inflammation is similar as in a child 2-12 years

2. Criteria for diagnosis (based on criteria developed by Hanifin & Rajka)

Major features (must have ≥ 3)

- Pruritus
- Typical morphology & distribution
- Facial and extensor involvement infants and children
- Flexural lichenification and linearity in adults
- Dermatitis – chronic or chronically relapsing
- Personal or family history of atopy: asthma, allergic rhinitis, atopic dermatitis

Minor / Less specific features (3 of 23 required)

- Cheilitis
- Hand/foot dermatitis
- Ichthyosis, hyperlinearity, keratosis pilaris
- Nipple eczema
- Perifollicular accentuation
- Xerosis
- Recurrent conjunctivitis
- Keratoconus
- Anterior subcapsular cataract
- Pityriasis alba
- White dermographism
- Dennie-Morgan infraorbital fold
- Facial pallor /facial erythema
- Anterior neck fold
- Itch when sweating
- Intolerance to wool and lipid solvents
- Food intolerance
- Course influenced by environmental /emotional factors
- Orbital darkening

- Immediate (type 1) skin test reactivity (radioallergosorbent, or prick test reactivity)
- Increased serum IgE
- Early age of onset
- Cutaneous infections (staphylococcus aureus, herpes simplex)

1. Investigation

- Blood tests to check for other causes of the rash
- Skin biopsy of the rash or lesion
- Skin biopsy and blood tests, including IgE levels, are not used routinely in the evaluation of patients with suspected atopic dermatitis and are not recommended. However, in selected patients, histologic examination of a skin biopsy or other laboratory tests (patch testing, genetic testing, KOH.) may be helpful to rule out other skin conditions.

VI. DIFFERENTIAL DIAGNOSIS

1. Children/infants

- Seborrheic dermatitis (greasy, yellowish scales on scalp and face)
- Diaper Dermatitis (localized to diaper area with erythema and scaling)
- Tinea corporis (annular plaques with central clearing and scaly active border)
- Scabies (burrows and papules, especially on palms, soles, and intertriginous areas)
- Nummular Eczema (coin-shaped, eczematous plaques; less common in infants)
- Psoriasis (well-demarcated erythematous plaques with silvery scales on extensor surfaces)

2. Adults

- Contact Dermatitis (localized to exposure sites, confirmed by patch testing)
- Psoriasis (well-demarcated erythematous plaques with silvery scales on extensor surfaces)
- Cutaneous T-cell lymphoma (CTCL): (persistent, erythematous patches often misdiagnosed as chronic atopic dermatitis)
- Xerotic eczema (dry, cracked, and erythematous skin, especially in older adults during winter)
- Lichen simplex chronicus (single, thickened lichenified plaques due to chronic rubbing or scratching)
- Nummular dermatitis (coin-shaped lesions on trunk and extremities; common in adults)

3. Both age groups

- Ichthyosis Vulgaris (fined generalized scaling on extensor surfaces; associated with atopy)
- Urticaria (transient wheals, pruritic but resolving within 24 hours)
- Keratosis pilaris (follicular papules on arms and thighs; minimal pruritus or inflammation)
- Dermatophytosis (annular plaques, confirmed by fungal culture or KOH preparation)

VII. MANAGEMENT

1. Objectives

- To help the patients to improve and cope with their symptoms
- To prevent exacerbations, and minimize therapeutic risks

- To improve their quality of life
2. Treating the underlying cause: if the cause can be identified, eliminate the cause. Management involved elimination of exacerbating factors, restoration of the skin barrier function and hydration of the skin, patient education, and pharmacologic treatment of skin inflammation.
 3. Non-pharmacological therapy
 - 3.1 Patient/caregiver education
 - Educate the patient about proper skin care (bathing, hydration and use of moisturizers)
 - Discuss that many factors probably contribute to flare-ups and usually a specific cause cannot be found
 - Explain the importance of therapy adherence, appropriate drug therapy and potential side effects of medications when used over extended periods of time
 - 3.2 Skin care
 - Bathing: helps cleanse the skin by removing scales, crusts, bacteria, allergens, and irritants
 - Moisturizers: Improves skin barrier function, reduces incidence of flares, increases time between flares and has steroid-sparing effect
 - 3.3 Avoidance of trigger factors
 - Identify and eliminate trigger factors: identify potential allergens by careful history and selective allergy tests
 - Common trigger factors: wool clothing, pollen, extreme temperature, sweat, products with perfumes or preservatives, foods, cigarette smoke and animal dander
 - Psychological factors: emotional factors (e.g. anxiety, stress, anger) cause disease exacerbation, induce immune activation and increase pruritus and scratching
 - 3.4 Photoprotection
 - Regular use of broad-spectrum sunscreens with SPF30+ and with inorganic UV filters (titanium dioxide, zinc oxide) is recommended
 - 3.5 Wet wrap therapy
 - May be used for chronic and refractory lesions or for moderate-severe weeping lesions
 - Cools inflamed skin, maintains hydration, and decreases scratching
 - Can help reduce water loss and disease severity in patients with moderate-severe atopic dermatitis
 - Combined with topical corticosteroid can be effective in treating refractory cases
 - 3.6 Phototherapy
 - Considered in patient with recalcitrant disease or when first-line therapy with topical agents has been unsuccessful
 - Broad-band UVB and UVA, narrow-band UVB and UVA-1 or combined UVA and UVB can be useful
 - Treats chronic lichenified forms of moderate-severe disease
 4. Pharmacological therapy

Topical Treatment:

Topical Corticosteroid

- Used as first-line treatment for mild to severe atopic dermatitis and non-pharmacological intervention have been unsuccessful
- Choice of products will depend on severity of flare-up, distribution and site of lesions, patient's age and preference, and other factors (e.g. humidity)
- Available in different potencies from mildly potent to very potent

Calcineurin Inhibitors – Topical Immunomodulators

- Steroid-sparing agents for acute and maintenance therapy, topical calcineurin inhibitors inhibit inflammatory cytokine transcription in activated T cell & other inflammatory cells through inhibition of calcineurin
- Second-line therapy for moderate to severe atopic dermatitis, 1st –line treatment over topical corticosteroids when atopic dermatitis is unresponsive to steroid therapy, with presence of atrophy or telangiectasia secondary to steroid use, when affected areas are either the face, anogenital area, and / or skin folds, and for long-term treatments
- May be used on all body locations for extended periods of time, especially the face, hand and feet
- Pimecrolimus
 - Safety and efficacy have been shown in children > 2 years of age & adult with mild-moderate atopic dermatitis
 - Pruritus relief has been seen as early as day 3 of use, does not cause atrophy
 - Prevents flare-ups & results in significant steroid-sparing effect when used for up to 12 months
 - When used in early stages of disease, it has shown to be therapeutically advantageous over typical moisturizers plus topical corticosteroids in long-term use
- Tacrolimus
 - Recommended for moderate to severe atopic dermatitis
 - Rapidly decreases the signs & symptoms of atopic dermatitis in adults & children >2 years of age
 - Studies have confirmed the efficacy of Tacrolimus 0.03% compared to low-potency topical corticosteroids in children & the efficacy of Tacrolimus 0.1% compared to mid-potency topical corticosteroids in adults

Systemic Therapy:

Indications for Systemic Therapy in Atopic Dermatitis

- **Severe Disease:**
 - Widespread lesions causing significant discomfort or impaired quality of life.
 - Intense pruritus leading to sleep disturbances, emotional distress, or secondary infections.
 - Chronic or recurrent disease unresponsive to optimized topical treatment and phototherapy.
- **Refractory Disease:**
 - Lack of improvement despite adherence to an adequate regimen of:
 - High-potency topical corticosteroids.
 - Topical calcineurin inhibitors (e.g., tacrolimus, pimecrolimus).
 - Moisturizers and skin barrier repair.
- **Frequent Flares:**

- Patients experiencing frequent, severe flares requiring repeated courses of systemic corticosteroids, which are not recommended for long-term control.
- **Associated Comorbidities:**
 - Coexistence of other atopic disorders (e.g., asthma, allergic rhinitis) that may worsen atopic dermatitis severity.
 - Psychological burden, including anxiety and depression related to the disease.
- **Contraindications to Topical Therapy:**
 - Significant skin barrier disruption, leading to pain or intolerance of topical agents.

Antihistamines

- Oral sedating antihistamine are typically used for drowsiness effect to facilitate sleep, to prevent increase of itch intensity at night, and as an adjunctive therapy to topical anti-inflammatory therapy
- Studies of oral non-sedating antihistamines have shown variable results in controlling pruritus; however, they may be useful in a small group of patients with associated urticarial

Systemic Corticosteroids

- Short-term treatment (up to 1 week) should only be considered in treatment-resistant atopic dermatitis, acute severe exacerbations and as a bridge therapy to other steroid-sparing systemic treatments
- Improve lesions but rebound flare-up usually occurs upon discontinuation
- Use short-term and decrease chance of rebound effect by tapering oral form slowly while increasing topical corticosteroid treatment & continuously hydrating the skin

Cyclosporin (Cyclosporin A)

- First-line option for short-term use in chronic, severe refractory disease in adults who require systemic immunosuppressive treatment
- Condition tends to return after discontinuation of therapy but not always at the original severity level
- May be used in children and adolescents with refractory or severe disease
- Long-term use is not justified because of risks of hypertension, renal and liver dysfunction
- Combination therapy with UV light (UVA, UVB, PUVA) is not recommended due to increased risk of skin cancer

Azathioprine

- Used for severe/ refractory disease unresponsive or with contraindications to Cyclosporin therapy
- Thiopurine methyltransferase (TMPT) levels should be monitored while patient is on azathioprine to test for myelosuppression
- Most children respond to low doses
- There is increasing evidence supporting its efficacy in severe refractory atopic dermatitis
- May cause nausea, fatigue, liver dysfunction & bone marrow depression in patients deficient in thiopurine methyltransferase

Methotrexate

- May be considered for severe or refractory disease unresponsive or with contraindication to Cyclosporin therapy
- Studies showed that efficacy of methotrexate for atopic dermatitis is comparable to that of Azathioprine and Cyclosporin

Retinoid

- Alitretinoin is a derivative of Isotretinoin with anti-inflammatory effects used for severe chronic hand atopic dermatitis unresponsive to first-line treatments

Phosphodiesterase Type-4 (PDE-4) Inhibitors

- **Apremilast**
 - May be considered in patients with moderate to severe atopic dermatitis unresponsive to standard treatments but further studies are needed to establish its efficacy and safety
- **Abrocitinib**
 - Oral Janus kinase 1 (JAK-1) inhibitor used for the treatment of refractory moderate to severe atopic dermatitis which is inadequately controlled with other systemic drugs (including biologics) or when use of those therapies is not advisable
 - May be used with or without topical corticosteroids
 - Not recommended to be used in combination with other JAK inhibitors, biologic immunomodulators or with other immunosuppressants
- **Baricitinib**
 - Oral selective JAK1 & JAK2 inhibitor which blocks signals of immune response and inflammation
 - May be given as an option for treatment of moderate to severe atopic dermatitis in adults who are unresponsive or contraindicated to at least one systemic immunosuppressant (e.g. Ciclosporin, Methotrexate, Azathioprine, Mycophenolate mofetil)
 - May be an alternative to Dupilumab with best supportive care and may be offered with topical corticosteroids
 - Studies showed Baricitinib, with or without corticosteroids, can reduce severity and symptoms of atopic dermatitis compared with placebo
 - Assess patient after 8 weeks of treatment if there is adequate response
 - Adequate response is defined as reduction of at least 50% in EASI-50 score & at least 4 points in Dermatology Life Quality Index (DLQI) from the start of the treatment
- **Mycophenolate mofetil**
 - May be considered for severe or refractory disease unresponsive or with contraindications to Ciclosporin therapy
- **Ruxolitinib**
 - Topical Ruxolitinib is a selective JAK1 and JAK2 inhibitor which is approved the United States Food and Drug Administration (US FDA) for short-term therapy of mild to moderate atopic dermatitis in immunocompetent patients more than 12 years old whose disease is not controlled with other topical therapies
- **Upadacitinib**
 - An oral selective and reversible inhibitor of JAK enzymes (preferentially JAK1 or JAK1/3), which are intracellular enzymes involved in the stimulation of hematopoiesis and immune cell function through a signaling pathway
 - Used in patients over 12 years old with refractory moderate to severe atopic dermatitis which is inadequately controlled with other systemic drug (including biologics) or when use of those therapies is not advisable
 - Not recommended to be used in combination with other JAK inhibitors, biologic immunomodulators or with other immunosuppressants
- **Dupilumab**

- A human immunoglobulin (Ig) G4 monoclonal antibody approved for use in patients ≥ 12 years of age with moderate to severe atopic dermatitis uncontrolled by first-line treatment and who are candidates for systemic therapy
- Binds to the interleukin-4Ra (IL-4Ra) subunit thereby inhibiting IL-4 and 13 cytokine-induced responses which includes proinflammatory cytokine, chemokines and IgE release
- Combination with emollients and as-needed topical anti-inflammatory agents is recommended
- **Tralokinumab**
 - Recommended for adult patients with moderate to severe atopic dermatitis uncontrolled by topical therapies and are candidates for systemic therapy
 - A human, high-affinity IgG4 monoclonal antibody which acts by neutralizing IL13
 - Combination treatment with topical corticosteroids, topical calcineurin inhibitors, and UV light is possible

Other Biologicals

- Further studies are needed to prove the efficacy of Nemolizumab, Rituximab, Mepolizumab, Omalizumab, and Ustekinumab for atopic dermatitis
- Trial treatment with Mepolizumab may be considered in select patients unresponsive to standard therapies
- The following biological agents are being studied for atopic dermatitis: Lebrikizumab, Nemolizumab, Tralokinumab, Tezepelumab
- Concomitant treatment with topical corticosteroids, topical calcineurin inhibitors and emollients is possible

Allergen-specific Immunotherapy

- May be considered for select patients with severe disease w/ house dust mite, birch or grass pollen sensitization, and concomitant history of clinical exacerbation after exposure or positive atopy patch test

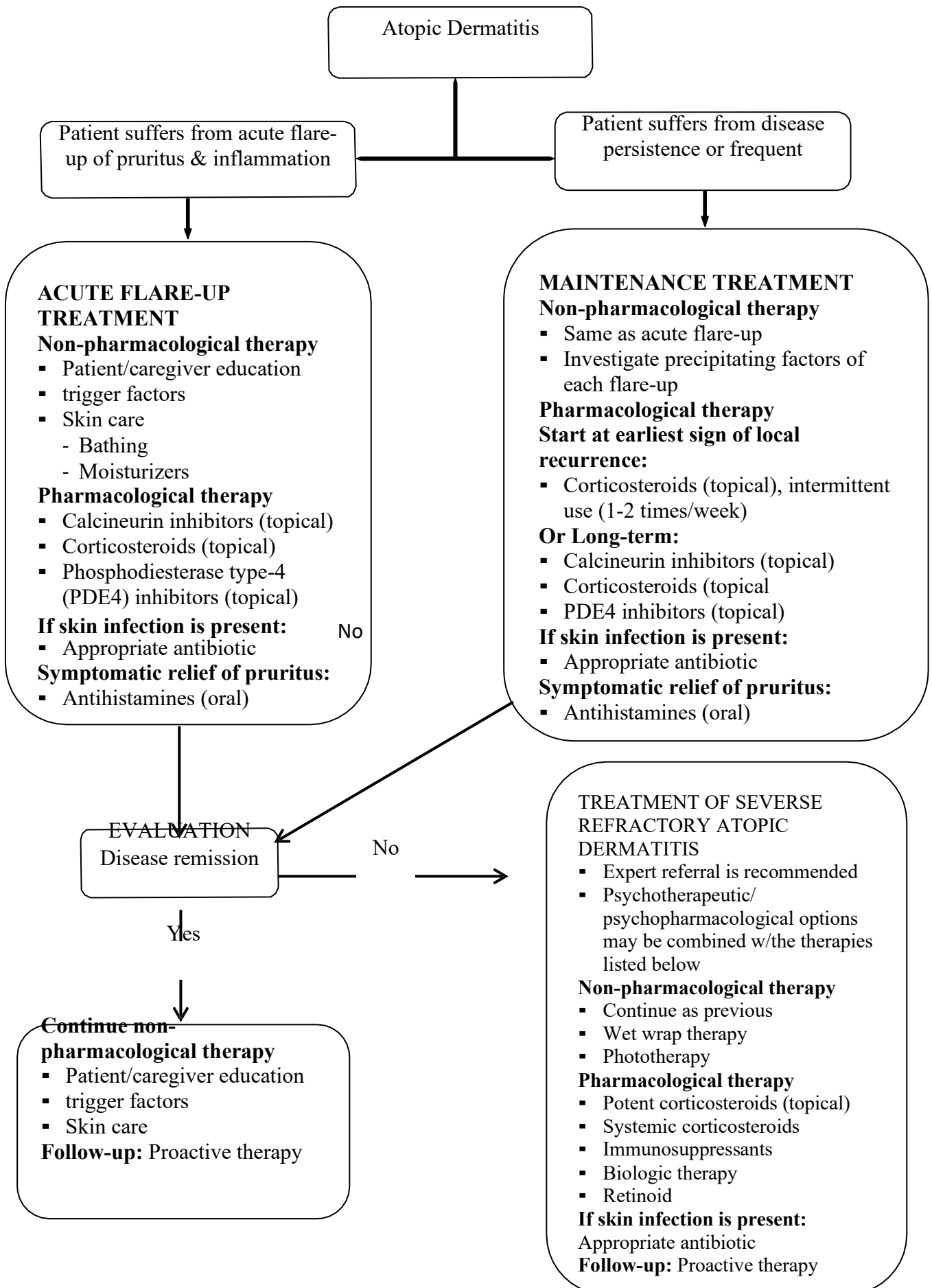
Other Pharmacotherapeutic Options

- Further studies are needed to prove the efficacy of probiotics, topical sodium cromoglycate, vitamin D, Tofacitinib, mast cell stabilizers & leukotriene antagonists (ex: Montelukast) in atopic dermatitis
- Vitamin D supplementation and use of prebiotics/probiotics may reduce symptoms but are not routinely recommended

Follow-up

- **Proactive Therapy**
 - In patients with moderate atopic dermatitis, intermittent application of a class II-III topical corticosteroid (1-2times/week) or calcineurin (Tacrolimus) (2-3 times/week) as maintenance therapy on areas that commonly flare may help prevent disease relapse.

5. Algorithm



VIII.SPECIAL CASES

Severe or Special Cases of Atopic Dermatitis Requiring Specialist Referral

1. Severe or Refractory Disease

- Criteria:
 - Failure to respond to optimized topical therapy (e.g., high-potency corticosteroids, calcineurin inhibitors) and phototherapy.
 - Persistent disease with significant impact on quality of life (e.g., sleep disturbances, missed work/school).
- Specialist Role:
 - Consideration for systemic immunosuppressive agents (e.g., cyclosporine, methotrexate, or biologics like dupilumab).

2. Complications of Atopic Dermatitis

- Infection-Related Complications:
 - Eczema herpeticum: rapidly spreading vesiculopustular lesions due to herpes simplex virus; requires urgent systemic antiviral therapy.
 - Staphylococcus aureus superinfection: extensive weeping, crusted lesions requiring systemic antibiotics.
- Other Complications:
 - Persistent secondary infections or recurrent episodes despite treatment.

3. Diagnostic Uncertainty

- Atypical presentations of atopic dermatitis or overlapping features with other dermatoses (e.g., psoriasis, cutaneous T-cell lymphoma).
- Cases requiring biopsy to rule out other conditions.

4. Severe Pruritus

- Persistent, debilitating itching unresponsive to standard anti-pruritic measures, leading to excoriations, lichenification, and impaired daily functioning.

5. Special Populations

- Infants and young children:
 - Severe atopic dermatitis interfering with growth and development.
 - Uncontrolled flares despite parental adherence to treatment.
- Pregnant or breastfeeding women:
 - Concerns about medication safety during pregnancy/lactation.
- Elderly patients:
 - Complex comorbidities or increased susceptibility to systemic treatment side effects.

6. Psychosocial or quality of life impact

- Significant psychological burden (e.g., depression, anxiety, social isolation) due to visible lesions or pruritus.
- Need for multidisciplinary management, including mental health support.

7. Frequent Systemic Corticosteroid Use

- Patients requiring recurrent courses of systemic corticosteroids, which are not ideal for long-term disease control.

8. Specialist role in management

- Initiating advanced therapies:
 - Systemic immunosuppressants: cyclosporine, methotrexate, azathioprine, or mycophenolate mofetil.
 - Biologic agents like dupilumab or emerging treatments (e.g., JAK inhibitors).
- Comprehensive evaluation:
 - Patch testing for contact allergens.

- Testing for underlying infections or immunodeficiencies.
- Multidisciplinary approach:
 - Coordination with allergists, psychologists, or other specialists for complex cases.

IX. EVOLUTION AND PROGNOSIS

Most atopic dermatitis starting in childhood is mild in severity, and a review of birth cohort studies found that 80% of cases remitted, at least temporarily, by 10 years of age. Symptoms of AD, however, may persist or reemerge in adulthood. The prognosis for patients with AD is generally favorable, with most children outgrowing the condition by early adolescence. However, patients with severe, widespread disease and concomitant atopic conditions, such as asthma and allergic rhinitis, are likely to experience poorer outcomes.

X. COMPLICATION

- Bacterial infection
- Viral infection
- Fungal infection
- Ocular problems
- Hand dermatitis
- Exfoliative dermatitis

XI. PREVENTION AND RECOMMENDATION

Skin barrier protection from birth may prevent or modify the development of AD. The use of petrolatum early in life may be protective against AD development. Emollients are effective at prevention flares in established AD. Emollients, either creams or ointments, improve barrier function by supplying the stratum corneum with water and lipids.

Dosage Guidelines (1)

ANTI-HISTAMINES (ORAL)		
Drug	Dosage	
	Pediatric	Adult
First Generation		
Chlorpheniramine	1-2 yr: 1mg PO 12hrly Max dose: 4mg/day 2-5 yr: 1mg PO 4-6 hrly Max dose: 6mg/day 6-12 yr: 2mg/day Max dose: 12mg/day >12 yr: same as adult dose	4mg PO 4-6 hrly or 8-12mg PO 12 hrly Max dose: 24mg/day
Hydroxyzine	6 mo-6yr: 30-50mg/day PO in divided doses >6yr: 50-100mg/day PO in divided doses	25mg PO at night, May be increased to 25mg PO 6-8hrly
Promethazine	2-5 yr: 5-15mg PO 24 hrly >5-10 yr: 10-25mg PO 24 hrly	10-20mg PO 8- 12 hrly Max dose: 60mg/day
Second Generation		
Bilastine	4-11 yr (w/body wt $\geq$ 16kg): 10mg orodispersible PO 24hrly $\geq$ 12yr: 20mg PO 24hrly	20mg PO 24 hrly
Cetirizine	6 mo – < 2yr: 2.5mg PO 12-24hrly 2-5 yr: 5mg PO 24 hrly or 2.5mg PO 12hrly 6-12 yr: 10mg PO 24 hrly or 5mg PO 12 hrly	10mg PO 24hrly or 5mg PO 12hrly
levocetirizine	6 mo – 5yr: 1.25mg PO 24hrly 6-11 yr: 2.5mg PO 24 hrly $\geq$ 12 yr: 2.5-5mg PO 24 hrly	2.5-5 mg PO 24 hrly
Desloratadine	6 mo – 11mo: 1mg PO 24hrly 1-5yr: 1.25mg PO 24hrly 6-11 yr: 2.5mg PO 24 hrly $\geq$ 12 yr: 5mg PO 24 hrly	5mg PO 24 hrly
fexofenadine	6-23mo: 15mg PO 12 hrly 2-11 yr: 30mg PO 12hrly $\geq$ 12yr: 60mg PO 12hrly or 120-180mg PO 24hrly	60mg PO 12 hrly or 120-180 mg PO 24 hrly
Loratadine	$\geq$ 2-6yr: 5mg PO 24 hrly >6yr: 10mg PO 24 hrly	10mg PO 24 hrly

Dosage Guidelines (2)

CORTICOSTEROIDS (SYSTEMIC)	
Drug	Dosage
Dexamethasone	Child: 0.02-0.3 mg/kg/day IV/IM/PO divided 6 hrly or 12 hrly Adult: 4-20 mg/day given in a single dose or 2-4 divided doses PO/IM/IV
Hydrocortisone	Child: 2.5-10 mg/kg/day PO divided 6-8 hrly or 1-5 mg/kg/day IM/IV divided 12-24 hrly Adult: 20-240mg/day PO, 100-500 mg IV over 30sec-10 min, may be repeated 2-6 hrly
Methylprednisolone	Child: 0.11-1.6mg/kg/day PO/IV/IM in 3-4 divided doses Adult: 16-64mg/day or in divided doses PO, 40-125mg/day IV
Triamcinolone	Child: 0.11-1.6mg/kg/day IM divided 3-4hrly Adult: 60 mg IM, adjust dose to a range of 40-80mg
Prednisolone	Child: 0.1-2 mg/kg/day in divided doses 1-4 times daily PO Adult: 10-60 mg/day PO or in 2-4 divided doses

CORTICOSTEROIDS (TOPICAL)		
Drug	Available Strength	Dosage
Very Potent		
Clobetasol propionate	0.05% cream, gel, oint, lotion	Apply 12-24 hrly
Potent		
Beclomethasone dipropionate	0.025% cream	Apply 8-24 hrly
Betamethasone dipropionate	0.05% cream, oint, soln	Apply 8-24 hrly
Fluocinolone acetonide	0.02% cream	Apply 8-24 hrly
Mometasone furoate	0.1% cream, oint, gel, lotion	Apply 24 hrly
Moderately Potent		
Clobetasone butyrate	0.05% cream, oint	Apply up 6 hrly
Triamcinolone acetonide	0.02% cream, 0.1% cream, lotion	Apply 6-12 hrly
Mildly Potent		
Hydrocortisone	1% cream, lotion, oint	Apply 8-24 hrly

CALCINEURIN INHIBITORS (TOPICAL)		
Tacrolimus	0.03%, 0.1% oint	Child 2-15yr: apply a thin layer of 0.03% oint to affected skin 12 hrly Adult & child $\geq$ 16yr: apply a thin layer of 0.03% or 0.1% oint to affected area 12 hrly
Pimecrolimus	1% cream	Adult & child $\geq$ 2yr: apply a thin layer to affected skin 12hrly

Dosage Guidelines (3)

IMMUNOSUPPRESSANTS (ORAL)	
Drug	Dosage
Ciclosporin (Cyclosporin A, Cyclosporin)	2.5-5mg/kg/day PO divided 12 hrly for a max of 8 wk May increase to max dose of 5mg/kg/day if necessary after 2wk
Janus Kinase (JAK) Inhibitors	
Abrocitinib	100mg PO 24 hrly May increase to up to 200mg PO 24 hrly CYP2C19 poor metabolizer: 50mg PO 24 hrly May increase to up to 100mg PO 24 hrly
Baricitinib	2-4 mg PO 24hrly Adjust dose depending on patient's response
Upadacitinib	Children $\geq$ 12 yr w/wt 40kg & adults < 65 yr: 15mg PO 24 hrly

EMOLLIENTS, CLEANSERS & SKIN PROTECTIVES (TOPICAL)			
Drug	Available	Indication	Administration
Ceramide	Cream, bar, lotion, oint, wash	- Dry/sensitive skin - Symptomatic relief of mild to moderate atopic dermatitis & skin rashes	- Use as soap substitute - Moisturizer & / or for treatment: apply 8-12hrly - Bath: use as liquid cleanser
Dexpanthenol	5% oint	- Diaper rash - Dry/damaged skin	Apply once or as needed
Hyaluronic acid (Na hyaluronate)	Cream, lotion	- Manages & relieves itching, burning & pain caused by different dermatoses, relieves dry, waxy skin - Promotes wound healing	Apply to affected area 8 hrly & massage gently
panthenol	Cream, lotion	- For sensitive & dry skin - Prevents skin irritation	Use as a soap substitute
Urea	10%, 20% cream, lotion, shampoo	Emollient for hyperkeratotic or excessive dry skin conditions (eg xerosis, contact irritant dermatitis, pruritus)	- Apply sparingly & rub into the affected area 8-12hrly & as required - Shampoo: use daily as needed
Vitamin A	Cream, oint	- Diaper rash, chafing, minor burns, sunburn, small wounds & dry skin - Helps relieve burning, itching & pain	Apply 8-12 hrly

		Enhances skin's healing process	
Vitamin E	Cream, lotion, wash	- Dry skin - Protects skin from UV radiation	Apply as needed
Zinc oxide	7.5%, 10%, 32%, 310mg/g cream, 40%, 200mg/g oint, topical powd	- Prevention & treatment of diaper rash by preventing wetness from coming into direct contact w/ the skin - Promotes healing of minor skin irritations, non-infected wounds & burn - Soothes & protects in eczema & slight excoriation	- infants: apply at each diaper change especially at bedtime - Adults: use daily on skin, affected areas & into skin folds, smooth onto areas prone to chafing

OTHER DERMATOLOGICALS (PARENTERAL)	
Drug	Dosage
Dupilumab	Initial dose: 600mg (two 300mg mg inj) SC followed by 300mg SC every other wk
Interleukin-13 Antagonist	
Tralokinumab	Initial dose: 600mg SC (given as four 150mg inj) followed by 300mg (given as two 150mg inj) once every other wk

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ACNE VULGARIS

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I. DEFINITION

Acne Vulgaris is the most common form of acne with chronic inflammation of pilosebaceous unit which is notable for open and/or closed comedones (blackheads and whiteheads) and pleomorphic lesions including papules, pustules and nodules.

II. EPIDEMIOLOGY

Acne vulgaris, occurs most commonly in adolescents and young adults, though is not limited to these age groups. Adolescent acne is due to the hormonal surges as a child transition into adulthood while post-adolescent acne is due to many contributing factors.

Acne is one of the top three most common skin diseases, particularly in adolescents and young adults, in whom the prevalence is estimated at 85% (ages 12–25 years).

Acne has no predilection for ethnicity; thus, it is an important disease worldwide and is considered one of the top 10 most prevalent global diseases.

III. ETIOLOGY

Current understanding of acne is that it is a complex and multifactorial inflammatory disease.

Contributing Factors:

Medications: corticosteroids, anabolic steroids, Lithium, some oral contraceptives, etc

Androgen excess:

- Causes of androgen excess: polycystic ovary disease or adrenal, ovarian, or pituitary tumor
- Signs of androgen excess: precocious puberty, hirsutism, irregular menstruation, androgenetic alopecia

Mechanical:

- Physical occlusion: eg. cosmetics (comedogenic), head bands, mask-related acne, etc.

IV. PATHOPHYSIOLOGY

Recent studies are better defining the cellular and molecular mechanisms involved in acne and the importance of inflammation and the immune response. The pathogenesis of acne is multifaceted, and at least four factors have been identified.

These key elements are:

- (1) follicular epidermal hyperproliferation,
- (2) sebum production,
- (3) *Propionibacterium acnes*, and
- (4) inflammation and immune response.

Each of these processes are interrelated and under hormonal and immune influence. Follicular epidermal hyperproliferation results in the formation of a microcomedo.

The epithelium of the upper hair follicle, the infundibulum, becomes hyperkeratotic with increased cohesion of the keratinocytes, resulting in the obstruction of the follicular ostium, where keratin, sebum, and bacteria begin to accumulate in the follicle and cause dilation of the upper hair follicle, producing a microcomedo. Exactly what initiates and stimulates the hyperproliferation and increased adhesion of keratinocytes is unknown. Several proposed factors in keratinocyte hyperproliferation include androgen stimulation, decreased linoleic acid, increased IL-1- α activity, and effects of P. acnes. Dihydrotestosterone (DHT) is a potent androgen that may play a role in acne.

V. DIAGNOSIS

1. Clinical Findings

Diagnosis is usually based on the morphology of the lesions and their site of distribution.

Type of Acne Lesions:

- Comedone (non-inflammatory)
- Papule and pustule (inflammatory)
- Nodule (inflammatory), acne conglobata is a severe form of nodular acne where the lesions are firm, inflamed and painful measuring from 10 mm.

Site of distribution:

- Lesions typically appear in the centro-facial areas: forehead, nose, chin.
- May eventually involve other areas containing sebaceous glands: scalp, ears, neck, shoulders, chest, back, armpits, groins and buttock.

Acne Grading Systems: Severity Classification by Lehman

	Mild	Moderate	Severe
Comedone	<20	20-100	>100
Papule/ pustule	<15	15-50	>50
Nodule/ cyst	0	<5	>5
Total lesion count	<30	30-125	>125

2. Investigations

No tests are necessary for most patients with acne. Tests may be requested to look for infection, to investigate the cause of the disease, or to monitor treatment.

□ Tests for micro-organisms:

- Culture for pyogenic bacterial folliculitis due to Staphylococcus aureus
- Culture for Gram negative folliculitis

□ Hormonal tests:

- Testosterone
- Dehydroepiandrosterone sulfate (DHEAS)
- Luteinising hormone (LH) and follicular stimulating hormone (FSH)

VI. DIFFERENTIAL DIAGNOSIS

Formation of comedones is inherent to the diagnosis of acne vulgaris. When there is no evident of comedones, the following diagnoses may be considered:

- Folliculitis: is an inflammation of the hair follicle, usually caused by bacterial (commonly *Staphylococcus aureus*), fungal, viral infections, or even irritation from shaving or friction.
- Perioral dermatitis: appear as red, inflamed bumps or pustules around the edges of the mouth, and nose.
 - Rosacea: facial redness (erythema), especially on the cheeks, nose, forehead, and chin. Visible blood vessels (telangiectasia or broken capillaries), giving the skin a flushed appearance.
 - Keratosis pilaris: caused by the buildup of keratin, a protein in the skin that forms a plug in the hair follicle. It is often genetic and more common in people with dry skin. No pus or inflammation as seen in acne.
 - Miliaria: presents as small red, or clear blisters, bumps or pustules. It's often triggered by hot, humid weather, excessive sweating, or wearing tight clothing.

VII. MANAGEMENT

Management is based on severity of acne, patient history with all contributing factors identified and treated accordingly.

- Topical therapy is the primary form of treatment for mild acne.
- Topical and systemic therapies are the principal forms of treatment for acne which is unresponsive to topical treatment and moderate to severe acne.

1. Objective

- Induce clearance of acne lesions
- Prevent physical and psychological complications
- Maintain remission and prevent relapse

2. Out patient Management

Management of acne usually OPD.

Topical Agents:

Retinoids:

There are 4 generations of retinoid, namely tretinoin, adapalene, tazarotene and trifarotene.

Tretinoin:

- Preparation: cream as 0.025%, 0.05% and 0.1%, gel as 0.01% and 0.025%, apply once daily at bedtime.
- Effective as monotherapy in treatment of non-inflammatory and mild to moderate inflammatory acne, normalizes the keratinization process within the hair follicle and prevents formation of comedones.
- Patients should be warned that they may suffer pustular flare during the first few weeks of therapy.
- Childbearing status of female patient should be assessed prior to initiation of treatment.

Adapalene:

- Preparation: 0.1% cream or gel, apply once daily at bedtime.
- May be useful as monotherapy in mild to moderate forms of acne by reducing both inflammatory and non-inflammatory lesions.
- Potent modulator of keratinization, cellular differentiation and inflammatory process.

- Most tolerated topical retinoid, more effective and less irritating than Tretinoin for the treatment of inflammatory and non-inflammatory lesions.

Tazarotene

- Preparation: 0.1% cream, gel as 0.05% and 0.1%, apply once daily at bedtime.
- Has been shown to reduce the number of non-inflammatory lesions counts in mild to moderate acne, modifies the accumulation and cohesion of corneocytes, and the inflammation process.
- Associated with a higher incidence of erythema, burning, pruritus and peeling than Tretinoin or Adapalene.

Trifarotene

- Preparation: 0.05% cream, apply to affected area once daily at bedtime.
- Gamma-selective retinoid cream suitable for acne on the face and trunk comedolytic, anti-inflammatory and anti-pigmenting properties.

Benzoyl Peroxide :

- Preparation: cream as 2.5%, 4%, 5%, 8% and 10%, gel as 2.5%, 5% and 10%, apply once or twice daily, morning and evening. There are also preparations for washing as bar and lotion.
- Monotherapy is used in the treatment of comedonal or mild acne, usually given in combination with topical antibiotics or retinoids for mild to moderate acne, considered a mainstay in acne therapy.
- Bactericidal activity reduces *C. acnes* by producing reactive oxygen species in the sebaceous follicle after a short course of treatment.
- Mild comedolytic, antibacterial, and anti-inflammatory effects.

Azelaic acid

- Preparation: cream as 10% and 20%, gel as 15% and lotion as 20%, apply twice daily, morning and evening.
- For mild comedolytic and anti-inflammatory.
- Use as second choice for maintenance therapy, an alternative to retinoids
- Treatment option for pregnant women with acne or patients with acne and post inflammatory hyperpigmentation (PIH).
- Has bacteriostatic properties and may reduce follicular microbial colonization by more than 97% normalizes keratinization along with reducing inflammation.
- Local adverse effects are generally less than Benzoyl peroxide or Tretinoin.

Topical Antibiotics:

- Useful for mild to moderate and severe acne with inflammatory lesions in combination with other topical agents.
- Reduce *C. acnes* in sebaceous follicles and have anti-inflammatory properties.
- Because of the risk of antimicrobial resistance, the following should be considered:
 - Should not be used as monotherapy for moderate to severe acne or as maintenance therapy for acne.
 - Recommend that topical antibiotics be used in combination with Benzoyl peroxide or retinoid.
 - When possible, treatment duration is only up to 12 weeks.

Erythromycin

- Preparation: gel as 2% and 4%, 1% emulsion, ointment as 2%, solution as 2% and 4%, apply twice daily, morning and evening.
- Effective in treating moderate to severe acne.

Clindamycin

- Preparation: gel as 1% and 2%, lotion as 1% and 2%, apply twice daily, morning and evening.
- Equally effective as Erythromycin in treating moderate to severe acne.

Dapsone

- Preparation: gel as 5% and 7.5%, apply twice daily, morning and evening.
- Possesses both antimicrobial (bacteriostatic) and anti-inflammatory properties, is used as an option for patients with mild to moderate and severe acne preferably in combination with topical retinoid.

Salicylic acid

- Preparation: cleanser as 2%, is the most common preparation for face wash twice daily.
- agent when patients are intolerant of standard therapy.
- Commonly found in facial cleansers.
- Third choice for maintenance therapy.
- Has mild comedolytic and anti-inflammatory effects.

Clascoterone

- Preparation: cream as 1%, apply once daily at bedtime.
- Androgen receptor inhibitor which is approved for treatment of acne vulgaris in patients aged more than 12 years.
- Alternative topical therapy for patients who fail to respond sufficiently to initial therapies.

Oral Agents:

- Preferred treatment option for acne resistant to topical therapy, acne with high scarring potential or in patients with nodular acne.

Oral antibiotics:

- Typically used for moderate to severe inflammatory acne that does not respond to topical combinations, or for acne with high scarring potential.
- Choice of antibiotic is based on side effect profiles and local patterns of resistance.
- Maximum course of treatment should be 3-4 months but treatment response assessment should be at 6-8 weeks.

Tetracycline 500 mg orally twice daily, morning and evening.

- Considered as a first-line agent for moderate to severe inflammatory acne as an alternative to Doxycycline.

Doxycycline 100 mg PO orally twice daily, morning and evening.

- Considered as an excellent first-line agent for inflammatory and moderate to severe acne.

Lymecycline 300 mg/day, orally once daily for 10-15 days.

- Considered as first-line therapy in combination with other agents for moderate to severe acne

- Less frequent occurrence of gastrointestinal (GI) and dermatological side effects

Minocycline 50-100 mg, orally twice daily, morning and evening.

- May be used as second-line antibiotic treatment for inflammatory acne
- Fewer reports of *C. acnes* resistance compared to Tetracycline and Doxycycline

Erythromycin 250-500 mg, orally twice daily, morning and evening.

- Has direct anti-inflammatory effect by reducing neutrophil chemotactic factors and reactive oxygen species.
- Considered as alternative agent when other therapies have failed.
- Associated with less anti-inflammatory activity and more gastrointestinal side effects than Tetracycline.
- There tends to be more resistance of *C. acnes* to Erythromycin than to Tetracycline.
- Decreased risk of photosensitivity compared to Tetracycline.

Azithromycin 500 mg orally once daily, 3 days per week for 12 weeks are the most commonly used regimens.

- It is not used as first-line treatment for acne compared to cycline derivative.
- Use in moderate to severe acne that does not respond to other oral antibiotics.
- Used in patients with contraindications to Tetracyclines.

Clindamycin 150-300 mg, orally every 6 hours.

- GI effects (nausea, vomiting, abdominal pain and diarrhea).
- Use with caution in patients with colitis or other GI diseases.

Co-trimoxazole (Trimethoprim 160mg/ Sulfamethoxazole 800 mg), orally twice daily.

- Considered as third-line therapy when other therapies have failed.
- Effective in patients refractory to other oral antibiotics or for Gram-negative acne.
- Low but possible risk for major side effects should be considered.

Isotretinoin 0.5 - 1mg/kg/day, orally once or twice daily, may increase or decrease dose based on side effects and patient response with max dose of 2mg/kg/day.

- Recommended first-line therapy for very severe (cystic and conglobata) acne.
- Most effective agent for severe acne and nodulocystic acne, provides a prolonged remission period.
- Reduces sebum excretion, comedogenesis, ductal and surface *C.acnes* by 80%
- Shown to have anti-inflammatory properties.
- Treatment is usually for 20 weeks with continuation of improvement for up to 5 months after ending treatment.
- Relapse may occur in 15% of patients, factors for higher risk for relapse after treatment include severe seborrhea, young age, family history of acne, prepubertal and truncal acne.
- Contraindicated in pregnant patients because it is teratogenic, therefore, childbearing-aged females must be tested for pregnancy prior to therapy. Female patients should start using contraception 1 month prior to starting Isotretinoin treatment and continue for 1 month after discontinuation of Isotretinoin.
- Avoid in patients with renal or hepatic impairment, hyperlipidemia. Monitor liver function tests (LFTs) and lipids prior to therapy.
- Severe adverse effects may limit use. Low-dose Isotretinoin may be used to reduce the frequency and severity of adverse effects.

- Should not be combined with oral cyclines derivative because the chance of pseudotumor cerebri is increased.

Corticosteroids

- May be considered in patients with severe acne who suffer from acne flare after initiation of oral Isotretinoin therapy and in patients with acne fulminans.

Intralesional Agents:

Intralesional Corticosteroid Agents : can dramatically decrease the size of deep nodular lesion.

- Triamcinolone acetonide (2.5-10 mg/mL), inject 0.05 to 0.25 ml per lesion.
- May be used for severe, large nodulocystic lesions.

Adjunctive Hormonal Therapy:

Oral Contraceptives

- Combined oral contraceptive (COC) (estrogen and progestin) is an alternative for women with moderate to severe acne who do not respond to conventional acne therapies or with lesions limited to lower half of face.
- Progestin-only oral contraceptives are not effective and may aggravate acne.
- Responses may not be observed until after 3-6 months of treatment, some patients may experience a flare of symptoms during early cycle.
- Androgen-modulating properties are attributed to the estrogen component, decreases androgen production and reduces excess testosterone by increasing sex-hormone binding globulin production, thereby deactivating free androgen receptors, suppresses sebum formation.

Spirolactone: 25-100mg orally twice daily, morning and evening.

- Alternative antiandrogen therapy with potential to significantly improve acne severity and sebum production.
- Alternative for women with moderate to severe acne and unresponsive to conventional therapy or unwilling to take Isotretinoin.
- Decreases testosterone production, inhibits testosterone and dihydrotestosterone.

Flutamide:

- A nonsteroidal selective androgen receptor blocker being studied for use in the management of acne.
- Flutamide-induced fatal hepatitis limits the use of this agent for acne, unless benefit outweighs the risk.

Adjuvant Therapy:

- May reduce hyperpigmentation and skin superficial scarring.
- Found to be safe, effective and can significantly improve moderate acne in Asians.

Keratolytic Agents:

- Sulfur and Resorcinol.
- May be used as an adjunctive treatment together with other anti-acne agents.

Other Therapies for Acne

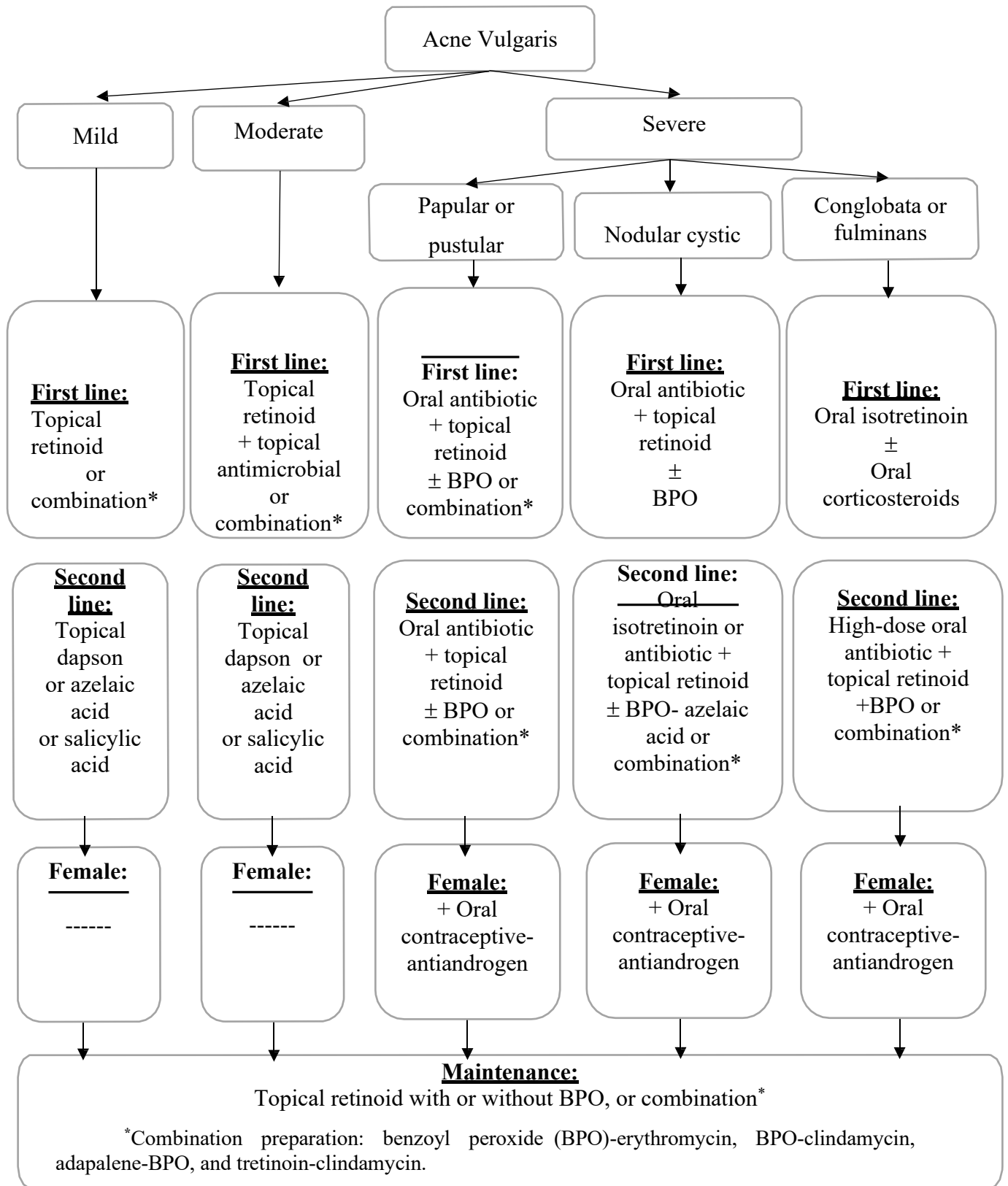
- Comedo removal: For treatment-refractory comedones.
- Tea tree oil (*Melaleuca alternifolia*): Reduces number of lesions in mild to moderate acne.

Chemical Peels (Other than Salicylic acid):

- Inhibit tyrosinase activity.

- Involve facial resurfacing wherein removal of epidermis stimulates re-epithelialization and skin rejuvenation.
 - Glycolic acid preparation: cream, gels and foams as 2% to 10%, apply once or twice daily. (Higher concentrations use as peeling by professionals : 20%).
 - Lactic acid: toner and serum as 5% to 10%, apply once or twice daily. (Higher concentrations use as peeling by professionals : 20%, 50%, 70%).
 - Trichloroacetic acid preparation: solution as 10% to 15% for acne treatment. (Higher concentrations as peeling or severe acne scarring, is used by professionals: 20%, 30%, 50%, 70% and 100%).

Treatment algorithm for Acne Vulgaris



VIII. SPECIAL CASES

Acne conglobata (AC) is a rare but severe form of nodulocystic acne. It usually presents with tender, disfiguring, double or triple interconnecting comedones, cysts, inflammatory nodules, and deep burrowing abscesses on the face, shoulders, back, chest, upper arms, buttocks, and thighs. The comedones often occur in groups of three, and the cysts often contain purulent, foul-smelling material that is discharged on the skin surface. It is a chronic inflammatory disease that inevitably leads to scar formation and disfigurement. Acne conglobata is part of the follicular occlusion tetrad, a group of related diseases involving dysfunction of the follicular unit, including dissecting cellulitis, pilonidal disease, and hidradenitis suppurativa (HS).

Hidradenitis suppurativa (HS) is an inflammatory disorder that is characterized by chronic deep-seated nodules, abscesses, fistulae, sinus tracts, and scars in the axilla, inguinal area, submammary folds, and perianal area. This disfiguring condition is accompanied by pain, embarrassment, and a significantly decreased quality of life.

Acne fulminans (AF) is a rare and severe form of inflammatory acne presenting clinically with inflammatory nodules with painful, hemorrhagic pustules and ulceration, that may or may not be associated with systemic symptoms, such as fever, polyarthritis, and laboratory abnormalities. It can be triggered by isotretinoin therapy or may occur spontaneously. It typically affects male teenagers with a pre-existing acne. The treatment significantly differs from severe acne according to severity of clinical presentation and possible systemic involvement.

If symptoms cannot be controlled with treatment protocols recommended in this guideline, specialist referral is indicated.

IX. EVOLUTION AND PROGNOSIS

The overall prognosis of acne is generally good with appropriate treatment. While acne may not be life-threatening, it can have significant and lasting psychosocial effects. Individuals with acne and acne scars often experience increased levels of anxiety and depression. The acne scars can be challenging to correct effectively.

X. COMPLICATION

The complications associated with acne vulgaris may include:

- Psychological effects, including depression, anxiety, and lowered self-esteem, may arise either from the visible appearance of the affected skin during an acne flare-up or from the resulting disfiguring scars.

XI. PREVENTION AND RECOMMENDATION

- Prevention (Maintenance phase): Use of mainly topical agents has been shown to control acne, prevent relapses and minimize sequelae
- Recommendation:
 - Patient Education**
 - Good communication between clinician and patient is important

- Noncompliance is the most common cause of treatment failure.
- Adolescents may be sensitive to actual or supposed lack of acceptance on the part of their physicians by giving the patient clear guidelines and realistic expectations, noncompliance can be avoided.
- Patient should be made aware that treatment will control acne, not cure it.
- Many acne sufferers expect to be disappointed with the results of treatment.
- Long-term therapy is required to control lesions.
- Acne tends to be less active as age increases.
- Let patient know that acne medication will take up to 8-12 weeks to work and there is no “quick-fix”.

Diet and Acne

- No specific recommendations are available for the correlation of diet and acne severity, therefore patient advice should be individualized.

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GONORRHEA

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MEY Sithach*

I. DEFINITION

Gonorrhea is a common sexually transmitted infection (STI) caused by the bacterium “*Neisseria gonorrhoeae*”. It typically spreads through vaginal, oral or anal sex. Gonorrhea is treatable and curable with antibiotics.

II. EPIDEMIOLOGY

Gonorrhea is the second most prevalent bacterial STD globally and has remained a major public health concern worldwide. WHO estimates that in 2005 there were 8.37 and 9.43 million cases for these South East Asia and West Pacific regions.

The prevalence and risk factor of gonorrhea are highest among vulnerable populations including men who have sex with men (MSM), sex workers, transgender women and adolescents, individual with previous gonorrhea or other STIs and those with concomitant STIs. Other factors include young age or early onset of sexual activity, multiple or new sexual partners, inconsistent condom use, drug use and commercial sex.

III. ETIOLOGY AND TRANSMISSION

Neisseria gonorrhoeae, a Gram-negative and aerobic diplococcus, is the causal agent of gonorrhea and belongs to the genus *Neisseria* in the family *Neisseriaceae*, which also includes *Neisseria meningitidis*.

N. gonorrhoeae is a pathogen restricted to humans as its only host and transmitted primarily by sexual contact. The most efficient transmission occurs by vaginal or anal intercourse where there is physical contact with the mucosal surface of a sexual partner with an asymptomatic or mildly symptomatic infection. An exception is indirect contact among prepubescent girls sharing contaminated objects. Vertical transmission can occur from an infected mother to her newborn during delivery, potentially leading to gonococcal conjunctivitis, pneumonia, or vulvovaginal infection. There is no evidence that gonococci can be spread through airborne droplets as is the case with meningococci.

IV. PHYSIOPATHOLOGY

The clinical manifestations of gonorrhea reveal multiple aspects of its pathogenesis:

- **Adherence:** Gonococci persist in the male urethra despite forces that would normally wash them away, indicating strong adherence mechanisms to mucosal surfaces by 2 primary adherence ligands, Pili and Opa proteins.
- **Survival Against Immune Responses:** Adheres to neutrophils primarily through Pili and certain Opa proteins, which enhance adherence but reduce phagocytosis and killing by neutrophils make Gonococci can survive despite close interaction with neutrophils, suggesting they possess mechanisms to withstand immune attacks from polymorphonuclear neutrophils.
- **Bloodstream Invasion and Dissemination:** Some gonococci can invade the bloodstream and evade plasma defenses (antibodies, complement, transferrin), which also implies an ability to breach mucosal barriers. Gonococcal Porin (Por) protein plays a role in invasion. It can be transferred from the bacteria to host cells, like red blood cells, and can alter the host cell's physiology, including calcium flux

and apoptosis. Por's ability to transfer and its effects on host cells might be important for gonococcal invasion and disease progression. Through lipooligosaccharide (LOS) modification, particularly by sialylation, which can mask immunogenic epitopes and hinder complement activation. And with the Rmp protein, expressed on the gonococcal surface, can bind to antibodies and block their access to other surface antigens, including LOS. This can interfere with complement activation and phagocytosis.

- **Tissue Damage:** Gonococci can damage host cells through various mechanisms. While they don't produce a classic extracellular toxin, their surface components, particularly lipooligosaccharide (LOS) and peptidoglycan, are toxic. LOS, specifically its lipid A component, can damage ciliated cells in the fallopian tubes, leading to impaired ciliary function. Peptidoglycan fragments can also contribute to tissue damage, potentially playing a role in inflammatory conditions like arthritis.

V. DIAGNOSIS

In many primary health care settings where diagnostic capacity for detecting *N. gonorrhoeae* is not available, a syndromic approach for case management is recommended.

1- Clinic

Gonorrhea can cause symptoms in the genitals, anus or throat. Men and women may experience different symptoms. Symptoms usually begin 1–14 days after sexual contact with an infected person.

Gonorrhea has a broad spectrum of clinical manifestations in both men and women, including asymptomatic infections, local symptomatic mucosal infections (with or without local complications), and systemic dissemination. Symptoms vary depending on the site of infection and strain of organism.

- Gonococcal in men: Acute anterior urethritis is the most common manifestation of gonococcal infection in men with dysuria and a urethral discharge that is typically purulent and profuse.
- Gonococcal in women: In about 50% of infected women, gonococcal infection is asymptomatic. However, the primary site of gonococcal infection is the endocervical canal, associated symptoms consisting of increased vaginal discharge, dysuria, intermenstrual bleeding, and menorrhagia and clinical inspection may shows purulent cervical discharge with erythema and edema; swabbing of the endocervical canal results in a highly characteristic yellow color. Urethral colonization is present in 70%–90% of infected women and is the usual site of infection in women who have had a hysterectomy. Occasionally, inflammation of the Bartholin glands is observed, with acute swelling of the labial folds and appearance of purulent discharge when pressure is applied to the gland.
- Extragenital Gonorrhea “Pharyngeal Gonorrhea”: This can occur in both men and women after oral sexual exposure. Because it is usually asymptomatic, the infection typically goes undetected and spontaneously resolves within a few weeks but some anecdotal reports suggest that gonococcal infection may cause acute pharyngitis or tonsillitis and occasionally is associated with fever or cervical lymphadenopathy.
- Extragenital Gonorrhea “Rectal Gonorrhea”: It is mainly seen in MSM and in heterosexual women who practice receptive anal intercourse. Rectal gonorrhea is asymptomatic in at least 50% of patients, but it may result in gonococcal proctitis that is accompanied by inflammation, rectal discharge, anal pruritus, bleeding, tenesmus, and constipation.

2- Investigation

Laboratory diagnosis is based on the identification of *N. gonorrhoeae* in secretions from infected mucous membranes by using:

- Stained Smear: It's a direct detection of Gram-negative diplococci within neutrophils in Gram- or methylene blue-stained smears. However, because of its lower sensitivity in asymptomatic individuals, a negative Gram stain should not be considered sufficient for excluding infection in such patients and they are not helpful for detection of rectal and pharyngeal gonorrhea due to the presence of a large number of other bacterias.
- Molecular biologic techniques "Nucleic acid amplification techniques (NAATs)": NAATs are used to detect gonococcal DNA or RNA, often in addition to chlamydial DNA or RNA, they are outperformed other tests that are currently available for the diagnosis of gonococcal infections. They are also more sensitive than culture for extragenital sites and when the infection is asymptomatic. Samples for NAATs can be obtained "non-invasively", using self-collected from vaginal or the opening of vagina swabs or just urine.
- Bacterial culture: It is reported that its specificity is ~95% and sensitivity is 80-95% for genital samples. We can use it to culture pustular skin lesions in suspected of gonococcemia. Beside of that, the advantage of isolating *N. gonorrhoeae* via culture is the ability to perform antimicrobial resistance (AMR) testing.
- All persons who receive a diagnosis of gonorrhea should be tested for other STIs, including chlamydia, syphilis, and HIV.

VI. DIFFERENTIAL DIAGNOSIS

The other causes of urethral and cervical discharge in men and women must be considered:

- Chlamydia trachomatis: a significant proportion of infections with *C. trachomatis* produce no discernible symptoms in both men and women. On women may visible only on clinical examination, or may complain of such symptoms as increased vaginal discharge, post coital/intermenstrual bleeding or lower abdominal pain and physical examination may reveal no abnormality of the cervix, or in some cases mucopurulent discharge associated with cervical friability. Women may also present with symptoms or signs related to urethral infection, namely urinary frequency, dysuria, or suprapubic pain. In men however, *C. trachomatis* is the most commonly identified cause of Non-Gonococcal Urethritis (NGU) and therefore may present with symptoms such as dysuria and/or urethral discharge. In some patients an erosive or papular, painless rash may be found on the penis in association with chlamydial infection and this is described as circinate balanitis. Urethral swabbing or urinalysis will confirm pyuria but no organisms will be demonstrated on gram stain or culture.
- Mycoplasma genitalium: often asymptomatic, but can be associated with similar symptoms and signs as chlamydia such as dysuria, urethral discharge, urethral discomfort. And in women may also have symptoms and signs of PID and cervicitis. In most situations, swabs from the urethra or cervix/vagina provide a slightly better specimen for mycoplasma detection than urine specimens and PCR is used to detect this species.
- Trichomonas vaginalis: men with trichomoniasis sometimes have symptoms of urethritis, epididymitis, or prostatitis, and women with trichomoniasis sometimes have vaginal discharge, which can be diffuse, malodorous, or yellow-green with or without vulvar irritation, and might have a strawberry-appearing cervix, which is observed more often on colposcopy than on physical examination. Wet-mount microscopy traditionally has been used as the preferred diagnostic test for *T. vaginalis* among women because it is inexpensive and can be performed at the Point of Care (POC);

however, it has low sensitivity (44%–68%) compared with culture (sensitivity of 44%–75% and specificity of <100%). NAATs are also highly sensitive (sensitivity of 95.3%–100% and specificity of 95.2%–100%) compared to wet mount microscopy and reliable samples include clinician-collected endocervical swabs, clinician-collected vaginal swabs, female urine specimens, and liquid Pap smear specimens. Lastly multiple FDA-cleared rapid tests are available for detecting *T. vaginalis* with improved sensitivities and specificities, compared with Wet-mount.

- *Candida albicans*: on women will often present with complaints of itching, burning, and a clumping discharge. Alternatively, there may be little vaginal discharge but instead erythema with or without scale on the vulva. While most infections are uncomplicated and limited, some women experience recurrent vulvovaginal candidiasis (4 or more episodes in 1 given year) without an obvious explanation. On men might present with erythema of the shaft of the penis or glans, sometimes appearing shiny and sometimes with erosions. Flecks of white curd-like material that are easily removed may be seen. Candidiasis can involve the scrotum, presenting with erythema and edema. Inguinal fold involvement tends to present with macerated or beefy red plaques with satellite papules and pustules.

VII. MANAGEMENT

1. Objective of treatment

The primary objectives in managing gonorrhea in men and women are as follows:

- Eradication of the infection and relief of symptoms
- Prevention of complication
- Prevention of transmission

2. Critical signs

In rare cases, disseminated gonococcal infection can occur and it is manifested as fever and infection in multiple organs of the body such as skin, heart, joints and meninges.

3. Risk factors for Disseminated Gonococcal Infection:

- Certain strains of *N. gonorrhoeae* that cause DGI can cause minimal genital inflammation, and urogenital or anorectal infections are often asymptomatic among DGI patients
- Women during menstruation
- Pregnancy
- Genetics complement deficiency
- Neonates

4. First-line treatment

Figure 1: Recommended Regimen for Uncomplicated Gonococcal Infection of the Cervix, Urethra, Rectum, or Oropharyngeal among adults and adolescents

Genital and anorectal gonococcal infections, Oropharyngeal gonococcal infections:

Ceftriaxone 500 mg* IM in a single dose for persons weighing <150kg

or

Gentamicin 240 mg IM in a single dose plus Azithromycin 2 g orally in a single dose

or

Cefixime* 800 mg orally in a single dose

* If chlamydial infection has not been excluded, providers should treat for chlamydia with doxycycline 100 mg orally 2 times/day for 7 days

Figure 2: Recommended Regimen for Gonococcal Conjunctivitis among adolescents and adults

Gonococcal Conjunctivitis Among Adolescents and Adults:

Ceftriaxone 1 g IM in a single dose

*Providers should consider one-time lavage of the infected eye with saline solution.

- Because gonococcal conjunctivitis is uncommon and data regarding treatment of gonococcal conjunctivitis among adults are limited, consultation with an infectious disease specialist should be considered.

Figure 3: Recommended Regimen for Gonococcal-Related Arthritis and Arthritis-Dermatitis Syndrome

Ceftriaxone 1 g IM or IV every 24 hours (for >7 days)

or

Cefotaxime 1 g IV every 8 hours (for >7 days)

or

Ceftizoxime 1 g every 8 hours

*If chlamydial infection has not been excluded, providers should treat for chlamydia with doxycycline 100 mg orally 2 times/day for 7 days.

- When treating for the arthritis-dermatitis syndrome, the provider can switch to an oral agent guided by antimicrobial susceptibility testing 24–48 hours after substantial clinical improvement, for a total treatment course of >7 days.
- Hospitalization and consultation with an infectious disease specialist are recommended for initial therapy, especially for persons who might not comply with treatment, have an uncertain diagnosis, or have purulent synovial effusions or other complications. Moreover, examination for clinical evidence of endocarditis and meningitis should be performed.

Figure 4: Recommended Regimen for Gonococcal Meningitis and Endocarditis

Ceftriaxone 1–2 g IV every 24 hours (once daily)

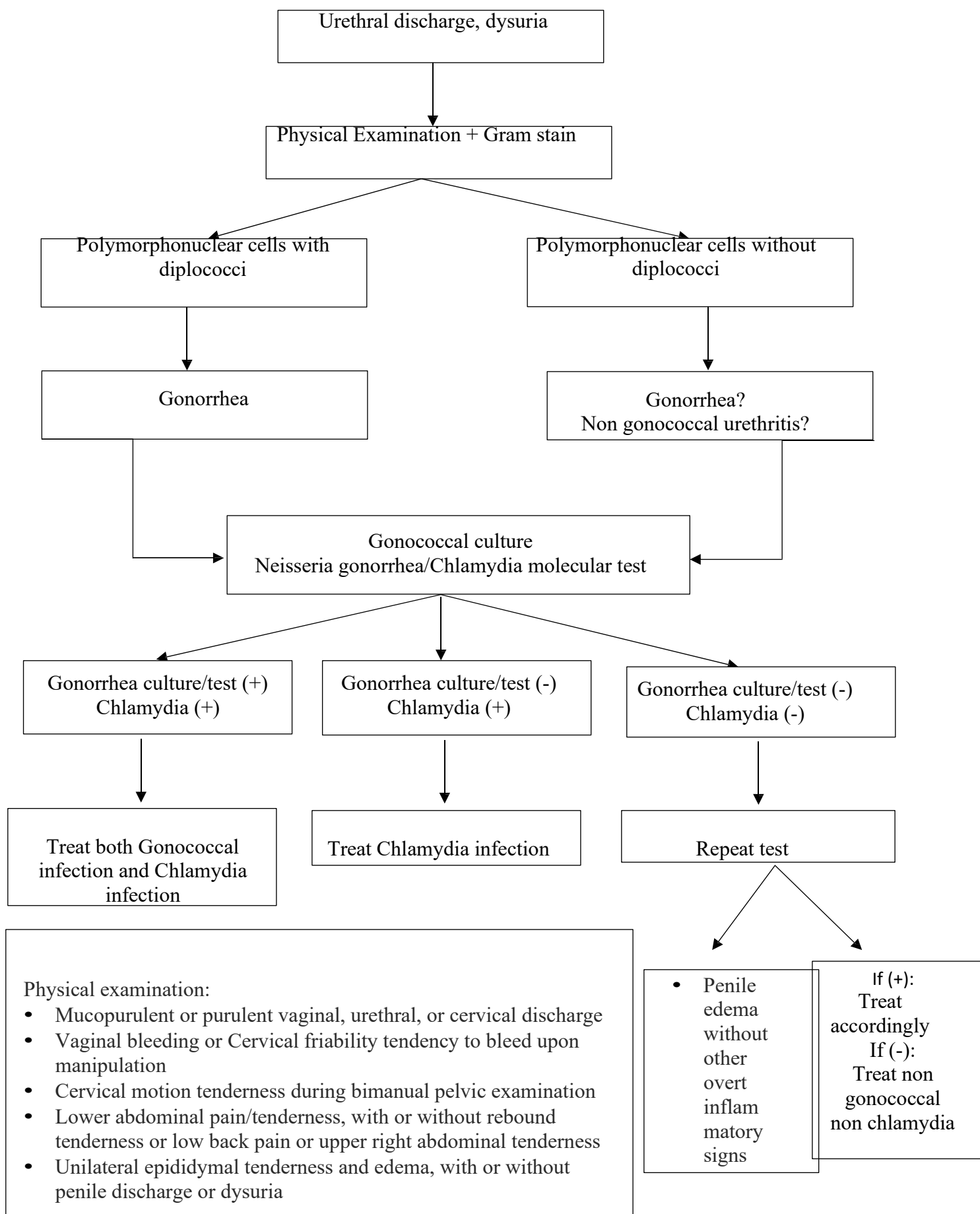
*If chlamydial infection has not been excluded, providers should treat for chlamydia with doxycycline 100 mg orally 2 times/day for 7 days

- Therapy for meningitis should be continued with recommended parenteral therapy for 10–14 days
- Parenteral antimicrobial therapy for endocarditis should be administered for >4 weeks
- However, the duration of treatment for DGI in these situations has not been systematically studied and should be determined in consultation with an infectious disease specialist and guided by the results of antimicrobial susceptibility testing.

Partner management

All contacts of an infected person, starting from the presumed time of infection for the index case, need to be informed about the possibility of infection. However, partner management is voluntary and needs to be performed with respect to human rights, social, cultural and religious behaviors, therefore, in many cases, not all contacts can be identified, nor notified. It is often difficult to determine the precise time of infection for the index case, some guideline like EADV has recommended to trace back in 3 months period. A contact/partner should be appropriately tested for the infection he/she has been exposed to. If infection is identified, the treatment should be started immediately and according to history and risk behaviors, the contact should be tested for other types of sexually transmitted infections.

5. Algorithm



IX. SPECIAL CASES

1. **Gonococcal Infection Among Neonates:** The most frequent manifestation is conjunctivitis (ophthalmia neonatorum). The disease presents as a mucopurulent to purulent discharge that literally pours out when the eyelids are separated. Both the eyes are nearly always affected. The conjunctiva becomes intensely inflamed, bright red, and swollen with marked chemosis. Prenatal screening and treatment of pregnant women for gonorrhea is the best method for preventing *N. gonorrhoeae* infection among neonates. Gonococcal infection among neonates results from perinatal exposure to the mother's infected cervix. It is usually an acute illness that manifests 2–5 days after birth. The most severe manifestations of *N. gonorrhoeae* infection among neonates are ophthalmia neonatorum and sepsis, which can include arthritis and meningitis. Less severe manifestations include rhinitis, vaginitis, urethritis, and scalp infection at sites of previous fetal monitoring.
 - For ocular prophylaxis, the WHO STI guideline suggests one of the following options for topical application to both eyes immediately after birth:
 - tetracycline hydrochloride 1% eye ointment
 - erythromycin 0.5% eye ointment
 - povidone iodine 2.5% solution (water-based)
 - silver nitrate 1% solution
 - chloramphenicol 1% eye ointment

Figure 5: Recommended Regimen for Gonococcal Ophthalmia Neonatorum

Ceftriaxone 25–50 mg/kg body weight IV or IM in a single dose, not to exceed 250 mg
Or
Cefotaxime 100 mg/kg body weight IV or IM as a single dose if the baby is unable to receive ceftriaxone because of simultaneous administration of IV calcium
Or
Kanamycin 25 mg/kg (maximum 75 mg) IM as a single dose
Or
Spectinomycin 25 mg/kg (maximum 75 mg) IM as a single dose

2. **Pregnancy:** Gonococcal infection may have a devastating impact on both the pregnant woman and the outcome of pregnancy. Prior to 7–12 weeks of gestation, there is a relatively high risk of developing salpingitis and PID. After this period, the chances of developing PID are less, mainly due to local cervical factors that are modified under the influence of progesterone. After the 12th week of gestation, gonococcal chorioamnionitis may occur which leads to septic abortion. It also increases the risk of premature rupture of membranes, preterm births, prematurity, perinatal mortality, or low birth weight. The prevalence of pharyngeal gonorrhea is also higher in pregnant women, as well as ophthalmia neonatorum, pharyngeal infections, and other syndromes in the newborn.

Figure 6: Recommended Regimen for Gonococcal infection in pregnancy

Ceftriaxone 500 mg in a single IM dose and treatment for chlamydia if infection has not been excluded

3. **Patient with penicillin allergy:** The risk for penicillin cross-reactivity is highest with first-generation cephalosporins but is rare (<1%) with 3rd generation cephalosporin (e.g., ceftriaxone and cefixime). And, if IgE-mediated penicillin allergy is strongly suspected, dual treatment with single doses of IM gentamicin 240 mg plus oral azithromycin 2 g can be administered.
4. **HIV Infection:** Persons who have gonorrhea and HIV infection should receive the same treatment regimen as those who do not have HIV.

X. EVOLUTIONS AND PROGNOSIS

With adequate early therapy, complete cure and return to normal function are the rule. Most gonococcal infections respond quickly to cephalosporin therapy. Late, delayed, or inappropriate therapy may lead to significant morbidity or, on rare occasions, death.

XI. COMPLICATIONS

Local Complication in Men:

- Epididymitis: Patients with acute epididymitis tend to present with unilateral testicular pain and swelling, and most patients with gonococcal epididymitis have also manifested urethritis when they present.
- Penile lymphangitis: Patients might present with penile edema (“bull-headed clap”).
- Post inflammatory urethral strictures were common complications of untreated gonorrhea.

Local Complication in Women

- Pelvic inflammatory disease (PID): In acute PID, the clinical syndrome comprised primarily of salpingitis, and frequently including endometritis, tubo-ovarian abscess, or pelvic peritonitis. PID is the most common of all complications of gonorrhea, as well as the most important in terms of public-health impact, because of both its acute manifestations and its long-term sequelae (infertility, ectopic pregnancy, and chronic pelvic pain).
- Bartholin’s gland abscess: had enlargement and tenderness of the gland

Systemic Complication:

- Disseminated gonococcal infection usually manifested by the acute arthritis-dermatitis syndrome. The most common clinical manifestations of DGI are joint pain and skin lesions. The “classic” skin lesion of gonococcal dermatitis is a tender, necrotic pustule on an erythematous base, in many patient’s skin lesions also may evolve from, or present as, macules, papules, pustules, petechiae, bullae, or ecchymoses. The skin lesions usually less than 30 in number and tend to be located on distal portions of the extremities. Joint pain in DGI most often involves wrist, metacarpophalangeal, ankle, and knee.
- Gonococcal endocarditis is an uncommon complication of gonococcal bacteremia, occurring in an estimated 1–3% of patients with DGI. The aortic valve appears to be infected most often in patients with gonococcal endocarditis, and in the pre-antibiotic era, most patients died of aortic valve incompetence and acute heart failure within 6 weeks.
- Gonococcal meningitis: Some case reports of this complication describe patients with typical presentations of acute bacterial meningitis, usually without typical findings of DGI. *N. gonorrhoeae* is indistinguishable from *N. meningitidis* on Gram’s stain of cerebrospinal fluid.

XII. PREVENTION AND RECOMMENDATION

- External condoms or male condom: when used consistently and correctly are effective in preventing the sexual transmission of HIV infection and other STIs disease including gonococcal infection.
- Internal condoms or female condoms can also provide protection from acquisition and transmission of STIs.
- Male circumcision reduces the risk for HIV infection and certain STIs among heterosexual men.
- Abstinence from non-protected oral, vaginal, and anal sex and participating in a long-term, mutually monogamous relationship with a partner known to be uninfected are also prevention approaches to avoid transmission of STIs.
- Annual screening for *N. gonorrhoeae* infection is recommended for all sexually active aged women at increased risk for infection (e.g., those aged ≥ 25 years who have a new sex partner, more than one sex partner, a sex partner with concurrent partners, or a sex partner who has an STI).

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This image illustrates Gonococcal urethritis with a purulent urethral discharge.

Source: Bologna JL, Schaffer JV, Cerroni L. *Dermatology*. 5th ed. Elsevier; 2024. p. [1475]



This image illustrates **Gonococemia (arthritis-dermatosis syndrome)**, showing a pustule with surrounding erythema on the toe. Source: Bologna JL, Schaffer JV, Cerroni L. *Dermatology*. 5th ed. Elsevier; 2024. p. [1475]



Annex: Courtesy of Gonorrhea patients at the dermatology department of Cambodia-China Friendship Preah Kossamak Hospital.

SEBORRHEIC DERMATITIS

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I. DEFINITION

Seborrheic Dermatitis is a common, chronic inflammatory skin condition as a relapsing eczema affecting seborrheic areas of the scalp, face, and upper trunk, ears and the diaper area. It is characterized by redness (erythema), flaky white or yellowish scales and itching. Patients usually complain of the cosmetic effects of scaling and redness with itching. Seborrheic dermatitis is a recurrent, waxing dermatosis.

II. EPIDEMIOLOGY

There are infantile and adult forms, with the former being self-limited and most prevalent the first 3–4 months of life, while the latter is chronic with a peak in the fourth to sixth decades. The prevalence of seborrheic dermatitis is estimated to be 5%, but its lifetime incidence is likely much higher. Men are affected more often than women. Extensive and therapy-resistant seborrheic dermatitis is an important cutaneous sign of HIV infection. It is more commonly observed in patients with neurologic diseases (Parkinson disease, cerebrovascular accidents) or who are immunosuppressed. Additional contributing factors include nutritional deficiencies, impaired epidermal barrier, and occasionally medications (neuroleptics, atypical antipsychotics).

III. ETIOLOGY

Unknown studies to uncover the pathogenesis of seborrheic dermatitis in adult and adolescent patients have been conducted, but the etiology has not been clearly identified yet. Multifactorial causes, including several endogenous and exogenous predisposing factors, are associated with seborrheic dermatitis. The role of sebaceous glands in pathogenesis of seborrheic dermatitis is notable considering time and lesion distribution of seborrheic dermatitis. The immunologic status of patients or the susceptibility to seborrheic dermatitis can be an important factor because seborrheic dermatitis is much more seen in those with certain underlying diseases such as AIDS and Parkinson disease. *Malassezia* may also be one of the causes of seborrheic dermatitis as antifungal medications are effective. The relationship of the seborrheic dermatitis lesions with seasonal fluctuations or sun exposure implies that multiple exogenous factors can contribute to the development of seborrheic dermatitis.

IV. PATHOPHYSIOLOGY

Although the pathogenesis of seborrheic dermatitis is not fully understood, it is linked to sebum overproduction (seborrhea) and the commensal yeast *Malassezia*. It results from a complex interaction of *Malassezia* overgrowth, altered skin lipids, microbial imbalance, and immune responses.

- **Malassezia Yeast and its Role**

Malassezia is a lipophilic yeast naturally found on human skin, depending on skin oil (sebum) for growth. While usually harmless, it can cause or worsen conditions such as, seborrheic dermatitis, pityriasis versicolor, folliculitis (inflammation of hair follicles), atopic dermatitis (in some cases).

- **Seborrheic Dermatitis and Malassezia**

Seborrheic dermatitis mainly occurs in areas with active sebaceous glands (e.g., scalp, face, chest). While *Malassezia* is more commonly found in seborrheic dermatitis lesions than in healthy skin, the number of yeast cells does not always correlate with disease severity. Even normal skin can carry similar amounts of *Malassezia* as affected skin.

- Sebum and Sebaceous Glands in seborrheic dermatitis often develops in areas with high sebum production, as sebum provides nutrients for *Malassezia* growth. However, it can also occur in individuals with normal sebum levels. This suggests that the composition of skin lipids, rather than their quantity, plays a key role. Patients with seborrheic dermatitis often have altered skin lipids, such as reduced free fatty acids, which normally have antimicrobial properties.
- Imbalance in Skin Microbiota
Seborrheic dermatitis is associated with an imbalance in skin microorganisms. *Malassezia* overgrowth occurs alongside a reduction in other bacteria, such as *Cutibacterium acnes*, which help produce protective free fatty acids.
- Immune Response and Inflammation
An abnormal immune response to *Malassezia* may cause inflammation. *Malassezia* produces toxic metabolites and enzymes (e.g., lipases) that irritate the skin, along with inflammatory molecules like reactive oxygen species (ROS), which worsen the condition. This immune response may also explain the higher prevalence of seborrheic dermatitis in individuals with weakened immune systems (HIV patients).

V. DIAGNOSIS OF SEBORRHEIC DERMATITIS

- Clinical findings
 - Duration of lesions symptoms develop over weeks and may persist or recur.
 - Skin symptoms typically asymptomatic or associated with mild itching (pruritus).
 - Physical examination present multiple erythematous patches scattered across affected areas, often with yellow greasy scales that are thick and adherent. Flaking or oily texture overlying plaques.
 - Commonly Affected Areas:
 - Scalp (dandruff or severe forms).
 - Face: especially central regions (nose, eyebrows, paranasal areas, and ears).
 - Upper chest and back.
 - In some cases
 - Axillae (armpits), groin, and interscapular areas.
 - Red, scaly plaques may also develop in the gluteal cleft or umbilicus.
 - In young children:
 - Tinea amiantacea
 - Thick, plate-like, yellowish-white scales tightly adhered to the scalp and hair.
 - Scales are usually 2-10 cm in size.
 - Persistent lesions may cause temporary hair loss or large adherent scales.
 - In adolescents and adults:
 - Classic seborrheic dermatitis:
 - Dandruff (fine, dry, white scalp scales with minor itching).
 - Scalp changes range from diffuse brawny scaling to thick, oily, yellow crusts overlying red, inflamed skin.
 - Severe cases:
 - Redness and scaling along the frontal hairline, eyebrows, nasolabial folds, and behind the ears (retro-auricular folds).
 - Marginal blepharitis: involvement of the eyelid margins.
 - External ears canal scaling.
 - Red, scaly plaques in the axillae and groin regions.
 - On the extremities, lesions may look more eczema-like, with less redness but still scaly and poorly defined.
 - Disease Severity:
 - The Seborrheic Dermatitis Area and Severity Index (SDASI) is a tool used to assess severity.

- The face is divided into four regions (each representing 25%).
- Each region is evaluated for:
 - Extent of involvement.
 - Pattern of presentation.
 - Degree of redness (erythema) and scaling.
- Investigations
 - **Wood's Lamp Examination:** Helps rule out fungal infections or other conditions.
 - **Dermoscopy** A non-invasive tool that shows unique patterns and features to differentiate seborrheic dermatitis from other scaling skin disorders.
 - **KOH Examination and Fungal Culture:** Useful in atypical or resistant cases, especially when fungal infection is suspected.
 - **Patch Testing:** Recommended if symptoms do not improve with standard treatments, to rule out contact allergies.
 - **Skin Biopsy:** Rarely needed but helpful to distinguish seborrheic dermatitis from other similar conditions (psoriasis). Typical findings in seborrheic dermatitis, Parakeratosis (retention of nuclei in the outer skin layers). Plugged follicular openings (ostia). Spongiosis (fluid between skin cells, causing swelling).

VI. DIFFERENTIAL DIAGNOSIS

1. Other Dermatitis

Atopic Dermatitis: Commonly affects flexural areas (elbows, knees), wrists, and face (especially around the eyes and mouth), unlike seborrheic dermatitis, which tends to affect sebaceous-rich areas (scalp, face, chest, and back). Atopic dermatitis is often associated with a personal or family history of asthma, hay fever, or allergic rhinitis. It is more intensely pruritic and can present with vesicles, oozing, and crusting in acute phases. It frequently starts in childhood, while seborrheic dermatitis can occur at any age.

Contact Dermatitis: Typically localized to areas of direct exposure to an irritant or allergen (hands, face, or other body parts in contact with products). Contact dermatitis has a clear relationship with exposure to an allergen or irritant. It often presents with erythema, scaling, vesicles, or oozing also history of exposure to a specific product or environmental irritant is key in diagnosis.

Nummular Dermatitis: Presents with round, coin-shaped lesions, often on the trunk or limbs, and may lack the greasy scale seen in seborrheic dermatitis. Typically, the lesions are intensely pruritic and may be dry or have a more eczematous appearance. Nummular dermatitis can be triggered by dry skin or cold weather, unlike seborrheic dermatitis, which tends to be oilier.

2. Scalp Psoriasis: Psoriasis can affect the scalp, but the distribution is often more extensive and may extend beyond the hairline. Unlike seborrheic dermatitis, psoriasis may also affect other areas such as elbows, knees, and lower back. Appearance Psoriatic plaques are well-demarcated, thick, and covered by silvery scales, while seborrheic dermatitis tends to have finer, greasy scales. History of Psoriasis is often familial, and patients may have a history of joint involvement (psoriatic arthritis) or other skin manifestations like nail pitting.

3. Systemic Lupus Erythematosus (SLE): Can cause a characteristic malar rash (butterfly-shaped), usually affecting the cheeks and nose, which can be confused with seborrheic dermatitis on the face. Appearance the lupus rash is typically erythematous, with a scaly surface but lacks the greasy appearance. It is also associated with photosensitivity. SLE can present with systemic features,

including fatigue, joint pain, and systemic involvement (renal, neurological, etc.). Laboratory tests (like ANA, anti-dsDNA) are crucial for diagnosis.

4. Pemphigus Foliaceus: Primarily causes blisters and erosions on the skin, often on the face and scalp, which can resemble seborrheic dermatitis if the erosions are crusted. Blisters quickly rupture, leaving erosions and crusts, whereas seborrheic dermatitis is characterized by scaling without large blister formation. Pemphigus foliaceus is an autoimmune condition, and patients may have a positive Nikolsky sign (skin shearing off with light pressure). Confirm by Direct immunofluorescence would show intercellular IgG deposits.

5. Pityriasis Rosea: Primarily affects the trunk and upper extremities, beginning with a herald patch followed by a “Christmas tree” pattern of smaller lesions. Lesions of pityriasis rosea have a fine scale with a “collarette” appearance at the periphery, which is different from the greasy scales of seborrheic dermatitis. The course usually resolves in 6–8 weeks without treatment, while seborrheic dermatitis tends to be chronic with flares and remissions.

6. Dermatophytosis (Tinea Capitis): Tinea capitis typically affects the scalp and can cause hair loss, unlike seborrheic dermatitis which does not result in significant hair loss. Tinea capitis presents with round, scaly patches, often with a central clearing and a raised border. It may be associated with pustules and scaling in the hair. It may be pruritic, and a positive fungal culture or microscopy would confirm the diagnosis.

7. Pityriasis Versicolor (Tinea Versicolor): Commonly affects the upper trunk (interscapular region) and other sebaceous areas. Lesions of pityriasis versicolor are hypopigmented or hyperpigmented with fine scaling and may be more subtle than seborrheic dermatitis. Unlike seborrheic dermatitis, pityriasis versicolor lesions have a more diffuse, non-inflammatory appearance. KOH preparation of skin scrapings may show characteristic Malassezia yeast cells, which can be differentiated from seborrheic dermatitis by the absence of greasy scaling.

8. Candidiasis: Often affects intertriginous areas (axillae, groin, under breasts), and in severe cases, it can involve the scalp. The skin in candidiasis appears red, inflamed, with a smooth, shiny surface and sometimes satellite lesions (small red pustules). Unlike seborrheic dermatitis, candidiasis does not have greasy scaling. Candidiasis is more often associated with warmth and moisture, and patients may have risk factors like diabetes, immunosuppression, or antibiotic use.

9. Impetigo: Typically affects the face (especially around the nose and mouth) and may also involve the scalp. Impetigo presents with honey-colored crusts and pustules, which can be mistaken for seborrheic dermatitis when the latter becomes crusted. However, the crusts in seborrheic dermatitis are often greasy, not honey-colored. Impetigo is caused by bacterial infection (often Staphylococcus aureus or Streptococcus pyogenes), and patients may have fever or a history of trauma or insect bites.

VII. MANAGEMENT

1. Objectives

Seborrheic dermatitis itself is generally not considered a life-threatening condition, may not be completely curable, treatment aims to improve quality of life by keeping the condition under control, focus on symptoms, reduce inflammation, and manage flare-ups, reducing redness, scaling and itching also important for preventing recurrence.

2. Critical Sign

- Infantile seborrheic dermatitis asymptomatic and clear spontaneously with month.
- Adolescent and childhood can clear also but recurrent course
- Scalp (Cradle cap)

- removal of the thick scale with mineral oil
- fine tooth comb off scale can provide symptomatic and cosmetic relief
- Shampoos containing selenium sulfide, Zinc pyrithione, tar, or ketoconazole used intermittently 2 to 3 time a week. It is important to instruct the shampoo and let it sit on the scalp for 5 minutes before rinsing
- Face and Diaper area
 - Cream containing ketoconazole help full and safe
 - Low potency topical steroids preparation 2.5% hydrocortisone cream, Fluticasone cream twice daily for 2 to 3 days for acute flare

3. Mortality risk factor

Seborrheic dermatitis itself is generally not considered a life-threatening condition, and it does not have a direct impact on mortality. It is primarily a chronic inflammatory skin disorder that causes discomfort, including itching, redness, and flaking, but it does not significantly affect over all health or lead to death. however, mortality risk factors related to seborrheic dermatitis may arise indirectly:

3.1 Underlying Conditions: Seborrheic dermatitis is more common in individuals with certain medical conditions, such as HIV/AIDS, Parkinson's disease, and other immunocompromised states. In these cases, the condition may be a marker of poorer overall health and could contribute to complications in these populations.

3.2 Infection Risk: Severe or untreated seborrheic dermatitis can cause skin damage, which might increase susceptibility to secondary bacterial or fungal infections. In rare cases, these infections, especially in immunocompromised individuals, could potentially lead to more serious health issues.

3.3 Quality of Life Impact: Chronic discomfort and the psychological burden of visible skin issues (like embarrassment or stress) might contribute to mental health challenges, but these are not direct causes of mortality.

4. Outpatients Management

Outpatient management of seborrheic dermatitis focuses on controlling symptoms, reducing flare-ups, and maintaining long-term skin health. The approach is usually tailored to the severity and location of the condition, and it may involve topical treatments, lifestyle modifications, and patient education. Management involves a combination of topical therapies (antifungal, corticosteroids, or calcineurin inhibitors), to reduce symptoms and minimize flare-ups usually responds satisfactorily to bathing and application of emollients Ketoconazole cream (2%) is indicated in more extensive or persistent. Short courses of low-potency topical corticosteroids may be used initially to suppress inflammation. Mild shampoos are recommended for the removal of scalp scales and crusts. Avoidance of irritation (the use of strong keratolytic shampoos or mechanical measures to remove the scales from the scalp) is important typical.

5. Treatment

- Treatment type of seborrheic Dermatitis
- First line therapy
 - Scalp and Beard preparation of 2% ketoconazole shampoo or selenium sulphide shampoo twice a week for a month, then once or twice for symptom control and can also be effective for scaling and inflammation.
 - Face and Body in adults 2% cream one daily/twice daily, 1% Clotrimazole Cream twice daily/three time daily, 1% Econazole Cream twice daily., 2% Miconazole Cream twice daily use as above for at least 4 weeks, then less frequently.
 - In case sever review diagnosis, consider specialist referral, HIV testing.

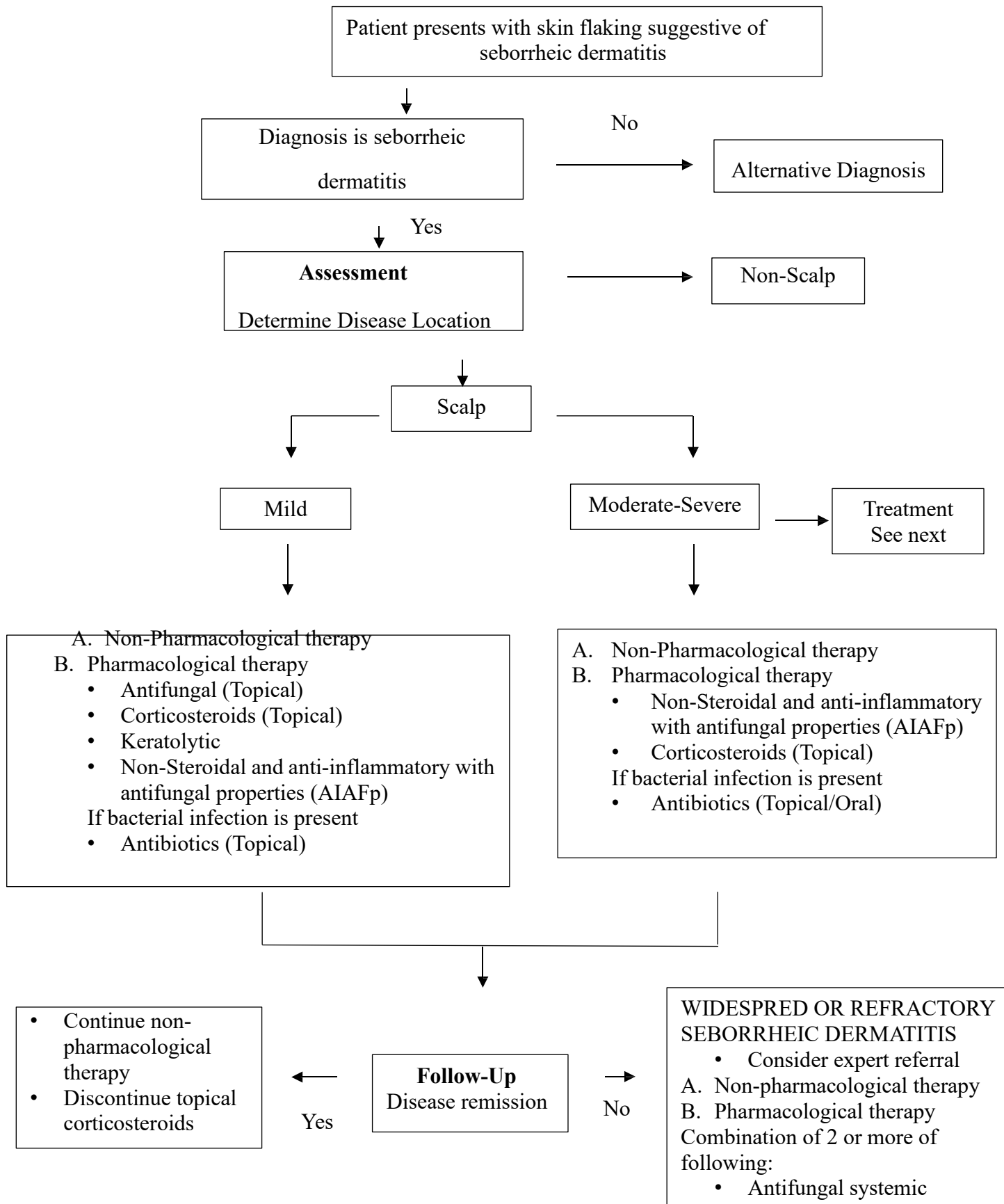
- In infants Removal of scalp crust with baby shampoo and gentle brushing. Overnight soak of petroleum jelly or warmed vegetable oil if needed. Daily bathing with soap substitute
- Second line therapy
 - Scalp and Beard Medicated shampoo with zinc pyrithione, Coal tar or salicylic acid
 - Face and Body in adults' mild topical corticosteroids for 1-2 weeks (hydrocortisone 1%) may be used for short-term flare-ups on the face, scalp, or body.
 - Body in adults' moderate topical corticosteroids for 1-2 weeks twice daily as 0.025% betamethasone cream can be used for more stubborn or widespread lesions but should be used cautiously to avoid skin thinning, especially on the body.
 - In case severe **Topical Calcineurin Inhibitors** (for sensitive areas like the face), **Tacrolimus** or **pimecrolimus** can be used in sensitive areas (face, eyelids) to reduce inflammation without the risk of skin thinning associated with steroids.
 - In infant's topical imidazole cream as 1% clotrimazole cream twice daily / three times daily 1% Econazole cream twice daily 2% miconazole cream twice daily.
- Additional therapy
 - Scalp topical keratolytic or mineral/olive oil for the removal of scale and crust
 - Face and body in adults' potent topical corticosteroid scalp application scalp application for 4 weeks if there is severe scalp itch.
 - Antifungal shampoo, 2% ketoconazole, as a body wash.
 - Hygiene measure for eyelid involvement using cotton buds moistened with baby shampoo.
 - In infants' topical corticosteroids not routinely advised but may be used for certain infants with nappy rash.
- **Lifestyle Modifications**
 - **Gentle Skin Care** Use mild, fragrance-free cleansers to avoid irritating the skin. Avoid harsh scrubbing or over-washing, as this can worsen irritation and dryness. **Moisturization** Regular use of non-comedogenic, fragrance-free moisturizers to help maintain skin hydration and protect the skin barrier.
 - Sun Protection** Moderate sun exposure may help improve symptoms in some individuals, but sun protection (SPF 30+) is important to prevent skin damage.
- **Patient Education**
 - **Trigger Identification** Educate the patient on potential triggers for seborrheic dermatitis, such as stress, certain skin products, cold and dry weather, or hormonal changes. **Consistency of Treatment.** Encourage regular and consistent use of prescribed treatments. Even after symptoms improve, continue maintenance treatment to prevent flare-ups. **Avoiding Overuse of Steroids.** Advise against the long-term use of topical corticosteroids, especially on the face, to prevent side effects like skin thinning.

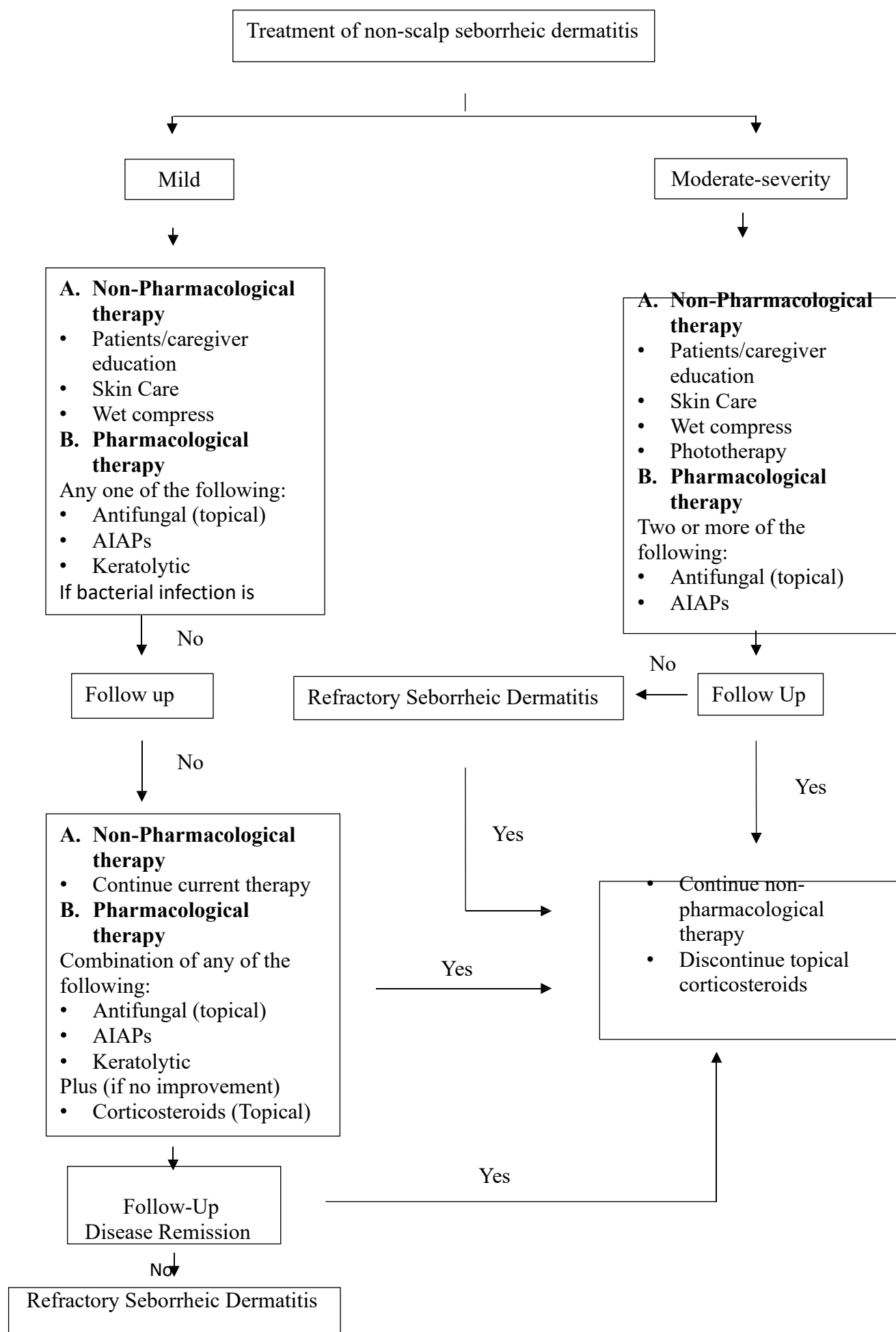
- **Follow-Up**
 - **Regular Monitoring:** Regular follow-up visits to assess treatment efficacy and adjust the management plan as needed, especially if new flare-ups occur. **Adjusting Treatment:** If the patient does not respond to initial treatments or experiences frequent relapses, further evaluation of the underlying causes or consideration of stronger treatments may be need.

6. Treatment adjustment

- **Phototherapy** in rare cases where the condition is resistant to topical treatments, light therapy (narrowband UVB) may be considered, typically under dermatological supervision.
- **Antifungals** – Oral Itraconazole 200mg/day PO x 7 days. Prophylaxis 200mg PO 24 hours for 1st 2 day of each month until 11 months.
- **Caution of Adverse Reactions** GI effects N/V, Diarrhea, abdominal pain, anorexia CNS effect headache, dizziness, somnolence dermatologic effect rash, pruritus, other effect decreased libido, malaise.

7. Algorithm





NON-PHARMACOLOGICAL THERAPY

Patient/Caregiver Education

- Cradle cap is a harmless condition that usually goes away on its own within the first year of life, without treatment.
- Assure the caregiver that seborrheic dermatitis doesn't cause permanent hair loss.
- Explain that treatment doesn't cure seborrheic dermatitis permanently; it may need to be repeated when symptoms return.
- Show the caregiver how to apply topical treatments correctly.

Scalp and Skin Care

- Wash affected areas regularly with soap to remove oils and reduce seborrhea.
- After washing, apply moisturizing creams to keep the skin hydrated.
- To remove thick scales on the scalp:
 - Apply warm mineral oil or olive oil to the scalp, then rinse it out a few hours later or tar shampoo.
 - Alternatively, leave a coal tar-keratolytic cream or phenol-saline solution on overnight, then shampoo it off in the morning.
 - For seborrheic blepharitis (eyelash involvement), gently clean the eyelashes with baby shampoo and a cotton swab.
 - Avoid using hair sprays, gels, and excessive sun exposure.

Wet Compress

- Apply a wet compress to moist or cracked areas before using topical corticosteroids.

Phototherapy

- Phototherapy (light therapy) can be an alternative treatment for widespread or hard-to-treat seborrheic dermatitis.
- UVA and UVB light can help slow the growth of *Malassezia* (the fungus that contributes to seborrheic dermatitis), and many people notice improvements in seborrhea during summer.

VIII. SPECIAL CASES

For severe or treatment-resistant seborrheic dermatitis, HIV test is warranted.

HIV-Associated Seborrheic Dermatitis symptoms more severe, widespread, and treatment-resistance lesion.

Seborrheic Dermatitis in Patients with Parkinson's Disease also Seborrheic Dermatitis in immunocompromised patients (post-transplant, Chemotherapy)

Explain to patients that the condition is chronic, relapsing and topical treatment should be reinstated during flares.

IX. EVOLUTION AND PROGNOSIS

Seborrheic dermatitis is a chronic, relapsing condition with a generally good prognosis when treated appropriately. While it is not curable, most individuals can manage their symptoms effectively with topical treatments and lifestyle adjustments. The severity of the disease, the presence of underlying conditions, and the patient's ability to manage triggers can all influence the course and outcome of the condition. Long-term control is possible, but flare-ups are common and can occur throughout life.

1. Onset seborrheic dermatitis often starts in early adulthood, though it can also appear in infants (as cradle cap) or later in life. It commonly affects areas rich in sebaceous (oil) glands, such as the scalp, face (especially around the eyebrows, nose, and ears), chest, and back. In infants, seborrheic dermatitis tends to resolve spontaneously within the first year of life. Cradle cap is the most common manifestation in infants.
2. Flare-ups and Remissions: The condition tends to follow a chronic, relapsing course, with symptoms fluctuating between flare-ups (when symptoms worsen) and remission (when symptoms improve or resolve completely for a time). Flare-ups may be triggered by factors

- such as stress, seasonal changes (especially winter with dry, cold weather), hormonal changes, illness, medication changes, or use of harsh skin products.
3. The severity of seborrheic dermatitis can range from mild (mild dandruff or flaking) to severe (widespread erythema, scaling, and inflammation, especially in areas like the scalp or face). Chronic, untreated seborrheic dermatitis can result in thickened skin or hyperpigmentation in the affected areas, particularly on the face or scalp.
 4. Affected Populations Infants: In most cases, seborrheic dermatitis in infants resolves by the age of 1 without long-term effects. Adults: In adults, seborrheic dermatitis may persist over the years and can be more challenging to manage. Those with underlying conditions (e.g., HIV/AIDS, Parkinson's disease, immunocompromised states) may experience more severe or persistent forms. Older adults: The condition can also affect older individuals, especially those with dry skin or those undergoing immunosuppressive therapy.

X. COMPLICATION

Potential Complications:

Secondary Infections: Persistent scratching or inflammation can lead to secondary bacterial or fungal infections in the affected areas.

Hyperpigmentation or Hypopigmentation: Prolonged inflammation may result in skin discoloration, particularly in darker-skinned individuals, due to post-inflammatory hyperpigmentation.

Scarring: rarely, chronic inflammation or severe cases may lead to scarring or thickening of the skin, especially in cases involving the scalp or other areas with more significant inflammation.

XI. REVENTION AND RECOMMENDATION

Management of Stress Reduction: Stress is a well-known trigger for seborrheic dermatitis flare-ups. Incorporating relaxation techniques, such as yoga, mindfulness, meditation, or regular physical activity, can help lower stress levels and reduce the frequency of flare-ups. Adequate Sleep Getting enough sleep is important for maintaining a strong immune system and managing stress, both of which can influence the severity of seborrheic dermatitis. Scratching or picking at affected areas can further irritate the skin, lead to secondary infections, or exacerbate the condition.

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Fig. 13.1 Infantile seborrheic dermatitis.

- A- Involvement of the scalp with thick adherent yellow scale overlying mild inflammation is often referred to as “cradle cap”.
- B- Glistening red plaques of the neck, axillary and inguinal folds as well as the penis and umbilicus. Note disseminated lesions on the trunk and extremities



Fig.1: Adult seborrheic dermatitis of the face, ear and scalp.

- A- Rather sharply demarcated pink plaque with flaky white and greasy scale. Note the fissure in the retro auricular sulcus.
- B- Sharply demarcated pink–orange thin plaques with yellow, greasy scale, especially in the nasolabial fold. When this degree of severity is seen, the possibility of underlying HIV infection needs to be considered.
- C- Symmetric red–brown to violet plaques of the central forehead, nasal bridge, and medial cheeks with an associated hypopigmented peripheral and figurate rim.
- D- Involvement of the upper eyelids, medial canthal regions, and infraorbital creases; seborrheic dermatitis confined to the major crease in the outer canthal region can also be seen.



PSORIASIS

SUONG Sinath, PROM Phanit, LOK Bunthay, SONG Nymul, CHHIT Kunthy, ENG Nirath, HONG Kettya, NONG Reaksmey, SARUN Senghour, DENG Angkirboss, CHAN Vicheth

I. DEFINITION

- Psoriasis is a genetically determined, systemic immune-mediated chronic inflammatory disease that affects primarily the skin and joints.
- The most characteristic lesions consist of red, scaly, sharply demarcated, indurated plaques, present particularly over extensor surfaces and scalp. The disease is enormously variable in duration, periodicity of flares and extent. Morphological variants are common.
- In general, the skin lesions appear well before the psoriatic arthritis becomes clinically apparent. However, 10-15% psoriatic arthritis appeared first.
- Psoriasis is now considered to be a systemic disease that is more than skin deep, involving joints as well as extra-articular sites such as the gastrointestinal tract and eye, etc.
- Psoriatic disease is an umbrella term used to define a unique illness involving several different anatomical sites in the same patient either simultaneously or in different sequential stages. It is a multi-factorial illness with a complex genetic basis showing an interplay of multiple genes situated on different loci, triggered by a variety of environmental factors including infections (*Streptococcus pyogenes*, *Malassezia*, *Candida albicans*, *Staphylococcus aureus*), trauma, stress, alcohol, smoking, stressful life events, dry weather conditions, drugs (*Lithium*, *B-blockers*, *antimalarials*, *non-steroidal anti-inflammatory drugs*, *tetracyclines*) and obesity.

II. EPIDEMIOLOGY

Psoriasis occurs worldwide. Psoriasis affects 1-3% of the adult population with various extra cutaneous manifestations. The incidence of psoriasis is low in Asians (0.4%). Psoriasis may be at any age, but it is uncommon under the age of 10 years. It is most likely to appear between the ages of 15 and 30 years. Possession of certain HLA class I antigens, particularly HLA, Cw6, is associated with an earlier age of onset and with a positive family history. Up to 90% of patients have a family history of psoriasis. Psoriasis is a common chronic inflammatory cutaneous disease affecting 0.5% to 2%

of children and adolescence. The disease affects 4% of all children younger than 16 years with all types of dermatologic disorders.

Males and females are equally affected by psoriasis vulgaris. Many studies indicate that age of onset is younger in females. Thus, one German study demonstrated a peak age of onset of 22 years in males and 16 years in females in early-onset disease.

III. ETIOLOGY

Psoriasis is an immune-mediated disease, which means that your body's immune system starts overacting and causing problems. Scientists do not fully understand what triggers the faulty immune cell activation, but they know that it involves a combination of genetics and environmental factors.

IV. PATHOPHYSIOLOGY

Psoriasis is a chronic inflammatory skin disease, with a strong genetic basis, characterized by complex alterations in epidermal growth and differentiation and multiple biochemical, immunologic, and vascular abnormalities, and a poorly understood relationship to nervous system function. Historically, psoriasis was widely considered to be a primary disorder of keratinocytes.

Psoriasis is a hyperproliferative skin disease with increase rate of epidermal turnover. Pathology is characterized by uniform elongation of the rete ridges, with dilated blood vessels, thinning of the suprapapillary plate, and intermittent parakeratosis. Epidermal and perivascular dermal infiltrates of lymphocytes, with neutrophils occasionally in aggregates in the epidermis.

Genome-wide association studies report that the histocompatibility complex HLA-C*06:02 (previously known as HLA-Cw6) is associated with early-onset psoriasis and guttate psoriasis. This major histocompatibility complex is not associated with psoriatic arthritis, nail psoriasis, or late-onset psoriasis.

Immune factors and inflammatory cytokines (messenger proteins) such as IL1 β and TNF α , IL-23 and IL-17 are responsible for the clinical features of psoriasis. These have therefore become targets for biological drugs and have led to success in drug management.

V. DIAGNOSIS

Psoriasis is diagnosed clinically; however biopsy may occasionally be needed to confirm cases with atypical presentations.

A. Clinical Findings:

The characteristic lesion is well-defined with a sharply demarcated erythematous plaques with silvery white scales with or without sterile pustules. The colour, “salmon pink” has a particular depth of hue not normally seen in eczema, seborrheic dermatitis or lichen simplex. This quality of colour is of special diagnosis value in lesions on the palms, soles and scalp. However, in dark-skinned races, the quality of the colour is lost (Fig. 2).

Commonly are on scalp, elbows and knees, hands, feet, nails and trunk including intergluteal fold. Less often, nearly all of the body surface is involve “Erythrodermic Psoriasis” or numerous, small, widely disseminated papules and plaques are seen “Guttate Psoriasis”; if pustules “Generalized Pustular Psoriasis and Pustulosis of the palms and soles.

The patterns of onset in childhood, including congenital psoriasis. Acute guttate attacks characteristically occur in this age group.

The successive removal of psoriatic scales usually reveals an underlying smooth, glossy red membrane with small bleeding points where the thin suprapapillary epithelium is torn off “Auspitz’s sign”.

Psoriasis can demonstrate the Koebner phenomenon. This involves the generation of new lesions on the skin that has been damaged or irritated such as by injury. Psoriasis is one of several conditions in which various types of trauma may elicit the disease in previously uninvolved skin. The Koebner reaction usually occurs 7-14 days after injury. Clearing of existing psoriasis following injury has been observed and termed the reverse Koebner reaction.

The psoriatic plaque may be encircled by a clear peripheral zone, the halo or ring or Woronoff. The term elephantine psoriasis might be used to described unusual but very persistent, thickly scaling, large plaques that sometimes occur on the back, limbs, hips or elsewhere.

Involution of a lesion usually starts in its centers, resulting in annular psoriatic lesions. Lesions often clear with postinflammatory hypo- or hyperpigmentation that can be expected to fade over several months.

A conventional classification is as follows:

- Chronic Plaque Psoriasis (CPP) or Psoriasis Vulgaris is the commonest form of psoriasis: It affects 80%-90% of patients. Plaques can appear anywhere on the body as raised patches of inflamed, itchy, and painful skin with scales or sometimes the skin may be red with silvery-white scales. Plaques are irregular, round to oval in shape and tend to be symmetrically distributed.
- Guttate Psoriasis: it affects roughly 8% of people living with psoriasis. It looks like small 2-5mm in diameter, round, red, drop-shaped scaly spots and often affects children and young adults. It often appears on the arms, legs, and torso; however, it can affect any area of the body. Usually occurs after an upper respiratory infection with group A beta-hemolytic Streptococci. However, it may be the first manifestation of psoriasis in a healthy individual or an acute exacerbation of long-standing plaque.
- Erythrodermic Psoriasis: is rare, affecting about 2% of people living with psoriasis. This type of psoriasis can cause intense redness or discoloration and shedding of skin layers in large sheets. It often affects nearly the whole body and can be life-threatening.
- Pustular Variants: it affects about 3% of people living with psoriasis. It has small, pus-filled bumps on top of plaques that may be surrounded by inflamed or reddened/discolored skin.

◆ Generalized Pustular Psoriasis (GPP)

- Acute Generalized Pustular Psoriasis (von Zumbusch) may develop de novo or from

preexisting plaque-type psoriasis and is characterized by fiery-red, irregular patches with round, arcuate, serpiginous borders, over which are seen tens of thousands of 1mm to 2mm superficial pustules. These tend to occur in flexural or skin-fold area but may occur anywhere. The pustules coalesce into lakes of pus, desquamate, and form new pustules as the border moves in waves every 24 to 72 hours. Most patients have fever, leukocytosis, hypocalcemia, and hypoalbuminemia. The incidence is equal in men and women.

- Generalized Pustular Psoriasis of Pregnancy also known as impetigo herpetiformis, is a rare variant of generalized pustular psoriasis that occurs in pregnancy. It typically appears in the third trimester of pregnancy and resolves rapidly after delivery. It presents as symmetrical annular or polycyclic erythematous patches and plaques, with the formation of sterile pustules at the periphery, beginning in skin folds and spreading to the extremities and trunk. The pustules rapidly merge to form large plaques with desquamation and red-brown pigmentation. The face, palms, and soles are usually spared.
- Infantile and Juvenile Pustular Psoriasis is extremely rare and thought to account for approximately 0.6% of pustular psoriasis cases. It is a severe variant of psoriasis characterized by scattered pustules on top of underlying erythematous skin, distributed across the trunk and extremities.
- centrifugum type or Lapiere Psoriasis, represents a rare variety of generalized pustular psoriasis clinically characterized by rapid onset of annular circinate lesions with micro-pustules at the borders without classic plaque psoriasis manifestations.
- Localized forms of GPP (not hands and feet)

- ◆ Localized Pustular Psoriasis the eruption is chronic and recurring and is recognizable by yellowish pustules on a background of redness and scaling on the palm or on the instep of the sole and side of the heel, or on both areas. Lesions are observed in all stages of development, including vesicles, vesico-pustules, frank pustules, and dried brown maculopapules. This is a female predominance.
- Palmoplantar Pustulosis: usually presents hyperkeratotic and erythematous plaques with symmetrical distribution on the palmar and/or plantar regions, in association with painful fissures and nail involvement, such as pitting, subungual hyperkeratosis, and longitudinal ridging
- Acrodermatitis continua of Hallopeau with acral pustule formation; chronic recurrent pustulation of nail folds, nail beds, and distal fingers leading to nail loss. Occurs alone or with generalized pustular psoriasis.
- Special Locations:
 - Scalp psoriasis or psoriasis capitis represents as mild scaling to crusting on the entire scalp sometimes extending onto the forehead, around the nose, in the beard area, or behind or inside the ears.
 - Mucosal lesions, sometimes geographic tongue are associated to psoriasis
 - Flexural psoriasis or psoriasis inversa: It affects one- quarter of people living with psoriasis. This type appears in your skin folds include inflamed deep-red or darkened skin that is smooth and not scaly.
 - Nail Psoriasis: this is seen in association with all types of psoriasis. 20-50% of all cases. Fingernails and toenails frequently involved. Nail changes include pitting, onycholysis, and oil spot (pathognomonic).
 - Psoriatic arthritis is the major associated systemic manifestation; most commonly presentation is asymmetric

oligoarthritis of small joints of hands and feet.

B. Investigation:

There is no constantly present laboratory abnormality in uncomplicated psoriasis. May be necessary for complicated or atypical disease. There are no serologic markers or pathognomonic lab tests for psoriasis.

Baseline Studies:

- Complete Blood Count (CBC), Platelet, Creatinine, Electrolytes, Liver Function Test (LFT), Hepatitis panel screening, latent tuberculosis (TB) screening, HIV screening.
- Primary providers should monitor cholesterol, blood sugars, and blood pressure because psoriatic patients are at risk for cardiovascular disease and metabolic syndrome.
- Throat cultures and Antistreptolysin O (ASO) titers should be performed in patients with guttate flares.
- Wet mounts and bacterial cultures should be performed in patients with inverse psoriasis or in patients who flare despite adequate topical or systemic therapy.
- In severe recalcitrant cases, evaluate for infection with HIV.
- Advised prior to starting, resuming or increasing therapy with systemic and immunosuppressive agents.

VI. DIFFERENTIAL DIAGNOSIS

The characteristics already defined are usually sufficient to enable the diagnosis to be made, but doubt may arise in atypical cases, in particular sites, and when psoriasis is complicated by or alternates with other diseases.

- **Seborrheic Dermatitis:** Most often, the scales of psoriasis look thicker and drier than the scales of seborrheic dermatitis. Psoriasis tends to extend beyond the hairline. And psoriasis usually affects more than one area of the body.
- **Hyperkeratotic Eczema of the palms and soles:** Hyperkeratotic eczema often starts with sudden itching on the palms, sides of the fingers, or soles of the feet. Blisters may then appear, causing more intense itching and pain. Eczema is usually “characterized by poorly demarcated red, dry areas with fine scale,” while psoriasis appears as “red, well-circumscribed lesions (called plaques) with heaped-up, white scaling.

- **Sezary syndrome:** This disease is characterized by some reddish patches or plaques all over the skin which extends to the whole body into erythroderma, lymphadenopathy. It is also indicated by the presence of atypical lymphocytes called Sezary cells.
- **Pityriasis Rubra Pilaris (PRP):** is a relatively uncommon skin disease, clinically characterized by follicular keratotic plugs, red to orange plaques, and palmoplantar hyperkeratosis.
- **Drugs Reactions:** Psoriasiform eruptions are the most common cutaneous consequence of beta-blocker therapy, seen more frequently in patients with no past or family history of psoriasis. Clinical improvement after withdrawal of the implicated drug is the distinguishing feature in many cases suggesting drug-induced psoriasis.
- **Small plaque parapsoriasis:** the plaques or scaly patches are thinner than they are in the most typical form of psoriasis. It is defined by lesions < 5 cm in diameter. Lesions are typically asymptomatic or accompanied by mild pruritus and are predominantly located on the trunk.
- **Pityriasis Lichenoids Chronica (PLC):** PLC causes small, round, scaly lesions that are often pink, red, or brown. The lesions flatten out over time, leaving a brown discoloration that fades slowly. Psoriasis causes red, scaly patches that can be itchy and painful. The patches can crack and bleed, and can look pink, red, brown, purple, or gray.
- **Secondary Syphilis:** Psoriasis is characterized by scaly, itchy patches of skin that can be pink, red, brown, purple, or gray. Secondary syphilis can cause a rash of rough, red, or reddish brown spots on the palms of the hands and the bottoms of the feet. Other symptoms of secondary syphilis include flu-like symptoms, weight loss, and patchy hair loss.
- **Pityriasis Rosea:** A temporary condition that usually clears up on its own within a few weeks to months. It's thought to be caused by a virus and often appears as a large, raised patch on the chest, back, or abdomen, followed by smaller patches that spread out. The rash usually has a "Christmas tree" pattern and doesn't leave permanent marks.
- **Tinea Corporis or Cruris:** Ringworm rashes are often circular with a clear center, and the itchiness is often more intense than psoriasis. Look closely at the red patches on your skin. If there is a silvery look to them, you might have psoriasis. If they look like circles or rings, it's more likely you have a fungal infection.

- **Pemphigus Foliaceus:** Pemphigus foliaceus produces superficial blisters confined to the skin, without involvement of mucous membranes.
- **Intertrigo:** Inverse psoriasis usually doesn't appear as scaly as in other forms of psoriasis. You may notice patches of red, inflamed skin. These areas can become more irritated if you perspire or if there is skin-on-skin rubbing. Inverse psoriasis is distinguished by its appearance in the folds of your skin. These patches typically form: under the breasts, in the armpits, between the crease of the buttocks, around the genitals, on any other dark, moist fold of skin.
- **Bowenoid Papulosis:** A rare, sexually transmitted infection caused by the human papillomavirus (HPV). It usually appears as small, pigmented papules on the genital or pubic area. Bowenoid papulosis is a precancerous lesion that closely resembles squamous cell carcinoma-in-situ (Bowen disease). Bowen disease is a slow-growing skin malignancy that can appear as red or brown, scaly patches on the skin. It can occur anywhere on the body, but it's most common in areas exposed to the sun.
- **Contact Dermatitis (CD):** Psoriasis is characterized by squamous, well-demarcated red plaques of variable size. In contrast, CD is mainly characterized by diffuse dryness, erythema, papules, and vesicles followed by scaling. It causes an intense itch.
- **Acute Generalized Exanthematous Pustulosis (AGEP):** AGEP is a severe cutaneous adverse reaction that is also characterized by non-follicular sterile pustules. In pustular psoriasis, the rash is usually monomorphic, lasts for a longer period of time and is recurrent. It is less flexural than AGEP. Many times it is associated with arthritis and regresses in a period of 10 to 14 days.
- **Mycosis Fungoides** variant of cutaneous T-cell lymphoma (CTCL): Mycosis fungoides is easily confused with psoriasis in the early stages. Unlike psoriasis, mycosis fungoides is often accompanied by persistently swollen lymph nodes. Pancreas and liver enlargement are also common.

VII. EVALUATION

Severity measurement of psoriasis can be performed with different tools.

- Defined by subjective and objective qualitative assessment based on Body Surface Area (BSA) involvement, location, severity and number of lesions, response to topical treatments, associated physical disability, presence or absence of psoriatic arthritis and other comorbidities, psychosocial effects

and impact on quality of life of the patient.

- BSA: measures percentage of the body surface affected by psoriasis based on “rule of 9” or taking patient’s one palm-size as 1%.

- BSA:

- < 3% Mild
 - ≥ 3% but < 10% Moderate
 - ≥ 10% Severe

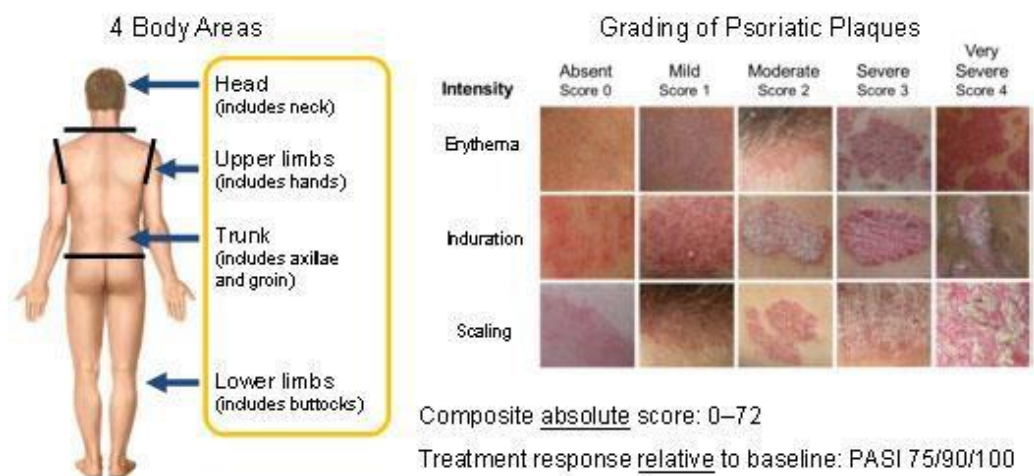
- Assessment of severity may also done using Psoriasis Area Severity Index (PASI), Physician’s Global Assessment (PGA), Patient’s Global Assessment or Dermatology Life Quality Index (DLQI).

- PGA measures psoriasis severity and response to treatment based on erythema, induration and scaling.

Moreover, BSA, PGA, BSA and PGA can be used in clinical practice, PASI is a tool recommended for clinical trials and not for clinical practice. Additionally, subjective symptoms and quality of life assessment tools such as DLQI, PASI and pruritus assessment can provide useful information about the patients quality of life.

Psoriasis Area Severity Index (PASI)

PASI is more specific in quantifying the severity of psoriasis with scores ranging from 0 to 72.



Dermatology Life Quality Index (DLQI)

The Dermatology Life Quality Index questionnaires is designed for use in adults, i.e. patients over the age of 16. It is self explanatory and can be simply

handed to the patient who is asked to fill it in without the need for detailed explanation.

SCORING

The scoring of each question is as follows:

Very much	Scored 3
A lot	Scored 2
A little	Scored 1
Not at all	Scored 0
Not relevant	Scored 0
Question 7, prevented work or studying	Scored 3

The DLQI is calculated by summing the score of each question resulting in a maximum of 30 and a minimum of 0. The higher the score, the more quality of life is impaired.

How to interpret meaning of DLQI scores?

0-1	no effect at all on patient's life
2-5	small effect on patient's life
6-10	moderate effect on patient's life
11-20	very large effect on patient's life
21-30	extremely large effect on patient's life

DERMATOLOGY LIFE QUALITY INDEX (DLQI)

Hospital No.:

Date:

Name:

Score:

Address:

Diagnosis:

The aim of this questionnaire is to measure how much your skin problem has affected your life OVER THE LAST WEEK. Please tick (✓) one box for each question.

- | | | |
|---|-------------------------------------|---------------------------------------|
| 1. Over the last week, how itchy, sore, painful or stinging has your skin been? | Very much ☐ | |
| | A lot ☐ | |
| | A little ☐ | |
| | Not at all ☐ | |
| 2. Over the last week, how embarrassed or self conscious have you been because of your skin? | Very much ☐ | |
| | A lot ☐ | |
| | A little ☐ | |
| | Not at all ☐ | |
| 3. Over the last week, how much has your skin interfered with you going shopping or looking after your home or garden ? | Very much ☐ | |
| | A lot ☐ | |
| | A little ☐ | |
| | Not at all ☐ | Not relevant ☐ |
| 4. Over the last week, how much has your skin influenced the clothes you wear? | Very much ☐ | |
| | A lot ☐ | |
| | A little ☐ | |
| | Not at all ☐ | Not relevant ☐ |
| 5. Over the last week, how much has your skin affected any social or leisure activities? | Very much ☐ | |
| | A lot ☐ | |
| | A little ☐ | |
| | Not at all ☐ | Not relevant ☐ |
| 6. Over the last week, how much has your skin made it difficult for you to do any sport ? | Very much ☐ | |
| | A lot ☐ | |
| | A little ☐ | |
| | Not at all ☐ | Not relevant ☐ |
| 7. Over the last week, has your skin prevented you from working or studying ? | Yes ☐ | |
| | No ☐ | Not relevant ☐ |
| If "No", over the last week how much has your skin been a problem at work or studying ? | A lot ☐ | |
| | A little ☐ | |
| | Not at all ☐ | |
| 8. Over the last week, how much has your skin created problems with your partner or any of your close friends or relatives ? | Very much ☐ | |
| | A lot ☐ | |
| | A little ☐ | |
| | Not at all ☐ | Not relevant ☐ |
| 9. Over the last week, how much has your skin caused any sexual difficulties ? | Very much ☐ | |
| | A lot ☐ | |
| | A little ☐ | |
| | Not at all ☐ | Not relevant ☐ |
| 10. Over the last week, how much of a problem has the treatment for your skin been, for example by making your home messy, or by taking up time? | Very much ☐ | |
| | A lot ☐ | |
| | A little ☐ | |
| | Not at all ☐ | Not relevant ☐ |

Please check you have answered EVERY question. Thank you.

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VII. ANAGEMENT

1. Objectives

Psoriasis is a chronic disease so the goal of treatment is to control the symptoms but not cure it. Treatment goals include a targeted final outcome of PASI ≤ 2 , PGA clear or almost clear, or DLQI < 2 . Treatment response is achieved if patient attains $\geq 75\%$ improvement from baseline PASI score.

2. Critical Sign

Psoriasis is associated with an increased risk for anxiety, depression, obesity, diabetes, hypertension, lymphoma, myocardial infarction, and stroke.

Flares of psoriasis may be triggered by local factors (skin injury, scratching), systemic factors (infections, especially Streptococcus and HIV), psychologic stress, medications (lithium, beta-blockers, interferon, and antimalarials). Factors that aggravate psoriasis including obesity, smoking, or excessive alcohol.

Aggressive mechanical removal of scale on the scalp may result in alopecia.

Guttate psoriasis is seen more commonly in children and young adults, who have been exposed to upper respiratory virus or Streptococcus pyogenes.

3. Mortality Risk Factor

Patients with psoriasis have an increased morbidity and mortality from cardiovascular events, particularly those with severe and long duration of psoriasis skin disease. Risk of myocardial infarction is particularly elevated in younger patients with severe psoriasis. Psoriasis patients have also been shown to have increased relative risk of lymphoma, particularly in patients with more severe disease.

4. Outpatient Management

Psoriasis is treated in a graded manner starting with topical therapy and progressing to phototherapy and finally to systemic therapy. Guidelines of care for the treatment of psoriasis with topical therapies have been developed by the American Academy of Dermatology.

Topical treatment is used for mild to moderate psoriasis. These include steroids, coal tar, dithranol, calcipotriol and moisturizers. These can be used alone, or in conjunction with other treatments.

Oral medication is an option for patients who have severe or extensive psoriasis but unable to have phototherapy or fail to respond to phototherapy. The commonly used tablets are methotrexate and cyclosporin. Other medications that are sometimes used include Acitretin.

Given the chronic relapsing nature of psoriasis, treatments are combined and rotated to minimize the toxicity of any one given treatment.

❖ Topical Agents:

Topical Corticosteroids (CS): 1st-line treatment for mild or localized psoriasis and for plaque psoriasis. Once-twice daily applications are recommended for most agents. Every other day or weekend application has been used to treat chronic conditions. Seven classes have been suggested based on potency. They range from the superpotent topical glucocorticosteroids in class 1 to the very low-potency topical glucocorticosteroids in class 7.

▪ Class 1 (Superpotent):

- Clobetasol propionate gel, ointment, cream and foam 0.05%
- Betamethasone dipropionate gel and ointment 0.05%
- Diflorasone diacetate ointment 0.05%
- Fluocinonide cream 0.1%
- Flurandrenolide tape 4ug/cm²
- Halobetasol propionate ointment and cream 0.05%
- **Class 2 (High Potency):**
 - Amcinonide ointment 0.1%
 - Betamethasone dipropionate cream and ointment 0.05%
 - Clobetasol propionate solution (scalp application) 0.05%
 - Desoximetasone ointment and cream 0.25% and gel 0.05%
 - Diflorasone diacetate ointment and cream 0.05%
 - Fluocinonide gel, ointment, cream and solution 0.05%
 - Halcinonide ointment and cream 0.1%
 - Mometasone furoate ointment 0.1%
- **Class 3 (High Potency):**
 - Amcinonide cream and lotion 0.1%
 - Betamethasone dipropionate cream and lotion 0.05%
 - Betamethasone valerate ointment 0.1%
 - Desoximetasone cream 0.05%
 - Diflorasone diacetate cream 0.05%
 - Fluticasone propionate ointment 0.005%
 - Triamcinolone acetonide ointment 0.1% and cream 0.5%
- **Class 4 (Medium Potency):**
 - Betamethasone valerate foam 0.12%
 - Fluocinolone acetonide ointment 0.025%
 - Flurandrenolide ointment 0.05%
 - Hydrocortisone valerate ointment 0.2%
 - Mometasone furoate cream and lotion 0.1%
 - Triamcinolone acetonide ointment and cream 0.1%
- **Class 5 (Medium Potency):**
 - Betamethasone dipropionate lotion 0.05%
 - Betamethasone valerate cream and lotion 0.1%
 - Clocortolone pivalate cream 0.1%
 - Fluocinolone acetonide cream 0.025% and oil 0.01%
 - Fluticasone propionate cream and lotion 0.05%
 - Flurandrenolide cream and lotion 0.05%
 - Hydrocortisone butyrate ointment, cream and lotion 0.1%
 - Hydrocortisone probutate cream 0.1%
 - Hydrocortisone valerate cream 0.2%
 - Prednicarbate ointment and cream 0.1%
 - Triamcinolone acetonide lotion 0.1%
- **Class 6 (Low Potency):**
 - Aclometasone dipropionate ointment and cream 0.05%
 - Triamcinolone acetonide cream 0.1%
 - Betamethasone valerate lotion 0.1%
 - Desonide gel, ointment, cream, lotion and foam 0.05%
 - Fluocinolone acetonide cream 0.01% and solution 0.05%
 - Triamcinolone acetonide cream and lotion 0.025%

- **Class 7 (Low Potency):**
 - Topicals with hydrocortisone, dexamethasone and prednisolone.
 - Commonly used in combination with vitamin D analogues, tar, topical retinoid to enhance their effectiveness and minimize side effects, such as skin thinning.
- **Dithranol (Anthralin) 0.02% cream** for patients with mild-moderate psoriasis. Apply on affected area twice daily.
- **Keratolytics:**
 - Salicylic Acid is the most commonly used keratolytic recommended for mild-moderate psoriasis. May be used alone or combination with other forms of therapy (corticosteroids and topical immunomodulators). Salicylic acid as 2-11.8% cream, lotion, ointment, shampoo: apply 1-4 times a day.
 - Sulfur 10% ointment apply once or twice daily.
 - Urea 10% cream, lotion or 20% cream: apply 2-3 times a day.
- **Retinoids:**
 - **Tazarotene** 0.05% cream, 0.1% gel or cream: apply thinly to the lesions once daily in the evening.
 - For mild-moderate psoriasis, including palmo-plantar psoriasis and nails psoriasis.
- **Coal Tar:**
 - Effective for use in chronic plaques in mild-moderate psoriasis.
 - May be used if vitamin D analogue and corticosteroid is ineffective or not tolerated.
 - Most popularly used as Scalp treatment with corticosteroid or combined with UVB
 - Preparation:
 - 0.5%, 1.5% lotion, 0.5%-2% gel, 2% ointment: apply 1-4 times a day.
 - 5% emulsion: apply 2-3 times a day; 6% cream: apply 1-2 times a day.
- **Vitamin D Analogues (Calcipotriol, Maxacalcitol, Tacalcitol):**
 - Indicated in chronic plaque psoriasis especially for long therapy and for patients with mild-moderate psoriasis.
 - Calcipotriol or Calcipotriene:
 - Preparation: 50mcg/g cream, gel, ointment: apply 1-2 times a day. Max dose 15g/day or 100g/week.
 - Preparation: 50mcg/ml, scalp solution: apply twice daily. Max dose 60ml/week.
 - Max treated area: 30% BSA Recommended treatment period.
 - Calcipotriol/Betamethasone Dipropionate:
 - Preparation: 50mcg/500mcg ointment, gel or foam
 - Apply once daily; *max dose 15g/day or 100g/week*. Max treated area 30% BSA. Recommended treatment period: ointment: 4weeks; gel: 8 weeks.
 - Calcitriol:
 - Preparation: 3mcg/g ointment, apply twice daily; *max dose 30g/day up to 35% BSA*.
- **Triamcinolone Acetonide (TA) injections (5-10mg)** effectively induce remission in many difficult-to-control psoriatic plaques. More severely

affected nails may be treated with TA into the nail bed. Generally, nail disease requires systemic therapy for improvement.

❖ **Phototherapy/Photochemotherapy**

Phototherapy is generally used in patients with moderate-severe psoriasis, with psoriasis in vulnerable areas that is unresponsive to topical therapy. Phototherapy may be used as solo therapy or in combination with topical or systemic agents.

- Broadband Ultraviolet B (BB-UVB)
- Narrow Band Ultraviolet B (NB-UVB)
- Psoralen plus UVA (PUVA)
- Soak/Bath PUVA
- 308nm Excimer Light

The most common treatment is narrowband ultraviolet B (311-313nm), which has been shown to be highly effective. Patients require 3 treatments per week for 1 to 3 months. Many patients go on and off phototherapy for years. Tar, Vitamin D analogs, retinoids (Tazarotene gel or cream), and topical steroids enhance the effectiveness of phototherapy.

The excimer laser is effective for limited plaque psoriasis and is administered 2 to 3 times per week.

❖ **Systemic Therapy**

Current systemic therapy may be divided into traditional and biologic therapy. Methotrexate, Cyclosporine, and Acitretin are the most widely used traditional systemic treatments for psoriasis. Cyclosporine is a calcineurin inhibitor and effectively improves psoriasis quickly such as Pustular Psoriasis and Erythrodermic Psoriasis. Acitretin is a systemic retinoid that is very effective for all forms of psoriasis, but is used mostly for erythrodermic, pustular, and palmo-plantar psoriasis.

Additionally, there have been new approvals for oral and topical therapies for psoriasis, including a topical first-in-class mechanism of action for psoriatic skin lesions. In terms of new and exciting topical therapies for patients with psoriasis, Roflumilast cream was approved by the FDA in 2022 for management of plaque psoriasis in adults and adolescents. Roflumilast is a topical PDE4 inhibitor. PDE4 degrades its substrate cyclic adenosine monophosphate (cAMP) to adenosine monophosphate (AMP), which leads to the production of proinflammatory mediators. Therefore, inhibitors of PDE4 work by blocking the degradation of cAMP, which leads to a reduction in inflammation.

Recently approved topical therapies for pediatric psoriasis includes Roflumilast cream, which just received FDA approval in October 2023 for pediatric patients 6 years or older. This is exciting news because there were no previously FDA-approved nonsteroidal topical therapies approved for pediatric patients younger than 12 years. Many parents and clinicians are cautious and wary of topical corticosteroids for the management of psoriasis in pediatric patients given the risk of irreversible skin striae and skin atrophy associated with long-term topical steroid use. Given that psoriasis is a chronic disease and many pediatric patients have only mild disease that is easily managed with topical therapy, a nonsteroidal topical option is very attractive to both clinicians and pediatric patients because of the nonexistent risk of skin striae and skin

atrophy. Furthermore, roflumilast cream is an ideal topical therapy for pediatric patients because it may be used on any skin site, for any severity of disease, and with no limitation on duration or use. There are no warnings or contraindications in the FDA package insert (with the exception of the fact that it should be avoided in patients with moderate to severe liver impairment). Liver impairment is extremely rare in pediatric patients, so in clinical practice, there are essentially no relative or absolute contraindications for the use of roflumilast cream for the management of pediatric psoriasis.

- **Hydroxycarbamide (Hydroxyurea)** 500mg PO 1-3 times a day.
- **Methotrexate**
 - 7.5-25mg/week. 3 divided oral doses 12h apart weekly, weekly single doses orally,
 - or single weekly dose of 2.5-25mg/week subcutaneously, intramuscularly or intravenously.
 - May increase dosage gradually by 2.5mg/week; *Max Dose: 30mg/week*.
 - Once a single dose exceeds 25mg PO, subcutaneous injections are recommended. Max dose 50mg/week.
 - 7.5-25mg PO/IV/IM once weekly; *Max Dose: 50mg/week*.
- **Apremilast**
 - **Initial dose:**
 - Day1: 10mg PO 12 hourly in the morning;
 - Day2: 10mg PO 12 hourly (morning & evening);
 - Day3: 10mg PO in the morning and 20mg in the evening;
 - Day4: 20mg PO 12 hourly (morning and evening);
 - Day5: 20mg PO in the morning and 30mg in the evening.
 - **Maintenance dose:** Day6 onwards 30mg PO 12 hourly (morning & evening).
- **Ciclosporin (Cyclosporin A, Cyclosporine)**
 - **Initial dose:** 2.5-5mg/kg/day PO once or twice daily for 1 month; may increase by 0.5-1mg/kg/month; *Max dose: 5mg/kg/day*.
- **Dimethyl fumarate**
 - Week1: 30mg PO 24hourly;
 - Week2: 30mg PO 12 hourly;
 - Week3: 30mg PO 8 hourly;
 - Week4: 120mg PO 24hourly;
 - Week5 120mg 12hourly;
 - Week6: 120mg PO 8hourly;
 - Week7: 120mg PO in AM & midday; 240mg PO in PM;
 - Week8 240mg PO in AM, 120mg PO in midday, 240mg PO in PM;
 - Week9 onwards: 240mg PO 8hourly; *Max dose: 720mg/day*.
- **Mycophenolic acid (Mycophenolate mofetil):**
 - Preparation: 250mg capsules, 500mg tablets, 200mg/ml oral suspension, and 500mg vials for intravenous injection.
 - Dose: 2-3g PO per day for 12 weeks. Many patients achieve long remissions but may take up to 12 weeks to see maximal effects.
- **Methoxsalen**
 - 10-70mg PO given 1.5-2hr prior to UVA exposure.
 - Usually given twice a week, most patients clear in 20-25 treatments, but maintenance treatment is needed.

□ **Acitretin**

- **Initial dose:** 25-30mg/day PO for 2-4weeks. Adjust dose based on patient response and adverse reactions.
- **Maintenance dose:** 25-50mg/day PO for 6-8 weeks; *Max dose 75mg/day.*

5. Immunosuppressants-Parenteral

Biologic treatments are immune-modulators that target the T cell and cytokines involved in psoriasis. These agents are highly effective but are very expensive and thus are not accessible to many patients.

Etanercept, Infliximab, and Adalimumab target tumor necrosis factor-alpha; Ustekinumab targets p40 subunit of interleukin-12/23; Secukinumab and Ixekizumab target interleukin-17A. Brodalumab binds to the IL-17 receptor. Guselkumab targets interleukin-23. There are many targeted immunologic modulators in the Pipeline for psoriasis.

There are now more than 13 FDA-approved biologic options for moderate to severe psoriasis. The newest biologic agent for psoriasis, Bimekizumab (Bimzelx), was approved by the FDA on October 18, 2023, for the management of moderate to severe psoriasis. Bimekizumab is unique in that it blocks both the IL-17A and IL-17F cytokines.

Although not considered a new biologic, Adalimumab will greatly affect the 2024 psoriasis treatment landscape because generic versions are approved for the management of adult plaque psoriasis. These include the 9 following FDA-approved agents: Adalimumab-aacf (Idacio), Adalimumab-fkjp (Hulio), Adalimumab-adbm (Cyltezo), Adalimumab-bwwd (Hadlima), Adalimumab-adaz (Hyrimoz), Adalimumab-aaty (Yuflyma), Adalimumab-aqvh (Yisumry), Adalimumab-afzb (Abrilada), and Adalimumab-atto (Amjevita).

- **Adalimumab** 80mg SC initially, followed by 40mg SC given every other week starting 1 week after initial dose.
- **Brodalumab** 210mg SC at week0, 1 and 2 followed by 210mg SC every 2 weeks.
- **Certolizumab**
 - 400mg SC (given as 2 SC inj of 200mg each) every 2 weeks
 - < 90kg: initial dose of 400mg SC (given as 2 SC inj of 200mg each) on treatment week 0, 2 & 4 then reduce to 200mg every 2 weeks starting on treatment week 6.
 - Maintenance dose: 200mg SC every 2 weeks or 400mg SC every 2 weeks in patients with insufficient response.
- **Efalizumab** 0.7mg/kg SC as a single dose; followed by 1mg/kg SC weekly for 12 week; *Max dose 200mg/dose.*
- **Etanercept**
 - **Initial dose:** 25mg SC 2x/week or 50mg SC weekly or 50mg SC 2x/week at intervals of 3-4 days x 3 months then reduce to 25mg SC 2x/week or 50mg SC weekly; continue treatment until remission is achieved for up to 24 weeks, discontinue if no response after 12 weeks.
- **Guselkumab** 100mg SC at week0 & 4, followed by maintenance dose every 8 weeks.
- **Infliximab** 5mg/kg IV infusion over 2hours at week0, 2 & 6 then every 8 weeks thereafter.

- **Ixekizumab**
 - 160mg SC inj (2SC inj of 80mg) at week0 then 80mg SC inj at week2, 4, 6, 8, 10 and 12, followed by maintenance dose of 80mg SC every 4 weeks.
- **Risankizumab**
 - 150mg SC inj (2 SC inj of 75mg) at week0 and 4 then every 12 weeks thereafter.
- **Secukinumab**
 - 300mg SC inj (2 SC inj) of 150mg at week0, 1, 2, 3 and 4, followed by monthly; maintenance dosing starting at week4.
- **Tildrakizumab** 100mg SC at week0 and 4, followed by maintenance dose every 12 weeks.
- **Ustekinumab** 45mg SC at week0 and 4, then every 12 weeks thereafter or patient > 100kg 90mg may be used.

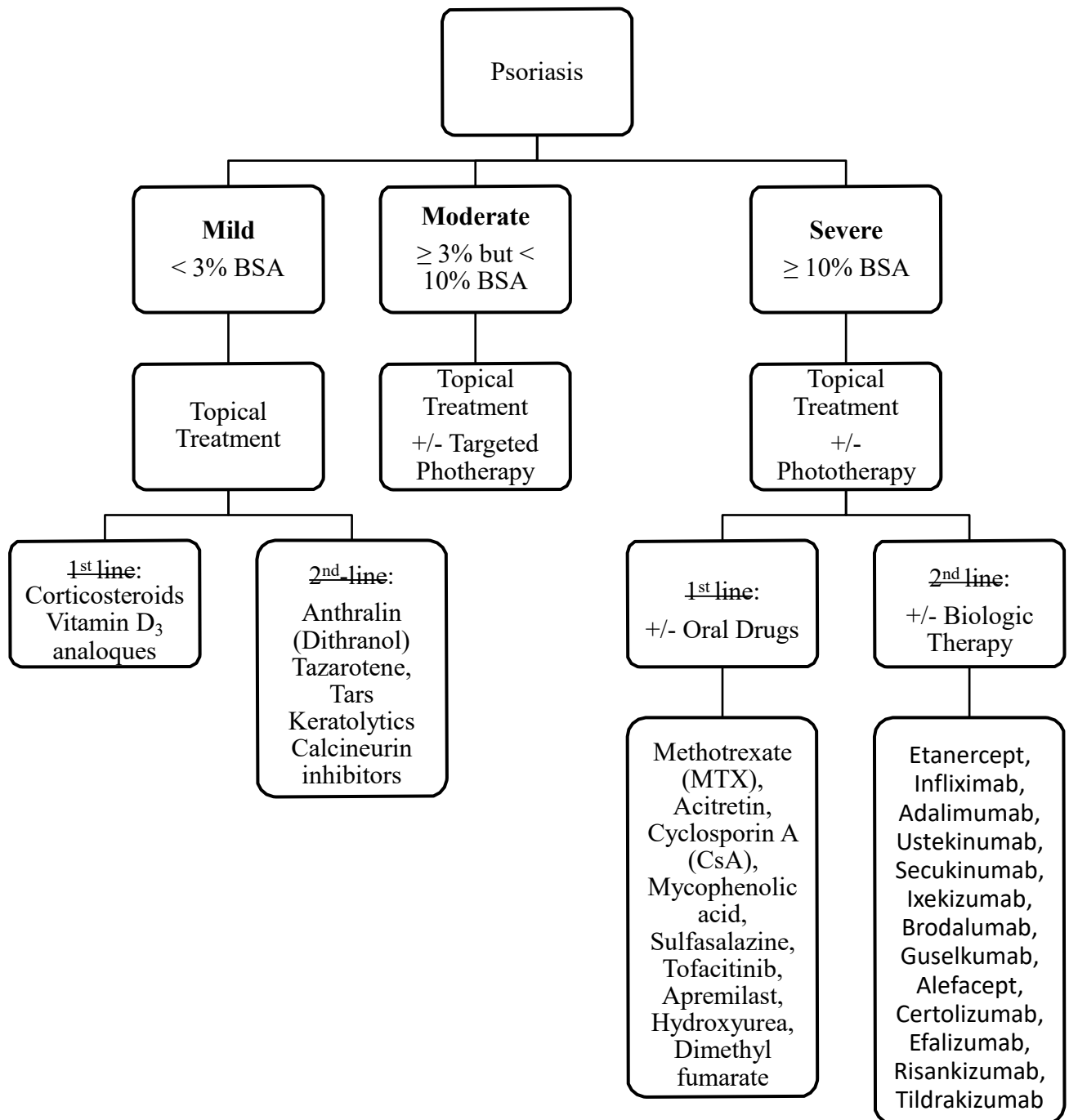
Other non-biologics that are discussed in the guideline for the treatment of psoriasis but are not FDA approved for that indication include: Tofacitinib, Hydroxyurea, Mycophenolate mofetil, Azathioprine, Leflunomide, Tacrolimus, and Thioguanine.

Fumaric acid esters are approved in the United States for treatment of relapsing forms of multiple sclerosis and can be considered for psoriasis.

6. Triggering Factors:

Triggering factors, both external and systemic, can elicit psoriasis in genetically predisposed individuals. External Factors: Cutaneous injury, sunburn; etc. Systemic Factors: Infections, HIV, endocrine factors, psychogenic stress, drugs, alcohol, smoking and obesity; etc.

6. Algorithm



Remark: BSA = Body Surface Area

Not all products are available or approved for above use in all countries.

VIII. SPECIAL CASES

- a. Generalized Pustular Psoriasis of Pregnancy (Impetigo herpetiformis): Flexural erythema, studded with pustules, often occurs initially, followed by a generalized pustular flare and increasing toxicity.
Psoriasis improves in 55%, worsen in 23% and remains static in 21% of patient during pregnancy. Psoriasis frequently flares in the immediate post-partum period.
- b. Associations between psoriasis and other diseases:
 - Atopic Dermatitis, Asthma, Urticaria, Contact Dermatitis, Lichen Simplex, Seborrheic Dermatitis or Sebopsoriasis
 - Infections:
 - Seldomly secondarily infected by bacteria
 - Mycotic infections are more complex; in flexural psoriasis, concomitant Candida infections are frequently observed; Candida species, but not dermatophytes, are more commonly found in psoriatic nails than in normal nails.
 - Associations with internal diseases.

IX. EVOLUTION AND PROGNOSIS

It is impossible to say, in any particular case, how long the disease will last, whether a relapse will occur, or for what period of time the patient will remain free from psoriasis.

Guttate psoriasis is often a self-limited disease, lasting from 12 to 16 weeks without treatment. It has been estimated that one-third to two-thirds of these patients later develop the chronic plaque type of psoriasis. In contrast, chronic plaque psoriasis is in most cases a lifelong disease, manifesting at unpredictable intervals. Spontaneous remissions, lasting for variable periods of time, may occur in the course of psoriasis in up to 50% of patients. The duration of remission ranges from 1 year to several decades. Erythrodermic and generalized pustular psoriasis have a poorer prognosis, with the disease tending to be severe and persistent. Small round to oval papules and plaques appears suddenly on the torso and extremities and have a tendency to resolve over a 6- to 12-month period.

Early onset and a family history of the disease appear to worsen the prognosis.

X. COMPLICATIONS

There are uncommon: Infection, Itching, Arthritis, Alcoholism, Nephritis and renal failure, Hepatic failure, Apical pulmonary fibrosis and Amyloidosis. Up to 30% of patients with psoriasis will develop psoriatic arthritis. Arthritis is more common in patients with widespread cutaneous involvement.

Erythrodermic Psoriasis, patients may have skin tenderness, fever, weight loss, protein loss, and lymphadenopathy.

XI. PREVENTION AND RECOMMENDATION

There is no known prevention form psoriasis.

Advise patients to follow up yearly especially for the first 10 years from initial diagnosis.

Special caution needs to be exercised when treating women of child-bearing potential and during pregnancy.

Patient education: lifestyle modifications such as weight loss, lipid profile reduction, and tight control of glucose levels reduce the mortality from

cardiovascular events in patients with psoriasis. Patients with psoriasis should be well-informed about their skin condition and its treatment. Recommendations include smoking cessation, safe limits for alcohol consumption and maintaining optimal weight.

Vaccinations:

- May consider administration of influenza, hepatitis A and B, tetanus-diphtheria and pneumococcal vaccines prior to therapy.
- Once immunosuppressive therapy is initiated, live-attenuated vaccines should be avoided.
- Avoid live vaccinations in infants until 6 months of age if mothers have received biologic therapy beyond 16 weeks' gestation.

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ANNEXES: LIST OF NATIONAL PROTOCOLS/NATIONAL GUIDELINES

CENAT NATIONAL GUIDELINES

1. សេចក្តីណែនាំបច្ចេកទេសស្តីពីការប្រយុទ្ធនឹងជំងឺរបេង បោះពុម្ពលើកទី២ កម្មវិធីជាតិកំចាត់រោគ
របេង ខែកញ្ញា ឆ្នាំ២០១៦ (Technical guideline on fight TB, 2nd edition, CENAT, September,
2016)
2. ឯកសារកម្មវិធីបណ្តុះបណ្តាលបំប៉នស្តីពីការស្រាវជ្រាវនិងព្យាបាលជំងឺរបេងសម្រាប់មណ្ឌលសុខភាព
និងមន្ទីរពេទ្យបង្អែក បោះពុម្ពលើកទី២ ដោយមជ្ឈមណ្ឌលជាតិកំចាត់រោគរបេងនិងហង់សិន ខែ
តុលា២០១៦ (Refresher Training Manual on screening and treatment of TB for health
centers and referral hospitals, 2nd edition, CENAT, October 2016)
3. ឯកសារបណ្តុះបណ្តាលជំងឺរបេងកុមារសម្រាប់មន្ទីរពេទ្យបង្អែក ដោយមជ្ឈមណ្ឌលជាតិកំចាត់រោគ
របេងនិងហង់សិន ខែកក្កដា ឆ្នាំ២០១៩ (Training manual on TB in children for referral
hospitals, CENAT, July, 2019)
4. និយាមបែបបទសម្រាប់អនុវត្តកម្មវិធីរបេងស្បាំថ្នាំ ដោយមជ្ឈមណ្ឌលជាតិកំចាត់រោគរបេងនិងហង់
សិន ខែ សីហា ឆ្នាំ ២០២១ (Standards for TB drug resistance, CENAT, August, 2021)
5. រូបមន្តស្តង់ដារព្យាបាលជំងឺរបេង ដោយមជ្ឈមណ្ឌលជាតិកំចាត់រោគរបេងនិងហង់សិន សីហា
២០២២ (Standard TB treatment guidelines, CENAT, August, 2022)
6. គោលការណ៍ណែនាំនិងពិធីសារស្រាវជ្រាវជំងឺរបេង ខែកញ្ញាឆ្នាំ២០២៣ (National protocol and
guideline for TB screening, CENAT, September, 2023)
7. គោលការណ៍ណែនាំព្យាបាលជំងឺរបេងកបថ្នាំ សម្រាប់មនុស្សពេញវ័យនិងក្មេងជំទង់នៅប្រទេសកម្ពុ
ជា ខែ ធ្នូ ឆ្នាំ ២០២៤ (Drug susceptible tuberculosis treatment guidelines for adolescents
and adults in Cambodia December 2024).
8. ឯកសារណែនាំស្តីពីជំងឺឃ្លង់ ឬ ជំងឺហង់សិនក្នុងសេវាសុខភាពសាធារណៈ

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(National guideline for clinical management of dengue, CNM, 2018)
10. គោលការណ៍គោលការណ៍ណែនាំព្យាបាលជំងឺគ្រុនចាញ់ ក្រសួងសុខាភិបាល (National guideline
for clinical management of malaria, CNM)
11. សៀវភៅមគ្គុទេសក៍ព្យាបាលជំងឺគ្រុនឈឺក ក្រសួងសុខាភិបាល ២០២០ National guideline for
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12. Summarized guideline for deworming, CNM, September, 2014

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National Consolidated Guidelines on HIV Testing Services in Cambodia, 2024
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15. និយាមបែបបទសម្រាប់ការអនុវត្ត (SOP) វិធីលេបថ្នាំបង្ការជាមុន នៅពេលប្រព្រឹត្តប្រឈមនឹងការឆ្លងមេរោគអេដស៍ (PreP) នៅប្រទេសកម្ពុជា, ខែមករា ឆ្នាំ២០២២. Standard Operating Procedure (SOP) For HIV Pre-Exposure Prophylaxis (PrEP) Implementation in Cambodia, January 2022 https://www.nchads.gov.kh/wp-content/uploads/2022/01/Finalizedwithsigned-SOP_on_PrEP_implementation_in_Cambodia_KH-1.pdf
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18. និយាមបែបបទសម្រាប់អនុវត្តយុទ្ធវិធីជំរុញខ្លាំងក្លាការបង្ការ ថែទាំ និងព្យាបាលបន្តដល់ក្រុមដែលប្រឈមមុខខ្ពស់នឹងការឆ្លងមេរោគអេដស៍ នៅព្រះរាជាណាចក្រកម្ពុជា
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24. សេចក្តីកែតម្រូវគោលការណ៍ណែនាំជាតិស្តីពីការគ្រប់គ្រងគ្លីនិក និងព្យាបាលជំងឺអេដស៍ សម្រាប់មនុស្សពេញវ័យ និងក្មេងជំទង់ ទំព័រទី១០៩ ដល់ ១១៧ ត្រង់ជំពូកទី១២ ចំនុច ១២.៣ អំពីរូបមន្តព្យាបាលដោយឌីសថប្រឆាំងមេរោគអេដស៍ជួរទី២, ១៣ ខែឧសភា ឆ្នាំ២០២២
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https://www.nchads.gov.kh/wp-content/uploads/2022/12/SOP-POC_Final-Khmer-version_singed.pdf
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https://www.nchads.gov.kh/wp-content/uploads/2022/10/SOP_for_Key_Populations_final_041022_kh.pdf
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NATIONAL IMMUNIZATION PROGRAM (NIP) NATIONAL GUIDELINES

33. គោលការណ៍ណែនាំស្តីពីការតាមដានស្រាវជ្រាវជំងឺដែលអាចការពារបានដោយវ៉ាក់សាំង កម្មវិធីជាតិផ្តល់ថ្នាំបង្ការ ពិនិត្យនិងកែសម្រួលលើកទី៣ឆ្នាំ២០២៤ (National guideline for Vaccine Preventable Disease Surveillance National Immunization Program Third edition, June 2024)

NATIONAL MATERNAL AND CHILD HEALTH CENTER (NMCHC) NATIONAL GUIDELINES

34. National Safe Motherhood Clinical Management protocols for Referral Hospitals, 3rd edition, June, NMCHC, 2020
https://nmchc.gov.kh/wp-content/uploads/2022/02/Safe-Motherhood-Protocol_June-2020-20210125-Final.pdf
35. National Protocol Comprehensive Abortion Care, NMCHC, December, 2010
<https://nmchc.gov.kh/wp-content/uploads/2021/07/National-Protocol-Comprehensive-Abortion-Care-2010-EN.pdf>

សមាសភាពក្រុមការងារបច្ចេកទេសមគ្គុទេសក៍ព្យាបាលគ្លីនិកវេជ្ជសាស្ត្រទូទៅ

Technical Working Group of Clinical Practice Guidelines for General Medicines

១	ឯកឧត្តមសាស្ត្រ	កុយ	ចាន់នី	រដ្ឋលេខាធិការក្រសួងសុខាភិបាល	ប្រធាន
២	ឯកឧត្តមសាស្ត្រ	សុខ	ជូន	អគ្គនាយករងបច្ចេកទេសសុខាភិបាល	អនុប្រធាន
៣	ឯកឧត្តមសាស្ត្រ	ឃួន	ពិជិត	ទីប្រឹក្សាក្រសួងសុខាភិបាល	អនុប្រធាន
៤	លោកសាស្ត្រ	នី	ចន្ទី	នាយករងមន្ទីរពេទ្យមិត្តភាពកម្ពុជា-ចិន ព្រះកុសុមៈ	អនុប្រធាន
៥	លោកសាស្ត្រ	ចាន់	មិច	អនុ.ការិ.បច្ចេកទេសមន្ទីរពេទ្យមិត្តភាពខ្មែរ-សូវៀត	អនុប្រធាន

I- ក្រុមការងារឯកទេស សង្គ្រោះបន្ទាន់ (Emergency Specialist working group)

៦	ឯកឧត្តមសាស្ត្រ	ឈុយ	ម៉េង	ទីប្រឹក្សាក្រសួងសុខាភិបាល	សមាជិក
៧	លោកវេជ្ជ.ឯកទេស	ខែត	សុបញ្ញា	នាយករងមន្ទីរពេទ្យជាតិគេជោសន្តិភាព	សមាជិក
៨	លោកសាស្ត្រ	ឈ	ណារ៉េត	មន្ទីរពេទ្យកាល់ម៉ែត	សមាជិក
៩	លោកវេជ្ជ.ឯកទេស	តាំង	សុវណ្ណា	មន្ទីរពេទ្យកាល់ម៉ែត	សមាជិក
១០	លោកវេជ្ជ.	ហ៊ុន	វណ្ណាសុឡា	មន្ទីរពេទ្យកាល់ម៉ែត	សមាជិក
១១	លោកវេជ្ជ.	តី	សុភ័ក្ត្រ	មន្ទីរពេទ្យកាល់ម៉ែត	សមាជិក
១២	លោកស្រីវេជ្ជ.	ស៊ី	ខេមរី	មន្ទីរពេទ្យកាល់ម៉ែត	សមាជិក
១៣	លោកវេជ្ជ.ឯកទេស	លី	ឈុន	មន្ទីរពេទ្យមិត្តភាពកម្ពុជា-ចិន ព្រះកុសុមៈ	សមាជិក
១៤	លោកវេជ្ជ.ឯកទេស	រ៉ា	វណ្ណៈ	មន្ទីរពេទ្យមិត្តភាពកម្ពុជា-ចិន ព្រះកុសុមៈ	សមាជិក
១៥	លោកវេជ្ជ.ឯកទេស	ធឿ	សុវណ្ណរដ្ឋា	មន្ទីរពេទ្យមិត្តភាពខ្មែរ-សូវៀត	សមាជិក
១៦	លោកវេជ្ជ.ឯកទេស	ទេស	សុវណ្ណារ៉ា	មន្ទីរពេទ្យមិត្តភាពខ្មែរ-សូវៀត	សមាជិក
១៧	លោកវេជ្ជបណ្ឌិត	អិម	សុធី	ប្រធានផ្នែកដាក់ថ្នាំសន្លប់នៃមន្ទីរពេទ្យគន្ធបុប្ផា	សមាជិក
១៨	លោកវេជ្ជបណ្ឌិត	ឆាង	តក្កី	មន្ទីរពេទ្យគន្ធបុប្ផា(ភ្នំពេញ)	សមាជិក
១៩	លោកវេជ្ជបណ្ឌិត	អ៊ឹម	វិរ័យ៉ា	មន្ទីរពេទ្យគន្ធបុប្ផា(ភ្នំពេញ)	សមាជិក
២០	លោកវេជ្ជបណ្ឌិត	ម៉ែន	ចលគុន	ប្រធានមន្ទីរពេទ្យបង្អែកខេត្តសៀមរាប	សមាជិក

II- ក្រុមការងារឯកទេស ជំងឺបេះដូង (Cardiologist working group)

២១	ឯកឧត្តមសាស្ត្រ.	ឈុន	ឡាយ	អគ្គនាយករងមន្ទីរពេទ្យកាល់ម៉ែត្រ	សមាជិក
២២	លោកវេជ្ជ.ឯកទេស	ច័ន្ទ	សុខា	មន្ទីរពេទ្យកាល់ម៉ែត	សមាជិក
២៣	លោកវេជ្ជ.ឯកទេស	ថេង	យុជានីន	មន្ទីរពេទ្យកាល់ម៉ែត	សមាជិក

២៤	លោកវេជ្ជ.ឯកទេស	សាន	សំរ៉ានី	មន្ទីរពេទ្យកាល់ម៉ែត	សមាជិក
២៥	លោកវេជ្ជបណ្ឌិត	លីម	ស៊ីវុតថា	មន្ទីរពេទ្យមិត្តភាពខ្មែរ-សូវៀត	សមាជិក
២៦	លោកវេជ្ជ.ឯកទេស	ស៊ុយ	សុវណ្ណារ៉ា	មន្ទីរពេទ្យមិត្តភាពខ្មែរ-សូវៀត	សមាជិក
២៧	លោកវេជ្ជ.ឯកទេស	សាំង	លាង	មន្ទីរពេទ្យមិត្តភាពកម្ពុជា-ចិន ព្រះកុសុមៈ	សមាជិក
២៨	លោកវេជ្ជ.ឯកទេស	លីម	វិហារស្រីន	មន្ទីរពេទ្យមិត្តភាពកម្ពុជា-ចិន ព្រះកុសុមៈ	សមាជិក
២៩	លោកវេជ្ជបណ្ឌិត	គីរី	សំណាង	មន្ទីរពេទ្យបង្អែកខេត្តសៀមរាប	សមាជិក
៣០	លោកវេជ្ជបណ្ឌិត	ភុយ	សុណារ៉ុក្ខ	មន្ទីរពេទ្យបង្អែកខេត្តសៀមរាប	សមាជិក

III- ក្រុមការងារឯកទេស ជំងឺសួត (Pneumologist working group)

៣១	លោកសាស្ត្រ	នី	ចន្ទី	អគ្គនាយករងមន្ទីរពេទ្យមិត្តភាពកម្ពុជា-ចិន ព្រះកុសុមៈ	សមាជិក
៣២	លោកសាស្ត្រ	ចាន់	មិច	អនុ.ការិ.បច្ចេកទេសមន្ទីរពេទ្យមិត្តភាពខ្មែរ-សូវៀត	សមាជិក
៣៣	លោកវេជ្ជ.ឯកទេស	តាន់	ម៉ុនលី	អនុ.ការិ.បច្ចេកទេសមន្ទីរពេទ្យមិត្តភាពកម្ពុជា-ចិន ព្រះកុសុមៈ	សមាជិក
៣៤	លោកសាស្ត្រ	សាំង	សាលី	អនុ.ការិ.សិក្សាបណ្តុះបណ្តាល ស.វ.ស.	សមាជិក
៣៥	លោកសាស្ត្រ	ចាន់	សារិន	មន្ទីរពេទ្យកាល់ម៉ែត	សមាជិក
៣៦	ឯកឧត្តមវេជ្ជ.ឯកទេស	ន	ម៉ុនម៉ូល	មន្ទីរពេទ្យកាល់ម៉ែត	សមាជិក
៣៧	កញ្ញាវេជ្ជ.ឯកទេស	នន	សុឆារី	មន្ទីរពេទ្យតេជោសន្តិភាព	សមាជិក

IV- ក្រុមការងារឯកទេស ជំងឺឆ្អឹង-ក្រពះ-ពោះទៀន (Hepato-Gastro-Enterologost working group)

៣៨	លោកវេជ្ជ.ឯកទេស	ស៊ី	ស៊ីវណ្ណារ៉ា	នាយកមន្ទីរពេទ្យហ្លួងម៉ែ	សមាជិក
៣៩	លោកវេជ្ជ.ឯកទេស	អ៊ុក	សារ៉េម	នាយករងមន្ទីរពេទ្យមិត្តភាពកម្ពុជា-ចិន ព្រះកុសុមៈ	សមាជិក
៤០	លោកវេជ្ជ.ឯកទេស	ជ័យ	វិឡារិទ្ធី	នាយករងមន្ទីរពេទ្យមិត្តភាពខ្មែរ-សូវៀត	សមាជិក
៤១	លោកវេជ្ជ.ឯកទេស	គី	វុត្តា	មន្ទីរពេទ្យកាល់ម៉ែត	សមាជិក
៤២	លោកវេជ្ជ.ឯកទេស	ម៉ក់	សុភក្តិ	មន្ទីរពេទ្យកាល់ម៉ែត	សមាជិក
៤៣	លោកវេជ្ជ.ឯកទេស	សាន	ចាន់ណា	មន្ទីរពេទ្យកាល់ម៉ែត	សមាជិក
៤៤	លោកសាស្ត្របារ្យ	ចាន់	សុវណ្ណជេត	អនុប្រធានមន្ទីរពេទ្យបង្អែកខេត្តកំពង់ចាម	សមាជិក
៤៥	លោកវេជ្ជ.ឯកទេស	អ៊ុន	កែវសិលា	មន្ទីរពេទ្យមិត្តភាពខ្មែរ-សូវៀត	សមាជិក
៤៦	លោកវេជ្ជ.ឯកទេស	នាយ	គឹមប៉ាវ	មន្ទីរពេទ្យមិត្តភាពខ្មែរ-សូវៀត	សមាជិក
៤៧	លោកវេជ្ជ.ឯកទេស	ថា	អ៊ុន	មន្ទីរពេទ្យមិត្តភាពកម្ពុជា-ចិន ព្រះកុសុមៈ	សមាជិក
៤៨	លោកវេជ្ជ.ឯកទេស	តូច	សុរិណា	មន្ទីរពេទ្យមិត្តភាពកម្ពុជា-ចិន ព្រះកុសុមៈ	សមាជិក
៤៩	លោកស្រីវេជ្ជ.ឯកទេស	ចង	ច័ន្ទសិណា	មន្ទីរពេទ្យកាល់ម៉ែត	សមាជិក

V- ក្រុមការងារឯកទេស ជំងឺតម្រងនោម (Nephroloist working gruop)

៥០	លោកវេជ្ជ.ឯកទេស	លឹម	វឌ្ឍនា	មន្ទីរពេទ្យកាល់ម៉ែត	សមាជិក
៥១	លោកវេជ្ជ.ឯកទេស	នីត	រត្នាវីរៈ	មន្ទីរពេទ្យកាល់ម៉ែត	សមាជិក
៥២	លោកវេជ្ជ.ឯកទេស	ស៊ីន	បូរណ៍	មន្ទីរពេទ្យកាល់ម៉ែត	សមាជិក
៥៣	លោកវេជ្ជ.ឯកទេស	លឹម	សុជន	មន្ទីរពេទ្យកាល់ម៉ែត	សមាជិក
៥៤	លោកវេជ្ជ.ឯកទេស	ហ៊ុ	ចន្ទសិលា	មន្ទីរពេទ្យកាល់ម៉ែត	សមាជិក
៥៥	លោកវេជ្ជ.ឯកទេស	វិទូ	សុវណ្ណារ៉ា	មន្ទីរពេទ្យមិត្តភាពខ្មែរ-សូវៀត	សមាជិក
៥៦	លោកស្រីវេជ្ជ.	ធីម	ពេជ្រធីតា	មន្ទីរពេទ្យកុមារជាតិ	សមាជិក
៥៧	លោកវេជ្ជ.ឯកទេស	កើត	ច័ន្ទសុផល	មន្ទីរពេទ្យមិត្តភាពកម្ពុជា-ចិន ព្រះកុសុមៈ	សមាជិក

VI- ក្រុមការងារឯកទេស ជំងឺសន្លាក់ឆ្អឹង (Rheumatologist working group)

៥៨	លោកវេជ្ជបណ្ឌិត	ឃុន	វិទ្យា	មន្ទីរពេទ្យព្រះសីហនុ មណ្ឌលនៃក្តីសង្ឃឹម	សមាជិក
៥៩	លោកវេជ្ជបណ្ឌិត	សែ	សិរីវុឌ្ឍ	មន្ទីរពេទ្យកាល់ម៉ែត	សមាជិក
៦០	លោកវេជ្ជបណ្ឌិត	កៅ	ផេងអេន	មន្ទីរពេទ្យកាល់ម៉ែត	សមាជិក
៦១	លោកស្រីវេជ្ជបណ្ឌិត	ហេង	ស្នី	មន្ទីរពេទ្យកាល់ម៉ែត	សមាជិក

VII- ក្រុមការងារឯកទេស ជំងឺសរសៃប្រសាទ (Neurologist working group)

៦២	លោកសាស្ត្រ	ជុំ	ណាតុប	មន្ទីរពេទ្យមិត្តភាពខ្មែរ-សូវៀត	សមាជិក
៦៣	លោកស្រីសាស្ត្រ	រស់	ស៊ីណា	មន្ទីរពេទ្យកាល់ម៉ែត	សមាជិក
៦៤	លោកវេជ្ជ.	ចាន់	វិជ្ជា	មន្ទីរពេទ្យកាល់ម៉ែត	សមាជិក
៦៥	លោកវេជ្ជ.ឯកទេស	ឈូ	ចាន់ណារ៉ា	មន្ទីរពេទ្យមិត្តភាពខ្មែរ-សូវៀត	សមាជិក
៦៦	លោកស្រីវេជ្ជ.ឯកទេស	សន	ផលនិកា	មន្ទីរពេទ្យមិត្តភាពខ្មែរ-សូវៀត	សមាជិក
៦៧	លោកវេជ្ជ.ឯកទេស	ឆាវ	ចាសនា	មន្ទីរពេទ្យមិត្តភាពខ្មែរ-សូវៀត	សមាជិក
៦៨	លោកវេជ្ជបណ្ឌិត	ម៉ែន	ពុទ្ធិ	មន្ទីរពេទ្យមិត្តភាពខ្មែរ-សូវៀត	សមាជិក
៦៩	លោកវេជ្ជបណ្ឌិត	អេម	ណាតុប	មន្ទីរពេទ្យតេជោសន្តិភាព	សមាជិក
៧០	លោកស្រីវេជ្ជបណ្ឌិត	អ៊ុន	ច័ន្ទស្នី	មន្ទីរពេទ្យកាល់ម៉ែត	សមាជិក
៧១	លោកវេជ្ជបណ្ឌិត	ស៊ីម	សុផាណា	មន្ទីរពេទ្យកាល់ម៉ែត	សមាជិក
៧២	លោកវេជ្ជបណ្ឌិត	អ៊ុន	បុត្តា	មន្ទីរពេទ្យកាល់ម៉ែត	សមាជិក
៧៣	លោកវេជ្ជបណ្ឌិត	កែវ	ចាសនា	មន្ទីរពេទ្យមិត្តភាពខ្មែរ-សូវៀត	សមាជិក

VIII- ក្រុមការងារឯកទេស ជំងឺក្រពេញ (Endocrinologis woring group)

៧៤	លោកសាស្ត្រ	ទូច	យុន	នាយកមន្ទីរពេទ្យមិត្តភាពកម្ពុជា-ចិន ព្រះកុសុមៈ	សមាជិក
៧៥	លោកវេជ្ជ.ឯកទេស	ចិន	សុភា	មន្ទីរពេទ្យមិត្តភាពខ្មែរ-សូវៀត	សមាជិក
៧៦	លោកវេជ្ជ.ឯកទេស	ស៊ី	សត្វា	មន្ទីរពេទ្យកាល់ម៉ែត	សមាជិក
៧៧	លោកវេជ្ជ.ឯកទេស	អែម	សាឡើង	មន្ទីរពេទ្យមិត្តភាពខ្មែរ-សូវៀត	សមាជិក
៧៨	លោកវេជ្ជ.ឯកទេស	សោ	ហុកបាង	មន្ទីរពេទ្យកាល់ម៉ែត	សមាជិក
៧៩	លោកវេជ្ជ.ឯកទេស	យ៉ាត	ស៊ីឡុង	មន្ទីរពេទ្យកាល់ម៉ែត	សមាជិក
៨០	លោកវេជ្ជ.ឯកទេស	សៀង	ឡុងឌី	មន្ទីរពេទ្យកាល់ម៉ែត	សមាជិក
៨១	លោកស្រីវេជ្ជ.ឯកទេស	សរ	សុផារី	មន្ទីរពេទ្យមិត្តភាពកម្ពុជា-ចិន ព្រះកុសុមៈ	សមាជិក
៨២	លោកវេជ្ជ.ឯកទេស	ជា	វិឌីម	មន្ទីរពេទ្យមិត្តភាពកម្ពុជា-ចិន ព្រះកុសុមៈ	សមាជិក
៨៣	លោកវេជ្ជ.ឯកទេស	ចន្ទា	គុណ	មន្ទីរពេទ្យមិត្តភាពកម្ពុជា-ចិន ព្រះកុសុមៈ	សមាជិក
៨៤	លោកវេជ្ជ.ឯកទេស	ធី	ហេង	មន្ទីរពេទ្យមិត្តភាពខ្មែរ-សូវៀត	សមាជិក
៨៥	លោកវេជ្ជ.ឯកទេស	ថេង	ហៀក	មន្ទីរពេទ្យមិត្តភាពខ្មែរ-សូវៀត	សមាជិក

IX- ក្រុមការងាររងករណ៍ ជំងឺ (Infections Disease specialist working group)

៨៦	លោកវេជ្ជ.ឯកទេស	ហូរី	សុផារិន្ទ	មន្ទីរពេទ្យកាល់ម៉ែត	សមាជិក
៨៧	លោកស្រីវេជ្ជ.	កុយ	សុផានី	មន្ទីរពេទ្យកាល់ម៉ែត	សមាជិក
៨៨	លោកស្រីវេជ្ជ.	ចន្ទា	ពិសី	មន្ទីរពេទ្យកាល់ម៉ែត	សមាជិក
៨៩	លោកវេជ្ជបណ្ឌិត	ផេ	ថុន	សមាគមវេជ្ជសាស្ត្រកម្ពុជា	សមាជិក
៩០	លោកស្រីវេជ្ជ.	លីម	ក្រុយ	មន្ទីរពេទ្យព្រះសីហនុ មណ្ឌលនៃក្តីសង្ឃឹម	សមាជិក
៩១	លោកវេជ្ជបណ្ឌិត	ទូន	គឹមចេង	មន្ទីរពេទ្យព្រះសីហនុ មណ្ឌលនៃក្តីសង្ឃឹម	សមាជិក
៩២	លោកស្រីវេជ្ជ.	ជា	សិរីវឌ្ឍនា	មន្ទីរនាយកដ្ឋានមន្ទីរពេទ្យ	សមាជិក

X- ក្រុមការងាររងករណ៍ ជំងឺឈាម និងជំងឺមហារីក (Hematologist and Oncologists working group)

៩៣	លោកវេជ្ជ.ឯកទេស	ទូច	សុជាតិ	នាយកដ្ឋានមន្ទីរពេទ្យហ្លួងម៉ែ	សមាជិក
៩៤	លោកស្រីវេជ្ជ.ឯកទេស	ត្រៃ	ហុតា	មន្ទីរពេទ្យកាល់ម៉ែត	សមាជិក
៩៥	លោកសាស្ត្រ	គុយ	សំណាង	មន្ទីរពេទ្យហ្លួងម៉ែ	សមាជិក
៩៦	លោកវេជ្ជ.ឯកទេស	ហេង	វិរដ្ឋ	មន្ទីរពេទ្យមិត្តភាពខ្មែរ-សូវៀត	សមាជិក
៩៧	លោកវេជ្ជ.ឯកទេស	កៅ	ចន្ទារិន្ទ	មន្ទីរពេទ្យកាល់ម៉ែត	សមាជិក
៩៨	លោកវេជ្ជ.ឯកទេស	អ៊ុំ	កក្កដា	មន្ទីរពេទ្យកាល់ម៉ែត	សមាជិក
៩៩	លោកវេជ្ជ.ឯកទេស	ទូច	ខាន់ណារ៉ា	មន្ទីរពេទ្យកាល់ម៉ែត	សមាជិក

១០០	លោកវេជ្ជ.ឯកទេស	វិទ្យា	សុភារិន	មន្ទីរពេទ្យមិត្តភាពខ្មែរ-សូវៀត	សមាជិក
១០១	លោកស្រីវេជ្ជ.ឯកទេស	ជិន	រៀបរា	មន្ទីរពេទ្យមិត្តភាពខ្មែរ-សូវៀត	សមាជិក

XI- ក្រុមការងារឯកទេស ជំងឺស្បែក (Dermatologist working group)

១០២	លោកសាស្ត្របារ្យ	ចាន់	វិចិត្រ	នាយកដ្ឋានមន្ទីរពេទ្យមិត្តភាពខ្មែរ-សូវៀត	សមាជិក
១០៣	លោកសាស្ត្របារ្យ	ឌី	ស៊ីថាប	នាយកដ្ឋានមន្ទីរពេទ្យមិត្តភាពកម្ពុជា-ចិន ព្រះកុសុមៈ	សមាជិក
១០៤	លោកវេជ្ជបណ្ឌិត	ឡុក	ប៊ុនថៃ	អនុ.កា.វិ.បច្ចេកទេសមន្ទីរពេទ្យមិត្តភាពខ្មែរ-សូវៀត	សមាជិក
១០៥	លោកវេជ្ជ.ឯកទេស	ប៊ិន	ហុរិ	មន្ទីរពេទ្យមិត្តភាពកម្ពុជា-ចិន ព្រះកុសុមៈ	សមាជិក
១០៦	លោកវេជ្ជ.ឯកទេស	ហុកម៉ុល	សិរីវិសា	មន្ទីរពេទ្យមិត្តភាពកម្ពុជា-ចិន ព្រះកុសុមៈ	សមាជិក
១០៧	លោកវេជ្ជបណ្ឌិត	ប៉ុក	សាឡើង	មជ្ឈមណ្ឌលជាតិប្រយុទ្ធនឹងជំងឺអេដស៍ ស្បែក និងកាមរោគ	សមាជិក
១០៨	លោកវេជ្ជបណ្ឌិត	ឡា	ភឿន	មន្ទីរពេទ្យអង្គឌួង	សមាជិក
១០៩	លោកវេជ្ជ.ឯកទេស	ថៃ	តេនៈ	មន្ទីរពេទ្យកាល់ម៉ែត	

XII- ក្រុមការងារឯកទេស ជំងឺផ្លូវចិត្ត (Psychiatrist working group)

១១០	លោកវេជ្ជបណ្ឌិត	ឈិត	សុផល	ប្រធាននាយកដ្ឋានសុខភាពផ្លូវចិត្ត និងការបំពានគ្រឿងញៀន	សមាជិក
១១១	លោកសាស្ត្រ	យីម	សុបុត្រា	មន្ទីរពេទ្យមិត្តភាពខ្មែរ-សូវៀត	សមាជិក
១១២	លោកសាស្ត្រ	ពៅ	ប៊ុនឡើង	មន្ទីរពេទ្យមិត្តភាពខ្មែរ-សូវៀត	សមាជិក
១១៣	លោកសាស្ត្រ	ស	សុធាវិទ្យា	មន្ទីរពេទ្យមិត្តភាពខ្មែរ-សូវៀត	សមាជិក
១១៤	លោកស្រីសាស្ត្រ	ចក	វិសា	សមាគមសុខភាពផ្លូវចិត្តកម្ពុជា	សមាជិក
១១៥	លោកស្រីវេជ្ជ.ឯកទេស	ការ	មីការហៀរ	មន្ទីរពេទ្យមិត្តភាពកម្ពុជា-ចិន ព្រះកុសុមៈ	សមាជិក
១១៦	លោកស្រីវេជ្ជ.ឯកទេស	ថា	ជានី	មន្ទីរពេទ្យមិត្តភាពកម្ពុជា-ចិន ព្រះកុសុមៈ	សមាជិក
១១៧	លោកវេជ្ជ.ឯកទេស	គីម	សុភា	មន្ទីរពេទ្យមិត្តភាពកម្ពុជា-ចិន ព្រះកុសុមៈ	សមាជិក
១១៨	លោកវេជ្ជ.ឯកទេស	ធីយ	ចាន់ថុល	អនុប្រធានមន្ទីរពេទ្យបង្អែកខេត្តសៀមរាប	សមាជិក
១១៩	លោកវេជ្ជ.ឯកទេស	ធី	ញីល	អនុប្រធានមន្ទីរពេទ្យបង្អែកខេត្តបាត់ដំបង	សមាជិក
១២០	អង្គការដៃគូអភិវឌ្ឍ			HHO, CHAI, Caritas-CCAMH, ACCESS, KHANA ect...	សមាជិក